THE FIELD-STRENGTH OSTERWALDER–SCHRADER AXIOMS FOR QED IN THREE DIMENSIONS

Abstract. We construct non-Gaussian laws of closed two-form fields from noncompact lattice QED with chamber fermions. Their Schwinger functions satisfy the Osterwalder–Schrader axioms, including clustering.

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1. Introduction

The observable in this paper is the field strength. We regard it as a real tempered two-form on $\mathbb{R}^3$, and write $F(f)$ for its pairing with a test two-form f. The regularization uses a noncompact

gauge field and four-component chamber fermions, with all flux sectors included. The order of limits is part of the construction: the volume tends to infinity at a fixed lattice spacing, and the spacing then tends to zero.

Theorem 1.1 (Field-strength OS theorem)**.** *Fix $0 < \rho < 1$ and $0 < \bar{m} \le 1$, the chamber Wilson parameter and reference mass parameter in physical units. Use the standard chamber weights $(6, 6\sqrt{2}, 6\sqrt{2}, 6)$. There are an odd scale factor $L \ge 5$, numbers $e_{\mathrm{phys}}, \sigma > 0$, and real volume-independent bare mass shifts $m_{0,N}(e)$ for $N \in \mathbb{Z}_{\ge 0}$ and $0 < e \le e_{\mathrm{phys}}$, with $|m_{0,N}(e)| < 1/8$, with the following property.*

Fix a real $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ such that $f = dd^*\beta$ has

$$b(f) := -6 \operatorname{tr}\big[(S_{\bar{m}}\, i\gamma \cdot d^*\beta)^4\big] > 0, \qquad S_{\bar{m}} = (\bar{m} + \gamma \cdot \partial)^{-1}. \tag{1.1}$$

Such probes exist. There is $e_\beta \in (0, e_{\mathrm{phys}}]$ such that, for $0 < e \le e_\beta$, the all-flux chamber laws on the coordinate cubic tori of unit side $n = R/\eta$, with periodic gauge fields and fermions antiperiodic in all three coordinate directions, and with

$$\eta = L^{-N}, \quad R = 8hL^T, \quad h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \quad q = e\eta^{1/2}, \quad \boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e) \tag{1.2}$$

with $N, T \in \mathbb{Z}_{\ge 0}$ have volume-first limit points. Let μ be any such limit point and $\mu = \int \mu_\xi \, \mathrm{d}\nu(\xi)$ its decomposition into translation-ergodic laws. For ν-almost every ξ, μ_ξ is carried by closed tempered two-forms and satisfies

$$\int e^{F(g)} \, \mathrm{d}\mu_\xi(F) \le e^{\|g\|_{L^2}^2/2} \qquad (g \in \mathcal{S}(\mathbb{R}^3; \Lambda^2) \text{ real}). \tag{1.3}$$

Its Schwinger functions $S_n^\xi(g_1 \otimes \cdots \otimes g_n) = \int \prod_{a=1}^n F(g_a) \, \mathrm{d}\mu_\xi(F)$ extend to the complex tensor Schwartz spaces $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{3n}; (\Lambda^2 \mathbb{C}^3)^{\otimes n})$ and obey the following conditions, with $S_0^\xi = 1$.

(E0′) *For $n \ge 1$ and $\Psi \in \mathcal{S}_n$,*

$$|S_n^\xi(\Psi)| \le 2\, 66^n \sqrt{n!}\, q_{6n}(\Psi), \qquad q_m(\Psi) = \max_{|\alpha| \le m} \sup_y (1 + |y|^2)^{m/2} |\partial^\alpha \Psi(y)|, \tag{1.4}$$

where the maximum is also over tensor components.

(E1) *For every Euclidean motion $g(x) = Qx + a$, $Q \in O(3)$, $S_n^\xi(g \cdot \Psi) = S_n^\xi(\Psi)$, where*

$$(g \cdot \Psi)(x_1, \dots, x_n) = (\Lambda^2 Q)^{\otimes n} \Psi(g^{-1}x_1, \dots, g^{-1}x_n).$$

(E2) *Let r be reflection in a plane through the origin and H either closed half-space bounded by that plane. If $\Psi_0 \in \mathbb{C}$ and $\Psi_k \in \mathcal{S}_k$, $1 \le k \le M$, have $\operatorname{supp} \Psi_k \subset H^k$, then*

$$\sum_{k,l=0}^{M} S_{k+l}^\xi\big(\Theta_k \Psi_k \otimes \Psi_l\big) \ge 0, \qquad \Theta_k \Psi_k = \overline{r \cdot \Psi_k}, \quad \Theta_0 c = \bar{c}. \tag{1.5}$$

(E3) *S_n^ξ is unchanged by simultaneous permutation of the arguments and their tensor indices.*

(E4) *For $\Psi \in \mathcal{S}_k$ and $\Phi \in \mathcal{S}_l$,*

$$S_{k+l}^\xi(\Psi \otimes \tau_a^{\otimes l} \Phi) \longrightarrow S_k^\xi(\Psi) S_l^\xi(\Phi) \quad \text{as } |a| \longrightarrow \infty, \tag{1.6}$$

where

$$(\tau_a^{\otimes l}\Phi)(x_1, \dots, x_l) = \Phi(x_1 - a, \dots, x_l - a).$$

Moreover, for a constant C_f independent of the limit point and of ξ,

$$\big|\kappa_4^{\mu_\xi}(f) - e^4 b(f)\big| \le C_f e^{4+\sigma}, \qquad \kappa_4^{\mu_\xi}(f) \ge \tfrac{1}{2} e^4 b(f) > 0. \tag{1.7}$$

In particular, μ_ξ is non-Gaussian.

Here $\kappa_4^\lambda(f)$ is the fourth cumulant of $F(f)$ under a law λ. For the Gaussian-dominated laws considered here, $F(f)$ has mean zero, so $\kappa_4^\lambda(f) = S_4^\lambda(f^{\otimes 4}) - 3S_2^\lambda(f^{\otimes 2})^2$. The charge threshold may depend on the fixed probe. The parent law μ satisfies (E0′)–(E3). The factorization in (E4) is a conclusion for its ergodic components.

The tensor action in (E1) is essential to (E2). For reflection in $x_1 = 0$, the components F_{12} and F_{13} change sign, while F_{23} does not. Replacing this action by scalar reflection changes the positivity assertion. The growth condition is stated on all tensor tests, including tests meeting the diagonals; bounds on products of separately smeared fields alone do not state (E0′).

1.1. **The two parts of the proof.** There are two principal tasks. The first is to obtain a Euclidean law from the chamber regularization. The second is to show that the law is non-Gaussian. Gaussian domination and reflection positivity supply the compactness and analytic bounds used in the first task. The renormalization analysis controls the fourth cumulant in the second.

The chamber sites lie away from the lattice mirrors. Every bond crossing a mirror is normal to it, and its fermionic crossing coefficients are $w_i(1 \pm \rho)/2$. This gives reflection positivity on tori adapted to the mirror. Coordinate tori are adapted to the axis mirrors, but not to the diagonal mirrors. The comparison of these geometries is therefore a substantive step. The fold inequality bounds a cubic-torus expectation by an adapted-torus expectation plus a vanishing pressure difference divided by the perturbation strength. The volume limit must therefore precede removal of the perturbation. Amplification controls local insertions; a two-layer reduction and elimination then prove reflection and translation invariance of the comparison states. Here the two-layer reduction is obtained by eliminating the outer layers, using the fermion equations of motion and an imaginary shift of the gauge integration contour; the remaining error tends to zero with the perturbation strength. Translation removes the restriction to an observable inside one cell, and convex separation gives diagonal positivity of the original limit. Taking volume before spacing preserves this input in the continuum.

The nine mirror directions do not generate all rotations. Their reflection-positive reconstructions give instead the sharp bound $H_j \geq |\mathsf{P}_i|$, $i \neq j$. This spectral bound and Gaussian domination control correlations in a complex tube after smearing in a transverse coordinate. At two points their continuation in the rotation angle t is entire, with growth at most $C \exp(2|\operatorname{Im} t|)$. Only angular frequencies of modulus at most two can occur, and the Bianchi identity eliminates the noncovariant terms.

At higher orders let $\mathfrak{a}(n)$ measure the largest deviation from the required tensor rotation weight. Reflection positivity gives convexity of this non-negative integer index; positivity on regular polygons gives a divisibility condition. With the linear angular bound, these yield

$$\mathfrak{a}(2k) = k\mathfrak{a}(2), \qquad \mathfrak{a}(n) \leq \tfrac{n}{2}\mathfrak{a}(2).$$

The two-point result gives $\mathfrak{a}(2) = 0$, so every order has the required rotation weight. Covariance is first obtained for separated planar projections; an analytic extension across coincidences of projected coordinates gives it for disjoint supports. Locality and vanishing on gradients then exclude contact terms, allowing passage to general tests. This proves full covariance of the law, and hence reflection positivity at every plane.

For clustering, the sharp cone identifies invariance under one coordinate translation with invariance under all translations. The local correlation estimate then yields factorization in an ergodic law. The required bounds and mirror positivity are first proved for almost every component; the rotation argument is applied to each such component. Rotation invariance of the parent mixture alone would not justify this step.

The non-Gaussianity argument uses only two reference volumes for its renormalization estimates. The chamber operator has a four-component light sector and a gapped heavy sector.

Exponential decay and Hölder bounds for its local inverses control the summability and locality of the polymer activities. Extraction of the vacuum and mass terms leaves an irrelevant activity that contracts under block averaging; the tuned flow gives a mass assignment independent of volume. The reference analysis controls fourth-order response to an external connection. Its transfer to cutoff volumes gives, beyond charge-dependent cutoff thresholds,

$$\big|\kappa_4(X_{\eta,R}(f)) - e^4 b(f)\big| \le C_f\big(e^{4+\sigma} + \eta^2 + e^4\eta^{1/4}\big),$$

uniformly in the volume; $X_{\eta,R}(f)$ is the lattice field smeared with f. The last term includes the chamber-to-continuum error in the free fourth coefficient. A separate continuum calculation produces a positive probe: writing $a = d^*\beta$, its large-mass quartic term is a positive multiple of $\int |\operatorname{curl} a|^4$, and dilation gives positivity at the fixed mass $\bar{m}$. Chessboard estimates also transfer products of translated Wick powers. These product bounds control invariant conditional expectations and are what makes the cumulant estimate hold in almost every component. A fourth-cumulant bound on the parent law by itself would be insufficient.

1.2. **Relation to earlier work.** Theorem 1.1 states the tensor form of the axioms of Osterwalder and Schrader [1, 2]. For reflection-positive lattice gauge theories and their transfer matrices, see Osterwalder and Seiler [3]; the general chessboard framework is developed by Fröhlich, Israel, Lieb, and Simon [4]. The construction-specific steps here are the chamber crossings, the comparison that gives diagonal positivity for the original limiting law, and the angular argument for covariance at every order.

Dimock [5] proves ultraviolet stability at weak coupling in a fixed finite volume using Bałaban's renormalization group. Our block-integration analysis adapts this framework to chamber fermions and the two reference tori used in the field-strength estimates. The volume is then taken to infinity before the spacing is removed; the conclusion concerns field-strength laws and their fourth cumulants.

1.3. **Conventions.** The components of a two-form are indexed by $1 \le i < j \le 3$; their L^2 inner product is the sum of the three component inner products. We complexify test spaces when needed, while the random field is real. The adjoint d^* is taken in this Euclidean inner product. The matrices $\gamma_i = \operatorname{diag}(\sigma_i, -\sigma_i)$ use the Pauli matrices, and satisfy $\gamma_i\gamma_j + \gamma_j\gamma_i = 2\delta_{ij}$.

For a Euclidean motion $g(x) = Qx + a$, its action on a one-argument test is $(g \cdot f)(x) = (\Lambda^2 Q) f(g^{-1}x)$ and on a distribution is $(gF)(f) = F(g^{-1} \cdot f)$. Translations follow this convention: $(\tau_a f)(x) = f(x - a)$. Observable reflection includes complex conjugation. The Hodge dual of a two-form transforms as an axial vector, by $\det(Q)Q$; this convention will also be used in the rotation argument.

1.4. **Scales and notation.** The physical and level coordinates have different unit lengths. The following dictionary collects the conventions used in the renormalization estimates.

Physical cutoff	$\eta = L^{-N}$, $R = 8hL^T$ and unit-lattice side $n = R/\eta$.
Level j	$a_j = L^{j-N}$, $\eta_j = \eta/a_j$ and $\ell_j = R/a_j$; one level unit has physical length a_j.
Charges	e is the physical charge, $q = e\sqrt{\eta}$ the fine unit charge, and $e_j = e\sqrt{a_j}$ the running charge.
Masses	$\mu_j = \bar{m}a_j$ is the positive reference mass and m_j the mass interaction. The total fine mass is $\boldsymbol{\mu}_N = \mu_0 + m_{0,N}$, which is real.
Dirac operators	$\mathcal{D}_A$ is the finite matrix including the mass; $D^{\mathrm{ch}}[B]$ is the unit-mesh massless matrix with bond phase B; D_A is the massless operator at the level mesh; $D_j(A)$ is the blocked coarse operator.
Curvatures	F^{u} is the unit lattice curvature, $F^{\sharp}$ its rescaled step field, and F the limiting tempered two-form; see Section 6.
Reference cells	R^* is the reference side, $\ell_c = R^*/(2L)$ its cell side, and $N_* = 8L^3$ the number of cells.

2. The chamber regularization

The photon field is a real one-cochain on the cubic lattice. The fermion has one site in each chamber of the lattice mirrors. This choice has a specific consequence: a fermion bond crossing a mirror joins a site to its reflected site and is perpendicular to the mirror. The hopping term then admits a positive reflected expansion. We prove this expansion and the resulting finite-volume reflection positivity below. The construction also requires a nonzero partition function when the bare mass has undetermined sign. For this purpose we identify a nonzero coefficient of the fermion determinant, independent of the mass.

Throughout this section, $0 < \rho < 1$, $q > 0$, and the mass parameter M is real. The hopping weights $w_0, \dots, w_3$ are positive. The estimates and identities below apply, in particular, to the standard weights used in the continuum construction.

2.1. **The photon field and the flux sum.** We orient each link (x, μ) from x to $x + e_\mu$ and each plaquette $(x; \mu\nu)$, $\mu < \nu$, by the ordered pair (e_μ, e_ν). The real k-cochains are denoted by C^k, and the integer cochains by $C^k(\mathbb{Z})$. The coboundary d and its adjoint d^* use these orientations. Thus

$$(dA)_{\mu\nu}(x) = A_\mu(x) + A_\nu(x + e_\mu) - A_\mu(x + e_\nu) - A_\nu(x). \tag{2.1}$$

An oriented link may also be written as a pair of its endpoints; $A(y, x) = -A(x, y)$ and $A(x, x) = 0$. Pullback by a lattice isometry includes the orientation of the image cell. In particular, pullback commutes with d. For a coordinate reflection, this gives a minus sign on the curvature components containing the normal direction and a plus sign on the tangential component. These are the two-form reflection signs used throughout.

Let $\Gamma \subset \mathbb{Z}^3$ be a rank-three subgroup with

$$\min\{|g|_\infty : g \in \Gamma \setminus \{0\}\} \geq 4.$$

We write $C^k(\Gamma)$ for the cochains on $\mathbb{Z}^3/\Gamma$ and identify them with their periodic extensions. The torus has $V = [\mathbb{Z}^3 : \Gamma]$ photon sites, $3V$ links and $3V$ plaquettes. The inner product on real cochains is the sum over positively oriented cells, and $\|\cdot\|$ denotes its norm.

A photon configuration is a class in

$$\mathrm{Conf}(\Gamma) = \{(A, z) : A \in C^1(\Gamma),\ z \in C^2(\Gamma; \mathbb{Z}),\ dz = 0\}/\sim, \qquad (A, z) \sim \left(A + \frac{2\pi}{q}k, z - dk\right), \tag{2.2}$$

where $k \in C^1(\Gamma;\mathbb{Z})$. Its curvature and link phases are

$$F = dA + \frac{2\pi}{q} z, \qquad u_b = \exp(iqA(b)). \tag{2.3}$$

Both are independent of the representative, and $dF = 0$. The integer cochain is required to be closed; there is no restriction on its flux. The curvature therefore remains a real, unbounded field.

Put $r_q = 2\pi/q$. Each class has a unique representative with $A(b) \in [0, r_q)$ at every link. The measure on the quotient is defined by

$$\int_{\mathrm{Conf}(\Gamma)} \Phi(A,z)\,\mathrm{d}(A,z) = \sum_{\substack{z \in C^2(\Gamma;\mathbb{Z}) \\ dz=0}} \int_{[0,r_q)^{3V}} \Phi(A,z)\,\mathrm{d}A. \tag{2.4}$$

This formula holds for nonnegative measurable functions and for absolutely integrable complex functions. Any measurable fundamental domain may be used instead. Indeed, the changes of representative preserve Lebesgue measure in A and counting measure in z.

We will repeatedly use the finite number

$$C_q(\varepsilon) = \sup_{y \in \mathbb{R}} \sum_{j \in \mathbb{Z}} e^{-\varepsilon(y + r_q j)^2}, \qquad \varepsilon > 0. \tag{2.5}$$

It satisfies

$$C_q(\varepsilon) \le 2 + \frac{q}{2\pi}\sqrt{\frac{\pi}{\varepsilon}}.$$

To see this, retain the two terms nearest the origin and bound each remaining term by r_q^{-1} times the Gaussian integral over the adjacent interval of length r_q closer to the origin. Those intervals have disjoint interiors. Write $C_q = C_q(1/2)$. Dropping the constraint $dz = 0$ in the flux sum gives

$$0 < \int_{\mathrm{Conf}(\Gamma)} e^{-\|F\|^2/2}\,\mathrm{d}(A,z) \le (r_q C_q)^{3V}. \tag{2.6}$$

The lower bound follows already from $z = 0$; for the upper bound, sum each z_σ separately at fixed A.

The periodic extensions of the curvatures lie in

$$\Omega = \{F \in C^2 : dF = 0\},$$

with the topology of pointwise convergence on plaquettes. This is a closed subspace of a countable product of copies of $\mathbb{R}$. Local observables depend on finitely many plaquette values. For a finitely supported real two-cochain β, the character $\exp(i\langle \beta, F\rangle)$ is bounded and continuous on Ω.

2.2. **Chambers and the fermion graph.**

The lattice mirrors are

$$\{x_\mu = c\}, \quad c \in \tfrac{1}{2}\mathbb{Z}, \qquad \{x_\mu \pm x_\nu = c\}, \quad c \in \mathbb{Z}, \quad \mu < \nu. \tag{2.7}$$

They form three coordinate families and six diagonal families. Let $\mathcal{L}$ be the corresponding linear forms ℓ and write $\ell(x) = n_\ell \cdot x$. The admissible positions are $\mathrm{Adm}(x_\mu) = \frac{1}{2}\mathbb{Z}$ and $\mathrm{Adm}(x_\mu \pm x_\nu) = \mathbb{Z}$. The reflection in a mirror $\Pi = (\ell, c)$ is

$$\theta_\Pi x = x - \frac{2(\ell(x) - c)}{|n_\ell|^2} n_\ell. \tag{2.8}$$

Its upward normal is $n_\ell/|n_\ell|$. Coordinate mirrors at integer positions are called site planes, and those at half-integer positions are called link planes.

The group generated by these reflections is

$$W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}, \qquad O_{\mathrm{cub}} = O(3) \cap \mathrm{GL}(3,\mathbb{Z}).$$

Here O_{cub} consists of the 48 signed permutation matrices. A fundamental chamber is

$$C_0 = \{0 < x_3 < x_2 < x_1 < \tfrac{1}{2}\}.$$

Its wall reflections are

$$s_0(x) = (x_1, x_2, -x_3), \qquad s_1(x) = (x_1, x_3, x_2),$$
$$s_2(x) = (x_2, x_1, x_3), \qquad s_3(x) = (1 - x_1, x_2, x_3).$$

Denote the walls by Π_i and their inward unit normals by

$$n_0 = e_3, \quad n_1 = (e_2 - e_3)/\sqrt{2}, \quad n_2 = (e_1 - e_2)/\sqrt{2}, \quad n_3 = -e_1. \tag{2.9}$$

Fix $v_0 = (v_1, v_2, v_3) \in C_0$. The fermion sites and their neighbours are

$$\mathcal{X} = Wv_0, \qquad (gv_0)^{(i)} = gs_iv_0. \tag{2.10}$$

The unordered bond $\{x, x^{(i)}\}$ has type i. A site has the unique form $x = k + rv_0$, with $k \in \mathbb{Z}^3$ and $r \in O_{\mathrm{cub}}$; put $\pi(x) = k$ and $y(x) = rv_0$. Thus $\pi(x)$ is its nearest photon site.

Proposition 2.1. *The chambers are precisely gC_0, $g \in W$, and W acts on them simply transitively. No fermion site lies on a mirror. Every bond crosses exactly one mirror, its chamber wall, and crosses it orthogonally at its midpoint. Consequently a bond crossing Π is $\{x, \theta_\Pi x\}$.*

The projection π is W-equivariant. Bonds of types $0, 1, 2$ have the same photon site at both ends; a type-3 bond joins two nearest photon sites. Exactly eight fermion bonds lie over each photon link. If a mirror does not cut the link and its open segment lies on one side, all eight bonds lie on that side. The graph is connected and bipartite, and each site has one neighbour of each type.

Proof. Coordinate reflections and coordinate transpositions generate O_{cub}, and $\theta_{x_\mu,1/2}\theta_{x_\mu,0} = \tau_{e_\mu}$ gives the translations. Conversely, (2.8) is an affine signed permutation with integer translation, so the mirror group is W.

For a point on no mirror, subtract the unique nearest lattice point. The resulting three coordinates are nonzero, have distinct absolute values, and lie in $(-1/2, 1/2)$. There is exactly one signed permutation making them positive and decreasing. This places the point in exactly one translate of C_0. These translates are disjoint open convex sets, meet no mirror, and cover its complement. They are therefore the chambers, with the asserted free transitive action.

For the four bonds from v_0, the midpoint is, respectively,

$$(v_1, v_2, 0), \quad (v_1, (v_2 + v_3)/2, (v_2 + v_3)/2), \quad ((v_1 + v_2)/2, (v_1 + v_2)/2, v_3), \quad (1/2, v_2, v_3).$$

Each belongs to the relative interior of its wall and to no other mirror. Each half of the bond, except its midpoint, lies in one of the two adjacent chambers. Transport by $g \in W$ proves the bond assertion at every site.

In the coordinates $x = k + rv_0$, the first three neighbours are $k + rs_iv_0$, whereas the fourth is $k + re_1 + rr_{e_1}v_0$, where r_{e_1} changes the sign of the first coordinate. This proves the statements about π and its equivariance. Over (k, μ) the lower ends are exactly the eight sites for which $y_\mu = v_1$: the remaining two absolute values can be ordered in two ways, with four choices of their signs. Their type-3 neighbours are the upper ends of these eight bonds.

The bond over $b = (k, \mu)$ has the segment $k + te_\mu + y_\perp$, $v_1 \le t \le 1 - v_1$, where the other two coordinates of $y_\perp$ have absolute values v_2, v_3. For a diagonal height involving x_μ, its varying part is $t \pm y_\nu \in (0, 1)$, the same interval between integers as on the open photon link. For heights not involving x_μ, the changes are y_ν or $y_\nu \pm y_{\nu'}$, whose absolute values are smaller than $1/2$ or 1, respectively. Away from a mirror containing b these preserve its side. The coordinate height x_μ gives the same side except at the unique link plane through the midpoint, which cuts b. This proves the assertion about photon links.

The generators s_0, s_1, s_2 generate the signed permutations, and $s_3 r_{e_1} = \tau_{e_1}$; conjugation gives all coordinate translations. Thus the four wall reflections generate W, proving connectivity. The two parts are the sites gv_0 with $\det r = 1$ and with $\det r = -1$, since each neighbour changes this determinant. □

The geometric location is useful for reflection, while the coordinates (k, r) are useful for the renormalisation analysis. If $d_i = \operatorname{dist}(v_0, \Pi_i)$, then

$$x^{(i)} - x = -2d_i\, r n_i, \qquad \frac{x^{(i)} - x}{|x^{(i)} - x|} = -r n_i.$$

The directions, bond types and projected endpoints in these coordinates do not depend on v_0. Hence neither the operator below nor its determinant depends on v_0 after this identification. The standard weights are

$$w_i = 48|\Pi_i \cap \overline{C}_0|, \qquad (w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6). \tag{2.11}$$

Their inward normals satisfy $\sum_i w_i n_i = 0$.

2.3. **The chamber operator and the finite-volume law.** Let σ_μ be the Pauli matrices and

$$\gamma_\mu = \operatorname{diag}(\sigma_\mu, -\sigma_\mu), \qquad \gamma_v = \sum_\mu v_\mu \gamma_\mu.$$

They form a Hermitian Clifford triple. For a unit vector n, put $P_n^\pm = (1 \pm \gamma_n)/2$. Both projections have rank two, and

$$\gamma_n \gamma_v \gamma_n = -\gamma_{r_n v}, \qquad r_n v = v - 2(v \cdot n) n. \tag{2.12}$$

This follows by multiplying $\gamma_n \gamma_v + \gamma_v \gamma_n = 2n \cdot v$ by γ_n.

For an oriented bond (x, x') of type i define

$$U_A(x, x') = \exp(iqA(\pi(x), \pi(x'))), \qquad w(x, x') = \frac{w_i}{2}\left(\rho - \gamma_{(x'-x)/|x'-x|}\right). \tag{2.13}$$

The chamber operator is

$$(\mathcal{D}_A \psi)(x) = M\psi(x) - \sum_{i=0}^{3} w(x, x^{(i)}) U_A(x, x^{(i)}) \psi(x^{(i)}). \tag{2.14}$$

Its lattice mass is

$$m = M - \frac{\rho}{2} \sum_i w_i. \tag{2.15}$$

For the standard weights, the weighted sum of the Clifford terms in the differences $\psi(x) - U_A(x, x^{(i)})\psi(x^{(i)})$ vanishes on the diagonal. Thus (2.14), in coordinates (k, r), is exactly the matrix $D^{\mathrm{ch}}[qA] + m$, where $D^{\mathrm{ch}}[B]$ is the unit-mesh massless chamber matrix with external hopping phase $\exp(iB(b))$. At a continuum cutoff this means $m = \bar{m}\eta + m_{0,N}(e)$ and $M = \bar{m}\eta + m_{0,N}(e) + \frac{\rho}{2}\sum_i w_i$.

A fermion boundary condition is a character $\chi : \Gamma \to \{1, -1\}$. Write $\mathbb{T} = (\Gamma, \chi)$. Its spinors satisfy $\psi(x + g) = \chi(g)\psi(x)$ for $g \in \Gamma$. The restriction $\mathcal{D}_A^{\mathbb{T}}$ acts on a space of dimension $192V$, with the inner product obtained by summing over representatives of $\mathcal{X}/\Gamma$. For each site introduce generators $\psi_\alpha(x), \bar{\psi}_\alpha(x)$, $1 \le \alpha \le 4$, obeying the same boundary conditions. Their Grassmann algebra is denoted by $\mathrm{Gr}_{\mathbb{T}}$. Its Berezin integral is normalized by

$$\int_{\mathbb{T}} \prod_{x \in \mathcal{X}/\Gamma} \prod_{\alpha=1}^{4} \psi_\alpha(x) \bar{\psi}_\alpha(x) = 1. \tag{2.16}$$

Products of pairs are even, so their ordering is immaterial. With $S_{\mathbb{T}}(A) = \sum_x \bar{\psi}(x)(\mathcal{D}_A \psi)(x)$, this normalization gives $\int_{\mathbb{T}} e^{-S_{\mathbb{T}}(A)} = \det \mathcal{D}_A^{\mathbb{T}}$.

The unnormalized functional is

$$Z_{\mathbb{T}}[G] = \int_{\mathrm{Conf}(\Gamma)} e^{-\|F\|^2/2} \int_{\mathbb{T}} e^{-S_{\mathbb{T}}(A)} G(A,z)\, \mathrm{d}(A,z). \tag{2.17}$$

For a scalar G the inner integral is $G(A,z)\det \mathcal{D}_A^{\mathbb{T}}$. For a Grassmann-valued integrand, absolute convergence is understood coefficient by coefficient. Bounded measurable coefficients suffice: $e^{-S_{\mathbb{T}}(A)}$ has finitely many coefficients, each a polynomial in the link phases, and (2.6) supplies an integrable majorant.

Proposition 2.2. *For every real A and M,*

$$\det \mathcal{D}_A^{\mathbb{T}} \geq 0, \qquad \|\mathcal{D}_A^{\mathbb{T}}\| \leq |M| + \frac{1+\rho}{2}\sum_i w_i.$$

Moreover $\Re \mathcal{D}_A^{\mathbb{T}} = m + \frac{\rho}{2}\nabla_A^*\nabla_A$, where

$$(\nabla_A \psi)(x,i) = \sqrt{w_i/2}\,\big(\psi(x) - U_A(x,x^{(i)})\psi(x^{(i)})\big).$$

The determinant is not identically zero as a function of the link phases, for every real mass parameter M.

Proof. Write

$$H_A\psi(x) = \sum_i w_i U_A(x,x^{(i)})\psi(x^{(i)}), \qquad J_A\psi(x) = \sum_i w_i \gamma_{(x^{(i)}-x)/|x^{(i)}-x|} U_A(x,x^{(i)})\psi(x^{(i)}).$$

Reversing a bond conjugates its phase and reverses its direction. Reindexing by the involution $x \mapsto x^{(i)}$ therefore gives $H_A^* = H_A$ and $J_A^* = -J_A$. Hence

$$\mathcal{D}_A = M - \frac{\rho}{2}H_A + \frac{1}{2}J_A, \qquad \nabla_A^*\nabla_A = \sum_i w_i - H_A.$$

This proves the real-part identity. In the splitting $\mathbb{C}^4 = \mathbb{C}^2 \oplus \mathbb{C}^2$, the two blocks of H_A coincide, and the two blocks of J_A have opposite signs. It follows that

$$\mathcal{D}_A^{\mathbb{T}} = \mathrm{diag}(\check{\mathcal{D}}_A^{\mathbb{T}}, (\check{\mathcal{D}}_A^{\mathbb{T}})^*), \qquad \det \mathcal{D}_A^{\mathbb{T}} = |\det \check{\mathcal{D}}_A^{\mathbb{T}}|^2. \tag{2.18}$$

Each hopping matrix has norm $w_i(1+\rho)/2$, while the associated translation with phase is unitary. Their sum gives the norm bound.

For the last assertion we view the determinant as a Laurent polynomial $P_{\mathbb{T}}$ in the independent variables u_b. Only type-3 bonds contain a variable, with u_b at the lower end and u_b^{-1} at the upper end. Fix a coordinate direction μ. The coefficient of

$$\prod_{b \| e_\mu} u_b^{32} \tag{2.19}$$

will be computed without using M.

Partition the sites modulo Γ into six layers according to the increasing values

$$y_\mu = -v_1, -v_2, -v_3, v_3, v_2, v_1.$$

Each layer has $8V$ sites. Each site has one upward neighbour, one downward neighbour, and two neighbours in its own layer. The upward map is a permutation of the sites and visits successive layers modulo six. Its bond types are

$$(i_1, i_2, i_3, i_4, i_5, i_6) = (2,1,0,1,2,3). \tag{2.20}$$

For layers $1,\dots,5$ the upward phase is one; in layer six it is u_b for the link $(\pi(x),\mu)$. These facts follow by changing the smallest sign or exchanging successive absolute values in the neighbour formulas above. The sixth step changes v_1 to $-v_1$ and increases the photon coordinate by one.

There are eight lower and eight upper fermion ends over each photon link, each carrying four matrix rows. An exponent 32 at a link therefore requires every lower row to take the upward type-3 block and every upper row to avoid the reverse block. Imposing this at all links in direction μ forces every row in every layer to take its upward block. Indeed, layer-six rows fill all columns in layer one. A layer-one row cannot then use a column in its own layer, and the reverse block is excluded by the exponent condition; it must go upward to layer two. Inductively the columns in the current and preceding layers are already occupied, so layers two through five must also go upward. The selected terms use no other link variables.

The remaining freedom is a permutation of the four spin indices at each site. The site permutation, repeated four times, has sign one. Summing over the spin permutations therefore gives the product of the 4×4 determinants of the selected hopping blocks. The boundary signs have fourth power one, and

$$\det w(x, x^{\uparrow}) = (w_{i_k}/2)^4(1-\rho^2)^2$$

at a site in layer k. Consequently the coefficient of (2.19) is C_*^V, where

$$C_* = (w_0/2)^{32}(w_1/2)^{64}(w_2/2)^{64}(w_3/2)^{32}(1-\rho^2)^{96} > 0. \tag{2.21}$$

This coefficient is independent of M. In particular the determinant polynomial, and its restriction to the phase torus, are nonzero. □

The real-part estimate alone would require control of the sign of m to prove invertibility. The coefficient computation instead proves that the determinant has positive integral at every real M, which is all that is needed to define the law.

Proposition 2.3. *The partition function satisfies* $0 < Z_{\mathbb{T}}[1] < \infty$. *More precisely,*

$$r_q^{3V} e^{-6r_q^2 V} C_*^V \le Z_{\mathbb{T}}[1] \le (r_q C_q)^{3V} \left(|M| + \frac{1+\rho}{2} \sum_i w_i \right)^{192V}. \tag{2.22}$$

Thus $\mathbb{E}_{\mathbb{T}}[G] = Z_{\mathbb{T}}[G]/Z_{\mathbb{T}}[1]$ *defines a probability law on scalar observables. Its curvature law* $\mu_{\mathbb{T}}$ *is the law of the periodic extension* $\widetilde{F}$ *on* Ω.

Proof. For a nonnegative Laurent polynomial P on the phase torus, every Fourier coefficient satisfies

$$|\widehat{P}(\lambda)| = \left| \int P(u)\overline{u^\lambda}\,\mathrm{d}u \right| \le \int P(u)\,\mathrm{d}u,$$

where Haar measure is normalized. By the preceding proposition and (2.21),

$$\int_{[0,r_q)^{3V}} \det \mathcal{D}_A^{\mathbb{T}}\,\mathrm{d}A \ge r_q^{3V} C_*^V.$$

On this box $|(dA)_\sigma| \le 2r_q$ and $\|dA\|^2 \le 12Vr_q^2$. The $z=0$ term in (2.17) gives the lower bound in (2.22). For the upper bound use $|\det \mathcal{D}_A^{\mathbb{T}}| \le \|\mathcal{D}_A^{\mathbb{T}}\|^{192V}$ and (2.6). The density $e^{-\|F\|^2/2} \det \mathcal{D}_A^{\mathbb{T}}$ is nonnegative. After division by its strictly positive integral it is a probability density, whose image under $(A,z) \mapsto \widetilde{F}$ is the asserted law. □

2.4. **Symmetries and reflected Grassmann products.** The field strength transforms as a two-cochain, with the orientation of each plaquette included. For an antiperiodic cubic torus the law is invariant under W and under $F \mapsto -F$. We record the corresponding fermionic identities, since reflection positivity uses their signs.

Gauge covariance follows directly from

$$U_{A-d\lambda}(x,x') = e^{iq\lambda(\pi(x))} U_A(x,x') e^{-iq\lambda(\pi(x'))}.$$

For lattice covariance set $\gamma_5 = \left(\begin{smallmatrix} 0 & 1_2 \\ 1_2 & 0 \end{smallmatrix}\right)$. It anticommutes with the Clifford triple, and

$$(\gamma_n\gamma_5)\gamma_v(\gamma_n\gamma_5)^{-1} = \gamma_{r_n v}.$$

If the linear part of $g \in W$ is $r_{n_1} \cdots r_{n_k}$, the corresponding product of these spin matrices, followed by the permutation $x \mapsto g^{-1}x$, intertwines $\mathcal{D}_A^{\mathbb{T}}$ and $\mathcal{D}_{(g^{-1})^* A}^{g\mathbb{T}}$, where $g\mathbb{T} = (r\Gamma, \chi \circ r^{-1})$ and r is the linear part of g. Taking determinants and changing variables in (2.17) proves

$$\mu_{g\mathbb{T}} = ((g^{-1})^*)_* \mu_{\mathbb{T}}.$$

The configuration measure is preserved because pullback permutes cochain coordinates up to signs and maps one fundamental domain to another. All translations preserve $\mathbb{T}$, and all elements of W preserve an antiperiodic cubic torus, proving the stated invariance.

For charge conjugation let $C = \sigma_2 \oplus \sigma_2$. The Pauli identity $C\gamma_v^\top C^{-1} = -\gamma_v$, together with reversal of the oriented bond, gives

$$C(\mathcal{D}_A^{\mathbb{T}})^\top C^{-1} = \mathcal{D}_{-A}^{\mathbb{T}}.$$

Thus the determinant is unchanged by $A \mapsto -A$. The substitution $(A, z) \mapsto (-A, -z)$ preserves the configuration measure and sends F to $-F$, proving charge symmetry of the curvature law.

At a mirror Π with upward normal n, the Grassmann reflection Θ_Π is antilinear and reverses products. In components it is

$$\Theta_\Pi \psi_\alpha(x) = -\sum_\beta \bar{\psi}_\beta(\theta_\Pi x)(\gamma_n)_{\beta\alpha}, \qquad \Theta_\Pi \bar{\psi}_\alpha(x) = -\sum_\beta (\gamma_n)_{\alpha\beta}\psi_\beta(\theta_\Pi x). \tag{2.23}$$

On scalar coefficient functions it acts as $\Phi(A, z) \mapsto \overline{\Phi(\theta_\Pi^* A, \theta_\Pi^* z)}$. The Clifford identities give $\Theta_\Pi^2 = 1$ and

$$\Theta_\Pi(\bar{\psi}(x) Y \psi(x')) = \bar{\psi}(\theta_\Pi x')\gamma_n Y^* \gamma_n \psi(\theta_\Pi x). \tag{2.24}$$

For a hopping term, $\gamma_n w(x, x')\gamma_n = w(\theta_\Pi x', \theta_\Pi x)$, by (2.12). Reflection therefore permutes the oriented terms of the action, including the real mass terms.

If Y is a finite set of sites, then

$$\int_{\theta_\Pi Y} \Theta_\Pi \xi = \overline{\int_Y \xi}. \tag{2.25}$$

Only the top-degree coefficient needs to be checked. At one site reflection maps $\prod_\alpha \psi_\alpha \bar{\psi}_\alpha$ to the same product at the reflected site, multiplied by $(\det \gamma_n)^2 = 1$. Products from different sites are even. This proves (2.25), and a measure-preserving change of photon variables gives

$$Z_{\theta_\Pi \mathbb{T}}[\Theta_\Pi G] = \overline{Z_{\mathbb{T}}[G]} \tag{2.26}$$

for bounded coefficient integrands, and for the integrable Gaussian-growth classes considered below.

On even integrands, compositions of these reflections define an action of W. The ambiguity of a reflection word acts only by the common sign on all fermion generators. To verify this, compose two words representing the same lattice isometry. The resulting even word has trivial geometric action, and its spin matrix is $\mathrm{diag}(u, u)$ with u a product of an even number of Pauli matrices σ_n. It commutes with all Pauli matrices, hence is scalar; its determinant is one, so $u = \pm 1_2$. Its action on an even monomial is consequently the identity. We write G^g for this action when folding even integrands. For a real scalar observable it is $G^g(F) = G(g^* F)$.

2.5. **The fermion crossing at a mirror.** Let a bond of type j cross Π, with lower endpoint x and upper endpoint $x' = \theta_\Pi x$. Its direction is the upward normal n. The two hopping matrices are

$$\begin{aligned} w(x,x') &= \tfrac{w_j}{2}\big((1+\rho)P_n^- - (1-\rho)P_n^+\big), \\ w(x',x) &= \tfrac{w_j}{2}\big((1+\rho)P_n^+ - (1-\rho)P_n^-\big). \end{aligned} \tag{2.27}$$

Choose orthonormal bases $e^\pm_{n,1}, e^\pm_{n,2}$ of the two projected spaces. For each $k = 1, 2$ introduce four crossing modes. Their coefficients and phases are displayed together:

	ζ	ω	u
1	$(e^-_{n,k})^*\psi(x')$	$\frac{w_j}{2}(1+\rho)$	$U_A(x,x')$
2	$(e^+_{n,k})^*\psi(x')$	$\frac{w_j}{2}(1-\rho)$	$U_A(x,x')$
3	$\bar\psi(x')e^+_{n,k}$	$\frac{w_j}{2}(1+\rho)$	$\overline{U_A(x,x')}$
4	$\bar\psi(x')e^-_{n,k}$	$\frac{w_j}{2}(1-\rho)$	$\overline{U_A(x,x')}$

(2.28)

All coefficients are positive in the range $0 < \rho < 1$.

Proposition 2.4. *Let J be a finite collection of bonds crossing Π, and let $S^\times_\Pi(J)$ be their part of the action. Index all their crossing modes by a finite set $\mathcal{I}$. Then*

$$-S^\times_\Pi(J) = \sum_{i\in\mathcal{I}} \omega_i u_i \Theta_\Pi(\zeta_i)\zeta_i. \tag{2.29}$$

Choose an order on $\mathcal{I}$ and, for $I \subset \mathcal{I}$, put $\zeta_I = \prod_{i\in I}\zeta_i$, $\omega_I = \prod_{i\in I}\omega_i$ and $u_I = \prod_{i\in I} u_i$. Then

$$e^{-S^\times_\Pi(J)} = \sum_{I\subset\mathcal{I}} \omega_I u_I \Theta_\Pi(\zeta_I)\zeta_I. \tag{2.30}$$

At site and diagonal mirrors all u_i are one. At a link plane, groups one and two have charge $+1$, groups three and four charge -1, and their phases are $e^{iq\mathfrak{q}(i)A(b)}$ at the cut photon link b.

Proof. The reflections of the four modes in (2.28) are, respectively,

$$\bar\psi(x)e^-_{n,k}, \quad -\bar\psi(x)e^+_{n,k}, \quad -(e^+_{n,k})^*\psi(x), \quad (e^-_{n,k})^*\psi(x).$$

Multiplying each reflected mode by the original mode and using anticommutation gives the four projected terms in (2.27), with the indicated phases. Summation over k uses $P^\pm_n = \sum_k e^\pm_{n,k}(e^\pm_{n,k})^*$ and gives the negative of the two bond terms. Summation over bonds proves (2.29).

Each $\Theta_\Pi(\zeta_i)\zeta_i$ is even and has square zero. These elements commute. Reversal of the order under Θ_Π gives

$$\prod_{i\in I}\Theta_\Pi(\zeta_i)\zeta_i = \Theta_\Pi(\zeta_I)\zeta_I,$$

with no remaining sign. Expanding the product of the finite exponentials proves (2.30). Only type-3 bonds carry a photon phase, and they cross link planes. Their projected endpoints are the oriented endpoints of the cut link, giving the last assertion. □

2.6. **Adapted tori and the second mirror.** An adapted torus for $\ell \in \mathcal{L}$ has the form

$$\Gamma = \Gamma_\parallel \oplus \mathbb{Z}v, \quad \Gamma_\parallel = \Gamma \cap \{\ell = 0\}, \quad v = \frac{p}{|n_\ell|^2} n_\ell \in \mathbb{Z}^3, \quad \chi(v) = -1, \tag{2.31}$$

where $p = \ell(v) > 0$ is the height period. For an admissible c set

$$c' = c + p/2, \quad \Pi = (\ell, c), \quad \Pi' = (\ell, c'), \quad U_+ = \{c < \ell < c'\}, \quad U_- = \{c - p/2 < \ell < c\}.$$

The two mirrors are admissible: for coordinate families $p \geq 4$, while for diagonal families p is even and $p \geq 8$. Their reflections satisfy $\theta_{\Pi'} = \tau_v\theta_\Pi$ and induce the same geometric reflection on the torus, exchanging its two halves. An antiperiodic coordinate torus is adapted to each

coordinate family. The torus with periods $ae_1 - ae_2$, $be_1 + be_2$ and ce_3, antiperiodic on these periods, is adapted to the corresponding two diagonal families and the third coordinate family. Cubic images give adapted tori for all nine families.

Choose representatives $\mathcal{X}_+$ in U_+ and $\mathcal{X}_- = \theta_\Pi \mathcal{X}_+$ in U_-. They partition the fermion sites of the torus. Let S_+ contain the mass terms and oriented bonds entirely in U_+, and set $\theta = \theta_\Pi$, $\Theta = \Theta_\Pi$; the second boundary mirror induces the same geometric reflection on the torus. The reflected part is ΘS_+. Its coefficient functions involve only links in the closed upper slab.

The crossing modes at Π lie in $\mathrm{Gr}(\mathcal{X}_+)$. At Π' their upper endpoints are outside U_+; write $\check{\zeta}_{i'} = \Theta_{\Pi'}\zeta_{i'}$, so the checked modes lie in $\mathrm{Gr}(\mathcal{X}_+)$. The boundary condition gives

$$\Theta_{\Pi'} = \chi(v)^{\deg}\Theta_\Pi \quad \text{on } \mathrm{Gr}_{\mathbb{T}}\,. \tag{2.32}$$

In the crossing term at Π' we therefore have

$$\Theta_{\Pi'}(\zeta_{i'})\zeta_{i'} = -\chi(v)\Theta(\check{\zeta}_{i'})\check{\zeta}_{i'}.$$

The minus sign is from exchanging the two odd factors. Since $\chi(v) = -1$, it is canceled by antiperiodicity. Both boundary crossings thus have the same positive coefficients, and

$$e^{-S_{\mathbb{T}}} = e^{-S_+}\Theta(e^{-S_+})\sum_{I,I'}\omega_I\omega_{I'}u_I u_{I'}\Theta(\zeta_I)\zeta_I\Theta(\check{\zeta}_{I'})\check{\zeta}_{I'}. \tag{2.33}$$

Here I and I' range over the finite crossing-mode sets at the two mirrors.

The reflected Berezin integral separates the two halves:

$$\int_{\mathbb{T}}\Theta(\xi')\xi = \overline{\int_{\mathcal{X}_+}\xi'}\int_{\mathcal{X}_+}\xi, \qquad \xi,\xi' \in \mathrm{Gr}(\mathcal{X}_+). \tag{2.34}$$

Indeed, the top monomial is the product of the two half-top monomials, and reflection maps one to the other by (2.25). We must still separate the photon flux constraints; these couple the halves at the cut cubes.

2.7. **The photon crossing and Gauss' law.** Let $\mathcal{M}$ be the two boundary mirrors, understood periodically. Classify a cell by its relative interior: it is upper or lower if that interior is in the respective half, lies on the mirrors if the whole cell does, and is cut if its relative interior meets a mirror without being contained in it. Write $B_+, B_-, B_0, B_\times$ for the four sets of links, $P_+, P_-, P_0, P_\times$ for the plaquettes, and $Q_+, Q_-, Q_\times$ for the cubes.

Reflection exchanges the upper and lower cells. It fixes a cell lying on the mirrors with orientation sign $+1$, and a cut cell with orientation sign -1. Each cell meets at most one boundary mirror: its height variation is at most $|n_\ell|^2$, smaller than $p/2$. To justify the orientation assertion, a point in the intersection of the relative interior with the mirror is fixed and lies in the relative interior of the reflected cell. Relative interiors of different cells are disjoint, so the cell is fixed. Reflection is the identity on its affine hull in the first case and a reflection of that affine hull in the cut case.

An upper cell has only upper faces or faces lying on the mirrors; a cut cell has only upper, lower or cut faces. Moreover a plaquette containing a cut link is cut, and a cube containing a cut plaquette is cut. An upper face can touch the boundary at a relative interior point only when its affine height is constant. A face of a cut cell cannot lie in the mirror: reflection in the supporting plane of a face would not preserve that cell. Finally, the height takes both signs on a cut face, hence on the relative interior of its coface. In particular, the incidences at a cut plaquette and a cut cube give

$$(dA)_\sigma = (d_+A)_\sigma - (d_+\theta^*A)_\sigma + (d_\times A)_\sigma, \qquad (dz)(Q) = (d_+z)(Q) - (d_+\theta^*z)(Q) + (d_\times z)(Q), \tag{2.35}$$

where d_+ and $d_\times$ retain the incidences from upper and cut faces, respectively. For example, $(d_+A)_\sigma = \sum_{b\in B_+}[\sigma : b]A(b)$. The minus sign in the lower term follows from orientation reversal of the cut cell and the identity $d\theta^* = \theta^* d$.

Use representatives

$$A = (A_+, A_-, A_0, a), \qquad z = (z_+, z_-, z_0, z_\times).$$

Choose the interval $[0, r_q)$ for upper and mirror links, its signed reflection for lower links, and $[-r_q/2, r_q/2)$ for cut links. This is a fundamental domain. Reflection maps the lower variables measure-preservingly to an independent copy of the upper variables, fixes (A_0, z_0), and changes the sign of a.

The upper and mirror photon weights are

$$W_+ = \prod_{\sigma\in P_+} e^{-F_\sigma^2/2} \prod_{Q\in Q_+} 1_{\{(dz)(Q)=0\}}, \qquad W_0 = \prod_{\sigma\in P_0} e^{-F_\sigma^2/2}. \tag{2.36}$$

The full Maxwell weight with the closedness constraint is $W_0W_+\Theta(W_+)$ times the Gaussian and constraint factors at the cut plaquettes and cubes. The half weights contain only upper and mirror variables. Their integrals are bounded uniformly in the mirror variables by

$$\int W_+ \,\mathrm{d}(A_+, z_+) \le r_q^{|B_+|} C_q^{|P_+|}, \qquad \int W_0 \,\mathrm{d}(A_0, z_0) \le r_q^{|B_0|} C_q^{|P_0|}.$$

The proof drops the cube constraints and uses (2.5) on each plaquette.

At a cut cube impose integer closedness by the Fourier multiplier

$$1_{\{(dz)(Q)=0\}} = \int_0^{2\pi} e^{i\alpha_Q (dz)(Q)} \frac{\mathrm{d}\alpha_Q}{2\pi}.$$

For $\alpha \in [0, 2\pi)^{Q_\times}$ and $k \in \mathbb{Z}^{P_\times}$ define

$$\alpha'_\sigma = \sum_{Q\in Q_\times} [Q : \sigma]\alpha_Q, \quad \beta_\sigma = \frac{q}{2\pi}(\alpha'_\sigma - 2\pi k_\sigma),$$

$$u_{\alpha,k} = \exp\left(i\sum_Q \alpha_Q (d_+ z)(Q) - i\sum_\sigma \beta_\sigma (d_+A)_\sigma \right). \tag{2.37}$$

The function $u_{\alpha,k}$ depends only on the upper variables and has modulus one.

Poisson summation, in the normalization needed here, reads

$$\sum_{j\in\mathbb{Z}} e^{-(y+r_q j)^2/2 + ij\alpha} = \frac{q}{\sqrt{2\pi}} \sum_{k\in\mathbb{Z}} e^{-\beta_k^2/2 - i\beta_k y}, \qquad \beta_k = \frac{q}{2\pi}(\alpha - 2\pi k). \tag{2.38}$$

It follows by Fourier transforming the Gaussian under $t = y + r_q s$. Insert the cube multipliers and apply this formula at each cut plaquette. Splitting dA and dz by (2.35) gives the exact identity

$$\begin{aligned} &\sum_{z_\times} \prod_{\sigma\in P_\times} e^{-F_\sigma^2/2} \prod_{Q\in Q_\times} 1_{\{(dz)(Q)=0\}} \\ &= \int \prod_{Q\in Q_\times} \frac{\mathrm{d}\alpha_Q}{2\pi} \sum_{k\in\mathbb{Z}^{P_\times}} \prod_{\sigma\in P_\times} \frac{q}{\sqrt{2\pi}} e^{-\beta_\sigma^2/2}\, u_{\alpha,k}\Theta(u_{\alpha,k}) \exp\left(iq \sum_{b\in B_\times} a_b (d^*k)(b) \right). \end{aligned} \tag{2.39}$$

Here k is extended by zero outside $P_\times$. To obtain the final factor, observe that at a cut link b,

$$\sum_{\sigma\in P_\times} [\sigma : b]\alpha'_\sigma = \sum_{Q\in Q_\times} \alpha_Q \sum_\sigma [Q : \sigma][\sigma : b] = 0.$$

The restriction to cut plaquettes may be dropped because any plaquette containing a cut link is cut; the last sum is zero by $d^2 = 0$. Thus the part of β involving α' cancels from the cut-link factor.

All the weights before the two reflected factors on the right of (2.39) are positive. Their total integral is at most $C_q^{|P_\times|}$. Indeed, with $y = 0$ in (2.38),

$$\frac{q}{\sqrt{2\pi}} \sum_k e^{-\beta_k^2/2} \le \sum_j e^{-(r_q j)^2/2} \le C_q$$

at each cut plaquette. The original Gaussian sum also has an absolute bound $C_q^{|P_\times|}$, justifying insertion of the Fourier multipliers and the exchanges of their sums and integrals.

Cut links occur only at coordinate link planes. A link fixed by a mirror reflection but not contained in its plane has its endpoints exchanged; hence it is perpendicular to the mirror, whose normal is a coordinate vector. A diagonal mirror has cut plaquettes and cubes, but no cut links.

For crossing-mode subsets I, I' let $\mathfrak{q}(I \cup I')(b)$ be the sum of their charges at modes lying over b. The fermionic phases in (2.33) are

$$u_I u_{I'} = \exp\left(iq \sum_{b \in B_\times} a_b \mathfrak{q}(I \cup I')(b) \right).$$

Integration of the cut links over one period gives

$$\int_{[-r_q/2, r_q/2)^{B_\times}} e^{iq \sum_b a_b ((d^* k)(b) + \mathfrak{q}(I \cup I')(b))} \, \mathrm{d}a = r_q^{|B_\times|} 1_{\{d^* k + \mathfrak{q}(I \cup I') = 0 \text{ on } B_\times\}}. \tag{2.40}$$

The quantities in the exponent are integers. This is precisely Gauss' law at the cut links, with a nonnegative multiplier. When there are no cut links the condition is empty.

2.8. **The reflected form on an adapted torus.** Let $\mathcal{B}^+$ be the Grassmann-valued integrands whose generators are in $\mathcal{X}_+$ and whose bounded measurable coefficient functions depend only on the upper and mirror curvatures and link phases. For the later operator estimates we also use $\mathcal{G}^+$: its coefficients Φ satisfy, for some $C < \infty$ and $\varepsilon > 0$,

$$W_+ |\Phi| \le C \prod_{\sigma \in P_+} e^{-\varepsilon F_\sigma^2}. \tag{2.41}$$

This condition is imposed on the upper and mirror variables and places no growth factor on mirror plaquettes. The bound is relevant only where the upper cube constraints hold. Such half-configurations can be extended to full closed configurations by reflecting their upper z, keeping z_0, and taking $z_\times = 0$; the cut-cube identity (2.35) proves closedness. Thus these coefficient functions have an unambiguous reading wherever $W_+ \neq 0$. In particular $\mathcal{B}^+ \subset \mathcal{G}^+$.

Theorem 2.5. *For a torus adapted to ℓ and $G, G' \in \mathcal{G}^+$, the integral $Z_{\mathbb{T}}[\Theta(G')G]$ converges absolutely, coefficient by coefficient, and admits a representation*

$$Z_{\mathbb{T}}[\Theta(G')G] = \int \overline{\widehat{G'}(s)} \widehat{G}(s) \, \mathrm{d}\mathfrak{m}(s), \tag{2.42}$$

where $\mathfrak{m}$ is a finite positive measure and $G \mapsto \widehat{G}$ is linear with bounded measurable values. This holds for even, odd and mixed-parity integrands. In particular $Z_{\mathbb{T}}[\Theta(G)G] \ge 0$.

For every bounded scalar observable Φ depending only on curvatures of plaquettes wholly contained in U_+, the curvature law satisfies

$$\int_\Omega \overline{\Phi(\theta^* F)} \Phi(F) \, \mathrm{d}\mu_{\mathbb{T}}(F) \ge 0. \tag{2.43}$$

It is invariant under this reflection.

Proof. The measure space has coordinates

$$s = (A_0, z_0, \alpha, k, I, I'),$$

with mirror variables (A_0, z_0), cut-cube variables $\alpha \in [0, 2\pi)^{Q_\times}$, integers $k \in \mathbb{Z}^{P_\times}$, and the two finite mode subsets. Define

$$\begin{aligned} \mathrm{d}\mathfrak{m}(s) = W_0 \, \mathrm{d}(A_0, z_0) \prod_{Q \in Q_\times} \frac{\mathrm{d}\alpha_Q}{2\pi} \prod_{\sigma \in P_\times} \frac{q}{\sqrt{2\pi}} e^{-\beta_\sigma^2/2} \\ \cdot \omega_I \omega_{I'} r_q^{|B_\times|} 1_{\{d^* k + \mathfrak{q}(I \cup I') = 0 \text{ on } B_\times\}}, \end{aligned} \tag{2.44}$$

where z_0, k, I, I' carry counting measure. Put

$$\widehat{G}(s) = \int W_+ u_{\alpha,k} \int_{\mathcal{X}_+} G e^{-S_+} \zeta_I \check{\zeta}_{I'} \, \mathrm{d}(A_+, z_+). \tag{2.45}$$

The action S_+ has bounded coefficients depending only on upper and mirror links. Thus the integrand in this formula is a half-integrand of the stated kind.

We first prove the identity for bounded coefficients and definite parities. Inserting (2.33) into the reflected functional combines each term with the half factors

$$\xi = G e^{-S_+} \zeta_I \check{\zeta}_{I'}, \qquad \xi' = G' e^{-S_+} \zeta_I \check{\zeta}_{I'}.$$

The sign from this rearrangement is $(-1)^{(|G|+|G'|)(|I|+|I'|)}$. It is one when the parities agree. When they disagree, one of ξ, ξ' is odd, so its half Berezin integral vanishes; the corresponding full integral also vanishes by (2.34). The sign consequently disappears from the integrated formula in all cases.

Now insert (2.39) and integrate the cut links. Formula (2.40) supplies the last factor in (2.44). At fixed s, the remaining factors are

$$\Theta\big(W_+ u_{\alpha,k} \xi'\big) W_+ u_{\alpha,k} \xi.$$

The lower photon variables map measure-preservingly onto an independent set of upper variables. Combining this change of variables with (2.34) gives $\overline{\widehat{G'}(s)} \widehat{G}(s)$, proving (2.42) once the exchanges of integration have been justified.

For that justification let C bound all the finitely many half Berezin coefficients of $G e^{-S_+} \zeta_I \check{\zeta}_{I'}$ and of the corresponding expression with G'. The scalar phase factors have modulus one. Before the cut-link integration, an absolute majorant therefore has total integral at most

$$C^2 (r_q C_q)^{3V} \prod_{i \text{ at } \Pi} (1 + \omega_i) \prod_{i' \text{ at } \Pi'} (1 + \omega_{i'}). \tag{2.46}$$

The upper, lower and mirror variables are bounded by their Gaussian sums; the cut-link intervals contribute their lengths; the dual photon weights are bounded by $C_q^{|P_\times|}$; and the finite mode sums give the two products. This proves absolute integrability and all required uses of Fubini's theorem. The same bounds without the half factors show that $\mathfrak{m}$ is finite. It is positive by construction. Formula (2.45) is measurable and bounded uniformly in s by $C r_q^{|B_+|} C_q^{|P_+|}$. Mixed parities follow by sesquilinearity.

For $G, G' \in \mathcal{G}^+$, use (2.41) with exponents ε and ε'. The algebraic identities above are unchanged. The absolute majorant now replaces the upper and lower Gaussian sums by $C_q(\varepsilon)^{|P_+|}$ and $C_q(\varepsilon')^{|P_-|}$, respectively, while the mirror and cut sums retain exponent $1/2$. These quantities are finite by (2.5). This proves coefficientwise integrability, the same representation, and the boundedness of $\widehat{G}$. Taking $G' = G$ proves positivity.

For a scalar Φ with this plaquette dependence, divide this positivity by $Z_{\mathbb{T}}[1] > 0$ and use the definition of the curvature law. Reflection invariance follows from (2.26), since the adapted torus is preserved by reflection. This proves (2.43) and completes the theorem. □

The same form remains positive after multiplication of the weight by $\Phi_+\Theta(\Phi_+)$, with Φ_+ bounded and scalar on the upper half: apply the theorem to $G\Phi_+$. This observation will be used for the reflected arrays of observables. It uses positivity of the joint reflected form before integrating out the fermion. The determinant's pointwise nonnegativity by itself does not imply reflection positivity.

3. Slab operators and finite-volume comparison

A cubic torus admits the coordinate reflection-positive representations of the preceding section. A diagonal mirror requires a different period lattice. To compare the two geometries, we express the functional as a trace of operators associated with slabs. Reflection gives the adjoint of a slab operator; repeated reflected copies give its Schatten norm. The resulting comparison changes the closing vector, and, when needed, the fermion character. Three such changes connect the cubic torus with the diagonal torus and with a second coordinate torus of the same volume. Their pressures differ by a quantity tending to zero. This is the finite-volume input for diagonal reflection positivity.

We also prove Gaussian domination on coordinate tori. Its trace formula uses real slice connections and a sum over magnetic fluxes. A real external two-cochain enters through unitary electric translations and magnetic multiplication operators. Schatten's inequality first removes the electric translations. A second choice of time direction removes the remaining magnetic perturbation.

Throughout, $q > 0$, $0 < \rho < 1$, $w_i > 0$, and $M \in \mathbb{R}$. The mass may depend on the mesh, but it is fixed when the volume tends to infinity. We use $r_q = 2\pi/q$ and the counting Berezin convention (2.16). The determinant, and hence every curvature law below, agrees with the convention used in the renormalization argument. More precisely, there are $192V$ fermion modes on a torus of photon volume V. With the pairing

$$\langle\bar\Psi, O\Psi\rangle_{\mathrm{ch}} = \frac{1}{48}\sum_x \bar\Psi(x)(O\Psi)(x), \qquad \mathrm{D}\Psi = 48^{192V}\int_{\mathbb{T}},$$

one has

$$\int \mathrm{D}\Psi\, e^{-\langle\bar\Psi, O\Psi\rangle_{\mathrm{ch}}} = \det O.$$

The factor 48^{192V} cancels the factor from the determinant of the weighted quadratic form. Thus no additional normalization of the curvature law is needed at this interface. The slab modes in this section use the counting convention.

3.1. Local observables and their lifts.

We need local observables that may contain fermion variables and whose coefficients grow with the curvature. Their definition must make sense in every flux sector. Let $B = y + [-r, r]^3$, and write $B^{(j)}$ for the oriented lattice j-cells contained in B. The fermion sites assigned to B are

$$\mathcal{X}(B) = \{x \in \mathcal{X} : \pi(x) + [-\tfrac12, \tfrac12]^3 \subset B\}.$$

This support condition includes the photon site at which the fermion transforms under gauge transformations.

Let $\mathcal{G}(B)$ consist of the Borel maps $G : \mathbb{R}^{B^{(1)}} \to \mathrm{Gr}(\mathcal{X}(B))$ with gauge-invariant values and with the following growth bound. If $G(\widehat A) = \sum_\Psi a_\Psi(\widehat A)\Psi$, then, for some $C < \infty$ and $\varepsilon > 0$,

$$|a_\Psi(\widehat A)| \le C \exp\left((\tfrac12 - \varepsilon)\sum_{\sigma\in B^{(2)}} (d\widehat A)_\sigma^2\right). \tag{3.1}$$

Here gauge invariance means

$$a_\Psi(\widehat A - d\lambda)e^{iq\langle\lambda, \mathfrak{q}(\Psi)\rangle} = a_\Psi(\widehat A) \tag{3.2}$$

for every real function λ on the photon sites of B; $\mathfrak{q}(\Psi)$ counts the ψ generators minus the $\bar{\psi}$ generators at fermion sites projecting to each photon site. We denote the even subspace by $\mathcal{G}^{\mathrm{ev}}(B)$.

A local lift of a configuration (A, z) on B is a real one-cochain $\widehat{A}$ on $B^{(1)}$ satisfying

$$d\widehat{A} = F \quad \text{on } B^{(2)}, \qquad e^{iq\widehat{A}(b)} = e^{iqA(b)} \quad (b \in B^{(1)}). \tag{3.3}$$

We evaluate G at this lift and then impose the fermion identifications of the torus or cylinder. We now verify that this procedure is Borel and independent of the lift.

Replace B by its integer box $\prod_\mu [a_\mu, b_\mu]$, where $a_\mu = \lceil y_\mu - r \rceil$ and $b_\mu = \lfloor y_\mu + r \rfloor$. If this box has no sites, use the unique empty lift. Otherwise its axial spanning tree consists of all links in direction 3, the links in direction 2 on $x_3 = a_3$, and the links in direction 1 on $x_2 = a_2$, $x_3 = a_3$. Prescribe a one-cochain on this tree. Its values on the remaining links are determined by

$$\begin{aligned}
\widehat{A}_1(x + e_2) &= \widehat{A}_1(x) + \widehat{A}_2(x + e_1) - \widehat{A}_2(x) - F_{12}(x), && x_3 = a_3, \\
\widehat{A}_1(x + e_3) &= \widehat{A}_1(x) + \widehat{A}_3(x + e_1) - \widehat{A}_3(x) - F_{13}(x), \\
\widehat{A}_2(x + e_3) &= \widehat{A}_2(x) + \widehat{A}_3(x + e_2) - \widehat{A}_3(x) - F_{23}(x).
\end{aligned} \tag{3.4}$$

First use the first line in increasing x_2, then the last two lines in increasing x_3. Each link outside the tree occurs as an unknown once. The construction is linear in F and the tree values. It gives the prescribed 13- and 23-curvatures and the 12-curvature on the bottom face. If $dF = 0$, induction over the cubes gives the remaining 12-curvatures: on each cube the two closed two-cochains F and $d\widehat{A}$ agree on five faces, hence on the sixth.

On the tree choose $\widehat{A}(b) = \arg_q(e^{iqA(b)})$, where $\arg_q$ takes values in $[0, r_q)$. This is a Borel choice. Subtract the recursion for A from (3.4). The tree differences lie in $r_q\mathbb{Z}$, and every subsequent difference is a signed sum of earlier ones and of a value of $r_q z$. Thus $\widehat{A} - A$ is integer-valued in units of r_q on every link. This proves (3.3), with Borel dependence on F and the tree phases.

Two local lifts differ by $r_q k$, where k is an integer closed one-cochain. Sum k along the tree paths from its bottom corner to define an integer function λ. Then $d\lambda = k$ on the tree. The zero-curvature recursion shows that $k - d\lambda = 0$ on every link. Equation (3.2), applied with $-r_q\lambda$, gives $a_\Psi(\widehat{A} + r_q d\lambda) = a_\Psi(\widehat{A})$, since $\langle \lambda, \mathfrak{q}(\Psi) \rangle$ is integer. The value of G is therefore independent of the lift.

For an open set U, let $\mathcal{A}(U)$ be the linear span of the maps

$$\widehat{A} \longmapsto \Phi(d\widehat{A}) \prod_{j=1}^{r} G_j(\widehat{A}|_{B_j^{(1)}}), \tag{3.5}$$

where the boxes $B_j \subset U$ are pairwise disjoint, $G_j \in \mathcal{G}^{\mathrm{ev}}(B_j)$, and Φ is a bounded Borel function of finitely many curvature values at plaquettes contained in U. A box B embeds in a period group Γ when the sets $B + [-1, 1]^3 + \gamma$, $\gamma \in \Gamma$, are pairwise disjoint. Boxes $B_1, \dots, B_r$ embed disjointly in Γ when each embeds and all sets $B_j + \gamma$, indexed by distinct pairs (j, γ), are pairwise disjoint. A product is read on the quotient only when its boxes embed disjointly.

It remains to check that equal maps (3.5) have equal values on configurations. This requires one lift for every box in a representation. On $\mathbb{Z}^3$, a closed integer two-cochain z has the integer primitive

$$\begin{aligned}
k_1(x) &= 0, \\
k_2(x) &= \sum_{s=0}^{x_1 - 1} z_{12}(s, x_2, x_3),
\end{aligned}$$

$$k_3(x) = \sum_{s=0}^{x_1-1} z_{13}(s, x_2, x_3) + \sum_{s=0}^{x_2-1} z_{23}(0, s, x_3).$$

The sums are oriented for negative endpoints. The first two components of $dk = z$ follow by taking differences. In the third, $dz = 0$ changes the summand into a difference of consecutive z_{23} values, so the sum telescopes. Therefore $\widehat{A} = A + r_q k$ is a lift on the whole lattice. It need not be periodic, even when (A, z) is periodic.

The same conclusion holds on a closed slab. Choose a coordinate μ_0 absent from the normal n_ℓ. The slab cell complex is a planar strip times $\mathbb{Z}$ in direction μ_0. Let $\mu < \nu$ be the other coordinates, denoted by (a, r). Write $Q(a, r)$ for the unit square with lower corner (a, r) and I_r for the interval of a such that $[(a, r), (a, r + 1)]$ lies in the strip. When I_r is nonempty choose $a_r \in I_r$. On the base sheet set the μ-link values of k to zero and put

$$k([(a, r), (a, r + 1)] \times \{0\}) = \sum_{a_r \le s < a} z(Q(s, r) \times \{0\}).$$

All squares in this oriented sum lie in the strip because I_r is an interval. Set $k = 0$ on the μ_0-links. For a horizontal link b in direction $\nu' \in \{\mu, \nu\}$, extend by

$$k(b \times \{j\}) = k(b \times \{0\}) + s_{\nu'} \sum_{0 \le j' < j} z(b \times [j', j' + 1]), \qquad s_{\nu'} = \begin{cases} -1, & \nu' < \mu_0, \\ +1, & \nu' > \mu_0. \end{cases}$$

The sums with reversed endpoints have the opposite sign. The resulting primitive has $dk = z$ on all vertical plaquettes and on the base sheet. The closed two-cochain $z - dk$ then has vanishing vertical components; closedness makes its horizontal component constant between sheets. It vanishes on the base sheet and hence everywhere. Again the primitive is taken on the lifted strip, not on its periodic quotient.

Evaluation at this common lift, followed by the quotient homomorphism on fermion generators, agrees with the separate local evaluations. It is linear in the map G, so it is independent of any representation of G as a sum of products. Moreover, different box plaquettes have different representatives on the quotient. Multiplying (3.1) by the Maxwell density therefore leaves

$$C \exp(-\varepsilon_G \|F\|^2), \qquad \varepsilon_G = \min(\tfrac{1}{2}, \varepsilon_1, \dots, \varepsilon_r) > 0. \tag{3.6}$$

The finite Grassmann expansion of the action has bounded coefficients for real A. The Gaussian sum (2.5) now proves absolute integrability, coefficient by coefficient. This also proves the required Borel measurability on the quotient.

3.2. **Boundary spaces and slab kernels.** Fix a mirror family ℓ and a cross-section $\Xi = (\Gamma_\parallel, \chi_\parallel)$, where $\Gamma_\parallel \subset \mathbb{Z}^3 \cap \{\ell = 0\}$ has rank two and systole at least four. Here and below, the systole of a period lattice Γ is $\min_{\gamma \in \Gamma \setminus \{0\}} |\gamma|_\infty$. Work on the cylinder $\mathbb{Z}^3/\Gamma_\parallel$. The admissible mirror positions are $c \in \frac{1}{2}|n_\ell|^2\mathbb{Z}$. Write $\mathcal{M}^j(c)$ for the j-cells contained in $\{\ell = c\}$, $\mathcal{X}^j(c)$ for those cut through their relative interiors by this plane, and $\mathcal{B}^j(c, c')$ for the j-cells contained in $\{c \le \ell \le c'\}$ and in neither bounding plane. In particular a slab cell may touch either boundary. All these sets are finite on a bounded slab.

For a chain of admissible mirrors, the cylinder cells partition into the cells in its mirrors, those cut by them, and those lying between consecutive mirrors. A cell cut by c has its height range $[c - |n_\ell|^2/2, c + |n_\ell|^2/2]$. Its non-cut faces lie on one side of the plane; the other faces are cut by the same plane. This partition applies also when two mirrors are consecutive. A fermion bond crosses at most one mirror of such a chain, and no fermion site lies in a mirror.

A closed-slab configuration $\varpi = (A, z)$ assigns real link values and integer plaquette values to all cells of the closed slab, with $dz = 0$ on its cubes. We integrate its interior values using

$A(b) \in [0, r_q)$ and counting measure for z. Its interior photon weight is

$$W_{c,c'}(\varpi) = \prod_{\sigma \in \mathcal{B}^2(c,c')} e^{-F_\sigma^2/2} \prod_{Q \in \mathcal{B}^3(c,c')} \mathbf{1}[(dz)(Q) = 0]. \tag{3.7}$$

We use the larger class $\mathcal{I}_\Xi[c, c']$ of Borel Grassmann-valued functions of closed-slab configurations, with fermion support $c < \ell < c'$, invariant under $(A, z) \mapsto (A + r_q k, z - dk)$, whose coefficients obey

$$|a_\Psi(\varpi)| \le C_G \exp\left((\tfrac{1}{2} - \varepsilon_G) \sum_{\sigma \in \mathcal{B}^2(c,c')} F_\sigma^2 \right), \qquad 0 < \varepsilon_G \le \tfrac{1}{2}. \tag{3.8}$$

The local products supported in the open slab give elements of this class by the lift construction above. Products in consecutive slabs remain in the class, with the smaller of the two positive ε's. Reflection and translation also preserve the class.

At the mirror c, let $\mathcal{J}(c)$ be the ordered set of crossing modes from (2.28), taken modulo $\Gamma_\parallel$. Let $\mathcal{F}(c)$ be the exterior space with orthonormal basis e_I, $I \subset \mathcal{J}(c)$. Put $\omega_c e_I = \omega_I e_I$ and $U_{\mathrm{f}} e_I = (-1)^{|I|} e_I$. Translation of the modes includes both the character $\chi_\parallel$ and the sign required to reorder their product:

$$U_v e_I = s_v(I) e_{r_v(I)}, \qquad \tau_v \zeta_I^{(c)} = s_v(I) \zeta_{r_v(I)}^{(c+\ell(v))}.$$

The U_v are unitary, satisfy the translation law, and commute with parity. For an integer translation parallel to the normal, choose the orders compatibly so that $s_v(I) = 1$.

For a Grassmann element H supported in the open slab define

$$N_{c,c'}[H]_{I',I} = \int_{(c,c')} \zeta_I^{(c)} H \Theta_{c'}(\zeta_{I'}^{(c')}). \tag{3.9}$$

An operator Υ on $\mathcal{F}(c)$ is represented at the mirror by

$$\mathrm{Ins}_c[\Upsilon] = \sum_{I,J} \Upsilon_{IJ} \Theta_c(\zeta_J^{(c)}) \zeta_I^{(c)}.$$

The particular insertion $\mathrm{Ins}_c[\omega_c \,\mathrm{diag}(u_I(A))]$ is the crossing exponential (2.30). The ordering in these definitions gives

$$N_{c,c''}[H \,\mathrm{Ins}_{c'}[\Upsilon] H'] = N_{c',c''}[H'] \Upsilon N_{c,c'}[H]. \tag{3.10}$$

Indeed, an entry of the left side is a sum of products $[\zeta_I H \Theta_{c'} \zeta_J][\zeta_K H' \Theta_{c''} \zeta_{I''}]$ in disjoint sets of slab generators. Berezin factorization gives the entries on the right without moving any of these factors. Reflecting the integrand reverses their order and conjugates the Berezin coefficient; with $v = 2(c' - c) n_\ell / |n_\ell|^2$ this gives

$$N_{c',2c'-c}[\Theta_{c'} H] = U_v N_{c,c'}[H]^*. \tag{3.11}$$

We next include the photon crossing in the boundary data. The cut incidence maps d_c and d_c^+ are the restrictions of the coboundary to cut faces and to upper faces, as in (2.35). Introduce one angle ϑ_Q for each cut cube and one integer k_σ for each cut plaquette, and set

$$\begin{aligned} \varkappa_\sigma &= r_q^{-1}((d_c^* \vartheta)_\sigma - 2\pi k_\sigma), \\ \mathrm{d}\mu_c^\times &= \prod_{Q \in \mathcal{X}^3(c)} \frac{\mathrm{d}\vartheta_Q}{2\pi} \prod_{\sigma \in \mathcal{X}^2(c)} \frac{q}{\sqrt{2\pi}} e^{-\varkappa_\sigma^2/2}, \\ e_c((\vartheta, k); \varpi) &= e^{i\langle \vartheta, d_c^+ z \rangle - i \langle \varkappa, d_c^+ A \rangle}, \\ D_c(k, I) &= r_q^{|\mathcal{X}^1(c)|} \mathbf{1}[d_c^* k + \mathfrak{q}(I) = 0]. \end{aligned} \tag{3.12}$$

Here $\mathfrak{q}(I)(b)$ is the number of modes of groups $1,2$ over the cut link b, minus the number of groups $3,4$ over it. There are no cut links at diagonal mirrors. At a coordinate site plane there are no cut cells at all; its Fourier space is a point.

The one-mirror photon identity proved by the Fourier and Poisson calculation of the preceding section is

$$\sum_{z^\times}\int \mathrm{d}A^\times \prod_{\sigma\in\mathcal{X}^2(c)} e^{-F_\sigma^2/2} \prod_{Q\in\mathcal{X}^3(c)} \mathbf{1}[(dz)(Q)=0]\, u_I(A^\times) = \int \mathrm{d}\mu_c^\times\, D_c(k,I) e_c((\vartheta,k);\varpi_+)\overline{e_c((\vartheta,k);\theta_c^*\varpi_-)}. \tag{3.13}$$

The upper and lower data on the right have the same values on the mirror. Their upper and reflected lower faces provide the non-cut faces on the left. Cube Fourier multipliers enforce closedness, Poisson summation gives the factors $e^{-\varkappa^2/2}$, and compact cut-link integration gives $D_c(k,I)$. This identity is used in the direction shown when constructing a kernel and in the opposite direction when gluing two kernels.

Mirror variables are $\xi=(\mathfrak{m},\vartheta,k,I)$, with $\mathfrak{m}=(A_0,z_0)\in[0,r_q)^{\mathcal{M}^1(c)}\times\mathbb{Z}^{\mathcal{M}^2(c)}$. On their space Y_c put

$$\mathrm{d}\mu_c(\xi)=\mathrm{d}A_0 \prod_{\sigma\in\mathcal{M}^2(c)} e^{-F_\sigma(\mathfrak{m})^2/2}\mathrm{d}\mu_c^\times(\vartheta,k) D_c(k,I), \qquad \mathcal{H}(c)=L^2(Y_c,\mu_c). \tag{3.14}$$

The integer and mode variables carry counting measure. This is a finite positive measure. Summing the mirror plaquette Gaussians gives $C_q^{|\mathcal{M}^2(c)|}$, while Poisson summation at fixed angles gives

$$\frac{q}{\sqrt{2\pi}}\sum_{k_\sigma} e^{-\varkappa_\sigma^2/2} \le \sum_{j\in\mathbb{Z}} e^{-r_q^2 j^2/2}.$$

The remaining factors are bounded by $r_q^{|\mathcal{M}^1(c)|+|\mathcal{X}^1(c)|}2^{|\mathcal{J}(c)|}$. The induced signed translations $V_v:\mathcal{H}(c)\to\mathcal{H}(c+\ell(v))$ are unitary and satisfy the group law. The parity operator multiplies the component I by $(-1)^{|I|}$.

Let $S_{(c,c')}(A)$ be the fermion action containing precisely the mass and bond terms with fermion endpoints in the open slab. For $G\in\mathcal{I}_\Xi[c,c']$, define the kernel

$$\mathfrak{S}[G](\xi',\xi)=\sqrt{\omega_{I'}\omega_I}\int \mathrm{d}\varpi^\circ\, W_{c,c'}(\varpi) e_c((\vartheta,k);\varpi)\overline{e_{c'}((\vartheta',k');\theta_{c'}^*\varpi)} \cdot N_{c,c'}[G(\varpi)e^{-S_{(c,c')}(A)}]_{I',I}, \qquad \varpi=\mathfrak{m}\vee\varpi^\circ\vee\mathfrak{m}'. \tag{3.15}$$

Set the whole integrand to zero where $W_{c,c'}(\varpi)=0$; $G(\varpi)$ is evaluated only on the closed configurations in its domain. The associated integral operator maps $\mathcal{H}(c)$ to $\mathcal{H}(c')$. The square roots of the mode weights belong in the kernel; the measure (3.14) contains none of these weights. This places one full crossing weight at a glued mirror.

Theorem 3.1. *Every $\mathfrak{S}[G]$ is Hilbert–Schmidt and depends linearly on G. If G has constants C_G,ε_G in* (3.8)*, then*

$$\begin{aligned} |\mathfrak{S}[G](\xi',\xi)| &\le C_{c,c'}C_G C_q(\varepsilon_G)^{|\mathcal{B}^2(c,c')|},\\ \|\mathfrak{S}[G]\|_2 &\le \sqrt{\mu_c(Y_c)\mu_{c'}(Y_{c'})}C_{c,c'}C_G C_q(\varepsilon_G)^{|\mathcal{B}^2(c,c')|}. \end{aligned} \tag{3.16}$$

The constant depends on the fixed cross-section, mirror positions, q,M,ρ,w_i, and chamber base point. For consecutive slabs,

$$\mathfrak{S}[G']\mathfrak{S}[G]=\mathfrak{S}[GG'],$$

$$\mathfrak{S}[\tau_v G] = V_v \mathfrak{S}[G] V_v^{-1}, \qquad \mathfrak{S}[\Theta_{c'} G] = V_{2(c'-c)n_\ell/|n_\ell|^2} \mathfrak{S}[G]^*. \tag{3.17}$$

If $\mathbb{T} = (\Gamma, \chi)$ has cross-section Ξ, $g \in \Gamma$ is a closing vector with minimal positive height $p = \ell(g)$, and $c_0 < \cdots < c_r = c_0 + p$ are admissible, $r \geq 2$, and $G_j \in \mathcal{I}_\Xi[c_{j-1}, c_j]$ for every j, then

$$Z_{\mathbb{T}}[G_1 \cdots G_r] = \mathrm{Tr}_{\mathcal{H}(c_0)} \left(V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{S}[G_r] \cdots \mathfrak{S}[G_1] \right). \tag{3.18}$$

These identities apply to arbitrary parities. For homogeneous G, $U_{\mathrm{f}} \mathfrak{S}[G] = (-1)^{\deg G} \mathfrak{S}[G] U_{\mathrm{f}}$.

Proof. The coefficient of the finite Grassmann polynomial $e^{-S_{(c,c')}(A)}$ is bounded uniformly in real A. Each coefficient in the matrix in (3.15) is consequently bounded by a fixed multiple of the majorant for G. The phases have modulus one. After multiplication by the photon weight, the absolute integrand is bounded by

$$C C_G \prod_{\sigma \in \mathcal{B}^2(c,c')} e^{-\varepsilon_G F_\sigma^2}.$$

Drop the cube constraints and sum the integers separately. Each sum is at most $C_q(\varepsilon_G)$; each compact link integral contributes r_q. The mode factors and finite matrix dimension enter $C_{c,c'}$. This proves the pointwise bound in (3.16). The boundary measures are finite, so the kernel is in $L^2(Y_{c'} \times Y_c)$, with the stated norm. These bounds also justify all the gluing integrals below before any cancellation.

For composition, multiply the two kernels and integrate over the common boundary. Their two square roots give $\omega_{I'}$ for the middle mode. The two phases and its boundary measure are precisely the right side of (3.13). Applying that identity restores the photon variables on the cut cells, their Maxwell weights, their cube constraints, and the phase $u_{I'}$ of the crossing modes. Sum the middle modes by (3.10), with $\Upsilon = \omega_{c'} \operatorname{diag}(u_{I'})$. This restores the crossing fermion action. The plaquettes in the middle mirror come from its boundary measure, and the cell partition restores the full weight on the combined slab. No cube lies in a mirror, so every closedness constraint is present once. This is the kernel of $\mathfrak{S}[GG']$.

Translation acts by a signed relabeling of the modes and an orientation-preserving relabeling of cells, proving covariance. For reflection use (3.11); the two boundary phases exchange their roles and the interior weight is preserved. Reflection may reverse a compact integration interval, but integer shift invariance makes any interval of length r_q equivalent to $[0, r_q)$. The reflected upper boundary is the translate of the original lower one by $2(c'-c)n_\ell/|n_\ell|^2$. This gives the last identity of (3.17). Parity follows because a Berezin integral of odd total degree vanishes.

For the trace, composition reduces the claim to two slabs. A product of two Hilbert–Schmidt kernels is trace class, and its trace is the absolutely convergent integral of their cyclic product. The closing operator identifies the upper mirror data with the translate by g of the lower data. Its contribution to this cyclic kernel is

$$(-\chi(g))^{|I_0|} s_g(I_0).$$

Here s_g is the mode relabeling sign; $(-\chi(g))^{|I_0|}$ is the remaining Grassmann closing sign. It is 1 for antiperiodic closure and the fermion parity for periodic closure. To see this sign directly, translate the lower crossing product to the upper boundary. Each of its $|I_0|$ generators contributes $\chi(g)$, while moving it across the adjacent upper product contributes $(-1)^{|I_0|}$ in every term of even total degree. Their product is $(-\chi(g))^{|I_0|}$.

Apply (3.13) at the two mirrors. For fixed photon variables the remaining mode sum is

$$\mathrm{tr}_{\mathcal{F}(c_0)}(U_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} N_2 \Upsilon_1 N_1 \Upsilon_0), \qquad \Upsilon_j = \omega_{c_j} \operatorname{diag}(u_I(A)).$$

Berezin composition and closure replace this trace by the full torus Berezin integral. The two open-slab actions and the two crossing actions include each mass and hopping term once. The photon variables likewise include every torus link and plaquette once, and the cut and interior cube constraints impose $dz = 0$ everywhere. Their measure is therefore the measure in (2.17). This proves (3.18) for even total degree. With odd total degree both sides vanish, the operator under the trace anticommuting with parity. Finally, expand nonhomogeneous entries into their even and odd parts. □

3.3. **Reflected folds and Schatten estimates.** For an operator Y between Hilbert spaces, write $\|Y\|_s = (\operatorname{Tr}(Y^*Y)^{s/2})^{1/s}$. We use Schatten's inequality in the following form. If $1 \le s_j < \infty$, $Y_j : H_{j-1} \to H_j$ belongs to the Schatten class of exponent s_j, $H_r = H_0$, and $\sum_j s_j^{-1} = 1$, then the cyclic product is trace class and

$$|\operatorname{Tr}(Y_r \cdots Y_1)| \le \prod_{j=1}^{r} \|Y_j\|_{s_j}. \tag{3.19}$$

The statement for different spaces follows from the usual inequality on their orthogonal direct sum: extend each Y_j by zero on all other summands. The nonzero block of the product is the displayed cyclic product, and the singular values of each extension are those of Y_j. A unitary closing identification can be absorbed into Y_r without changing its Schatten norm.

Let $\mathbb{T}$ have height period p in the family ℓ. Its straightened torus $\mathbb{T}^0$ has the same cross-section and the closing vector

$$g^0 = \frac{p}{|n_\ell|^2} n_\ell, \qquad \Gamma^0 = \Gamma_\parallel \oplus \mathbb{Z} g^0, \qquad \chi^0(g^0) = -1. \tag{3.20}$$

We use this torus only when $g^0 \in \mathbb{Z}^3$ and its lattice has systole at least four. Its volume is the volume of $\mathbb{T}$: both lattices have the same covolume in the mirror and the same normal height period.

Divide one period into $2k$ slabs of equal height h', with all $c_j = c_0 + jh'$ admissible and $p = 2kh'$. For an integrand G in one of these slabs, form its fold by reflecting it successively in the bounding planes, taking all $2k$ copies. This is an integrand on $\mathbb{T}^0$. Its fermion boxes are disjoint: they lie strictly between different consecutive mirrors; within each slab the cross-section embedding separates their translates.

Proposition 3.2. *For integrands $G_j \in \mathcal{I}_\Xi[c_{j-1}, c_j]$,*

$$Z_{\mathbb{T}^0}[\operatorname{fold}_\ell G_j] = \|\mathfrak{S}[G_j]\|_{2k}^{2k},$$

$$|Z_{\mathbb{T}}[G_1 \cdots G_{2k}]| \le \prod_{j=1}^{2k} Z_{\mathbb{T}^0}[\operatorname{fold}_\ell G_j]^{1/(2k)}. \tag{3.21}$$

In particular every folded factor on the right is nonnegative.

Proof. Translate the starting slab, if necessary, to $[c_0, c_1]$, and put $Y = \mathfrak{S}[G]$. Let $v = 2h' n_\ell / |n_\ell|^2$, which is integer because both mirror positions are admissible. The first reflected pair has product $V_v Y^* Y$ by (3.17). Its translate by sv has product $V_{(s+1)v} Y^* Y V_{sv}^{-1}$. Multiplying all k pairs telescopes to

$$V_{kv} (Y^* Y)^k.$$

Since $kv = g^0$ and $\chi^0(g^0) = -1$, the closing translation in (3.18) cancels V_{g^0} and introduces no parity. The trace is $\operatorname{Tr}(Y^*Y)^k$, proving the first identity. Translation of a different starting slab changes neither this trace nor the singular values.

For the original torus, the operator $V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2}$ is unitary. Apply (3.19) to its trace formula, with every exponent equal to $2k$. The first identity identifies the resulting norms with folded

partition functions. The Hilbert–Schmidt estimate ensures all operators belong to the required Schatten classes. □

We next apply this inequality to cells. A frame is a triple of the mirror families with mutually orthogonal normals n_1, n_2, n_3. The two frames needed below are (x_1, x_2, x_3) and $(x_1 - x_2, x_1 + x_2, x_3)$. Choose admissible offsets o_a and an integer height width $h > 0$. The open cells are

$$Q_\kappa = \{x : o_a + h\kappa_a < \ell^a(x) < o_a + h(\kappa_a + 1),\ a = 1, 2, 3\}, \qquad |Q| = \frac{h^3}{|n_1||n_2||n_3|}. \tag{3.22}$$

The reflections in the planes $\ell^a = o_a + hj$ generate a product of three infinite dihedral groups. Exactly one element of this group carries a given open cell to any specified open cell. Its action on even Grassmann-valued functions is generated by the reflections Θ and translations used above. Denote the resulting copy of $G \in \mathcal{A}(Q_\kappa)$ in $Q_{\kappa'}$ by $G^{[\kappa']}$.

A rectangular torus in this frame has periods $\gamma_a = p_a n_a/|n_a|^2 \in \mathbb{Z}^3$, with $\chi(\gamma_a) = -1$. If each $p_a \in 2h\mathbb{Z}$, the copies fit its period lattice. Let $N = |\mathbb{T}|/|Q|$ be the number of cells, and define

$$\mathrm{Fold}_{\mathbb{T}}\, G = \prod_{\kappa' \bmod \Gamma} G^{[\kappa']}, \qquad \|G\|_{\mathbb{T}} = Z_{\mathbb{T}}[\mathrm{Fold}_{\mathbb{T}}\, G]^{1/N}. \tag{3.23}$$

This is an unnormalized seminorm. Its defining partition function is nonnegative, by the first identity of (3.21) applied to one layer of the folded array. The value is independent of the chosen representatives of the cells.

Theorem 3.3. *If a rectangular torus $\mathbb{T}$ carries the cells* (3.22)*, then*

$$\left| Z_{\mathbb{T}} \left[\prod_\kappa G_\kappa \right] \right| \le \prod_\kappa \|G_\kappa\|_{\mathbb{T}}, \qquad G_\kappa \in \mathcal{A}(Q_\kappa). \tag{3.24}$$

Let $C(a, b, c)$ have coordinate periods ae_1, be_2, ce_3, all with character -1. Let $e_- = e_1 - e_2$, $e_+ = e_1 + e_2$, and let $D(a, b, c)$ have periods ae_-, be_+, ce_3, all with character -1. Let $D'(a, b, c)$ have the same periods and characters, except for $\chi(be_+) = +1$. For the diagonal frame and $n \in 2h\mathbb{Z}$, $n \ge 8$, the cells have representatives

$$0 \le \kappa_1 < n/h, \qquad 0 \le \kappa_2 < 2n/h, \qquad 0 \le \kappa_3 < n/h.$$

They represent the cells of $C(n, n, n)$, $D'(n/2, n, n)$, and $D(n/2, n, n)$, and

$$\left| Z_{\mathbb{T}} \left[\prod_\kappa G_\kappa \right] \right| \le \prod_\kappa \|G_\kappa\|_{D(n/2,n,n)}, \quad \mathbb{T} = C(n, n, n) \text{ or } D'(n/2, n, n). \tag{3.25}$$

For coordinate cells of width h, $a, c \in 2h\mathbb{Z}$, $a, c \ge 4$, one has

$$\left| Z_{D(a,2a,c)} \left[\prod_\kappa G_\kappa \right] \right| \le \prod_\kappa \|G_\kappa\|_{C(a,4a,c)}. \tag{3.26}$$

Proof. Group the cell integrands into layers perpendicular to the first normal, and apply (3.21). Its right side consists of the arrays obtained by retaining one layer and reflecting it throughout the straightened torus. Next apply the same inequality to each such array in direction two, and then in direction three. The reflected array after these three steps is the full fold of the integrand from one original cell. The three exponents multiply to

$$\frac{h}{p_1}\frac{h}{p_2}\frac{h}{p_3} = \frac{1}{N}.$$

Thus the last product consists of one factor $\|G_\kappa\|_{\mathbb{T}_3}$ for each original cell, where $\mathbb{T}_3$ is the final rectangular torus.

We verify that all three arrays have consistent periods and that the same cells represent them. If Γ_0 is the initial lattice, then straightening successively in the frame gives

$$\Gamma_1 = (\Gamma_0 \cap \{\ell^1 = 0\}) \oplus \mathbb{Z}\gamma_1,$$
$$\Gamma_2 = (\Gamma_0 \cap \{\ell^1 = \ell^2 = 0\}) \oplus \mathbb{Z}\gamma_1 \oplus \mathbb{Z}\gamma_2,$$
$$\Gamma_3 = \mathbb{Z}\gamma_1 \oplus \mathbb{Z}\gamma_2 \oplus \mathbb{Z}\gamma_3, \qquad \gamma_a = p_a n_a/|n_a|^2.$$

The kernel in the second line is a rank-one group along n_3; its positive generator is γ_3, by the definition of the third height period. Characters already fixed at γ_1, γ_2 are retained in subsequent cross-sections, so all three final characters are -1. The brick $0 \le \kappa_a < p_a/h$ represents Γ_3. It represents Γ_2 as well; then its section $\kappa_2 = 0$ represents the second cross-section, and its section $\kappa_1 = 0$ represents the first one. Restoring the normal coordinate in each case gives representatives for the preceding lattice. These groups all have the same covolume.

We also check the stronger box embedding condition used in the local classes. If $B = y + [-r, r]^3$ lies in an open cell, then $2|n_a|^2 r < h$ for every a, by taking the range of ℓ^a on B. For a nonzero period γ, at least one height is nonzero and $|\ell^a(\gamma)| \ge 2h$. Consequently $|\gamma|_\infty > 4r$, and the systole also gives $|\gamma|_\infty \ge 4$. These inequalities imply $|\gamma|_\infty > 2r + 2$: use the second if $r < 1$, the first if $r \ge 1$. Thus translates of $B + [-1, 1]^3$ are disjoint. Boxes in different cell orbits cannot meet after period translations, since the open cells themselves are disjoint. It follows that every layer belongs to its slab class. This justifies the three applications.

For a rectangular torus, every straightening leaves the torus unchanged, which proves (3.24). For a cubic torus with even side n, the diagonal operations are

$$C(n,n,n) \xrightarrow{\ell_-} D'(n/2,n,n) \xrightarrow{\ell_+} D(n/2,n,n), \qquad D(n/2,n,n) \xrightarrow{x_1} C(n/2,2n,n). \tag{3.27}$$

For the first operation the cross-section is generated by ne_+, ne_3, with characters $+1, -1$; its height period is n, and straightening replaces the closing vector ne_1 by $(n/2)e_-$. For the second, the cross-section is generated by $(n/2)e_-, ne_3$, both with character -1. Its closing vector ne_+ already points along the normal, and the operation changes its character from $+1$ to -1. For the third, the cross-section is generated by $2ne_2, ne_3$, both with character -1; the height period is $n/2$ and the straight closing vector is $(n/2)e_1$. The lattices have systole at least four.

The first two diagonal operations, followed by the unchanged x_3 operation, have height periods $(n, 2n, n)$. Hence they give (3.25); starting at D' instead, use the order (ℓ_+, x_3, ℓ_-) to obtain the same final fold. Likewise, starting at $D(a, 2a, c)$ and using coordinate directions gives periods $(a, 4a, c)$ and ends at $C(a, 4a, c)$, proving (3.26). □

3.4. **Pressure and the comparison error.** Write $C(n) = C(n,n,n)$ and

$$f(n_1,n_2,n_3) = \frac{1}{n_1 n_2 n_3} \log Z_{C(n_1,n_2,n_3)}[1].$$

The mass and charge in this function are held fixed. The following error will compare local expectations in the cubic and diagonal geometries.

Proposition 3.4. *The pressure f is nonincreasing in each side, and*

$$f(n_1,n_2,n_3) \longrightarrow f_\infty := \inf f \in \mathbb{R} \quad \text{as } \min_\mu n_\mu \longrightarrow \infty.$$

For every even $n \ge 8$,

$$Z_{C(n)}[1] \le Z_{D'(n/2,n,n)}[1] \le Z_{D(n/2,n,n)}[1] \le Z_{C(n/2,2n,n)}[1]. \tag{3.28}$$

With $\Delta(n) = f(n/2, 2n, n) - f(n,n,n)$,

$$0 \le \Delta(n) \le f(n/2,n/2,n/2) - f_\infty \longrightarrow 0. \tag{3.29}$$

The logarithms of any two partition functions in (3.28) *differ in absolute value by at most $n^3\Delta(n)$.*

Proof. Fix the cross-section with sides n_2, n_3 and let $Y = \mathfrak{S}[1] : \mathcal{H}(0) \to \mathcal{H}(1/2)$ be the coordinate half-step. The first identity of (3.21), applied to $2n_1$ half-slabs, gives

$$Z_{C(n_1,n_2,n_3)}[1] = \operatorname{Tr}(Y^*Y)^{n_1}. \tag{3.30}$$

The positive trace-class operator Y^*Y is independent of n_1 and is nonzero, since the partition function is positive. Consequently

$$f(n_1, n_2, n_3) = \frac{1}{n_2 n_3} \log \|Y^*Y\|_{n_1}.$$

Schatten norms decrease as their exponent increases, so f decreases in n_1. Permuting the time direction proves the other two claims. The same trace identity shows separate convexity of $\log Z$ in the side lengths: Hölder's inequality gives $\operatorname{Tr} A^{(1-t)s+ts'} \le (\operatorname{Tr} A^s)^{1-t}(\operatorname{Tr} A^{s'})^t$ for $A = Y^*Y$.

The lower partition bound (2.22) gives the volume-independent floor

$$f \ge \log C_* + 3 \log r_q - 6 r_q^2.$$

Together with the upper bound in the same equation, it makes f_∞ finite. Monotonicity now gives convergence in any manner with all sides tending to infinity: choose a finite triple at which $f < f_\infty + \delta$ and then exceed its three sides.

Apply (3.21) with constant integrands to each of the three operations (3.27). A fold of the constant is the constant, so $Z_{\mathbb{T}}[1] \le Z_{\mathbb{T}^0}[1]$. This proves the chain. Each torus in it has volume n^3, so its first and last logarithms differ by $n^3\Delta(n)$. Side monotonicity gives $f(n/2, 2n, n) \le f(n/2, n/2, n/2)$, while $f(n, n, n) \ge f_\infty$, proving (3.29). All intermediate logarithms lie between the endpoints, which proves the last assertion. □

Thus the estimate concerns the logarithm per unit volume. It does not give a bounded ratio of partition functions: the retained bound is $e^{n^3\Delta(n)}$, with $\Delta(n) \to 0$. The local comparison divides the logarithmic error bound $n^3\Delta(n)$ by the number of copies of the observable and by the perturbation strength.

3.5. **Coordinate slices and the all-flux trace.** For Gaussian domination we use a slice representation with real spatial connections. Take direction 1 as time on $C_{\mathbf{n}} = C(n_1, n_2, n_3)$, $\mathbf{n} = (n_1, n_2, n_3)$, $n_\mu \ge 4$, and let $\Sigma = \mathbb{Z}/n_2\mathbb{Z} \times \mathbb{Z}/n_3\mathbb{Z}$. Its links and plaquettes are $\Sigma^{(1)}$ and $\Sigma^{(2)}$, its coboundary is d_s, and $v = |\Sigma| = n_2 n_3$. A connection has spatial slices A_j and temporal links a_j, and its curvature has the parts

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_s a_j + r_q z_j^{\mathrm{el}}, \qquad F_j^{\mathrm{mag}} = d_s A_j + r_q z_j^{\mathrm{mag}}. \tag{3.31}$$

Closedness reads $z_{j+1}^{\mathrm{mag}} - z_j^{\mathrm{mag}} = d_s z_j^{\mathrm{el}}$.

We first unfold the integer variables. Put $k_0 = 0$, $k_{j+1} = k_j + z_j^{\mathrm{el}}$. Closedness is equivalent to

$$z_j^{\mathrm{mag}} = z_0^{\mathrm{mag}} + d_s k_j, \qquad d_s k_{n_1} = 0.$$

The integer shift with spatial values k_j and zero temporal values removes all electric integers except the last one, k_{n_1}, and changes every spatial slice to $A_j + r_q k_j$. Summing k_j while integrating A_j over its compact box unfolds the intermediate slices to all of $\mathbb{R}^{\Sigma^{(1)}}$. The last integer remains as a winding vector

$$g = r_q k_{n_1} \in \Gamma_s, \qquad \Gamma_s = r_q(\mathbb{Z}^{\Sigma^{(1)}} \cap \ker d_s).$$

Fix a plaquette σ_0 of Σ and write $k = \sum_\sigma z_0^{\mathrm{mag}}(\sigma)$. Then $z_0^{\mathrm{mag}} - k\mathbf{1}_{\sigma_0} = d_s k'$ for an integer one-cochain k'. Integer exactness follows from the dual graph whose vertices are the plaquettes. A link of Σ is a dual edge directed toward the plaquette where its coefficient in d_s is positive. The dual graph is connected. Fix a root and a spanning tree. Unit signed flow along the path from the root to a vertex σ has divergence $\delta_\sigma - \delta_{\sigma_0}$. Summing these paths with the integer coefficients of

a zero-sum two-cochain gives precisely that cochain as its divergence, hence as $d_s k'$. Two choices of k' differ by an integer element of $\ker d_s$.

Shift every slice by $r_q k'$. The magnetic integers now equal $k\mathbf{1}_{\sigma_0}$ in every slice. Summation over k' modulo closed integer cochains replaces the compact box for A_0 by a fundamental domain of Γ_s. We arrive at paths

$$A_0 \in \mathbb{R}^{\Sigma^{(1)}}/\Gamma_s, \qquad A_j \in \mathbb{R}^{\Sigma^{(1)}}\ (0 < j < n_1), \qquad A_{n_1} = A_0 + g, \quad g \in \Gamma_s,$$

with $a_j \in [0, r_q)^{\Sigma}$ and flux $k \in \mathbb{Z}$. Their curvatures are

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_s a_j, \qquad F_j^{\mathrm{mag}} = d_s A_j + r_q k \mathbf{1}_{\sigma_0}. \tag{3.32}$$

Thus integration over the original configuration quotient equals summation over k, g and integration over these paths. For a nonnegative integrand the identity follows by Tonelli's theorem. For an absolutely integrable complex integrand it follows by applying that identity to its modulus and then using Fubini's theorem. Each change above preserves link phases and curvature, so it applies to the determinant and to all curvature observables used here.

There are two finite Fock spaces in this representation: $\mathcal{F}'$ at a coordinate site plane and $\mathcal{F}$ at a link plane. The matrix of the open half-slab gives a map $N(A) : \mathcal{F}' \to \mathcal{F}$, and we include both crossing weights by writing

$$\widehat{N}(A) = \omega^{1/2} N(A) (\omega')^{1/2}.$$

Let $u(a)$ be the diagonal unitary on $\mathcal{F}$ with entries $e^{iq\langle a, \mathfrak{q}(I)\rangle}$. The fermion trace is

$$\det \mathcal{D}_A^{C_{\mathbf{n}}} = \mathrm{tr}_{\mathcal{F}'} \prod_{j=n_1-1,\dots,0} \big(\widehat{N}(A_{j+1})^* u(a_j) \widehat{N}(A_j)\big). \tag{3.33}$$

This formula also holds on an unfolded path, since $A_{n_1} - A_0 \in \Gamma_s$ leaves $\widehat{N}$ unchanged.

To derive (3.33), split the torus at each site and link plane. The half-slab above a link plane is the reflected half-slab below it, hence its matrix is $N(A_{j+1})^*$. The crossing at a site plane has no link phase and gives ω'. At a link plane it gives $\omega^{1/2} u(a_j) \omega^{1/2}$. Compose the matrices and close antiperiodically; the closing parity is the identity. Cyclicity moves half of the final ω' to the front, giving (3.33). Every entry of $\widehat{N}(A)$ is a polynomial in the link phases: the half-slab action is quadratic in finitely many generators, its exponential terminates, and Berezin integration extracts a coefficient. Consequently

$$C_N := \sup_{A \in \mathbb{R}^{\Sigma^{(1)}}} \|\widehat{N}(A)\| < \infty, \qquad \widehat{N}(A + r_q k') = \widehat{N}(A) \quad (k' \in \mathbb{Z}^{\Sigma^{(1)}}). \tag{3.34}$$

On real slice connections the two Hilbert spaces are

$$H^s = \bigoplus_{k \in \mathbb{Z}} L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s; \mathcal{F}'), \qquad H^l = \bigoplus_{k \in \mathbb{Z}} L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s; \mathcal{F}).$$

All quotient measures are induced by Lebesgue measure. Define commuting self-adjoint electric operators E_b by $e^{-i\langle x, E\rangle}\Phi(A) = \Phi(A - x)$. On H^l let

$$U_G(a)\Phi(A) = u(a)\Phi(A - d_s a), \qquad P_G = r_q^{-v} \int_{[0,r_q)^{\Sigma}} U_G(a)\, \mathrm{d}a, \qquad K = r_q^v e^{-|E|^2/2} P_G. \tag{3.35}$$

The integer mode charges make U_G a representation of the compact gauge group $\mathbb{R}^{\Sigma}/r_q\mathbb{Z}^{\Sigma}$. Its average P_G is the orthogonal projection onto invariant vectors. It commutes with every function of E, because $u(a)$ acts on the Fock index and all spatial translations commute. Hence

$$K^{1/2} = r_q^{v/2} e^{-|E|^2/4} P_G. \tag{3.36}$$

For a real slice two-cochain b, define the multiplication operator $M_b : H^s \to H^l$ by

$$(M_b\Phi)_k(A) = e^{-\|d_sA + r_qk\mathbf{1}_{\sigma_0} + b\|^2/4}\widehat{N}(A)\Phi_k(A), \qquad X_b = K^{1/2}M_b. \tag{3.37}$$

The magnetic weight is bounded and $\widehat{N}$ is bounded, so M_b is bounded. We must prove that X_b is Hilbert–Schmidt; the real quotient is not compact and there are infinitely many flux sectors.

The periodized Gaussian

$$\widetilde{g}_{1/2}([x]) = \sum_{\gamma\in\Gamma_s}\prod_{b\in\Sigma^{(1)}} \pi^{-1/2}e^{-(x_b+\gamma_b)^2}$$

has integral one on the quotient and is bounded. Indeed $\Gamma_s \subset r_q\mathbb{Z}^{\Sigma^{(1)}}$, so the series is dominated by the product of one-dimensional Gaussian sums on that larger lattice. It is therefore in $L^1 \cap L^\infty \subset L^2$. The kernel of $K^{1/2}$ is

$$K^{1/2}(A', A) = r_q^{-v/2}\int_{[0,r_q)^\Sigma} \widetilde{g}_{1/2}([A' - A - d_sa])u(a)\,\mathrm{d}a.$$

Its operator norm is bounded by the nonnegative function

$$\widehat{g}([x]) = r_q^{-v/2}\int_{[0,r_q)^\Sigma} \widetilde{g}_{1/2}([x - d_sa])\,\mathrm{d}a,$$

which has finite L^1 and L^∞ norms and hence finite L^2 norm. Integrating the square of the kernel of X_b over A' now gives

$$\|X_b\|_2^2 \le (\dim\mathcal{F}')C_N^2\|\widehat{g}\|_2^2\sum_{k\in\mathbb{Z}}\int_{\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s} e^{-\|d_sA + r_qk\mathbf{1}_{\sigma_0} + b\|^2/2}\,\mathrm{d}A. \tag{3.38}$$

To bound this last sum, split the quotient as

$$(\ker d_s/\Gamma_s) \times (\ker d_s)^\perp.$$

The first factor is compact: $\ker d_s$ is the kernel of an integer matrix, hence has an integer basis, and Γ_s has full rank in it. On the second factor d_s is injective, so $\|d_sA\|^2$ is positive definite. Moreover $\ker d_s^*$ consists of constant plaquette functions. This follows from the two equations

$$(d_s^*b)(y,2) = b(y) - b(y - e_3), \qquad (d_s^*b)(y,3) = b(y - e_2) - b(y).$$

Thus $\operatorname{ran} d_s$ is the orthogonal complement of the constant two-cochain. The orthogonal constant component of $r_qk\mathbf{1}_{\sigma_0} + b$ has squared norm $v^{-1}(r_qk + \sum_\sigma b_\sigma)^2$. Translate the variable in $(\ker d_s)^\perp$ to remove the remaining component. We obtain

$$\sum_k \int e^{-\|d_sA + r_qk\mathbf{1}_{\sigma_0} + b\|^2/2}\,\mathrm{d}A = C_s\sum_{k\in\mathbb{Z}} e^{-(r_qk + \sum_\sigma b_\sigma)^2/(2v)} < \infty, \tag{3.39}$$

where $C_s = \operatorname{vol}(\ker d_s/\Gamma_s)\int_{(\ker d_s)^\perp} e^{-\|d_sA\|^2/2}\,\mathrm{d}A < \infty$. This constant is independent of k and b. Equations (3.38)–(3.39) prove the Hilbert–Schmidt property for every real magnetic perturbation. The bounds here depend on the fixed cross-section; their purpose is to justify the trace argument.

For a real external two-cochain β on $C_{\mathbf{n}}$, put

$$Z_{C_{\mathbf{n}}}(\beta) = \int_{\mathrm{Conf}(C_{\mathbf{n}})} e^{-\|F+\beta\|^2/2}\det\mathcal{D}_A^{C_{\mathbf{n}}}\,\mathrm{d}(A,z). \tag{3.40}$$

Let β_j^{el} and β_j^{mag} be its electric and magnetic slices. Then

$$Z_{C_{\mathbf{n}}}(\beta) = (2\pi)^{n_1n_2n_3}\operatorname{Tr}_{H^s}\prod_{j=n_1-1,\dots,0}(X_{j+1}^*U_jX_j), \quad X_j = X_{\beta_j^{\mathrm{mag}}}, \quad U_j = e^{i\langle\beta_j^{\mathrm{el}},E\rangle}, \quad X_{n_1} = X_0. \tag{3.41}$$

Here is the conversion from the unfolded path integral. The electric Maxwell factors are the kernels

$$e^{-|A_{j+1}-A_j-d_s a_j+\beta_j^{\mathrm{el}}|^2/2} = (2\pi)^v g_1^{\otimes 2v}(A_{j+1}-A_j-d_s a_j+\beta_j^{\mathrm{el}}).$$

After summing the winding vector they are periodized on Γ_s. Gaussian convolution with a translated kernel is $e^{i\langle\beta_j^{\mathrm{el}},E\rangle}e^{-|E|^2/2}$. Integration over a_j includes the matrix $u(a_j)$ from (3.33); it is exactly the gauge average in (3.35). The magnetic Gaussian in slice j splits into the two quarter-Gaussians in M_j and M_j^*. The determinant half-slab matrices supply their other factors. Thus the cyclic path integral is the trace on H^l of

$$\prod_{j=n_1-1,\dots,0}(K^{1/2}U_jK^{1/2}M_jM_j^*).$$

Each time slice contributed $(2\pi)^v$, giving the constant in (3.41). Move the final M_0^* to the front and regroup the products as $X_{j+1}^*U_jX_j$. The Hilbert–Schmidt property above and the absolute Gaussian path majorant justify this cyclicity and every interchange. In particular the external-field integral is finite; alternatively this follows directly by absorbing a linear factor into a smaller Gaussian.

We spell out the path and kernel identification used here. On a fundamental domain write every intermediate real slice as $A_j = A_j^\flat + g_j$, $g_j \in \Gamma_s$, and set $g_0 = 0$ and $g_{n_1} = A_{n_1} - A_0$. Each unperiodized electric kernel then has the argument $A_{j+1}^\flat - A_j^\flat + g_{j+1} - g_j - d_s a_j$. The map $(g_1,\dots,g_{n_1}) \mapsto (g_{j+1}-g_j)_{0\le j<n_1}$ is a bijection of $\Gamma_s^{n_1}$. Summing these differences independently is therefore precisely periodization of every cyclic kernel. The magnetic factors and $\widehat{N}$ are unchanged by the shifts g_j. The finite Fock trace is changed from $\mathcal{F}'$ to $\mathcal{F}$ by moving its last $\widehat{N}(A_0)^*$ to the front; periodicity identifies it with $\widehat{N}(A_{n_1})^*$.

For absolute convergence replace every matrix factor by its norm, and bound each shifted electric Gaussian by a fixed constant times $e^{-|A'-A|^2/4}$ and each shifted magnetic factor by a constant times $e^{-\|F^{\mathrm{mag}}\|^2/4}$. This defines nonnegative scalar kernels. Their periodized convolutions are bounded and in L^2, and multiplication by the magnetic Gaussian is square integrable after summing fluxes by (3.39). Hence each scalar majorant operator is Hilbert–Schmidt. The integral of their cyclic product is finite: group the first factor against the product of the others and use the Hilbert–Schmidt Cauchy–Schwarz inequality. Tonelli first gives this conclusion for the nonnegative majorants, and Fubini then permits the kernel expansions and periodization changes for the original integrand. This proves the trace identity without a formal interchange of the infinite flux sum and the operator trace.

3.6. **Gaussian domination.**

Theorem 3.5. *For every coordinate torus with sides at least four and every real two-cochain β,*

$$0 < Z_{C_{\mathbf{n}}}(\beta) \le Z_{C_{\mathbf{n}}}(0). \tag{3.42}$$

Consequently its normalized curvature law obeys

$$\mathbb{E}_{C_{\mathbf{n}}} e^{\langle\beta,F\rangle} \le e^{\|\beta\|^2/2}. \tag{3.43}$$

For real two-cochains β_i and complex numbers z_i,

$$\left|\mathbb{E}_{C_{\mathbf{n}}} e^{\sum_i z_i\langle\beta_i,F\rangle}\right| \le \exp\left(\tfrac{1}{2}\left\|\sum_i (\operatorname{Re} z_i)\beta_i\right\|^2\right). \tag{3.44}$$

The constants are independent of the volume, the charge, and the mass.

Proof. The determinant is nonnegative and positive on a set of positive configuration measure, since $Z_{C_{\mathbf{n}}}(0) > 0$. The shifted photon density is strictly positive everywhere, proving the strict lower bound in (3.42).

In (3.41) move X_0^* through the trace to obtain a trace on H^l:

$$(2\pi)^{-n_1 n_2 n_3} Z_{C_{\mathbf{n}}}(\beta) = \operatorname{Tr}_{H^l}(U_{n_1-1} X_{n_1-1} X_{n_1-1}^* \cdots U_0 X_0 X_0^*).$$

Every U_j is unitary. Apply Schatten's inequality with all exponents equal to n_1; the trace on the left is nonnegative. Since $X_j X_j^*$ and $X_j^* X_j$ have the same nonzero eigenvalues, this gives

$$Z_{C_{\mathbf{n}}}(\beta) \le \prod_{j=0}^{n_1-1} \left((2\pi)^{n_1 n_2 n_3} \operatorname{Tr}(X_j^* X_j)^{n_1}\right)^{1/n_1} = \prod_{j=0}^{n_1-1} Z_{C_{\mathbf{n}}}(\beta^{(j)})^{1/n_1}. \tag{3.45}$$

Here $\beta^{(j)}$ has zero electric part and the same magnetic part β_j^{mag} in every time slice. The equality uses the trace formula again, with the same X_j in each slice and all unitary factors equal to the identity.

Each $\beta^{(j)}$ is supported only on 23-plaquettes. Interchange directions 1 and 2. The torus becomes $C(n_2, n_1, n_3)$ and the perturbation is supported only on 13-plaquettes, so its magnetic part for the new time direction is zero. The reduction (3.45) on this torus bounds its partition function by the unperturbed one. Lattice isometry covariance of the determinant and photon measure identifies these partition functions with $Z_{C_{\mathbf{n}}}(\beta^{(j)})$ and $Z_{C_{\mathbf{n}}}(0)$. Substitution in (3.45) proves the upper bound in (3.42).

Replacing β by $-\beta$ and completing the square gives (3.43). For complex parameters, take the absolute value inside the probability integral and apply that bound to $\sum_i (\operatorname{Re} z_i)\beta_i$, proving (3.44). □

We record the consequences used below. Charge conjugation centers $F(\beta) = \langle \beta, F \rangle$. The Gaussian bound and the nonnegative Taylor coefficients of cosh give

$$\begin{aligned} \mathbb{E}F(\beta)^2 &\le \|\beta\|^2, \\ \mathbb{E}F(\beta)^{2r} &\le (2r \exp(1) \|\beta\|^2)^r, \qquad r \ge 1, \\ \mathbb{E}F(\beta)^4 &\le \tfrac{3}{2} \exp(2) \|\beta\|^4. \end{aligned} \tag{3.46}$$

For the second moment let the parameter in the quadratic term tend to zero. For the $2r$-th moment use $(2r)! s^{-2r} e^{s^2\|\beta\|^2/2}$ and choose $s^2 = 2r/\|\beta\|^2$. The fourth-moment constant uses $4! = 24$ before estimating the factorial. Hölder and Cauchy–Schwarz in the fourth-cumulant identity then give

$$|\kappa_4(F(\beta_1), \dots, F(\beta_4))| \le 15 \prod_{i=1}^{4} \|\beta_i\|. \tag{3.47}$$

For nonzero β_i with disjoint supports and thresholds $\xi_i \ge 0$, Markov's inequality applied to $\sum_i \pm \xi_i \beta_i / \|\beta_i\|$, for each choice of signs, gives

$$\Pr\{|F(\beta_i)| \ge \xi_i \|\beta_i\| \text{ for all } i\} \le \prod_i 2e^{-\xi_i^2/2}. \tag{3.48}$$

Disjointness makes the squared norm of this test equal to $\sum_i \xi_i^2$. Finally, if a finite-support lattice test embeds in the torus, its periodization has the same norm and the same pairing with the periodic curvature. Thus (3.43) also applies to these local tests on the lifted field law.

3.7. **The finite-volume fold inequality.** We give the local form of the comparison that will be used to obtain diagonal positivity. Fix diagonal cells of even height width h, and let O be a real trigonometric polynomial in finitely many curvature values at plaquettes contained in the base cell Q_0. Thus O is bounded and depends only on this open cell. Let $n \geq 8$ be divisible by $4h$, and abbreviate $D_n = D(n/2, n, n)$. Its cell count, and that of the cubic torus, is

$$N_n = \frac{n^3}{|Q|} = \frac{2n^3}{h^3}.$$

The translations between cells form the integer lattice

$$\Lambda_h = \frac{h}{2}\mathbb{Z}e_- + \frac{h}{2}\mathbb{Z}e_+ + h\mathbb{Z}e_3.$$

On $C(n)$ form the translated copies, and on D_n the reflected copies:

$$\widetilde{O}_n(F) = \sum_{v \in \Lambda_h/n\mathbb{Z}^3} O(\tau_v^* F), \qquad O_n^{\mathrm{fold}}(F) = \sum_{\kappa \bmod \Gamma_{D_n}} O(g_\kappa^* F). \tag{3.49}$$

Each sum has N_n terms. The first sum is read on cubic-periodic curvatures, the second on D_n-periodic curvatures. Here g_κ is the reflected map from the base cell to Q_κ.

Let $\mu_{C(n)}$ and μ_{D_n} be the normalized curvature laws. For $t > 0$ define the tilted diagonal law

$$\mu_{D_n}^t(\Phi) = \frac{\mu_{D_n}(e^{tO_n^{\mathrm{fold}}}\Phi)}{\mu_{D_n}(e^{tO_n^{\mathrm{fold}}})}. \tag{3.50}$$

The folded array is invariant under the group generated by the cell walls: that group permutes the cells of D_n and their reflected copies. Thus the tilt retains these symmetries.

Proposition 3.6. *For the preceding data and every $t > 0$,*

$$\mu_{C(n)}(O) \leq \mu_{D_n}^t(O) + \frac{|Q|\Delta(n)}{t}. \tag{3.51}$$

The error is the fixed-parameter pressure error of (3.29).

Proof. Jensen's inequality and translation invariance on $C(n)$ give

$$\log \frac{Z_{C(n)}[e^{t\widetilde{O}_n}]}{Z_{C(n)}[1]} \geq t\mu_{C(n)}(\widetilde{O}_n) = tN_n\mu_{C(n)}(O). \tag{3.52}$$

The diagonal chessboard inequality gives

$$Z_{C(n)}[e^{t\widetilde{O}_n}] \leq \|e^{tO}\|_{D_n}^{N_n} = Z_{D_n}[e^{tO_n^{\mathrm{fold}}}]. \tag{3.53}$$

To check that its N_n factors have the same seminorm, translate the base observable by $v \in \Lambda_h$. Translation permutes the cells and conjugates their reflected maps, so the fold of the translated observable is the translate of the original full fold. The law on D_n is translation invariant, hence its partition function and the seminorm are unchanged. All observables here are bounded scalar functions supported in distinct open cells, so the chessboard inequality applies directly.

Set $L(t) = \log \mu_{D_n}(e^{tO_n^{\mathrm{fold}}})$. Boundedness permits differentiation, giving $L'(t) = \mu_{D_n}^t(O_n^{\mathrm{fold}})$. Convexity and $L(0) = 0$ give $L(t) \leq tL'(t)$. Wall-group invariance of the tilted law makes the expectation of every reflected copy equal to that of O; hence

$$\log \frac{Z_{D_n}[e^{tO_n^{\mathrm{fold}}}]}{Z_{D_n}[1]} \leq tN_n\mu_{D_n}^t(O). \tag{3.54}$$

Combining (3.52)–(3.54) leaves, in addition, the normalization difference

$$\log Z_{D_n}[1] - \log Z_{C(n)}[1] \leq n^3\Delta(n).$$

Division by tN_n proves (3.51). □

For fixed O, its cell geometry, and $t > 0$, the error in (3.51) tends to zero as $n \to \infty$ along the admissible sides. The estimate compares the cubic law with a tilted diagonal law. The next step must control these tilted laws as $t \downarrow 0$; the pressure estimate alone does not identify their local limits.

4. Diagonal reflection positivity

The finite-volume fold inequality compares a cubic expectation with an expectation in a tilted adapted law. At each wall of the chosen cell structure, the tilted functional is reflection positive for observables in the corresponding open positive half-period slab. The walls are separated by the cell width, and this positivity does not yet place the original volume limit among the diagonal-positive laws. We must first show that removing the tilt also removes its residual dependence on layers. The argument uses local operators inserted into a coordinate crossing and then eliminates the fields outside its two adjacent fermion layers. The resulting comparison states have enough translation invariance to apply the fold inequality to an arbitrary local observable. Convex separation will then give the required diagonal positivity.

Fix $q > 0$, $0 < \rho < 1$, positive weights w_i and a real mass M. All volume limits in this section keep these parameters fixed. We use cubic sides $n_i \to \infty$ such that eight divides all sufficiently large sides and the set

$$\mathcal{D} = \{m \in \mathbb{N} : m \mid n_i \text{ for all sufficiently large } i\}$$

is infinite. The set $\mathcal{D}$ is closed under divisors and least common multiples. The cutoff sides $8L^{a+T+N}$ have this property at fixed N, since $8L^j \in \mathcal{D}$ for each $j \geq 0$.

4.1. **The tilted law at its walls.** Fix diagonal cells with offsets (c_0, c_+, c_3), even width h, and $c_0, c_+ \in \mathbb{Z}$, $c_3 \in \mathbb{Z} + \frac{1}{2}$, and $4h \in \mathcal{D}$. Let O be the real trigonometric polynomial used in the finite fold comparison. Its support is contained in a box of radius r_O, and we require $h > 8r_O + 8$. Such widths can be chosen because $\mathcal{D}$ contains arbitrarily large multiples of eight. For large divisible sides, put $D_i = D(n_i/2, n_i, n_i)$ and $\mu_i^t = \mu_{D_i}^t$, as in (3.50). Write

$$O(F) = \sum_\beta a_\beta e^{i\langle \beta, F\rangle}, \qquad |O|_1 = \sum_\beta |a_\beta|.$$

The sum is over the distinct exponents in the chosen finite expansion. In particular $|O| \leq |O|_1$. Let $V_i = O_{n_i}^{\mathrm{fold}}$ and $N_i = n_i^3/|Q|$. For an admissible Grassmann integrand we use the same notation for the normalized functional $\mu_i^t(G) = Z_{D_i}[e^{tV_i}G]/Z_{D_i}[e^{tV_i}]$. On bosonic integrands it agrees with integration against the tilted probability law. The normalizing denominator satisfies

$$Z_{D_i}[e^{tV_i}] \geq e^{-tN_i|O|_1} Z_{D_i}[1] > 0, \qquad t \geq 0. \tag{4.1}$$

This uses only the positive bosonic measure; it makes no positivity assertion about a Grassmann integrand.

At a wall $\Pi = (\ell^a, o_a + jh)$, the adapted torus has height period p_a, with $(p_1, p_2, p_3) = (n_i, 2n_i, n_i)$. The number p_a/h is even. Split the cells into the two half-slabs adjacent to this wall, and define

$$\Phi_+(F) = \exp\left(t \sum_{Q_\kappa \subset \{o_a + jh < \ell^a < o_a + jh + p_a/2\}} O(g_\kappa^* F) \right),$$

with representatives in one half-period. Reflection in the wall maps these cells onto the other half-period modulo the normal closing vector. Because the reflected copies of O form one wall-invariant

array, on periodic curvatures

$$e^{tV_i} = \Phi_+ \Theta_\Pi \Phi_+. \tag{4.2}$$

The function Φ_+ is bounded, real, and supported strictly within the upper half-slab. The support lies there because each plaquette of a copy of O is contained in its open cell.

Apply the adapted reflected representation to $\Phi_+ G$ and $\Phi_+ G'$. For local even integrands G, G' in the upper half-slab,

$$\mu_i^t(\Theta_\Pi G' G) = \frac{\mathrm{Tr}(\mathfrak{S}[\Phi_+ G']^* \mathfrak{S}[\Phi_+ G])}{Z_{D_i}[e^{tV_i}]}. \tag{4.3}$$

The numerator is the Hilbert–Schmidt inner product of the two slab operators. Thus, on D_i, the tilted functional is reflection positive at each cell wall for local even integrands supported in the corresponding open positive half-period slab. Wall reflection also gives $\mu_i^t(\Theta_\Pi G) = \overline{\mu_i^t(G)}$: the unperturbed functional is reflection covariant, the scalar weight e^{tV_i} is invariant, and its denominator is real and positive.

The bosonic law is invariant under the full group generated by the cell walls. Indeed, that group preserves D_i and permutes the reflected copies in V_i, while the unperturbed bosonic law is invariant under its lattice isometries. In particular the tilt is invariant under translations by he_-, he_+ and $2he_3$. These are the periods of the reflected cell array, rather than all lattice translations.

To compare weak limits we use the Polish space

$$\Omega = \{F \in \mathbb{R}^{\{\text{oriented plaquettes of } \mathbb{Z}^3\}} : dF = 0\},$$

with its product topology. The pullback by a lattice reflection or translation is continuous, so invariance passes to weak limits. Positivity passes for each bounded continuous local positive-side test once its support lies in the finite positive half-period slab for all large volumes. Positivity against bounded Borel local tests follows once reflection invariance has passed: on their finite-dimensional marginal approximate the test in L^2 by compactly supported continuous functions. The reflected-form difference is at most

$$\|\Phi - \Phi_k\|_2 (\|\Phi\|_2 + \|\Phi_k\|_2),$$

because reflection preserves the L^2 norm. The Gaussian domination bound also passes to weak limits, by truncating $e^{\langle \beta, F\rangle}$ and then using monotone convergence. These facts will be applied after obtaining tightness of the tilted laws; their extensive weight alone gives no uniform moment estimate.

4.2. **Local operators at a coordinate link plane.** An insertion must carry an operator norm that is uniform in the cross-section and in electric characters. Fix a finite set S of links of $\mathbb{Z}^2$ and a nonempty finite set Λ of sites containing their endpoints. Let $\mathcal{F}_\Lambda$ be the Fock space of crossing modes over Λ, and put

$$H_S = L^2(\mathbb{R}^S) \otimes \mathcal{F}_\Lambda, \qquad E_b = -i\partial_{A(b)}.$$

The diagonal unitary $u(a)$ on $\mathcal{F}_\Lambda$ has integer mode charges. An operator Υ has charge $z \in \mathbb{Z}^\Lambda$ when $u(a)\Upsilon u(a)^{-1} = e^{iq\langle a, z\rangle}\Upsilon$.

Let $\mathfrak{A}(S)$ be the span of the functions $e^{-Q(\xi_{S'}) + \langle \lambda, \xi_{S'}\rangle}$, where $S' \subset S$, Q is real and positive definite on $\mathbb{R}^{S'}$, and $\lambda \in \mathbb{C}^{S'}$. The empty subset gives the constant one. These functions are bounded and uniformly continuous. Their span is closed under products, conjugation and real translations: products add the positive quadratic forms on the union of their supports, while translation completes the same quadratic exponent.

For $\delta \in \mathbb{R}^S$, extended by zero to the full slice, require $d_s^*\delta = qz(\delta)$ with $z(\delta) \in \mathbb{Z}^\Lambda$. The local operator algebra consists of the finite sums

$$X = \sum_\delta e^{i\langle \delta, A\rangle} \Upsilon_\delta(E_S), \qquad \Upsilon_\delta \in \mathfrak{A}(S) \otimes \mathrm{End}_{z(\delta)}(\mathcal{F}_\Lambda). \tag{4.4}$$

We denote it by $\mathfrak{W}(S,\Lambda)$. The Weyl identity gives

$$\begin{aligned} (e^{i\delta A}\Upsilon(E_S))(e^{i\delta' A}\Upsilon'(E_S)) &= e^{i(\delta+\delta')A}\Upsilon(E_S+\delta')\Upsilon'(E_S), \\ (e^{i\delta A}\Upsilon(E_S))^* &= e^{-i\delta A}\Upsilon^*(E_S-\delta). \end{aligned} \tag{4.5}$$

Here δA abbreviates the real pairing. These formulas make $\mathfrak{W}(S,\Lambda)$ a unital $*$-algebra. The normal form is unique. In Fourier variables its terms send a function supported near ξ_0 to functions supported near $\xi_0 + \delta$; small enough neighborhoods separate the finitely many distinct exponents. If the sum vanishes as an operator, every continuous symbol vanishes on each such neighborhood and hence everywhere.

Every X commutes with the local gauge transformations $\Phi(A) \mapsto u(a)\Phi(A - d_s a)$. The phase change $e^{-i\langle \delta, d_s a\rangle}$ cancels the Fock charge phase. The total charge of $z(\delta)$ is zero, since it is the divergence of a finitely supported cochain. Each Fock term is therefore even. For $\beta \in \mathbb{R}^S$, define

$$\alpha_\beta X = e^{i\langle \beta, A\rangle} X e^{-i\langle \beta, A\rangle} = \sum_\delta e^{i\langle \delta, A\rangle}\Upsilon_\delta(E_S - \beta). \tag{4.6}$$

This is an isometric $*$-automorphism on H_S.

For a rectangular cross-section Ξ containing the smallest integer rectangle spanned by Λ, the same formulas define X_Ξ on the real slice quotient space H^l of §3.5. The Fock operators are localized into the full crossing Fock space by ι. The localization is the tensor inclusion for the exterior space, with the reordering sign and cross-section character attached to each representative; it is a unital injective $*$-homomorphism. The phase $e^{i\langle \delta, A\rangle}$ is well defined on the quotient. Indeed, on the support hull an integer closed one-cochain is an integer coboundary, by the box Poincare argument. Thus for $\gamma \in \Gamma_s$, $\langle \delta, \gamma\rangle \in 2\pi\mathbb{Z}$, using $d_s^*\delta = qz(\delta)$. The local Weyl formulas therefore remain valid on the quotient. Uniqueness of the normal form makes the representation independent of its presentation. It commutes with P_G and with functions of the electric variables outside S.

Define the norm needed for characters by

$$\|X\|_\# = \sup_{\Xi,\beta} \|(\alpha_\beta X)_\Xi\|. \tag{4.7}$$

The support hull must embed in the cross-section; the supremum is over such cross-sections and all $\beta \in \mathbb{R}^S$. The electric spectrum of a quotient can be smaller than $\mathbb{R}^S$, so its unshifted operator norm need not equal the local norm.

Proposition 4.1. *For $X \in \mathfrak{W}(S,\Lambda)$,*

$$\|X\|_\# = \|X\|_{\mathcal{L}(H_S)}.$$

The norm is unchanged by enlarging S, Λ. In particular it is a C^*-norm, $\|1\|_\# = 1$, and every quotient realization satisfies $\|(\alpha_\beta X)_\Xi\| \le \|X\|_\#$.

Proof. First remove the Fock localization. Its tensor extension is an injective $*$-homomorphism and hence isometric; we may work on $L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s) \otimes \mathcal{F}_\Lambda$ with the original finite Fock operators. Call this realization X^0_Ξ. Introduce an additional variable $z \in \mathbb{R}^S$, with electric operators $E' = -i\partial_z$, on

$$L^2(\mathbb{R}^S_z) \otimes L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s) \otimes \mathcal{F}_\Lambda.$$

The operator $X(z, E')$ has norm $\|X\|$, since it is X tensored with the identity. Choose a Borel representative $A^\flat$ of the quotient and use the two unitaries

$$(U_1\Phi)(z,[A]) = \Phi(z + A^\flat|_S, [A]), \qquad (U_2\Phi)(z,[A]) = \Phi(z,[A-z]),$$

where z is extended by zero outside S. The first conjugation multiplies $e^{i\delta z}$ by $e^{i\delta A}$. The second replaces E' by $E' + E_S$ and cancels the factor $e^{i\delta z}$. Hence

$$U_2U_1X(z,E')(U_2U_1)^{-1} = \sum_\delta e^{i\delta A}\Upsilon_\delta(E_S + E').$$

Fourier transform in z. This becomes a direct integral of $T_p = (\alpha_{-p}X)^0_\Xi$, $p \in \mathbb{R}^S$. Uniform continuity of the symbols makes $p \mapsto T_p$ norm continuous. Its direct-integral norm is therefore $\sup_p \|T_p\|$: the reverse inequality follows by using a unit vector nearly maximizing T_{p_0} on a small ball about p_0. It follows that

$$\|X\| = \sup_p \|(\alpha_{-p}X)^0_\Xi\| = \sup_p \|(\alpha_{-p}X)_\Xi\|.$$

This holds for each admissible cross-section and proves the asserted supremum identity. Enlarging S tensors with an identity on the additional L^2 factor; enlarging Λ uses the same isometric Fock inclusion. Both preserve the norm. □

4.3. **Insertions and a coordinate trace bound.** Let $c = j + \frac{1}{2}$. Write Ins_c for the crossing insertion map, and ω for its diagonal mode weights. For an even matrix kernel $k(A', A)$ on $\mathbb{R}^S$, set

$$\begin{aligned} G[k](\widehat{A}) = \mathrm{Ins}_c\Big[\omega^{1/2}k(\widehat{A}_{j+1}|_S, (\widehat{A}_j + d_s\widehat{a}_j)|_S)u(\widehat{a}_j)\omega^{1/2}\Big] \\ \cdot \prod_{b\in S} g_1(F^{\mathrm{el}}_j(b))^{-1}\exp(S^\times_c(\Lambda)). \end{aligned} \tag{4.8}$$

Here $S^\times_c(\Lambda)$ is the part of the fermion action crossing the plane over Λ. The two factors on the second line remove the original electric and fermion crossing weights on the support of k. Thus $G[k]$ inserts k in their place. Its definition uses the infinite lattice and is independent of a torus.

We require

$$\begin{aligned} u(a)^{-1}k(A' + d_sa, A + d_sa)u(a) = k(A', A), \\ \|k(A', A)\| \le C\exp(-\epsilon|A' - A|^2), \qquad \epsilon > 0. \end{aligned} \tag{4.9}$$

Choose a box containing the two copies of S, the temporal links over Λ, and the unit cubes centered at (j, y) and $(j + 1, y)$, $y \in \Lambda$. In what follows we use the cube $B^\#(c, S, \Lambda)$ obtained by taking twice the largest half-width of the smallest rectangular box containing these sets, with the same center. This choice is covariant under coordinate translations and reflections. The insertion reads only these links and the crossing fields, so it is supported in $B^\#$.

The insertion is an even, gauge invariant local integrand. Indeed, an even kernel has entries only between occupation vectors of the same parity, so every monomial $\Theta_c\zeta_J\,\zeta_I$ in its insertion is even. If the gauge parameter is λ, its upper and lower slice values change the matrix in (4.8) to

$$u(\lambda_{j+1})^{-1}k(\widehat{A}_{j+1}, \widehat{A}_j + d_s\widehat{a}_j)u(\widehat{a}_j)u(\lambda_j).$$

The matrix entry acquires precisely the inverse of the phase of $\Theta_c\zeta_J\,\zeta_I$. The electric curvature and the crossing action are gauge invariant. Finally, the kernel estimate and $g_1(x)^{-1} = \sqrt{2\pi}e^{x^2/2}$ bound every Grassmann coefficient by

$$C'\exp\left((\tfrac{1}{2} - \epsilon')\sum_{b\in S} F^{\mathrm{el}}_j(b)^2\right), \qquad \epsilon' = \min(\epsilon, \tfrac{1}{2}). \tag{4.10}$$

The other factors have bounded coefficients. This is the local integrand bound of §3.1, since its electric plaquettes lie in the box.

For $X \in \mathfrak{W}(S,\Lambda)$ let $k_X^{(s)}$ be the continuous kernel of $e^{-s|E_S|^2} X e^{-s|E_S|^2}$, with $s = \frac{1}{4}$ or $\frac{1}{8}$, and write

$$G_X = G[k_X^{(1/4)}], \qquad G'_X = G[k_X^{(1/8)}]. \tag{4.11}$$

Both kernels satisfy (4.9). To see their decay, consider one normal-form term $e^{i\delta A}\Upsilon(E)$ with $\Upsilon(\xi) = e^{-Q(\xi)+\lambda\xi}\Upsilon_0$. Identify Q with its symmetric matrix, so $Q(\xi) = \langle \xi, Q\xi\rangle$, extending Q by zero to the other links. With $Q_s = 4sI + 2Q$ and $\lambda_s = \lambda - 2s\delta$, its kernel is

$$\begin{aligned} k_X^{(s)}(A',A) = e^{i\delta A'}(2\pi)^{-|S|/2}(\det Q_s)^{-1/2} e^{-s|\delta|^2} \\ \cdot \exp\left(\tfrac{1}{2}\langle \lambda_s + i(A'-A), Q_s^{-1}(\lambda_s + i(A'-A))\rangle\right)\Upsilon_0. \end{aligned} \tag{4.12}$$

The real part of this exponent has a strictly negative quadratic form in $A' - A$, with a fixed linear term. It gives the required Gaussian decay after decreasing its exponent. Gauge covariance follows by conjugating the regularized operator by the gauge unitary; equality of their continuous kernels gives the first identity in (4.9). Finite sums preserve these properties. The identity operator has kernel $\prod_{b\in S} g_1(A'(b) - A(b))$ when $s = \frac{1}{4}$, so $G_1 = 1$; unused links and modes cancel in the same way.

Reflection acts on these insertions by adjoints:

$$\Theta_c G_X = G_{X^*}, \qquad \Theta_c G'_X = G'_{X^*}. \tag{4.13}$$

The kernel is changed to $k(A,A')^*$. At the same time the two slices exchange, $\widehat{a}_j$ changes sign, and covariance moves $u(\widehat{a}_j)$ to the required side of the kernel. This gives (4.13), including its field signs. We also need coordinate translations and spatial reflections to preserve the operator norm, since each coordinate fold contains such images.

For a spatial reflection θ in the second or third coordinate, the spin matrix exchanges the two eigenspaces of γ_1. Its unitary matrix v on crossing modes preserves their weights and reverses their charges. On exterior products the reordering sign is $r_I = (-1)^{|I|(|I|-1)/2}$; hence the Fock unitary is $U = (\bigwedge v) r$. The anticommutation of the two field reflection maps contributes the sign $(-1)^{\deg}$, and moving the two mode products past one another contributes $(-1)^{|I||J|}$. For the parity-matched entries of an even kernel these signs leave exactly $r_I r_J$. Consequently

$$\Theta_\theta \operatorname{Ins}_c[\Upsilon(\widehat{A})] = \operatorname{Ins}_c[U\overline{\Upsilon(\theta^*\widehat{A})}U^*].$$

The antiunitary $(J\Phi)(A) = U\overline{\Phi(\theta_s^* A)}$ gives this kernel transformation. It sends an oriented electric operator to the negative of the reflected electric operator, and sends $e^{i\delta A}$ to $e^{-i\langle\theta_s^*\delta, A\rangle}$. Thus it preserves the charge constraint, the local algebra and the heat factors $e^{-s|E|^2}$. Translations act by the corresponding unitary relabeling. Together with adjoints at the link plane, these facts show that each translate or signed coordinate reflection of G_X is an insertion G_{X^g} with $\|X^g\| = \|X\|$. The same holds for G'_X. The box $B^\#$ transforms with the insertion.

An even local integrand G in a box B satisfies the coordinate bound $C \geq 0$ if, on every rectangular coordinate torus $T = (\Gamma,\chi) = C(n_1,n_2,n_3)$ with $n_\mu \geq 4$, for every nonempty finite family of images $g_a B$ under lattice translations and signed coordinate reflections that embeds disjointly in Γ, and every $\beta \in C^2(\Gamma)$,

$$\left| Z_T\left[\varphi_\beta \prod_{a=1}^{r} G^{g_a}\right]\right| \leq C^r Z_T[1], \qquad \varphi_\beta(F) = e^{i\langle\beta,F\rangle}, \tag{4.14}$$

provided that, in each coordinate direction μ, there is a plaquette σ orthogonal to e_μ such that none of the boxes or their period translates contains any plaquette $\sigma + ke_\mu$, $k \in \mathbb{Z}$. This condition places the flux representative used in the slice trace on plaquettes read by none of the insertions.

When the coordinate cell structure cuts T and its walls are at half-integers, boxes contained in its open cells satisfy the free-column condition. Choose a plaquette orthogonal to the chosen direction that meets a wall normal to another direction. Period translations permute the open cells, while every plaquette in this column meets that wall. Thus none of these plaquettes is contained in a box or any of its period translates.

Proposition 4.2. *The insertion G_X satisfies* (4.14) *with $C = \|X\|$. The insertion G'_X satisfies it with $C = 2^{|S|/2}\|X\|$. A bounded bosonic integrand satisfies it with $C = \|G\|_\infty$. These bounds survive box enlargement, multiplication by a curvature character, and transport by a signed permutation of the coordinates.*

Proof. We give the trace calculation, including the support condition. Choose the time direction to be the first coordinate. By the covariance just proved, the images of the insertions have the same operator norms and are at link planes $j + \frac{1}{2}$. The free column lets us choose the magnetic flux plaquette σ_0 so that no plaquette in its column is contained in an insertion box. At a fixed step j their spatial supports and modes are disjoint. Let S_j be the union of the supports of the primed insertions at that step.

The unfolded path variables are genuine local lifts on each $[j, j+1] \times \mathrm{hull}(\Lambda_a)$. On electric plaquettes their coboundary is $A_{j+1} - A_j - d_s a_j$. On magnetic plaquettes it is $d_s A_j$; the flux correction is supported at σ_0, which is not a plaquette of the local hull. Two local lifts with these curvatures and phases differ by an integer exact cochain $\frac{2\pi}{q} d\lambda$. Gauge invariance therefore allows the insertion to be read directly at the path variables. This is why the free-column hypothesis is needed.

At each crossing, the factors $e^{S_c^\times(\Lambda_a)}$ remove the crossing weights on the local modes. The remaining crossing factors, together with their insertions, form

$$\mathrm{Ins}_c \left[\omega^{1/2} \left(\prod_a \iota(k_a(A_{j+1}, A_j + d_s a_j)) \right) u(a_j) \omega^{1/2} \right].$$

The even localized matrices commute because their mode sets are disjoint. The factors g_1^{-1} similarly remove the electric Gaussian densities on the local links. The slice determinant factorization of §3.5 now gives the path product with these modified crossing kernels. Its Gaussian majorant is integrable: every modified kernel has strict decay in its electric difference, all remaining links retain g_1, and the magnetic multiplier has its Gaussian weight. Hence unfolding, periodization and the path trace are valid for this product.

Separate the curvature character into electric factors $e^{i\langle \beta_j, F_j^{\mathrm{el}} \rangle}$ and magnetic unitaries u_j. Multiplying a regularized kernel by $e^{i\langle \beta_j, A' - A \rangle}$ changes the corresponding operator to

$$e^{-s|E-\beta_j|^2} (\alpha_{\beta_j} X) e^{-s|E-\beta_j|^2}.$$

Fourier transformation of the disjoint link kernels therefore gives

$$Z_T \left[\varphi_\beta \prod_a G_a \right] = (2\pi)^{|T|} \operatorname{Tr}_{H^l} \prod_{j=n_1-1,\dots,0} \left(K^{1/2}_{\beta_j, S_j} B_j K^{1/2}_{\beta_j, S_j} M_0 u_j M_0^* \right), \tag{4.15}$$

where

$$B_j = \prod_{a : j_a = j} (\alpha_{\beta_j} X_a)_\Xi, \qquad K_{\beta,S} = r_q^v e^{-|E-\beta|^2/2 + |E-\beta|_S^2/4} P_G.$$

Here $r_q^v = (2\pi/q)^{|\Sigma|}$ is the gauge-volume factor in (3.35); the primed kernels change the electric Gaussian variance only on S. Both the kernels and the local operators commute with P_G. In particular $\|B_j\| \le \prod_{a:j_a=j} \|X_a\|$ by the norm identity above.

Set $T_j = K^{1/2}_{\beta_j,S_j} M_0$. These are Hilbert–Schmidt by the same quotient Gaussian calculation as (3.38). Move the first square root cyclically to the end of the trace. The product then has alternating factors $B_j T_j u_j T_{j-1}^*$. Schatten Hölder, with $2n_1$ equal exponents, bounds its absolute trace by

$$\prod_j \|B_j\| \prod_j \|T_j\|^2_{2n_1}.$$

The slice trace applied to $(T_j^* T_j)^{n_1}$ gives exactly

$$\mathrm{Tr}(T_j^* T_j)^{n_1} = (2\pi)^{-|T|} Z_T \left[\prod_{l=0}^{n_1-1} e^{i\langle \beta_j, F_l^{\mathrm{el}} \rangle} \prod_{l=0}^{n_1-1} \prod_{b\in S_j} \sqrt{2} e^{-F_l^{\mathrm{el}}(b)^2/2} \right]. \tag{4.16}$$

The second product changes g_1 into $g_{1/2}$ on those links. The positive bosonic law bounds the absolute value by $2^{|S_j| n_1/2}(2\pi)^{-|T|} Z_T[1]$. Consequently

$$\prod_j \|T_j\|^2_{2n_1} \le 2^{\frac{1}{2}\sum_j |S_j|} (2\pi)^{-|T|} Z_T[1].$$

Together with (4.15) this proves the two operator bounds. For insertions in another time direction use an orientation-preserving coordinate permutation. The scalar bound follows directly from the positive law. Multiplication by a character merely changes β, and enlargement restricts the allowed placements. A signed permutation transports the torus, the free columns and the placements; conjugation of a character still has modulus one. These observations prove the remaining assertions. □

4.4. **Amplification in two cell structures.** A diagonal fold exchanges the first and second time directions. Its copies of a coordinate insertion therefore cannot all be put in one coordinate trace. We will place these copies in different coordinate cells and apply a second chessboard estimate. Each resulting coordinate fold contains copies of only one insertion, all with the same time direction, so (4.14) applies.

The placement requires two scales depending on the support radius r, but not its center or the tilt. Choose $h'' \in h\mathbb{Z}$ and $k \in \mathbb{N}$ such that

$$h'' \ge 2r + 6h, \qquad kh \ge 16h'', \qquad kh \in h''\mathbb{Z}, \qquad 4h'', 2kh \in \mathcal{D}. \tag{4.17}$$

These choices exist: take $4h''$ to be the least common multiple of $4h$ and an element of $\mathcal{D}$ exceeding $4(2r+6h)$, then take $2kh$ to be the least common multiple of $4h''$ and an element exceeding $32h''$. Fix choices depending only on h, r and the side sequence. For all sufficiently large i, the needed divisibility conditions then hold simultaneously.

Here is the geometry of the placement. Let $B = y + [-r, r]^3$. Choose coarse diagonal offsets $c'_\pm \in c_\pm + h\mathbb{Z}$ (with $c_- = c_0$) and $c'_3 \in c_3 + h\mathbb{Z}$ so that

$$\ell_-(y) - c'_- \in [2h'', 2h'' + h), \quad \ell_+(y) - c'_+ \in [4h'', 4h'' + h), \quad y_3 - c'_3 \in [h'', h'' + h). \tag{4.18}$$

The box lies strictly in the coarse cell Q'_0 of width kh. Every element of its wall group has the form

$$g = \tau_{kh(z_- e_- + z_+ e_+ + 2z_3 e_3)} \theta_-^{a_-} \theta_+^{a_+} \theta_3^{a_3}, \qquad z \in \mathbb{Z}^3, \quad a \in \{0,1\}^3,$$

where the reflections are at the lower walls of Q'_0. From (4.18) one obtains

$$gy = y + h''\kappa(g) + s(a), \qquad |s(a)|_\infty < 2h, \tag{4.19}$$

with

$$\kappa(g) = (-2a_- - 4a_+,\, 2a_- - 4a_+,\, -2a_3) + \frac{kh}{h''}(z_- + z_+,\, z_+ - z_-,\, 2z_3).$$

The eight first vectors are distinct, and their pairwise differences have size at most six. Since $kh/h'' \geq 16$, equality of two indices first forces equality of their a's and then of their z's. Thus $g \mapsto \kappa(g)$ is injective.

Choose half-integer coordinate offsets o_μ in

$$I_\mu = (y_\mu + r + 2h - h'',\ y_\mu - r - 2h).$$

These intervals have length at least $2h$. Equation (4.19) places every gB strictly inside coordinate cell $\kappa(g)$ of width h''. We can also choose the offsets so that all these coordinate walls avoid every reflected copy of $\operatorname{supp} O$. Under the original diagonal wall group, the centers of the copies of a box containing O have at most four residue classes modulo h in either of the first two coordinates, and at most two in the third. Its radius r_O therefore forbids at most $4(2r_O+1) = 8r_O+4$ half-integer classes. Since $h > 8r_O+8$, an allowed class exists. Each I_μ contains a representative, and $h'' \in h\mathbb{Z}$ keeps every wall in the allowed class.

For large i, the adapted torus D_i carries the original and coarse diagonal structures; its period lattice lies in the translation subgroup of the coarse wall group. The coordinate structure of width h'' cuts D_i and is carried by $T_i = C(n_i/2, 2n_i, n_i)$. Formula (4.19) also shows that coarse diagonal cells distinct modulo the periods have their boxes in coordinate cells distinct modulo those periods. Indeed a period translation changes κ by that translation divided by h''; injectivity then gives the converse. Thus the counts are

$$N_i' = \frac{n_i^3}{|Q|k^3}, \qquad N_i'' = \frac{n_i^3}{(h'')^3}. \tag{4.20}$$

The first is the number of coarse diagonal cells in D_i; the second is the number of coordinate cells in either torus. The shortest periods are at least $n_i/2 \geq kh \geq 16h''$, so the boxes, with the margins used for local lifts, embed. Folding one such box in the coordinate structure gives a disjoint placement with free columns, since every image stays inside its open cell.

Theorem 4.3. *Put $\Delta_i = \Delta(n_i)$, with Δ defined in* (3.29)*. It tends to zero as $i \to \infty$, with the charge, mass and weights fixed. If G satisfies the coordinate bound C in a box of radius r, then, for k as above and all sufficiently large i,*

$$|\mu_i^t(G)| \leq C\gamma_{t,i}(k), \qquad \gamma_{t,i}(k) = \exp(2t|O|_1 k^3 + |Q|k^3\Delta_i), \qquad t \geq 0. \tag{4.21}$$

The threshold and k depend only on the side sequence, h and r. In particular they are uniform in the location of G.

Proof. Each coarse diagonal cell contains k^3 original cells. Put $P = \exp(t\sum_{Q_\kappa \subset Q_0'} O^{[\kappa]})$. The tilt e^{tV_i} is exactly the product of the reflected copies of P over the coarse cells: their walls are original walls, so each reflection carries the original array to the same array. Apply the adapted chessboard estimate to the integrands PG in one coarse cell and P in the others. The fold of P is e^{tV_i}. After dividing its powers out, this gives

$$|\mu_i^t(G)|^{N_i'} \leq \frac{Z_{D_i}[e^{tV_i}\prod_{a=1}^{N_i'} G^{g_a}]}{Z_{D_i}[e^{tV_i}]}. \tag{4.22}$$

The right side is nonnegative by the reflected fold form. This step uses reflection positivity of even integrands and does not require G itself to be a positive scalar function.

Expand e^{tV_i} in curvature characters. The coefficient sum is at most $\exp(tN_i|O|_1)$: the character algebra is closed under multiplication, and the exponential series is absolutely summable in its coefficient norm. Every plaquette in a character of this expansion lies strictly within a coordinate cell, because the chosen walls avoid the support of every copy of O. For each character term, collect its factors by coordinate cells. Each cell then contains a character alone or a character

times one G^{g_a}; precisely N_i' cells have the latter form. Period translations can be used to place the integrands in a chosen coordinate brick without changing their value on D_i.

Use the chessboard estimate from the adapted to the coordinate frame, (3.26). A cell with a character alone has fold bounded by $Z_{T_i}[1]$. A cell with a copy of G has a fold consisting of a character and its coordinate images; the geometry above supplies the free columns and disjoint embedding, so its fold is bounded by $C^{N_i''}Z_{T_i}[1]$. Taking the chessboard roots and multiplying the cell contributions gives, for every character term,

$$\left| Z_{D_i}\left[\varphi_\beta \prod_{a=1}^{N_i'} G^{g_a}\right]\right| \le C^{N_i'} Z_{T_i}[1].$$

The coefficient sum therefore bounds the numerator of (4.22) by $e^{tN_i|O|_1}C^{N_i'}Z_{T_i}[1]$. The denominator bound (4.1) contributes a second $e^{tN_i|O|_1}$. Finally, the pressure comparison gives $Z_{T_i}[1]/Z_{D_i}[1] \le e^{n_i^3\Delta_i}$. The root of order N_i' gives (4.21). □

We also need exponential moments, for which a coordinate operator bound is unnecessary. Repeat (4.22) with $G = e^{\langle\beta,F\rangle}$ supported in B. It is an admissible local integrand, since $\langle\beta,F\rangle \le \|\beta\|^2 + \frac{1}{4}\sum_{\operatorname{supp}\beta} F_\sigma^2$. Its diagonal copies have disjoint supports. Remove the tilt from the numerator by its pointwise bound, and apply the second chessboard estimate to their coordinate cell integrands. The fold of one exponential has N_i'' disjoint copies of β, and hence squared norm $N_i''\|\beta\|^2$. Gaussian domination on T_i bounds its chessboard root by $e^{\|\beta\|^2/2}Z_{T_i}[1]^{1/N_i''}$. There are N_i' nonempty cells. The same denominator and pressure bounds therefore yield

$$\mu_i^t(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2}\gamma_{t,i}(k). \tag{4.23}$$

For each fixed t this proves tightness on Ω. Indeed, Markov's inequality applied to $\beta = \pm s1_\sigma$ gives a uniform coordinate tail $2e^{-sR+s^2/2}\gamma_{t,i}(k(h,1))$. Choose radii R_σ with these tails summable over plaquettes; the intersection of Ω with their product of compact intervals is compact. The finitely many initial laws are tight individually. Any weak limit ν_t consequently satisfies

$$\nu_t(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2+2t|O|_1k^3}. \tag{4.24}$$

This uses lower semicontinuity of the exponential integral, as explained after (4.3).

4.5. **The two adjacent fermion layers.** The remaining comparison argument uses gauge invariant polynomials

$$P(\widehat{A}) = \sum_a b_a e^{i\langle\delta_a,\widehat{A}\rangle}\Psi_a, \qquad d^*\delta_a = q\mathfrak{q}(\Psi_a),$$

where the cochains have finite support and the Ψ_a are ordered Grassmann monomials. Denote their class by $\mathcal{P}$. The charge condition is exactly gauge invariance. It also makes each term even: the total divergence is zero, so its numbers of ψ and $\bar\psi$ factors agree. The expansion is unique after equal pairs (δ_a,Ψ_a) have been collected, because the monomials and the distinct characters are linearly independent. Each such polynomial is an admissible local integrand with bounded coefficients.

At a coordinate link plane $c = j+\frac{1}{2}$, a two-layer polynomial reads photon links in the planes j and $j+1$ and between them, and fermion fields only at the two ends of the bonds crossing c. These are the thin layers $\mathcal{X}_{j,6}$ and $\mathcal{X}_{j+1,1}$; the other thin layers of the two chambers do not occur. Call this subclass $\mathcal{P}_1(c)$. For a monomial $Y = e^{i\delta\widehat{A}}\Psi$, write its exponent as $(\delta_0,\delta_1,\delta_\perp)$ on the lower spatial, upper spatial and temporal links. If $\mathfrak{q}_\pm$ are the lower and upper charge densities, gauge invariance gives

$$d_s^*\delta_0 - \delta_\perp = q\mathfrak{q}_-, \qquad d_s^*\delta_1 + \delta_\perp = q\mathfrak{q}_+. \tag{4.25}$$

Let $S = \operatorname{supp}\delta_0 \cup \operatorname{supp}\delta_1$ and choose a nonempty Λ containing their endpoints, the support of $\delta_\perp$ and all sites over which the fields of Ψ occur.

Every such Y is a kernel insertion, but the operator between the usual quarter-heat factors need not be bounded. We give the different regularization that permits approximation by bounded operators. The crossing modes and their reflected modes form a basis of the two-layer generators. Since all their weights are positive when $0 < \rho < 1$, there is a unique finite matrix Υ_Y with

$$\Psi e^{-S_c^{\times}(\Lambda)}\Big|_{\widehat{a}_j=0} = \mathrm{Ins}_c[\omega^{1/2}\Upsilon_Y\omega^{1/2}]. \tag{4.26}$$

To obtain it, expand the left side in the basis $\Theta_c\zeta_J\zeta_I$ and multiply the resulting matrix by $\omega^{-1/2}$ on both sides. Its nonzero entries have $\mathfrak{q}(I)-\mathfrak{q}(J) = \mathfrak{q}_- + \mathfrak{q}_+$. Thus Υ_Y has charge $q^{-1}d_s^*(\delta_0+\delta_1)$ and is even.

Set

$$k_Y^0(A', A) = e^{i\delta_1 A'}\prod_{b\in S} g_1(A'(b) - A(b))e^{i\delta_0 A}\Upsilon_Y. \tag{4.27}$$

It is the continuous kernel of $e^{i(\delta_0+\delta_1)A}e^{-|E_S+\delta_0|^2/2}\Upsilon_Y$. We also write k_Y^0 for this integral operator. The charge identity and unchanged Gaussian difference prove its covariance. Moreover

$$Y = G[k_Y^0]. \tag{4.28}$$

Indeed, the Gaussian factors in the insertion cancel. Applying the automorphism that multiplies the lower fields by their charge phases to (4.26) replaces its matrix by $\omega^{1/2}\Upsilon_Y u(\widehat{a}_j)\omega^{1/2}$ and its left side by $e^{-iq\langle\mathfrak{q}_-,\widehat{a}_j\rangle}\Psi e^{-S_c^{\times}(\Lambda)}$. The remaining temporal exponent in $G[k_Y^0]$ is $d_s^*\delta_0 - q\mathfrak{q}_- = \delta_\perp$ by (4.25), proving the identity.

Put $\widetilde{E} = E_S + \delta_0$, $D = \delta_1 - \delta_0$ and $a = (|\delta_0|^2 + |\delta_1|^2)/8$. Define the bounded operator

$$X'_Y = e^{i(\delta_0+\delta_1)A}e^{-\frac{1}{4}|\widetilde{E}|^2+\frac{1}{4}\langle\widetilde{E},D\rangle+a}\Upsilon_Y. \tag{4.29}$$

It belongs to $\mathfrak{W}(S,\Lambda)$, and completion of the quadratic exponent gives

$$e^{-|E_S|^2/8}X'_Y e^{-|E_S|^2/8} = k_Y^0, \qquad \|X'_Y\| = \|\Upsilon_Y\|e^{|D|^2/16+a}.$$

For the first identity move the left heat factor through the phase using the Weyl rule. Its resulting squares are $\frac{1}{8}|\widetilde{E}+\delta_1|^2+\frac{1}{8}|\widetilde{E}-\delta_0|^2$; their linear and constant terms cancel those in (4.29). Therefore $Y = G'_{X'_Y}$ and it satisfies the coordinate bound $2^{|S|/2}\|X'_Y\|$. This bound will also control the approximation error uniformly in the torus.

For $\epsilon > 0$ let $P_\epsilon = e^{-\epsilon|E_S|^2}$ and

$$X_{Y,\epsilon} = e^{i(\delta_0+\delta_1)A}\exp\Big(-\epsilon|\widetilde{E}+\delta_1|^2 - \epsilon|\widetilde{E}-\delta_0|^2 + \tfrac{1}{2}\langle\widetilde{E},D\rangle + 2a\Big)\,\Upsilon_Y. \tag{4.30}$$

The quadratic part is strictly negative, so this too is in $\mathfrak{W}(S,\Lambda)$. The same Weyl calculation gives

$$\begin{aligned} e^{-|E_S|^2/4}X_{Y,\epsilon}e^{-|E_S|^2/4} &= P_\epsilon k_Y^0 P_\epsilon, \\ Y - G_{X_{Y,\epsilon}} &= G'_{X'_Y - P_\epsilon X'_Y P_\epsilon}. \end{aligned} \tag{4.31}$$

The second equality follows from the first and the primed representation, since their continuous kernels agree. Its coordinate bound is $2^{|S|/2}\xi_Y(\epsilon)$, where

$$\begin{aligned} \xi_Y(\epsilon) = \|\Upsilon_Y\| \sup_{x\in\mathbb{R}^S} & e^{-|x|^2/4+\langle x,D\rangle/4+a} \\ & \cdot (1 - e^{-\epsilon|x+\delta_1|^2-\epsilon|x-\delta_0|^2}) \le C_Y\epsilon. \end{aligned} \tag{4.32}$$

The last inequality uses $1 - e^{-u} \le u$ and the boundedness of a Gaussian times a quadratic polynomial. By contrast,

$$\|X_{Y,\epsilon}\| = \|\Upsilon_Y\| \exp\left(\frac{(1/2 - 2\epsilon)^2 |D|^2}{8\epsilon} + (1/4 - \epsilon)(|\delta_0|^2 + |\delta_1|^2) \right). \tag{4.33}$$

It can grow exponentially in $1/\epsilon$. This forces the order of the limits in the next result.

Reflection changes $(\delta_0, \delta_1, \delta_\perp)$ to $(-\delta_1, -\delta_0, \delta_\perp)$ and Υ_Y to Υ_Y^*. The latter follows by reflecting (4.26); $\mathrm{Ins}_c[\omega]$ is even and invariant. Taking adjoints in the two heat identities then gives $X'_{\Theta_c Y} = (X'_Y)^*$ and $X_{\Theta_c Y,\epsilon} = X^*_{Y,\epsilon}$; the heat operator is injective with dense range, so these identities determine the bounded middle operators uniquely. All constructions extend linearly to two-layer polynomials. Coordinate permutations give the same assertions at every coordinate link plane.

Proposition 4.4. *For every $Y \in \mathcal{P}_1(c)$ at a coordinate link plane,*

$$\lim_{t \downarrow 0} \limsup_{i \to \infty} |\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| = 0. \tag{4.34}$$

The assertion concerns the finite-volume functionals on fermion polynomials; it does not presume that they converge to fermion laws.

Proof. For a fixed support, $o_i^t(X) = \mu_i^t(G_X)$ is a unital linear functional on the local operator algebra with $|o_i^t(X)| \le \gamma \|X\|$, where $\gamma = \gamma_{t,i}(k)$. A unital functional this close in norm to one is nearly Hermitian. If $X_0 = X_0^*$ and $\|X_0\| \le 1$, write $o_i^t(X_0) = u + iv$. For each real s,

$$(v + s)^2 \le |o_i^t(X_0 + is)|^2 \le \gamma^2 \|X_0 + is\|^2 \le \gamma^2 (1 + s^2).$$

For $\gamma > 1$ take $s = v/(\gamma^2 - 1)$; this gives $|v| \le \sqrt{\gamma^2 - 1}$. For $\gamma = 1$ the same conclusion follows by letting s tend to either infinity. Decomposing a general X into its real and imaginary self-adjoint parts yields

$$|o_i^t(X^*) - \overline{o_i^t(X)}| \le 4\sqrt{\gamma^2 - 1}\,\|X\|. \tag{4.35}$$

Use (4.13) to interpret this as reflection covariance of the insertion expectation.

Apply (4.35) to $X_{Y,\epsilon}$ and use (4.31) for Y and its reflection. The amplification theorem controls the two errors and gives

$$\begin{aligned} |\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| \le{}& 4\sqrt{\gamma^2 - 1}\,\|X_{Y,\epsilon}\| \\ &+ 2 \cdot 2^{|S|/2} \gamma \xi_Y(\epsilon). \end{aligned} \tag{4.36}$$

The support box and its coarse scales do not depend on ϵ. First take $i \to \infty$, and then $t \downarrow 0$ with ϵ fixed. Since $\Delta_i \to 0$, the first term vanishes and the remaining upper bound is $2 \cdot 2^{|S|/2} \xi_Y(\epsilon)$. Let $\epsilon \downarrow 0$ using (4.32). Finite sums and coordinate transport give the stated result. □

4.6. **Removing fields from a thin layer.** The fermion graph has six thin layers in each thick layer j, at heights $j - v_1, j - v_2, j - v_3, j + v_3, j + v_2, j + v_1$. Each site has one neighbor in the next thin layer, one in the preceding one and two in its own layer. The forward matching bonds have types $2, 1, 0, 1, 2, 3$ in succession. This follows directly from the chamber reflections: type zero reverses the coordinate of modulus v_3, types one and two interchange the adjacent moduli, and type three reverses the coordinate of modulus v_1 and shifts its nearest photon site by the corresponding unit vector. The bonds of types zero, one and two carry no photon phase. Type three bonds within thin layers $2, \dots, 5$ carry spatial phases in plane j, while those between layer six and the next layer one carry temporal phases between j and $j + 1$. These facts control which planes the recursion can read.

For a photon site y, let $\mathcal{P}^{\pm}(y)$ consist of the same exponential monomials as $\mathcal{P}$, with $d^*\delta = q(\mathfrak{q}(\Psi) \mp 1_y)$. They transform with gauge phase $e^{\pm iq\lambda(y)}$ and are odd. Removing a $\bar\psi(x)$ from an invariant monomial leaves an element of $\mathcal{P}^+(\pi(x))$; removing a $\psi(x)$ leaves one in $\mathcal{P}^-(\pi(x))$.

Multiplication by a hopping phase transports the specified charge to the other endpoint. These assertions are the identities

$$d^*1_{(y,y')} = 1_{y'} - 1_y, \qquad U_A(x,x') = e^{iq\langle 1_{(\pi(x),\pi(x'))}, A\rangle}.$$

They show that every term of the following replacements stays in $\mathcal{P}$.

Berezin integration by parts, with left derivatives, gives

$$\begin{aligned} Z_T[\Phi(\bar\psi\mathcal{D}_A)_\alpha(x')G] &= -Z_T[\Phi\partial_{\psi_\alpha(x')}G], \\ Z_T[\Phi(\mathcal{D}_A\psi)_\alpha(x')G] &= Z_T[\Phi\partial_{\bar\psi_\alpha(x')}G]. \end{aligned} \tag{4.37}$$

The appropriate charge of G makes each integrand invariant. These identities first hold at fixed complex A: the derivatives of $e^{-S_T(A)}$ are respectively $(\bar\psi\mathcal{D}_A)_\alpha e^{-S_T(A)}$ and $-(\mathcal{D}_A\psi)_\alpha e^{-S_T(A)}$, and the Berezin integral annihilates a derivative. Integrating against any bounded bosonic factor $\Phi(F)$ then proves (4.37). The enlarged local box must embed so that all generators involved are distinct torus generators and the phases can be read on the local lift.

Let x' be the neighbor of x in the next thin layer. Its matching hopping matrix $w(x,x')$ has eigenvalues $\frac{w_i}{2}(\rho \mp 1)$ and is invertible for our fixed $0 < \rho < 1$. Solve the equation at x' for the field at x. With the replaced field moved to the left of the monomial, the two rules are

$$\begin{aligned} \bar\psi(x) &\longmapsto \left(M\bar\psi(x') - \sum_{x''\sim x',\, x''\neq x} \bar\psi(x'')w(x'',x')U_A(x'',x') + \partial_{\psi(x')} \right) w(x,x')^{-1}U_A(x',x), \\ \psi(x) &\longmapsto U_A(x,x')w(x',x)^{-1} \left(M\psi(x') - \sum_{x''\sim x',\, x''\neq x} w(x',x'')U_A(x',x'')\psi(x'') - \partial_{\bar\psi(x')} \right). \end{aligned} \tag{4.38}$$

The derivatives act on the monomial to their right. Multiplying by the inverse matching matrix and its inverse phase in (4.37) proves equality of the expectations before and after either replacement, for every bosonic factor Φ. This equality is independent of a choice of time direction or a symmetry of the torus.

For a thin layer $\mathbb{L}$ define a linear map $R^+_{\mathbb{L}}$ by recursion on the number of its fields in an ordered monomial. Select the first such field, move it to the front with its Grassmann sign, and apply (4.38) through its matching neighbor in $\mathbb{L}^+$. That neighbor has only one neighbor in $\mathbb{L}$, the field just removed. All other fields created by the rule lie in $\mathbb{L}^+$ or $\mathbb{L}^{++}$; its derivatives remove fields in $\mathbb{L}^+$. Thus the number in $\mathbb{L}$ falls by exactly one at every recursive call. The map is well-defined, preserves $\mathcal{P}$, removes all fields in $\mathbb{L}$, and creates no new fields outside its next two layers. A downward map $R^-_{\mathbb{L}}$ is obtained using the preceding matching neighbor.

If the initial polynomial is read in a box B, the entire one-layer recursion is read in $B+[-1,1]^3$. The fields being removed stay in the original box, and their next neighbors and the other neighbors of those sites have photon projections equal or nearest to the original projection: among two successive different bonds at most one has type three. Hence these sites and every added phase are in the same enlarged box, irrespective of the number of recursive calls. Induction therefore gives

$$Z_T[\Phi P] = Z_T[\Phi R^\pm_{\mathbb{L}} P] \tag{4.39}$$

whenever this enlarged box embeds. The new photon links are only those of type three bonds with an endpoint in $\mathbb{L}^\pm$. The recursion commutes with multiplication by a curvature character, since it differentiates only fields and its choices depend only on the monomial.

We first use this recursion to bound an arbitrary polynomial; here its photon links need not yet be reduced. Move its fields upwards through all thin layers below $\mathcal{X}_{j,6}$ and downwards through those above $\mathcal{X}_{j+1,1}$. The last downward replacement can reach $\mathcal{X}_{j,6}$, and cannot reach a lower

layer. We obtain, on every sufficiently large torus and with every bounded bosonic factor, an equality

$$Z_T[\Phi P] = \sum_a b_a Z_T[\Phi \varphi_{\beta_a} Y_a], \qquad Y_a \in \mathcal{P}_1(c). \tag{4.40}$$

Here the Y_a can be chosen with integer exponents qj_a. Indeed each remaining monomial has integer charge of total zero on the two adjacent photon planes. In their connected rectangular support graph choose paths from a fixed vertex to each charged vertex; the charge-weighted sum of these paths is an integer cochain j_a with the required divergence. Then $u_a = \delta_a - qj_a$ is divergence free in the same integer box and can be written $d^*\beta_a$ there.

The filling must stay inside the support box for the polynomial bound below. In $\prod_\mu [a_\mu, a'_\mu]$, set electric components $\beta^{\mathrm{el}}_j = -\sum_{l=a_1}^{j} (u_a)_l$ for $a_1 \le j < a'_1$. Subtracting $d^*\beta$ removes all spatial links below the top plane; divergence zero then removes the temporal links. On that top plane a second partial-sum sweep in the second coordinate removes the third-direction links below its top edge. Divergence zero removes the second-direction links and leaves only a terminal segment in direction three. Its divergence equation and vanishing boundary value make it zero. The two sweeps use only plaquettes of the box. This proves the asserted filling and thus the splitting $\delta_a = d^*\beta_a + qj_a$ in (4.40).

Apply the coordinate bound to each Y_a, absorb the extra character and use amplification. For any fixed box B_0 there are constants C_P, k_P and a volume threshold with

$$\sup_{\operatorname{supp}\beta \subset B_0} |\mu_i^t(\varphi_\beta P)| \le C_P \gamma_{t,i}(k_P), \qquad t \ge 0. \tag{4.41}$$

One may take $C_P = \sum_a |b_a| 2^{|S_a|/2} \|X'_{Y_a}\|$, with the summand one for $Y_a = 1$. All support boxes in the finite normal form and B_0 fit in one fixed box; its radius determines k_P. The size of β does not enter the bound. Reflected polynomials have the same kind of bound, since the normal form identity can be reflected on the reflected torus and the coordinate bounds are preserved by reflection.

4.7. **The contour shift and an outer plane.** A fermion recursion removes fields from a plane but leaves its photon exponent. An imaginary shift of links in the next plane removes that exponent together with the outgoing temporal one. The all-flux measure is essential for this shift: summation over exact integer fluxes unfolds the compact link intervals to real lines on which the contour can move.

Let u be a real cochain supported on finitely many links S of one coordinate plane, and let $P \in \mathcal{P}$. Choose a box B reading P and containing $b + [-1,1]^3$ for every $b \in S$. Put $P^{(u)}(\widehat{A}) = P(\widehat{A} - iu)$ by its entire exponential expansion. Define

$$\begin{aligned} Q_u = \sum_{b \in S} \sum_{(x,x') \text{ over } b} \big[& (e^{qu_b} - 1)\bar\psi(x) w(x,x') U_A(x,x') \psi(x') \\ & + (e^{-qu_b} - 1)\bar\psi(x') w(x',x) U_A(x',x) \psi(x) \big], \qquad V_u = e^{Q_u}. \end{aligned} \tag{4.42}$$

The bonds are oriented with b. Each term is invariant and even, and the exponential is a finite polynomial by nilpotence. It reads only those links of S with $u_b \neq 0$ and their overlying bonds. For a plane normal to e_1, those fields are in thin layers $2, \dots, 5$ of its thick layer. The change in the action is $S_T(A - iu) - S_T(A) = -Q_u$, so $e^{-S_T(A-iu)} = e^{-S_T(A)} V_u$.

If $\Phi(F)$ is entire and bounded on each tube of bounded imaginary curvature, then on every torus in which B embeds we have

$$Z_T[\Phi P] = e^{\|du\|^2/2} Z_T[\Phi(F - idu)\varphi_{du}(F) V_u P^{(u)}]. \tag{4.43}$$

The norm here is the lattice norm, equal to the torus norm because its support embeds. We supply the contour justification. For each b choose its electric plaquette leaving the plane in the normal direction. Its only side in S is b, with coefficient ±1. Consequently d is injective on cochains supported in S, and the action $z \mapsto z + dk$, $k \in \mathbb{Z}^S$, on closed integer fluxes is free. Choose orbit representatives z_0. The compact A_S intervals and the sum over k become the integral over $\mathbb{R}^S$, since $(A, z_0 + dk)$ represents the same configuration as $(A + \frac{2\pi}{q}k, z_0)$.

For fixed z_0 and the other real links, choose an integer primitive of z_0 in the local box. Adding it to A gives a local lift affine in the complex variables $\zeta = A_S$. The Berezin coefficients of $e^{-S_T(A)}$ are polynomials in the link phases, so the full integrand extends to an entire function $f(\zeta)$. On the strip $|\Im\zeta_b| \le \|u\|_\infty$ the selected plaquettes give

$$|f(\zeta)| \le C \prod_{b\in S} e^{-(\Re\zeta_b - a_b)^2/2}. \tag{4.44}$$

Indeed each of their real curvatures is $\pm\Re\zeta_b + a'_b$, all imaginary curvatures are bounded in this finite strip, the factor Φ is bounded there, and each remaining exponential phase is bounded there. The constant can be chosen independent of the remaining real links and the orbit representative; only the real Gaussian centers vary.

Move one variable at a time along the rectangle with real endpoints $\pm R$ and imaginary displacement $-iu_b$. The vertical sides tend to zero by (4.44). Fubini then gives $\int f(\xi)\,\mathrm{d}\xi = \int f(\xi - iu)\,\mathrm{d}\xi$. At the shifted arguments the Maxwell exponent becomes

$$-\tfrac12\langle F - idu, F - idu\rangle = -\tfrac12\|F\|^2 + i\langle du, F\rangle + \tfrac12\|du\|^2.$$

The change in the fermion action is the polynomial V_u, and the observable becomes $P^{(u)}$. Both the initial integrand and the resulting real integrand are absolutely integrable by their local bounds. Hence we may sum the identities over z_0 and integrate the remaining links, which proves (4.43).

We now eliminate the lowest plane j read by P. This means that its fields have photon projection height at least j and its links have no endpoint below plane j. Apply the six upward thin-layer maps of this thick layer to obtain $P^\uparrow = \sum_a b_a e^{i\delta'_a \widehat{A}}\Psi_a$. It reads no fields of thick layer j and no links below it; the only new links are spatial links in plane j and temporal links from j to $j+1$. At each term, gauge invariance in plane j therefore says

$$(\delta'_a)^\perp_j = d^*_s(\delta'_a)_j. \tag{4.45}$$

For this term let u_a be supported on plane $j+1$ with $(u_a)_{j+1} = (\delta'_a)_j$. Define

$$P_a = b_a e^{\|du_a\|^2/2 + \langle \delta'_a, u_a\rangle} e^{i\langle \delta'_a + d^*du_a, \widehat{A}\rangle} V_{u_a}\Psi_a. \tag{4.46}$$

Then (4.39) and (4.43) give the exact relation

$$Z_T[\Phi P] = \sum_a Z_T[\Phi(F - idu_a)P_a]. \tag{4.47}$$

The construction is on the infinite lattice; the same relation holds for every Φ entire in the plaquette variables and bounded on each tube $|\Im F_\sigma| \le r$ for all σ, $r > 0$, and every torus containing the fixed enlarged support box.

Here is the cancellation of the eliminated exponent. Write $v = (\delta'_a)_j$. From the slice formulas $(du)^{\mathrm{el}}_l = u_{l+1} - u_l - d_s u^\perp_l$ and $(d^*\beta)_l = \beta^{\mathrm{el}}_{l-1} - \beta^{\mathrm{el}}_l + d^*_s\beta^{\mathrm{mag}}_l$, the only nonzero components of d^*du_a are

$$\begin{aligned}
(d^*du_a)_j &= -v, & (d^*du_a)^\perp_j &= -d^*_s v,\\
(d^*du_a)_{j+1} &= (2 + d^*_s d_s)v, & (d^*du_a)^\perp_{j+1} &= d^*_s v,\\
(d^*du_a)_{j+2} &= -v.
\end{aligned} \tag{4.48}$$

Thus the spatial exponent in plane j vanishes, and (4.45) cancels its outgoing temporal exponent as well. The polynomial V_{u_a} reads only thick layer $j+1$. Hence P_a has lowest plane at least $j+1$ and highest plane at most the maximum of $j+2$ and the previous highest plane. The six field recursions enlarge the box by six, and the shifted links with their plaquettes require two more. The whole step is read in $B+[-8,8]^3$. Elimination from the highest plane is obtained by the downward maps and a shift in plane $j-1$. The corresponding incoming temporal exponent is $-d_s^*(\delta'_a)_j$, and the same two components cancel. The highest plane falls by one, while the lowest can fall only to $j-2$.

The source of error in the tilted law is precisely the shifted bosonic factor. For a finite cochain η, only finitely many copies in the array V_i meet its support. Their change is the fixed trigonometric polynomial

$$H_\eta(F)=\sum_{\kappa,b} a_b(e^{\langle (g_\kappa^{-1})^*\beta_b,\eta\rangle}-1)\varphi_{(g_\kappa^{-1})^*\beta_b}(F), \tag{4.49}$$

where $O=\sum_b a_b\varphi_{\beta_b}$ and only contributing cells occur. At most $|\operatorname{supp}\eta|$ cells contribute; each lies in the support box of η enlarged by h. Its coefficient norm has the bound

$$|H_\eta|_1\le C(\eta):=|\operatorname{supp}\eta|\,|O|_1(e^{\max_b\|\beta_b\|_1\|\eta\|_\infty}-1). \tag{4.50}$$

If that enlarged box embeds, no two contributing cells are period equivalent and the same polynomial is the torus difference $V_i(F-i\eta)-V_i(F)$. Expand its exponential in characters:

$$e^{tV_i(F-i\eta)}=e^{tV_i(F)}\left(1+\sum_\alpha b_\alpha(t)\varphi_{\beta_\alpha}(F)\right),\qquad \sum_\alpha|b_\alpha(t)|\le e^{tC(\eta)}-1. \tag{4.51}$$

The coefficient norm is submultiplicative, so the exponential series proves absolute uniform convergence. All character supports lie in one fixed enlarged box, independent of i and t; sums of character exponents do not enlarge this support.

Set $P^\flat=\sum_a P_a$ in (4.47). Insert (4.51) with $\eta=du_a$ and divide by the positive denominator. Termwise integration is valid: at a fixed torus the functional on bounded bosonic factors multiplying P_a is continuous for their supremum norm, by the absolute integrability of its Berezin coefficients. We get

$$|\mu_i^t(P)-\mu_i^t(P^\flat)|\le(e^{Ct}-1)\sum_a\sup_{\operatorname{supp}\beta\subset B'}|\mu_i^t(\varphi_\beta P_a)|, \tag{4.52}$$

where $C=\max_a C(du_a)$ and B' is a fixed box enlarged by h. The a priori bound (4.41) makes this $O(t)$ uniformly for all large i and $0\le t\le 1$: each of the finitely many amplification factors is bounded once $\Delta_i\le 1$, and $e^{Ct}-1\le(e^C-1)t$ on this interval.

The same estimate applies after transporting the identity. This needs care because D_i need not be invariant under a coordinate permutation. Reflect the equality on the reflected torus and then reflect it back, using functional covariance. The conjugations change a shift η to $-\theta^*\eta$. Indeed, if $\Phi^\theta(F)=\overline{\Phi(\theta^*\overline{F})}$, then $(\Phi^\theta(\,\cdot-i\eta))^\theta=\Phi(\,\cdot+i\theta^*\eta)$. Thus (4.47) becomes an exact identity for ΘP with shifts $-\theta^*du_a$ and polynomials ΘP_a on the original torus. Iterating reflections transports it by any g, with shifts $\det(\vec g)(g^{-1})^*du_a$, where $\vec g$ is the linear part of g. Its support box is transported at the same time. Apply (4.51) and (4.41) on D_i itself to obtain the transported $O(t)$ estimate. No invariance of D_i or of its tilted array under g is used.

Theorem 4.5. *For each $P\in\mathcal{P}$ and coordinate link plane c there is $Y_P\in\mathcal{P}_1(c)$, independent of the torus and tilt, such that, for every fixed lattice isometry g,*

$$|\mu_i^t(P^g)-\mu_i^t((Y_P)^g)|\le C_{P,g}t\quad(0\le t\le 1, i\ge i_0(P,g)). \tag{4.53}$$

At a plane normal to either other coordinate the corresponding reduction and its reflected estimate hold as well.

Proof. For $c = j_0 + \frac{1}{2}$, eliminate lowest planes until the lowest is at least j_0. Every step raises it by at least one, so this takes finitely many steps. Then eliminate highest planes until the highest is at most $j_0 + 1$. During this phase an eliminated plane $j \geq j_0 + 2$ can create links only as low as $j - 2 \geq j_0$, preserving the first reduction. The resulting polynomial reads just the two thick layers and their connecting links. Apply the upward recursions in thin layers $1, \dots, 5$ of the lower thick layer, followed by the downward recursions in thin layers $6, \dots, 2$ of the upper one. New fields remain at or above the lower layer six and at or below the upper layer one. The new type three phases are spatial phases in these two planes or the temporal phases between them. Thus the result Y_P is in $\mathcal{P}_1(c)$.

There are finitely many outer-plane steps, each with (4.52) and its transported version; the final ten thin-layer recursions are exact with the unchanged bosonic factor and contribute no error. Take one volume threshold for all their fixed support boxes and a priori bounds, and sum the finite $O(t)$ errors. This proves (4.53). For another direction first apply the construction to $P^{g_\mu^{-1}}$ and then transport its exact identities by the cyclic coordinate permutation g_μ. The general transported estimate, evaluated on D_i, proves the claim there and its reflected version. □

4.8. **Comparison states and translation invariance.** A comparison state for the chosen cell structure and O is a law ν_0 obtained by the ordered limits

$$\mu_{i_m}^{t_k} \longrightarrow \nu_{t_k} \quad (m \to \infty) \text{ for each } k, \qquad \nu_{t_k} \longrightarrow \nu_0 \quad (k \to \infty), \qquad t_k > 0, \quad t_k \to 0. \tag{4.54}$$

All convergences here are weak convergences of bosonic laws on Ω, and the volume subsequence is common to all strengths. Given a prescribed subsequence of volumes, a further subsequence produces such a state. To see this, take strengths $t_k = 1/k$, extract nested convergent volume subsequences by the tightness proved above, and choose their diagonal subsequence. The limiting laws ν_{t_k} are themselves uniformly tight: applying (4.24) to $\beta = \pm 1_\sigma$ and using $t_k \leq 1$ bounds their exponential coordinate moments by a common constant. The same product-of-intervals argument gives a weakly convergent strength subsequence. This proves existence of (4.54) subordinate to any prescribed volume subsequence.

Every comparison state has Gaussian domination

$$\nu_0(e^{\langle \beta, F \rangle}) \leq e^{\|\beta\|^2/2}, \qquad \beta \text{ of finite support.} \tag{4.55}$$

This follows from (4.24) by letting $t_k \to 0$ and truncating the exponential in the second weak limit. Wall invariance and wall reflection positivity also pass through both limits. For each local positive-side test, its support is contained in the positive half-period slab once i_m is large, so (4.3) applies. The continuity and L^2 extension described in §4.1 give positivity for bounded Borel local tests. In particular ν_0 is invariant and reflection positive at $(x_1 - x_2, c_0)$.

Suppose the unperturbed cubic laws $\mu_{C(n_i)}$ converge to ν on the prescribed subsequence. For a comparison state subordinate to it, the finite fold inequality gives

$$\nu(O) \leq \nu_0(O). \tag{4.56}$$

Indeed fix $t_k > 0$, take the volume limit in

$$\mu_{C(n_{i_m})}(O) \leq \mu_{i_m}^{t_k}(O) + |Q|\Delta_{i_m}/t_k,$$

and use bounded continuity of O and $\Delta_{i_m} \to 0$. Then let $t_k \to 0$. The denominator t_k in the error precludes reversing this order.

We now obtain the extra invariance required to use (4.56) for observables at arbitrary positions. Fix a coordinate link reflection θ_c and a curvature character $P = \varphi_\beta$. Its polynomial representation is $e^{i\langle d^*\beta, \widehat{A}\rangle}$; on every sufficiently large torus this equals $e^{i\langle \beta, \widetilde{F}\rangle}$ on the local lift. Apply Theorem 4.5 to P and its reflection, and then Proposition 4.4 to its two-layer reduction. The finite errors are at most two $O(t)$ terms plus a finite sum of the terms in (4.36). Along the common volume

subsequence in (4.54), take volume to infinity, then $t_k \to 0$ with the regularization fixed, and finally remove the regularization. Only the character expectations are passed to weak limits. The conclusion is

$$\nu_0(\Theta_c\varphi_\beta) = \overline{\nu_0(\varphi_\beta)}. \tag{4.57}$$

There is no need to extract limiting fermion expectations in this argument.

For a bosonic character $\Theta_c\varphi_\beta = \varphi_{-\theta_c^*\beta}$, while positivity of the probability law gives $\overline{\nu_0(\varphi_\beta)} = \nu_0(\varphi_{-\beta})$. Thus (4.57), with $-\beta$ in place of β, is invariance under the tensor pullback $F \mapsto \theta_c^* F$. Characters determine all finite plaquette marginals by uniqueness of their Fourier transforms; these marginals determine the law because cylinder sets generate the Borel sigma algebra of Ω. Consequently every comparison state is invariant under all coordinate link reflections, in all three directions. The product of reflections at c and $c+1$ is translation by $2e_\mu$, so

$$(\tau_v^*)_*\nu_0 = \nu_0, \qquad v \in 2\mathbb{Z}^3. \tag{4.58}$$

We need only these even translations. The original wall periods alone would not have permitted the next placement.

4.9. **Convex separation and the nine mirror families.** Fix an integer c_0. Let $\mathcal{R}(c_0)$ be the set of laws on Ω satisfying Gaussian domination and both invariance and reflection positivity at $(x_1 - x_2, c_0)$. This is convex and weakly compact. Gaussian domination gives a common tail $\lambda(|F_\sigma| \geq R) \leq 2e^{-R^2/2}$ and hence uniform tightness. It is weakly closed by truncation of its exponential integrals. Mirror invariance and positivity against continuous local tests are also closed, and the L^2 approximation extends the latter to Borel tests. Each condition is affine or an affine inequality in the law, proving convexity.

We claim that, for every real local trigonometric polynomial O and every cubic limit ν along our divisible side sequence, some $\nu_0 \in \mathcal{R}(c_0)$ satisfies (4.56). The polynomial need not initially fit in a cell having its lower wall at c_0. Let $\operatorname{supp} O \subset y_0 + [-r, r]^3$, with $r \geq 1$, and choose an admissible even width $h > 8r + 8$. Choose an even integer a in

$$(c_0 + 2r - \ell_-(y_0),\ c_0 + h - 2r - \ell_-(y_0)).$$

The interval has length greater than two. For $v = ae_1$ and $y = y_0 + v$ its defining inequalities place the translated box between the two ℓ_- walls. Choose integer c_+ and half-integer c_3 so that

$$c_+ < \ell_+(y) - 2r < \ell_+(y) + 2r < c_+ + h, \qquad c_3 < y_3 - r < y_3 + r < c_3 + h.$$

The available intervals have length greater than one. The box therefore lies in the open diagonal cell, since $|\ell_\pm(x) - \ell_\pm(y)| \leq 2r$ on it. Form the tilted law with the translated polynomial O^{τ_v}.

Choose a comparison state subordinate to the volume subsequence defining ν. It belongs to $\mathcal{R}(c_0)$, and the fold inequality gives $\nu(O^{\tau_v}) \leq \nu_0(O^{\tau_v})$. The original cubic laws are invariant under every lattice translation, and this invariance passes to ν. Equation (4.58) gives the same invariance under this particular v for ν_0. Hence

$$\nu(O) = \nu(O^{\tau_v}) \leq \nu_0(O^{\tau_v}) = \nu_0(O). \tag{4.59}$$

The comparison state may depend on O. Convex separation is what turns these individual inequalities into reflection positivity of the original law.

Lemma 4.6. *Let $\mathcal{R}$ be a convex weakly compact set of laws on Ω. If for every real local trigonometric polynomial O some $\lambda \in \mathcal{R}$ satisfies $\nu(O) \leq \lambda(O)$, then $\nu \in \mathcal{R}$.*

Proof. Give the real vector space of finite signed measures on Ω the weak topology defined by integration against these polynomials. It is locally convex and Hausdorff: if every such integral of a signed measure vanishes, its finite-coordinate marginals have zero Fourier transform, so its positive and negative parts agree on cylinder sets and hence on every Borel set. Its continuous

linear functionals are precisely integration against the test space; a continuous linear functional for a topology defined by these seminorms depends on finitely many of them, and their finite real span is still that space. Since the tests are bounded and continuous, the inclusion from the usual weak topology of laws is continuous. Thus $\mathcal{R}$ is a compact convex subset of this locally convex Hausdorff vector space. It is nonempty by the hypothesis at $O = 0$.

If ν were outside $\mathcal{R}$, strict separation of a compact convex set and a disjoint closed point would give a continuous linear functional with $\max_{\lambda\in\mathcal{R}} \lambda(O) < \nu(O)$ for some test O. Its maximum exists by compactness. This contradicts the assumed inequality for that O. □

Theorem 4.7. *At fixed $q > 0$, $0 < \rho < 1$, positive chamber weights and real mass, the cubic curvature laws along the divisible side sequence are tight on Ω. Every weak limit satisfies Gaussian domination, is invariant under lattice translations, signed coordinate permutations and $F \mapsto -F$, and is reflection positive at all mirrors*

$$x_\mu = c, \quad c \in \tfrac{1}{2}\mathbb{Z}; \qquad x_\mu \pm x_\nu = c, \quad c \in \mathbb{Z}, \quad \mu < \nu.$$

Thus it has all nine lattice mirror families.

Proof. The coordinate Gaussian domination theorem gives the uniform one-plaquette tails and hence tightness. Any finitely supported β embeds without overlap once the side is large, so its periodization has unchanged norm. Truncation passes Gaussian domination to every limit. The laws are supported on the closed subspace $dF = 0$. Finite-volume translation, signed permutation and charge-conjugation invariances pass to the limit under their continuous actions on Ω.

For a coordinate mirror $x_\mu = c$, a local positive-side continuous test lies in its positive half-period slab for all large sides. The cubic torus is adapted to this mirror, and its reflected slab trace gives nonnegativity of the reflected form. Weak convergence and the L^2 extension give coordinate reflection positivity of the limit. For the diagonal mirror $(x_1 - x_2, c_0)$, (4.59) holds for every real trigonometric polynomial. Lemma 4.6 therefore gives $\nu \in \mathcal{R}(c_0)$, proving its diagonal positivity. This is valid for every integer c_0.

A signed permutation transports this diagonal mirror and its positive half-space to any plane $x_\mu \pm x_\nu = c$. The corresponding reflection is the conjugate of the original one. Pull a positive-side test back by this isometry; invariance of ν identifies its reflected form with the already nonnegative one. This gives the six diagonal families and completes the proof. □

The fixed-parameter qualification is used throughout: the pressure error and every support-dependent amplification scale are allowed to depend on q, M and the weights. At spacing $\eta = L^{-N}$ the construction has $q = e\eta^{1/2}$ and $M = \bar{m}\eta + m_{0,N}(e) + \frac{\rho}{2}\sum_i w_i$, by (2.15). The side sequence in the volume parameter has the divisibility assumed here. Applying this theorem at fixed N therefore proves positivity for the volume limits used in the construction. The subsequent mesh limit must preserve these volume limits; the argument above supplies no estimate permitting simultaneous variation of spacing and volume.

5. Gaussian domination and tensor moments

The Gaussian bound is used at three different points: to obtain compactness of the curvature laws, to control products of field insertions, and to pass moments to limiting and conditional laws. We record its consequences in a form independent of the regulator. The tensor extension below supplies the stated growth axiom on general tests, rather than only on tensor products.

5.1. **Random linear forms.** Let λ be a probability law on the real tempered two-forms, with its cylinder σ-algebra, and suppose

$$\mathbb{E}_\lambda e^{F(f)} \le e^{\|f\|_2^2/2} \qquad (f \in \mathcal{S}(\mathbb{R}^3; \Lambda^2) \text{ real}). \tag{5.1}$$

Set $X(f) = F(f)$ and extend X complex linearly. For $p > 0$ put $c_p = 2\,2^{p/2}\Gamma(1 + p/2)$.

Proposition 5.1. *The map X extends uniquely and continuously to $L^2(\mathbb{R}^3;\Lambda^2)$ with values in $L^p(\lambda)$ for every $p \geq 1$. For real f and $u > 0$,*

$$\mathbb{E}_\lambda X(f) = 0, \qquad \mathbb{E}_\lambda X(f)^2 \leq \|f\|_2^2, \tag{5.2}$$

$$\lambda(|X(f)| \geq u) \leq 2e^{-u^2/(2\|f\|_2^2)}, \qquad \mathbb{E}_\lambda |X(f)|^p \leq c_p \|f\|_2^p \quad (p > 0). \tag{5.3}$$

If $\|f\|_2 = 0$, the tail bound is interpreted as zero for $u > 0$. For real $f_1, \dots, f_n$,

$$\mathbb{E}_\lambda \prod_{j=1}^n |X(f_j)| \leq c_n \prod_j \|f_j\|_2, \qquad c_n \leq 2\sqrt{2^n n!}. \tag{5.4}$$

For complex tests the same bound holds with an additional factor $2^{n/2}$. The generating function of any finite family of tests is entire, and

$$\left|\mathbb{E}_\lambda e^{X(f)}\right| \leq e^{\|\operatorname{Re} f\|_2^2/2}. \tag{5.5}$$

Proof. For a real test, (5.1) applied to tf gives $\mathbb{E}_\lambda e^{tX(f)} \leq e^{t^2\|f\|_2^2/2}$. Also

$$\mathbb{E}_\lambda e^{t|X(f)|} \leq \mathbb{E}_\lambda e^{tX(f)} + \mathbb{E}_\lambda e^{-tX(f)} \leq 2e^{t^2\|f\|_2^2/2}.$$

Thus the exponential series can be differentiated at zero. Comparing its linear and quadratic terms with the Gaussian upper bound, for both signs of t, gives (5.2). Markov's inequality with $t = u/\|f\|_2^2$ gives the upper tail; applying it to $-f$ gives the lower tail. Integrating this bound yields

$$\mathbb{E}_\lambda |X(f)|^p = p \int_0^\infty u^{p-1} \lambda(|X(f)| > u)\, \mathrm{d}u \leq 2\, 2^{p/2} \Gamma(1 + p/2) \|f\|_2^p.$$

Hölder's inequality gives (5.4); log-convexity of Γ gives $\Gamma(1 + n/2) \leq \sqrt{n!}$. For $f = g + ih$, the triangle inequality in $L^n(\lambda)$ and $\|g\|_2 + \|h\|_2 \leq \sqrt{2}\|f\|_2$ give the complex bound. These estimates applied to differences extend X from real Schwartz tests to L^2, consistently for all $p \geq 1$.

On a compact polydisc in the variables of a finite generating function, its integrand is bounded by a product of exponentials of $|X(\operatorname{Re} f_j)|$ and $|X(\operatorname{Im} f_j)|$. Hölder's inequality and the preceding exponential bound make this majorant integrable. Differentiation under the integral proves the entire extension. Taking the modulus of the integrand and applying (5.1) proves (5.5). □

For fixed n, the moment bound controls differences of products by expanding one factor at a time. In particular, $S_n(f_1, \dots, f_n) = \mathbb{E}_\lambda \prod_j X(f_j)$ is a bounded multilinear form on complex L^2, symmetric in its arguments and real on real tests.

5.2. **Generating functions and cumulants.** Near the origin the logarithm of a generating function is well defined uniformly in the law. If $f = g + ih$, the elementary remainder bound $|e^z - 1 - z| \leq e^{|z|} - 1 - |z|$, convexity, and $\mathbb{E}_\lambda X(f) = 0$ give

$$\left|\mathbb{E}_\lambda e^{X(f)} - 1\right| \leq e^{2\|f\|_2^2} - 1. \tag{5.6}$$

Indeed, writing $\varphi(t) = e^t - 1 - t$, one has $\varphi(t) \leq 2(\cosh t - 1)$ for $t \geq 0$, and

$$\begin{aligned} \mathbb{E}_\lambda \varphi(|X(f)|) &\leq \tfrac{1}{2}\mathbb{E}_\lambda \varphi(2|X(g)|) + \tfrac{1}{2}\mathbb{E}_\lambda \varphi(2|X(h)|) \\ &\leq e^{2\|g\|_2^2} - 1 + e^{2\|h\|_2^2} - 1 \leq e^{2\|f\|_2^2} - 1. \end{aligned}$$

Consequently

$$W(z) = \log \mathbb{E}_\lambda \exp\Bigl(\sum_{j=1}^n z_j X(f_j)\Bigr)$$

is analytic and satisfies $|W(z)| \le \log 2$ on $\sum_j |z_j| \|f_j\|_2 \le 2/5$, using the branch with $W(0) = 0$. Cauchy's formula on the polydisc of radii $2/(5n\|f_j\|_2)$ gives

$$|\partial_1 \cdots \partial_n W(0)| \le (\log 2)(5n/2)^n \prod_j \|f_j\|_2. \tag{5.7}$$

If a test has zero norm, its random variable vanishes and the corresponding derivative is zero. These derivatives are the joint cumulants used in the fourth-order response calculation.

The entire extension also gives moment determinacy. Two laws obeying (5.1) and having the same moments have the same entire generating functions for every finite family of real tests. Their characteristic functions, and hence their finite-dimensional distributions, coincide. Cylinder sets generate the chosen σ-algebra, so the two laws coincide. We will use this after proving rotation covariance of all moments.

5.3. **Extension to tensor tests.** Let $A = 1 + |x|^2 - \Delta$ and let $h_\kappa = h_k e_{ij}$, $k \in \mathbb{Z}^3_{\ge 0}$, $i < j$, be the real Hermite basis of $L^2(\mathbb{R}^3; \Lambda^2)$. Its eigenvalues are $\lambda_\kappa = 2|k|_1 + 4$. For a tensor test write

$$\Psi = \sum_{\boldsymbol{\kappa}} c_{\boldsymbol{\kappa}}(\Psi) h_{\kappa_1} \otimes \cdots \otimes h_{\kappa_n}.$$

The Hermite expansion converges in Schwartz space. The estimate we need is

$$\sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \le 4^{-n} (34\pi\sqrt{3})^n q_{6n}(\Psi). \tag{5.8}$$

To see the derivative order and the dependence on n, set $Z_4 = \sum_\kappa \lambda_\kappa^{-4} < 1/16$. Weighted Cauchy–Schwarz and Parseval give

$$\sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \le Z_4^{n/2} \|(A^2)^{\otimes n} \Psi\|_2.$$

With $M(x) = 1 + |x|^2$,

$$A^2 = M^2 - 2M\Delta - 4x \cdot \nabla - 6 + \Delta^2.$$

The sum of the moduli of the differential coefficients of A^2 is bounded by $34M^2$. Applying this expression in each of the n arguments gives the componentwise bound

$$|(A^2)^{\otimes n} \Psi(x)| \le 34^n q_{6n}(\Psi) \prod_{a=1}^n (1 + |x_a|^2)^{-1}.$$

Since $\int_{\mathbb{R}^3} (1+|x|^2)^{-2} \, \mathrm{d}x = \pi^2$ and there are 3^n components, (5.8) follows. The bound on Z_4 follows by counting Hermite eigenvalues:

$$Z_4 = \frac{3}{32} \sum_{d \ge 0} \frac{d+1}{(d+2)^3} \le \frac{3}{32} \Big(\frac{\pi^2}{6} - 1 \Big) < \frac{1}{16}.$$

Define the random tensor evaluation by the $L^2(\lambda)$ sum

$$Q_\Psi = \sum_{\boldsymbol{\kappa}} c_{\boldsymbol{\kappa}}(\Psi) \prod_{a=1}^n X(h_{\kappa_a}). \tag{5.9}$$

The product moment bound, now applied to $2n$ real tests of norm one, shows that this sum converges absolutely in $L^2(\lambda)$ and

$$\|Q_\Psi\|_{L^2(\lambda)} \le \sqrt{c_{2n}} \sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)|.$$

For a decomposable test it equals $\prod_a X(f_a)$, by the Hermite expansion of each f_a and the same product bounds. Thus $S_n(\Psi) := \mathbb{E}_\lambda Q_\Psi$ is the unique continuous extension of the moments to $\mathcal{S}_n$. Moreover,

$$\begin{aligned}|S_n(\Psi)| &\le c_n \sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \\ &\le 2\sqrt{n!}\left(\frac{34\pi\sqrt{6}}{4}\right)^n q_{6n}(\Psi) \le 2\,66^n\sqrt{n!}\,q_{6n}(\Psi).\end{aligned}$$

This is (E0′). The construction also provides the L^2 tensor evaluations needed when passing the cluster estimate from local observables to Schwinger functions.

5.4. **The topology of the limits.** The Hilbert scale used for the limit laws is

$$\|F\|_{-s}^2 = \sum_\kappa \lambda_\kappa^{-2s} |F(h_\kappa)|^2, \qquad H_{-s} = \{F \in \mathcal{S}' : \|F\|_{-s} < \infty\}.$$

A closed bounded ball of H_{-3} is compact in H_{-4}. In Hermite coordinates, the tail of the squared H_{-4} norm on the indices $\lambda_\kappa > M$ is bounded by M^{-2} times the squared H_{-3} norm; the remaining coordinates form a finite-dimensional space. A diagonal subsequence argument proves compactness.

The lattice application will give the uniform bound $\mathbb{E}\|F^\sharp\|_{-3}^2 \le 61$. Its consequence is

$$P_{N,T}(\|F\|_{-3} > r) \le 61 r^{-2},$$

so the laws are tight on the separable Hilbert space H_{-4}. Prokhorov's theorem gives subsequential weak limits, and the inclusion $H_{-4} \hookrightarrow \mathcal{S}'$ is continuous for the strong distribution topology.

For fixed N, let $\mathcal{V}_N$ be the set of weak limits in H_{-4} obtained as $T \to \infty$. A volume-first limit is a weak limit of laws $\lambda_i \in \mathcal{V}_{N_i}$ with $N_i \to \infty$. The uniform compactness bound makes each $\mathcal{V}_N$ nonempty. The H_{-3} balls are closed in H_{-4}, so the portmanteau theorem passes their probability lower bound to every fixed-spacing volume limit. The family $\bigcup_{N \ge 0} \mathcal{V}_N$ therefore obeys the same tightness bound on H_{-4}, giving volume-first limit points. Their additional diagonal positivity depends on taking the volume limit at each fixed spacing; compactness alone would not supply it.

6. Volume and continuum limits

Gaussian domination gives compactness of the curvature laws at all cutoffs. Diagonal reflection positivity uses a further identification: at a fixed spacing, a volume limit is the step-field image of an infinite lattice law. This permits the nine-mirror theorem to be applied before the spacing is removed.

6.1. **The test map and its bounds.** Write $\eta = L^{-N}$, $R = 8hL^T$, and $n = R/\eta$, as in (1.2). A plaquette of the unit covering lattice is $\sigma = (x; \mu\nu)$, with $\mu < \nu$ and centre $y_\sigma = x + (e_\mu + e_\nu)/2$. Its averaging cube is

$$Q_\eta(\sigma) = \eta\big(y_\sigma + [-1/2, 1/2)^3\big).$$

For each fixed orientation these cubes partition $\mathbb{R}^3$. If u is a real lattice curvature, its physical step field is $\eta^{-3/2} u(\sigma)$ on $Q_\eta(\sigma)$ in component $\mu\nu$. For a Schwartz two-form f, define

$$f_{[\eta]}(\sigma) = \eta^{-3/2} \int_{Q_\eta(\sigma)} f_{\mu\nu}(y)\,\mathrm{d}y, \tag{6.1}$$

$$f^R(y) = \sum_{k \in \mathbb{Z}^3} f(y + Rk),$$

$$f_\eta([\sigma]) = \eta^{-3} \int_{Q_\eta(\sigma)} f^R_{\mu\nu}(y)\,\mathrm{d}y. \tag{6.2}$$

Here $[\sigma]$ is the plaquette class modulo $n\mathbb{Z}^3$. The torus cochain norm is $|v|_\eta^2 = \eta^3 \sum_{[\sigma]} |v([\sigma])|^2$. The covering cochain norm $\|\cdot\|$ is the unweighted ℓ^2 norm. For the periodic extension $\widetilde{u}$ of u,

$$F^\sharp(f) = \eta^{3/2} \sum_{[\sigma]} f_\eta([\sigma]) u(\sigma) = \sum_\sigma f_{[\eta]}(\sigma)\widetilde{u}(\sigma). \tag{6.3}$$

The normalization in the first equality combines the cell volume η^3 with the physical field factor $\eta^{-3/2}$.

The estimates needed below are

$$\|f_{[\eta]}\| \le \|f\|_2, \tag{6.4}$$

$$\sum_\sigma |f_{[\eta]}(\sigma)|(1+|x(\sigma)|_\infty) \le \eta^{-3/2} \int_{\mathbb{R}^3} \sum_{\mu<\nu} |f_{\mu\nu}(y)|(2+|y|_\infty/\eta)\,\mathrm{d}y, \tag{6.5}$$

$$|f_\eta|_\eta \le \|f^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)} \le p_R(f), \tag{6.6}$$

where, with $Q_R = [-R/2, R/2)^3$,

$$p_R(f) = \|f\|_2 + 26R^{-2}\||y|^2 f\|_{L^2(\mathbb{R}^3\setminus Q_R)}. \tag{6.7}$$

One also has $|f_\eta|_\eta \le \|f\|_2$ if the support of f lies in a translate of an open cube of side R. For $R \ge 1$, $p_R(f) \le p_1(f)$, and $p_R(f) \to \|f\|_2$ as $R \to \infty$. The seminorm p_1 is continuous on Schwartz space.

The first bound follows from Cauchy–Schwarz on each averaging cube. On that cube $1 + |x(\sigma)|_\infty \le 2 + |y|_\infty/\eta$, which gives the second. Absolute convergence permits unfolding:

$$\eta^{3/2} f_\eta([\sigma]) = \sum_{k\in\mathbb{Z}^3} f_{[\eta]}(\sigma + nk).$$

This proves (6.3). Applying the cube Cauchy–Schwarz inequality to f^R gives the first inequality in (6.6). If the support lies in one open R-cube, its distinct $R\mathbb{Z}^3$ translates are disjoint, so $\|f^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)} = \|f\|_2$.

For the remaining estimate put $a_k = \|f\|_{L^2(Q_R+Rk)}$. Minkowski's inequality bounds the periodized norm by $\sum_k a_k$. On $Q_R + Rk$, $k \ne 0$, one has $|y| \ge R|k|_\infty/2$; hence

$$\sum_{k\ne 0} a_k \le 4R^{-2} \left(\sum_{k\ne 0} |k|_\infty^{-4}\right)^{1/2} \||y|^2 f\|_{L^2(\mathbb{R}^3\setminus Q_R)}.$$

The number of points with $|k|_\infty = j$ is $24j^2 + 2$, so the sum in parentheses is $4\pi^2 + \pi^4/45 < 41.65$ and its square root times four is less than 26. This proves (6.6) and (6.7). The comparison with p_1 and the limit of p_R follow by inclusion of the cubes and integrability of $|y|^4|f(y)|^2$.

For the Hermite two-forms of Section 5,

$$p_1(h_\kappa) \le 31\lambda_\kappa. \tag{6.8}$$

Here is the weighted estimate behind this bound. Let $\mathsf{M} = 1 + |y|^2$, $P_j = -i\partial_j$, and $\mathsf{A} = \mathsf{M} - \Delta$. Since $[P_j, [P_j, \mathsf{M}]] = -2$,

$$\|\mathsf{A}v\|_2^2 = \|\mathsf{M}v\|_2^2 + \|\Delta v\|_2^2 + 2\sum_j \langle P_j v, \mathsf{M}P_j v\rangle - 6\|v\|_2^2 \ge \|\mathsf{M}v\|_2^2 - 6\|v\|_2^2.$$

The Hermite eigenvalues satisfy $\lambda_\kappa \ge 4$, and therefore $\|\mathsf{M}v\|_2^2 \le (11/8)\|\mathsf{A}v\|_2^2$. Taking $v = h_\kappa$ gives $p_1(h_\kappa) \le 1 + 26\sqrt{11/8}\lambda_\kappa \le 31\lambda_\kappa$.

The test map respects the affine cubic symmetries. For a signed permutation and lattice translation g of the unit lattice, let $\mathfrak{g}(y) = \eta g(y/\eta)$. With the exterior-square action on two-forms and the cochain pull-back,

$$(\mathfrak{g}\cdot f)_{[\eta]} = (g^{-1})^* f_{[\eta]}, \qquad (\mathfrak{g}\cdot f)_\eta = (g^{-1})^* f_\eta. \tag{6.9}$$

Indeed, g sends centres to centres and the averaging cubes to averaging cubes up to null boundaries. Its orientation sign is the sign of $\wedge^2 \vec{g}$ on the corresponding plaquette. Changing variables gives the first formula; periodization and the bijection $k \mapsto \vec{g}k$ give the second.

6.2. **Compactness and convergence of moments.** Let $P_{N,T}$ be the law of $F^\sharp$ for the all-flux chamber measure. The mass assignment may be any real sequence independent of T. Theorem 3.5 and (6.3) give

$$\mathbb{E}_{N,T} e^{F^\sharp(f)} \le e^{|f_\eta|_\eta^2/2} \le e^{p_R(f)^2/2} \qquad (f \text{ real}). \tag{6.10}$$

The random-linear-form proof of Proposition 5.1 applies here with the Hilbert seminorm $f \mapsto |f_\eta|_\eta$. In particular its variance bound and (6.8) give

$$\mathbb{E}_{N,T}\|F^\sharp\|_{-3}^2 \le \sum_\kappa \lambda_\kappa^{-6} p_1(h_\kappa)^2 \le 961 \sum_\kappa \lambda_\kappa^{-4} < 61. \tag{6.11}$$

Tonelli's theorem justifies the first sum. Every finite lattice cochain has a step field in H_{-3}: Cauchy–Schwarz in (6.3) bounds its κ-th Hermite coordinate by $31\lambda_\kappa \|u\|_{\ell^2(C(n)^{(2)})}$, with the unweighted norm on the unit-lattice torus plaquettes. The step map from this finite-dimensional cochain space to H_{-3} is consequently continuous.

The compactness argument at the end of Section 5 now applies. The laws $P_{N,T}$ are uniformly tight on H_{-4}, and their fixed-spacing volume limits are uniformly tight there. Thus the sets $\mathcal{V}_N$ and the set of volume-first limits are nonempty.

For a joint sequence $N_k, T_k \to \infty$ suppose $P_{N_k,T_k} \to \mu$ weakly on H_{-4}. Truncating $e^{F(f)}$ above by a constant, passing to the weak limit, and then removing the truncation gives

$$\mathbb{E}_\mu e^{F(f)} \le e^{\|f\|_2^2/2}. \tag{6.12}$$

The limiting constant follows from $p_{R_k}(f) \to \|f\|_2$. For a fixed finite family of complex tests $f_1, \dots, f_s$, define

$$Z_k(z) = \mathbb{E}_{N_k,T_k} \exp\left(\sum_a z_a F^\sharp(f_a)\right), \qquad Z(z) = \mathbb{E}_\mu \exp\left(\sum_a z_a F(f_a)\right).$$

The evaluations $F \mapsto (F(f_a))_a$ are continuous on H_{-4}. Moreover,

$$\mathbb{E}_{N_k,T_k}\left|\exp\left(\sum_a z_a F^\sharp(f_a)\right)\right|^2 \le \exp\left(2p_1\left(\operatorname{Re}\sum_a z_a f_a\right)^2\right).$$

This bound is uniform in k and on compact sets of z. Uniform integrability therefore gives pointwise convergence $Z_k \to Z$. The same exponential bounds give local uniform bounds on the entire functions Z_k. Cauchy's estimates make them equi-Lipschitz on compact sets, so their convergence is locally uniform. Cauchy's formula then passes every derivative to the limit. In particular all joint moments converge.

On each compact subset of $\{\sum_a |z_a|\|f_a\|_2 < 2/5\}$, the test-map bounds give $\sum_a |z_a||(f_a)_{\eta_k}|_{\eta_k} < 2/5$ for all large k. The zero-free estimate (5.6), with this seminorm, bounds $|Z_k - 1|$ by $1/2$. The principal logarithms therefore converge locally uniformly near zero, and their derivatives give convergence of every joint cumulant.

6.3. **Closedness and the affine cubic action.** The step fields are closed on the lattice. To pass this identity to distributions, let $\xi \in C_c^\infty(\mathbb{R}^3;\mathbb{R})$ represent the three-form $\xi\,\mathrm{d}y_1\wedge\mathrm{d}y_2\wedge\mathrm{d}y_3$. Put

$$\partial_j^\eta\xi(y) = \eta^{-1}\big(\xi(y+\eta e_j/2) - \xi(y-\eta e_j/2)\big)$$

and define the two-form v^η in the component order $23, 13, 12$ by

$$(v_{23}^\eta, v_{13}^\eta, v_{12}^\eta) = (-\partial_1^\eta\xi, \partial_2^\eta\xi, -\partial_3^\eta\xi).$$

It satisfies

$$F^\sharp(v^\eta) = 0, \qquad \|v^\eta - d^*\xi\|_2 \le \frac{\eta^2}{24}\sum_{j=1}^3 \|\partial_j^3\xi\|_2. \tag{6.13}$$

For the exact identity, assign to each unit cube the number $\xi_{[\eta]}(x) = \eta^{-5/2}\int_{\eta(x+[0,1)^3)}\xi(y)\,\mathrm{d}y$. The three plaquette coefficients of $v_{[\eta]}^\eta$ are

$$-(\xi_{[\eta]}(x) - \xi_{[\eta]}(x-e_1)), \quad \xi_{[\eta]}(x) - \xi_{[\eta]}(x-e_2), \quad -(\xi_{[\eta]}(x) - \xi_{[\eta]}(x-e_3)).$$

In the pairing with $\widetilde{u}$ these regroup to $\sum_x \xi_{[\eta]}(x)(d\widetilde{u})(x) = 0$. The weighted summability in (6.5) justifies regrouping; at finite volume $\widetilde{u}$ is bounded. For the error bound, the Fourier multiplier of $\partial_j^\eta - \partial_j$ is $i(2\eta^{-1}\sin(\eta p_j/2) - p_j)$, whose modulus is at most $\eta^2|p_j|^3/24$. Plancherel's theorem gives (6.13).

For all large k, $d^*\xi - v^{\eta_k}$ is supported in one open R_k-cube. The variance bound and (6.13) then give

$$\mathbb{E}_{N_k,T_k}F^\sharp(d^*\xi)^2 \le \|d^*\xi - v^{\eta_k}\|_2^2 \le \frac{\eta_k^4}{24^2}\left(\sum_j \|\partial_j^3\xi\|_2\right)^2.$$

Passing bounded truncations of the square to the weak limit shows $F(d^*\xi) = 0$ almost surely. Choose a countable dense family of compactly supported smooth three-form tests. Continuity of each tempered distribution extends the identity from this family to all tests, so $\mu(dF = 0) = 1$.

The finite laws are invariant under lattice translations, signed coordinate permutations and $u \mapsto -u$. By (6.9) their step-field characteristic functionals have the corresponding invariances. Signed permutations and change of sign pass directly to the weak limit. For a real translation b, choose $b_k \in \eta_k\mathbb{Z}^3$ with $|b_k - b|_\infty \le \eta_k/2$. The variance bound gives

$$\left|\mathbb{E}_{N_k,T_k}e^{iF^\sharp(\tau_b f)} - \mathbb{E}_{N_k,T_k}e^{iF^\sharp(\tau_{b_k} f)}\right| \le p_1(\tau_b f - \tau_{b_k} f) \longrightarrow 0.$$

Thus the limiting characteristic functional is invariant under every translation. Characteristic functionals determine the law, so μ has these invariances.

6.4. **Identifying a fixed-spacing volume limit.** Fix N, hence η, and take any $\mu^{(N)} \in \mathcal{V}_N$. Choose a defining sequence $T_j \to \infty$ with $P_{N,T_j} \to \mu^{(N)}$ weakly on H_{-4}. Let ν_j be the law on Ω of the periodically extended unit-lattice curvature. We shall identify a product-topology limit of these laws and prove

$$\nu_j \longrightarrow \nu^{(N)}, \qquad \mu^{(N)} = (\sharp_\eta)_*\nu^{(N)}. \tag{6.14}$$

The step map in this identity is only Borel; the identification requires more than the continuous mapping theorem.

For a closed cochain u put

$$\Omega' = \left\{u \in \Omega : \sup_\sigma \frac{|u(\sigma)|}{1+|x(\sigma)|_\infty} < \infty\right\}.$$

On Ω' the componentwise step field $\sharp_\eta u$ is tempered, and $(\sharp_\eta u)(f) = \sum_\sigma f_{[\eta]}(\sigma)u(\sigma)$ is absolutely convergent by (6.5). Let $\Omega^{(\eta)}$ be the subset on which this distribution has finite H_{-3} norm. Regard $\sharp_\eta$ as a map $\Omega \to H_{-4}$ by setting it to zero outside $\Omega^{(\eta)}$. Both subsets and the map are Borel. Indeed, the growth bound is a countable supremum; each test evaluation is a pointwise limit of finite coordinate sums; and the squared H_{-3} norm is a countable sum of nonnegative evaluations. The Hermite evaluations generate the Borel sigma algebra of H_{-4}.

For each plaquette choose a real smooth bump b^σ, with only its corresponding two-form component nonzero, supported in the interior of $Q_\eta(\sigma)$ and of integral $\eta^{3/2}$. Then $(b^\sigma)_{[\eta]} = \mathbf{1}_\sigma$. The map

$$\mathcal{J}_\eta : H_{-4} \longrightarrow \mathbb{R}^{\{\text{plaquettes}\}}, \qquad \mathcal{J}_\eta(F)(\sigma) = F(b^\sigma) \tag{6.15}$$

is continuous for the product topology, and $\mathcal{J}_\eta \sharp_\eta u = u$ on $\Omega^{(\eta)}$. The finite-volume identities are $P_{N,T_j} = (\sharp_\eta)_* \nu_j$ and $\nu_j = (\mathcal{J}_\eta)_* P_{N,T_j}$. Thus the continuous mapping theorem gives $\nu_j \to \nu^{(N)} := (\mathcal{J}_\eta)_* \mu^{(N)}$ in the product space. Closedness of Ω and portmanteau show that $\nu^{(N)}$ is supported on Ω.

This lattice law satisfies Gaussian domination for every finitely supported cochain β. Once n_j is large, the support of β embeds without overlap in the cubic torus, and its periodization has unchanged ℓ^2 norm. Theorem 3.5, weak convergence and bounded truncations of $e^{\langle \beta, u\rangle}$ give

$$\int_\Omega e^{\langle \beta, u\rangle} \, \mathrm{d}\nu^{(N)}(u) \le e^{\|\beta\|^2/2}. \tag{6.16}$$

For any law ν on Ω with this bound, the one-coordinate tail estimate is $\nu(|u(\sigma)| \ge t) \le 2e^{-t^2/2}$. There are at most $3(2k+1)^3$ plaquettes with base point of sup norm k, and hence

$$\sum_\sigma \nu\big(|u(\sigma)| \ge 1 + |x(\sigma)|_\infty\big) \le 6 \sum_{k \ge 0} (2k+1)^3 e^{-(k+1)^2/2} < \infty.$$

Borel–Cantelli gives $\nu(\Omega') = 1$. Truncate $f_{[\eta]}$ to a finite set of plaquettes. Absolute convergence on Ω' and Fatou's lemma pass (6.16) to this cochain, giving

$$\int e^{(\sharp_\eta u)(f)} \, \mathrm{d}\nu(u) \le e^{\|f_{[\eta]}\|^2/2} \le e^{\|f\|_2^2/2}.$$

Its variance bound and Tonelli's theorem yield

$$\int \|\sharp_\eta u\|_{-3}^2 \, \mathrm{d}\nu(u) \le \sum_\kappa \lambda_\kappa^{-6}.$$

In particular $\nu(\Omega^{(\eta)}) = 1$.

It remains to recover $\mu^{(N)}$ from $\nu^{(N)}$. Choose a countable Schwartz-dense family $\mathcal{D}_0$ of real compactly supported smooth two-forms. For $f \in \mathcal{D}_0$ set

$$f' = f - \sum_\sigma f_{[\eta]}(\sigma) b^\sigma.$$

The sum is finite and $f'_{[\eta]} = 0$. Consequently $F^\sharp(f') = 0$ at every finite volume. The hyperplane $\{F : F(f') = 0\}$ is closed in H_{-4}, so it has full $\mu^{(N)}$ measure. Intersecting these sets for $f \in \mathcal{D}_0$ with the full-measure set $\mathcal{J}_\eta(F) \in \Omega^{(\eta)}$ gives

$$F(f) = \sum_\sigma f_{[\eta]}(\sigma) F(b^\sigma) = (\sharp_\eta \mathcal{J}_\eta(F))(f) \qquad (f \in \mathcal{D}_0)$$

on one set of full measure. Both sides are tempered distributions, so density extends this equality to every test. This proves (6.14).

6.5. **Nine mirror families in the continuum.** At fixed N the charge $q = e\eta^{1/2}$ and the chamber mass parameter $M = \bar{m}\eta + m_{0,N} + \rho\sum_i w_i/2$ are independent of volume. The sides $n_j = 8hL^{T_j+N}$ satisfy the divisibility condition of Theorem 4.7: every required power of L eventually divides the side, and 8 divides it. That theorem applies to $\nu^{(N)}$. It is invariant under the affine cubic group and change of sign, and reflection positive at the three coordinate and six diagonal mirror families. The identification (6.14) transports its characteristic functional to $\mu^{(N)}$.

Write $\ell(y) = n_\ell \cdot y$ for one of $y_j, y_j - y_k, y_j + y_k$, $j < k$. The admissible unit-lattice heights are $\frac{1}{2}\mathbb{Z}$ for a coordinate family and $\mathbb{Z}$ for a diagonal family. If c is an admissible physical height, the cochain test map intertwines the lattice reflection with $r_{\ell,c}$ by (6.9). Suppose that real $f_1, \dots, f_s \in C_c^\infty$ have support in $\{\ell \geq c + \delta\}$ with $\delta > 2\eta$. If $f_{a,[\eta]}(\sigma) \neq 0$, its averaging cube meets this support. Every corner ηx of the plaquette is within sup distance η of a point y in that intersection. Since $|n_\ell|_1 \leq 2$,

$$\ell(\eta x) \geq \ell(y) - 2\eta > c.$$

Thus $f_{a,[\eta]}$ has finite support strictly on the positive lattice side. Lattice reflection positivity gives

$$\sum_{a,b} \overline{z_a} z_b \mathbb{E}_{\mu^{(N)}} e^{iF(f_b - r_{\ell,c}\cdot f_a)} \geq 0. \tag{6.17}$$

This is the reflected form of the character sum $\sum_b z_b e^{i\langle f_{b,[\eta]}, u\rangle}$.

Proposition 6.1. *Every volume-first limit of the chamber cutoff laws is a translation-invariant law of closed tempered two-forms with Gaussian domination. It is invariant and reflection positive at every plane in the nine families $y_j = c$ and $y_j \pm y_k = c$, $j < k$, $c \in \mathbb{R}$. All finite-family moments and cumulants converge along a joint cutoff sequence realizing that limit.*

Proof. Suppose $\mu^{(N_i)} \to \mu$ weakly on H_{-4}, with $N_i \to \infty$ and $\mu^{(N_i)} \in \mathcal{V}_{N_i}$. Fix a metric for weak convergence of probability laws on H_{-4}. For each i, choose a volume index $T_i \geq i$ along the defining volume subsequence such that the distance between P_{N_i,T_i} and $\mu^{(N_i)}$ is less than $1/i$. Then $P_{N_i,T_i} \to \mu$ with both indices tending to infinity. The preceding compactness, moment convergence, closedness and invariance arguments apply to this joint sequence.

Fix ℓ and first take $c \in \mathbb{Z}[L^{-1}]$. Then $c \in \eta_i\mathbb{Z}$ for all large i, and hence it is an admissible height for each lattice mirror family. Any finite family of compact tests in $\{\ell > c\}$ has a common support margin $\delta > 0$. For all large i, $2\eta_i < \delta$, so (6.17) holds for $\mu^{(N_i)}$. The functions in this finite sum are bounded and continuous on H_{-4}; weak convergence passes its nonnegativity to μ.

For an arbitrary height c, choose $c' \in \mathbb{Z}[L^{-1}]$ and $v = (c - c')|n_\ell|^{-2}n_\ell$. Then $r_{\ell,c} = \tau_v r_{\ell,c'} \tau_{-v}$. Translate the test forms by $-v$; they are now supported on the positive side of $\ell = c'$. Translation invariance identifies the reflected form at c with that at c'. Thus (6.17) holds at every height. Character sums are dense in the corresponding positive-side L^2 space, and the reflected form is bounded there by Cauchy–Schwarz. Its nonnegativity therefore extends to that space. Finally $r_{\ell,0}$ is a signed permutation, and $r_{\ell,c}$ is its composition with a translation. The invariances already proved give invariance at each mirror. □

7. Reflection-positive reconstruction, the spectral cone, and clustering

The diagonal mirrors supply an energy–momentum bound with constant one. We prove this directly by comparing two reflection-positive reconstructions. This avoids using rotational covariance in the step that will later provide its spectral input.

7.1. **The reflection-positive Hilbert space.** For an open set O, let Σ_O be generated by $F(f)$ with real $f \in C_c^\infty(O; \Lambda^2)$. Let λ be a translation-invariant law and let r_j reflect in $x_j = 0$. Suppose that λ is invariant and reflection positive at this plane. For a Euclidean motion g

write $\widehat{g}A(F) = A(g^{-1}F)$ and $U(a) = \widehat{\tau_a}$. For a reflection r put $\mathcal{R}_r = \widehat{r}$. Write $\mathcal{R}_j = \mathcal{R}_{r_j}$ and $\Theta_j A = \overline{\mathcal{R}_j A}$. On $\mathcal{E}_j^+ = L^2(\Sigma_{\{x_j>0\}}, \lambda)$ the form

$$b_j(A,B) = \mathbb{E}_\lambda[(\Theta_j A)B]$$

is positive semidefinite and $b_j(A,A) \le \|A\|^2_{L^2(\lambda)}$. Let $\mathcal{H}_j$ be its quotient completion, $[A]_j$ the quotient map, and $\Omega_j = [1]_j$. Reflection positivity stated on the characters $e^{iF(f)}$, with real $f \in C_c^\infty(\{x_j > 0\}; \Lambda^2)$, extends to this form: their linear span is dense in the positive-side L^2 space, and the form is bounded by the L^2 norm.

Proposition 7.1. *There are a non-negative self-adjoint H_j and a strongly continuous unitary representation V_j of $e_j^\perp$ on $\mathcal{H}_j$ such that*

$$e^{-tH_j} V_j(y)[A]_j = [U(te_j + y)A]_j, \qquad t \ge 0, \quad y \perp e_j. \tag{7.1}$$

The two representations commute, and $H_j\Omega_j = 0$.

Proof. Translations are strongly continuous on $L^2(\lambda)$. On characters this follows from continuity of translation of a Schwartz test and bounded convergence; their density gives it on all of $L^2(\lambda)$. Positive time translations preserve the positive-side algebra. With $|A|_b = b_j(A,A)^{1/2}$, reflection and translation invariance give

$$|U(te_j)A|_b^2 = b_j(A, U(2te_j)A) \le |A|_b\, |U(2te_j)A|_b.$$

Iteration yields

$$|U(te_j)A|_b \le |A|_b^{1-2^{-k}} \|A\|_{L^2(\lambda)}^{2^{-k}}.$$

Letting $k \to \infty$ proves that $[A]_j \mapsto [U(te_j)A]_j$ is a well-defined contraction. It is symmetric, obeys the semigroup law, and is strongly continuous by the contraction of the quotient map. Writing it as $T_j(t)$, one has $T_j(t) = T_j(t/2)^2 \ge 0$. If $T_j(t)\psi = 0$, repeated square roots and strong continuity give $\psi = 0$. Spectral calculus therefore defines $H_j = -\log T_j(1) \ge 0$; uniqueness of positive roots gives $T_j(t) = e^{-tH_j}$ first at dyadic times and then at all times.

For $y \perp e_j$, $U(y)$ preserves the half-space and commutes with reflection, so it is an isometry for b_j, with inverse $U(-y)$. Its quotient extension is the unitary $V_j(y)$. Strong continuity and commutation with $T_j(t)$ follow from the corresponding Euclidean translation identities. The constant observable is fixed throughout. □

Write $V_j(se_i) = e^{is\mathsf{P}_i}$ for $i \ne j$. The joint spectral measure E_j of H_j and these commuting momenta lies on $[0,\infty) \times e_j^\perp$. For $\psi \in \mathcal{H}_j$ let $\varrho_\psi(A) = \langle \psi, \mathsf{E}_j(A)\psi\rangle$. Then

$$\langle \psi, e^{-tH_j} V_j(y)\psi\rangle = \int e^{-tE + ip\cdot y}\, \mathrm{d}\varrho_\psi(E,p). \tag{7.2}$$

7.2. **A diagonal mirror.**

Proposition 7.2. *Let $i \ne j$, and suppose in addition that λ is invariant and reflection positive at $x_j + x_i = 0$. Then*

$$\mathsf{E}_j\{(E,p) : E < |p_i|\} = 0. \tag{7.3}$$

Thus $H_j \ge |\mathsf{P}_i|$ in the joint spectral calculus.

Proof. Fix $D > 0$ and $A \in L^2(\Sigma_B, \lambda)$, where $B = \{0 < x_j < D,\ |x_i| < D\}$, and set $\psi = [A]_j$. Let ϱ be the image of ϱ_ψ under $(E,p) \mapsto (E, p_i)$. This is a finite positive measure on the right half-plane, identified by $z = E - ip_i$. For $a, c \ge 0$ consider

$$K(a,c) = \mathbb{E}_\lambda\big[(\Theta_j A) U(2De_j + a(e_j + e_i) + c(e_j - e_i))A\big].$$

The axis reconstruction gives

$$K(a,c)=\int e^{-2DE}e^{-az-c\bar z}\,\mathrm{d}\varrho(E,p_i). \tag{7.4}$$

There is a second representation with the diagonal as time direction. Put $n_\pm=(e_j\pm e_i)/\sqrt{2}$, $y=D(e_j+e_i)/2$, and let r_+ reflect in $x_j+x_i=0$. The observables

$$A_1=\mathcal{R}_{r_+}U(-y)\mathcal{R}_jA,\qquad A_2=U(2De_j-y)A$$

are supported on the positive side of this diagonal. Indeed, r_jB-y lies in its negative side and $B+2De_j-y$ in its positive side. Their L^2 norms equal $\|A\|_2$. Writing $\Theta_+A=\overline{\mathcal{R}_{r_+}A}$, translation of the original expectation by $-y$ gives

$$K(a,c)=\mathbb{E}_\lambda\big[(\Theta_+A_1)U(a(e_j+e_i)+c(e_j-e_i))A_2\big].$$

Choose $Q\in O(3)$ carrying e_j,e_i to n_+,n_- and apply Proposition 7.1 to the transformed law $(Q^{-1})_*\lambda$. If E' is its joint spectral measure, set $\psi_a=[\widehat{Q^{-1}A_a}]'$, $a=1,2$, and let ϱ' be the image of $\langle\psi_1,\mathsf{E}'(\cdot)\psi_2\rangle$ under $(E,p)\mapsto(\sqrt{2}E,\sqrt{2}p_i)$. This gives

$$K(a,c)=\int e^{-aE'+icp'}\,\mathrm{d}\varrho'(E',p'),\qquad \|\varrho'\|_{\mathrm{TV}}\le\|A\|_2^2, \tag{7.5}$$

where $E'\ge0$. The variation bound is spectral Cauchy–Schwarz applied to the two quotient vectors of A_1,A_2. The factors $\sqrt{2}$ from the diagonal coordinates have been absorbed into E',p'.

For $h\in L^1(0,\infty)$ write $\widetilde{h}(z)=\int_0^\infty h(a)e^{-az}\,\mathrm{d}a$. Integrating the two representations against $h(a)\overline{h(c)}$ yields the decisive bound

$$\int e^{-2DE}|\widetilde{h}(E-ip_i)|^2\,\mathrm{d}\varrho\le\|A\|_2^2\sup_{u\ge0}|\widetilde{h}(u)|\sup_{v\in\mathbb{R}}|\widetilde{h}(iv)|. \tag{7.6}$$

All integrations are justified by the finite variation of the spectral measures and the L^1 assumption on h.

We choose transforms whose two axis suprema have bounded product, but whose modulus grows exponentially outside $|\arg z|\le\pi/4$. For $s=\pm1$ and $n\ge1$ use the principal-power functions

$$\widetilde{h}_n(z)=e^{-n}z^{1-4isn/\pi}(1+z)^{-3},\qquad \widetilde{h}_n(0)=0. \tag{7.7}$$

Their suprema on the positive and imaginary axes are at most e^{-n} and e^{n}, respectively. Each is the Laplace transform of an L^1 function. Here is the factorization that verifies this point, including the boundary of the half-plane. For $\nu=1-4isn/\pi$,

$$z^\nu(1+z)^{-3}=\big(1-(1+z)^{-1}\big)^\nu(1+z)^{-(3-\nu)}.$$

The second factor is the transform of $a^{2-\nu}e^{-a}/\Gamma(3-\nu)$, an integrable function. The binomial coefficients of the first factor are absolutely summable, since $|\binom{\nu}{k}|=O(k^{-2})$. It is therefore the transform of the finite complex measure

$$\delta_0+\sum_{k\ge1}(-1)^k\binom{\nu}{k}\frac{a^{k-1}e^{-a}}{(k-1)!}\,\mathrm{d}a.$$

Convolution gives $h_n\in L^1$. Absolute convergence and $\operatorname{Re}\nu=1$ also give the stated boundary value at $z=0$.

Fix $\delta>0$ and the sector $S_\delta=\{z\neq0:\operatorname{Re}z\ge0,\ s\arg z\ge\pi/4+\delta\}$. On this sector,

$$|\widetilde{h}_n(z)|^2\ge e^{8\delta n/\pi}|z|^2|1+z|^{-6}.$$

Substitution in (7.6) gives

$$e^{8\delta n/\pi}\int_{S_\delta} e^{-2DE}|z|^2|1+z|^{-6}\,\mathrm{d}\varrho \le \|A\|_2^2.$$

The integrand is positive on S_δ, so letting $n\to\infty$ shows $\varrho(S_\delta)=0$. Taking both signs and rational $\delta>0$ gives $|\arg z|\le\pi/4$ on its support, or $E\ge|p_i|$.

Finite linear combinations of characters with compact positive-side tests lie in some such B and have dense quotient classes in $\mathcal{H}_j$. The spectral projection of $\{E<|p_i|\}$ vanishes on this dense set and hence vanishes identically. □

Applying this result to the nine-mirror laws gives $H_j\ge|\mathsf{P}_i|$ for every ordered pair $i\ne j$. This coordinate cone is the input for the analytic rotation argument and for the identification of invariant projections used in clustering.

7.3. **Invariant observables and reflection.** Let P_j be the orthogonal projection in $L^2(\lambda)$ onto the observables fixed by all $U(te_j)$, and let P project onto those fixed by all translations. These are the corresponding conditional expectations. The averages $T^{-1}\int_0^T U(te_j)\,\mathrm{d}t$ converge strongly to P_j. The projections commute with one another and with every translation.

An observable fixed by translations along e_j can be represented in either half-space bounded by any plane $x_j=c$. Indeed, approximate it in L^2 by finite sums of characters with compact tests, and translate each approximant far enough into the chosen half-space. Translation fixes the limiting observable and preserves the approximation error. The half-space L^2 subspace is closed.

The linear geometric reflection $\mathcal{R}_j$ restricts to a positive self-adjoint involution on this invariant subspace. Positivity follows from its inclusion in the positive-side space and reflection positivity. Its spectrum is therefore $\{1\}$, so

$$\mathcal{R}_jA=A\qquad(A\in\operatorname{ran}P_j). \tag{7.8}$$

Thus the quotient map is an isometry there. Its range is precisely $\ker H_j$. To identify it, average

$$\langle[A]_j,e^{-tH_j}[B]_j\rangle=\langle\mathcal{R}_jA,U(te_j)B\rangle_{L^2(\lambda)}$$

over $0\le t\le T$. Spectral dominated convergence on the left and the mean ergodic theorem on the right give

$$P_0[B]_j=[P_jB]_j,\qquad P_0=\mathbf{1}_{\{0\}}(H_j). \tag{7.9}$$

The image of $\operatorname{ran}P_j$ is closed because the map is isometric, and contains the dense subspace $P_0[\mathcal{E}_j^+]_j$ of $\ker H_j$. This proves the asserted identification.

For a law invariant and reflection positive at the nine mirrors, the cone forces the joint spectral measure at zero energy to be concentrated at zero momentum. Hence $V_j(y)$ fixes $\ker H_j$. If $A\in\operatorname{ran}P_j$ and $y\perp e_j$, then

$$[U(y)A-A]_j=0.$$

The difference is again in $\operatorname{ran}P_j$, so the isometry gives $U(y)A=A$. Consequently

$$P_1=P_2=P_3=P. \tag{7.10}$$

7.4. **Correlations in all directions.**

Proposition 7.3. *Suppose λ is translation invariant and invariant and reflection positive at the nine mirrors. For all $A,B\in L^2(\lambda)$,*

$$\langle A,U(a)B\rangle\longrightarrow\langle A,PB\rangle\quad\text{as }|a|\longrightarrow\infty. \tag{7.11}$$

If λ is translation ergodic, the limit is $\overline{\mathbb{E}_\lambda A}\,\mathbb{E}_\lambda B$ and $\ker H_j=\mathbb{C}\Omega_j$ for every j.

Proof. First fix j. For $A = \mathcal{R}_j A'$ and $A', B \in \mathcal{E}_j^+$, (7.9) gives, for $t \geq 0$ and $y \perp e_j$,

$$\begin{aligned} |\langle A, U(te_j + y)(1 - P_j)B\rangle| &= |\langle [A']_j, V_j(y)e^{-tH_j}(1 - P_0)[B]_j\rangle| \\ &\leq \|[A']_j\| \, \|e^{-tH_j}(1 - P_0)[B]_j\|. \end{aligned}$$

The last norm tends to zero by spectral dominated convergence; the bound is uniform in y. Approximate general A, B by local character sums. Translate their supports into opposite half-spaces, at a cost of replacing t by $t-2r$ for one fixed r. The correlation error is bounded uniformly in t, y by $\|A - A_1\|_2\|B\|_2 + \|A_1\|_2\|B - B_1\|_2$. The adjoint identity gives the same conclusion as $t \to -\infty$. Thus

$$\lim_{|t|\to\infty} \sup_{y \perp e_j} |\langle A, U(te_j + y)(1 - P_j)B\rangle| = 0.$$

By (7.10), $P_j = P$ for all j. For any sequence $|a| \to \infty$, a subsequence has one fixed coordinate tending in modulus to infinity. The displayed uniform bound applies with that coordinate, while $U(a)PB = PB$. Every subsequence has such a further subsequence, proving (7.11). In an ergodic law P is the projection onto constants. The isometry onto $\ker H_j$ then gives its one-dimensionality. □

7.5. **Passing the hypotheses to ergodic components.** The conditional laws in an ergodic decomposition require their own reflection and Gaussian bounds. We prove these from invariant weights; covariance of a mixture alone does not give covariance of its individual components.

Let $\mathcal{I}$ be the translation-invariant event sigma algebra. The space of tempered distributions, with the sigma algebra generated by its evaluations, is standard Borel. It therefore admits regular conditional laws λ_F given $\mathcal{I}$, with

$$\lambda_F(\Phi) = \mathbb{E}_\lambda[\Phi \mid \mathcal{I}](F), \qquad \lambda = \int \lambda_F \, \mathrm{d}\lambda(F). \tag{7.12}$$

The first equality is understood almost surely for each integrable Φ (and for nonnegative Φ by monotone convergence). The image measure on laws is the translation-ergodic decomposition. We recall why a single full-measure set of ergodic laws can be used.

For an open set O, $\mathcal{S}(\overline{O}; \Lambda^2)$ denotes the subspace of $\mathcal{S}(\mathbb{R}^3; \Lambda^2)$ consisting of forms supported in $\overline{O}$, with the induced Schwartz topology. Fix a countable real test space $\mathcal{D}_0 \subset C_c^\infty(\mathbb{R}^3; \Lambda^2)$ over $\mathbb{Q}$, stable under rational translations and cubic isometries, with $\mathcal{D}_0 \cap C_c^\infty(O; \Lambda^2)$ dense in $\mathcal{S}(\overline{O}; \Lambda^2)$ for $\mathbb{R}^3$ and the six coordinate half-spaces O. Take the rational span of translated cubic images of countable Schwartz-dense compact-test families in these seven open sets to obtain it. Let $\mathcal{A}_0$ consist of finite sums of $e^{iF(f)}$, $f \in \mathcal{D}_0$, with coefficients in $\mathbb{Q} + i\mathbb{Q}$. This same countable family is dense in L^2 for every law on the evaluation sigma algebra and determines such laws. For density, condition an observable orthogonal to its span on finitely many evaluations. Rational combinations of the tests show that the Fourier transform of the resulting finite complex measure vanishes on rational frequencies. Continuity and Fourier uniqueness make it zero. Conditional expectations on the increasing finite evaluation sigma algebras converge to the observable, proving density. The same finite-dimensional Fourier argument proves determination of laws. For the elements A_p, choose cube-average times $T_k \to \infty$ so that

$$\|M_{T_k} A_p - PA_p\|_2^2 \leq 2^{-k} \quad (p \leq k), \qquad M_T = T^{-3} \int_{[0,T]^3} U(a) \, \mathrm{d}a.$$

Equation (7.12) and summability give convergence in $L^2(\lambda_F)$ for every p on one full-measure set of F. Moreover PA_p has zero conditional variance, and thus equals the constant $\lambda_F(A_p)$ in that space. The conditional laws are translation invariant: test the parent invariance against $\mathcal{I}$-measurable bounded weights, first for rational translations and the countable characters, then use continuity and Fourier uniqueness. The mean ergodic theorem for each such law now shows

that its invariant projection is the constant projection, first on the dense algebra and then on all $L^2(\lambda_F)$. Thus these conditional laws are ergodic. This also gives uniqueness of their image measure: the same cube averages converge pointwise along a subsequence and in L^2 under almost every law in any ergodic decomposition, so they determine the same character expectations and hence the same component law.

Fix an axis mirror. A bounded invariant weight h satisfies $\mathcal{R}_j h = h$ by (7.8) and has a representative in either half-space. If $h \geq 0$ and $A \in \mathcal{E}_j^+$, reflection positivity applied to $\sqrt{h}A$ gives

$$\mathbb{E}_\lambda[h(\Theta_j A)A] \geq 0. \tag{7.13}$$

Reflection invariance similarly gives $\mathbb{E}_\lambda[h\mathcal{R}_j A] = \mathbb{E}_\lambda[hA]$. Taking all bounded $\mathcal{I}$-measurable weights in these identities proves that the corresponding conditional reflection form is nonnegative and the conditional character expectations are reflection invariant. Countably many positive-side character sums suffice; density extends the statements to the full positive-side algebra in each conditional law. Translation invariance moves the mirror to every real height.

This argument also transfers any further fixed mirror normal n. Choose $Q \in O(3)$ with $Qe_1 = n$ and apply the axis argument to the transported law $(Q^{-1})_*\lambda$. The transported law is reflection positive at $x_1 = 0$; it need not equal λ. The map Q_* sends translation-ergodic laws to translation-ergodic laws and transports their reflection forms. By uniqueness of the ergodic decomposition, its image decomposition is that of λ. Applying this to the six diagonal normals, and intersecting the resulting nine full-measure sets, transfers all required mirrors.

Suppose now that the parent law satisfies (5.1). To transfer this Gaussian domination with the same constant, fix real $f \in \mathcal{S}$ and put $G = e^{F(f)}$, $c = e^{\|f\|_2^2/2}$, $\phi = PG$. All moments of G are finite. For each integer $n \geq 2$, Tonelli's theorem and the parent bound give

$$\mathbb{E}_\lambda[(M_T G)^n] \leq T^{-3n} \int_{([0,T]^3)^n} \exp\left(\frac{n}{2}\|f\|_2^2 + \frac{1}{2}\sum_{p\neq q}\langle f, \tau_{a_p - a_q} f\rangle\right) \mathrm{d}a. \tag{7.14}$$

The overlaps tend to zero as $|a_p - a_q| \to \infty$ and are bounded by $\|f\|_2^2$. Fix $\epsilon > 0$. Outside the set where one pair is within distance δ, the right side is at most $c^n e^{n(n-1)\epsilon/2}$ once δ is sufficiently large. The exceptional set has relative volume at most $\binom{n}{2}(2\delta)^3 T^{-3}$ and its integrand is at most $e^{n^2\|f\|_2^2/2}$. Since $M_T G \to \phi$ in L^2, and the averages and ϕ are uniformly bounded in L^{2n}, interpolation gives convergence in L^n. Sending $T \to \infty$ and then $\epsilon \to 0$ in (7.14) yields $\|\phi\|_n \leq c$ for every n, hence $\phi \leq c$ almost surely. By (7.12),

$$\lambda_F(e^{F(f)}) \leq e^{\|f\|_2^2/2} \quad \text{almost surely.} \tag{7.15}$$

Choose a countable Schwartz-dense real test family and intersect these full-measure sets. For any other test, pointwise convergence of the evaluations under each conditional law and Fatou's lemma extend (7.15) to it. Thus all real tests obey the bound on one full-measure set of component laws.

Finally a measurable set of full parent measure has full measure in almost every component by (7.12). In particular closedness passes to the components; its event is measurable by testing $dF = 0$ on a countable dense family of three-form tests. The nine reflections generate the cubic group, so each of these components is also cubically invariant. Proposition 7.3 applies to each component and gives clustering and $\ker H_j = \mathbb{C}\Omega_j$. The rotation theorem below will be applied to these same component laws.

8. Rotational covariance

The rotation argument starts with a field insertion separated from the two observables in a reflection form. Gaussian domination bounds the insertion in the reconstructed Hilbert space,

independently of the number of factors in either observable. The spectral cone then allows complex translations. Together these bounds give bounded operators for insertions concentrated at a point of the (x_1, x_2)-plane. Closedness removes the extra angular frequencies at two points. At higher orders, reflection positivity gives a positive kernel for the highest angular coefficient. Polygon divisibility and a convexity estimate then reduce every order to the two-point result.

Let μ be a translation-invariant, Gaussian-dominated law, invariant and reflection positive at $x_1 = 0$. We first work with this reconstruction, writing $\mathcal{H} = \mathcal{H}_1$, $H = H_1$, and $U_t = U(te_1)$. Let $\mathcal{A}_+$ denote the positive-side measurable functions belonging to every $L^p(\mu)$, $p < \infty$. It is an algebra. Their classes are dense in $\mathcal{H}$, since $\mathcal{A}_+$ contains the bounded positive-side functions.

Use the Hodge isometry $*f = (f_{23}, -f_{13}, f_{12})$ to write $\mathbb{B}(b) = F(*^{-1}b)$, complex linearly. Proposition 5.1 extends this map to vector-valued L^2 tests. The action on these tests is

$$(g \cdot b)(x) = \det(g)\, g\, b(g^{-1}x) \quad (g \in O(3)), \qquad (\tau_a b)(x) = b(x - a).$$

In particular $\Theta_1 \mathbb{B}(b) = \mathbb{B}(\overline{r_1 \cdot b})$ and $\Theta_1 U_t = U_{-t}\Theta_1$. Write $L^2_{[a,b]}$ for the vector tests vanishing almost everywhere outside $a \le x_1 \le b$.

8.1. **The insertion estimate.** For $\ell > 0$ and $h \in L^2_{[0,\ell]}$, consider $M_\ell(h)\Phi = e^{\mathbb{B}(h)} U_\ell \Phi$ for bounded positive-side Φ. This function lies in $\mathcal{A}_+$. We first prove

$$\|[M_\ell(h)\Phi]_1\| \le e^{\|\Re h\|_2^2/2} \|[\Phi]_1\|. \tag{8.1}$$

Set $h^\# = \tau_{\ell e_1}(\overline{r_1 \cdot h} + h)$. Reflection and translation invariance give

$$\|[M_\ell(h)\Phi]_1\|^2 = \langle [\Phi]_1, [M_{2\ell}(h^\#)\Phi]_1 \rangle. \tag{8.2}$$

The two summands in $h^\#$ have disjoint slab interiors, and $\tau_{2\ell e_1} \overline{r_1 \cdot h^\#} = h^\#$.

For a source h' on $[0, \ell']$ with $\tau_{\ell' e_1} \overline{r_1 \cdot h'} = h'$, let $h'^{(n)} = \sum_{j=0}^{n-1} \tau_{j\ell' e_1} h'$. Doubling $h'^{(n)}$ gives $h'^{(2n)}$. Thus, with $m_a = \|[M_{2^a \ell'}(h'^{(2^a)})\Phi]_1\|$,

$$m_a^2 \le \|[\Phi]_1\| m_{a+1}, \qquad m_0 \le \|[\Phi]_1\|^{1-2^{-a}} m_a^{2^{-a}}.$$

On the other hand, the integrand in the doubling identity is bounded in modulus by

$$\|\Phi\|_\infty^2 e^{\mathbb{B}(\Re h'^{(2^{a+1})})}.$$

The real translates have disjoint interiors, so Gaussian domination gives

$$m_a^2 \le \|\Phi\|_\infty^2 e^{\,2^a \|\Re h'\|_2^2}.$$

Taking the indicated powers and sending $a \to \infty$ proves (8.1) for symmetric sources, including $\|[\Phi]_1\| = 0$. Apply this to $h^\#$, whose real part has squared norm $2\|\Re h\|_2^2$, in (8.2). Cauchy–Schwarz proves the bound for arbitrary h.

Proposition 8.1. *For $\ell > 0$ and $f \in L^2_{[0,\ell]}$ there is a unique bounded operator $\mathcal{B}_\ell(f)$ on $\mathcal{H}$ satisfying*

$$\mathcal{B}_\ell(f)[\Phi]_1 = [\mathbb{B}(f) U_\ell \Phi]_1 \quad (\Phi \in \mathcal{A}_+).$$

It is complex linear in f, and

$$\|\mathcal{B}_\ell(f)\| \le k_1 \|f\|_2, \qquad k_1 = \sqrt{2\exp(1)}. \tag{8.3}$$

For real f, the constant is $\exp(1/2)$. *If $f \in L^2_{[c,c']}$ with $0 \le c < c' \le \ell$, then*

$$\mathcal{B}_\ell(f) = e^{-cH} \mathcal{B}_{c'-c}(\tau_{-ce_1} f) e^{-(\ell - c')H}. \tag{8.4}$$

For $f \in L^2_{[-d,d]}$, set $\mathcal{B}^{\mathrm{c}}_d(f) = \mathcal{B}_{2d}(\tau_{de_1} f)$. Its matrix elements and adjoint are

$$\langle [\Phi']_1, \mathcal{B}^{\mathrm{c}}_d(f)[\Phi]_1 \rangle = \mathbb{E}_\mu[(\Theta_1 U_d \Phi')\mathbb{B}(f) U_d \Phi], \tag{8.5}$$

$$\mathcal{B}^{\mathrm{c}}_d(f)^* = \mathcal{B}^{\mathrm{c}}_d(\overline{r_1 \cdot f}).$$

For $r \geq |t|$ and $a \in \mathbb{R}$, one also has

$$e^{-(r+t)H+ia\mathsf{P}_2}\mathcal{B}^{\mathrm{c}}_d(f)e^{-(r-t)H-ia\mathsf{P}_2} = \mathcal{B}^{\mathrm{c}}_{d+r}(\tau_{te_1+ae_2}f). \tag{8.6}$$

Proof. For real f and bounded Φ, the map $z \mapsto [e^{z\mathbb{B}(f)}U_\ell\Phi]_1$ is entire. Indeed its exponential series converges locally uniformly in $L^2(\mu)$, dominated on $|z| \leq R$ by $\|\Phi\|_\infty e^{R|\mathbb{B}(f)|}$. The quotient map is a contraction. On $|z| = \|f\|_2^{-1}$, (8.1) bounds its norm by $\exp(1/2)\|[\Phi]_1\|$. Cauchy's coefficient estimate gives the real insertion bound. Applying it to $\Re f$ and $\Im f$, and using $\|\Re f\|_2 + \|\Im f\|_2 \leq \sqrt{2}\|f\|_2$, gives (8.3). The estimate sends null classes to null classes, so it defines the operator on the dense classes of bounded functions and extends it uniquely to $\mathcal{H}$.

For $\Phi \in \mathcal{A}_+$, its bounded truncations converge in L^4. Since $\mathbb{B}(f) \in L^4$, their products with the inserted field converge in L^2 by Hölder's inequality. This proves the asserted formula on $\mathcal{A}_+$ and complex linearity in f.

Equation (8.4) follows by applying U_c to $\mathbb{B}(\tau_{-ce_1}f)U_{\ell-c}\Phi$ and using (7.1). Translation by $-de_1$ in the reflection form gives (8.5). Applying reflection and conjugation to that integrand gives the adjoint formula. For (8.6), apply the two exterior semigroups to a class $[\Phi]_1$. The resulting function is

$$\mathbb{B}(\tau_{(d+r)e_1+te_1+ae_2}f)\, U_{2(d+r)}\Phi.$$

The test $\tau_{te_1+ae_2}f$ is supported in $[-d-r, d+r]$, since $|t| \leq r$. The displayed function therefore defines its centered insertion at $d+r$, which is the right side. Equality on these dense classes proves the operator identity. □

In particular, for $\Phi_1, \Phi_2 \in \mathcal{A}_+$ measurable on $x_1 > d$,

$$|\mathbb{E}_\mu[(\Theta_1\Phi_1)\mathbb{B}(f)\Phi_2]| \leq k_1\|f\|_2\, \|[U_{-d}\Phi_1]_1\|\|[U_{-d}\Phi_2]_1\|. \tag{8.7}$$

This is (8.5) applied to the two translated observables. The constant does not depend on their degrees.

8.2. **Complex translations and point plates.** From now on assume also the sharp cone $\mathsf{E}_1\{(E, p_\perp) : E < |p_2|\} = 0$. For the laws under construction, Proposition 7.2 supplies this from the diagonal mirror.

For $\xi = (\xi^0, \xi^1) \in \mathbb{C}^2$, put

$$d_{\mathcal{W}}(\xi) = \Re\xi^0 - |\Im\xi^1|, \qquad \mathcal{W} = \{d_{\mathcal{W}} > 0\}, \qquad \xi^* = (\overline{\xi^0}, -\overline{\xi^1}), \quad n_0 = (1, 0).$$

The cone (7.3) gives contractions

$$T(\xi) = \int e^{-\xi^0 E + i\xi^1 p_2}\, \mathrm{d}\mathsf{E}_1(E, p_\perp) \quad (\xi \in \overline{\mathcal{W}}).$$

They obey $T(\xi)T(\xi') = T(\xi+\xi')$, $T(\xi)^* = T(\xi^*)$, and $T(\xi)\Omega_1 = \Omega_1$. They are strongly continuous on the closed tube by spectral dominated convergence. In its interior they are holomorphic in operator norm: on $d_{\mathcal{W}} \geq \epsilon > 0$ the first and second derivatives of the multiplier are bounded by constants times $Ee^{-\epsilon E}$ and $E^2e^{-\epsilon E}$. Taylor's formula therefore gives a uniform quadratic remainder. For real $y = (y^0, y^1)$ with $y^0 \geq 0$, $T(y)[\Phi]_1 = [U(y^0e_1 + y^1e_2)\Phi]_1$.

Write $x = (y, x_3)$ with $y = (x_1, x_2)$. Set

$$e^+ = e_1 + ie_2, \quad e^- = e_1 - ie_2, \quad e^3 = e_3,$$
$$w(+) = 1, \quad w(-) = -1, \quad w(3) = 0,$$
$$s(+) = s(-) = 1, \quad s(3) = -1.$$

A label $\lambda = (\sigma, \chi)$ has $\sigma \in \{+, -, 3\}$ and real $\chi \in L^2(\mathbb{R})$. For $\alpha \in L^2(\mathbb{R}^2; \mathbb{C})$ its plate is $(\alpha \otimes \lambda)(x) = \alpha(y)\chi(x_3)e^{\sigma}$. Thus $\|\alpha \otimes \lambda\|_2 \le \sqrt{2}\|\alpha\|_2\|\chi\|_2$. For rotations R_t in the (x_1, x_2)-plane and the coordinate reflection r_1, direct multiplication gives

$$R_t \cdot (\alpha \otimes \lambda) = e^{-iw(\sigma)t}(\alpha \circ R_{-t}) \otimes \lambda, \tag{8.8}$$
$$\overline{r_1 \cdot (\alpha \otimes \lambda)} = s(\sigma)(\overline{\alpha} \circ r_1) \otimes \lambda.$$

Proposition 8.2. *For each label λ there are bounded operators $\mathcal{B}_\lambda(\xi, \xi')$, $\xi, \xi' \in \mathcal{W}$, holomorphic in operator norm and real linear in χ, such that*

$$\|\mathcal{B}_\lambda(\xi, \xi')\| \le \frac{k_2\|\chi\|_2}{\min(d_{\mathcal{W}}(\xi), d_{\mathcal{W}}(\xi'))}, \qquad k_2 = \frac{8}{\pi}\exp(1/2). \tag{8.9}$$

For $\kappa, \kappa' \in \overline{\mathcal{W}}$,

$$T(\kappa)\mathcal{B}_\lambda(\xi, \xi')T(\kappa') = \mathcal{B}_\lambda(\kappa + \xi, \xi' + \kappa'), \tag{8.10}$$
$$\mathcal{B}_\lambda(\xi, \xi')^* = s(\sigma)\mathcal{B}_\lambda(\xi'^*, \xi^*).$$

If $\alpha \in L^2(\mathbb{R}^2)$ is compactly supported in $|y^0| < d$, then

$$\mathcal{B}^{\mathrm{c}}_d(\alpha \otimes \lambda) = \int \alpha(y)\mathcal{B}_\lambda(dn_0 + y, dn_0 - y)\,\mathrm{d}y. \tag{8.11}$$

The integral is a Bochner integral in operator norm. When $\operatorname{supp}\alpha \subset \{|y^0| \le d'\}$, $d' < d$, its norm is at most $k_2\|\chi\|_2\|\alpha\|_1/(d - d')$.

Proof. Let $j_\delta = (\pi\delta^2)^{-1}\mathbf{1}_{\{|y|<\delta\}}$. Set $A_\delta = \mathcal{B}^{\mathrm{c}}_\delta(j_\delta \otimes \lambda)$ and, when both tube distances exceed δ, set

$$\mathcal{B}^\delta_\lambda(\xi, \xi') = T(\xi - \delta n_0)A_\delta T(\xi' - \delta n_0).$$

These operators obey the shift and adjoint identities in (8.10). Here j_δ is real and reflection invariant, so $A^*_\delta = s(\sigma)A_\delta$. The real translation identity (8.6) also gives, for $r \ge |y^0|$,

$$\mathcal{B}^\delta_\lambda((r + \delta)n_0 + y, (r + \delta)n_0 - y) = \mathcal{B}^{\mathrm{c}}_{\delta+r}((\tau_y j_\delta) \otimes \lambda). \tag{8.12}$$

We prove a bound uniform as $\delta \downarrow 0$. Fix ξ, ξ' and set $d_0 = \min(d_{\mathcal{W}}(\xi), d_{\mathcal{W}}(\xi'))$, $r = (d_0 - \delta)/2 > 0$. For unit vectors ψ, ψ',

$$h(z) = \langle \psi, \mathcal{B}^\delta_\lambda(\xi + z, \xi' - z)\psi' \rangle$$

is holomorphic on $|\Re z^0| < r$, $|\Im z^1| < r$. Write $z = y + i(t, t')$. Factoring the exterior translations gives

$$h(y + i(t, t')) = \langle \psi_1, \mathcal{B}^{\mathrm{c}}_{\delta+r}((\tau_y j_\delta) \otimes \lambda)\psi'_1 \rangle,$$

where $\psi_1 = T(\xi - (\delta + r)n_0 + i(t, t'))^*\psi$ and $\psi'_1 = T(\xi' - (\delta + r)n_0 - i(t, t'))\psi'$. For $|t'| < r$ these are contractions of the original vectors; they are independent of y. For $\eta \in L^2((-r, r)^2)$, extended by zero, the integral of $\eta(y)\tau_y j_\delta$ is $\eta * j_\delta$ in L^2. Its support lies in $|y^0| \le r + \delta$. The insertion bound and Young's inequality therefore give

$$\left| \int \eta(y)h(y + i(t, t'))\,\mathrm{d}y \right| \le 2\exp(1/2)\|\eta\|_2\|\chi\|_2.$$

Hence $\int_{(-r,r)^2} |h(y + i(t, t'))|^2\,\mathrm{d}y \le 4\exp(1)\|\chi\|_2^2$. Apply the mean-value inequality on a polydisc of radius $r' < r$ and bound its integral by that over the four-dimensional box. This yields

$$|h(0)|^2 \le (\pi r'^2)^{-2}(2r)^2\,4\exp(1)\|\chi\|_2^2.$$

Letting $r' \uparrow r$ and taking arbitrary unit vectors proves

$$\|\mathcal{B}^\delta_\lambda(\xi, \xi')\| \le \frac{k_2\|\chi\|_2}{d_0 - \delta}. \tag{8.13}$$

To identify the limit, put $r_0 = d_0/4$. For $\delta < r_0$ the functions $h_\delta(z) = \langle \psi, \mathcal{B}^\delta_\lambda(\xi + z, \xi' - z)\psi' \rangle$ form a bounded normal family on $|z^0|, |z^1| < r_0$ by (8.13). For $\eta \in L^2((-r_0, r_0)^2)$, the real translation identity and an additional equal time translation on both sides give

$$\int \eta(y) h_\delta(y)\,\mathrm{d}y = \langle \psi, T(\xi - 2r_0 n_0)\mathcal{B}^{\mathrm{c}}_{2r_0}((\eta * j_\delta) \otimes \lambda) T(\xi' - 2r_0 n_0)\psi' \rangle. \tag{8.14}$$

Since $\eta * j_\delta \to \eta$ in L^2, this has a unique limit. Any two holomorphic subsequential limits thus have the same integrals on the real square; continuity identifies them there, and the identity theorem identifies them on the polydisc. Consequently every matrix element at $z = 0$ converges. The bound (8.13) defines a bounded limiting sesquilinear form, represented by $\mathcal{B}_\lambda(\xi, \xi')$ with (8.9). This is weak operator convergence. The shift, adjoint, and real linearity identities pass to these limits.

For holomorphy, fix (ξ_0, ξ_0') and a positive ϵ smaller than both tube distances. Near this pair the shift identity factors the operator as

$$T(\xi - \xi_0 + \epsilon n_0)\, \mathcal{B}_\lambda(\xi_0 - \epsilon n_0, \xi_0' - \epsilon n_0)\, T(\xi' - \xi_0' + \epsilon n_0).$$

The two exterior factors are holomorphic in operator norm.

Finally choose $d' < d$ containing the time support of α. Integrating (8.12) with $r = d - \delta$ gives

$$\int \alpha(y) \mathcal{B}^\delta_\lambda(dn_0 + y, dn_0 - y)\,\mathrm{d}y = \mathcal{B}^{\mathrm{c}}_d((\alpha * j_\delta) \otimes \lambda).$$

The right side converges in norm to the left side of (8.11). For matrix elements on the left, (8.13) gives the integrable majorant $2k_2 \|\chi\|_2 |\alpha(y)|/(d - d')$ for sufficiently small δ. Dominated convergence identifies the limiting matrix elements. Holomorphy makes the limiting integrand norm continuous on the support, and (8.9) bounds its norm. It is therefore Bochner integrable, proving the formula and its stated norm bound. □

8.3. **Ordered field kernels.** The intermediate arguments of a product of point plates matter only through their sum. Indeed, for $\kappa_1, \kappa_2 \in \mathcal{W}$ and sufficiently small $\epsilon > 0$, the shift identity gives

$$\mathcal{B}_\lambda(\xi, \kappa_1)\mathcal{B}_{\lambda'}(\kappa_2, \xi') = \mathcal{B}_\lambda(\xi, \epsilon n_0) T(\kappa_1 + \kappa_2 - 2\epsilon n_0) \mathcal{B}_{\lambda'}(\epsilon n_0, \xi').$$

The same expression results from any other decomposition of $\kappa_1 + \kappa_2$ into two points of $\mathcal{W}$. An exterior argument adjacent to the vacuum can likewise be changed arbitrarily in $\mathcal{W}$, since $T(\kappa)\Omega_1 = T(\kappa)^*\Omega_1 = \Omega_1$.

For $N \geq 2$ and $\xi_1, \dots, \xi_{N-1} \in \mathcal{W}$, define

$$G_N(\xi; \lambda) = \langle \Omega_1, \mathcal{B}_{\lambda_1}(\kappa, \xi_1/2) \left(\prod_{a=2}^{N-1} \mathcal{B}_{\lambda_a}(\xi_{a-1}/2, \xi_a/2) \right) \mathcal{B}_{\lambda_N}(\xi_{N-1}/2, \kappa)\Omega_1 \rangle, \qquad \kappa \in \mathcal{W}. \tag{8.15}$$

Set $G_1(\lambda) = \langle \Omega_1, \mathcal{B}_\lambda(\kappa, \kappa)\Omega_1 \rangle$. Both definitions are independent of κ. For $x_1, x_{a+1} - x_a \in \mathcal{W}$, define the vector

$$\Psi(x; \lambda) = \mathcal{B}_{\lambda_1}(x_1, (x_2 - x_1)/2) \left(\prod_{a=2}^{n-1} \mathcal{B}_{\lambda_a}((x_a - x_{a-1})/2, (x_{a+1} - x_a)/2) \right) \mathcal{B}_{\lambda_n}((x_n - x_{n-1})/2, \kappa)\Omega_1. \tag{8.16}$$

Products here and below are ordered with increasing index from left to right; an empty product is the identity. For $n = 1$ this means $\mathcal{B}_{\lambda_1}(x_1, \kappa)\Omega_1$; for $n = 0$ it means Ω_1. It too is independent of κ. Put $s(\lambda) = \prod_a s(\sigma_a)$, $w(\lambda) = \sum_a w(\sigma_a)$, and let $\bar{\lambda}$ exchange the $+$ and $-$ labels.

Proposition 8.3. *The kernels and vectors are holomorphic on their stated domains. For $N \geq 2$,*

$$|G_N(\xi;\lambda)| \leq \prod_{a=1}^{N} (2k_2\|\chi_a\|_2) \left(\min_a d_{\mathcal{W}}(\xi_a)\right)^{-N}. \tag{8.17}$$

Let $\alpha_1, \dots, \alpha_N \in L^2(\mathbb{R}^2)$ have compact time-ordered supports: $\sup_{\operatorname{supp}\alpha_a} y^0 < \inf_{\operatorname{supp}\alpha_{a+1}} y^0$. Then

$$\mathbb{E}_\mu \prod_a \mathbb{B}(\alpha_a \otimes \lambda_a) = \int \prod_a \alpha_a(y_a)\, G_N(y_2 - y_1, \dots, y_N - y_{N-1}; \lambda)\, \mathrm{d}Y. \tag{8.18}$$

If the supports are also in $y^0 > 0$, their product class is the Bochner integral of $\prod_a \alpha_a(y_a)\Psi(Y;\lambda)$. One has $G_1 = 0$, $\overline{G_N(\xi;\lambda)} = G_N(\bar{\xi};\bar{\lambda})$, and

$$\begin{aligned}\langle \Psi(x;\lambda), \Psi(z;\lambda')\rangle = s(\lambda) G_{n+n'}((x_n - x_{n-1})^*, \dots, (x_2 - x_1)^*,\\ x_1^* + z_1, z_2 - z_1, \dots, z_{n'} - z_{n'-1}; \lambda_n, \dots, \lambda_1, \lambda'_1, \dots, \lambda'_{n'})\end{aligned} \tag{8.19}$$

when $n, n' \geq 1$. With a vacuum in the first slot the inner product is $G_{n'}$ on the successive differences of z; with a vacuum in the second slot it is $s(\lambda)G_n$ on the reversed starred differences and reversed labels. Furthermore $T(\kappa')\Psi(x;\lambda) = \Psi(x + \kappa';\lambda)$ for $\kappa' \in \overline{\mathcal{W}}$, where every point is translated.

Proof. Holomorphy follows from that of the factors. In the definition of G_N, choose $\kappa = tn_0$ with $t \geq \frac{1}{2}\min_a d_{\mathcal{W}}(\xi_a)$ and apply (8.9) to every factor. This proves (8.17).

To prove the density formula, first put the supports in $y^0 > 0$. Choose disjoint time slabs $[a_j, b_j]$, in order, with each support strictly inside its slab, $a_1 \geq 0$, and $b_j \leq a_{j+1}$. Let $f_j = \alpha_j \otimes \lambda_j$, $c_j = (a_j + b_j)/2$, $d_j = (b_j - a_j)/2$, and $I_j = \mathcal{B}^{\mathrm{c}}_{d_j}(\tau_{-c_j e_1} f_j)$. The insertion identity and its localization give

$$\left[\prod_j \mathbb{B}(f_j)\right]_1 = T(a_1 n_0) I_1 T((a_2 - b_1)n_0) I_2 \cdots I_N \Omega_1. \tag{8.20}$$

For precision, the backward induction uses $\Phi' = \prod_{j>j_0} \mathbb{B}(\tau_{-b_{j_0}e_1} f_j)$, which belongs to $\mathcal{A}_+$. The tail product is $U_{b_{j_0}}\Phi'$, so insertion of f_{j_0} is $\mathcal{B}_{b_{j_0}}(f_{j_0})[\Phi']_1$. Localization supplies $T(a_{j_0} n_0) I_{j_0}$; the induction on the translated tail supplies the next nonnegative gap $a_{j_0+1} - b_{j_0}$ and the remaining factors.

Use (8.11) to write

$$I_j = \int \alpha_j(y_j) \mathcal{B}_{\lambda_j}(y_j - a_j n_0, b_j n_0 - y_j)\, \mathrm{d}y_j.$$

On the compact supports all these operators are norm bounded; the α_j are integrable. The product in (8.20) is therefore one Bochner integral. Absorb each gap semigroup into a neighbouring point plate. The inner arguments between plates j and $j+1$ then add to $y_{j+1} - y_j$. Replace them by equal halves using the product identity above. The initial argument becomes y_1, and the terminal argument is removed against the vacuum. The integral is exactly the stated vector formula. Pairing with Ω_1 proves (8.18). A common real time translation puts any time-ordered supports in the positive half-plane, without changing their differences or their moment, proving the general formula. For $N = 1$, this formula and the vanishing mean field give $G_1 = 0$.

For the Gram identity, take the adjoint of the product defining $\Psi(x;\lambda)$. It reverses the order, stars the arguments, and multiplies by $s(\lambda)$. In the middle of the two products the arguments x_1^* and z_1 add to $x_1^* + z_1$; replace them by equal halves. This gives (8.19), including the stated vacuum cases. The translation identity follows by shifting the first argument.

Conjugating (8.18) replaces the plates by $\bar\alpha_a \otimes \bar\lambda_a$. Testing in small ordered product neighbourhoods identifies $\overline{G_N(\xi;\lambda)} = G_N(\xi;\bar\lambda)$ on real $\xi \in \mathcal{W}^{N-1}$. Both sides, with ξ conjugated on the first, are holomorphic, and this domain is connected. The identity theorem proves the claimed conjugation formula. □

8.4. **The two-point angular estimate.** The operator bound for N point plates gives angular growth of order N. At two points a positive spectral measure gives the precise growth order two. Closedness will remove the unwanted endpoint frequencies.

Fix a coordinate reconstruction j. For scalar $u \in L^1 \cap L^2(\mathbb{R}^3)$ supported in $x_j \ge 0$, put

$$\widetilde{u}(E, p_\perp) = \int u(x) e^{-x_j E + i p_\perp \cdot x_\perp}\, \mathrm{d}x.$$

There are complex Radon measures $\varrho_{ab}^{(j)}$, $a, b \in \{1,2,3\}$, with

$$\langle [\mathbb{B}_a(u)]_j, [\mathbb{B}_b(u')]_j \rangle = \int \overline{\widetilde{u}}\, \widetilde{u}'\, \mathrm{d}\varrho_{ab}^{(j)}. \tag{8.21}$$

They have a common positive Radon measure ϱ_0 such that $\varrho_{ab}^{(j)} = r_{ab}\varrho_0$ and the matrix r is positive semidefinite and Hermitian almost everywhere. In particular, for vector coefficients c, c', the measures $\varrho_{cc'} = \sum_{a,b} \bar c_a c'_b \varrho_{ab}$ satisfy

$$\int |G_1 G_2|\, \mathrm{d}|\varrho_{cc'}| \le \left(\int |G_1|^2\, \mathrm{d}\varrho_{cc} \right)^{1/2} \left(\int |G_2|^2\, \mathrm{d}\varrho_{c'c'} \right)^{1/2}. \tag{8.22}$$

All these measures are carried by the joint spectral cone.

Here is the construction, including the fact that the measures need only be locally finite. Convolution and (7.1) give

$$[\mathbb{B}_a(u * \alpha)]_j = \widetilde{u}(H_j, \mathsf{P})[\mathbb{B}_a(\alpha)]_j$$

for $\alpha \in L^2$ supported in $x_j \ge 0$. This follows by applying the bounded field and quotient maps to the L^2 Bochner integral $u*\alpha = \int u(x)\tau_x \alpha\, \mathrm{d}x$; Fubini applies to its scalar spectral matrix elements since $u \in L^1$. Let $P_1(v) = (\pi(1+v^2))^{-1}$, whose Fourier transform with either sign is $e^{-|p|}$, and choose

$$\alpha_0(x) = \mathbf{1}_{(0,1)}(x_j) \prod_{i \ne j} P_1(x_i), \qquad a_0(E, p_\perp) = \widetilde{\alpha}_0 = \frac{1 - e^{-E}}{E} e^{-|p_\perp|_1} > 0.$$

The quotient at $E = 0$ is one. With $\psi_a = [\mathbb{B}_a(\alpha_0)]_j$ and $m_{ab} = \langle \psi_a, \mathsf{E}_j(\cdot)\psi_b \rangle$, commutativity of convolution gives

$$a_0(H_j, \mathsf{P})[\mathbb{B}_a(u)]_j = \widetilde{u}(H_j, \mathsf{P})\psi_a. \tag{8.23}$$

The finite matrix measure m is positive. If $m_0 = \sum_a m_{aa}$, spectral Cauchy–Schwarz shows $m_{ab} \ll m_0$. Its density matrix is positive Hermitian on one common full-measure set: test its quadratic form on the countable rational complex vectors and then use continuity. Define $\varrho_{ab} = a_0^{-2} m_{ab}$ and $\varrho_0 = a_0^{-2} m_0$. The strictly positive continuous a_0 has a positive lower bound on bounded sets, so these measures are Radon. Pointwise matrix Cauchy–Schwarz followed by its integral version gives (8.22).

On $A_n = \{a_0 > 1/n\}$, which is bounded, one may divide (8.23) by a_0 using bounded spectral calculus. This proves (8.21) with the measures restricted to A_n. As A_n increase to the whole spectral space, the diagonal identity follows by monotone convergence; its integral is $\|[\mathbb{B}_a(u)]_j\|^2 \le \|u\|_2^2$. Equation (8.22) then makes the mixed integral absolutely convergent and gives the unrestricted identity by dominated convergence. Cone support follows from that of E_j.

Now use $j = 1$, and take real profiles $\chi, \chi' \in C_c^\infty(\mathbb{R})$. Their transverse transforms in this spectral convention are $\widehat{\chi}_+(p_3) = \int \chi(x_3) e^{ip_3 x_3}\,\mathrm{d}x_3$. For labels $\lambda = (\sigma, \chi)$ and $\lambda' = (\sigma', \chi')$, set

$$\varrho_{\lambda\lambda'} = \sum_{a,b} \overline{e_a^\sigma} e_b^{\sigma'} \overline{\widehat{\chi}_+} \widehat{\chi'}_+ \varrho_{ab}^{(1)}.$$

In this formula $\widehat{\chi'}_+$ means the same positive-sign transform of χ'. These are Radon measures on (E, p_2, p_3), and $\varrho_{\lambda\lambda} \geq 0$. For $\alpha, \alpha' \in L^1 \cap L^2(\mathbb{R}^2; \mathbb{C})$ vanishing almost everywhere on $y^0 < 0$, put $\alpha^L(E, p_2) = \int \alpha(y) e^{-y^0 E + iy^1 p_2}\,\mathrm{d}y$. Equation (8.21) gives

$$\langle [\mathbb{B}(\alpha \otimes \lambda)]_1, [\mathbb{B}(\alpha' \otimes \lambda')]_1 \rangle = \int \overline{\alpha^L} (\alpha')^L \,\mathrm{d}\varrho_{\lambda\lambda'}. \tag{8.24}$$

The crucial estimate is

$$I_\lambda(t) := \int e^{-tE}\,\mathrm{d}\varrho_{\lambda\lambda} \leq \frac{36\|\chi\|_2^2}{\pi t^2} \quad (t > 0). \tag{8.25}$$

Indeed, for $c > 0$ take $\alpha_c(y) = c^{-2}\mathbf{1}_{(c,2c)}(y^0) P_1(y^1/c)$. On $E \geq |p_2|$ its transform is bounded below by $e^{-2cE - c|p_2|} \geq e^{-3cE}$. Since $\|\alpha_c\|_2^2 = (2\pi c^2)^{-1}$, the variance bound and the quotient contraction in (8.24) give $I_\lambda(6c) \leq \|\chi\|_2^2/(\pi c^2)$. Choosing $c = t/6$ proves the estimate. Its mixed version is

$$\int e^{-tE}\,\mathrm{d}|\varrho_{\lambda\lambda'}| \leq \sqrt{I_\lambda(t) I_{\lambda'}(t)}. \tag{8.26}$$

Define, for $y^0 > 0$,

$$G^{[1]}_{\lambda\lambda'}(y) = s(\sigma) \int e^{-y^0 E + iy^1 p_2}\,\mathrm{d}\varrho_{\lambda\lambda'}.$$

It is smooth, since on $y^0 \geq t > 0$ every derivative of order q is bounded in modulus by $E^q e^{-tE} \leq q!(2/t)^q e^{-tE/2}$ on the cone. It computes the two-point function on time-ordered compact plane supports. To see this, translate the supports to opposite sides of $y^0 = 0$ and put $\alpha^\theta = \bar{\alpha} \circ r_1$. Then

$$\mathbb{E}_\mu[\mathbb{B}(\alpha \otimes \lambda)\mathbb{B}(\alpha' \otimes \lambda')] = s(\sigma) \int \overline{(\alpha^\theta)^L} (\alpha')^L \,\mathrm{d}\varrho_{\lambda\lambda'}.$$

Expanding the transforms gives the integral against $G^{[1]}_{\lambda\lambda'}(y' - y)$. Fubini is justified by

$$\|\alpha\|_1 \|\alpha'\|_1 e^{-\delta E},$$

where $\delta > 0$ is the time gap, and (8.26). Thus this kernel agrees with G_2 on real positive-time differences.

For $r > 0$ and $n_\zeta = (\cos\zeta, \sin\zeta)$, it has the holomorphic continuation

$$h^{[1]}_{\lambda\lambda'}(\zeta) = s(\sigma) \int e^{-rE\cos\zeta + irp_2 \sin\zeta}\,\mathrm{d}\varrho_{\lambda\lambda'} \quad (|\Re\zeta| < \pi/2).$$

If $\zeta = t + i\beta$, the real part of the exponent is $-r\cos t(E\cosh\beta + p_2 \sinh\beta)$. The cone bounds this above by $-r\cos t\, e^{-|\beta|}E$. On every compact substrip this gives an integrable exponential majorant; dominated convergence and Morera's theorem give holomorphy. Equations (8.25)–(8.26) give

$$|h^{[1]}_{\lambda\lambda'}(t + i\beta)| \leq \frac{36\|\chi\|_2 \|\chi'\|_2}{\pi(r\cos t)^2}\, e^{2|\beta|}. \tag{8.27}$$

8.5. **Closedness removes the unwanted two-point frequencies.** Assume in addition that μ is closed and cubically invariant. In particular, its two-point form is invariant under $R_{\pi/2}$. We use only this quarter-turn consequence here.

For $\upsilon \in (\pi/2)\mathbb{Z}$ and $n_\upsilon \cdot y > 0$, define

$$G^{(\upsilon)}_{\lambda\lambda'}(y) = e^{iw\upsilon} G^{[1]}_{\lambda\lambda'}(R_{-\upsilon} y), \qquad w = w(\sigma) + w(\sigma').$$

By quarter-turn invariance and (8.8), this is the two-point density for supports ordered along n_υ. Two such densities agree on overlaps: test them on small product neighbourhoods ordered in both directions and use continuity. The four half-planes cover $\mathbb{R}^2 \setminus \{0\}$, so they define one smooth kernel $G_{\lambda\lambda'}$ there. It computes the two-point function for any two disjoint compact plane supports. Indeed, partition them into sufficiently small half-open squares. Each pair of squares is ordered along one of the four coordinate directions, and there are only finitely many pairs. Sum the ordered density identities.

For fixed $r > 0$, the functions

$$h^{(\upsilon)}(\zeta) = e^{iw\upsilon} h^{[1]}_{\lambda\lambda'}(\zeta - \upsilon) \quad (|\Re\zeta - \upsilon| < \pi/2)$$

agree on adjacent strips, by the identity theorem applied on their real overlap. They give an entire function h with

$$h(\vartheta) = G_{\lambda\lambda'}(r n_\vartheta), \qquad h(\zeta + \pi/2) = i^w h(\zeta), \qquad |h(t + i\beta)| \le C_{\chi,\chi'} r^{-2} e^{2|\beta|}, \tag{8.28}$$

where $C_{\chi,\chi'} = 72\|\chi\|_2\|\chi'\|_2/\pi$. For the bound, choose a strip centre within $\pi/4$ of t and apply (8.27).

The entire function $e^{-iw\zeta} h(\zeta)$ has period $\pi/2$, so it is $H(e^{4i\zeta})$ for a holomorphic H on $\mathbb{C} \setminus \{0\}$. The Laurent series of H gives frequencies $k = w + 4m$, $m \in \mathbb{Z}$. Cauchy's formula on the circle parametrized by $e^{4i(t+i\beta)}$, $0 \le t \le \pi/2$, bounds the coefficient $a^k(r)$ by $C_{\chi,\chi'} r^{-2} e^{2|\beta| + k\beta}$. Sending β to the appropriate infinity kills $|k| > 2$. Consequently

$$G_{\lambda\lambda'}(r n_\vartheta) = \sum_{\substack{|k| \le 2 \\ k - w \in 4\mathbb{Z}}} a^k(r) e^{ik\vartheta}, \qquad |a^k(r)| \le C_{\chi,\chi'} r^{-2}. \tag{8.29}$$

These coefficients are smooth in r and real bilinear in the profiles, by their Fourier integral formula.

Every pair of labels has only its expected frequency $k = w$, except for the pairs $(+,+)$ and $(-,-)$, whose windows contain both 2 and -2. To eliminate the opposite frequency, use closedness in its vector form $\mathbb{B}(\nabla u) = 0$. Take $\eta \in C^\infty_c(\mathbb{R}^2; \mathbb{C})$ and $\chi_0 \in C^\infty_c(\mathbb{R}; \mathbb{R})$. With $z = y^0 + iy^1$ and $\partial_z = (\partial_{y^0} - i\partial_{y^1})/2$,

$$\nabla(\eta(y)\chi_0(x_3)) = (\partial_z \eta) \otimes (+, \chi_0) + (\partial_{\bar z}\eta) \otimes (-, \chi_0) + \eta \otimes (3, \chi_0').$$

Pair this gradient with a second plate whose plane support is disjoint. Apply the density identities and integrate by parts. Testing on small disjoint product neighbourhoods gives

$$\partial_z G_{(+,\chi_0)\lambda'} + \partial_{\bar z} G_{(-,\chi_0)\lambda'} + G_{(3,\chi_0')\lambda'} = 0 \quad \text{on } \mathbb{R}^2 \setminus \{0\}. \tag{8.30}$$

There is no diagonal contribution in this equation. In polar coordinates,

$$\partial_z(f(r)e^{ik\vartheta}) = \tfrac{1}{2} e^{i(k-1)\vartheta}(f'(r) + kf(r)/r),$$
$$\partial_{\bar z}(f(r)e^{ik\vartheta}) = \tfrac{1}{2} e^{i(k+1)\vartheta}(f'(r) - kf(r)/r).$$

For $\lambda' = (+, \chi')$, the frequency -3 in (8.30) occurs only in the derivative of $a^{-2}_{(+,\chi_0)(+,\chi')}$. It obeys $f' = 2f/r$. For $\lambda' = (-, \chi')$, the frequency 3 likewise gives $f' = 2f/r$ for $a^{2}_{(-,\chi_0)(-,\chi')}$. Each solution is $f(r) = f(1)r^2$. The bound $|f(r)| \le C_{\chi_0,\chi'} r^{-2}$ for all $r > 0$ forces $f(1) = 0$ by sending $r \to \infty$. Thus

$$G_{\lambda\lambda'}(R_t y) = e^{i(w(\sigma) + w(\sigma'))t} G_{\lambda\lambda'}(y). \tag{8.31}$$

The plate action cancels this phase in the two-point density formula, proving rotational covariance for disjoint compact plane supports.

8.6. **Extending identities across coincident plane supports.** The preceding density argument does not yet address coincident plane supports. We use the following L^2 argument, which will also be applied at every higher order.

Proposition 8.4. *Let $U \subset \mathbb{R}^2$ be open, and write L^2_U for the vector fields supported in $U \times \mathbb{R}$. Let $D : (L^2_U)^N \to \mathbb{C}$ be a bounded multilinear form. Suppose it vanishes when the entries are supported over pairwise disjoint compact subsets of U, and whenever one entry is ∇u with $u \in C^\infty_c(U \times \mathbb{R})$. Then $D = 0$.*

Proof. We argue simultaneously for all open U by induction on N. At $N = 1$, fields with compact plane support are dense, so the first assumption suffices.

At $N = 2$, represent the bilinear form as $D(f_1, f_2) = \int f_1(x) \cdot (Tf_2)(x)\,\mathrm{d}x$ for a bounded operator T on L^2_U. The pairing here is bilinear. For disjoint compact $K, K' \subset U$ the first assumption gives $\mathbf{1}_K T \mathbf{1}_{K'} = 0$. Inner regularity and strong convergence of the corresponding multiplication projections extend this to disjoint Borel sets. Using A and $U \setminus A$ on both sides gives $T\mathbf{1}_A = \mathbf{1}_A T$ for every Borel A. Uniform approximation by simple functions then gives $Th = hT$ for every bounded plane multiplier h. Hence

$$D(h\nabla u, f_2) = D(\nabla u, hf_2) = 0.$$

Fix a compact $A \subset U$, choose $u_0 \in C^\infty_c(U)$ equal to one near A, and let $\chi_0 \in C^\infty_c(\mathbb{R})$. On $A \times \mathbb{R}$, the gradients of $u_0\chi_0$, $u_0 y^0 \chi_0$, and $u_0 y^1 \chi_0$ are respectively

$$\chi_0' e_3, \qquad \chi_0 e_1 + y^0 \chi_0' e_3, \qquad \chi_0 e_2 + y^1 \chi_0' e_3.$$

The preceding identity with multiplier $h\mathbf{1}_A$ first annihilates $h\mathbf{1}_A \chi_0' e_3$. Applying it with $hy^0\mathbf{1}_A$ and $hy^1\mathbf{1}_A$ then annihilates $h\mathbf{1}_A\chi_0 e_1$ and $h\mathbf{1}_A\chi_0 e_2$. These multipliers are bounded since A is compact. Such plane-transverse products span a dense subspace. For the transverse derivative family, its orthogonal complement in $L^2(\mathbb{R})$ has distributional derivative zero, hence is a constant and therefore zero. Thus these three vector families are total in L^2_U, and $D = 0$.

Suppose $N \geq 3$ and the assertion is proved at every smaller order. Two consequences of the induction hypothesis separate unequal plane cells. First, if one entry is supported over a compact $K \subset U$ and the others over $U \setminus K$, fix that entry. The form on the remaining $N - 1$ slots satisfies both assumptions on the open set $U \setminus K$ and is zero. Second, if the slots divide into nonempty groups I, J, with the J entries supported over a compact K_J and the I entries over its complement, fix the J entries. The form on the I slots inherits the gradient assumption. When those slots have pairwise disjoint compact supports, the first consequence, applied to any one of them, gives zero. The induction hypothesis on $U \setminus K_J$ therefore makes this form zero as well.

By density put all entries over one compact $K_0 \subset U$. Partition the plane into half-open squares Q of side d and expand D by multilinearity. A tuple of squares which are not all equal has zero contribution. To justify compact separation, replace each square by its concentric closed square of side $d - 2\epsilon$. Separate the slots whose square equals the first from the remaining slots, use the second consequence with $K_J = K_0$ intersected with the union of their closed squares, and send $\epsilon \downarrow 0$ in L^2. Only the equal-square terms remain, and Cauchy–Schwarz gives

$$|D(f_1, \ldots, f_N)| \leq \|D\| \max_Q \|\mathbf{1}_Q f_3\|_2 \|f_1\|_2 \|f_2\|_2 \prod_{i=4}^{N} \|f_i\|_2.$$

Indeed, sum the first two square norms by Cauchy–Schwarz and bound the others by their whole norms. The maximum tends to zero as $d \downarrow 0$: its square is the integral over an area-d^2 cell of

the fixed integrable function $y \mapsto \int |f_3(y, x_3)|^2 \, \mathrm{d}x_3$. Absolute continuity of the integral proves the assertion. □

Apply this proposition with $N = 2$ to

$$D(b, b') = \mathbb{E}_\mu[\mathbb{B}(R_t b)\mathbb{B}(R_t b')] - \mathbb{E}_\mu[\mathbb{B}(b)\mathbb{B}(b')],$$

where $R_t b$ means $R_t \cdot b$. Gaussian domination makes it a bounded bilinear form. Finite sums of plates with smooth real transverse profiles are dense in the fields over any fixed compact plane set: simple plane-transverse tensors are dense in L^2, and e^+, e^-, e^3 span $\mathbb{C}^3$. Thus (8.31) supplies the disjoint-support assumption for general fields. Closedness supplies the gradient assumption, since rotations carry gradients to gradients. Consequently the two-point function is invariant under every R_t, on all L^2 tests.

8.7. **The angular expansion at every order.** Fix $N \geq 2$ and a tuple of labels λ, with total weight $w = w(\lambda)$. Put $\mathcal{W}_\upsilon = R_\upsilon \mathcal{W}$ for $\upsilon \in (\pi/2)\mathbb{Z}$. For a permutation $\mathfrak{s}$, let

$$\mathcal{Z}(\mathfrak{s}, \upsilon) = \{Z = (z_1, \dots, z_N) : z_{\mathfrak{s}(a+1)} - z_{\mathfrak{s}(a)} \in \mathcal{W}_\upsilon, \ 1 \leq a < N\}.$$

On this open convex set define

$$G_\upsilon^{\mathfrak{s}}(Z; \lambda) = e^{iw\upsilon} G_N\left((R_{-\upsilon}(z_{\mathfrak{s}(a+1)} - z_{\mathfrak{s}(a)}))_{a=1}^{N-1}; \mathfrak{s}\lambda\right).$$

Here $(\mathfrak{s}\lambda)_a = \lambda_{\mathfrak{s}(a)}$. Moment symmetry, quarter-turn invariance, and (8.18) identify this as the moment density on compact product supports in its real part. Two such holomorphic functions agree on their intersection. Indeed, the intersection is convex and is preserved by taking real parts, so a nonempty intersection contains an open set of real configurations. Testing both densities there identifies them by continuity, and the holomorphic identity theorem identifies them on the full intersection. They therefore define one holomorphic function $\mathcal{G}_N(Z; \lambda)$ on $\mathcal{Z}_N = \bigcup_{\mathfrak{s},\upsilon} \mathcal{Z}(\mathfrak{s}, \upsilon)$.

The real set $\mathcal{O}_N = \mathcal{Z}_N \cap (\mathbb{R}^2)^N$ consists of configurations whose first coordinates are all distinct or whose second coordinates are all distinct. Its complement is the finite union of the codimension-two subspaces $\{y_a^0 = y_b^0, \ y_k^1 = y_l^1\}$. Thus $\mathcal{O}_N$ is open, dense, and connected. The functions $\mathcal{G}_N$ are symmetric under simultaneous permutation of points and labels, invariant under common translations, and satisfy

$$\mathcal{G}_N(R_{\pi/2} Z; \lambda) = i^w \mathcal{G}_N(Z; \lambda), \qquad \overline{\mathcal{G}_N(Z; \lambda)} = \mathcal{G}_N(\bar{Z}; \bar{\lambda}).$$

These identities follow from the defining densities and the corresponding identities for G_N.

For $\beta \in \mathbb{R}$ write $\Lambda_\beta = R_{i\beta}$. In the coordinates $\xi^\pm = \xi^0 \pm i\xi^1$, $\mathcal{W} = \{\Re\xi^+ > 0, \Re\xi^- > 0\}$ and $\Lambda_\beta \xi^\pm = e^{\mp\beta} \xi^\pm$. Thus every tube $\mathcal{Z}(\mathfrak{s}, \upsilon)$ is preserved by Λ_β. Moreover $d_{\mathcal{W}}(\Lambda_\beta \xi) \geq e^{-|\beta|} d_{\mathcal{W}}(\xi)$ on the closed tube.

Proposition 8.5. *There are holomorphic coefficients $c_m(Z; \lambda)$ on $\mathcal{Z}_N$, vanishing unless $|w + 4m| \leq N$, such that*

$$\mathcal{G}_N(\Lambda_\beta Z; \lambda) = \sum_m c_m(Z; \lambda) e^{-(w+4m)\beta} \quad (\beta \in \mathbb{R}). \tag{8.32}$$

They are unique, symmetric under simultaneous permutation, and invariant under common translations. They satisfy

$$\begin{aligned} c_m(\Lambda_\beta Z; \lambda) &= e^{-(w+4m)\beta} c_m(Z; \lambda), \\ c_m(R_{\pi/2} Z; \lambda) &= i^w c_m(Z; \lambda), \qquad \sum_m c_m(Z; \lambda) = \mathcal{G}_N(Z; \lambda), \\ \overline{c_m(X; \lambda)} &= c_{-m}(X; \bar{\lambda}) \quad (X \in \mathcal{O}_N). \end{aligned} \tag{8.33}$$

Proof. Fix $0 < \vartheta_0 < \pi/4$ and first consider real configurations whose consecutive differences have positive lengths and angles $|\vartheta_a| < \vartheta_0$ from n_0. For $\zeta = t + i\beta$,

$$\Re(R_{\zeta-\upsilon}\xi_a)^{\pm} = |\xi_a| e^{\mp\beta} \cos(\vartheta_a + t - \upsilon).$$

The strips $|t - \upsilon| < \pi/2 - \vartheta_0$ cover the complex ζ-plane. Thus $h_X(\zeta) = \mathcal{G}_N(R_\zeta X; \lambda)$ is entire. Choosing υ within $\pi/4$ of t bounds every gap distance below by $r_X e^{-|\beta|} \cos(\vartheta_0 + \pi/4)$, where $r_X = \min_a |\xi_a|$. Equation (8.17) gives $|h_X(t+i\beta)| \le C_{X,\lambda} e^{N|\beta|}$. Its quarter-period is $h_X(\zeta+\pi/2) = i^w h_X(\zeta)$. The Laurent coefficient argument used at two points, with growth exponent N, gives the finite frequencies $k = w + 4m$, $|k| \le N$.

To extend this statement to a whole tube, let $\mathcal{M} = \{m : |w+4m| \le N\}$ and $P(u) = \prod_{m\in\mathcal{M}}(u + w + 4m)$. The functions $Z \mapsto P(\partial_\beta) G^{\mathfrak{s}}_\upsilon(\Lambda_\beta Z; \lambda)$ are holomorphic for each real β. They vanish on the nonempty open real subset obtained by rotating and permuting the configurations just used. The identity theorem gives zero on the entire convex tube. The resulting constant-coefficient ordinary differential equation has distinct characteristic roots $-(w + 4m)$. Its coefficients in the exponential expansion are fixed linear combinations of its derivatives at $\beta = 0$, and hence are holomorphic in Z. On overlapping tubes the exponentials are linearly independent, so the coefficients agree and define c_m on $\mathcal{Z}_N$. This also proves uniqueness. Comparing coefficients after a permutation, common translation, or addition of the parameter β gives the stated identities. For the conjugation identity use $\overline{\Lambda_\beta X} = \Lambda_{-\beta} X$ and $w(\bar\lambda) = -w$. □

The coefficients have exact real rotation phases whenever both configurations are in $\mathcal{O}_N$:

$$c_m(R_t X; \lambda) = e^{i(w+4m)t} c_m(X; \lambda). \tag{8.34}$$

Locally in one tube, the allowed angles form an interval: each ordered difference has a positive projection precisely when $\cos(\vartheta_a + t) > 0$. The identity theorem in $t + i\beta$ extends (8.32) from the imaginary axis to that interval. Comparing the exponential coefficients gives (8.34) locally.

To pass through changes of ordering, first take distinct points with no two distinct chords perpendicular. Such configurations are a dense open set, the complement of finitely many nonzero polynomial zero sets. Every rotation of one of them belongs to $\mathcal{O}_N$: simultaneous coincidences of first and second coordinates would give perpendicular chords, or identical points. The local phase identity therefore continues along its whole rotation orbit. Approximation proves it whenever $X, R_t X$ are in $\mathcal{O}_N$.

For an arbitrary configuration X of distinct plane points, choose t so that all first coordinates of $R_t X$ are distinct. Only finitely many angles modulo π are excluded. Define

$$\widetilde{c}_m(X; \lambda) = e^{-i(w+4m)t} c_m(R_t X; \lambda).$$

The phase identity makes this independent of the choice. Locally one may keep t fixed, so it is real analytic on the connected space $(\mathbb{R}^2)^N_{\neq}$ of distinct configurations. It inherits permutation symmetry, translation invariance, and

$$\widetilde{c}_m(R_t X; \lambda) = e^{i(w+4m)t} \widetilde{c}_m(X; \lambda).$$

Put $\widetilde{G}_N = \sum_m \widetilde{c}_m$. It is real analytic and agrees with $\mathcal{G}_N$ on $\mathcal{O}_N$.

Define $\mathfrak{a}(0) = \mathfrak{a}(1) = 0$, and, for $N \ge 2$, let $\mathfrak{a}(N)$ be the largest $|m|$ for which c_m is not identically zero for some tuple of N labels; set it to zero if all coefficients vanish. Since $|w| \le N$ and $|w + 4m| \le N$,

$$\mathfrak{a}(N) \le N/2. \tag{8.35}$$

A nonzero coefficient can be detected on a strictly time-ordered real configuration after a quarter turn and permutation: the identity theorem on each tube and the coefficient identities above give this reduction. Conjugation then shows that if $\mathfrak{a}(N) > 0$, both indices $\mathfrak{a}(N)$ and $-\mathfrak{a}(N)$ occur.

8.8. **Reflection bounds for the angular coefficients.** We next prove the convexity estimate

$$2\mathfrak{a}(a+b) \le \mathfrak{a}(2a) + \mathfrak{a}(2b) \quad (a,b \ge 0). \tag{8.36}$$

Let $N = a + b$ and $M = \mathfrak{a}(N) > 0$. Choose a strictly time-ordered $X = (y_1, \dots, y_N)$ and labels with $c_{-M}(X;\lambda) \neq 0$. Translate its time cut so that the first a points have negative times and the remaining b have positive times. Reflect the first group and reverse its order to obtain x, with reversed labels λ^x; call the second group z, with labels λ^z. Their weights satisfy $w = w_x + w_z$. The Gram identity gives

$$\langle \Psi(\Lambda_\beta x;\lambda^x), \Psi(\Lambda_\beta z;\lambda^z)\rangle = s(\lambda^x)\mathcal{G}_N(\Lambda_\beta X;\lambda).$$

An empty group gives the vacuum. The squared norm of the first vector is, up to the sign $s(\lambda^x)$, the kernel of its reflected double: the configuration $(r_1x_a, \dots, r_1x_1, x_1, \dots, x_a)$ with the same labels in reverse then forward order. Its weight is $2w_x$. The other squared norm has weight $2w_z$. These are direct instances of (8.19); the reflection-conjugation keeps the labels.

Apply Cauchy–Schwarz to this matrix element and expand the three kernels by (8.32). Multiplication by $e^{2w\beta}$ cancels their weights and gives

$$\left|\sum_m c_m(X;\lambda)e^{-4m\beta}\right|^2 \le \left|\sum_{|m|\le\mathfrak{a}(2a)} d_m e^{-4m\beta}\right| \left|\sum_{|m|\le\mathfrak{a}(2b)} d'_m e^{-4m\beta}\right|.$$

The coefficients d_m, d'_m belong to the two specified doubles. A vacuum factor is one. As $\beta \to +\infty$, the left side has a nonzero leading term of rate $8M$, and the right side is bounded by a constant times $e^{4(\mathfrak{a}(2a)+\mathfrak{a}(2b))\beta}$. This proves (8.36). The cases $M = 0$ or $N \le 1$ already satisfy it.

There is also a positive kernel for the highest coefficient. For $a \ge 1$, set $M = \mathfrak{a}(2a)$. On labelled configurations of a distinct points in $y^0 > 0$, define

$$K_a((x;\lambda),(z;\lambda')) = s(\lambda)\widetilde{c}^{(2a)}_{-M}(r_1x, z;\lambda,\lambda'). \tag{8.37}$$

The superscript indicates the number of points. This kernel is Hermitian and positive semidefinite. First suppose the times in each configuration are distinct and order them increasingly. By the Gram formula,

$$\langle \Psi(\Lambda_\beta x;\lambda), \Psi(\Lambda_\beta z;\lambda')\rangle = s(\lambda) \sum_{|m|\le M} \widetilde{c}^{(2a)}_m(r_1x, z;\lambda,\lambda')e^{-(w(\lambda)+w(\lambda')+4m)\beta}.$$

Reversing the reflected points is harmless by labelled symmetry. Multiply this Gram kernel by the positive factors $e^{w(\lambda)\beta}e^{w(\lambda')\beta}e^{-4M\beta}$. Its pointwise limit as $\beta \to +\infty$ is K_a. Positive semidefiniteness passes to this limit on every finite matrix. Approximating configurations with tied times by ones with distinct times proves the same assertion for all the stated configurations.

The exact phase of $\widetilde{c}_{-M}$ transports this consequence to any line L in the plane. If x, z lie on one open side of L and r_L is its reflection, translation to a point of L and rotation of its normal to n_0 express $\widetilde{c}_{-M}(r_Lx, z)$ as K_a times a unit-modulus factor. Taking absolute values of the positive-kernel Cauchy–Schwarz inequality gives

$$|\widetilde{c}^{(2a)}_{-M}(r_Lx, z;\lambda,\lambda')|^2 \le |\widetilde{c}^{(2a)}_{-M}(r_Lx, x;\lambda,\lambda)||\widetilde{c}^{(2a)}_{-M}(r_Lz, z;\lambda',\lambda')|. \tag{8.38}$$

In particular a zero reflected diagonal forces every such off-diagonal coefficient to vanish.

8.9. **Regular polygons and the reduction to two points.** We prove the arithmetic restriction

$$a \mid 2\mathfrak{a}(2a) \quad (a \geq 1). \tag{8.39}$$

Fix a and put $M = \mathfrak{a}(2a)$, $c_* = \widetilde{c}^{(2a)}_{-M}$. The coefficients depend only on the labelled set of points. If a rotation through t permutes such a set without changing its labels, its exact phase gives $(1 - e^{i(w-4M)t})c_* = 0$. For a regular $2a$-gon with the same label (σ, χ) at each vertex, use $t = \pi/a$ and $w = 2aw(\sigma)$. The label weight cancels, leaving $e^{-4\pi i M/a}$. Thus c_* vanishes on this polygon whenever a fails to divide $2M$.

That selection rule concerns a special configuration. The positive kernel propagates its zero to arbitrary configurations by the following halving argument. For a labelled set S of at most $2a$ distinct points, let $\mathcal{N}(S)$ mean that c_* vanishes on every $2a$-point labelled configuration containing S. Let S' have $j' \leq a$ points on one open side L_+ of a line L, and suppose $S \subset r_L S' \cup S'$. Then

$$\mathcal{N}(S) \implies \mathcal{N}(S') \text{ and } \mathcal{N}(r_L S'). \tag{8.40}$$

To see this, add $a - j'$ arbitrary points R in L_+ to S', forming $x = S' \cup R$. The reflected double $r_L x \cup x$ contains S, so its diagonal coefficient is zero. Equation (8.38) implies $c_*(r_L x \cup z) = 0$ for every a-point labelled set z in L_+. Thus, with $r_L S'$ fixed, the remaining $a - j'$ points on L_- and a points on L_+ may vary in an open set.

For each choice of their labels, the space of their distinct positions, avoiding the fixed points, is the complement of finitely many codimension-two affine subspaces and is connected. The coefficient is real analytic there. Its zero on the nonempty open set just obtained extends to all positions. Every $2a$-point configuration containing $r_L S'$ is obtained this way. This proves $\mathcal{N}(r_L S')$. Interchange the sides of L and repeat the argument to obtain $\mathcal{N}(S')$.

Suppose now $a \nmid 2M$. Then $M > 0$ and $a \geq 3$. Fix an arbitrary label and any regular $2a$-gon with vertices P_k, $k \in \mathbb{Z}/(2a)\mathbb{Z}$, in cyclic order. Let $\mathcal{N}_j$ assert $\mathcal{N}(S)$ for every arc of j consecutive vertices, with this label, in every such polygon. Selection gives $\mathcal{N}_{2a}$. For $j \geq 2$, put $j' = \lceil j/2 \rceil \leq a$ and take an arc $S' = \{P_{k_0+1}, \ldots, P_{k_0+j'}\}$. Reflect in the perpendicular bisector of $[P_{k_0}, P_{k_0+1}]$. The a vertices $P_{k_0+1}, \ldots, P_{k_0+a}$ lie on one open side: their angles from the edge-midpoint radius are $(2k-1)\pi/(2a) \in (0, \pi)$. Their images satisfy $r_L P_{k_0+k} = P_{k_0+1-k}$. Consequently $r_L S' \cup S'$ is an arc of $2j'$ consecutive vertices and contains a j-vertex arc. Equation (8.40) gives $\mathcal{N}_j \Rightarrow \mathcal{N}_{j'}$.

Iteration reaches $j = 1$. Any plane point is a vertex of some regular $2a$-gon, so the coefficient vanishes whenever a configuration has even one point with the fixed label. Since the label was arbitrary, c_* vanishes identically for all label tuples. This contradicts the occurrence of the index $-M$, proving (8.39).

Now set $\nu(a) = \mathfrak{a}(2a)$. It is a nonnegative integer sequence, with $\nu(0) = 0$, $\nu(a) \leq a$, and $2\nu(a) \leq \nu(a-1) + \nu(a+1)$ by (8.36). Its increments $d_a = \nu(a) - \nu(a-1)$ are nondecreasing integers. The linear bound keeps them bounded above, so they are eventually constant, say d. Thus $\nu(a) = da - c$ for large a, where $c = \sum_k (d - d_k) \geq 0$ is an integer. Divisibility (8.39) makes every large a divide $2c$, forcing $c = 0$. All increments equal $d = \nu(1)$, and

$$\mathfrak{a}(2a) = a\mathfrak{a}(2), \qquad 2\mathfrak{a}(N) \leq N\mathfrak{a}(2).$$

For the second inequality split $N = \lfloor N/2 \rfloor + \lceil N/2 \rceil$ in (8.36).

The two-point result gives $\mathfrak{a}(2) = 0$. For smooth transverse profiles this follows from (8.31) and the ordered density formula: near any strictly ordered two-point configuration, only the frequency w occurs. Linear independence of the finite angular exponentials makes $c_m = 0$ there for $m \neq 0$, and the identity theorem on each tube gives zero throughout. For arbitrary L^2 profiles, the all-L^2 two-point covariance proved above gives the same local density identity. Hence

$$\mathfrak{a}(N) = 0 \quad \text{for every } N, \qquad \widetilde{G}_N(R_t X; \lambda) = e^{iw(\lambda)t} \widetilde{G}_N(X; \lambda). \tag{8.41}$$

8.10. **Moment densities on all distinct configurations.** The analytic continuation $\widetilde{G}_N$ must still be identified as the moment density at distinct configurations outside $\mathcal{O}_N$. This is a separate step; continuation of a density does not by itself exclude an additional singular part on that smaller set.

We first record the analytic displacement used for this step. If polynomial field observables Φ' and Φ'' have test supports in $x_1 < c - \delta$ and $x_1 > c + \delta$, respectively, then

$$s \longmapsto \mathbb{E}_\mu[\Phi' U(se_2)\Phi'']$$

extends holomorphically to $|\Im s| < 2\delta$. Translate c to zero and put $\psi' = [U_{-\delta}\Theta_1\Phi']_1$, $\psi'' = [U_{-\delta}\Phi'']_1$. The quotient and semigroup identities express the real correlation as $\langle \psi', T((2\delta, s))\psi''\rangle$. This is holomorphic on the stated strip by the cone.

For compact, pairwise disjoint plane supports we claim

$$\mathbb{E}_\mu \prod_a \mathbb{B}(\alpha_a \otimes \lambda_a) = \int \prod_a \alpha_a(y_a)\widetilde{G}_N(Y;\lambda)\,\mathrm{d}Y. \tag{8.42}$$

The integral is absolutely convergent: the product support is a compact subset of $(\mathbb{R}^2)^N_{\neq}$, the kernel is bounded there, and each $\alpha_a \in L^2$ is integrable on its compact support. Call the difference of the two sides $D(\alpha_1, \dots, \alpha_N)$ on these tuples.

We show local vanishing near each distinct configuration X by induction on the number of equal-second-coordinate pairs. When there are none, $X \in \mathcal{O}_N$ and the ordered tube density gives the identity in small product discs. Otherwise take a pair with equal second coordinates and different times. Choose a time cut between them, different from all point times, and split the points into groups I, J below and above the cut. Move the J group by se_2. The points remain distinct for every real s, since the groups are separated in time. A sufficiently small positive s_0 removes every second-coordinate coincidence across the cut and creates none: take it smaller than every nonzero initial difference of second coordinates. There are fewer coincident pairs at $X(s_0)$.

The induction hypothesis gives a radius $2r$ near $X(s_0)$ on which D vanishes. Reduce r so that the original discs of radius r are disjoint and preserve the time gap. Fix tests in these original discs, and translate those in group J. The resulting difference $F(s)$ is real analytic on $\mathbb{R}$. For its moment term this follows from the separated-cluster strip argument above. For its density term, change variables to integrate $\widetilde{G}_N(Y + sv_J;\lambda)$ against the fixed product of tests, where $(v_J)_a = e_2$ for $a \in J$ and zero otherwise. Over any bounded real s-interval these configurations form a compact set of distinct points. Finite local holomorphic extensions of the real analytic kernel give a common radius $\epsilon > 0$ and a bound C there. Cauchy's estimates bound the integrated derivative of order q by $Cq!\epsilon^{-q}\prod_a \|\alpha_a\|_1$. Taylor's formula therefore gives real analyticity of this integral.

For $|s - s_0| < r$ the translated tests remain in the discs of radius $2r$ at $X(s_0)$, so $F(s) = 0$. The real analytic identity theorem gives $F(0) = 0$. This proves local vanishing near X and completes the induction. For general disjoint compact supports, finitely many of these product-disc neighbourhoods cover their product. Partition each support into sufficiently small half-open squares. Every product of nonzero pieces lies in one of the larger vanishing neighbourhoods: choose a point of the product support in it and make each square diameter smaller than the available half-radius. Multilinearity and finite summation prove (8.42).

8.11. **Euclidean covariance and all reflection planes.**

Theorem 8.6. *Let μ be a Gaussian-dominated law of closed tempered two-forms. Suppose it is invariant under translations and the cubic group, and reflection positive at the nine planes $x_j = 0$ and $x_j \pm x_i = 0$, $i < j$. Then it is invariant under all Euclidean motions and reflection positive at every plane, on the full L^2 space of either open side.*

Proof. Proposition 7.2 supplies the sharp coordinate cones. For the plane $(1, 2)$, fix t and define the bounded N-linear form

$$D(b_1, \dots, b_N) = \mathbb{E}_\mu \prod_a \mathbb{B}(R_t \cdot b_a) - \mathbb{E}_\mu \prod_a \mathbb{B}(b_a).$$

Its boundedness follows from Proposition 5.1. For plates over pairwise disjoint compact plane supports, (8.42) and (8.41) prove $D = 0$: the factor $e^{-iw(\lambda)t}$ from the plate action cancels the factor $e^{iw(\lambda)t}$ from the kernel. Density of finite sums of plates gives the same assertion for arbitrary L^2 fields over those compact sets. The form also vanishes on gradients, by closedness and the fact that rotations carry gradients to gradients. Proposition 8.4 therefore gives $D = 0$ on all L^2 entries, for every N.

The entire generating function of each finite real Schwartz test family is determined by these moments. Thus its characteristic function is unchanged by R_t, and Fourier uniqueness on the evaluation sigma algebra gives $(R_t)_*\mu = \mu$. Permuting coordinates applies the same argument in the other coordinate planes: the nine mirrors give their cones, and cubic invariance gives their quarter turns. These plane rotations generate $SO(3)$. A coordinate reflection already preserves μ, so the full $O(3)$ acts invariantly. Translation invariance gives the Euclidean group.

Let g carry $x_1 = 0$ to any chosen plane and its positive side to the chosen open side. The composition operator of g is unitary on $L^2(\mu)$, maps these side spaces onto each other, and conjugates the axis reflection to the chosen reflection. Transporting its nonnegative reflection form proves the final assertion. □

Theorem 8.6 applies separately to the component laws of Section 7.5. They retain Gaussian domination, closedness, the cubic group, and the nine mirror forms. Together with their all-direction clustering, it supplies the hypotheses for the tensor OS verification.

9. Chamber propagators

The block integration requires bounds for the chamber Dirac operator which are uniform in the mesh, the volume, and the running mass. There are two sources of coercivity. The internal chamber graph controls the part of a spinor whose mean over a cluster vanishes. The averaging penalty controls the remaining four components. The first of these bounds also identifies the singular part of the free propagator. This permits a local regularity estimate with constants independent of the mesh, which is needed when the inverses on small tori are combined into an inverse on the full torus.

9.1. **The cluster decomposition.** Let $a > 0$ be a mesh. A fermion site is written $x = (c, g)$, where $c \in a\mathbb{Z}^3$ and $g \in O_{\mathrm{cub}}$. In these coordinates the four neighbours are

$$x^{(i)} = (c, gs_i) \quad (0 \le i \le 2), \qquad x^{(3)} = (c + age_1, gr_{e_1}).$$

Here s_0 changes the sign of the third coordinate, s_1 interchanges the second and third coordinates, s_2 interchanges the first and second coordinates. The matrix r_{e_1} changes the sign of the first coordinate; it is the linear part of the affine wall reflection s_3 used in the regulator. The inward normals of the four walls are

$$n_0 = e_3, \qquad n_1 = \frac{e_2 - e_3}{\sqrt{2}}, \qquad n_2 = \frac{e_1 - e_2}{\sqrt{2}}, \qquad n_3 = -e_1.$$

Thus $\widehat{d}_i(c,g) = -gn_i$ and

$$M_i(c,g) = \frac{w_i}{2}(\rho - \gamma\cdot\widehat{d}_i(c,g)), \qquad (w_0,w_1,w_2,w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6).$$

Throughout this section $0 < \rho < 1$ is fixed.

The discussion applies on the covering lattice and on a torus of side $R \in a\mathbb{N}$, with any unitary character $f(c + Re_\mu, g) = -\exp(i\theta_\mu) f(c,g)$. Pairings on the torus are formed over a period cell. The notation $d(c,c')$ denotes the sup distance modulo $R\mathbb{Z}^3$, or the covering sup distance when $R = \infty$. They use the measure

$$\int \mathrm{d}x\, v(x) = \frac{a^3}{48}\sum_c \sum_g v(c,g), \qquad \|f\|_2^2 = \int \mathrm{d}x\, |f(x)|_2^2.$$

On four-component spinors depending only on c, the measure is $a^3\sum_c$. The cluster injection and its adjoint are

$$(J\xi)(c,g) = \xi(c), \qquad (J^*f)(c) = \frac{1}{48}\sum_g f(c,g).$$

Set $P = JJ^*$ and $Q = 1 - P$. The decomposition $f = J\xi + h$, with $\xi = J^*f$ and $h = Qf$, is orthogonal. In particular,

$$\|f\|_2^2 = \|\xi\|_2^2 + \|h\|_2^2. \tag{9.1}$$

The normalization by 48 makes J an isometry. Kernels on the chamber lattice are taken with respect to $\mathrm{d}x$, so the kernel of the identity is $48a^{-3}\mathbf{1}[x = x']1_4$.

Let A be a connection and write $u_A(c,c') = \exp(ie_a aA(c,c'))$ on a photon bond. The charge e_a in this notation is the charge at the level under consideration. The phases on the internal bonds are one. Put

$$R_i(A)f(x) = U_A(x, x^{(i)})f(x^{(i)}), \qquad \mathcal{L}_A = \sum_{i=0}^{3} w_i(1 - R_i(A)).$$

The massless chamber operator is

$$D_A f(x) = -a^{-1}\sum_{i=0}^{3} M_i(x)\big(U_A(x,x^{(i)})f(x^{(i)}) - f(x)\big). \tag{9.2}$$

At level k, $a = \eta_k$ and dilation to the unit lattice gives

$$D_A = \eta_k^{-1} D^{\mathrm{ch}}[B_k], \qquad B_k(c/\eta_k, c'/\eta_k) = e_k\eta_k A(c,c'). \tag{9.3}$$

The reference mass is added separately as $D_A + \mu_k$. For real A, the involutions $R_i(A)$ are unitary and self-adjoint, and the bond expansion gives

$$\mathrm{Re}(f, D_A f) = \frac{\rho}{2a}(f, \mathcal{L}_A f), \qquad D_A^* = \gamma_5 D_A \gamma_5, \qquad \gamma_5 = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}. \tag{9.4}$$

Here γ_5 is Hermitian, $\gamma_5^2 = 1$, and it anticommutes with each γ_μ. Each unoriented bond contributes $a^3 w_i |U_A(x,x^{(i)})f(x^{(i)}) - f(x)|_2^2/48$ to the energy.

On one cluster let

$$\mathcal{L}_{\mathrm{int}} = \sum_{i=0}^{2} w_i(1 - R_i), \qquad R_i v(g) = v(gs_i).$$

Its kernel consists of the constants: the three reflections s_0, s_1, s_2 generate O_{cub}. Denote its least positive eigenvalue by λ_{int} and set

$$g_{\mathrm{h}} = \frac{\rho\lambda_{\mathrm{int}}}{2}, \qquad c_* = \frac{\min(w_3, \lambda_{\mathrm{int}})}{18}, \qquad c_{\mathrm{lh}} = \sqrt{3}(1+\rho).$$

No numerical evaluation of λ_{int} is needed.

Proposition 9.1. *For real A and $\xi = J^*f$,*

$$(f, \mathcal{L}_A f) \geq \lambda_{\mathrm{int}} \|Qf\|_2^2, \qquad a^{-2}(f, \mathcal{L}_A f) \geq c_* \|\nabla_A \xi\|_2^2, \tag{9.5}$$

where $\nabla_{A,\mu}\xi(c) = a^{-1}(u_A(c, c+ae_\mu)\xi(c+ae_\mu) - \xi(c))$. The same inequalities hold on a finite set E of photon sites, with the energy restricted to bonds whose endpoints lie over E and the photon differences restricted to bonds in E. Moreover,

$$\|QD_A J\xi\|_2 \leq c_{\mathrm{lh}} \|\nabla_A \xi\|_2, \qquad |(\xi, J^* D_A Q f)| \leq c_{\mathrm{lh}} \|\nabla_A \xi\|_2 \|f\|_2. \tag{9.6}$$

On $\operatorname{ran} Q$*, for $t > -g_{\mathrm{h}}/a$,*

$$\|(QD_A Q + t)^{-1}\|_{2\to 2} \leq (t + g_{\mathrm{h}}/a)^{-1}. \tag{9.7}$$

All constants are independent of a, R, the character, and the real connection.

Proof. The internal bonds over c give $a^3(f(c,\cdot), \mathcal{L}_{\mathrm{int}} f(c,\cdot))$ and hence at least $\lambda_{\mathrm{int}} a^3 \|Qf(c,\cdot)\|^2$. Summing over $c \in E$ proves the first bound in (9.5).

For the second bound, consider a photon bond $c' = c + av$. The eight labels with $ge_1 = v$ correspond to the eight chamber bonds crossing it. Write $\mathcal{O}_v(c)$ for this octet. Since their endpoints form $\mathcal{O}_{-v}(c')$, the difference of the cluster means is the sum of the mean of the eight bond differences and two means of Qf:

$$\begin{aligned} u_A(c,c')\xi(c') - \xi(c) = {} & \frac{1}{8} \sum_{x \in \mathcal{O}_v(c)} \big(U_A(x, x^{(3)}) f(x^{(3)}) - f(x) \big) \\ & - \frac{u_A(c,c')}{8} \sum_{x' \in \mathcal{O}_{-v}(c')} Qf(x') + \frac{1}{8} \sum_{x \in \mathcal{O}_v(c)} Qf(x). \end{aligned}$$

The square is bounded by $3/8$ times the sum of the squared terms in these three octets. Distinct photon bonds use distinct chamber bonds; each octet of cluster remainders occurs at most once. Consequently, with $\mathcal{E}_3$ and $\mathcal{E}_{\mathrm{int}}$ denoting the two parts of the energy,

$$a^2 \|\nabla_A \xi\|_{2,E}^2 \leq 18 \big(w_3^{-1} \mathcal{E}_3 + \|Qf\|_{2,E}^2 \big) \leq 18 \big(w_3^{-1} \mathcal{E}_3 + \lambda_{\mathrm{int}}^{-1} \mathcal{E}_{\mathrm{int}} \big).$$

This proves the local bound and therefore the global one.

The internal hops vanish on $J\xi$. Thus

$$(D_A J\xi)(c,g) = -3(\rho - \gamma \cdot ge_1) \nabla_{A, ge_1} \xi(c).$$

Each of the six directions occurs eight times. The norm of the matrix in this formula is $3(1+\rho)$, and the two directed differences on a photon bond have equal norms. It follows that $\|D_A J\xi\|_2^2 \leq 3(1+\rho)^2 \|\nabla_A \xi\|_2^2$. Projection gives the first bound in (9.6); the second follows from (9.4), since J and Q commute with γ_5.

For $Qf = f$, (9.4) gives $\operatorname{Re}(f, (QD_A Q + t)f) \geq (t + g_{\mathrm{h}}/a)\|f\|_2^2$. The same lower bound holds for the adjoint. The operator has closed range and its adjoint has zero kernel, so it is onto, also on the infinite lattice. This proves (9.7). □

9.2. **The free symbol and its Schur complement.** Use lattice units for the moment. Let $\mathcal{D}_m(p)$ be the symbol of $D_0 + m$, $0 \leq m \leq 1$, on $V = \ell^2(O_{\mathrm{cub}}; \mathbb{C}^4)$ with measure $48^{-1} \sum_g$:

$$(\mathcal{D}_m(p)v)(g) = mv(g) + \sum_{i=0}^{2} M_i(g)(v(g) - v(gs_i)) + M_3(g)(v(g) - \exp(ip \cdot ge_1) v(gr_{e_1})).$$

Let $D_{\mathrm{ll}}, D_{\mathrm{lh}}, D_{\mathrm{hl}}, D_{\mathrm{hh}}$ be its blocks in $V = \operatorname{ran} J \oplus \operatorname{ran} Q$. The mass is included in the diagonal blocks. Set

$$H_m(p) = D_{\mathrm{hh}}(p)^{-1}, \qquad B_m(p) = D_{\mathrm{ll}}(p) - D_{\mathrm{lh}}(p) H_m(p) D_{\mathrm{hl}}(p), \qquad t(p) = \sum_{\mu} (1 - \cos p_\mu).$$

The compression of the full operator is the Wilson operator:

$$D_{\mathrm{ll}}(p) = m + i\gamma \cdot \sin p + \rho t(p).$$

Its Schur complement also contains the contribution from the heavy part. In particular, adding a mass to the chamber operator changes H_m as well as D_{ll}.

Proposition 9.2. *There are positive constants $\theta_0, c_{\mathrm{e}}, \kappa_{\mathrm{strip}}$, depending on ρ only, such that H_m and B_m are holomorphic in a neighbourhood of $\{|\operatorname{Im} p_\mu| \le \theta_0\}$ and $2\pi\mathbb{Z}^3$-periodic. On the real cube $[-\pi, \pi]^3$,*

$$\|H_m(p)\| \le (g_{\mathrm{h}} + m)^{-1}, \tag{9.8}$$

$$\|D_{\mathrm{lh}}(p)\| + \|D_{\mathrm{hl}}(p)\| \le 2c_{\mathrm{lh}}(2t(p))^{1/2}, \tag{9.9}$$

$$B_m(p) = m + i\gamma \cdot \sin p + E_m(p), \qquad \|E_m(p)\| \le K_\rho t(p), \tag{9.10}$$

$$\operatorname{Re}(v, B_m(p)v) \ge (m + \rho c_* t(p))|v|_2^2, \tag{9.11}$$

$$|B_m(p)v|_2 \ge c_{\mathrm{e}}(|p|_2^2 + m^2)^{1/2}|v|_2, \tag{9.12}$$

where $K_\rho = \rho + 2c_{\mathrm{lh}}^2/g_{\mathrm{h}}$. For every multi-index ν,

$$\|\partial_p^\nu B_m(p)^{-1}\| \le C_\nu(\rho)(|p|_2^2 + m^2)^{-(1+|\nu|)/2}. \tag{9.13}$$

If $m > 0$, the inverse exists also on $|\operatorname{Im} p_\mu| \le \kappa_{\mathrm{strip}} m$, where the same bound holds with $\operatorname{Re} p$ on the right, for $\operatorname{Re} p \in [-\pi, \pi]^3$. At $m = 0$ the real derivative bound is asserted away from $p = 0$.

Proof. Only the fourth hop depends on p. For $p = p' + ip''$,

$$\|\mathcal{D}_m(p) - \mathcal{D}_m(p')\| \le 3(1 + \rho)(\exp(|p''|_\infty) - 1).$$

Choose θ_0 so that the right side is $g_{\mathrm{h}}/2$ when $|p''|_\infty = \theta_0$. The heavy gap then gives $\operatorname{Re} D_{\mathrm{hh}}(p) \ge m + g_{\mathrm{h}}/2$ on this strip. Invertibility holds on a slightly larger strip, which proves holomorphy up to the closed strip. For real p, the off-diagonal bounds in Proposition 9.1 give $\|D_{\mathrm{lh}}\|, \|D_{\mathrm{hl}}\| \le c_{\mathrm{lh}}(2t(p))^{1/2}$. This proves (9.10).

To retain the real-part bound after eliminating the heavy part, put $u = Jv - H_m D_{\mathrm{hl}} v$. Then $J^* u = v$, $Q\mathcal{D}_m u = 0$, and $\mathcal{D}_m u = J B_m v$. Therefore

$$\operatorname{Re}(v, B_m v) = \operatorname{Re}(u, \mathcal{D}_m u) = m\|u\|^2 + \frac{\rho}{2}(u, \mathcal{L}_0(p)u) \ge (m + \rho c_* t(p))|v|_2^2.$$

This proves (9.11).

The ellipticity estimate uses the Dirac part near zero and the real part away from zero. Put $t_0 = (4K_\rho^2)^{-1}$. One has $K_\rho \ge 48/\lambda_{\mathrm{int}} > 1$, since $\lambda_{\mathrm{int}} \le 12 + 24\sqrt{2} < 48$. Hence $t_0 \le 1/4$. If $t(p) \le t_0$, then $|\sin p|_2^2 \ge t(p)$ and $\|E_m(p)\| \le \frac{1}{2} t(p)^{1/2}$. Since

$$|(m + i\gamma \cdot \sin p)v|_2^2 = (m^2 + |\sin p|_2^2)|v|_2^2,$$

it follows that $|B_m(p)v|_2 \ge \frac{1}{2}(m^2 + t(p))^{1/2}|v|_2$. On $[-\pi, \pi]^3$, $t(p) \ge 2|p|_2^2/\pi^2$. This proves (9.12) in this region. If $t(p) \ge t_0$, then (9.11) gives $|B_m(p)v|_2 \ge (m + \rho c_* t_0)|v|_2$. The inequality $(|p|_2^2 + m^2)^{1/2} \le m + \pi\sqrt{3}$ closes the estimate with, for example,

$$c_{\mathrm{e}} = \min\left(0.22, \frac{\rho c_* t_0}{\pi\sqrt{3}}\right).$$

The strip bounds give uniform bounds on the derivatives of B_m by Cauchy's formula. Differentiating $B_m B_m^{-1} = 1$ and inducting on $|\nu|$ proves (9.13). For the complex strip, let K_s bound B_m on the fixed strip. On half that strip, $\|\partial_\mu B_m\| \le 2K_s/\theta_0$. Choose

$$\kappa_{\mathrm{strip}} = \min\left(\frac{\theta_0}{2}, \frac{c_{\mathrm{e}}\theta_0}{12K_s}\right).$$

Integration along the segment from p' to $p' + ip''$ shows that the change in B_m is at most $c_{\mathrm{e}}m/2$. The lower bound is thus at least $c_{\mathrm{e}}(|p'|_2^2 + m^2)^{1/2}/2$. Differentiating the inverse identity once more gives the asserted complex derivative bounds. □

9.3. **Free kernels and coercivity.** Solving the heavy equation gives the inverse symbol

$$\mathcal{D}_m^{-1} = \begin{pmatrix} B_m^{-1} & -B_m^{-1} D_{\mathrm{lh}} H_m \\ -H_m D_{\mathrm{hl}} B_m^{-1} & H_m + H_m D_{\mathrm{hl}} B_m^{-1} D_{\mathrm{lh}} H_m \end{pmatrix}. \tag{9.14}$$

The first term is independent of the cluster labels after insertion by J and J^*. Every other term has a weaker singularity at $p = 0$: D_{lh} and D_{hl} vanish there, whereas H_m and all its derivatives are uniformly bounded.

Proposition 9.3. *On the unit covering lattice let*

$$G_m(c, g; c', g') = \int_{[-\pi,\pi]^3} \frac{\mathrm{d}p}{(2\pi)^3} \exp(ip \cdot (c - c'))[\mathcal{D}_m(p)^{-1}](g, g').$$

The bracket is the kernel on V relative to the label measure $48^{-1}\sum_g$, consistently with the chamber kernel convention. It is well-defined also at $m = 0$. It decomposes as $G_m(x, x') = G_m^{\mathrm{l}}(c - c') + G_m^{\mathrm{h}}(x, x')$, where

$$|G_m^{\mathrm{l}}(n)|_2 \le C_G(1 + |n|_\infty)^{-2}, \tag{9.15}$$

$$|G_m^{\mathrm{l}}(n + v) - G_m^{\mathrm{l}}(n)|_2 \le C_G(1 + |n|_\infty)^{-3}, \quad v \in \{\pm e_\mu\}, \tag{9.16}$$

$$|G_m^{\mathrm{h}}(x, x')|_2 \le C_G(1 + |c - c'|_\infty)^{-3}. \tag{9.17}$$

Consequently $|G_m(x, x')|_2 \le 2C_G(1 + |c - c'|_\infty)^{-2}$ and

$$|G_m(x^{(i)}, x') - G_m(x, x')|_2 \le 20C_G(1 + |c - c'|_\infty)^{-3}. \tag{9.18}$$

For $m > 0$ these bounds have the additional factor $\exp(-\kappa_{\mathrm{strip}} m|c - c'|_\infty)$. The constants are uniform for $0 \le m \le 1$.

Proof. The Fourier estimate used here can be proved directly. If a periodic symbol T satisfies $|\partial^\nu T(p)| \le A|p|_2^{-q-|\nu|}$, where $q < 3$ and $|\nu| \le N > 3 - q$, then its Fourier coefficients are bounded by $CA(1 + |n|_\infty)^{-(3-q)}$. For $n \ne 0$, put $r = |n|_\infty^{-1}$ and split the integral using a cutoff equal to one for $|p|_2 \le r$ and zero for $|p|_2 \ge 2r$. The inner integral is bounded by CAr^{3-q}. On the complement integrate N times in a coordinate with $|n_\mu| = |n|_\infty$. The differentiated integrand has integral at most CAr^{3-q-N}; the prefactor is r^N. Derivatives of the cutoff are supported in $r \le |p|_2 \le 2r$ and have the same bound. Boundary terms cancel by periodicity.

Apply this estimate with $(q, N) = (1, 3)$ to B_m^{-1}. Multiplication by $\exp(ip \cdot v) - 1$ lowers its singularity to order zero, for which $(q, N) = (0, 4)$ applies. In (9.14), the two off-diagonal terms have order zero and the term with two off-diagonal factors has order -1. The latter can also be bounded by the order-zero estimate. The remaining term H_m is holomorphic on the fixed strip, and its Fourier coefficients decay exponentially. The kernel normalization costs a factor 48 for operator-valued symbols on V; this is absorbed in C_G.

For $m > 0$, shift one coordinate of the contour by $i\kappa_{\mathrm{strip}} m \operatorname{sgn}(n_\mu)$, where $|n_\mu| = |n|_\infty$. The complex derivative bounds of Proposition 9.2 give the same Fourier estimates on the shifted

contour. This supplies the exponential factor. The fixed-strip bound for H_m is stronger than the required one.

An internal hop leaves c fixed, so its difference cancels G_m^{l}. An external hop replaces c by $c+ge_1$; use the light difference bound and the two heavy bounds at these points. The inequalities $1+|n+v|_\infty \geq (1+|n|_\infty)/2$ and $\exp(\kappa_{\mathrm{strip}} m) < \sqrt{3}$ give the factor 20 in (9.18). □

The Green kernel at mesh a, for $0 \leq \mu a \leq 1$, is $a^{-2}G_{\mu a}(x/a, x'/a)$. Thus

$$|G(x,x')|_2 \leq 2C_G(a+|c-c'|_\infty)^{-2}, \qquad |G(x^{(i)},x') - G(x,x')|_2 \leq 20 C_G a (a+|c-c'|_\infty)^{-3}. \tag{9.19}$$

Fourier multiplication gives both inverse identities on finitely supported functions. The factor 48 in the identity kernel is consistent with the measure $\mathrm{d}x$ above.

For distinct sites put $d_X(x,\widetilde{x}) = \max(|c-\widetilde{c}|_\infty, a)$. The chamber graph has paths from x to $\widetilde{x}$ of at most $39 d_X(x,\widetilde{x})/a$ bonds, with photon sites in the coordinate box spanned by $c, \widetilde{c}$. Indeed, the generators s_0, s_1, s_2 are the simple reflections of the signed permutation group, with nine positive roots. Their Coxeter length is at most nine [6, Theorem 22.14], so any two labels in one cluster are joined by at most nine internal bonds. A monotone photon path has at most $3|c-\widetilde{c}|_\infty/a$ steps. Before each step move to a label with the required external direction, and after the last step move to the final label. This uses at most $10 \cdot 3|c-\widetilde{c}|_\infty/a + 9 \leq 39 d_X/a$ bonds. If d_X is less than half the distance to c', sum (9.19) along such a path. In the other case use the two size bounds. The resulting interpolation gives, for $0 \leq \alpha \leq 1$,

$$|G(x,x') - G(\widetilde{x},x')|_2 \leq 6240 C_G d_X(x,\widetilde{x})^\alpha \left((a+|c-c'|_\infty)^{-2-\alpha} + (a+|\widetilde{c}-c'|_\infty)^{-2-\alpha} \right). \tag{9.20}$$

The following coercivity estimate is separate from the real-part bound. It is needed to recover mesh-independent control of the block remainder.

Proposition 9.4. *There is $c_b = c_b(\rho) > 0$ such that*

$$\|D_0 f\|_2^2 \geq c_b a^{-2} (f, \mathcal{L}_0 f). \tag{9.21}$$

For a cube E of side ℓ_E, let P_E be the mean over $E \times O_{\mathrm{cub}}$. Then

$$\|f - P_E f\|_{2,E}^2 \leq C_P \ell_E^2 a^{-2} (f, \mathcal{L}_0 f)_E, \qquad C_P = \frac{1}{8c_*} + \frac{1}{\lambda_{\mathrm{int}}}. \tag{9.22}$$

Proof. The Schur estimate at zero mass implies $|B_0(p)v| \geq c_{\mathrm{l}} t(p)^{1/2} |v|$, with $c_{\mathrm{l}} = \rho c_*/(2K_\rho)$: use the Dirac part for $t(p) \leq t_0$ and the real part for $t(p) \geq t_0$. Write $u = Jv + h$ in a fibre and $f = \mathcal{D}_0(p)u$. The heavy equation and then the light equation give

$$h = H_0(f_{\mathrm{h}} - D_{\mathrm{hl}} v), \qquad B_0 v = f_{\mathrm{l}} - D_{\mathrm{lh}} H_0 f_{\mathrm{h}}.$$

Since $t(p) \leq 6$, these imply $(2t(p))^{1/2}|v| \leq A_1 \|f\|$ and $\|h\| \leq A_2 \|f\|$, where

$$A_1 = \frac{\sqrt{2}}{c_{\mathrm{l}}} \left(1 + \frac{6(1+\rho)}{g_{\mathrm{h}}} \right), \qquad A_2 = g_{\mathrm{h}}^{-1}(1 + c_{\mathrm{lh}} A_1).$$

The bond decomposition also gives

$$(u, \mathcal{L}_0(p) u) \leq 6 t(p) |v|^2 + (30 + 24\sqrt{2}) \|h\|^2.$$

For the internal bonds the light part cancels, leaving at most $2(w_0+w_1+w_2)\|h\|^2$. In an external bond difference, split off the photon difference of v and the two heavy endpoint values, and bound the square by three times their squared norms. The octet count then gives $6t(p)|v|^2 + 3w_3\|h\|^2$. This proves the displayed upper bound. This yields (9.21) in each fibre with $c_b = (3A_1^2 + (30 + 24\sqrt{2})A_2^2)^{-1}$. Fourier decomposition gives the assertion for all characters on the torus and on the covering lattice, restoring the factor a^{-2}.

For (9.22), first apply the scalar lattice Poincaré bound $\|\xi - P_E\xi\|_{2,E}^2 \leq \ell_E^2 \|\nabla_0 \xi\|_{2,E}^2/8$. In one dimension this follows by writing each difference as a sum of nearest-neighbour differences;

the coefficient of a bond at position j is $j(n-j)/2 \le n^2/8$. In three dimensions use the three commuting line averages, whose successive remainders are orthogonal. Now $f - P_E f = J(\xi - P_E\xi) + Qf$. Its two parts are orthogonal, and the local bounds (9.5) give (9.22). The heavy term uses $\ell_E \ge a$. □

9.4. **Averaging penalties and the inverse at zero field.** Fix an odd $L \ge 5$ and put $a = \eta_k = L^{-k}$. Write e_k for the charge e_a at this mesh. The level data k, μ, R, e_k are independent, with $0 < e_k \le e_0$; the flow will specify their relation later. Unit blocks have side one, and the next blocks have side L. In this and the following subsections a finite period R is a positive integer multiple of L. For $z \in \mathbb{Z}^3$ the unit block is $\Delta_z = \{c \in a\mathbb{Z}^3 : |c - z|_\infty < 1/2\}$. For $z' \in L\mathbb{Z}^3$ the L-block is $E_{z'} = \{c \in a\mathbb{Z}^3 : |c - z'|_\infty < L/2\}$. These partitions are taken modulo $R\mathbb{Z}^3$ on the torus. Since L is odd, no fine site lies on a block boundary. Their means, including the cluster mean, are denoted Π_1 and Π_L. At nonzero connection the means use the gauge phases specified below. Let

$$b = 1, \qquad b_L = b/L, \qquad b_- = b(1 - L^{-1}), \qquad b_k = \frac{b(1 - L^{-1})}{1 - L^{-k}} \quad (k \ge 1).$$

For $k \ge 1$, $y \in \mathbb{R}$, define

$$\alpha_y = b_k i\gamma_5 y(b_k + i\gamma_5 y)^{-1}, \qquad \beta_y = b_k^2 b_L (b_k + b_L + i\gamma_5 y)^{-1}(b_k + i\gamma_5 y)^{-1},$$

and set $\alpha_\infty = b_k$, $\beta_\infty = 0$. The blocked operators at zero connection are

$$F_y(0) = D_0 + \mu + \alpha_y \Pi_1 + \beta_y \Pi_L \quad (k \ge 1), \qquad F_y(0) = D_0 + \mu + b_L \Pi_L + i\gamma_5 y \quad (k = 0). \tag{9.23}$$

Here $0 \le \mu \le 1$. In the first formula $y \in \mathbb{R} \cup \{\infty\}$; in the second $y \in \mathbb{R}$. We use $-\infty = \infty$ when an adjoint identity reverses this parameter. The notation F_y denotes an operator in this section, not the curvature.

The penalty bounds follow by diagonalizing γ_5. On its eigenspace of sign ϵ, the two eigenvalues and their sum are

$$\frac{b_k i\epsilon y}{b_k + i\epsilon y}, \quad \frac{b_k^2 b_L}{(b_k + b_L + i\epsilon y)(b_k + i\epsilon y)}, \quad \frac{b_k(b_L + i\epsilon y)}{b_k + b_L + i\epsilon y}.$$

For the first eigenvalue v_1 and the sum v_2, direct calculation gives

$$|v_1|^2 = b_k \operatorname{Re} v_1, \qquad |v_2|^2 \le b_k \operatorname{Re} v_2, \qquad \operatorname{Re} v_2 \ge \frac{b_k b_L}{b_k + b_L} \ge \frac{b_-}{L}.$$

Also $|\alpha_y| \le b_k \le b$ and $|\beta_y| \le b_L$. On an L-block the decomposition $\mathrm{Pen} = \alpha_y(\Pi_1 - \Pi_L) + (\alpha_y + \beta_y)\Pi_L$ has orthogonal ranges. The scalar inequalities on the two spin eigenspaces therefore give

$$\operatorname{Re}(f, \mathrm{Pen}\, f)_E \ge \frac{b_-}{L}\|P_E f\|_{2,E}^2, \qquad \|\mathrm{Pen}\, f\|_{2,E}^2 \le b \operatorname{Re}(f, \mathrm{Pen}\, f)_E. \tag{9.24}$$

The kernel on E is $\alpha_y \mathbf{1}[c, c' \text{ in the same unit block}] + \beta_y L^{-3}$, and its norm is at most $2b$.

Proposition 9.5. *There are $C_I, C_5, \gamma_5' > 0$, depending on L, ρ and a fixed $\alpha \in (0,1)$ only, such that $F_y(0)$ is invertible and*

$$\|F_y(0)^{-1}\|_{2\to 2} \le LC_I \qquad (k \ge 1).$$

If f is supported over one unit block $\Delta_{z'}$ and $v = F_y(0)^{-1} f$, then for x over Δ_z and $0 < d_X(x, \widetilde{x}) \le 1$,

$$|v(x)|_2 + d_X(x, \widetilde{x})^{-\alpha}|v(x) - v(\widetilde{x})|_2 \le C_5 \exp(-\gamma_5' d(z, z'))\|f\|_\infty. \tag{9.25}$$

This includes pairs in one cluster. The constants are uniform in k, R, the character, μ, and y.

Proof. First suppose $k \geq 1$. Put $f = F_y(0)u$ and $X = \mathrm{Re}(u, f)$. The real-part identity and (9.24) imply

$$\sum_E \|P_E u\|_{2,E}^2 \leq \frac{L}{b_-} X, \qquad \|\operatorname{Pen} u\|_2^2 \leq bX, \qquad \|\mu u\|_2^2 \leq X.$$

Therefore $\|D_0 u\|_2 \leq \|f\|_2 + (1+\sqrt{b})X^{1/2}$. Use (9.21) for $D_0 u$ and (9.22) on each L-block. With $K_1 = \sqrt{L} + \sqrt{Lb}$ this gives

$$\|u\|_2^2 \leq L\left(\frac{1}{b_-} + \frac{2C_P K_1^2}{c_b}\right)\|u\|_2\|f\|_2 + \frac{2C_P}{c_b} L^2 \|f\|_2^2.$$

Hence $\|u\|_2 \leq LC_I\|f\|_2$, where $C_I = b_-^{-1} + 2C_P K_1^2/c_b + (2C_P/c_b)^{1/2}$. The identity $F_y(0)^* = \gamma_5 F_{-y}(0)\gamma_5$ gives the same lower bound for the adjoint. Thus $F_y(0)$ has closed, dense range and is onto, including on the infinite lattice.

To obtain decay, let ψ be a bounded real periodic function with $|\psi(c) - \psi(c')| \leq a_0|c - c'|_1/L$. The mass commutes with this weight. The external hopping term gives $\|\exp(\psi)[D_0, \exp(-\psi)]\|_{2\to 2} \leq 3(1+\rho)\exp(1)a_0/L$. The penalty kernel is supported in one L-block, has norm at most $2b$, and $|\psi(c) - \psi(c')| \leq 3a_0$ there. Its contribution is at most $6L^4 b \exp(1)a_0/L$. Put

$$c_w = \exp(1)(3(1+\rho) + 6L^4 b), \qquad a_0 = \min(1/3, (2C_I c_w)^{-1}).$$

Applying the L^2 inverse bound and absorbing the commutator gives

$$\|\exp(\psi)F_y(0)^{-1} f\|_2 \leq 2LC_I \|\exp(\psi) f\|_2. \tag{9.26}$$

Take truncated multiples of the ℓ^1 distance, divided by L, to the source L-block. Letting the truncation increase proves an exponentially weighted L^2 bound.

The passage from L^2 to the sup norm uses the free kernel. On the covering lattice choose $c^\circ \in a\mathbb{Z}^3$ and $r \geq 8a$, and put

$$B = \{|c - c^\circ|_\infty < r\}, \quad B^+ = \{|c - c^\circ|_\infty < r + a\}, \quad B' = \{|c - c^\circ|_\infty \leq r/4\}.$$

Suppose u is defined over B^+ and $(D_0 + \mu)u = h$ over B, with $0 \leq \mu a \leq 1$. Both evaluation points below lie over B', and C depends on the fixed $\alpha \in (0,1)$ and ρ. Let χ equal one on the concentric half cube and zero outside the cube, with Lipschitz constant $2/r$. Then

$$\chi u = G(\chi h) + G([D_0, \chi]u).$$

The commutator contains only the external hop. Its rows are supported where $|c - c^\circ|_\infty > r/2 - a$. It satisfies $\|[D_0, \chi]u\|_1 \leq 45 r^{1/2}\|u\|_{2,B^+}$. For a point of the quarter cube, this support is at distance at least $r/4 - a \geq r/8$. The bounds (9.19) and (9.20) consequently give

$$\begin{aligned} |u(x)|_2 &\leq C\big(r\|h\|_{\infty,B} + r^{-3/2}\|u\|_{2,B^+}\big), \\ |u(x) - u(\widetilde{x})|_2 &\leq C\left(\frac{d_X(x,\widetilde{x})}{r}\right)^{\alpha}\big(r\|h\|_{\infty,B} + r^{-3/2}\|u\|_{2,B^+}\big). \end{aligned} \tag{9.27}$$

For the first term, sum $(a + |c - c'|_\infty)^{-2}$ over the cube; the result is $O(r)$. For the difference, the analogous sum with exponent $2+\alpha$ is $O(r^{1-\alpha}/(1-\alpha))$. The cluster average in $\mathrm{d}x'$ contributes one to both sums. For the commutator term, its support separation gives the size bound; summing bond differences along the chamber path gives the difference bound. These estimates explain why the constants do not grow when $a \downarrow 0$.

For $u = F_y(0)^{-1} f$, use $h = f - \operatorname{Pen} u$ in (9.27), with $r = L/4$. The enlarged cube meets at most eight L-blocks. On each, the penalty kernel and Cauchy–Schwarz give $\|\operatorname{Pen} u\|_\infty \leq 2bL^{3/2}\|u\|_{2,E}$. The weighted bound (9.26) therefore bounds both terms in (9.27) by $CL\exp(-a_0 d_1(x, E'))\|f\|_\infty$, where $d_1(x, E')$ is the ℓ^1 distance to the source L-block divided by L. This applies for $k \geq 2$. For $k = 1$, the pointwise L^2 bound costs only $\sqrt{48}L^{3/2}$; L is fixed. Larger Hölder separations are

controlled by the two sup bounds. Finally, $d_1(x, E') \geq d(z, z')/L - 3$, and the source unit block lies in E'. This proves (9.25) with $\gamma_5' = a_0/L$. The argument uses only periodic moduli and so applies to every unitary character. □

The first level has a different resolvent parameter. On an L-block the real-part bound and Poincaré inequality imply

$$\mathrm{Re}(u, F_y(0)u) \geq \frac{c_r}{L}\|u\|_2^2, \qquad c_r = \min(b_-, \rho/(2C_P L)).$$

The term $i\gamma_5 y$ has zero real part. Conjugation by a weight with Lipschitz constant a_z changes the hopping and penalty terms by at most $\exp(1)(6+3b)a_z$. Choose

$$a_z \leq \min\big((3L)^{-1}, c_r/(2\exp(1)L(6+3b))\big).$$

The weighted inverse then has L^2 norm at most $2L/c_r$. On the other hand its part other than $i\gamma_5 y$ has norm at most $60 + b$. For $|y| \geq 2(60+b)$, the inverse norm is at most $2/|y|$. Combining the two ranges and converting to a kernel, with its factor 48, gives

$$|F_y(0)^{-1}(x, x')|_2 \leq C_{00}(1+|y|)^{-1}\exp(-a_z d(c, c')). \tag{9.28}$$

This factor in y is needed when the determinant is represented by an integral of resolvents.

9.5. **Small connections and local inverses.** The covariant block means are defined by the same nested blocks. Let $\tau A(c, c')$ be the arithmetic average of the oriented bond sums over the six coordinate-order paths from c to c', and let $c = c_0, c_1, \ldots, c_k = z$ be the centres of the nested blocks containing c. Put

$$\varphi_A(z, c) = \exp\left(ie_k\eta_k \sum_{j<k} \tau A(c_{j+1}, c_j)\right).$$

The unit-block average on four-component spinors and its inverse injection are

$$(Q_k(A)\xi)(z) = \eta_k^3 \sum_{c\in\Delta_z} \varphi_A(z, c)\xi(c), \qquad (Q_k^T(-A)v)(c) = \varphi_A(z, c)^{-1}v(z).$$

For the next average set, on the unit lattice,

$$(Q(A)v)(z') = L^{-3} \sum_{|z-z'|_\infty < L/2} U_A(z', z)v(z), \qquad (Q^T(-A)v')(z) = U_A(z', z)^{-1}v'(z').$$

On the chamber space, for $k \geq 1$, put

$$\Pi_1(A) = JQ_k^T(-A)Q_k(A)J^*, \qquad \Pi_L(A) = JQ_k^T(-A)Q^T(-A)Q(A)Q_k(A)J^*.$$

At $k = 0$ only $\Pi_L(A) = JQ^T(-A)Q(A)J^*$ is used. This defines $F_y(A)$ by (9.23). The formulas permit complex connections; for real A the idempotents are orthogonal projections.

For a pair of sites define

$$U_A(x, x') = \exp(ie_k\eta_k\tau A(c, c')), \qquad (\delta_{\alpha,A}v)(x, x') = \frac{U_A(x, x')v(x') - v(x)}{d_X(x, x')^\alpha}.$$

Only pairs with $0 < d_X < 1$ enter the local norms. If $c = c'$ then $U_A = 1$, so these quotients include variations between the labels in one cluster. There are no such short pairs when $k = 0$.

For an operator on the torus define

$$N_\gamma(T) = \sup_z \sum_{z'} \exp(\gamma d(z, z'))\|\mathbf{1}_{\Delta_z} T \mathbf{1}_{\Delta_{z'}}\|_{\infty\to\infty}.$$

For operators with values on pairs, the left indicator restricts the first point. Inserting the block partition between two factors and using the triangle inequality for d proves $N_\gamma(TT') \leq$

$N_\gamma(T)N_\gamma(T')$. Write $w_k(y) = 1$ for $k \geq 1$ and $w_0(y) = (1+|y|)^{-1}$. Enlarge C_5, C_{00} to at least one, and choose

$$\gamma = \tfrac{1}{2}\min(\gamma_5', a_z), \qquad a' = \max(C_5, C_{00}) \sum_{n\in\mathbb{Z}^3} \exp(-\gamma|n|_\infty).$$

The preceding estimates imply

$$N_\gamma(F_y(0)^{-1}) \leq a' w_k(y), \qquad N_\gamma(\delta_{\alpha,0} F_y(0)^{-1}) \leq a'. \tag{9.29}$$

At $k = 0$ the analogous bound holds in the kernel row norm N_γ^{ker}, obtained by replacing the localized operator norm with the supremum of $|T(x,x')|_2$ over the two clusters.

Proposition 9.6. *Let B be a complex connection with $e_k\|B\|_\infty \leq \delta$, where $3L\delta \leq 1$ and $2C_V a'\delta \leq 1$, with $C_V = 2\exp(1)(42+6b)$. Then $F_y(B)$ is invertible and*

$$N_\gamma(F_y(B)^{-1}) \leq 2a' w_k(y), \qquad N_\gamma(\delta_{\alpha,B} F_y(B)^{-1}) \leq 18a'. \tag{9.30}$$

At $k = 0$ the first bound holds also in N_γ^{ker}.

Proof. Write $V = F_y(B) - F_y(0)$. The internal hops, the mass, and the resolvent term at level zero cancel. For the external hop, $|u_B - 1| \leq \eta_k\delta\exp(\eta_k\delta)$; its weighted row sum is at most $42\exp(\gamma)\exp(\delta)\delta$. The two averaging penalties contribute respectively $3b\delta\exp(3\delta)$ and $3b\delta\exp(\gamma(L-1))\exp(3L\delta)$. Indeed, the nested paths have total length at most $3/2$, and the path to an L-block centre has length at most $3L/2$. Summing their normalized block kernels removes the numbers of sites. The bounds $\gamma L \leq \log 2$ and $3L\delta \leq 1$ give $N_\gamma(V) \leq C_V\delta$.

Thus $N_\gamma(VF_y(0)^{-1}) \leq 1/2$. The Neumann series gives

$$F_y(B)^{-1} = F_y(0)^{-1}\sum_{n\geq 0}(-VF_y(0)^{-1})^n = \sum_{n\geq 0}(-F_y(0)^{-1}V)^n F_y(0)^{-1}.$$

It proves the first bound in (9.30). The first form gives $N_\gamma(\delta_{\alpha,0}F_y(B)^{-1}) \leq 2a'$. For distinct photon sites in a short pair, $d_X^{-\alpha}|U_B - 1| \leq 3\delta\exp(3\delta) \leq 1/4$; for intracluster pairs the expression is zero. Evaluation at the second point of a short pair uses at most 27 neighbouring blocks, and costs at most $27\exp(\gamma)$. Its contribution to the covariant quotient is at most $16a'$, proving the second bound. At level zero, the second form of the Neumann series keeps the kernel norm in the last factor, so the kernel estimate follows as well. □

Fix a positive exponent ε; its choice in the flow will be made with the other renormalization parameters. The connection domain used by the RG is the following. Write $\ell = R$ for the side, and assume $\ell/(LM)$ is odd and at least 13. Here M is a power of L with $M \geq L^2$. The M-cubes have centres $z_\square \in M\mathbb{Z}^3/\ell\mathbb{Z}^3$ and are the periodic images of $\{c : |c - z_\square|_\infty < M/2\}$. The differential of a scalar in a connection is normalized by $(d\lambda)(c,c') = \eta_k^{-1}(\lambda(c') - \lambda(c))$. A connection belongs to $\widetilde{\mathcal{G}}_k$ if $A = A_0 + A_1$, where A_0 is real, $\|A_1\|_\infty < e_k^{-1+\varepsilon}$, and on every $11M$ enlargement of an M-cube, $A_0 = B_0 + d\lambda$, with B_0, λ real and $\|B_0\|_\infty < e_k^{-1+\varepsilon}$. Consequently, on the $3M$ enlargement $W_\square$,

$$A = B + d\lambda, \qquad e_k\|B\|_\infty < 2e_k^\varepsilon, \qquad e_k\|\operatorname{Im} A\|_\infty < e_k^\varepsilon. \tag{9.31}$$

This follows by restricting $B_0 + A_1$ and λ to that cube. The domain is open: a sufficiently small complex perturbation can be added to A_1. It is invariant under real gauge transformations.

For a lift $z_\square$ of the cube centre put $W_\square = \{|c - z_\square|_\infty < 3M/2\}$ and $W_\square^- = \{|c - z_\square|_\infty < 3M/2 - \eta_k\}$. Embed $W_\square$ in the positive chamber of the periodic torus of side $6M$, whose lattice is translated by $\eta_k(1,1,1)/2$. Its level-j block centres are $L^j\eta_k(1,1,1)/2 + L^j\eta_k\mathbb{Z}^3$ for $0 \leq j \leq k+1$. Translation by $L(1,1,1)/2$ identifies these blocks with the unshifted nested blocks, because L^{k+1-j} is odd. The zero-field and small-connection estimates therefore apply on this torus with the same

constants. Fold its photon coordinates across the planes at multiples of $3M$. Adjacent sites fold either to adjacent sites in $W_\square$ or to one site; in the latter case assign phase one. Pull back A by this fold to obtain A^W. The fold preserves the supremum norm and commutes with d on scalar functions. It also preserves the nested blocks, since $3M/(L^j\eta_k)$ is odd for $j \le k+1$.

Let ext extend a function over $W_\square$ by zero in the seven other chambers, and let res restrict to its embedded copy. Labels are unchanged in these two maps. Define

$$S_\square(A) = \operatorname{res} F_y(A^W)^{-1}\operatorname{ext}.$$

For $A = B + d\lambda$ as in (9.31), $A^W = B^W + d\lambda^W$, with λ^W real. Gauge conjugation by $\exp(ie_k\lambda^W)$ reduces the inverse to the small connection B^W. If $6Le_0^\varepsilon \le 1$ and $4C_V a' e_0^\varepsilon \le 1$, Proposition 9.6 applies to every $0 < e_k \le e_0$. Restriction, extension, and unitary gauge factors do not increase the localized norms. Thus

$$N_\gamma(S_\square(A)) \le 2a' w_k(y), \qquad N_\gamma(\delta_{\alpha,A} S_\square(A)) \le 18a'. \tag{9.32}$$

The second operator here is formed on the local cube before extension to the original torus. It does not differentiate a discontinuous extension by zero.

The local inverse has both identities needed for patching. If f is defined over $W_\square$, then

$$F_y(A) S_\square(A) f = f \quad \text{over } W_\square^-. \tag{9.33}$$

If f is supported over $W_\square^-$, then $F_y(A)f$ is supported over $W_\square$ and

$$S_\square(A) F_y(A) f = f. \tag{9.34}$$

For (9.33), the rows at an interior site use only its internal neighbours, its external neighbour, and its L-block. These lie in $W_\square$, where the original and folded operators agree. For (9.34), the support margin ensures that applying F_y cannot leave $W_\square$. After embedding, applying the folded operator to the extension of f agrees with extending $F_y(A)f$. Apply the inverse and restrict. For an antiperiodic spinor on the original torus, use one lift and then its signed translates; the enlarged cubes do not overlap their translates. These arguments also show that $S_\square$ reads A only in $W_\square$ and is analytic in those connection variables.

9.6. **The parametrix and the walk expansion.** Choose the product cutoff $h_\square$ with centre $z_\square$ and one-dimensional factor

$$h_0(t) = \begin{cases} 1, & |t| \le 1/4, \\ \cos(\pi(2|t| - 1/2)/2), & 1/4 \le |t| \le 3/4, \\ 0, & |t| \ge 3/4. \end{cases}$$

On a lift of the cube set

$$h_\square(c) = \prod_{\mu=1}^{3} h_0\big((c_\mu - z_{\square,\mu})/M\big), \qquad h_\square(c,g) = h_\square(c).$$

Extend this with period R and set it to zero away from the periodic images of the lifted cube neighbourhood. The cutoffs satisfy $\sum_\square h_\square^2 = 1$, $\operatorname{supp} h_\square \subset \{|c - z_\square|_\infty < 3M/4\}$, and $|h_\square(c) - h_\square(c')| \le \pi|c - c'|_1/M$. Put $C_\square = -[F_y(A), h_\square]$. Internal hops commute with $h_\square$, and the penalty has range at most L. The phase bounds and the same normalized sums used in the perturbation estimate give

$$N_\gamma(C_\square) \le \pi C_V/M. \tag{9.35}$$

The kernel is $(h_\square(c) - h_\square(c'))F_y(A)(x,x')$. Its two photon sites lie within $3M/4 + L$ of $z_\square$. If $M > 4L$, both lie in $W_\square^-$; multiplication by $h_{\square'}$ can give a nonzero operator only when $d(z_\square, z_{\square'}) \le M$.

Using (9.33) and (9.34) gives

$$F_y h_\square S_\square h_\square = h_\square^2 - C_\square S_\square h_\square, \tag{9.36}$$

$$h_\square S_\square h_\square F_y = h_\square^2 + h_\square S_\square C_\square. \tag{9.37}$$

Let $S^{\mathrm{p}} = \sum_\square h_\square S_\square h_\square$ and $R^{\mathrm{p}} = \sum_\square C_\square S_\square h_\square$. At most 27 cubes contribute to one row of R^{p}. Consequently,

$$N_\gamma(R^{\mathrm{p}}) \le q_0 := \frac{54\pi C_V a'}{M}.$$

Choose $M \ge 108\pi C_V a'$. Equation (9.36) now gives a right inverse by a convergent series:

$$S_y(A) = F_y(A)^{-1} = S^{\mathrm{p}} \sum_{n\ge 0} (R^{\mathrm{p}})^n. \tag{9.38}$$

The finite-dimensional operator $F_y(A)$ is therefore invertible. The left identity gives the second form with $R^{\mathrm{p}\prime} = -\sum_\square h_\square S_\square C_\square$.

The terms in (9.38) have a local description. A walk of length n is a sequence $\omega = (\square_0, \dots, \square_n)$ with successive centres at distance at most M. Its term is

$$S_\omega = (h_0 S_0 h_0)(C_1 S_1 h_1) \cdots (C_n S_n h_n),$$

where the subscript i denotes the cube $\square_i$. Other sequences give zero because of the support of the commutator. Its connection variables lie in $\bigcup_{i=0}^n W_{\square_i}$.

Proposition 9.7. *For $p \ge 1$ with $q(p) = 54\pi C_V a' p/M \le 1/2$,*

$$\sup_z \sum_{z'} \exp(\gamma d(z,z')) \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} S_\omega \mathbf{1}_{\Delta_{z'}}\| \le 16 a' p w_k(y) q(p)^n, \tag{9.39}$$

$$\sup_z \sum_{z'} \exp(\gamma d(z,z')) \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} \delta_{\alpha,A} S_\omega \mathbf{1}_{\Delta_{z'}}\| \le 152 a' p q(p)^n. \tag{9.40}$$

The norms are sup operator norms. At $k = 0$ the first bound holds also with the supremum of the normalized kernel over the two clusters. In particular, $S_y(A)$ is analytic on $\widetilde{\mathcal{G}}_k$ and

$$N_\gamma(S_y(A)) \le 32 a' w_k(y), \qquad N_\gamma(\delta_{\alpha,A} S_y(A)) \le 304 a'. \tag{9.41}$$

These constants are independent of the level, the side, the running mass in $[0,1]$, and the connection in the stated domain.

Proof. The first factor in a walk has norm at most $2a' w_k(y)$. Every additional factor has norm at most $2\pi C_V a'/M$ by (9.32) and (9.35). For a fixed output block, at most eight initial cutoffs contribute; there are at most 27 choices at each further step. Apply submultiplicativity row by row and sum these $8 \cdot 27^n$ bounds. This proves (9.39).

For the covariant quotient, only the factor at the left end of the product is differentiated. The product identity is

$$\delta_{\alpha,A}(h_\square v)(x,x') = h_\square(c) \delta_{\alpha,A} v(x,x') + d_X^{-\alpha}(h_\square(c') - h_\square(c)) U_A(x,x') v(x').$$

The first contribution has localized norm at most $18a'$. The second has coefficient at most $3\pi \exp(1)/M$, and evaluation at the second point costs at most $27\exp(\gamma)$. With the stated choice of M this contribution is at most a'. Thus $N_\gamma(\delta_{\alpha,A} h_\square S_\square) \le 19a'$. For intracluster pairs the second term vanishes, since $c = c'$. There are again at most eight possible initial cubes, including when the second point of a short pair enters the cutoff support. The count $8 \cdot 27^n$ now gives (9.40).

At level zero keep the kernel norm in the rightmost local inverse; the other factors are controlled in operator norm. The same submultiplicative row estimate proves the kernel assertion. Summing the geometric bounds with $p = 1$ gives (9.41). For a fixed torus the sum of the operator norms over all walks is finite; its tail of lengths at least j is bounded by $4a'(\ell/M)^3 2^{-j}$. Each walk term

is analytic in the finitely many connection variables it reads. The uniform tail bound therefore proves analyticity of the sum. The local bounds in (9.39) contain no factor depending on ℓ. □

The walk expansion provides both the inverse and the localization needed by the block integration. Its geometric decay survives summing over the intermediate cubes. The covariant Hölder estimate treats the short pairs within one chamber cluster on the same footing as pairs of distinct photon sites. These are the two bounds used to control the chamber fields and the local fermion activities.

10. Photon block integration

The photon block transformation uses a Landau representative for the background and an axial representative for the fluctuation. The background minimizer must have uniform local derivative bounds. The fluctuation form must be uniformly positive on the axial slice, and its inverse square root must decay between the local fluctuation coordinates. These estimates also permit analytic weakening parameters to be inserted without losing locality or the field domains of the renormalization step.

10.1. **Coarse averages and the Landau minimizer.** Let $L \geq 5$ be odd, $\eta = \eta_k = L^{-k}$, and let ℓ be a positive integer divisible by L. The photon fields are periodic cochains on $T_k = \eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with the pairing $\eta^3 \sum_c$ on each component. Here the differences are normalized by the mesh:

$$\partial_\mu f(x) = \eta^{-1}(f(x+\eta e_\mu) - f(x)), \qquad (d\lambda)_\mu = \partial_\mu \lambda, \qquad (d\mathscr{A})_{\mu\nu} = \partial_\mu \mathscr{A}_\nu - \partial_\nu \mathscr{A}_\mu.$$

We put $\delta = d^T$ and $\Delta_{\mathrm{sc}} = \delta d$ on scalars. Fix $0 < \alpha < 1$. The Hölder quotient $\delta_\alpha \partial$ uses pairs of lattice sites at distance at most one, as in (10.16). The unit lattice is $T_k^0 = \mathbb{Z}^3/\ell\mathbb{Z}^3$ and the unit blocks are $\Delta_z = \{c : |c - z|_\infty < 1/2\}$. Set $N = L^k$. The cochain averages are denoted by $\mathbf{Q}_k$; on scalars, bonds, and plaquettes they are respectively

$$\begin{aligned}
(\mathbf{Q}_k \lambda)(z) &= \eta^3 \sum_{c \in \Delta_z} \lambda(c), \\
(\mathbf{Q}_k \mathscr{A})_\mu(z) &= \eta^3 \sum_{c \in \Delta_z} \sum_{j=0}^{N-1} \eta \mathscr{A}_\mu(c + j\eta e_\mu), \\
(\mathbf{Q}_k^{(2)} F)_{\mu\nu}(z) &= \eta^3 \sum_{c \in \Delta_z} \sum_{i,j=0}^{N-1} \eta^2 F_{\mu\nu}(c + i\eta e_\mu + j\eta e_\nu).
\end{aligned} \tag{10.1}$$

All sums are on the periodic covering lattice. These averages preserve constants and the mean of a bond field, are onto on bond fields, and satisfy

$$\mathbf{Q}_k d = d\mathbf{Q}_k, \qquad d\mathbf{Q}_k = \mathbf{Q}_k^{(2)} d, \qquad \|\mathbf{Q}_k^{(2)} F\|_2 \leq \|F\|_2. \tag{10.2}$$

On scalars $\mathbf{Q}_k^T \lambda$ is constant with value $\lambda(z)$ on Δ_z, and $\mathbf{Q}_k \mathbf{Q}_k^T = 1$. The differential identities follow by telescoping along the segments in (10.1). The plaquette average has total weight one. Jensen's inequality and summation over coarse sites count each fine plaquette with total weight η^3, proving its contraction. To see surjectivity on bonds, give the fine bond leaving a unit block in direction μ the value $NA_\mu(z)$, and give the other bonds value zero. Every segment of length one starting in Δ_z crosses exactly one such boundary bond, and its average is $A_\mu(z)$.

Let $\mathcal{R}_k$ be the orthogonal projection on $\Delta_{\mathrm{sc}}(\ker \mathbf{Q}_k)$ in the scalar space. Denote by $H_k^{\mathrm{ph}} A$ the minimizer of $\|d\mathscr{A}\|_2^2$ subject to

$$\mathbf{Q}_k \mathscr{A} = A, \qquad \mathcal{R}_k \delta \mathscr{A} = 0. \tag{10.3}$$

The superscript distinguishes it from the fermion field map $H_k(A)$. Define the coarse photon form

$$\Delta_k = (dH_k^{\mathrm{ph}})^T dH_k^{\mathrm{ph}}. \tag{10.4}$$

Proposition 10.1. *The minimizer exists uniquely. It is real and linear, and* $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$. *It preserves constant fields; if* $dA = 0$, *then* $dH_k^{\mathrm{ph}} A = 0$. *At* $k = 0$ *it is the identity. Its characterization is*

$$\mathbf{Q}_k \mathscr{A} = A, \qquad \delta d\mathscr{A} \in \operatorname{ran} \mathbf{Q}_k^T \text{ on bonds}, \qquad \Delta_{\mathrm{sc}} \delta \mathscr{A} \in \operatorname{ran} \mathbf{Q}_k^T \text{ on scalars}. \tag{10.5}$$

Proof. Surjectivity makes the averaging constraint nonempty. The possible curvatures form an affine subspace of the finite lattice cochain space, on which the squared norm has a minimum. Differentiating along the kernel of $\mathbf{Q}_k$ shows that a minimizer satisfies

$$\langle \delta d\mathscr{A}, B \rangle = 0;$$

this is the middle condition in (10.5). The parallelogram identity shows that all minimizers have the same curvature.

For a closed periodic bond field the discrete Hodge decomposition is $B = c + d\lambda$, where c is its mean. One can see this directly in Fourier variables: at nonzero momentum, closedness says that the components of $\widehat{B}$ are proportional to the forward-difference symbol, and the proportionality factor is $|\widehat{d}|^{-2}\widehat{d}^* \cdot \widehat{B}$. At zero momentum only the constant remains. Exact fields have mean zero. Hence a closed $B \in \ker \mathbf{Q}_k$ is $d\lambda$ with $\lambda \in \ker \mathbf{Q}_k$: average the decomposition, use (10.2), and subtract the constant value of $\mathbf{Q}_k \lambda$. Thus the unconstrained minimizers are $\mathscr{A}_1 + d(\ker \mathbf{Q}_k)$.

The scalar Laplacian is injective on $\ker \mathbf{Q}_k$. Indeed, $\Delta_{\mathrm{sc}} \lambda = 0$ gives $d\lambda = 0$, so λ is constant and its average forces it to be zero. For $\lambda \in \ker \mathbf{Q}_k$,

$$\mathcal{R}_k \delta(\mathscr{A}_1 + d\lambda) = \mathcal{R}_k \delta \mathscr{A}_1 + \Delta_{\mathrm{sc}} \lambda.$$

Since the first term lies in the range $\Delta_{\mathrm{sc}}(\ker \mathbf{Q}_k)$, exactly one such gauge correction imposes the Landau constraint. Moreover $\mathcal{R}_k f = 0$ is equivalent to $\Delta_{\mathrm{sc}} f \perp \ker \mathbf{Q}_k$; this proves the last condition of (10.5). All these equations are real and linear, so uniqueness proves the reality and linearity of H_k^{ph}.

Constants satisfy the equations. If $A = c + d\lambda$ is closed, then $c + d\mathbf{Q}_k^T \lambda$ is a closed fine field with average A, so the minimum curvature energy is zero. This proves preservation of closed fields. Finally $\mathbf{Q}_0 = 1$ gives the level-zero assertion. □

10.2. **Axial coercivity on a block.** On the unit lattice, let $B(y) = \{z \in \mathbb{Z}^3 : |z - y|_\infty < L/2\}$, with centres in $L\mathbb{Z}^3$. The one-step bond average and axial functional are

$$(\mathbf{Q}A)(y, y + Le_\mu) = L^{-4} \sum_{z \in B(y)} A(\Gamma_{z, z+Le_\mu}), \qquad (\tau A)(y, z) = \tfrac{1}{6} \sum_\pi A(\Gamma^\pi(y, z)).$$

Here the first path is straight, and Γ^π changes the coordinates in the order π. The sums $A(\Gamma)$ are oriented bond sums. For an interior bond $b = (z, z + e_\nu)$, subtract the two paths from y to its endpoints and append b. For each order this loop bounds the strips belonging to the directions that come after ν. Their total number of plaquettes is $\sum_{\mu \text{ after } \nu} |z_\mu - y_\mu|$. If $\tau A = 0$, averaging over the six orders gives

$$|A(b)| \le \tfrac{1}{2} \sum_{\mu \neq \nu} |z_\mu - y_\mu| \sup_{p \subset B(y)} |dA(p)| \le \tfrac{L-1}{2} \sup_{p \subset B(y)} |dA(p)|. \tag{10.6}$$

For a face between $B(y)$ and $B(y + Le_\nu)$, let $a(z)$ be its face bond values and let p_c be the maximum curvature on plaquettes in these two blocks and across their common face. A transverse face step is one plaquette plus the difference of two interior bond values. Thus $|a(z) - a(z')| \le$

$L|z-z'|_1 p_c$. The defining average $\mathbf{Q}A$ counts each face bond L times and $L^3(L-1)$ interior bond occurrences. Therefore its face mean $\bar{a}$ satisfies

$$|\bar{a}| \le L|(\mathbf{Q}A)(c)| + \tfrac{1}{2}(L-1)^2 p_c.$$

The average transverse distance from a face site is at most $L-1$, so $|a(z)-\bar{a}| \le L(L-1)p_c$. Combining these gives

$$|A(b)| \le \tfrac{3}{2}L^2 p_c + L|(\mathbf{Q}A)(c)|. \tag{10.7}$$

For interior bonds use (10.6) and omit the coarse-average term. In particular, on the axial fluctuation space,

$$\mathbf{Q}Z = 0, \quad \tau Z = 0 \quad \Longrightarrow \quad \|Z\|_\infty \le \tfrac{3}{2}L^2\|dZ\|_\infty, \qquad \|Z\|_2^2 \le 9L^7\|dZ\|_2^2. \tag{10.8}$$

For the second inequality, square the local bound and replace each local curvature maximum by its sum of squares. An interior plaquette belongs to the neighbourhoods of at most $3L^3+3L^2 \le 4L^3$ bonds; a face plaquette to those of at most L^2 bonds. Summation gives $(9/4)L^4 \cdot 4L^3 = 9L^7$.

10.3. **The minimizer symbol and uniform regularity.** The Fourier calculation must separate the zero alias before the continuum mesh limit is taken. Set $h = (N-1)/2$ and, for $s \in \mathbb{C}$,

$$u(s) = N^{-1}\sum_{j=-h}^{h} e^{isj/N}, \qquad s_N(s) = 2N\sin(s/(2N)), \qquad c(s) = 2\sin(s/2).$$

The identity $u(s)s_N(s) = c(s)$ follows by telescoping the geometric sum. For $q \in \mathbb{C}^3$ define

$$d_\mu(q) = N(e^{iq_\mu/N}-1), \quad d_\mu^*(q) = N(e^{-iq_\mu/N}-1), \quad D(q) = \sum_\mu s_N(q_\mu)^2, \quad U(q) = \prod_\mu u(q_\mu),$$

and

$$V_\mu(q) = e^{i(1-N^{-1})q_\mu/2}u(q_\mu), \quad V_\mu^*(q) = V_\mu(-q), \quad \rho_\mu(p) = e^{ip_\mu}-1, \quad \rho_\mu^*(p) = e^{-ip_\mu}-1.$$

The stars in these formulas denote analytic symbols. At real arguments they are complex conjugates. The aliases are $q_m = p + 2\pi m$, $m \in \{-h,\dots,h\}^3$, and

$$V_\mu(q_m)d_\mu(q_m) = \rho_\mu(p), \qquad V_\mu^*(q_m)d_\mu^*(q_m) = \rho_\mu^*(p), \qquad V_\mu V_\mu^* = u(q_{m,\mu})^2. \tag{10.9}$$

The scalar averaging symbol is $U(q_m)$; on bonds it is $U(q_m)V_\mu(q_m)$. Thus the Euler equations (10.5) can be solved separately in each alias class.

Write $D_0 = D(p)$, $U_0 = U(p)$, $u_\mu = u(p_\mu)$, and

$$\begin{aligned} X_\mu &= \sum_{m\neq 0}\frac{U(q_m)^2u(q_{m,\mu})^2}{D(q_m)}, & Y &= \sum_{m\neq 0}\frac{U(q_m)^2}{D(q_m)^2}, \\ N_\mu &= U_0^2u_\mu^2 + D_0X_\mu, & N_G &= U_0^2 + D_0^2Y, \\ \mathcal{N} &= \prod_\mu N_\mu, & \Pi &= \mathcal{N} - \sum_\mu s_N(p_\mu)^2X_\mu\prod_{\nu\neq\mu}N_\nu. \end{aligned}$$

The coarse form symbol and the scalar multiplier for the Landau condition are

$$\begin{aligned} \Lambda_{\mu\nu} &= \delta_{\mu\nu}\frac{D_0}{N_\mu} - \frac{\rho_\mu\rho_\nu^*U_0^2\mathcal{N}}{N_\mu N_\nu\Pi}, \\ \omega_\nu &= \frac{\rho_\nu^*D_0U_0^2\mathcal{N}}{N_\nu N_G\Pi}, \\ h_{\mu\nu}(p,m) &= \frac{U(q_m)V_\mu^*(q_m)\Lambda_{\mu\nu}}{D(q_m)} + \frac{d_\mu(q_m)U(q_m)\omega_\nu}{D(q_m)^2} \quad (m\neq 0). \end{aligned} \tag{10.10}$$

The zero-alias formula has its apparent pole cancelled:

$$h_{\mu\nu}(p,0) = \delta_{\mu\nu}\frac{U_0 V_\mu^*(p)}{N_\mu} + \frac{d_\mu(p)\rho_\nu^* U_0^3 (X_\mu - u_\mu^2 D_0 Y)\mathcal{N}}{N_\mu N_\nu N_G \Pi}. \tag{10.11}$$

Indeed, $N_\mu - u_\mu^2 N_G = D_0(X_\mu - u_\mu^2 D_0 Y)$, so (10.11) equals the preceding formula wherever $D_0 \neq 0$.

For real $p \in [-\pi,\pi]^3$, the nonzero aliases have $D(q_m) > 0$, and $X_\mu, Y \geq 0$. The positivity of u_μ gives $N_\mu, N_G > 0$. If $p \neq 0$, the identity

$$\Pi = \sum_\mu \frac{s_N(p_\mu)^2}{D_0} U_0^2 u_\mu^2 \prod_{\nu\neq\mu} N_\nu \tag{10.12}$$

shows that $\Pi > 0$; at zero $N_\mu = N_G = \Pi = 1$. The algebraic identity $U_0^2 \sum_\mu \rho_\mu \rho_\mu^*/N_\mu = D_0\Pi/\mathcal{N}$ implies $\sum_\mu \rho_\mu^* \Lambda_{\mu\nu} = 0$. Substitution in the alias equations now gives

$$\widehat{(H_k^{\mathrm{ph}} A)}_\mu(q_m) = \sum_\nu h_{\mu\nu}(p,m)\widehat{A}_\nu(p), \qquad \widehat{(\Delta_k A)}_\mu(p) = \sum_\nu \Lambda_{\mu\nu}(p)\widehat{A}_\nu(p).$$

To check the first formula, average the displayed alias expression. The sums including $m = 0$ are N_μ/D_0 and N_G/D_0^2. Its two terms give $\widehat{A}_\mu$ plus opposite multiples of ρ_μ, so the average is A. Its divergence is $U(q_m)\widehat{\omega}(p)/D(q_m)$; applying the scalar Laplacian gives the scalar range condition, and subtracting its gradient from $D(q_m)\widehat{H_k^{\mathrm{ph}} A}$ gives $U(q_m)V_\mu^*(q_m)\widehat{\Lambda}_\mu(p)$. These are the other two Euler conditions. At $p = 0$ only the zero alias survives, giving the constant field. Uniqueness of the real minimizer proves the formula. Pairing its bond Euler equation with $H_k^{\mathrm{ph}} A'$ and using $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$ identifies the symbol of Δ_k with Λ.

There is a strip of fixed width, independent of N, in which these symbols are analytic. Here is the estimate that provides that width and the derivative bounds. On $|\operatorname{Re} p_\mu| < 5\pi/4$, $|\operatorname{Im} p_\mu| < 1/4$, put $a = (8/(5\pi))\sin(5\pi/8)$ and $c_h = \cosh(1/8)$. The elementary complex sine identities give

$$\operatorname{Re} s_N(s)^2 \geq a^2(\operatorname{Re} s)^2 - c_h^2(\operatorname{Im} s)^2, \qquad |u(s)| \leq c_h$$

in the required alias range. Consequently, for $m \neq 0$,

$$\operatorname{Re} D(q_m) \geq |m|_2^2, \qquad |u(q_{m,\mu})| \leq C\langle m_\mu\rangle^{-1}, \qquad |d_\mu(q_m)| \leq C\langle m_\mu\rangle, \tag{10.13}$$

where $\langle t\rangle = \max(1,|t|)$. For the first estimate use $|\operatorname{Re} q_{m,\mu}| \geq (3\pi/4)|m_\mu|$ when $m_\mu \neq 0$; the lower bound before simplification is $a^2(3\pi/4)^2|m|_2^2 - 3c_h^2/16 \geq |m|_2^2$. The bound on u follows from $u = c/s_N$, and the bound on d_μ from its sine expression. Thus $X_\mu, Y, N_\mu, N_G, \Pi$ are uniformly bounded on the larger rectangle, since their alias sums are bounded by constant multiples of $\sum_m \prod_\mu \langle m_\mu\rangle^{-2}$.

On its real cube $u_\mu \geq a$, so $N_\mu \geq a^8$, $N_G \geq a^6$, and (10.12) gives $\Pi \geq a^{24}$. If an analytic function is bounded by B on that rectangle and is at least $c > 0$ on the slightly smaller real cube, apply Schwarz's lemma to $t \mapsto f(p' + itp'') - f(p')$ on the disk $|t| < 4B/c$. For $|p''|_\infty < c/(16B)$ its argument stays in the larger rectangle, and $|f(p) - f(p')| \leq c/2$. Apply this to the three denominators. It gives a fixed width $g_2 > 0$ on $|\operatorname{Re} p_\mu| < 5\pi/4 - 1/4$ where all remain bounded away from zero. In particular, the zero-alias formula is analytic there.

In this smaller strip the alias bounds imply

$$|h_{\mu\nu}(p,m)| \leq C\left(\langle m_\mu\rangle^{-1}|m|_2^{-2} + \langle m_\mu\rangle|m|_2^{-4}\right)\prod_\sigma \langle m_\sigma\rangle^{-1} \quad (m \neq 0), \tag{10.14}$$

and $|h(p,0)| \leq C$. For every $0 \leq \alpha < 1$ this proves

$$\sup_p \sum_m |h_{\mu\nu}(p,m)| \max\{1, |d_\lambda(q_m)|, |d_\lambda(q_m)||q_m|_1^\alpha\} \leq S_\alpha, \tag{10.15}$$

with S_α independent of N. In fact, after multiplication by $\langle m\rangle_\infty^{1+\alpha}$, the first term in (10.14) is bounded by

$$C\langle m_\mu\rangle^{-2}\prod_{\sigma\neq\mu}\langle m_\sigma\rangle^{-(3-\alpha)/2},$$

and the second by $C\prod_\sigma\langle m_\sigma\rangle^{-4/3}$. Both product sums converge; the first is where $\alpha<1$ is used.

For $x\in\eta\mathbb{Z}^3$, set $K_{\mu\nu}(p;x)=\sum_m e^{iq_m\cdot x}h_{\mu\nu}(p,m)$. It is 2π-periodic in p. To check periodicity, away from $D_0=0$ write $E_\mu=N_\mu/D_0$ and $G=N_G/D_0^2$ as sums over all aliases, and $r=\sum_\mu\rho_\mu\rho_\mu^*/E_\mu$. Then

$$\Lambda_{\mu\nu}=\delta_{\mu\nu}/E_\mu-\rho_\mu\rho_\nu^*/(E_\mu E_\nu r),\qquad \omega_\nu=\rho_\nu^*/(E_\nu G r).$$

Translation of p by 2π permutes the alias classes modulo $2\pi N\mathbb{Z}^3$, and every summand, including $e^{iq_m\cdot x}$, is periodic modulo this latter lattice. Thus K and Λ agree on real overlaps of the rectangles; analyticity extends the identity and defines them on the full strip. Define the covering kernel by $H^\infty_{\mu\nu}(x)=(2\pi)^{-3}\int_{[-\pi,\pi]^3}K_{\mu\nu}(p;x)\mathrm{d}p$. Shift the contour to $p+i\gamma_H\,\mathrm{sgn}(x)$, with $\gamma_H=g_2/2$. The vertical faces cancel by periodicity, and (10.15) gives

$$|H^\infty(x)|,|\partial H^\infty(x)|,|z|^{-\alpha}|\partial H^\infty(x+z)-\partial H^\infty(x)|\le CS_\alpha e^{-\gamma_H|x|_1},\qquad x,z\in\eta\mathbb{Z}^3,\quad 0<|z|\le 1.$$

The last estimate uses $|e^{iq\cdot z}-1|\le 2.5|z|^\alpha|q|_1^\alpha$ on the shifted contour. The torus kernel is $H_k^{\mathrm{ph}}(b,b')=\sum_{n\in\mathbb{Z}^3}H^\infty(c-z'+\ell n)$, with the corresponding direction indices. Summing the exponential bound over the lifts proves

$$|H_k^{\mathrm{ph}}(b,b')|,|\partial H_k^{\mathrm{ph}}(b,b')|,|\delta_\alpha\partial H_k^{\mathrm{ph}}((b,\tilde b);b')|\le C_H(\alpha)e^{-\gamma_H d(b,b')}.\tag{10.16}$$

Here $b,\tilde b$ have the same direction and their initial points are at distance at most one. The same contour argument for the bounded symbol Λ gives

$$|\Delta_k(b,b')|\le C_\Delta e^{-\gamma_H d(b,b')},\qquad \|\Delta_k\|_{2\to2}\le C_\Delta^{(2)}.\tag{10.17}$$

All three constants are independent of k and ℓ. For the operator norm, Parseval's formula reduces the estimate to $\sup_p\|\Lambda(p)\|$. A uniform bound A_1 for its nine entries gives $\|\Lambda(p)\|\le 3A_1$, so one may take $C_\Delta^{(2)}=3A_1$.

10.4. **Fluctuation coordinates and the positive square root.** The fluctuations in one block integration satisfy $\mathbf{Q}Z=0$ and $\tau Z=0$. We use coordinates in which the Gaussian measure becomes a product measure, while each coordinate still has bounded support.

For an L-block $B(y)$, let $\mathrm{Bd}(y)$ denote its interior bond functions, with the counting inner product. The functional τ_y has rank L^3-1: given values $f(x)$ for $x\neq y$, extend $f(y)=0$ and apply τ_y to the interior restriction of df. Every path sum telescopes to $f(x)$. Thus

$$\dim\ker\tau_y=3L^2(L-1)-(L^3-1)=2L^3-3L^2+1.$$

The three coordinate reflections preserve $\ker\tau_0$ and act as commuting orthogonal involutions. Choose an orthonormal basis e_i^0 of simultaneous eigenvectors and translate it to e_i^y. This choice depends only on L.

Each face between neighboring blocks has a central bond, on the line joining their centers. Let

$$V_k=\bigoplus_y\ker\tau_y\ \oplus\bigoplus_{b\text{ noncentral face bond}}\mathbb{R}\delta_b,\qquad \{\Upsilon\}=\{e_i^y\}\cup\{\delta_b\}.$$

Here δ_b is the point mass on the positively oriented bond b. These are orthonormal coordinates, and

$$\dim V_k=d_V\ell^3,\qquad d_V=2-2L^{-3}.\tag{10.18}$$

The support of a basis vector lies in one L-block or consists of the two ends of a bond.

For $z \in V_k$ define Cz to agree with z off the central bonds, and set, on the central bond b_c of the face $c = \langle y, y + Le_\mu \rangle$,

$$(Cz)(b_c) = -\sum_{\substack{b \text{ on } c \\ b \neq b_c}} z(b) - \sum_{b \text{ interior}} \frac{n_c(b)}{L} z(b). \tag{10.19}$$

The second sum is over the bonds of direction μ in the two adjacent blocks; $n_c(b)$ counts the straight averaging paths that contain b. Numbering the bonds on such a line by $t = 0, \dots, 2L-2$ gives $n_c(b) = \min(t+1, 2L-1-t)$. The face bond has multiplicity L; every interior bond has multiplicity at most $L-1$. Consequently

$$L^4(\mathbf{Q}Z)(c) = L \sum_{b \text{ on } c} Z(b) + \sum_b n_c(b) Z(b).$$

Only $Z(b_c)$ is unknown in the equation for this face. Central bonds occur in no axial path and in no other face equation. It follows that C is a real linear bijection

$$C : V_k \longrightarrow \{Z : \mathbf{Q}Z = 0,\ \tau Z = 0\}. \tag{10.20}$$

The following coordinate estimates will be used both for the covariance and for the small-field domain:

$$\|Cz\|_\infty \le c_C \sup_\Upsilon |z(\Upsilon)|, \qquad \|z\|_2 \le \|Cz\|_2 \le (5L^3)^{1/2} \|z\|_2, \qquad \sum_b |(C\Upsilon)(b)| \le c_1, \tag{10.21}$$

where $c_C = 3L^{9/2}$ and $c_1 = 14L^{3/2}$. To prove the first bound, normalize the coordinate supremum to one. An interior value is at most $(2L^3)^{1/2}$ by Cauchy–Schwarz, and a noncentral face value is at most one. Formula (10.19) bounds a central value by $L^2 + 2L^3(2L^3)^{1/2} \le 3L^{9/2}$. For the second bound, the new central values are supported away from z. A central row has at most $2L^3$ entries, of modulus at most one, and an entry contributes to at most two such rows. Cauchy–Schwarz therefore adds at most $4L^3\|z\|_2^2$. For a face coordinate, $C\delta_b$ has two entries of modulus one. For an interior unit vector, the interior ℓ^1 norm is at most $(3L^3)^{1/2}$, and the total contribution of the six adjacent central bonds is at most six times this norm. This proves the last estimate with $c_1 = 14L^{3/2}$. The same formula gives

$$d(x, x') \le L \quad (x \in \operatorname{supp} C\Upsilon,\ x' \in \operatorname{supp} \Upsilon). \tag{10.22}$$

Coordinate sign changes and translations by $M\mathbb{Z}^3$ permute $\{\pm\Upsilon\}$. For the interior vectors this follows from the simultaneous eigenvector choice; for face coordinates it follows from the preservation of central bonds. Hence the standard Gaussian μ_I on V_k and its normalized restriction $\widehat{\mu}_k$ to

$$\mathrm{Box}_k = \{z :\ |z(\Upsilon)| \le p_{0,k} \text{ for every } \Upsilon\}$$

are invariant under these transformations and under $z \mapsto -z$. Independence of the coordinates gives

$$\mu_I(\mathrm{Box}_k) = \exp(-\ell^3 \delta\varepsilon_k^0), \qquad \delta\varepsilon_k^0 = -d_V \log \Pr(|g| \le p_{0,k}) \le 12 \exp(-p_{0,k}^2/2), \tag{10.23}$$

when $p_{0,k} \ge 1$, where g is standard normal. Indeed, the Gaussian tail is at most

$$2(2\pi)^{-1/2} p_{0,k}^{-1} \exp(-p_{0,k}^2/2);$$

at $p_{0,k} = 1$ this is less than $1/2$, and $-\log(1-t) \le 2t$ for $t \le 1/2$ suffices.

Set

$$\widetilde{\Delta}_k = C^T \Delta_k C, \qquad C_k = \widetilde{\Delta}_k^{-1}.$$

The first of these definitions makes sense before invertibility has been proved. From the axial estimate, the contraction of the plaquette average, and $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$, we get

$$\|z\|_2^2 \le \|Cz\|_2^2 \le 9L^7 \|dCz\|_2^2 = 9L^7 \|\mathbf{Q}_k^{(2)} dH_k^{\mathrm{ph}} Cz\|_2^2 \le 9L^7 \langle z, \widetilde{\Delta}_k z\rangle.$$

The upper bound follows from (10.17) and (10.21). Thus, with

$$c_P = \max\{9L^7,\ 5L^3 C_\Delta^{(2)}\}, \qquad C_{\mathrm{Dt}} = c_1^2 C_\Delta \exp(2\gamma_H L),$$

we have

$$c_P^{-1} \le \widetilde{\Delta}_k \le c_P, \qquad |\widetilde{\Delta}_k(\Upsilon, \Upsilon')| \le C_{\mathrm{Dt}} \exp(-\gamma_H d(\Upsilon, \Upsilon')). \tag{10.24}$$

For the second estimate, expand $\langle C\Upsilon, \Delta_k C\Upsilon'\rangle$ as a double bond sum. Every contributing pair satisfies $d(\Upsilon, \Upsilon') \le d(b, b') + 2L$ by (10.22); the two ℓ^1 bounds then give the asserted constant.

The distance between supports has a triangle defect at most L. We record the weighted resolvent argument with this defect, since it is responsible for the uniform decay of the square root. Suppose that $T \ge t_0 > 0$ is a real symmetric matrix, that $|T(a,b)| \le C_T \exp(-g d(a,b))$, and that

$$d(a,c) \le d(a,b) + d(b,c) + \phi.$$

Choose

$$S \ge \exp(g\phi/2) \sup_a \sum_b (d(a,b) + \phi) \exp(-g d(a,b)/2), \qquad \theta = \min\{g/2,\ t_0/(2C_T S)\}.$$

Fix c and conjugate T by the diagonal matrix $W(a,a) = \exp(\theta d(a,c))$. The error $E = WTW^{-1} - T$ satisfies

$$|E(a,b)| \le \theta C_T \exp(g\phi/2)(d(a,b) + \phi) \exp(-g d(a,b)/2),$$

because $|d(a,c) - d(b,c)| \le d(a,b) + \phi$ and $|\exp(u) - 1| \le |u| \exp(|u|)$. The Schur test gives $\|E\| \le t_0/2$. Therefore $\|(T + x + E)^{-1}\| \le (t_0/2 + x)^{-1}$ for $x \ge 0$. Undoing the conjugation yields

$$|(T + x)^{-1}(a,c)| \le (t_0/2 + x)^{-1} \exp(-\theta d(a,c)).$$

The scalar integral $\pi^{-1} \int_0^\infty x^{-1/2}(t + x)^{-1}\, \mathrm{d}x = t^{-1/2}$ and finite-dimensional diagonalization now give

$$T^{-1/2} = \frac{1}{\pi} \int_0^\infty x^{-1/2} (T + x)^{-1}\, \mathrm{d}x, \qquad |T^{-1/2}(a,c)| \le (2/t_0)^{1/2} \exp(-\theta d(a,c)). \tag{10.25}$$

Apply this to $T = \widetilde{\Delta}_k$, $\phi = L$, $t_0 = c_P^{-1}$ and $g = \gamma_H$. The basis support count gives

$$\sup_\Upsilon \sum_{\Upsilon'} \exp(-g d(\Upsilon, \Upsilon')) \le c_{\mathrm{bas}}(g), \qquad c_{\mathrm{bas}}(g) = 2L^3 \exp(2gL)(1 + 6/g)^3.$$

For example, assign a basis vector to its block center or to an end of its face bond. At most $2L^3$ vectors are assigned to any site, and replacing support distance by point distance costs at most $2L$; the remaining lattice sum is bounded by $(1 + 6/g)^3$. Since $t \exp(-gt/4) \le 2/g$, we may take

$$S_P = \exp(\gamma_H L/2)(L + 2/\gamma_H) c_{\mathrm{bas}}(\gamma_H/4), \quad \gamma_P = \min\{\gamma_H/2,\ (2c_P C_{\mathrm{Dt}} S_P)^{-1}\}.$$

With

$$C_P^{\mathrm{ph}} = \max\{C_H(\alpha), C_{\mathrm{Dt}}, (2c_P)^{1/2}, 4\exp(2)\},$$

all the kernels in (10.16), (10.24), and the positive square root satisfy their respective bounds at rate γ_P with constant C_P^{ph}. In particular,

$$C_k^{1/2} = \widetilde{\Delta}_k^{-1/2}, \qquad |C_k^{1/2}(\Upsilon, \Upsilon')| \le C_P^{\mathrm{ph}} \exp(-\gamma_P d(\Upsilon, \Upsilon')). \tag{10.26}$$

At $k = 0$, $H_0^{\mathrm{ph}} = 1$; its first differences and their Hölder differences are supported within distance two and have entries of modulus at most two. Our choice $C_P^{\mathrm{ph}} \geq 4\exp(2)$ therefore includes this level as well.

10.5. **The exact block decomposition.** For a real unit bond field A' on the torus of side ℓ/L, put $A'_L(b) = L^{-1/2}A'(b/L)$. Let $A_k^{\min}(A')$ minimize $\langle A, \Delta_k A\rangle$ under

$$\mathbf{Q}A = A'_L, \qquad \tau A = 0.$$

This affine space is nonempty: set each face bond of the face c equal to $LA'_L(c)$, and set every interior bond to zero. Its direction is $C(V_k)$, and (10.24) makes the quadratic form positive definite on this direction. The minimizer is therefore unique and is characterized by

$$\langle A_k^{\min}(A'), \Delta_k C z\rangle = 0 \quad (z \in V_k). \tag{10.27}$$

This characterization is a real linear system with a unique solution, so $A_k^{\min}$ is real and linear.

Define $\mathrm{Sq}_k = CC_k^{1/2}$. It maps V_k bijectively onto the axial fluctuations, has zero block average and mean, and

$$\langle \mathrm{Sq}_k\zeta, \Delta_k \mathrm{Sq}_k\zeta'\rangle = \langle \zeta, C_k^{1/2}\widetilde{\Delta}_k C_k^{1/2}\zeta'\rangle = \langle \zeta, \zeta'\rangle.$$

Thus the exact Gaussian decomposition is

$$\langle A_k^{\min}(A') + \mathrm{Sq}_k\zeta, \Delta_k(A_k^{\min}(A') + \mathrm{Sq}_k\zeta)\rangle = \langle A_k^{\min}(A'), \Delta_k A_k^{\min}(A')\rangle + \|\zeta\|_2^2. \tag{10.28}$$

The block integration uses the standard Gaussian in these ζ coordinates, restricted to Box_k; the loss of normalization is exactly (10.23).

We also need the relation between the two backgrounds. With the scalar scaling $\lambda_L(x) = L^{1/2}\lambda(x/L)$, there is a unique real scalar $\omega_k(A')$ with $\mathbf{Q}_{k+1}\omega_k(A') = 0$ such that

$$H_k^{\mathrm{ph}} A_k^{\min}(A') = (H_{k+1}^{\mathrm{ph}} A')_L - d\omega_k(A'), \qquad \|dH_k^{\mathrm{ph}} A_k^{\min}(A')\|_2 = \|dH_{k+1}^{\mathrm{ph}} A'\|_2. \tag{10.29}$$

The harmonic part of A' is arbitrary. Here $\mathbf{Q}_{k+1} = \mathbf{Q}\mathbf{Q}_k$ is the average from the mesh-η_k lattice of side ℓ to its L-spaced lattice. The operator H_{k+1}^{ph} is formed at mesh η_{k+1} and side ℓ/L, before scaling by L.

Here is the minimizing argument. At mesh η, the scaled field $(H_{k+1}^{\mathrm{ph}} A')_L$ minimizes the curvature norm under the $(k+1)$-step constraint $\mathbf{Q}_{k+1}\mathscr{A} = A'_L$. This follows by scaling the defining problem; the curvature L^2 norm is unchanged. The field $\mathscr{B} = H_k^{\mathrm{ph}} A_k^{\min}(A')$ has the same average. If $\mathscr{A}$ is any competitor, set $A = \mathbf{Q}_k\mathscr{A}$. Choose the one-step restricted gauge

$$\mu(x) = (\tau A)(y, x) - L^{-3}\sum_{x' \in B(y)} (\tau A)(y, x'), \qquad x \in B(y).$$

Then $\mathbf{Q}\mu = 0$ and $\tau(A - d\mu) = 0$, by telescoping the exact-field path sums. Therefore

$$\|d\mathscr{A}\|_2^2 \geq \|dH_k^{\mathrm{ph}} A\|_2^2 = \|dH_k^{\mathrm{ph}}(A - d\mu)\|_2^2 \geq \|d\mathscr{B}\|_2^2.$$

Both backgrounds are minimizers of the same problem. The Landau-minimizer argument shows that their difference is the coboundary of a scalar with zero $(k+1)$-step average. This scalar is unique: a closed field with zero average is exact, and the only constant scalar with zero block average is zero. The two backgrounds are real and linear in A', hence so is ω_k. This proves (10.29) and identifies the first term on the right of (10.28) with $\|dH_{k+1}^{\mathrm{ph}} A'\|_2^2$.

10.6. **Curvature control in local gauges.** Assume from now on that $\ell/(LM)$ is odd and at least 13. For an M-cube $\square$, let $\square^\dagger$ be its enlargement by LM in each direction. Its side is $s_d = (2L+1)M \le 3\ell/13$. For a bond field B on a set W, write

$$\|B\|_{1,W} = \|B\|_{\infty,W}, \qquad \|B\|_{2,W} = \|\partial B\|_{\infty,W}, \qquad \|B\|_{3,W} = \|\delta_\alpha \partial B\|_{\infty,W}.$$

Only bonds and differences contained in W enter these norms. They are ordinary differences of the photon coefficients; no connection occurs in their definition.

Proposition 10.2. *For every real unit bond field A, including an arbitrary constant part, and every M-cube $\square$, there is a real scalar ν on $\square^\dagger$ such that*

$$\|H_k^{\mathrm{ph}} A - d\nu\|_{i,\square^\dagger} \le C_B \|dA\|_\infty, \qquad i = 1,2,3. \tag{10.30}$$

The constant C_B depends on L, M, α and is independent of k, ℓ and the harmonic part of A.

Proof. Let a be the mean of A, put $A^0 = A - a$, and choose the center x_0 of $\square$ as base point. On the canonical lifted period cell

$$W = x_0 + \{-(\ell-1)/2, \dots, (\ell-1)/2\}^3$$

define $\lambda(x)$ to be the line sum of A^0 along the rectilinear path from x_0 to x, in coordinate order $1,2,3$. This defines a scalar on the torus. Put $A'' = A^0 - d\lambda$ and $p_A = \|dA\|_\infty$. The path sums of A'' vanish. For a bond whose two ends lie in W without crossing its boundary, the strip identity used in the axial estimate gives

$$|A''(x, x+e_\mu)| \le \sum_{\nu>\mu} |x_\nu - x_{0,\nu}| p_A \le 2d(x,x_0) p_A. \tag{10.31}$$

Indeed, commuting the final bond through the later portions of the two base-point paths sweeps one strip in each direction $\nu > \mu$. Each strip contributes its plaquette sum, and both remaining path sums are zero.

A seam bond lies on a closed coordinate line with $\ell - 1$ inner bonds. Let $\mathrm{hol}(v)$ be the line sum of A^0 in that direction. Translating this line by one transverse step changes its sum by at most ℓp_A, by summing the intervening ℓ plaquettes. Its mean over the transverse two-torus is zero, since A^0 has mean zero. The mean distance in each coordinate circle is at most $\ell/4$, hence $|\mathrm{hol}(v)| \le \ell^2 p_A/2$. The line sum of $d\lambda$ around a closed line vanishes. Removing the inner bonds using (10.31) therefore gives

$$|A''(b')| \le |\mathrm{hol}(v)| + (\ell-1)\ell p_A \le \tfrac{3}{2}\ell^2 p_A \tag{10.32}$$

for the seam value. In particular this construction also gives a global real gauge with $\|A^0 - d\lambda\|_\infty \le 2\ell^2 \|dA\|_\infty$.

Since $H_k^{\mathrm{ph}} d\lambda$ is closed and has mean zero, it is exact, say $d\nu'$. On the nonwrapping cube $\square^\dagger$, let x denote lifted coordinates and set $\nu = a \cdot x + \nu'$. Then $H_k^{\mathrm{ph}} A - d\nu = H_k^{\mathrm{ph}} A''$ there. If $b \subset \square^\dagger$ and b' is inner, its initial point is at distance at most $d(b,b') + 2$ from an end of b. Thus (10.31) gives

$$|A''(b')| \le (2d(b,b') + 4 + s_d) p_A.$$

For a seam bond, $d(b,b') \ge (\ell - 1 - s_d)/2$.

Put $g = \gamma_P$ and $c_{\mathrm{bd}}(g) = 3\exp(3g)(1+6/g)^3$, which bounds the sum of $\exp(-gd(b,b'))$ over unit bonds. Summing any of the three kernel bounds against the inner bonds gives

$$C_P^{\mathrm{ph}} p_A \big[(s_d+4) c_{\mathrm{bd}}(g) + (2/g) c_{\mathrm{bd}}(g/2)\big],$$

using $2t\exp(-gt) \le (2/g)\exp(-gt/2)$. For the seam contribution split off half the decay. Since $s_d \le 3\ell/13$,

$$\exp(-gd(b,b')) \le \exp(g/4)\exp(-5g\ell/26)\exp(-gd(b,b')/2), \qquad \ell^2 \exp(-5g\ell/26) \le 15/g^2.$$

Together with (10.32), this bounds the seam contribution by $62C_P^{\mathrm{ph}}p_A g^{-2}c_{\mathrm{bd}}(g/2)$. The proposition follows with

$$C_B = C_P^{\mathrm{ph}}\big[(s_d+4)c_{\mathrm{bd}}(\gamma_P) + (2/\gamma_P + 62/\gamma_P^2)c_{\mathrm{bd}}(\gamma_P/2)\big]. \tag{10.33}$$

□

10.7. **Weakened photon kernels.** For $a = M$ or LM, let π_a consist of the closed cubes of half-side a centered at $a\mathbb{Z}^3/\ell\mathbb{Z}^3$. These are overlapping parameter cubes, distinct from the cubes that partition polymers. Two parameter cubes are linked when they contain a common lattice site; components below refer to chains of such links. For a set S of unit sites, let $\mathrm{Cub}_a(S)$ be all parameter cubes meeting S.

Choose the shortest rectilinear path $\mathrm{Pth}(p,p')$ in coordinate order $1,2,3$. Its signed coordinate offsets are uniquely determined because ℓ is odd. A site belongs to at most 27 parameter cubes. If S is connected by nearest-neighbor steps, then $\mathrm{Cub}_a(S)$ is linked: consecutive sites lie together in a cube, and all cubes containing either site are linked to this cube. Along a coordinate segment, at most nine transverse choices of centers occur. Counting the longitudinal centers for the three segments of $\mathrm{Pth}(p,p')$ gives

$$\#\mathrm{Cub}_a(\mathrm{Pth}(p,p')) \le 9(3d(p,p')/a + 9). \tag{10.34}$$

An L-block, and the ends of a unit bond, meet at most 27 cubes of either kind.

We smooth only in the row variable of the minimizer. Let

$$\vartheta(r) = \begin{cases} 0 & r \le 0, \\ 3r^2 - 2r^3 & 0 \le r \le 1, \\ 1 & r \ge 1, \end{cases} \qquad \varphi(t) = \vartheta(2(3/4 - |t|)), \qquad \chi_c(x) = \prod_{\nu=1}^{3} \varphi((x_\nu - c_\nu)/M).$$

The coordinates in the last expression are represented with modulus at most $\ell/2$. Put $\chi_c(b) = \chi_c(x_b)$, where x_b is the midpoint of b. The identity $\vartheta(r) + \vartheta(1-r) = 1$ implies $\sum_{j\in\mathbb{Z}} \varphi(t-j) = 1$, and therefore $\chi_c \ge 0$ and $\sum_c \chi_c = 1$ on the torus. Moreover φ' is bounded by three and is Lipschitz with constant 24. Tensor products and the averaged derivative formula for the discrete difference give, for $\tilde{b} = b + r$ with $|r| \le 1$,

$$|\partial_\nu \chi_c(b)| \le 3/M, \qquad |\chi_c(\tilde{b}) - \chi_c(b)| \le 9|r|/M, \qquad |\partial_\nu \chi_c(\tilde{b}) - \partial_\nu \chi_c(b)| \le 42|r|/M^2. \tag{10.35}$$

For the last bound the derivative factor contributes $24/M^2$, and the two remaining factors contribute $2 \cdot 9/M^2$. If $\chi_c(b) \ne 0$, both ends of b belong to the cube centered at c.

For a unit bond b' with ends p_1', p_2', define

$$P'(c,b') = \mathrm{Cub}_M(\mathrm{Pth}(c,p_1') \cup \mathrm{Pth}(c,p_2')), \qquad w(b,b';s) = \sum_c \chi_c(b) \prod_{a \in P'(c,b')} s_a.$$

For an interior basis vector set $\mathrm{St}(e_i^y) = B(y)$ and $\mathrm{Pt}(e_i^y) = \{y\}$; for δ_b, both sets are the two ends of b. Put

$$P''(\Upsilon,\Upsilon') = \mathrm{Cub}_{LM}\left(\mathrm{St}(\Upsilon) \cup \mathrm{St}(\Upsilon') \cup \bigcup_{\substack{p\in\mathrm{Pt}(\Upsilon)\\ p'\in\mathrm{Pt}(\Upsilon')}} \mathrm{Pth}(p,p')\right).$$

The weakened kernels are

$$\begin{aligned} H_k^{\mathrm{ph}}(s)(b,b') &= H_k^{\mathrm{ph}}(b,b')w(b,b';s), \quad (k \ge 1), \\ C_k^{1/2}(s)(\Upsilon,\Upsilon') &= C_k^{1/2}(\Upsilon,\Upsilon') \prod_{a\in P''(\Upsilon,\Upsilon')} s_a, \quad (k \ge 0). \end{aligned} \tag{10.36}$$

The two families have parameter sets π_M and π_{LM}, respectively. When both occur in one expression, s denotes their pair. Each entry is a polynomial, independent of the connection and real for real parameters. At $s = 1$ the exact operators are recovered. For general complex parameters, the second family is defined by the displayed kernel; positivity or the square-root identity is used only at $s = 1$.

Fix $0 < \alpha_0 \le 1/4$ and take

$$M = L^m \ge L^2, \qquad 2\alpha_0 n_P \log M \le \gamma_P M, \qquad n_P = 402. \tag{10.37}$$

Parameters range over the closed polydisc $|s_a| \le M^{\alpha_0}$. Such an M exists because $(\log M)/M \to 0$. If $\chi_c(\hat{b}) \ne 0$ for a translate of b by distance at most two, then $d(c, x_b) \le M + 1$; applying (10.34) to both paths gives

$$\#P'(c, b') \le 234 + 54 d(b, b')/M \le n_P(1 + d(b, b')/M).$$

At most 27 such centers occur: in each coordinate their representatives lie in an interval of length $3M/2 + 5 < 3M$. The set P'' has two endpoint sets and at most four paths. Since their endpoints are within L of the respective supports, the same count gives

$$\#P''(\Upsilon, \Upsilon') \le 378 + 216/M + 108 d(\Upsilon, \Upsilon')/(LM) \le n_P(1 + d(\Upsilon, \Upsilon')/M).$$

For either product, with corresponding distance D, we have

$$\left| \prod_{a \in P} s_a \right| \le M^{\alpha_0 n_P} \exp(\alpha_0 n_P D \log M/M) \le M^{\alpha_0 n_P} \exp(\gamma_P D/2). \tag{10.38}$$

This is the loss that the exact exponential decay absorbs.

Proposition 10.3. *Under* (10.37)*, put* $\gamma^P = \gamma_P/2$ *and* $C^P = 116 C_P^{\mathrm{ph}} M^{\alpha_0 n_P}$*. The three kernels of* $H_k^{\mathrm{ph}}(s)$ *satisfy*

$$|H_k^{\mathrm{ph}}(s)(b, b')|, |\partial H_k^{\mathrm{ph}}(s)(b, b')|, |\delta_\alpha \partial H_k^{\mathrm{ph}}(s)((b, \tilde{b}); b')| \le C^P \exp(-\gamma^P d(b, b')), \tag{10.39}$$

and

$$|C_k^{1/2}(s)(\Upsilon, \Upsilon')| \le C^P \exp(-\gamma^P d(\Upsilon, \Upsilon')). \tag{10.40}$$

These bounds hold on the closed polydiscs.

Proof. The root estimate follows at once from (10.26) and (10.38). For the minimizer, fix b' and abbreviate $H(b) = H_k^{\mathrm{ph}}(b, b')$, $w(b) = w(b, b'; s)$ and $b^+ = b + \eta e_\nu$. Put

$$\mathrm{Pm} = M^{\alpha_0 n_P} \exp(\gamma_P d(b, b')/2).$$

For $b, b^+, \tilde{b}, \tilde{b}^+$, and the intervening translates, (10.35) and the 27-center count give

$$|w| \le \mathrm{Pm}, \quad |\partial_\nu w| \le 9\mathrm{Pm}, \quad |w(\tilde{b}^+) - w(b^+)| \le 27|r|\mathrm{Pm}, \quad |\partial_\nu w(\tilde{b}) - \partial_\nu w(b)| \le 14|r|\mathrm{Pm}.$$

Here $\eta \le 1/3$ and $M \ge 9$; all translates are within two of the initial point. A rectilinear path between b and $\tilde{b} = b + r$ has at most $3|r|/\eta$ steps, at distance at most one from b. Summing the first-derivative bound for H along this path gives

$$|H(\tilde{b}) - H(b)| \le 3|r| C_P^{\mathrm{ph}} \exp(\gamma_P) \exp(-\gamma_P d(b, b')).$$

Use the discrete product rule

$$\partial_\nu (Hw)(b) = (\partial_\nu H)(b) w(b^+) + H(b) \partial_\nu w(b).$$

It bounds the zeroth and first kernels with factors one and ten. Subtracting this formula at b and $\tilde{b}$ gives four terms: the difference of $\partial_\nu H$, the difference of $w(b^+)$, the difference of H, and the difference of $\partial_\nu w$. Their bounds have factors $1, 27, 27\exp(\gamma_P), 14$, respectively, times

$|r|^{\alpha} C_P^{\mathrm{ph}} \mathrm{Pm} \exp(-\gamma_P d(b,b'))$. We used $|r| \le |r|^{\alpha}$ and $\gamma_P \le 1$. Their sum is at most 116 times this expression, proving (10.39). □

At level zero the row smoothing is replaced by the diagonal family

$$H_0^{\mathrm{ph}}(s)(b,b') = \delta_{bb'} \prod_{a \in \pi_M : b \subset a} s_a. \tag{10.41}$$

Every product has at most 27 factors and at least one. The first difference has two diagonal terms; the Hölder difference has four, supported within distance two. Thus their entries are bounded by $4\exp(2\gamma^P) M^{27\alpha_0} \exp(-\gamma^P d(b,b'))$, which is at most the right side of (10.39) by $C_P^{\mathrm{ph}} \ge 4\exp(2)$.

The weakening also separates components exactly. Suppose that $s_a = 0$ outside a set $\mathscr{Y}$ of parameter cubes, and let Y_1 be one linked component of their union. A linked path set P meeting this component either lies entirely in it, or contains a cube outside $\mathscr{Y}$ and its parameter product vanishes. Both P' and P'' are linked and contain every parameter cube meeting the respective column support; P'' also contains every cube meeting the row support. In the first family, every contributing row cube c contains the fine bond b and belongs to $P'(c,b')$. It follows that a row or column supported in Y_1 depends only on parameters in that component and has no nonzero entry at an index outside it. A nonzero row is supported in some component. The diagonal family has the same properties because all cubes containing b are linked through an end of b.

Finally, coordinate sign changes preserve the ordered shortest paths, and translations by the relevant center lattice preserve the parameter cubes. The exact kernels commute with these transformations: the averaging, axial and Landau constraints are preserved, and uniqueness identifies the transformed minimizers and covariance. The basis transforms by a signed permutation. The parameter products and χ_c transform accordingly. Hence $H_k^{\mathrm{ph}}(s)$ is covariant under coordinate sign changes and M translations, and $C_k^{1/2}(s)$ under coordinate sign changes and LM translations, with the parameters transformed as well.

10.8. **The domain bounds for block integration.** We make explicit the domain conditions used with these operators. Take $0 < \varepsilon \le 1/200$, $e_k \in (0,1)$, and put $t_{i,k} = e_k^{-3/4+i\varepsilon}$ for $i = 1,2,3$. The set R_k consists of complex fine bond fields with $\|B\|_i < t_{i,k}$ in the three norms. A field belongs to $c\widetilde{R}_k$, $0 < c \le 1$, if it has a decomposition $B = B_0 + B_1$, with $B_1 \in cR_k$ and B_0 real, such that on each $\square^{\dagger}$ the field B_0 is gauge equivalent by a real scalar to a real field with norms strictly below $ct_{i,k}$. The decomposition and gauges need only exist. For $\widetilde{R}_k^-$ the local real representative has the bounds $t_{i,k}$, while the global term B_1 has the bounds $(1 - e_k^{\varepsilon})t_{i,k}$. Finally,

$$\mathrm{Dom}_k = \{A \text{ real unit bond field} : \ H_k^{\mathrm{ph}} A \in \tfrac{1}{4}\widetilde{R}_k\}.$$

These definitions permit every real harmonic part.

Let $C_w \ge C^P$ and $0 < \gamma_w \le \gamma^P$, and choose c_γ to bound both sums

$$\sup_{\Upsilon} \sum_{\Upsilon'} \exp(-\gamma_w d(\Upsilon,\Upsilon')), \qquad \sup_{b} \sum_{b'} \exp(-\gamma_w d(b,b'))$$

over basis elements and unit bonds, respectively. The counts above make c_γ independent of k, ℓ. Let $p_{0,k} = (-\log e_k)^{p_0}$ and $p_k = (-\log e_k)^{p}$. Assume

$$\begin{aligned} p_{0,k} &\le e_k^{-\varepsilon/4}, \qquad 2C_w^2 c_\gamma^2 c_C e_k^{\varepsilon/4} \le 1, \\ 8C_B p_k &\le t_{3,k}, \qquad 8e_k^{3/4-7\varepsilon/2} \le 1, \qquad L^{3/4} \ge 8/3. \end{aligned} \tag{10.42}$$

Since the constants are fixed before the charge, these are conditions on an upper bound e_0, valid for every $0 < e_k \le e_0$.

Proposition 10.4. *Under* (10.42)*, every real unit bond field with* $\|dA\|_\infty \le p_k$ *belongs to* Dom_k*. For every* $\zeta \in \mathrm{Box}_k$ *and every pair of parameters in the closed polydiscs, the field*

$$\mathscr{Z}_k(s,\zeta) = H_k^{\mathrm{ph}}(s) C C_k^{1/2}(s)\zeta$$

has all three norms bounded by $\frac{1}{2}e_k^{-\varepsilon/2}$*. If* ℓ/L *is also an admissible side and* $e_{k+1} = L^{1/2}e_k$*, then*

$$A_k^{\min}(A') + \mathrm{Sq}_k\zeta \in \mathrm{Dom}_k \quad (A' \in \mathrm{Dom}_{k+1},\ \zeta \in \mathrm{Box}_k). \tag{10.43}$$

Proof. The first assertion follows from (10.30): its local real representative has each norm at most $C_B p_k \le t_{3,k}/8 < t_{i,k}/4$. Use $B_1 = 0$ in the domain decomposition.

For the fluctuation bound, the root estimate and the first summation bound give

$$\sup_{\Upsilon} |(C_k^{1/2}(s)\zeta)(\Upsilon)| \le C_w c_\gamma e_k^{-\varepsilon/4}.$$

The coordinate estimate for C holds also for complex coordinates: evaluating its real bound on the signs of the real row coefficients gives $\sum_\Upsilon |C(b,\Upsilon)| \le c_C$. Each norm of $H_k^{\mathrm{ph}}(s)v$ is bounded by $C_w c_\gamma \|v\|_\infty$, by the respective kernel sum. Consequently all three norms of $\mathscr{Z}_k(s,\zeta)$ are at most

$$C_w^2 c_\gamma^2 c_C e_k^{-\varepsilon/4} \le \tfrac{1}{2}e_k^{-\varepsilon/2}.$$

This proves the complex weakened assertion at every level, including the diagonal family at level zero.

For (10.43), write $\mathscr{A}' = H_{k+1}^{\mathrm{ph}} A'$. It is real and belongs to $\frac{1}{4}\widetilde{R}_{k+1}$. On any coarse enlarged cube there is a real gauge in which $\mathscr{A}'$ is represented by a real field A^c with i-th norm below $t_{i,k+1}/2$: add the real global summand to the local representative. For any fine M-cube $\square$, choose a coarse cube $\square'$ with $\square \subset L\square'$. Then $\square^\dagger \subset L(\square'^\dagger)$. The scaling of the three norms is $L^{-1/2}, L^{-3/2}, L^{-3/2-\alpha}$, respectively, and

$$t_{i,k+1} = L^{-3/8+i\varepsilon/2} t_{i,k}.$$

Thus $(A^c)_L$ has i-th norm strictly below $\frac{1}{2}L^{-7/8+3\varepsilon/2}t_{i,k} \le \frac{1}{2}L^{-3/4}t_{i,k} \le 3t_{i,k}/16$. By (10.29), the new background on $\square^\dagger$ is gauge equivalent to $(A^c)_L + \mathscr{Z}_k(1,\zeta)$. The latter fluctuation is real, and its norms satisfy

$$\tfrac{1}{2}e_k^{-\varepsilon/2} \le t_{3,k}/16 \le t_{i,k}/16$$

by (10.42). The sum has norms strictly below $t_{i,k}/4$. This is the required local representative, again with global summand zero. □

The weakened fluctuation also gives the complex radius used in the later Taylor expansions. If $A \in \widetilde{R}_k^-$ and

$$|t| \le \mathrm{rad}_1 := \tfrac{1}{2}e_k^{-3/4+9\varepsilon/2}, \tag{10.44}$$

then $A + t\mathscr{Z}_k(s,\zeta) \in \widetilde{R}_k$. Indeed the three added norms are at most $\frac{1}{4}e_k^{-3/4+4\varepsilon} < e_k^{\varepsilon} t_{3,k}$; this fits the margin in the global summand of $\widetilde{R}_k^-$. The fluctuation is a polynomial in the two parameter families and linear in ζ. This is the photon input to analyticity of the step integrand.

10.9. **Consistency of the exact operators on covers.** Let $\ell_2 = J\ell_1$ with J odd and with both sides admissible. Write pj^* for periodic pullback, also on the coarse unit lattices and on fluctuation coordinates. The common reflection eigenbasis in each block makes the last pullback well defined. Its image is the subspace of ℓ_1-periodic fields, and

$$\langle \mathrm{pj}^* f, \mathrm{pj}^* h\rangle_{\ell_2} = J^3 \langle f, h\rangle_{\ell_1}.$$

All exact photon operators commute with this pullback:

$$O^{(2)}\mathrm{pj}^* = \mathrm{pj}^* O^{(1)}, \qquad O = H_k^{\mathrm{ph}}, \Delta_k, C, \widetilde{\Delta}_k, (\widetilde{\Delta}_k + x)^{-1}, C_k^{1/2}, \mathrm{Sq}_k. \tag{10.45}$$

The superscript specifies the torus; $x \geq 0$. Also

$$(A_k^{\min})^{(2)}(\mathrm{pj}^* A') = \mathrm{pj}^*(A_k^{\min})^{(1)}(A'). \tag{10.46}$$

To prove these identities, first note that $d, \delta, \Delta_{\mathrm{sc}}, \mathbf{Q}_j, \mathbf{Q}_j^T, \tau, C$ have the same finite row formulas on the periodic lifts. They commute with pullback and with translations by $\ell_1\mathbb{Z}^3$. Pulling back the three Euler equations (10.5) therefore characterizes the larger-torus minimizer with the pulled-back datum. Uniqueness gives the identity for H_k^{ph}.

The exact operators intertwine the orthogonal translations by $\ell_1\mathbb{Z}^3$. Transposing these relations shows that their adjoints also intertwine the translations and preserve the periodic subspaces. If $O^{(2)}\mathrm{pj}^* = \mathrm{pj}^* O^{(1)}$, pairing with two pullbacks and canceling J^3 gives the same identity for O^T. Products give the identities for Δ_k and $\widetilde{\Delta}_k$. Invertibility gives the resolvent identity; integrating it by (10.25) gives the root identity, and hence the identity for Sq_k.

Finally let $A = (A_k^{\min})^{(1)}(A')$ and test its pullback against a larger-torus axial fluctuation $\widetilde{Z}$. Sum the translates of $\widetilde{Z}$ over $\ell_1\mathbb{Z}^3/\ell_2\mathbb{Z}^3$ to obtain $\mathrm{pj}^* Z$. Its average and axial sums vanish, so $Z \in C(V_k)$ on the smaller torus. The field $\Delta_k^{(2)}\mathrm{pj}^* A$ is periodic; consequently

$$\langle \mathrm{pj}^* A, \Delta_k^{(2)} \widetilde{Z}\rangle_{\ell_2} = J^{-3}\langle \mathrm{pj}^* \Delta_k^{(1)} A, \mathrm{pj}^* Z\rangle_{\ell_2} = \langle A, \Delta_k^{(1)} Z\rangle_{\ell_1} = 0.$$

Its average and axial constraints also pull back. The characterization (10.27) proves (10.46).

11. Local fermion integration

The walk expansion also provides the localization required by the renormalization step. A parameter is attached to each M-cube traversed by a walk, together with its one-layer neighbourhood. Setting parameters to zero then separates the operator into components. The coarse covariance is obtained from this expansion by an exact resolvent identity. At parameter value one it agrees with the exact covariance; its local terms expand the determinant and the interaction. We work at mesh $\eta_k = L^{-k}$ on the torus of side ℓ; this is the side denoted by R in the local inverse construction. Throughout this section the mass satisfies $0 < \mu \leq 1$, the range of the renormalization data. Transposition is with respect to the bilinear lattice pairings: for $O : E \to E'$, $\langle v, Ou\rangle_{E'} = \langle O^T v, u\rangle_E$. These pairings use the counting, chamber, or coarse lattice measure of the respective space. No complex conjugation enters this operation.

11.1. **Weakened walks and polymer propagators.** The resolvent parameter ranges over $Y_0 = \mathbb{R}$ at level zero and $Y_k = \mathbb{R} \cup \{\infty\}$ at $k \geq 1$, with $-\infty = \infty$ as in the preceding resolvent construction.

Let π_k be the partition into M-cubes and let $\mathcal{P}_k$ be the set of nonempty unions of these cubes connected through faces. For $X \in \mathcal{P}_k$ write $|X|_M$ for its number of cubes. With the net and length conventions of Appendix A, set

$$M d_M(X) = \inf\{\mathrm{length}(\mathcal{N}) : \mathcal{N} \text{ is a net meeting every closed cube of } X\}.$$

For a face-connected union of cubes, joining successive cube centres gives $d_M(X) \leq |X|_M - 1$.

For a walk $\omega = (\square_0, \dots, \square_n)$ from §9.6, put

$$X_\omega = \bigcup_{i=0}^{n} \square_i^{\sim 1}, \qquad s_\omega = \prod_{\square \subset X_\omega} s_\square, \qquad |s_\square| \leq M^{\alpha_0}, \quad \alpha_0 = 1/54.$$

The union X_ω is a polymer and

$$|X_\omega|_M \leq 27(n+1), \qquad d_M(X_\omega) \leq 27(n+1). \tag{11.1}$$

Indeed, each one-layer neighbourhood contains 27 cubes. Successive centres differ by at most M in each coordinate, so their neighbourhoods are joined through faces. The preceding bound on

d_M applies. Every walk, including a walk of length zero, carries the parameters of all cubes in X_ω.

The left parametrix gives terms

$$S'_{y,\omega} = (h_n S_n)(-C_n h_{n-1} S_{n-1}) \cdots (-C_1 h_0 S_0) h_0.$$

Here, as for $S_{y,\omega}$, S_i is the local inverse on $\square_i$ and $C_i = -[F_y, h_i]$. Define

$$S_y(s,A) = \sum_\omega s_\omega S_{y,\omega}(A), \qquad S_y(X,A) = \sum_{X_\omega = X} S_{y,\omega}(A), \tag{11.2}$$
$$S'_y(s,A) = \sum_\omega s_\omega S'_{y,\omega}(A), \qquad S'_y(X,A) = \sum_{X_\omega = X} S'_{y,\omega}(A).$$

Their averages are denoted by S_y^{sy}. The superscript # below means any of these three choices. Put $\kappa' = \kappa_{\mathrm{w}} + \kappa_0$, where $\kappa_{\mathrm{w}}, \kappa_0 > 0$ are fixed polymer decay rates. Choose M large enough that

$$q(M^{27\alpha_0}) \le \tfrac{1}{2}, \qquad q(e^{27\kappa'}) \le \tfrac{1}{2}, \qquad q(p) = 54\pi C_V a' p/M. \tag{11.3}$$

Proposition 11.1. *For $A \in \widetilde{\mathcal{G}}_k$, $y \in Y_k$, and $s \in \prod_{\pi_k} \{|s_\square| \le M^{\alpha_0}\}$, the series* (11.2) *converge absolutely in N_γ, uniformly in (s,A), and*

$$N_\gamma(S_y^\#(s,A)) \le 32 a' M^{27\alpha_0} w_k(y), \tag{11.4}$$
$$N_\gamma(\delta_{\alpha,A} S_y^\#(s,A)) \le 304 a' M^{27\alpha_0},$$
$$\|\mathbf{1}_{\Delta_z} S_y^\#(X,A) \mathbf{1}_{\Delta_{z'}}\| \le 32 a' e^{27\kappa'} w_k(y) e^{-\gamma d(z,z') - \kappa' d_M(X)}. \tag{11.5}$$

The norm in the last line is the sup operator norm. At $k = 0$ it can also be replaced by the supremum of the normalized kernel over the two clusters, with $w_0(y) = (1+|y|)^{-1}$. At parameter value one,

$$S_y^\#(1,A) = S_y(A) = \sum_{X \in \mathcal{P}_k} S_y^\#(X,A). \tag{11.6}$$

The weakened operators are continuous on the closed parameter polydisk, jointly analytic in its interior and in A, and analytic in A for each fixed parameter value, including boundary values. For finite real y the families are continuous in y.

The operator $S_y^\#(X,A)$ is supported on X in both arguments and depends on A only on bonds with endpoints in X. If parameters vanish off a union Y of M-cubes, then each component of Y is preserved by $S_y^\#(s,A)$. On such a component the operator depends only on its parameters and its connection. Its row at $x = (c,g)$ vanishes unless all M-cubes meeting $K_c = \{c' : |c'-c|_\infty \le 2L\}$ have active parameters; the corresponding assertion holds for columns.

Proof. By (11.1), $|s_\omega| \le p_s^{n+1}$ with $p_s = M^{27\alpha_0}$. Apply the walk row estimate with this weight. The length-n contribution is bounded by $16 a' p_s w_k(y) 2^{-n}$; its Hölder row estimate is bounded by $152 a' p_s 2^{-n}$. Summing these bounds proves the first two estimates. For a fixed polymer use

$$\mathbf{1}[X_\omega = X] \le e^{-\kappa' d_M(X)} p_X^{n+1}, \qquad p_X = e^{27\kappa'}.$$

The same row estimate with p_X proves (11.5). The left walk estimates follow from the left parametrix by the same weighted product estimate; taking the average preserves all bounds.

On a fixed torus, summing the row bounds over its finitely many unit blocks also bounds the sum of the N_γ norms of the individual terms. The tails decay geometrically, uniformly in (s,A). The series may therefore be regrouped by X_ω. At $s = 1$ they are the right and left inverse expansions, so both equal $S_y(A)$. This proves (11.6). Each finite sum is analytic in its connection variables and polynomial in s. The uniform tail estimates give the stated analyticity and continuity. The local inverse is continuous at every real resolvent parameter. Its walk terms

inherit this property, and their tails are uniform in y since $w_k(y) \le 1$. The sums are therefore continuous in real y.

A walk term is a product of local factors on the neighbourhoods $\square_i^{\sim 1}$. Its row and column cutoffs, its commutators, and all connection variables consequently lie in X_ω. If the first row is nonzero, its photon site satisfies $|c - z_{\square_0}|_\infty < 3M/4$. Since $3M/4 + 2L < 3M/2$, every M-cube meeting K_c is in $\square_0^{\sim 1}$, and its parameter occurs in s_ω. The last cutoff gives the column assertion; the order is reversed for a left walk. Finally, a surviving walk has every cube of X_ω in Y. Since X_ω is face-connected, it lies in a single component of Y. Its local factors use only the connection in that component. This proves all the locality statements. □

The lattice symmetries used in this section have coordinate sign changes and translations in $M\mathbb{Z}^3$. Their spin lifts lie in the group generated by $\gamma_1, \gamma_2, \gamma_3, \gamma_5$; they act on (c, g) by $(rc + a, rg)$, where $S\gamma_\mu S^{-1} = r_{\mu\mu}\gamma_\mu$ determines r. Write $\widetilde{g} = (S, a)$ for such a symmetry. The step group restricts a to $LM\mathbb{Z}^3$. A lift commuting with γ_5 preserves y; a lift anticommuting with it sends y to $-y$ at nonneutral values. Gauge covariance follows term by term: the conjugating phases of adjacent factors cancel. A lattice symmetry permutes cubes and walks, and hence acts on the parameters by $(\widetilde{g}s)_\square = s_{\widetilde{g}^{-1}\square}$. Transposition reverses the order of the local factors and interchanges the right and left expansions. More precisely, with $\mathsf{C} = i\gamma_2$ one has $F_y(A)^T = \mathsf{C}^{-1}F_{-y}(-A)\mathsf{C}$. Therefore

$$S_y(s, A)^T = \mathsf{C}^{-1}S'_{-y}(s, -A)\mathsf{C}. \tag{11.7}$$

Thus at the neutral values $y = 0, \infty$ for $k \ge 1$, and $y = 0$ for $k = 0$, the symmetrized family has the same transpose covariance as the exact inverse. This is the reason for using it in the coarse covariance and determinant.

11.2. **The fine and coarse resolvents.** For $k \ge 1$ let

$$\mathcal{Q}_k = Q_k(A)J^*, \qquad \mathcal{Q}_k^\sharp = JQ_k^T(-A), \qquad P_L = Q^T(-A)Q(A).$$

Here Q_k averages the fine lattice onto the unit lattice, and Q averages the unit lattice onto the lattice of spacing L. The notation $\sharp$ denotes the displayed algebraic transpose with reversed connection; for complex A it is not a Hermitian adjoint. Here $B(z') = E_{z'}$ is the L-block from the preceding section. To specify the phases, if $c \in \Delta_z$ and $z \in B(z')$, set

$$\varphi_A(z, c) = e^{ie_k\eta_k(\tau_k A)(z,c)}, \qquad U_A(z', z) = e^{ie_k\eta_k(\tau A)(z',z)}.$$

The path average τA is the arithmetic mean of the oriented bond sums in the six coordinate orders, and $\tau_k A$ is their sum along the nested block centres. Then

$$\begin{aligned}
(\mathcal{Q}_k f)(z) &= \eta_k^3 \sum_{c \in \Delta_z} \varphi_A(z, c)\frac{1}{48}\sum_g f(c, g),\\
(\mathcal{Q}_k^\sharp v)(c, g) &= \varphi_A(z, c)^{-1}v(z),\\
(P_L v)(z) &= U_A(z', z)^{-1}L^{-3}\sum_{z'' \in B(z')} U_A(z', z'')v(z'').
\end{aligned}$$

The nesting of the odd blocks and cancellation of the phases give

$$\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1, \quad P_L^2 = P_L, \quad \mathcal{Q}_k^\sharp\mathcal{Q}_k = \Pi_1(A), \quad \mathcal{Q}_k^\sharp P_L\mathcal{Q}_k = \Pi_L(A). \tag{11.8}$$

All these operators commute with constant spin matrices.

Write $S_k = S_\infty(A)$ and define maps on the unit spinors by

$$\begin{aligned}
H_k &= b_k S_k \mathcal{Q}_k^\sharp, & \mathfrak{H}_k &= b_k\mathcal{Q}_k S_k,\\
D_k &= b_k - b_k^2\mathcal{Q}_k S_k\mathcal{Q}_k^\sharp, & M_k &= b_k - D_k.
\end{aligned} \tag{11.9}$$

For real y put

$$\mathcal{B}_{k,y} = (b_k + i\gamma_5 y)^{-1}(1 - P_L) + (b_k + b_L + i\gamma_5 y)^{-1} P_L.$$

These 4×4 inverses exist since $(t + i\gamma_5 y)^{-1} = (t - i\gamma_5 y)/(t^2 + y^2)$ for $t > 0$. The idempotent decomposition and (11.8) give

$$\mathcal{B}_{k,y}^{-1} = b_k + b_L P_L + i\gamma_5 y, \qquad F_y = F_\infty - b_k^2 \mathcal{Q}_k^\sharp \mathcal{B}_{k,y} \mathcal{Q}_k. \tag{11.10}$$

Proposition 11.2. *For $k \geq 1$, $A \in \widetilde{\mathcal{G}}_k$, and real y, the operator $D_k + b_L P_L + i\gamma_5 y$ is invertible and its inverse is*

$$\Gamma_{k,y} = \mathcal{B}_{k,y} + b_k^2 \mathcal{B}_{k,y} \mathcal{Q}_k S_y \mathcal{Q}_k^\sharp \mathcal{B}_{k,y}. \tag{11.11}$$

Moreover,

$$S_y = S_k + H_k \Gamma_{k,y} \mathfrak{H}_k. \tag{11.12}$$

In particular, $\Gamma_k = \Gamma_{k,0}$ exists and $\delta Z_k(A) = \det(D_k + b_L P_L) \neq 0$.

Proof. Set $Z = b_k^2 \mathcal{Q}_k^\sharp \mathcal{B}_{k,y} \mathcal{Q}_k$. The two fine inverses exist by the preceding section, and (11.10) gives both identities

$$S_y - S_k = S_k Z S_y = S_y Z S_k.$$

Since $D_k + b_L P_L + i\gamma_5 y = \mathcal{B}_{k,y}^{-1} - b_k^2 \mathcal{Q}_k S_k \mathcal{Q}_k^\sharp$, multiplication by the right side of (11.11) gives the identity: the additional term is $b_k^2 \mathcal{Q}_k (S_y - S_k - S_k Z S_y) \mathcal{Q}_k^\sharp \mathcal{B}_{k,y} = 0$. The product in the other order uses $S_y - S_k - S_y Z S_k = 0$. This proves the inverse formula without assuming that the coarse operator is invertible. Substituting the formula in $H_k \Gamma_{k,y} \mathfrak{H}_k$ gives

$$S_k Z S_k + S_k Z S_y Z S_k = S_k Z (S_k + S_y Z S_k) = S_k Z S_y = S_y - S_k.$$

At $y = 0$ the coarse operator is an invertible map on the finite dimensional unit spinor space, so its determinant is nonzero. □

At level zero this algebra is replaced by a single chamber integration. The operators b_0, S_∞, $\mathcal{B}_{0,y}$, and M_0 are not defined. Instead,

$$F_0 = D_A + \mu + b_L J P_L J^*, \qquad \Gamma_0 = F_0^{-1}, \qquad \delta Z_0(A) = \det F_0 \neq 0. \tag{11.13}$$

Here Γ_0 acts on 192 components per photon site, whereas Γ_k for $k \geq 1$ acts on four components per unit site.

11.3. **Localized coarse covariances.** The weakened families for $k \geq 1$ are

$$\begin{aligned} H_k(s, A) &= b_k S_\infty(s, A) \mathcal{Q}_k^\sharp, \\ \Gamma_k(s, A) &= \mathcal{B}_{k,0} + b_k^2 \mathcal{B}_{k,0} \mathcal{Q}_k S_0^{\mathrm{sy}}(s, A) \mathcal{Q}_k^\sharp \mathcal{B}_{k,0}, \\ M_k(X, A) &= b_k^2 \mathcal{Q}_k S_\infty^{\mathrm{sy}}(X, A) \mathcal{Q}_k^\sharp. \end{aligned} \tag{11.14}$$

The unprimed polymer propagators $S_{k,y}(X, A) = S_y(X, A)$ are used in the determinant integral. At level zero let $\mathrm{Cub}(c)$ be the M-cubes having a lattice site in common with K_c, and set

$$\begin{aligned} (H_0(s, A)u)(c, g) &= \left(\prod_{\square \in \mathrm{Cub}(c)} s_\square \right) u(c, g), \\ \Gamma_0(s, A) &= S_0^{\mathrm{sy}}(s, A), \qquad S_{0,y}(X, A) = S_y(X, A). \end{aligned} \tag{11.15}$$

At $s = 1$ these are the exact operators; also $\sum_X M_k(X, A) = M_k(A)$. The exponential formulas for the averaging phases and Proposition 11.1 give analyticity on the same domains.

For use in the estimates, let $\mathfrak{i}(A) = e_k \| \operatorname{Im} A\|_\infty$. The small-connection domain and the charge bound give $\mathfrak{i}(A) \le 1/(6L)$; reduce γ, if necessary, so that $\gamma L \le 1/6$. The path lengths imply

$$|\varphi_A(z,c)^{\pm 1}| \le e^{3\mathfrak{i}(A)/2}, \qquad \|\mathcal{Q}_k\|_{\infty\to\infty}, \|\mathcal{Q}_k^\sharp\|_{\infty\to\infty} \le e^{3\mathfrak{i}(A)/2}. \tag{11.16}$$

The sup norms here are formed with the Euclidean spinor norm. Each unit block has chamber measure one. In particular, for a fine operator O,

$$|(\mathcal{Q}_k O \mathcal{Q}_k^\sharp)(z,z')|_2 \le \exp(1)\, \|\mathbf{1}_{\Delta_z} O \mathbf{1}_{\Delta_{z'}}\|. \tag{11.17}$$

Indeed, apply the two averaging bounds to a unit spinor supported at z'; their product is $e^{3\mathfrak{i}(A)} \le \exp(1)$.

Proposition 11.3. *The families* (11.14) *satisfy*

$$\|H_k(s,A)\|_{\infty\to\infty} \le 32\exp(1) b_k a' M^{27\alpha_0}, \tag{11.18}$$

$$\|\delta_{\alpha,A} H_k(s,A)\|_{\infty\to\infty} \le 304\exp(1) b_k a' M^{27\alpha_0},$$

$$|\Gamma_k(s,A)(z,z')|_2 \le C_\Gamma M^{27\alpha_0} e^{-\gamma d(z,z')}, \tag{11.19}$$

$$|M_k(X,A)(z,z')|_2 \le 32\exp(1) b_k^2 a' e^{27\kappa'} e^{-\gamma d(z,z') - \kappa' d_M(X)},$$

where

$$C_\Gamma = \frac{2(1+2\exp(1))}{b_-} + 128\exp(1)(1+2\exp(1))^2 a'.$$

The Hölder estimate is on short pairs, including the distinct sites of one cluster. The families $H_k(s,A)$ and $\Gamma_k(s,A)$ preserve every component of the active M-cubes and depend there only on its parameters and connection. The row of $H_k(s,A)$ at (c,g) vanishes unless all cubes meeting K_c are active. The polymer $M_k(X,A)$ is supported on the unit sites of X and depends only on the connection in X.

At level zero the normalized kernel of $\Gamma_0(s,A)$ is bounded by $32a'M^{27\alpha_0}e^{-\gamma d(c,c')}$, and (11.5) *bounds $S_{0,y}(X,A)$ with its factor $(1+|y|)^{-1}$. The multiplier $H_0(s,A)$ has sup norm at most $M^{8\alpha_0}$ and equals the identity at $s=1$. Its row at (c,g) vanishes unless every cube meeting K_c is active. There are no short pairs of distance less than one at this level.*

Proof. The first two estimates follow by composing the weakened fine bounds with (11.16). The third and fourth use (11.17). For the covariance it remains to control the two factors $\mathcal{B}_{k,0}$. Its kernel is

$$\mathcal{B}_{k,0}(z,z') = b_k^{-1}\mathbf{1}[z=z']1_4 - \frac{b_L}{b_k(b_k+b_L)} P_L(z,z').$$

It vanishes unless z, z' belong to one L-block. Since $|P_L(z,z')|_2 \le \exp(1)L^{-3}$ on that block, its row and column sums are at most $(1+2\exp(1))/b_k$. In a nonzero term of $\mathcal{B}_{k,0}G\mathcal{B}_{k,0}$, where $G = \mathcal{Q}_k S_0^{\mathrm{sy}}(s,A)\mathcal{Q}_k^\sharp$, the two intermediate sites are within L of the exterior sites. Therefore

$$e^{-\gamma d(z_1,z_2)} \le e^{2\gamma L} e^{-\gamma d(z,z')} \le 4 e^{-\gamma d(z,z')}.$$

The fine bound and (11.17) give $|G(z_1,z_2)|_2 \le 32\exp(1)a'M^{27\alpha_0}e^{-\gamma d(z_1,z_2)}$. Multiplying the row and column sums cancels the factor b_k^2 and gives $128\exp(1)(1+2\exp(1))^2 a' M^{27\alpha_0}$. The local term $\mathcal{B}_{k,0}$ is bounded by $2(1+2\exp(1))b_-^{-1}e^{-\gamma d(z,z')}$, proving (11.19).

The odd block partitions are nested, and every unit block and every L-block belongs to a single M-cube. Moreover, the paths entering $\varphi_A(z,c)$ lie in Δ_z, and the paths entering $U_A(z',z)$ lie in the fine block of side L centred at z': each coordinate of a rectilinear path remains between those of its endpoints. Consequently the averaging maps and $\mathcal{B}_{k,0}$ preserve every union of M-cubes and depend only on its connection. Composing them with the component-local fine families proves the locality assertions. This applies also to the parameter-independent term $\mathcal{B}_{k,0}$ in $\Gamma_k(s,A)$. For the

row of H_k, the terminal average does not change the first row index of $S_\infty(s,A)$, so the stronger vanishing statement is preserved. At level zero the assertions for Γ_0 follow from the fine kernel bound and locality with $\eta_0 = 1$.

For H_0, lift c and K_c to the lattice. In each coordinate K_c is an interval of $4L+1$ sites. Since $M \geq L^2$ and $L \geq 5$, this interval meets at most two consecutive M-blocks. The cubes meeting K_c are the product of these three sets of blocks. They number at most eight, form a face-connected set, and include the cube of c. Thus the defining product is a monomial of degree at most eight, bounded by $M^{8\alpha_0}$. If one of its cubes is inactive, its factor is zero. If all are active, their connectedness puts them in the active component containing c; hence changing parameters outside that component has no effect on the row or column at (c,g). The multiplier is scalar in spin and chamber labels and independent of the connection. These observations give its analyticity, gauge covariance, and component locality. □

These families have the gauge and lattice covariance of their exact counterparts. For Γ_k and M_k, transposition follows from (11.7) at a neutral value and the scalar averaging phases. The barred field map is $\bar{H}_k(s,A) = \mathsf{C}^{-1} H_k(s,-A)\mathsf{C}$. Its bounds follow from those for H_k because C is unitary and $\mathsf{C}\gamma_\mu\mathsf{C}^{-1} = -\gamma_\mu^T$. When component sup norms are used in the Grassmann substitution, the Euclidean operator bounds above acquire at most the factor two from $|v|_c \leq |v|_2 \leq 2|v|_c$.

11.4. **Weighted Grassmann integration.** Let $\phi_i(t)$, $1 \leq i \leq p$, be independent Grassmann families on finite ordered index sets. If $\phi_i(I_i)$ is the ordered product over I_i, write

$$F = \sum_{I_1,\dots,I_p} c_F(I_1,\dots,I_p)\phi_1(I_1)\cdots\phi_p(I_p), \qquad \|F\|_{\mathsf{w}} = \sum_I |c_F(I)| \prod_i \mathsf{w}_i^{|I_i|}.$$

The positive measures $\nu_i(t)$ on the index sets enter the kernels, not this coefficient norm. Thus an antisymmetric kernel expansion

$$F = \sum_n \prod_i \frac{1}{n_i!} \int \mathrm{d}t\, K_n(t) \prod_i \prod_{j=1}^{n_i} \phi_i(t_j^i)$$

has norm

$$\|F\|_{\mathsf{w}} = \sum_n \prod_i \frac{\mathsf{w}_i^{n_i}}{n_i!} \int \mathrm{d}t\, |K_n(t)|. \tag{11.20}$$

For arbitrary kernels the right side is an upper bound. Products of ordered monomials vanish on repeated generators and otherwise multiply their weights, so

$$\|FG\|_{\mathsf{w}} \leq \|F\|_{\mathsf{w}}\|G\|_{\mathsf{w}}, \qquad \|e^F - 1\|_{\mathsf{w}} \leq e^{\|F\|_{\mathsf{w}}} - 1 \quad (F \text{ even}). \tag{11.21}$$

For an additional finite nonempty mode set $\mathcal{I}$ with positive weights $\nu(t)$, introduce $W(t), \bar{W}(t)$ and set

$$\langle \bar{W}, OW\rangle_\nu = \sum_{t,t'} \nu(t)\nu(t')\bar{W}(t)O(t,t')W(t'), \qquad \mathrm{Top} = \prod_{t\in\mathcal{I}} W(t)\bar{W}(t).$$

The kernel represents O with measure ν; its matrix entry is $O(t,t')\nu(t')$. Define $\int \mathrm{D}W$ as extraction of the coefficient of Top, multiplied by $\prod_t \nu(t)^{-1}$, and define

$$\int \mathrm{d}\mu_I(W) F = \int \mathrm{D}W\, e^{-\langle \bar{W}, W\rangle_\nu} F, \tag{11.22}$$

$$\int \mathrm{d}\mu_C(W) F = (\det C) \int \mathrm{D}W\, e^{-\langle \bar{W}, C^{-1}W\rangle_\nu} F \quad (C \text{ invertible}).$$

These are algebra-valued functionals when other Grassmann generators are present. The order of modes in Top is immaterial because its factors are even.

Proposition 11.4. *Put* $w_W = \max_t \nu(t)^{-1/2}$. *Then*

$$\left\| \int \mathrm{d}\mu_I(W) F \right\|_{\mathsf{w}} \le \|F\|_{(\mathsf{w}, w_W)}. \tag{11.23}$$

The integral factors for polynomials involving disjoint sets of modes. For an invertible covariance,

$$\int \mathrm{d}\mu_C(W) F(\bar W, W) = \int \mathrm{d}\mu_I(W) F(\bar W, CW), \tag{11.24}$$

$$\int \mathrm{d}\mu_C(W) W(t) \bar W(t') = C(t,t'),$$

$$\int \mathrm{D}W \, e^{-\langle \bar W, OW\rangle_\nu} = \det O \quad \textit{for every } O. \tag{11.25}$$

Finally, consider a linear substitution $\phi_i \mapsto \sum_j O_{ij}\theta_j$, *and put*

$$\|O_{ij}\|_{1,\infty} = \sup_t \sum_s |O_{ij}(t,s)| \nu_j'(s).$$

If $\sum_j \|O_{ij}\|_{1,\infty} \mathsf{w}_j' \le \mathsf{w}_i$ *for each* i, *then its image* F' *obeys* $\|F'\|_{\mathsf{w}'} \le \|F\|_{\mathsf{w}}$. *Only rows corresponding to generators occurring in* F *need satisfy this bound.*

Proof. The identity density factors as

$$e^{-\langle \bar W, W\rangle_\nu} = \prod_t (1 + \nu(t) W(t) \bar W(t)).$$

An ordered monomial $W(J)$ contributes to its integral only if J contains both generators of every mode it meets. In that case its integral is its reordering sign times $\prod_{(t,0)\in J} \nu(t)^{-1}$; otherwise it is zero. Its modulus is therefore at most $\prod_{u\in J} \nu(u)^{-1/2} \le w_W^{|J|}$. Apply this coefficientwise to obtain (11.23). For disjoint mode sets a nonzero integral requires each monomial separately to be paired. Their degrees in W are then even, so the signs from exchanging external generators cancel. The paired-mode formula gives factorization. It also shows that modes absent from the integrand may be removed.

A linear change $W \mapsto OW$ multiplies the coefficient of the top monomial by $\det O$, since the matrix of the change is $O(t,t')\nu(t')$. Applied to the identity density this proves (11.25), without an invertibility assumption. For invertible C, use $W \mapsto CW$, leaving $\bar W$ fixed, in the density with covariance C. This proves (11.24) and normalization. Its two-point function is

$$\sum_s C(t,s)\nu(s) \int \mathrm{d}\mu_I(W) W(s) \bar W(t') = \sum_s C(t,s)\nu(s) \frac{\mathbf{1}[s=t']}{\nu(t')} = C(t,t').$$

Thus it is the kernel, rather than the matrix entry.

Under the stated substitution, the norm of the image of $\phi_i(t)$ is at most

$$\sum_j \mathsf{w}_j' \sum_s |O_{ij}(t,s)| \nu_j'(s) \le \mathsf{w}_i$$

by hypothesis. Submultiplicativity bounds the image of each monomial by its original weight; summing its coefficients proves the last assertion, with the same proof on the subalgebra generated by the rows that occur. □

For linear combinations $v_1, \dots, v_{2r}$ of the $W, \bar W$, Wick's formula gives

$$\int \mathrm{d}\mu_I(W) v_1 \cdots v_{2r} = \operatorname{Pf}\left(\int \mathrm{d}\mu_I(W) v_i v_j \right)_{i,j}. \tag{11.26}$$

Odd products integrate to zero. To fix the signs, let ∂_u be the left Grassmann derivative. It satisfies $\partial_u(vG) = (\partial_u v)G - v\partial_u G$, and its Berezin integral is zero. Differentiating the factored Gaussian density gives

$$\int \mathrm{d}\mu_I(W)W(t)G = \nu(t)^{-1}\int \mathrm{d}\mu_I(W)\partial_{\bar{W}(t)}G,$$
$$\int \mathrm{d}\mu_I(W)\bar{W}(t)G = -\nu(t)^{-1}\int \mathrm{d}\mu_I(W)\partial_{W(t)}G.$$

Consequently, if $I(v) = \int \mathrm{d}\mu_I(W)v$,

$$I(v_1\cdots v_q) = \sum_{j=2}^{q}(-1)^j I(v_1v_j)I(v_2\cdots\widehat{v}_j\cdots v_q).$$

This is the expansion of the Pfaffian along its first row and proves (11.26) by induction. Formula (11.24) gives the corresponding Wick rule for μ_C.

In the renormalization step the modes have weight one for $k \geq 1$. At $k = 0$ they are (c, g, β) with weight $1/48$. Thus the integration weight is

$$w_W = 1 \quad (k \geq 1), \qquad w_W = \sqrt{48} \quad (k = 0). \tag{11.27}$$

In particular the first-step identity covariance has two-point function $48\mathbf{1}[(c, g, \beta) = (c', g', \beta')]$. The pair index used for a Hölder field has measure $(\eta_k^3/48)^2$. In a substitution involving $\delta_{\alpha,A}H_k$, its scalar component row sum is computed with the input measure, as in Proposition 11.4. Internal-cluster differences are therefore controlled by the Hölder bound with the same field weight as other pairs.

11.5. **The Gaussian block identity.** For this identity use a common notation for the integration space: at $k \geq 1$ let $\mathcal{D}_k = D_k$ and $\mathcal{A}_k(A) = Q(A)$; at $k = 0$ let $\mathcal{D}_0 = D_A + \mu$ and $\mathcal{A}_0(A) = Q(A)J^*$. The integration measure is respectively counting measure or the chamber measure $1/48$. The target is the four-component spinor space on the lattice of spacing L, with measure L^3. Set

$$\mathcal{A}_k^\sharp(A) = \mathcal{A}_k(-A)^T, \qquad K_k = \mathcal{D}_k + b_L\mathcal{A}_k^\sharp\mathcal{A}_k, \qquad \Gamma_k = K_k^{-1}.$$

Thus $K_k = D_k + b_L P_L$ for $k \geq 1$ and $K_0 = F_0$. Define the quadratic operator on the coarse lattice by

$$D_{k+1}^0 = b_L - b_L^2\mathcal{A}_k\Gamma_k\mathcal{A}_k^\sharp.$$

Let $\Phi, \bar{\Phi}$ be external coarse Grassmann fields and put

$$\Psi^c = b_L\Gamma_k\mathcal{A}_k^\sharp\Phi, \qquad \bar{\Psi}^c = b_L\Gamma_k^T\mathcal{A}_k(A)^T\bar{\Phi}.$$

For $\Psi = \Psi^c + W'$ and $\bar{\Psi} = \bar{\Psi}^c + \bar{W}'$, completion of the square is the exact identity

$$\langle\bar{\Psi}, \mathcal{D}_k\Psi\rangle + b_L\langle\bar{\Phi} - \mathcal{A}_k(-A)\bar{\Psi}, \Phi - \mathcal{A}_k(A)\Psi\rangle_L = \langle\bar{\Phi}, D_{k+1}^0\Phi\rangle_L + \langle\bar{W}', K_kW'\rangle. \tag{11.28}$$

Indeed, write $s = b_L\mathcal{A}_k^\sharp\Phi$ and $\bar{s} = b_L\mathcal{A}_k(A)^T\bar{\Phi}$. The left side is

$$\langle\bar{\Psi}, K_k\Psi\rangle - \langle\bar{\Psi}, s\rangle - \langle\bar{s}, \Psi\rangle + b_L\langle\bar{\Phi}, \Phi\rangle_L.$$

Substitute the means $\Gamma_k s, \Gamma_k^T\bar{s}$. Both cross terms cancel, and the remaining constant is $b_L\langle\bar{\Phi}, \Phi\rangle_L - \langle\bar{s}, \Gamma_k s\rangle = \langle\bar{\Phi}, D_{k+1}^0\Phi\rangle_L$. All factors remain in their displayed order; no conjugation of Grassmann fields is involved.

Translation by external odd fields preserves the top coefficient in the integration variables. Applying this to (11.28) and then using (11.24) gives, for any Grassmann polynomial F, the block integration formula

$$\int \mathrm{D}\Psi\, e^{-\langle\bar\Psi,\mathcal{D}_k\Psi\rangle - b_L\langle\bar\Phi-\mathcal{A}_k(-A)\bar\Psi,\Phi-\mathcal{A}_k(A)\Psi\rangle_L} F(\bar\Psi,\Psi) \\ = \delta Z_k(A) e^{-\langle\bar\Phi, D^0_{k+1}\Phi\rangle_L} \int \mathrm{d}\mu_I(W) F(\bar\Psi^c + \bar W, \Psi^c + \Gamma_k W). \tag{11.29}$$

At the first step this is a single integration over all chamber components. The determinant and the integration weight are those of (11.13) and (11.27); the heavy components are included in both.

11.6. **An integral for the determinant.** The determinant in (11.29) has a local logarithm obtained by inserting the polymer propagators into a resolvent integral. We first derive the integral at real connections. Put $n_k = \ell^3$ and

$$c_k^d = 4\big((1-L^{-3})\log b_k + L^{-3}\log(b_k+b_L)\big) \quad (k\ge 1), \qquad c_0^d = 0.$$

For real $A \in \widetilde{\mathcal{G}}_k$ and $0 < \mu \le 1$,

$$\delta Z_k(A) = \exp\left\{ c_k^d n_k - i b_k^2 \int_0^\infty \mathrm{d}y\ \mathrm{Tr}\big[\gamma_5 \mathcal{B}_{k,y} \mathcal{Q}_k S_y \mathcal{Q}_k^\sharp \mathcal{B}_{k,y}\big] \right\} \quad (k\ge 1), \tag{11.30}$$

$$\delta Z_0(A) = \exp\left\{ \int_1^\infty \frac{\mathrm{d}y}{y}\, \mathrm{Tr}[F_0 S_y] - i\int_0^1 \mathrm{d}y\ \mathrm{Tr}[\gamma_5 S_y] \right\}. \tag{11.31}$$

The trace in the first line is on the unit spinor space. At level zero it is the normalized chamber trace $\mathrm{Tr}\, O = 48^{-1}\sum_{c,g} \mathrm{tr}\, O((c,g),(c,g))$.

To prove these identities, write $K = K_k$ as in the preceding subsection. At real A, the fine adjoint identity $D_A^* = \gamma_5 D_A \gamma_5$, the reality of μ, b_k, b_L, and the orthogonal averaging projections give $K^* = \gamma_5 K \gamma_5$. Hence $K\gamma_5$ is Hermitian and invertible. Since $\det\gamma_5 = 1$ on each four-component spinor, its determinant is $\det K$. Let λ_j be its real nonzero eigenvalues, counted with multiplicity. There are $4n_k$ at positive levels and $192 n_0$ at level zero. Then

$$\mathrm{Tr}\big[\gamma_5 (K+i\gamma_5 y)^{-1}\big] = \sum_j (\lambda_j + iy)^{-1}, \qquad \mathrm{Tr}\big[K(K+i\gamma_5 y)^{-1}\big] = \sum_j \lambda_j(\lambda_j+iy)^{-1}.$$

For a real $\lambda \ne 0$, choose along $y \ge 0$ the logarithm with value $\log|\lambda| + i\pi \mathbf{1}[\lambda<0]$ at zero. Its derivative is $i/(\lambda+iy)$ and $\mathrm{Log}(\lambda+iy) - \log y \to i\pi/2$. Integration gives

$$-i\int_0^Y \frac{\mathrm{d}y}{\lambda+iy} = \mathrm{Log}\,\lambda - \mathrm{Log}(\lambda+iY),$$
$$\int_1^\infty \frac{\mathrm{d}y}{y}\frac{\lambda}{\lambda+iy} - i\int_0^1 \frac{\mathrm{d}y}{\lambda+iy} = \mathrm{Log}\,\lambda - i\pi/2.$$

Summing the second identity proves (11.31): its constant phase is $-96\pi i n_0 \in 2\pi i\mathbb{Z}$.

For $k \ge 1$, the ranks of P_L and $1-P_L$ are respectively $4L^{-3}n_k$ and $4(1-L^{-3})n_k$. The explicit formula for $\mathcal{B}_{k,y}$ therefore gives

$$-i\int_0^Y \mathrm{d}y\ \mathrm{Tr}[\gamma_5 \mathcal{B}_{k,y}] = -4n_k \log Y + c_k^d n_k + o(1) \quad (Y\to\infty).$$

Insert (11.11) in the first scalar integral. The terms $4n_k \log Y$ cancel. The remaining phase is $2\pi i n_k$, which disappears on exponentiation, and this proves (11.30). The integral converges absolutely: block compression bounds $\mathcal{Q}_k S_y \mathcal{Q}_k^\sharp$ uniformly, while each row and column sum of $\mathcal{B}_{k,y}$ is at most $(1+2\exp(1))(b_k^2+y^2)^{-1/2}$. Its trace is bounded by

$$128\exp(1)(1+2\exp(1))^2 a' n_k (b_k^2+y^2)^{-1}.$$

At level zero the spectral identities already give absolute convergence. The uniform kernel bounds below will give a bound independent of the eigenvalues.

11.7. **Local determinant terms.** Choose $\gamma_{\mathrm{w}} \le \gamma$ and $\kappa_{\mathrm{w}} = \kappa + \kappa_0 + 3$ with $\kappa, \kappa_0 > 0$. The propagator bounds above give a fixed constant C_w such that, in component sup norms, for $y \ge 0$,

$$\|\mathbf{1}_{\Delta_z} S_{k,y}(X,A)\mathbf{1}_{\Delta_{z'}}\| \le C_w e^{-\gamma_{\mathrm{w}} d(z,z') - \kappa_{\mathrm{w}} d_M(X)} \quad (k \ge 1),$$
$$|S_{0,y}(X,A)(x,x')|_c \le C_w (1+y)^{-1} e^{-\gamma_{\mathrm{w}} d(c,c') - \kappa_{\mathrm{w}} d_M(X)}.$$

Here $|\cdot|_c$ is the matrix norm induced by the component sup norm. We may use the same C_w after reducing the rates from γ, κ'. For $X \in \mathcal{P}_k$ define

$$\widetilde{E}_k(X,A) = -i b_k^2 \int_0^\infty \mathrm{d}y\ \mathrm{Tr}\big[\gamma_5 \mathcal{B}_{k,y} \mathcal{Q}_k S_{k,y}(X,A) \mathcal{Q}_k^\sharp \mathcal{B}_{k,y}\big] \quad (k \ge 1), \tag{11.32}$$
$$\widetilde{E}_0(X,A) = \int_1^\infty \frac{\mathrm{d}y}{y} \mathrm{Tr}[F_0 S_{0,y}(X,A)] - i \int_0^1 \mathrm{d}y\ \mathrm{Tr}[\gamma_5 S_{0,y}(X,A)],$$
$$E_k^d(X,A) = \tfrac{1}{4}\big(\widetilde{E}_k(X,A) + \widetilde{E}_k(X,-A) + \overline{\widetilde{E}_k(X,\overline{A})} + \overline{\widetilde{E}_k(X,-\overline{A})}\big),$$
$$E_k^{\det}(X,A) = E_k^d(X,A) - E_k^d(X,0).$$

Set $c_X = 54$ and

$$C_d = 16 c_X C_w \max\left\{\tfrac{\pi}{2} b \exp(1)(1+2\exp(1)),\ 2 + 58\exp(1) + b\exp(1)\right\}.$$

Proposition 11.5. *The integrals* (11.32) *converge absolutely for $A \in \widetilde{\mathcal{G}}_k$. They define analytic functions, depending only on bonds meeting X, with*

$$|\widetilde{E}_k(X,A)|, |E_k^d(X,A)| \le C_d M^3 e^{-(\kappa+1) d_M(X)}. \tag{11.33}$$

They satisfy

$$\delta Z_k(A) = e^{c_k^d n_k + \sum_X \widetilde{E}_k(X,A)}, \qquad \frac{\delta Z_k(A)}{\delta Z_k(0)} = e^{\sum_X E_k^{\det}(X,A)}. \tag{11.34}$$

The functions E_k^d are gauge invariant, even in A, and satisfy $\overline{E_k^d(X,\overline{A})} = E_k^d(X,A)$. They are covariant under the lattice step group.

Proof. For $k \ge 1$ write $N_y(X) = \mathcal{Q}_k S_{k,y}(X,A) \mathcal{Q}_k^\sharp$. It is supported on the unit sites of X, and each entry is bounded by $\exp(1) C_w e^{-\kappa_{\mathrm{w}} d_M(X)}$ by normalized block compression. To estimate the trace, cycle one factor $\mathcal{B}_{k,y}$ to the left and use

$$G_y = \mathcal{B}_{k,y}\gamma_5\mathcal{B}_{k,y} = \gamma_5\big((b_k + i\gamma_5 y)^{-2}(1-P_L) + (b_k + b_L + i\gamma_5 y)^{-2} P_L\big).$$

Its column sum in Euclidean matrix norm is at most $(1+2\exp(1))/(b_k^2+y^2)$: the identity contributes one and the two projection coefficients contribute at most $2\exp(1)$. Since $|\mathrm{tr}(mm')| \le 8|m|_2 \max_{i,j}|m'_{ij}|$ for four-component matrices, the integrand is bounded by

$$8\exp(1)(1+2\exp(1)) C_w M^3 |X|_M e^{-\kappa_{\mathrm{w}} d_M(X)} (b_k^2 + y^2)^{-1}. \tag{11.35}$$

Integrating and multiplying by b_k^2 leaves the factor $\pi b_k/2 \le \pi b/2$.

At $k = 0$, the formula for D_A and the averaging penalty give the scalar component row bound

$$\sup_{x,\beta} \sum_{\beta'} \int \mathrm{d}x'\, |F_0(A)(x,x')_{\beta\beta'}| \le R_F := 58\exp(1) + 1 + b\exp(1). \tag{11.36}$$

The diagonal and three internal hops contribute at most $41 + \mu$, the external hop at most $6e^{\mathrm{i}(A)}$, and the block penalty at most $b\exp(1)$. The bound follows from $\mu \le 1$ and $e^{\mathrm{i}(A)} \le \exp(1)$. Point evaluation has normalized kernel $48\mathbf{1}[x = x']$; its factor 48 cancels the chamber measure. The two traces in the level-zero integral are consequently bounded by

$$4R_F C_w M^3 |X|_M (1+y)^{-1} e^{-\kappa_{\mathrm{w}} d_M(X)}, \qquad 4C_w M^3 |X|_M (1+y)^{-1} e^{-\kappa_{\mathrm{w}} d_M(X)}.$$

The second bound uses that γ_5 has four entries of modulus one. Their integrals contribute respectively $\int_1^\infty (y(1+y))^{-1}\mathrm{d}y = \log 2$ and $\int_0^1 (1+y)^{-1}\mathrm{d}y = \log 2$.

The geometric bound (A.1) gives $|X|_M \le c_X(d_M(X)+1) \le c_X e^{d_M(X)}$. Applying this to the trace bounds, and using $\kappa_{\mathrm{w}} - 1 \ge \kappa + 1$, proves (11.33) for $\widetilde{E}_k$ and hence for its four-term mean. The majorants are independent of A and integrable in y. Each integrand is analytic in A; Cauchy's formula on a compact polydisk and the majorant permit its contour integrals to be exchanged with the y integral. The result is analytic.

Every phase in the compressed trace belongs to a unit block or an L-block meeting X, and that block is contained in X by nesting. The fine propagator is supported on X and uses only its connection. At level zero a nonzero external-hop trace uses a bond meeting X; the penalty uses the L-block of its row. This proves locality. Gauge transformations conjugate each product under the trace and hence leave it unchanged. A lattice transformation whose spin matrix commutes with γ_5 does the same. Every spatial action of the step group has such a lift: multiplication by $\gamma_1\gamma_2\gamma_3$ changes its commutation sign with γ_5 and leaves its spatial action fixed. This proves step covariance. The mean in (11.32) then makes E_k^d even and real in the stated analytic sense.

For real connections, sum the polymer propagators and use (11.30) or (11.31). This proves the first identity of (11.34) there. Both sides are analytic on $\widetilde{\mathcal{G}}_k$, which is open and star-shaped. They agree on a real neighbourhood of zero, so their Taylor series agree there and analytic continuation proves the identity throughout the domain.

To justify the symmetrized quotient, put $\mathcal{L}(A) = \sum_X \widetilde{E}_k(X, A)$. The charge-conjugation transpose and the complex-conjugation identity give $\delta Z_k(-A) = \delta Z_k(A) = \overline{\delta Z_k(\overline{A})}$. For the latter, the spin matrix $\varkappa = \gamma_5 \mathsf{C}$ conjugates $\overline{K_k(\overline{A})}$ to $K_k(-A)$. Thus $\mathcal{L}(-A) - \mathcal{L}(A)$ and $\overline{\mathcal{L}(\overline{A})} - \mathcal{L}(A)$ take values in $2\pi i\mathbb{Z}$. They are continuous on the connected domain, so they are constant. Subtracting their values at zero gives

$$\sum_X \big(E_k^d(X, A) - E_k^d(X, 0)\big) = \mathcal{L}(A) - \mathcal{L}(0).$$

This proves the second identity in (11.34) without a choice of logarithm branch for the complex determinant. □

11.8. **The small-field gain for the determinant.** For a photon connection write $A_\mu(c) = A(c, c + \eta_k e_\mu)$ and

$$\partial_\nu A_\mu(c) = \eta_k^{-1}(A_\mu(c + \eta_k e_\nu) - A_\mu(c)), \qquad \|\delta_\alpha \partial A\|_\infty = \sup_{\mu,\nu} \sup_{0<d(c,c')\le 1} \frac{|\partial_\nu A_\mu(c') - \partial_\nu A_\mu(c)|}{d(c, c')^\alpha}.$$

Local versions use only differences and bonds inside the given set. In this section the curl is mesh normalized: $(dA)_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Let R_k be the complex bond functions whose three norms $\|A\|_\infty$, $\|\partial A\|_\infty$, and $\|\delta_\alpha \partial A\|_\infty$ are less than $e_k^{-3/4+\varepsilon}$, $e_k^{-3/4+2\varepsilon}$, and $e_k^{-3/4+3\varepsilon}$, respectively. Let $\widetilde{R}_k$ consist of $A_0 + A_1$, with $A_1 \in R_k$ and A_0 real, such that on every $\square^\dagger = \square^{\sim L}$ one can write $A_0 = A' + d\lambda$, where λ, A' are real and the same three local bounds hold for A'. The external connection domain is

$$\mathfrak{X}_k = \{\mathscr{C} : \|\mathscr{C}\|_\infty < e_k^{-1+2\varepsilon}, \quad \|\partial\mathscr{C}\|_\infty < e_k^{-1+3\varepsilon}\}.$$

If $e_k^{1/4} + e_k^\varepsilon < 1$, both $\mathfrak{X}_k$ and $\widetilde{R}_k + \mathfrak{X}_k$ lie in $\widetilde{\mathcal{G}}_k$: on the local cubes the real part has a representative of size $e_k^{-3/4+\varepsilon}$, and the remaining part has size at most $(e_k^{1/4} + e_k^\varepsilon)e_k^{-1+\varepsilon}$.

Put $v_k = -\log e_k$. Assume

$$v_k \ge 9, \qquad e_k^\varepsilon \le \tfrac{1}{4}, \qquad 8M(v_k/4 + 4)e_k^\varepsilon \le 1, \qquad \ell \ge M v_k^2. \tag{11.37}$$

The last condition follows from the stop-scale volume condition $\ell \ge M r_k$ with $r_k \ge v_k^2$.

Proposition 11.6. *For $\mathscr{A} \in \widetilde{R}_k$, $\mathscr{C} \in \mathfrak{X}_k$, and $X \in \mathcal{P}_k$, under* (11.37),

$$|E_k^d(X, \mathscr{A}+\mathscr{C}) - E_k^d(X, \mathscr{C})| \le 8C_d M^3 e_k^{1/4} e^{-\kappa d_M(X)}. \tag{11.38}$$

Proof. If $d_M(X) > v_k/4$, bound the two terms separately by (11.33) and use $e^{-d_M(X)} \le e_k^{1/4}$. Suppose therefore that $d_M(X) \le v_k/4$. The polymer lift and diameter estimates give a non-wrapping lattice box W containing every bond meeting X, with each side of length at most $s_W = M(d_M(X)+3)$. Their applicability follows from $2M(d_M(X)+2) < \ell$ and $M(d_M(X)+4) < \ell$, which hold by (11.37).

Write $\mathscr{A} = A_0 + A_1$ as in the domain definition. On each plaquette, the local real representative gives $\|dA_0\|_\infty < 2e_k^{-3/4+2\varepsilon}$. Choose a fixed-order rectilinear path $\Gamma(o,c)$ from a site $o \in W$ and set $\lambda(c) = \eta_k A_0(\Gamma(o,c))$ on W. Then

$$\|A_0 - d\lambda\|_{\infty,W} \le 2s_W \|dA_0\|_{\infty,W} < 4s_W e_k^{-3/4+2\varepsilon}.$$

To see the factor $2s_W$, append one bond to $\Gamma(o,c)$ and subtract $\Gamma(o,c')$. After cancelling common bonds, the resulting loop bounds at most two rectangles of width η_k. Together they contain at most $2s_W/\eta_k$ plaquettes, each contributing $\eta_k dA_0$. The paths and rectangles remain in W because it is a product of intervals. Extend the real function λ by zero outside the projected box. On every bond meeting X, the bound is unchanged. By (11.37) it is at most $\frac{1}{2}e_k^{-3/4+\varepsilon}$.

Let $B = A_0 - d\lambda + A_1$ on bonds meeting X and zero elsewhere. Then $\|B\|_\infty < \frac{3}{2}e_k^{-3/4+\varepsilon}$. For $|t| \le \frac{1}{3}e_k^{-1/4}$,

$$\|tB + \mathscr{C}\|_\infty < (\tfrac{1}{2} + e_k^{\varepsilon})e_k^{-1+\varepsilon} \le \tfrac{3}{4}e_k^{-1+\varepsilon}.$$

Thus this connection belongs to the complex connection ball $\mathcal{G}_k = \{A : \|A\|_\infty < e_k^{-1+\varepsilon}\}$, which is contained in $\widetilde{\mathcal{G}}_k$, and $f(t) = E_k^d(X, tB+\mathscr{C})$ is analytic on this disk and bounded by the right side of (11.33). Gauge invariance and locality identify the difference in (11.38) with $f(1) - f(0)$. On the circle of radius $r = \frac{1}{4}e_k^{-1/4} > 2$, Cauchy's formula yields

$$f(1) - f(0) = \frac{1}{2\pi i}\oint_{|t|=r} \frac{f(t)}{t(t-1)}\,\mathrm{d}t, \qquad |f(1) - f(0)| \le \frac{\sup|f|}{r-1} \le 8e_k^{1/4}\sup|f|.$$

This proves the claimed gain, retaining the stronger decay $e^{-(\kappa+1)d_M(X)}$ in this case. □

12. Small-field renormalization

The renormalization step separates the vacuum and mass terms from an activity, transports its remaining quadratic part to local averages, and integrates the fluctuation fields. Two estimates are needed before this integration. Extraction must preserve the chamber symmetries and produce a scalar mass. Reconstruction of a coarse field must preserve the support of an activity, including the bonds that cross its boundary. The latter estimate uses cancellation in the chamber Dirac operator and curvature control of the averaging phases.

We use $\gamma_\mu = \sigma_3 \otimes \sigma_\mu$, $\gamma_5 = \sigma_1 \otimes 1_2$ and $\Sigma_3 = \sigma_3 \otimes 1_2$. The matrices acting on the Grassmann variables below are

$$\mathsf{C} = i\gamma_2, \qquad \varkappa = \gamma_5\mathsf{C} = \sigma_2 \otimes \sigma_2.$$

Both are real, $\mathsf{C}^T = \mathsf{C}^{-1} = -\mathsf{C}$, and $\varkappa^T = \varkappa = \varkappa^{-1}$. The regulator matrix in the reflection-positive construction is $C = \sigma_2 \oplus \sigma_2$; the relation in these spin conventions is $\mathsf{C} = i\Sigma_3 C$.

12.1. **Activities, norms and fields.** Fix a level k, mesh $\eta_k = L^{-k}$ and an admissible side ℓ, with $\ell/(LM)$ odd and at least 13. Let $\mathcal{P}_k$ be the collection of face-connected polymers of M-cubes on this torus. A polymer is small when $d_M(X) \le L$. A small polymer has an injective lift, unique up to a period translation, by the diameter and lifting estimates of Appendix A. Write $\mathrm{Vol}(X) = M^3|X|_M$.

The independent Grassmann generators are ψ, $\bar\psi$ on the chamber sites $x = (c, g)$ and χ, $\bar\chi$ on ordered pairs. For distinct chamber sites on the covering lattice, put $d_X(x, x') = \max(|c-c'|_\infty, \eta_k)$ and restrict pairs to $d_X \le M$. Pairs on the torus are simultaneous period classes; their shortest lifts are unique since $2M < \ell$. A pair is admissible for X when its photon sites and all six rectilinear paths between them lie in X, or when it is $(x, x^{(3)})$ and one of its photon sites belongs to X. Thus a pair may leave X only along a bond of type 3. Pairs inside a cluster have $d_X = \eta_k$ and are admissible when their common photon site belongs to X.

Let $\xi = (x, \beta, \omega)$ index an ordinary leg and $\zeta = (x, x', \beta, \omega)$ a Hölder leg, where $\omega = 0$ is unbarred and $\omega = 1$ is barred. We use the index measures

$$\nu(\xi) = \eta_k^3/48, \qquad \nu(\zeta) = (\eta_k^3/48)^2.$$

Integrals sum the spin and bar indices as well. An activity has a unique expansion with separately antisymmetric kernels,

$$\mathscr{E}(X; \mathscr{A}, \mathscr{C}) = \sum_{n,m} \frac{1}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, \mathscr{E}_{nm}(X; \mathscr{A}, \mathscr{C}; \xi; \zeta) \psi(\xi_1) \cdots \psi(\xi_n) \chi(\zeta_1) \cdots \chi(\zeta_m). \tag{12.1}$$

Barred generators are included in this notation. The generator weights, as distinct from the index measures, are

$$h_{k,1} = e_k^{-1/4}, \qquad h_{k,2} = e_k^{-1/4+\varepsilon}.$$

For an element F the weighted coefficient norm is

$$\|F\|_{h_k} = \sum_{n,m} \frac{h_{k,1}^n h_{k,2}^m}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, |F_{nm}|. \tag{12.2}$$

This is the coefficient norm already used for weighted fermion integration. In particular a generator has norm $h_{k,i}$; the weight ν enters only when coefficients are written as integral kernels.

An activity family in $K_k^{x,\mathbb{C}}$ is balanced: every monomial has as many barred as unbarred legs. Its kernels are holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, vanish unless all ordinary legs lie over X and all pairs are admissible for X, and read the two connections only on bonds meeting X. The norm is

$$\|\mathscr{E}\|_k^x = \sup_{X \in \mathcal{P}_k} \exp(\kappa d_M(X)) \sup_{(\mathscr{A}, \mathscr{C}) \in \widetilde{R}_k \times \mathfrak{X}_k} \|\mathscr{E}(X; \mathscr{A}, \mathscr{C})\|_{h_k}. \tag{12.3}$$

We require $\kappa \ge \kappa_0$.

The required symmetries act on both the connections and the generators. For $\tilde g = (S, a)$ with the coordinate sign action $r(S)$ and $a \in M\mathbb{Z}^3$, put

$$\tilde g(c, g) = (r(S)c + a, r(S)g), \qquad a_{\tilde g}\psi(x) = S\psi(\tilde g^{-1}x), \qquad a_{\tilde g}\bar\psi(x) = (S^{-1})^T \bar\psi(\tilde g^{-1}x).$$

The same spin matrices act on pair generators after transforming both endpoints. Local gauge transformations multiply an unbarred ordinary or pair leg by $\exp(ie_k\Lambda(c))$, using the photon site of its first point, and a barred leg by the inverse phase. Write this automorphism as G_Λ. Charge conjugation and the antilinear reality map act by

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar\psi, \qquad \mathrm{Cg}\,\bar\psi = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \qquad \mathbf{K}\bar\psi = \varkappa\bar\psi,$$

and by the same rules on the Hölder generators.

The covariance requirements on a family are

$$\begin{aligned} a_{\tilde g}\mathscr{E}(\tilde g X;\tilde g\mathscr{A},\tilde g\mathscr{C}) &= \mathscr{E}(X;\mathscr{A},\mathscr{C}),\\ \mathrm{Cg}\,\mathscr{E}(X;-\mathscr{A},-\mathscr{C}) &= \mathscr{E}(X;\mathscr{A},\mathscr{C}). \end{aligned} \tag{12.4}$$

For real scalar functions λ,λ' on the photon torus and two pairs in the domain which satisfy $\mathscr{A}^1=\mathscr{A}-d\lambda$ and $\mathscr{C}^1=\mathscr{C}-d\lambda'$ on bonds at distance at most one from X, the local gauge requirement is

$$G_{\lambda+\lambda'}\mathscr{E}(X;\mathscr{A}^1,\mathscr{C}^1)=\mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

The real subspace K_k^x has in addition

$$\mathbf{K}\mathscr{E}(X;-\overline{\mathscr{A}},-\overline{\mathscr{C}})=\mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

All requirements refer to the generators before evaluating fields. They are essential to the form of the counterterms.

The spaces $K_k^{\mathbb{C}},K_k$ without the second connection argument have the same definitions on $\widetilde R_k$. Restriction $\mathscr{E}|_0(X;\mathscr{A})=\mathscr{E}(X;\mathscr{A},0)$ maps $K_k^{x,\mathbb{C}}$ into $K_k^{\mathbb{C}}$ and has norm at most one. The real subspaces also restrict. Analyticity, locality and the symmetries follow by setting the second argument and its gauge function to zero; the norm bound follows from the smaller supremum.

For $k\ge 1$ the step is defined on balls with windows

$$w_k^{\mathrm m}=e_k^{3/4-8\varepsilon},\qquad w_k^{\mathrm E}=e_k^{1/4-7\varepsilon},\qquad \mathrm{Ball}_k^x(c)=\{(m,\mathscr{E}):|m|\le cw_k^{\mathrm m},\|\mathscr{E}\|_k^x\le cw_k^{\mathrm E}\}. \tag{12.5}$$

At level zero the ball is instead $\{(m,0):|m|\le cw_0^{\mathrm m}\}$. This restriction is part of the first-step construction.

When fields are evaluated, the total connection is $\mathscr{T}=\mathscr{A}+\mathscr{C}$ and

$$\psi=H_k(\mathscr{T})\Psi,\qquad \chi=\delta_{\alpha,\mathscr{T}}\psi,\qquad \bar\psi=\bar H_k(\mathscr{T})\bar\Psi,\qquad \bar\chi=\delta_{\alpha,-\mathscr{T}}\bar\psi.$$

For an intra-cluster pair the phase is one, so $\psi(x')-\psi(x)=\eta_k^\alpha\chi(x,x')$. Accordingly

$$(1-JJ^*)\psi(x)=-\frac{\eta_k^\alpha}{48}\sum_{x'\ne x}\chi(x,x').$$

The sum is within the cluster of x. This identity expresses the heavy part through the same ordinary and Hölder legs. The norm of a Hölder generator remains $h_{k,2}$, including for an intra-cluster pair; the factor η_k^α belongs to this evaluated identity.

12.2. **The total connection and its inverse domain.** The inverse domain $\widetilde G_k$ consists of fields B_0+B_1 with B_0 real and $\|B_1\|_\infty<e_k^{-1+\varepsilon}$, where on each $\square^{\sim5}$ the field B_0 has a real-gauge representative of supremum norm below $e_k^{-1+\varepsilon}$. The external domain is

$$\mathfrak{X}_k=\{\mathscr{C}:\|\mathscr{C}\|_\infty<e_k^{-1+2\varepsilon},\|\partial\mathscr{C}\|_\infty<e_k^{-1+3\varepsilon}\}.$$

The sets $R_k,\widetilde R_k,\widetilde R_k^-$ were defined with the photon estimates. Their real-gauge representatives may be chosen independently on each enlarged cube.

Assume $L\ge5$, $0<\varepsilon\le1/200$ and

$$e_k^\varepsilon\le\tfrac14,\qquad e_k^{1/4}+e_k^\varepsilon\le1,\qquad 4e_k^{1/4+2\varepsilon}+2e_k^{3\varepsilon}\le1. \tag{12.6}$$

Then

$$\widetilde R_k+\mathfrak{X}_k\subset\widetilde G_k. \tag{12.7}$$

Indeed, write $\mathscr{A} = A_0 + A_1$ with $A_1 \in R_k$ and A_0 locally real-gauge equivalent on $\square^\dagger$ to $A^b_\square$ with the three threshold bounds. Restrict the representative to $\square^{\sim 5} \subset \square^\dagger$. Its supremum norm is below $e_k^{-3/4+\varepsilon} \le e_k^{-1+\varepsilon}$, and

$$\|A_1 + \mathscr{C}\|_\infty < e_k^{-3/4+\varepsilon} + e_k^{-1+2\varepsilon} = (e_k^{1/4} + e_k^{\varepsilon}) e_k^{-1+\varepsilon} \le e_k^{-1+\varepsilon}.$$

This is the required decomposition of the total connection. Real gauge transformations can absorb any real constant part, locally on the nonwrapping cubes.

Put $\mathfrak{i}(B) = e_k \| \operatorname{Im} B\|_\infty$ and $q_d(B) = e_k \|dB\|_\infty$. The same decomposition gives

$$\mathfrak{i}(\mathscr{A} + \mathscr{C}) \le e_k^{1/4+\varepsilon} + e_k^{2\varepsilon} \le 1, \qquad q_d(\mathscr{A} + \mathscr{C}) \le 4 e_k^{1/4+2\varepsilon} + 2 e_k^{3\varepsilon} \le 1. \tag{12.8}$$

For the curvature, every plaquette lies in an enlarged cube, and $dA_0 = dA^b_\square$ there. This costs at most $2\|\partial A^b_\square\|_\infty$; the global small field costs another $2\|\partial A_1\|_\infty$, and the external field costs $2\|\partial \mathscr{C}\|_\infty$. The imaginary bound uses only that A_0 is real.

The domains are open: the strict inequalities define open sets for the global summand, which can absorb a sufficiently small complex perturbation while the local real representative is kept fixed. The norm-defined sets are invariant under sign and conjugation; the local real-gauge definitions are also invariant under real gauges and real constant additions. Coordinate sign changes and M translations preserve them. Their derivative and Hölder norms are preserved, with a possible reversal of a bond and translation of its base point. Thus (12.7) applies also at the signs, conjugates and real gauges used in the step.

12.3. **Vacuum and mass extraction.** For a balanced activity, its part of bidegree $(2,0)$ can be written uniquely as

$$\mathscr{E}^{[2,0]}(X; \mathscr{A}, \mathscr{C}) = \int \mathrm{d}x\, \mathrm{d}x'\, \bar\psi(x)^T K_X(\mathscr{A}, \mathscr{C}; x, x') \psi(x').$$

The matrix kernel is the ordinary antisymmetric kernel with the first bar index one and the second bar index zero. Antisymmetry accounts for the other ordering; in particular

$$\|\mathscr{E}^{[2,0]}(X; \mathscr{A}, \mathscr{C})\|_{h_k} = h_{k,1}^2 \sum_{\beta,\beta'} \int \mathrm{d}x\, \mathrm{d}x' |K_X(\mathscr{A}, \mathscr{C}; x, x')_{\beta\beta'}|. \tag{12.9}$$

The kernel is antiperiodic in each argument separately.

For a small X, choose one lift $\widetilde{X}$ and define

(12.10)

$$a_0(\mathscr{E}, X) = \operatorname{Vol}(X)^{-1} \mathscr{E}_{00}(X; 0, 0), \qquad a_2(\mathscr{E}, X) = \operatorname{Vol}(X)^{-1} \int_{\widetilde{X}} \mathrm{d}x \int_{\widetilde{X}} \mathrm{d}x'\, K_X(0,0; x, x').$$

Both cluster labels are averaged with measure $\eta_k^3/48$. The second expression is independent of the lift: simultaneous translation by a period multiplies its two arguments by the same antiperiodic sign, whose product is one.

For an M-cube $\square$ put

$$\varepsilon_{\mathrm{v}}(\mathscr{E}, \square) = - \sum_{\substack{X \supset \square \\ X \text{ small}}} a_0(\mathscr{E}, X), \qquad M(\mathscr{E}, \square) = - \sum_{\substack{X \supset \square \\ X \text{ small}}} a_2(\mathscr{E}, X). \tag{12.11}$$

Translations by $M\mathbb{Z}^3$ identify the small-polymer sets over different cubes. They leave the two extracted quantities unchanged. Thus the two sums do not depend on $\square$. Write their values as $\varepsilon_{\mathrm{v}}(\mathscr{E})$ and $M(\mathscr{E})$.

Proposition 12.1. *There is a complex-linear scalar map $m(\mathscr{E})$ such that $M(\mathscr{E}) = m(\mathscr{E})1$. Both m and ε_{v} are real on K_k^x, and*

$$|\varepsilon_{\mathrm{v}}(\mathscr{E})| \le K_0 M^{-3} \|\mathscr{E}\|_k^x, \qquad |m(\mathscr{E})| \le K_0 M^{-3} e_k^{1/2} \|\mathscr{E}\|_k^x. \tag{12.12}$$

They commute with restriction of the second connection argument to zero.

Proof. Substitution of the generator transformations in a bilinear, followed by uniqueness of its kernel, gives

$$\begin{aligned} a_2(\mathscr{E}, \tilde{g}X) &= S a_2(\mathscr{E}, X) S^{-1}, \\ a_2(\mathscr{E}, X) &= \mathsf{C}^{-1} a_2(\mathscr{E}, X)^T \mathsf{C}, \\ a_2(\mathscr{E}, X) &= \varkappa^{-1} \overline{a_2(\mathscr{E}, X)} \varkappa \quad (\mathscr{E} \in K_k^x). \end{aligned} \tag{12.13}$$

For the first relation, the two variables and their labels are transformed together, and the measure is unchanged. For charge conjugation, moving the unbarred leg past the barred leg cancels the minus sign in its transformation, and exchanges the two kernel variables. For the last relation, the reality map conjugates the coefficient and multiplies both legs by $\varkappa$; its inverse transpose is $\varkappa^{-1}$. The constant kernel has the corresponding scalar invariance and is real for a real family.

Let z be the center of $\square$ and take $S = \gamma_1, \gamma_2, \gamma_3$ or γ_5. The symmetry $(S, (1 - r(S))z)$ fixes $\square$ and permutes the small polymers containing it. The first relation in (12.13) therefore shows that $M(\mathscr{E})$ commutes with all four matrices. Their commutant is scalar. In the fixed Dirac basis, γ_3 and $\Sigma_3 = -i\gamma_1\gamma_2\gamma_3$ are diagonal with the four distinct pairs of diagonal entries $(1,1), (-1,1), (-1,-1), (1,-1)$. A commuting matrix is diagonal; commutation with γ_1 identifies its first two and last two entries, and commutation with γ_5 identifies these pairs. This proves the scalar assertion. The last relation in (12.13) makes this scalar real for a real family.

The coefficient norm gives, for each small X,

$$\begin{aligned} \mathrm{Vol}(X) |a_0(\mathscr{E}, X)| &\le \|\mathscr{E}(X; 0, 0)\|_{h_k}, \\ \mathrm{Vol}(X) \sum_{\beta,\beta'} |a_2(\mathscr{E}, X)_{\beta\beta'}| &\le h_{k,1}^{-2} \|\mathscr{E}^{[2,0]}(X; 0, 0)\|_{h_k}. \end{aligned} \tag{12.14}$$

Use $h_{k,1}^{-2} = e_k^{1/2}$ and $\mathrm{Vol}(X) \ge M^3$. Each contribution is bounded by the corresponding right side of (12.12), times $\exp(-\kappa d_M(X))$. The polymer sum over a fixed cube is at most K_0 by $\kappa \ge \kappa_0$. For the mass use one diagonal entry of the matrix sum, now known to be scalar. This proves the bounds. Linearity follows from uniqueness of the antisymmetric kernels. Restriction gives the same kernels at $(0, 0)$, proving the last assertion. □

Subtract these local terms only on small polymers:

$$(R\mathscr{E})(X; \mathscr{A}, \mathscr{C}) = \mathscr{E}(X; \mathscr{A}, \mathscr{C}) - a_0(\mathscr{E}, X) \,\mathrm{Vol}(X) - \int_X \mathrm{d}x \, \bar{\psi}(x)^T a_2(\mathscr{E}, X) \psi(x). \tag{12.15}$$

For large X set $R\mathscr{E} = \mathscr{E}$. Then

$$\sum_X \mathscr{E}(X; \mathscr{A}, \mathscr{C}) = -\varepsilon_{\mathrm{v}}(\mathscr{E}) \ell^3 - m(\mathscr{E}) \langle \bar{\psi}, \psi \rangle + \sum_X (R\mathscr{E})(X; \mathscr{A}, \mathscr{C}). \tag{12.16}$$

Indeed, decompose the integral and volume of each small polymer over its M-cubes, interchange the finite sums, and use (12.11) on each cube. Every chamber component receives the same scalar mass coefficient:

$$m(\mathscr{E}) \langle \bar{\psi}, \psi \rangle = m(\mathscr{E}) \frac{\eta_k^3}{48} \sum_c \sum_g \sum_{\beta=1}^{4} \bar{\psi}_\beta(c, g) \psi_\beta(c, g).$$

The diagonal bilinear of a constant matrix A on X has kernel $48\eta_k^{-3}1[x = x' \text{ over } X]A$ when both arguments are taken over the same lift. Its integral is $\mathrm{Vol}(X)A$, and its norm is

$$h_{k,1}^2 \,\mathrm{Vol}(X) \sum_{\beta,\beta'} |A_{\beta\beta'}|.$$

These identities and (12.14) show that

$$\begin{aligned} (R\mathscr{E})_{00}(X;0,0) &= 0 \quad (X \text{ small}), \\ \int_{\widetilde{X}} \mathrm{d}x\,\mathrm{d}x'\,(R\mathscr{E})_{20}(X;0,0;x,x') &= 0 \quad (X \text{ small}), \\ \|R\mathscr{E}\|_k^x &\le 2\|\mathscr{E}\|_k^x. \end{aligned} \tag{12.17}$$

For the bound, the two subtracted terms have distinct bidegrees and together have norm at most $\|\mathscr{E}(X;0,0)\|_{h_k}$. Apply the triangle inequality to (12.15), then the activity norm.

The map R preserves the activity spaces and restriction to zero. The subtracted kernels are independent of the connections, and their two legs lie over X, so analyticity and locality remain. A diagonal bilinear is invariant under every local gauge. Its lattice, charge and reality transformations follow from (12.13); the volume and constant term have the same symmetries. This verifies each requirement on the remainder, without assuming that an individual matrix $a_2(\mathscr{E}, X)$ is scalar.

For later comparisons, the extraction has a precise local meaning across different sides. Fix a lifted M-cube and consider all lifted small polymers containing it. Their projections are bijective onto the small polymers containing its image, and their d_M and volumes are unchanged. If two families have the same constant and quadratic kernels on every such lifted polymer at zero connections, then their vacuum and mass matrices over this cube agree term by term in (12.11). Volume independence of a flow will require this equality of kernels from the step comparison.

12.4. **Reconstruction by interior and boundary legs.** Let $k \ge 1$, $0 < \mu \le 1$ and $b_- \le b_k$. With $\mathcal{Q}_k(A) = Q_k(A)J^*$ and $\mathcal{Q}_k^\sharp(A) = JQ_k^T(-A)$, set

$$T_k(A) = b_k^{-1}\mathcal{Q}_k(A)\big(D_A + \mu + b_k\mathcal{Q}_k^\sharp(A)\mathcal{Q}_k(A)\big) = \mathcal{Q}_k(A)\big(b_k^{-1}D_A + 1 + \mu/b_k\big). \tag{12.18}$$

The second form follows from $\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1$. The fine inverse used in the definition of $H_k(A)$ exists on $\widetilde{G}_k$. Hence

$$T_k(A)H_k(A) = b_k^{-1}\mathcal{Q}_k(A)F_\infty(A)b_kF_\infty(A)^{-1}\mathcal{Q}_k^\sharp(A) = 1. \tag{12.19}$$

The barred identity follows at $-A$, by conjugation with C. At level zero all four field and reconstruction operators are the identity on the 192 chamber components per photon site. This first step uses the unblocked chamber spinors.

For $c \in \Delta_y$, let $c_j(c)$ be the center of its block of side $L^j\eta_k$, $0 \le j \le k$, so that $c_0(c) = c$ and $c_k(c) = y$. The averaging phase is

$$\varphi_A(y,c) = \exp\left(ie_k\eta_k \sum_{j=0}^{k-1} (\tau A)(c_{j+1}(c), c_j(c))\right). \tag{12.20}$$

The sum uses the average of the six rectilinear path sums. We write $(\tau_k A)(y,c)$ for this sum. All of these paths lie in Δ_y and their total length in bonds is at most $\frac{3}{2}(L^k - 1)$. Thus

$$\begin{aligned} |\varphi_A(y,c)^{\pm 1}| &\le \exp(3\mathfrak{i}(A)/2), \\ |u_A(c,c')^{\pm 1}| &\le \exp(\eta_k\mathfrak{i}(A)) \le \exp(\mathfrak{i}(A)). \end{aligned} \tag{12.21}$$

for a photon bond, with $u_A(c,c') = \exp(ie_k\eta_k A(c,c'))$.

The row form of (12.18) is

$$(T_k(A)\psi)(y) = \int_{\Delta_y} \mathrm{d}x\, \varphi_A(y,c)\Big\{(1+\mu/b_k)\psi(x) - (b_k\eta_k)^{-1}\sum_{i=0}^{3} M_i(x)[U_A(x,x^{(i)})\psi(x^{(i)}) - \psi(x)]\Big\}. \tag{12.22}$$

Write $c_3 = c + \eta_k g e_1$ for the photon base point of $x^{(3)}$. Here $U_A = 1$ for $i \le 2$ and $U_A(x,x^{(3)}) = u_A(c,c_3)$. The normalized chamber integral over Δ_y has total mass one. This row exists for every complex connection, regardless of invertibility; the inverse domain was needed only for (12.19).

The rows map antiperiodic fields to antiperiodic coarse fields: translation by a period preserves blocks, neighbors and phases, and supplies the single antiperiodic sign from the source field.

If Y is a union of M-cubes and $y \in Y$ is a unit site, then $\Delta_y \subset Y$: both M and L^k are odd, and the unit block is contained in the M-cube of its center. Define the two rows

$$\begin{aligned}(T_Y^{(1)}(A)\psi)(y) &= \int_{\Delta_y} \mathrm{d}x\, \varphi_A(y,c)\Big\{(1+\mu/b_k)\psi(x) - (b_k\eta_k)^{-1}\sum_{i:\, x^{(i)} \text{ over } Y} M_i(x)[U_A(x,x^{(i)})\psi(x^{(i)}) - \psi(x)]\Big\},\\ (T_Y^{(2)}(A)\chi)(y) &= -b_k^{-1}\eta_k^{\alpha-1}\int_{\Delta_y} \mathrm{d}x\, 1[c_3 \notin Y]\varphi_A(y,c)M_3(x)\chi(x,x^{(3)}).\end{aligned} \tag{12.23}$$

Rows outside Y are zero. The first row contains ordinary legs only over Y. The second row contains precisely the type-3 pairs leaving Y, which are admissible boundary pairs of length η_k. Their coefficients read A only on bonds with an endpoint in Δ_y. If $\chi = \delta_{\alpha,A}\psi$ on these boundary pairs, then

$$T_Y^{(1)}(A)\psi + T_Y^{(2)}(A)\chi = T_k(A)\psi \quad \text{on the unit sites of } Y. \tag{12.24}$$

This follows by replacing the missing type-3 difference in (12.22) by $\eta_k^{\alpha}\chi(x,x^{(3)})$. The intra-cluster differences remain in $T_Y^{(1)}$.

Proposition 12.2. *For any complex connection, write* $\mathfrak{i} = \mathfrak{i}(A)$ *and* $q_d = q_d(A)$. *In the norm obtained by taking the supremum of the Euclidean spinor norm,*

$$\begin{aligned}&\|T_Y^{(1)}(A)\|_{\mathrm{su}},\ \|T_k(A)\|_{\mathrm{su}} \le \exp(3\mathfrak{i}/2)\left[1 + b_-^{-1}(54Lq_d\exp(6q_d) + 12 + 6\exp(\mathfrak{i}) + \mu)\right],\\ &\|T_Y^{(2)}(A)\|_{\mathrm{su}} \le 6b_-^{-1}\exp(3\mathfrak{i}/2)\eta_k^{\alpha}.\end{aligned} \tag{12.25}$$

In particular, if $A \in \widetilde{R}_k + \mathfrak{X}_k$ *and* (12.6) *holds, the weighted coefficient row sums satisfy*

$$|T_Y^{(i)}(A)|_{1,\infty},\ |T_k(A)|_{1,\infty}, |\bar{T}_Y^{(i)}(A)|_{1,\infty},\ |\bar{T}_k(A)|_{1,\infty} \le C_T, \tag{12.26}$$

where $C_T = 2\exp(3/2)[1 + b_-^{-1}(54L\exp(6) + 12\exp(1) + 1)]$. *The bound is independent of* k, ℓ *and* Y.

Proof. Put $E = \Delta_y \times O_{\mathrm{cub}}$ and divide it into

$$E_1 = \{x \in E : c_3 \in \Delta_y\}, \quad E_2 = \{x \in E : c_3 \in Y \setminus \Delta_y\}, \quad E_3 = \{x \in E : c_3 \notin Y\}.$$

For each coordinate direction $a \in \{\pm e_\mu\}$ there are eight labels with $ge_1 = a$. The face of Δ_y through which $c + \eta_k a$ leaves the unit block has η_k^{-2} photon sites. Thus

$$\#E = 48\eta_k^{-3}, \qquad \#(E_2 \cup E_3) = 48\eta_k^{-2}. \tag{12.27}$$

The chamber hopping matrices satisfy

$$M_i(x^{(i)}) - M_i(x) = w_i\gamma\cdot\widehat{d}_i(x), \qquad \sum_{i=0}^{3} w_i\gamma\cdot\widehat{d}_i(x) = 0.$$

Also $\|M_3(x)\|_2 \le 6$ and $\|w_3\gamma\cdot\widehat{d}_3(x)\|_2 = 6$. Reindex each intra-cluster sum by $x \mapsto x^{(i)}$ and each type-3 sum in E_1 by $x \mapsto x^{(3)}$. Both are involutions of their summation sets. The intra-cluster sums become $\sum_{x\in E}\varphi_A(y,c)\sum_{i\le 2}(M_i(x^{(i)}) - M_i(x))\psi(x)$, which equals $-\sum_{x\in E}\varphi_A(y,c)w_3\gamma\cdot\widehat{d}_3(x)\psi(x)$. The part over E_1 cancels with the corresponding matrix difference from the reindexed type-3 sum. The resulting row is $(T_Y^{(1)}(A)\psi)(y) = R_0 + R_1 + R_2 + R_3$, with

$$\begin{aligned}
R_0 &= (1+\mu/b_k)\frac{\eta_k^3}{48}\sum_{x\in E}\varphi_A(y,c)\psi(x),\\
R_1 &= b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_2\cup E_3}\varphi_A(y,c)w_3\gamma\cdot\widehat{d}_3(x)\psi(x),\\
R_2 &= -b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_2}\varphi_A(y,c)M_3(x)[u_A(c,c_3)\psi(x^{(3)}) - \psi(x)],\\
R_3 &= -b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_1}[\varphi_A(y,c_3)u_A(c_3,c) - \varphi_A(y,c)]M_3(x^{(3)})\psi(x).
\end{aligned} \tag{12.28}$$

The face count and phase moduli bound the first three terms by

$$\exp(3\mathfrak{i}/2)\|\psi\|_{\mathrm{su}}\big[1+\mu/b_k + 6b_k^{-1} + 6b_k^{-1}(1+\exp(\mathfrak{i}))\big].$$

The fourth term requires the curvature estimate.

For neighboring photon sites $c, c' = c+\eta_k a$ in Δ_y, let $j(c,a)$ be the first scale at which their block centers agree. The centers at all larger scales then agree as well. For $j \ge 2$, such a c lies on a face of a block of side $L^{j-1}\eta_k$. There are $L^{3(k-j+1)}$ such blocks inside Δ_y, with $L^{2(j-1)}$ sites on the indicated face. For $j = 1$ use all sites. We obtain

$$\#\{c : c, c+\eta_k a \in \Delta_y,\ j(c,a) = j\} \le \eta_k^{-3}L^{-(j-1)}. \tag{12.29}$$

Write

$$\begin{aligned}
\varphi_A(y,c')u_A(c',c) - \varphi_A(y,c) &= \varphi_A(y,c)(\exp(v(c,a)) - 1),\\
v(c,a) &= ie_k\eta_k[(\tau_k A)(y,c') + A(c',c) - (\tau_k A)(y,c)].
\end{aligned}$$

The contributions above scale $j(c,a)$ cancel in this bracket. The remaining expression is the mean of the line sums over closed paths from their common scale-j center down to c', then across the bond to c, and back up the reversed paths. All these loops lie in the scale-j block, of side at most $L^j\eta_k$, and have at most $3L^j$ bonds. For a closed path of n bonds in a box of side s, the axial path estimate gives $|A(\Gamma)| \le 2ns\|dA\|_\infty$. To recall the reason, subtract the gauge obtained by line sums from one corner in coordinate order $1,2,3$. The difference on each bond is a sum over at most two strips of width one mesh and length at most s. Its modulus is at most $2s\|dA\|_\infty$, and the gauge sum around the loop vanishes. This argument applies to complex fields as well. Consequently, with $j = j(c,a)$,

$$|v(c,a)| \le 6L^{2(j-k)}q_d, \qquad |\exp(v(c,a)) - 1| \le 6L^{2(j-k)}q_d\exp(6q_d). \tag{12.30}$$

The factor $L^{2(j-k)}$ must be kept until after summing the locations of the first common block. By (12.29),

$$\eta_k^2 \sum_{\substack{c\in\Delta_y\\ c+\eta_k a\in\Delta_y}} |\exp(v(c,a))-1| \le 6q_d\exp(6q_d)L^{1-k}\sum_{j=1}^{k}L^j \le 9Lq_d\exp(6q_d).$$

The six directions, eight labels per direction, and $\|M_3\|_2\le 6$ now give

$$|R_3|_2 \le 54Lq_d\exp(6q_d)b_k^{-1}\exp(3\mathfrak{i}/2)\|\psi\|_{\mathrm{su}}.$$

Together with the first three terms this proves the first line of (12.25) for $T_Y^{(1)}$. Take Y to be the full torus to obtain the bound for T_k.

The second row is supported on E_3. Its definition and (12.27) give directly

$$|(T_Y^{(2)}(A)\chi)(y)|_2 \le b_k^{-1}\eta_k^{\alpha-1}\frac{\eta_k^3}{48}48\eta_k^{-2}\exp(3\mathfrak{i}/2)6\|\chi\|_{\mathrm{su}},$$

which is the second bound.

For the coefficient norm, fix a row and choose each source component to have the conjugate phase of its row coefficient. Its component supremum is at most one, and its Euclidean spinor supremum is at most two. The row coefficient sum is then at most twice the corresponding bound in (12.25). At a total connection, (12.8) gives $\mathfrak{i}, q_d \le 1$; also $\mu\le 1$ and $\eta_k^\alpha\le 1$. The resulting bound is C_T as stated. The normalized index measure is already in the row coefficients, so no extra factor 48 enters. The domain is invariant under sign, and the matrix C is monomial; conjugating $T_Y^{(i)}(-A)$ or $T_k(-A)$ by it permutes row and column moduli. This proves the barred bounds. □

The local reconstruction respects the symmetries of an activity. Telescoping the scalar gauge sum in the averaging paths gives

$$\varphi_{A-d\lambda}(y,c)\exp(ie_k\lambda(c)) = \exp(ie_k\lambda(y))\varphi_A(y,c),$$
$$u_{A-d\lambda}(c,c')\exp(ie_k\lambda(c')) = \exp(ie_k\lambda(c))u_A(c,c').$$

Insert these two identities in (12.23). Every term acquires the coarse factor $\exp(ie_k\lambda(y))$, where the boundary leg transforms with the phase of its first point. Thus, with λ^{u} the restriction to unit sites,

$$\begin{aligned} T_Y^{(1)}(A-d\lambda)\exp(ie_k\lambda)\psi &= \exp(ie_k\lambda^{\mathrm{u}})T_Y^{(1)}(A)\psi,\\ T_Y^{(2)}(A-d\lambda)\chi^\lambda &= \exp(ie_k\lambda^{\mathrm{u}})T_Y^{(2)}(A)\chi, \qquad \chi^\lambda(x,x') = \exp(ie_k\lambda(c))\chi(x,x'). \end{aligned} \tag{12.31}$$

The identity also holds for gauges specified locally on the bonds and sites read by the row.

A scheme symmetry $\tilde g = (S,a)$ maps nested blocks and their centers to their images, preserves the order of each coordinate path, and permutes the 48 labels. Moreover $M_i(\tilde g x)S = SM_i(x)$ and $(\tilde g x)^{(i)} = \tilde g(x^{(i)})$. Changing variables in the two rows therefore proves

$$T_{\tilde g Y}^{(i)}(\tilde g A)\tilde g = \tilde g T_Y^{(i)}(A). \tag{12.32}$$

Complex conjugation changes the phases to those at $-A$, and $\overline{M_i} = \varkappa^{-1}M_i\varkappa$, giving

$$\overline{T_Y^{(i)}(\overline{A})} = \varkappa^{-1}T_Y^{(i)}(-A)\varkappa. \tag{12.33}$$

The barred rows are $\bar T_Y^{(i)}(A) = \mathsf{C}^{-1}T_Y^{(i)}(-A)\mathsf{C}$. Their gauges have the opposite sign. For the lattice covariance, $(S^{-1})^T = \sigma(S)\mathsf{C}^{-1}S\mathsf{C}$ with $\sigma(S)\in\{\pm1\}$: both matrix maps are homomorphisms, and on each generating Dirac matrix they differ by a minus sign. This same sign occurs in the source and target spin actions and cancels in the barred intertwining identity. For reality,

C is real and $\varkappa\mathsf{C} = -\mathsf{C}\varkappa$; the two signs from commuting the conjugations cancel. Thus the barred versions of (12.32) and (12.33) hold with the inverse transpose spin action and the same $\varkappa$.

For $B \in \widetilde{R}_k + \mathfrak{X}_k$, equations (12.19) and (12.24) now define the local substitution used after fluctuation integration:

$$\Psi(y) \longmapsto (T_Y^{(1)}(B)\psi)(y) + (T_Y^{(2)}(B)\chi)(y), \qquad \bar{\Psi}(y) \longmapsto (\bar{T}_Y^{(1)}(B)\bar{\psi})(y) + (\bar{T}_Y^{(2)}(B)\bar{\chi})(y).$$

After setting $\psi = H_k(B)\Psi$ and $\chi = \delta_{\alpha,B}\psi$, and their barred analogues, this substitution is the identity on the coarse generators over Y. Its coefficients are holomorphic in B and satisfy the uniform row bounds above. The new legs lie over Y and its admissible boundary pairs. The substitution fixes any old generators, so the full polynomial has these support properties when its old legs already have them. The weighted linear-substitution bound therefore applies with these two row operators. The resulting degree weights will be fixed with the full step.

12.5. **Adjustment of the quadratic kernel.** The extraction makes the integral of the zero-field quadratic kernel vanish. For contraction we need the kernel itself to vanish. We achieve this by writing each ordinary leg as a transported average and a sum of Hölder differences. This changes the activity before evaluation and leaves its evaluated value unchanged.

Let X be small, choose a lift $\widetilde{X}$, and let $x = (c, g)$ and $\tilde{x} = (\tilde{c}, \tilde{g})$ lie over this lift. Take all shortest face chains of its M-cubes between their two cubes. For a chain $(\square_0, \dots, \square_n)$ with centers z_i, use the points

$$p_0 = x, \qquad p_{i+1} = (z_i, g) \quad (0 \le i \le n), \qquad p_{n+2} = \tilde{x}.$$

The initial label is kept at each intermediate center. Each distinct consecutive pair is admissible for X of the first kind. The paths in the first and last cube remain in those cubes; between adjacent centers all six paths are the straight segment across their common face. The last pair can have different labels, and can lie inside one cluster. For a coincident pair there is no difference term.

With $u_B(p, p') = U_B(p, p')$ and $u_B(p, p) = 1$, set

$$U_i^{\pm} = \prod_{j<i} u_B(p_j, p_{j+1})^{\pm 1}.$$

The chain has no repeated cube, and $n + 1 \le |X|_M \le c_X(L + 1)$. Hence

$$\sum_{i=0}^{n+1} |\pi p_i - \pi p_{i+1}|_1 \le (n+3)M \le 3c_X LM, \qquad \sum_{i:p_i \neq p_{i+1}} d_X(p_i, p_{i+1})^{\alpha} \le 3c_X LM. \tag{12.34}$$

The second estimate uses $d_X^{\alpha} \le \max(1, d_X) \le M$. The transport bound is $|u_B(p, p')^{\pm 1}| \le \exp(e_k \| \operatorname{Im} B\|_{\infty} |\pi p - \pi p'|_1)$, obtained by counting the bonds in each path.

For $B = \mathscr{A} + \mathscr{C}$, the domain decomposition and $e_k^{\varepsilon} \le 1/4$ give $e_k \| \operatorname{Im} B\|_{\infty} \le e_k^{\varepsilon}/2$. Assume in addition

$$90 c_X LM e_k^{\varepsilon} \le 1. \tag{12.35}$$

The modulus of every partial product $U_i^{\pm}$ is then at most $\exp(1)$, uniformly in the chain and mesh.

Let Mn be the mean over the shortest chains. Define the homomorphism $\Theta_{X,B}$ on ordinary generators by

$$\begin{aligned}\Theta_{X,B}\psi(x) &= \mathrm{Vol}(X)^{-1}\int_{\widetilde{X}} \mathrm{d}\tilde{x}\ \mathrm{Mn}\left[U^{+}_{n+2}\psi(\tilde{x}) - \sum_{i:p_i\neq p_{i+1}} U^{+}_{i}\, d_X(p_i,p_{i+1})^{\alpha}\chi(p_i,p_{i+1})\right],\\ \Theta_{X,B}\bar{\psi}(x) &= \mathrm{Vol}(X)^{-1}\int_{\widetilde{X}} \mathrm{d}\tilde{x}\ \mathrm{Mn}\left[U^{-}_{n+2}\bar{\psi}(\tilde{x}) - \sum_{i:p_i\neq p_{i+1}} U^{-}_{i}\, d_X(p_i,p_{i+1})^{\alpha}\bar{\chi}(p_i,p_{i+1})\right].\end{aligned} \tag{12.36}$$

The spin components are unchanged. Both displayed images are linear combinations of odd independent generators, so they anticommute and determine a unique homomorphism on the subalgebra of ordinary legs over X. Translation of the lift by a period preserves every transport and chain and multiplies each generator by the same antiperiodic sign. Thus the homomorphism is independent of this choice of lift.

If the generators are evaluated at fields satisfying their covariant Hölder relations, repeated substitution of

$$\psi(p_i) = u_B(p_i,p_{i+1})\psi(p_{i+1}) - d_X(p_i,p_{i+1})^{\alpha}\chi(p_i,p_{i+1})$$

along a chain gives exactly the first expression in (12.36); the barred relation uses u_B^{-1}. Taking the two means therefore reproduces the original leg. This proves equality after evaluation for every polynomial in the ordinary legs over X.

The adjusted family is

$$\mathscr{E}^{\mathrm{nat}}(X) = \mathscr{E}(X) - \mathscr{E}^{[2,0]}(X) + \Theta_{X,\mathscr{A}+\mathscr{C}}\mathscr{E}^{[2,0]}(X) \qquad (X \text{ small}), \tag{12.37}$$

and is unchanged on large polymers. The equality just proved shows that the adjusted and original families have equal evaluated values. Only bidegrees $(2,0),(1,1),(0,2)$ can change.

At zero connection all $U_i^{\pm}$ are one. The ordinary part of an adjusted bilinear with kernel K is therefore

$$\bar{\psi}^{T}_{\mathrm{av}}\left[\int_{\widetilde{X}} \mathrm{d}x\,\mathrm{d}x'\,K(x,x')\right]\psi_{\mathrm{av}}, \qquad \psi_{\mathrm{av}} = \mathrm{Vol}(X)^{-1}\int_{\widetilde{X}}\mathrm{d}x\,\psi(x).$$

Apply this to $R\mathscr{E}$, whose kernel integral vanishes. With $E = (R\mathscr{E})^{\mathrm{nat}}$, we obtain the stronger normalization

$$E_{00}(X;0,0) = 0, \qquad E_{20}(X;0,0;\cdot,\cdot) = 0 \quad (X \text{ small}). \tag{12.38}$$

The same chain estimate controls the norm of adjustment. Decompose $\Theta\psi = L_\psi + L_\chi$ by bidegree. For arbitrary generator weights (w_1,w_2),

$$\|L_\psi\|_w \le \exp(1)w_1, \qquad \|L_\chi\|_w \le \exp(1)(3c_X LM)w_2.$$

The normalized site and label mean has total mass one. Writing $s_K = \sum_{\beta,\beta'}\int|K_{\beta\beta'}|$ for the balanced quadratic kernel and $F = \Theta\mathscr{E}^{[2,0]}$ gives the three bounds

$$\begin{aligned}\|F^{[2,0]}\|_w &\le \exp(2)w_1^2 s_K,\\ \|F^{[1,1]}\|_w &\le 2\exp(2)(3c_X LM)w_1 w_2 s_K,\\ \|F^{[0,2]}\|_w &\le \exp(2)(3c_X LM)^2 w_2^2 s_K.\end{aligned}$$

Since $h_{k,2} = e_k^{\varepsilon}h_{k,1}$ and $3c_X LMe_k^{\varepsilon} \le 1/30$, summing these bounds yields

$$\|\mathscr{E}^{\mathrm{nat}}(X)\|_{h_k} \le \exp(2)(1+3c_X LMe_k^{\varepsilon})^2\|\mathscr{E}(X)\|_{h_k} \le 10\|\mathscr{E}(X)\|_{h_k}. \tag{12.39}$$

Subtracting the original quadratic part before adding its image uses additivity of the norm over bidegrees.

All new legs lie over X or at its admissible chain pairs, and all new transports read bonds with both ends in X. Thus adjustment preserves locality. It preserves analyticity because these transports are entire in the connection. For the lattice covariance, symmetries carry shortest chains, their centers and labels to the corresponding image chains. For a real gauge the factors along a partial chain telescope to the phase difference at its endpoints, giving the original leg phase. Charge conjugation replaces u_B by u_B^{-1}; the reality map uses $\overline{u_{-\overline{B}}} = u_B$. These identities on generators prove all four covariance identities for Θ, and hence the activity symmetries for adjustment. Restriction to $\mathscr{C} = 0$ commutes with adjustment because it leaves the same total connection and the same ordinary quadratic kernel. Therefore adjustment preserves the complex and real activity spaces.

12.6. **Scaling and reblocking.** For the rest of this group let both ℓ and ℓ/L be admissible sides. In particular, $\ell \geq 13L^2M$. Set $\bar{\varepsilon} = 1/200$ and assume

$$\begin{gathered} 0 < \varepsilon \leq \bar{\varepsilon}, \qquad 1 - \varepsilon < \alpha < 1, \qquad L^{1/8-3\bar{\varepsilon}/2} \geq 2400, \qquad e_k^{\varepsilon} \leq 1/4, \\ 160L^{1+3\varepsilon/2}Me_k^{\varepsilon} \leq 1, \qquad 9ML^{\varepsilon}e_k^{2\varepsilon} \leq 1. \end{gathered} \tag{12.40}$$

The adjustment condition (12.35) is also retained. These inequalities will control the analytic disk and the phases on a small polymer.

Write the three thresholds of R_k as

$$\theta_1 = e_k^{-3/4+\varepsilon}, \qquad \theta_2 = e_k^{\varepsilon}\theta_1, \qquad \theta_3 = e_k^{\varepsilon}\theta_2,$$

and the thresholds of $\mathfrak{X}_k$ as $d_1 = e_k^{-1+2\varepsilon}$ and $d_2 = e_k^{\varepsilon}d_1$. For a level $k+1$ bond function and gauge function define

$$A_L(b) = L^{-1/2}A(b/L), \qquad \lambda_L(c) = L^{1/2}\lambda(c/L). \tag{12.41}$$

Since $\eta_{k+1} = \eta_k/L$, the finite differences obey $(d\lambda)_L = d\lambda_L$ and

$$\|A_L\|_\infty = L^{-1/2}\|A\|_\infty, \qquad \|\partial A_L\|_\infty = L^{-3/2}\|\partial A\|_\infty, \qquad \|\delta_\alpha \partial A_L\|_\infty \leq L^{-3/2-\alpha}\|\delta_\alpha \partial A\|_\infty.$$

The last inequality uses only the coarse pairs at distance $\leq 1/L$, a subset of those in the coarse seminorm. The charge relation $e_{k+1} = L^{1/2}e_k$ therefore gives the following fine-scale bounds for the scaled complex small field, and for each scaled local real representative:

$$\begin{aligned} b_1 &= L^{-7/8+\varepsilon/2}\theta_1, \\ b_2 &= L^{-15/8+\varepsilon}\theta_2, \\ b_3 &= L^{-15/8+3\varepsilon/2}\theta_3. \end{aligned} \tag{12.42}$$

Here the additional factor $L^{-\alpha}$ in the third bound has been discarded. For the external argument the bounds are

$$\|\mathscr{C}_L\|_\infty < L^{-1+\varepsilon}d_1, \qquad \|\partial\mathscr{C}_L\|_\infty < L^{-2+3\varepsilon/2}d_2. \tag{12.43}$$

The enlarged coarse cube used to define a local representative, after multiplication by L, contains the enlarged fine cube. Scaling its real gauge thus gives the corresponding fine representative. All the ratios in (12.42) and (12.43) are at most $L^{-3/4}$, by (12.40). It follows that

$$\mathscr{A}_L \in L^{-3/4}\widetilde{R}_k \subset \widetilde{R}_k^-, \qquad \mathscr{C}_L \in L^{-3/4}\mathfrak{X}_k. \tag{12.44}$$

For the second inclusion in the first statement use $L^{-3/4} \leq 1/2 \leq 1 - e_k^{\varepsilon}$.

The Grassmann scaling homomorphism is

$$S_L\psi(x) = L^{-1}\psi'(x/L), \qquad S_L\chi(x,x') = L^{-1-\alpha}\chi'(x/L, x'/L), \tag{12.45}$$

with the same powers for the barred generators. Only the photon coordinate is divided by L; the chamber label and spin index are kept. Ordinary indices are mapped bijectively onto the coarse

ordinary indices. Pair indices are mapped bijectively onto the coarse pair indices with $d_X \le M/L$, since d_X is divided by L. This includes a pair inside one cluster: $d_X = \eta_k$ becomes $d_X = \eta_{k+1}$. The antiperiodic signs agree under this bijection.

Thus distinct basis monomials have distinct images, up to their ordering signs, and every monomial of bidegree (n, m) is multiplied by $L^{-n-(1+\alpha)m}$. There is no cancellation of coefficients. For arbitrary positive weights,

$$\|S_L F\|_{w_1,w_2} = \|F\|_{L^{-1}w_1,L^{-1-\alpha}w_2}. \tag{12.46}$$

As $h_{k+1,1}/h_{k,1} = L^{-1/8}$ and $h_{k+1,2}/h_{k,2} = L^{-1/8+\varepsilon/2}$, this gives

$$\|(S_L F)^{[n,m]}\|_{h_{k+1}} = L^{-p(n,m)}\|F^{[n,m]}\|_{h_k}, \qquad p(n,m) = n + (1+\alpha)m + \frac{n+m}{8} - \frac{m\varepsilon}{2}. \tag{12.47}$$

The powers are identical for all admissible pair types.

In kernel notation the same calculation reads

$$(S_L F)_{nm}(\xi';\zeta') = \begin{cases} L^{3n+6m-n-(1+\alpha)m} F_{nm}(L\xi'; L\zeta'), & d_X(\zeta'_j) \le M/L \text{ for every } j, \\ 0, & \text{otherwise.} \end{cases} \tag{12.48}$$

Indeed, the index weights satisfy $\eta_k^3/48 = L^3\eta_{k+1}^3/48$ and hence contribute L^{3n+6m} to the kernel. Antisymmetry and antiperiodicity are preserved, so uniqueness of the kernel expansion gives (12.48).

Let $\bar{X}^L$ be the union of LM-cubes containing the cubes of X. Define the reblocked family by

$$(\mathcal{L}F)(Y; \mathscr{A}, \mathscr{C}) = \sum_{X:\bar{X}^L = LY} S_L F(X; \mathscr{A}_L, \mathscr{C}_L). \tag{12.49}$$

The sum is finite on a fixed torus. It is local in Y if F is local in each X. Ordinary legs over X scale to legs over Y; the six paths for a pair of the first kind scale to paths in Y. For a boundary pair, $x^{(3)}/L = (x/L)^{(3)}$, and an endpoint in X scales to an endpoint in Y. Finally, every fine bond meeting X scales to a coarse bond meeting Y. Equality of both coarse connection arguments on these bonds consequently gives equality of the fine arguments read by each term. This proves both the support and the coefficient locality of (12.49).

12.7. **Global representatives on a small polymer.** The scaled fields belong to the fine domain, but this alone does not give an analytic gain for their vacuum and ordinary quadratic kernels. On a small polymer they admit a stronger representation, with a real gauge, two constant fields and global fields small enough for an analytic disk of radius $L^{7/4}$.

Proposition 12.3. *Assume* (12.40). *Let X be small and let $(\mathscr{A}, \mathscr{C}) \in \widetilde{R}_{k+1} \times \mathfrak{X}_{k+1}$. There are a real function λ_*, constant bond functions $\mathscr{K}_1, \mathscr{K}_2$, and global bond functions $\mathscr{A}^{\mathrm{b}}, \mathscr{C}^{\mathrm{b}}$ on the fine torus such that, on every bond with an endpoint at distance at most* 1 *from X,*

$$\mathscr{A}_L - d\lambda_* = \mathscr{A}^{\mathrm{b}} + \mathscr{K}_1, \qquad \mathscr{C}_L = \mathscr{C}^{\mathrm{b}} + \mathscr{K}_2. \tag{12.50}$$

They satisfy

$$\begin{aligned} \|\mathscr{A}^{\mathrm{b}}\|_\infty &\le 64 L^{-15/8+3\varepsilon/2}\theta_1, \\ \|\partial\mathscr{A}^{\mathrm{b}}\|_\infty &\le 137 L^{-15/8+3\varepsilon/2}\theta_2, \\ \|\delta_\alpha\partial\mathscr{A}^{\mathrm{b}}\|_\infty &\le 1382 L^{-15/8+3\varepsilon/2}\theta_3, \\ \|\mathscr{C}^{\mathrm{b}}\|_\infty < L^{-7/4}d_1, &\qquad \|\partial\mathscr{C}^{\mathrm{b}}\|_\infty < L^{-7/4}d_2, \end{aligned} \tag{12.51}$$

and

$$\|\mathscr{K}_1\|_\infty < b_1, \qquad \|\mathscr{K}_2\|_\infty < L^{-1+\varepsilon}d_1. \tag{12.52}$$

Consequently

$$(t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}}) \in R_k \times \mathfrak{X}_k \qquad (|t| \le L^{7/4}). \tag{12.53}$$

The construction uses a base point of X and a decomposition and local real representative of $\mathscr{A}$, and is independent of the activity. When $\mathscr{C} = 0$, it gives $\mathscr{C}^{\mathrm{b}} = \mathscr{K}_2 = 0$.

Proof. Choose a lift of X and a photon site c_* in it. Set $r = 2LM \ge 50$ and $K_{\mathrm{loc}} = K(c_*, 2r+5)$, with sup balls understood through their nonwrapping lifts. Choose the coarse M-cube containing c_*/L. If $\mathscr{A} = A_0 + A_1$ is its domain decomposition, its real representative A' and gauge λ on the enlarged cube give

$$a_0 = (A_0)_L = a_0' + d\lambda_L, \qquad a_0' = (A')_L, \qquad a_1 = (A_1)_L$$

on K_{loc}. To see that these functions are defined there, the scaled enlarged cube has side $(2L+1)LM$ and center z with $|c_* - z| < LM/2$. The inequality $4LM + 6 \le L^2M$ puts $K(c_*, 2r+6)$ inside this cube. Moreover $4r + 11 < \ell$, since $\ell \ge 13L^2M$. Thus the boxes and all derivative cells used below do not wrap.

We first record the cutoff estimate with its derivative support. Put

$$f(t) = \begin{cases} 1, & |t| \le 1, \\ 1 - 2(|t|-1)^2, & 1 \le |t| \le 3/2, \\ 2(2-|t|)^2, & 3/2 \le |t| \le 2, \\ 0, & |t| \ge 2. \end{cases}$$

For a bond $(c, c+\eta_k e_\mu)$ use the product of the three functions $f((c_\nu + \eta_k \delta_{\mu\nu}/2 - c_{*,\nu})/r)$ as its cutoff multiplier. Denote this operation by cut_r. The function f is continuously differentiable, $|f'| \le 2$, and f' is Lipschitz with constant 4. If a bond function b has local norms B_0, B_1, B_α on K_{loc}, extension by zero after multiplication gives

$$\begin{aligned} \|\operatorname{cut}_r b\|_\infty &\le B_0, \\ \|\partial \operatorname{cut}_r b\|_\infty &\le B_1 + 2B_0/r, \\ \|\delta_\alpha \partial \operatorname{cut}_r b\|_\infty &\le B_\alpha + 12B_1/r + 12B_0/r^2. \end{aligned} \tag{12.54}$$

Indeed, the multiplier has oscillation at most $6|c-c'|/r$, first difference at most $2/r$, and first-difference oscillation at most $12|c-c'|/r^2$. Apply the product rule to the first difference. In its Hölder difference the four terms are bounded by $B_\alpha|c-c'|^\alpha$, twice $6B_1|c-c'|/r$, and $12B_0|c-c'|/r^2$. Here $|c-c'| \le 1$ implies $|c-c'| \le |c-c'|^\alpha$. If any such term is nonzero, all its cells lie in $K(c_*, 2r+5)$: a nonzero multiplier or its difference requires a site within $2r+1$, the first-difference cell adds one mesh length, and the pair adds at most 1. The same bound covers the two-step cell at $k = 0$. Thus the local estimates suffice for the global ones. The cutoff equals one on bonds with both ends in $K(c_*, r)$.

For $b = a_0'$ or a_1, write $v = c - c_*$ and define

$$V[b]_{\nu\mu} = (\partial_\nu b_\mu)(c_*), \qquad \mathrm{Lin}[b]_\mu(c) = \sum_\nu V[b]_{\nu\mu} v_\nu, \qquad \mathrm{rem}[b] = b - b(c_*) - \mathrm{Lin}[b].$$

Its first difference is $\partial_\nu b_\mu(c) - \partial_\nu b_\mu(c_*)$, and its Hölder first-difference seminorm is at most b_3. Join the two sites by at most $2r+5$ simultaneous coordinate steps of length at most 1. Their difference is therefore at most $(2r+5)b_3 \le 3rb_3$. A rectilinear path to c_* has length at most $3(2r+5)$, and the remainder vanishes at c_*. Keeping the sharper derivative bound along this path and using $r \ge 50$ gives

$$\|\operatorname{rem}[b]\|_{K_{\mathrm{loc}}} \le 3(2r+5)^2 b_3 \le 18r^2 b_3, \qquad \|\partial \operatorname{rem}[b]\|_{K_{\mathrm{loc}}} \le 3rb_3.$$

Equation (12.54) consequently bounds the three global norms of its cutoff by $18r^2b_3$, $39rb_3$, and $253b_3$, respectively.

Only a real gauge is available. We remove the constant and symmetric linear part of a'_0, leaving its curvature and the complex linear part of a_1. Set

$$V^0 = V[a'_0], \qquad V^{\mathrm{s}} = (V^0 + (V^0)^T)/2, \qquad F = V^0 - (V^0)^T,$$

and use the real quadratic function

$$q_*(c) = \sum_{\mu} \Big[a'_{0,\mu}(c_*) - \frac{\eta_k}{2} V^{\mathrm{s}}_{\mu\mu} \Big] v_\mu + \frac{1}{2} \sum_{\nu,\mu} V^{\mathrm{s}}_{\nu\mu} v_\nu v_\mu. \tag{12.55}$$

Its finite difference is exactly $(dq_*)_\mu(c) = a'_{0,\mu}(c_*) + \sum_\nu V^{\mathrm{s}}_{\nu\mu} v_\nu$. The mesh term in (12.55) cancels the diagonal term from differencing the quadratic polynomial. With $\mathscr{K}_1 = a_1(c_*)$ and

$$P_\mu(c) = \sum_{\nu} \Big[\frac{1}{2} F_{\nu\mu} + V[a_1]_{\nu\mu} \Big] v_\nu,$$

we have the local identity

$$\mathscr{A}_L - d(\lambda_L + q_*) = \mathscr{K}_1 + P + \mathrm{rem}[a'_0] + \mathrm{rem}[a_1]. \tag{12.56}$$

To cut off P we need a bound that also uses the side ℓ. A global mean-zero periodic function whose oscillation between sites at distance at most 1 is at most B has sup norm at most ℓB. Any two sites can be joined by at most $\lceil \ell/2 \rceil \le \ell$ simultaneous coordinate steps of length at most 1; comparison with the mean gives the claim. Every $\partial_\nu a_{1,\mu}$ has mean zero, sup norm at most b_2, and this oscillation at most b_3. For a_0, the global curl agrees on each derivative cell with the curl of a local real representative. Its sup norm is at most $2b_2$, its oscillation at most $2b_3$, and it has mean zero. The cells of a pair at distance at most 1 lie in one enlarged cube, so that the same representative can be used for their comparison. It follows that every coefficient of P is bounded by

$$s = 2\min(b_2, \ell b_3). \tag{12.57}$$

Use a second radius

$$r' = \min(e_k^{-\varepsilon}, \ell/5).$$

The charge condition in (12.40) gives $r' \ge r$, and $4r' + 11 < \ell$. Thus its cutoff is also defined without wrapping. The three norms of P on its support are at most $6.3r's$, s, and 0. Equation (12.54), with r' in place of r, bounds the cutoff norms by $6.3r's$, $13.6s$, and $87.6s/r'$. If $r' = e_k^{-\varepsilon}$, use $s \le 2b_2$; in units of $(\theta_1, \theta_2, \theta_3)$ these are at most $(12.6, 27.2, 175.2)L^{-15/8+\varepsilon}$. If $r' = \ell/5 < e_k^{-\varepsilon}$, use $s \le 2\ell b_3$. Since $\ell e_k^{\varepsilon} < 5$, the corresponding three bounds are at most $(63, 136, 876)L^{-15/8+3\varepsilon/2}$.

Define the global field and real gauge by

$$\begin{aligned} \mathscr{A}^{\mathrm{b}} &= \mathrm{cut}_r \big(\mathrm{rem}[a'_0] + \mathrm{rem}[a_1] \big) + \mathrm{cut}_{r'} P, \\ \lambda_* &= \lambda_L + q_* \quad \text{on } K_{\mathrm{loc}}, \qquad \lambda_* = 0 \quad \text{elsewhere.} \end{aligned} \tag{12.58}$$

The two remainder cutoffs contribute $(36r^2, 78r, 506)b_3$. As $r = 2LM$, the charge condition $160L^{1+3\varepsilon/2}Me_k^{\varepsilon} \le 1$ bounds the first two by $L^{-15/8+3\varepsilon/2}\theta_1$ and $L^{-15/8+3\varepsilon/2}\theta_2$, respectively. Adding the linear contribution gives the constants 64, 137, and 1382 in (12.51).

In the external argument take

$$\mathscr{K}_2 = \mathscr{C}_L(c_*), \qquad \mathscr{C}^{\mathrm{b}} = \mathrm{cut}_r(\mathscr{C}_L - \mathscr{K}_2).$$

A rectilinear path to c_* bounds the undifferentiated field on K_{loc} by $6.3r\|\partial\mathscr{C}_L\|_\infty$. The cutoff bounds therefore give

$$\|\mathscr{C}^{\mathrm{b}}\|_\infty \le 6.3r\|\partial\mathscr{C}_L\|_\infty, \qquad \|\partial\mathscr{C}^{\mathrm{b}}\|_\infty \le 13.6\|\partial\mathscr{C}_L\|_\infty.$$

Use (12.43) and $d_2 = e_k^{\varepsilon} d_1$. The first charge condition involving 160 gives the desired bound for the sup norm; the large-L condition gives $13.6L^{-2+3\varepsilon/2} \le L^{-7/4}$ for the derivative. This proves the external bounds and the constant bounds.

The diameter estimate in Appendix A puts X in $K(c_*, M(L+2))$. Every bond with an endpoint within distance 1 of X has both ends in $K(c_*, r)$, since $M(L+2)+2 \le 2LM$. All cutoffs in the local identity are one there, and the gauge is $\lambda_L + q_*$ there. This proves (12.50). Finally, the large-L condition implies $1382L^{-15/8+3\varepsilon/2} < L^{-7/4}$. Multiplication by any $|t| \le L^{7/4}$ consequently leaves all five norms strictly below their domain thresholds, proving (12.53). □

Proposition 12.4. *Let $F(X)$ be a level-k Grassmann activity holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, with the kernel support and coefficient locality specified in §12.1, and covariant under the independent real gauges of its two connection arguments. For the fields of Proposition 12.3, its kernels satisfy*

$$|F_{nm}(X; \mathscr{A}_L, \mathscr{C}_L; \xi; \zeta)| \le \exp(2(n+m))|F_{nm}(X; \mathscr{A}^{\mathrm{b}}, \mathscr{C}^{\mathrm{b}}; \xi; \zeta)|. \tag{12.59}$$

For the vacuum kernel the two values are equal.

Proof. For each kernel argument a let $c(a)$ be the photon site of its ordinary leg, or the first photon site of its pair leg, and let $\omega(a) \in \{0,1\}$ distinguish unbarred and barred legs. On kernels the gauge homomorphism has the form

$$(G_\lambda F)_{nm}(\xi;\zeta) = F_{nm}(\xi;\zeta) \prod_a \exp\{(-1)^{\omega(a)} i e_k \lambda(c(a))\}. \tag{12.60}$$

This follows by applying the homomorphism to the kernel expansion; the multiplying product is symmetric in each set of arguments, and hence preserves their antisymmetry. The gauge relation (12.50) first gives

$$G_{\lambda_*} F(X; \mathscr{A}^{\mathrm{b}} + \mathscr{K}_1, \mathscr{C}^{\mathrm{b}} + \mathscr{K}_2) = F(X; \mathscr{A}_L, \mathscr{C}_L).$$

Both pairs lie in the domains: a constant changes only the sup norm, and $L^{-7/4} + L^{-7/8+\varepsilon/2} < 1$ and $L^{-7/4} + L^{-1+\varepsilon} < 1$. Every phase from this real gauge has modulus one.

We remove the constant fields by continuation of a real gauge identity. Write $\mathscr{K}_j = K_j^{\mathrm{R}} + iK_j^{\mathrm{I}}$ with real constant fields. On the connected domain

$$\Omega = \{z \in \mathbb{C}^4 : |\operatorname{Im} z_1| < 2,\ |\operatorname{Im} z_2| < 2,\ |z_3| < 2,\ |z_4| < 2\},$$

set

$$A(z) = \mathscr{A}^{\mathrm{b}} + z_1 K_1^{\mathrm{R}} + z_2 K_1^{\mathrm{I}}, \qquad C(z) = \mathscr{C}^{\mathrm{b}} + z_3 K_2^{\mathrm{R}} + z_4 K_2^{\mathrm{I}},$$

and set

$$\Lambda_z(c) = -(z_1 K_1^{\mathrm{R}} + z_2 K_1^{\mathrm{I}} + z_3 K_2^{\mathrm{R}} + z_4 K_2^{\mathrm{I}}) \cdot (c - c_*) \quad (c \in K(c_*, r)),$$

with value zero elsewhere. The first connection belongs to $\widetilde{R}_k$: its complex small summand has sup norm below $(L^{-7/4} + 4L^{-7/8+\varepsilon/2})\theta_1 < \theta_1$ and derivatives those of $\mathscr{A}^{\mathrm{b}}$, and its remaining constant summand is real. Invariance under real constant additions handles its unrestricted real part. The external connection has sup norm below $(L^{-7/4} + 4L^{-1+\varepsilon})d_1 < d_1$, and derivative that of $\mathscr{C}^{\mathrm{b}}$. The large-L condition implies these two strict inequalities.

For fixed kernel arguments the function

$$H(z) = F_{nm}(X; A(z), C(z); \xi; \zeta) \prod_a \exp\{(-1)^{\omega(a)} i e_k \Lambda_z(c(a))\}$$

is holomorphic on Ω. When z is real, split Λ_z into its first two and last two terms. These are independent real gauge functions for the two connections, with differentials equal to the negatives of their respective constants on $K(c_*, r)$. Every bond read by the local gauge condition lies in this ball. The real gauge identity and (12.60) therefore give $H(z) = H(0)$ for $z \in \mathbb{R}^2 \times (-2,2)^2$. All derivatives of $H - H(0)$ at zero vanish, so its holomorphic power series is zero there. The identity

theorem extends this equality to the connected domain Ω. Evaluating at $z = (1, i, 1, i)$ gives the identity for the two original complex constants.

It remains to bound the inverse phases. By locality, an ordinary argument lies over X. For a pair of the first kind its first photon site also lies in X. For a boundary pair that site has distance at most $\eta_k \le 1$ from X. Hence in all nonzero kernel terms

$$|c(a) - c_*|_1 \le 3(M(L+2)+1) \le 9LM.$$

The constant bounds and the charge conditions give

$$e_k|K_1^{\mathrm{I}} \cdot (c(a) - c_*)| \le 9LML^{-7/8+\varepsilon/2} e_k^{1/4+\varepsilon} \le 1,$$
$$e_k|K_2^{\mathrm{I}} \cdot (c(a) - c_*)| \le 9ML^{\varepsilon} e_k^{2\varepsilon} \le 1.$$

Each leg phase, and its inverse, thus has modulus at most $\exp(2)$. Combining this bound with the unit modulus phases from λ_* proves (12.59). The vacuum has no leg phases, which proves its equality. □

12.8. **Contraction of the normalized activity.** The ordinary quadratic kernel now vanishes at zero connection. The vacuum is also even in the connection, so its first derivative vanishes there. The analytic disk gives an additional gain for precisely these two terms; every other term is controlled directly by its Grassmann degree.

Proposition 12.5. *Assume* (12.35), (12.40), *and*

$$\kappa \ge 7\kappa_0 + 7, \qquad L(\kappa - 6\kappa_0 - 4) - 1 \ge \kappa. \tag{12.61}$$

For $\mathscr{E} \in K_k^{x,\mathbb{C}}(\ell)$ set $E = (R\mathscr{E})^{\mathrm{nat}}$. The linear map $\mathscr{E} \mapsto \mathcal{L}E$ takes $K_k^{x,\mathbb{C}}(\ell)$ into $K_{k+1}^{x,\mathbb{C}}(\ell/L)$, preserves the real subspaces, and satisfies

$$\|\mathcal{L}E\|_{k+1}^x \le c_{\mathrm{fl}} L^{-1/4+3\varepsilon/2} \|\mathscr{E}\|_k^x \le c_{\mathrm{fl}} L^{-1/4+2\varepsilon} \|\mathscr{E}\|_k^x, \qquad c_{\mathrm{fl}} = 61 c_{\mathrm{rb}}. \tag{12.62}$$

Relative to the activity windows,

$$\frac{\|\mathcal{L}E\|_{k+1}^x}{w_{k+1}^{\mathrm{E}}} \le c_{\mathrm{fl}} L^{-3/8+11\varepsilon/2} \frac{\|\mathscr{E}\|_k^x}{w_k^{\mathrm{E}}}. \tag{12.63}$$

Restriction to the zero external argument commutes with the map.

Proof. Write

$$N(X) = \sup_{\widetilde{R}_k \times \mathfrak{X}_k} \|E(X; \cdot, \cdot)\|_{h_k}.$$

Extraction and adjustment give

$$N(X) \le \begin{cases} 20\|\mathscr{E}\|_k^x \exp(-\kappa d_M(X)), & X \text{ small}, \\ \|\mathscr{E}\|_k^x \exp(-\kappa d_M(X)), & X \text{ large}. \end{cases} \tag{12.64}$$

For a small polymer this uses $\|R\mathscr{E}(X; \mathscr{A}, \mathscr{C})\|_{h_k} \le \|\mathscr{E}(X; \mathscr{A}, \mathscr{C})\|_{h_k} + \|\mathscr{E}(X; 0, 0)\|_{h_k}$ and (12.39). For a large polymer both operations leave the activity unchanged.

Fix a small X and a coarse connection pair, and use the global representatives above. Put $\rho = L^{7/4}$. The inverse image of the open set $R_k \times \mathfrak{X}_k$ under $t \mapsto (t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}})$ contains the closed disk $|t| \le \rho$, hence a slightly larger disk. The functions

$$f_{nm}(t) = E_{nm}(X; t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}})$$

are holomorphic there, with values in a finite-dimensional kernel space. Their weighted norms obey

$$\frac{h_{k,1}^n h_{k,2}^m}{n!m!} \|f_{nm}(t)\|_1 \le N(X) \qquad (|t| \le \rho).$$

Here $\|\cdot\|_1$ is the sum of kernel moduli with their index weights.

For the vacuum, (12.38) gives $f_{00}(0) = 0$. Charge conjugation fixes a scalar kernel and reverses both connections, so $f_{00}(-t) = f_{00}(t)$ and $f'_{00}(0) = 0$. Cauchy's formula gives

$$f_{00}(1) = \frac{1}{2\pi i} \oint_{|t|=\rho} \frac{f_{00}(t)}{t^2(t-1)}\,\mathrm{d}t.$$

The vacuum equality in Proposition 12.4 and $\rho \geq 2$ therefore give

$$\|(S_L E)^{[0,0]}(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \leq \frac{N(X)}{\rho(\rho-1)} \leq 2L^{-7/2} N(X). \tag{12.65}$$

For the ordinary quadratic kernel, normalization gives $f_{20}(0) = 0$ as a full antisymmetric kernel. Its balanced matrix components vanish by (12.38); their oppositely ordered components vanish by antisymmetry, and the two equally barred or equally unbarred components vanish by balance. Hence

$$f_{20}(1) = \frac{1}{2\pi i} \oint_{|t|=\rho} \frac{f_{20}(t)}{t(t-1)}\,\mathrm{d}t, \qquad \|f_{20}(1)\|_1 \leq \frac{\sup_{|t|=\rho} \|f_{20}(t)\|_1}{\rho - 1}.$$

The comparison of kernels costs $\exp(4)$ in this bidegree. Its scaling exponent is $p(2,0) = 9/4$. Multiplying the last inequality by $h_{k,1}^2/2$ and using (12.47) gives

$$\|(S_L E)^{[2,0]}(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \leq 2\exp(4) L^{-4} N(X). \tag{12.66}$$

For all remaining bidegrees use the original scaled connection pair, which belongs to the fine domain. Evenness leaves only $(1,1)$, $(0,2)$, and total degrees at least 4. The exponents satisfy

$$p(1,1) > \frac{13}{4} - \frac{3\varepsilon}{2}, \qquad p(0,2) > \frac{13}{4} - \frac{3\varepsilon}{2}, \qquad p(n,m) \geq n + m.$$

Consequently

$$\begin{aligned} \sum_{(n,m)\neq(0,0),(2,0)} \|(S_L E)^{[n,m]}(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} &\leq L^{-13/4+3\varepsilon/2} \sum_{n,m} \|E^{[n,m]}(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_k} \\ &\leq L^{-13/4+3\varepsilon/2} N(X). \end{aligned}$$

The sum is taken at one connection pair before bounding it by $N(X)$. The large-L condition ensures $2L^{-7/2} \leq L^{-13/4}$ and $2\exp(4)L^{-4} \leq L^{-13/4}$. Adding the three degree contributions yields

$$\|S_L E(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \leq 3L^{-13/4+3\varepsilon/2} N(X). \tag{12.67}$$

We now sum over the fine polymers in a coarse polymer Y. The hypotheses (12.61) imply $\kappa - 1 \geq \kappa_0$ and $L(\kappa - \kappa_0 - 2) \geq \kappa$. The reblocking bound (A.5) at rate κ and (12.64) therefore give

$$\sum_{\substack{X \text{ small} \\ \bar{X}^L = LY}} \|S_L E(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \leq 60 c_{\mathrm{rb}} L^{-1/4+3\varepsilon/2} \|\mathscr{E}\|_k^x \exp(-\kappa d_M(Y)).$$

For large X, scaling decreases the norm by (12.46) and the ratios of the two weights. Since $d_M(X) > L$,

$$\exp(-\kappa d_M(X)) \leq \exp(-L)\exp(-(\kappa-1) d_M(X)).$$

Use (A.5) at rate $\kappa - 1$ to obtain

$$\sum_{\substack{X \text{ large} \\ \bar{X}^L = LY}} \|S_L E(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \leq c_{\mathrm{rb}} L^3 \exp(-L) \|\mathscr{E}\|_k^x \exp(-\kappa d_M(Y)).$$

As $L^3 \exp(-L) \leq L^{-1/4}$, these estimates prove (12.62) after taking the two suprema in the activity norm. Division by $w_{k+1}^{\mathrm{E}} = L^{1/8-7\varepsilon/2} w_k^{\mathrm{E}}$ gives the relative factor $c_{\mathfrak{fl}} L^{-3/8+5\varepsilon}$, which is at most the

factor in (12.63). The additional parameter choice $c_{\mathrm{fl}} L^{-3/8+6\bar{\varepsilon}} \le 1/16$ makes its relative factor at most $1/16$.

The map is linear. Its kernels are finite sums of holomorphic fine kernels composed with the complex linear connection scaling. The locality was proved after (12.49). For lattice covariance a coarse group element $g' = (S, a')$ has the fine representative $g = (S, La')$. It gives a bijection of the polymers in the two reblocking sums, satisfies $(g'\mathscr{A})_L = g\mathscr{A}_L$, and intertwines the spin actions by $a_{g'} S_L = S_L a_g$. The same identity holds for $\mathscr{C}$. Thus covariance of each fine activity gives covariance of its sum. For the real gauges, every fine bond within distance 1 of X scales to a bond within distance $1/L \le 1$ of Y, while $G_\lambda S_L = S_L G_{\lambda_L}$ follows from $e_k \lambda_L(c) = e_{k+1}\lambda(c/L)$. Independent real gauges of the two arguments therefore pass through each term. Finally, scaling has real coefficients, preserves the bar index, and commutes with charge conjugation and the antilinear reality map. These facts prove membership in both stated activity spaces. Since extraction and adjustment commute with restriction and $0_L = 0$, reblocking also commutes with restriction to the zero external argument. □

The reblocked sum retains the evaluated normalized activity. Indeed, each fine polymer belongs to exactly one reblocking class, so

$$\sum_Y (\mathcal{L}E)(Y; \mathscr{A}, \mathscr{C}) = S_L \sum_X E(X; \mathscr{A}_L, \mathscr{C}_L). \tag{12.68}$$

If the coarse generators are evaluated at fields with $\chi' = \delta_{\alpha, \mathscr{A}+\mathscr{C}}\psi'$ and its barred counterpart, their scaled fields satisfy the fine relations. This follows from $d_X(x/L, x'/L) = d_X(x, x')/L$ and $U_{(\mathscr{A}+\mathscr{C})_L}(x, x') = U_{\mathscr{A}+\mathscr{C}}(x/L, x'/L)$: the path bijection and $e_k \eta_k L^{-1/2} = e_{k+1}\eta_{k+1}$ give the latter. Substitution in the difference quotient produces $L^{-1-\alpha}\chi'(x/L, x'/L)$, and the barred quotient uses the inverse transport. Adjustment consequently has its original value under this evaluation. Thus the evaluation of (12.68) is also the evaluation of

$$S_L \sum_X (R\mathscr{E})(X; \mathscr{A}_L, \mathscr{C}_L).$$

12.9. **Lifted formulas and comparison of local data.** For the later comparison of different torus volumes we need formulas that depend on a lift and local data alone. The required margin also distinguishes the pairs produced by scaling from the full set of coarse pairs.

For a finite photon set $\widetilde{U}$ on the covering lattice, let $\mathfrak{G}_k(\widetilde{U})$ be the algebra generated by the ordinary indices over $\widetilde{U}$ and pair indices whose two photon sites lie in $\widetilde{U}$. Its ordinary and pair measures remain $\eta_k^3/48$ and $(\eta_k^3/48)^2$. For a system of separately antisymmetric kernels $K = (K_{nm})$ define

$$\mathrm{El}_k(K) = \sum_{n,m} \frac{1}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, K_{nm}(\xi; \zeta)\psi(\xi_1)\cdots\psi(\xi_n)\chi(\zeta_1)\cdots\chi(\zeta_m).$$

The integrals range over these two finite index sets. The kernel expansion is a bijection between the systems of such kernels and this algebra: at increasing arguments its coefficient is the kernel times the positive product of index weights, as in (12.1).

Map each covering generator to the torus generator at its simultaneous period class, with the antiperiodic convention, and denote the homomorphism by cov_ℓ. If

$$\mathrm{diam}_\infty(\widetilde{U}) + \eta_k < \ell, \tag{12.69}$$

it is injective and preserves kernels when read on this lift. In fact two congruent photon sites in $\widetilde{U}$ must coincide. Choose a fundamental domain containing $\widetilde{U}$ and order its independent ordinary and pair indices so that they extend the covering orders. The homomorphism maps distinct covering basis monomials to distinct torus basis monomials with the same coefficients

and weights. This proves injectivity, preservation of bidegrees, and equality of the kernels at the lifted arguments. All torus kernels at indices outside these period classes vanish.

Now let $\widetilde{X}$ be a face-connected lift and put $\widetilde{U} = \widetilde{X}^{+\eta_k}$, its mesh neighborhood. The margin

$$\operatorname{diam}_\infty(\widetilde{X}) + 2\eta_k + M < \ell \tag{12.70}$$

gives a bijection of ordinary indices over $\widetilde{X}$ onto those over X, and of admissible pairs for $\widetilde{X}$ onto those for X. For a pair of the first kind, its endpoints and every point of its shortest paths lift inside $\widetilde{X}$: a lifted path point within distance M of its first endpoint differs from its representative in $\widetilde{X}$ by a period of norm at most $M + \operatorname{diam}_\infty(\widetilde{X}) < \ell$. That period is zero. For a boundary pair $x' = x^{(3)}$, lift the endpoint in X; the other is its prescribed nearest neighbor in $\widetilde{U}$. Two lifted pairs in the same class coincide because their first photon sites coincide. These arguments prove surjectivity, injectivity and preservation of pair kind. The margin also implies (12.69) for $\widetilde{U}$. By the nonwrapping lifting result, X is small exactly when $d_M(\widetilde{X}) \le L$: for either a small torus polymer or a small covering polymer the unique nonwrapping lift has the same net length.

Take data $(K^0, K, \widetilde{B})$ consisting of two kernel systems over $\widetilde{U}$ and a connection on bonds with both ends in $\widetilde{X}$. For $d_M(\widetilde{X}) \le L$, define

$$\begin{aligned} M^0 &= \operatorname{Vol}(\widetilde{X})^{-1} \int_{\widetilde{X}} \mathrm{d}x' \int_{\widetilde{X}} \mathrm{d}x''\, K^0_{20}(x', x''), \\ \Phi^R &= \operatorname{El}_k(K) - K^0_{00} - \int_{\widetilde{X}} \mathrm{d}x\, \bar\psi(x)^T M^0 \psi(x), \\ \Phi^E &= \Phi^R - (\Phi^R)^{[2,0]}_{\widetilde{X}} + \Theta_{\widetilde{X},\widetilde{B}} (\Phi^R)^{[2,0]}_{\widetilde{X}}. \end{aligned} \tag{12.71}$$

Here K^0_{20} is the balanced matrix component in the earlier convention, and the subscript $\widetilde{X}$ retains only quadratic terms whose two sites lie over $\widetilde{X}$. Thus the transport homomorphism is applied within its domain. For $d_M(\widetilde{X}) > L$, set $\Phi^R = \Phi^E = \operatorname{El}_k(K)$. In both cases define

$$\Phi^S = \widetilde{S}_L \Phi^E,$$

where $\widetilde{S}_L$ is the covering version of (12.45). These formulas use no side ℓ. Translation by $v \in M\mathbb{Z}^3$ preserves the cube geometry, measures and chain transports. Therefore

$$\Phi^R(T_v \text{data}) = T_v \Phi^R(\text{data}), \qquad \Phi^E(T_v \text{data}) = T_v \Phi^E(\text{data}), \qquad \Phi^S(T_v \text{data}) = T_{v/L} \Phi^S(\text{data}), \tag{12.72}$$

with the translated polymer understood on the left.

For an activity $\mathscr{E}$, let K^0 and K be its kernels at $(0,0)$ and at $(\mathscr{A}_L, \mathscr{C}_L)$, restricted to the lift, and let $\widetilde{B}$ be the restriction of $(\mathscr{A} + \mathscr{C})_L$ to the internal bonds. Then

$$\begin{aligned} (R\mathscr{E})(X; \mathscr{A}_L, \mathscr{C}_L) &= \operatorname{cov}_\ell \Phi^R, \\ (R\mathscr{E})^{\mathrm{nat}}(X; \mathscr{A}_L, \mathscr{C}_L) &= \operatorname{cov}_\ell \Phi^E, \\ S_L[(R\mathscr{E})^{\mathrm{nat}}(X; \mathscr{A}_L, \mathscr{C}_L)] &= \operatorname{cov}_{\ell/L} \Phi^S. \end{aligned} \tag{12.73}$$

To prove the first identity, locality and the index bijection reproduce the original activity from K. The vacuum subtraction is exactly K^0_{00}, and the two label-averaged mass integrals have the same weights and volume in the lift and on the torus. The diagonal bilinear is period invariant, since its two antiperiodic signs cancel. The subtraction is therefore also the same. For the second identity the ordinary quadratic terms of Φ^R lie over $\widetilde{X}$. Cubes, their shortest chains and their transports are the lifted versions of the torus construction. On each ordinary generator this gives $\operatorname{cov}_\ell \Theta_{\widetilde{X},\widetilde{B}} = \Theta_{X,(\mathscr{A}+\mathscr{C})_L} \operatorname{cov}_\ell$. The homomorphism identity proves the adjusted formula. Finally, on each generator

$$\operatorname{cov}_{\ell/L} \widetilde{S}_L = S_L \operatorname{cov}_\ell .$$

Both sides have the same scaled index and coefficient, with compatible antiperiodic signs. This proves the third identity.

The margin (12.70) suffices for the image of $\widetilde{S}_L$, whose pairs have $d_X \le M/L$. To identify $\mathrm{cov}_{\ell/L}$ with the quotient map on the full coarse local algebra, including pairs with $M/L < d_X \le M$, we use the stronger margin

$$\mathrm{diam}_\infty(\widetilde{U}) + LM < \ell. \tag{12.74}$$

It is equivalent to $\mathrm{diam}_\infty(\widetilde{U}/L) + M < \ell/L$. For a small polymer both margins hold, since

$$\mathrm{diam}_\infty(\widetilde{X}) + 2\eta_k + LM \le 2LM + 4M < \ell.$$

The torus expressions in (12.73) do not depend on the chosen lift. Translation of that lift by ℓw translates the data; their total even degree makes the product of their argument signs one. Equation (12.72) and the same evenness then cancel the generator signs under the quotient maps at both levels. Thus identical lifted data produce identical lifted extraction, adjustment and scaled kernels on different admissible tori, subject to the stated margins.

13. The exact block density

The block transformation has three factors: a new quadratic fermion density, a determinant quotient, and a finite integral over photon and fermion fluctuations. We derive this identity before expanding its last factor. The integration at level zero includes every chamber component and uses the chamber Berezin measure.

13.1. **Integration spaces and real backgrounds.** Fix $k \ge 0$, $\eta_k = L^{-k}$, odd $L \ge 5$, and $M = L^m \ge L^2$, with $\ell/(LM)$ odd and at least 13. The regulator parameter satisfies $0 < \rho < 1$. Keep $b = 1$ as a letter, and write $b_L = b/L$ and $b_k = b(1 - L^{-1})/(1 - L^{-k})$ for $k \ge 1$. Write $\mathscr{V}_k = \mathcal{S}(1, \ell)$ for $k \ge 1$, with four spin components and counting measure, and $\mathscr{V}_0 = \mathcal{S}^{\mathrm{ch}}(1, \ell)$, with 192 components per photon site and measure $1/48$. The one-step averaging operator into $\mathcal{S}(L, \ell)$ is

$$\mathcal{A}_k(B) = \begin{cases} Q(B), & k \ge 1, \\ Q(B)J^*, & k = 0, \end{cases} \qquad \mathcal{A}_k^\sharp(B) = \mathcal{A}_k(-B)^T.$$

The target measure is L^3. On these spaces set

$$\mathcal{D}_k(B) = \begin{cases} D_k(B), & k \ge 1, \\ D_B + \mu, & k = 0, \end{cases} \qquad K_k(B) = \mathcal{D}_k(B) + b_L \mathcal{A}_k^\sharp(B)\mathcal{A}_k(B). \tag{13.1}$$

Thus $K_k = D_k + b_L P_L$ for $k \ge 1$ and $K_0 = F_0 = D_B + \mu + b_L J P_L J^*$. We write $D_0(B) = D_B + \mu$, so that $\mathcal{D}_k(B) = D_k(B)$ at every level. When it exists, $\Gamma_k = K_k^{-1}$, and $\delta Z_k(B) = \det K_k(B)$. The transpose refers to the displayed bilinear measures. For a real B, the phases have modulus one and $\mathcal{A}_k^\sharp = \mathcal{A}_k^*$.

The normalized averages give $Q(B)Q^T(-B) = 1$. On a coarse block $B(z')$ the product is $L^{-3}\sum_{z \in B(z')} U_B(z', z)U_B(z', z)^{-1} = 1$. At level zero the additional cluster mean contributes $48/48 = 1$. Similarly $\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1$ for $k \ge 1$. In a unit block the photon count gives $\eta_k^3\eta_k^{-3} = 1$ and the cluster count again gives $48/48 = 1$. For real B the inverse phases in the embeddings are the complex conjugates of the averaging phases. The products $\Pi_1 = \mathcal{Q}_k^\sharp\mathcal{Q}_k$ and $\Pi_L = \mathcal{Q}_k^\sharp P_L \mathcal{Q}_k$ are therefore orthogonal projections; so are P_L and, at level zero, JP_LJ^*. Their scalar spin kernels commute with every constant spin matrix.

Proposition 13.1. *Let B be real and $\mu > 0$. The operators $D_B + \mu$, $\mathcal{D}_k(B)$ and $K_k(B)$ are invertible at every level at which they are defined. For $k \geq 1$,*

$$D_k(B)^{-1} = b_k^{-1} + \mathcal{Q}_k(B)(D_B + \mu)^{-1}\mathcal{Q}_k^\sharp(B), \qquad \operatorname{Re} D_k(B)^{-1} \geq b_k^{-1}. \tag{13.2}$$

Moreover $\operatorname{Re} D_k > 0$ and $\operatorname{Re} K_k > 0$ as quadratic forms for $k \geq 1$, whereas $\operatorname{Re} K_0 \geq \mu$. Each K_k satisfies $K_k^* = \gamma_5 K_k \gamma_5$. For every real y, $K_k + i\gamma_5 y$ is invertible, and for $y \neq 0$,

$$\|(K_k + i\gamma_5 y)^{-1}\|_{2\to 2} \leq |y|^{-1}. \tag{13.3}$$

For $k \geq 1$ the fine inverse $S_{k,y}$ also exists for every real y. In particular $\det D_k(0)$ and $\delta Z_k(0)$ are nonzero, including the determinants on $\mathscr{V}_0$.

Proof. At a real connection the chamber hops are self-adjoint unitary involutions. In the proof of (9.4), reindexing x by $x^{(i)}$ uses $U_B(x^{(i)}, x) = \overline{U_B(x, x^{(i)})}$. It gives

$$(f, \mathcal{L}_B f) = \frac{1}{2}\sum_i w_i \int \mathrm{d}x\, |R_i(B)f(x) - f(x)|_2^2 \geq 0.$$

The direction vector changes sign under the same involution; hence $\sum_i w_i(\gamma \cdot \widehat{d}_i)R_i(B)$ is anti-self-adjoint. The cancellation $\sum_i w_i \widehat{d}_i = 0$ yields $D_B^* = \gamma_5 D_B \gamma_5$ and $\operatorname{Re} D_B = (\rho/(2\eta_k))\mathcal{L}_B \geq 0$. Thus $\operatorname{Re}(D_B + \mu) \geq \mu$. For $k \geq 1$ adding the positive penalty $b_k\Pi_1$ gives $\operatorname{Re} F_\infty \geq \mu$. These inequalities imply injectivity and then invertibility in the finite-dimensional spaces.

Put $G = D_B + \mu$ and $S_k = (G + b_k\mathcal{Q}_k^\sharp\mathcal{Q}_k)^{-1}$. Multiplication of $b_k^{-1} + \mathcal{Q}_k G^{-1}\mathcal{Q}_k^\sharp$ by $D_k = b_k - b_k^2\mathcal{Q}_k S_k \mathcal{Q}_k^\sharp$ gives

$$1 + b_k\mathcal{Q}_k G^{-1}[1 - (G + b_k\mathcal{Q}_k^\sharp\mathcal{Q}_k)S_k]\mathcal{Q}_k^\sharp = 1.$$

This proves (13.2). For an invertible operator O, $\operatorname{Re}(u, Ou) = \operatorname{Re}(Ou, O^{-1}Ou)$. In particular $\operatorname{Re} O \geq 0$ implies $\operatorname{Re} O^{-1} \geq 0$. As $\mathcal{Q}_k^\sharp = \mathcal{Q}_k^*$ here, the second term of D_k^{-1} has nonnegative real part. For a nonzero v, consequently,

$$\operatorname{Re}(v, D_k v) = \operatorname{Re}(D_k v, D_k^{-1}D_k v) \geq b_k^{-1}\|D_k v\|_2^2 > 0.$$

Adding $b_L P_L$ preserves strict positivity. At $k = 0$ the same conclusion follows directly by adding $b_L J P_L J^*$ to $D_B + \mu$.

The projections commute with γ_5. It follows first that $F_\infty^* = \gamma_5 F_\infty \gamma_5$ and $S_k^* = \gamma_5 S_k \gamma_5$. Taking the adjoint in the formula for D_k and using $\mathcal{Q}_k^\sharp = \mathcal{Q}_k^*$ gives $D_k^* = \gamma_5 D_k \gamma_5$. The penalty has the same symmetry, proving it for K_k. For $k = 0$ this is already the symmetry of its three terms. Hence

$$(K_k + i\gamma_5 y)^*(K_k + i\gamma_5 y) = K_k^* K_k + y^2, \qquad \|(K_k + i\gamma_5 y)f\|_2^2 = \|K_k f\|_2^2 + y^2\|f\|_2^2.$$

For $y \neq 0$ this proves injectivity and the inverse bound; $y = 0$ is covered by strict positivity. For the fine inverse at $k \geq 1$, use the block matrix with diagonal blocks $b_k + b_L P_L + i\gamma_5 y$ and F_∞, and off-diagonal blocks $-b_k\mathcal{Q}_k$ and $-b_k\mathcal{Q}_k^\sharp$. The first diagonal block is invertible by its idempotent decomposition (11.10). Its two Schur complements are F_y and $K_k + i\gamma_5 y$. The latter is invertible, so the block matrix and then F_y are invertible. This proves existence of $S_{k,y}$. The asserted normalizing determinants are therefore nonzero at the real zero connection. □

13.2. **The next quadratic density.** Assume now that ℓ/L is also admissible, and let $B = B'_L$. At the outgoing level the mass is $L\mu$ and the charge is $e_{k+1} = L^{1/2}e_k$. The following identities hold whenever $S_k(B)$ and $\Gamma_k(B)$ exist for $k \geq 1$, or $\Gamma_0(B)$ exists for $k = 0$. This invertibility domain is open and invariant under $B \mapsto -B$: all elementary entries are entire functions of the bond values, and the transpose identities are

$$\mathcal{D}_k(B)^T = \mathsf{C}^{-1}\mathcal{D}_k(-B)\mathsf{C}, \qquad K_k(B)^T = \mathsf{C}^{-1}K_k(-B)\mathsf{C}, \qquad \Gamma_k(B)^T = \mathsf{C}^{-1}\Gamma_k(-B)\mathsf{C}. \tag{13.4}$$

For the first identity reindex each hopping term by $x^{(i)}$, using $M_i(x^{(i)})^T = \mathsf{C}^{-1}M_i(x)\mathsf{C}$ and the reverse phase at $-B$. Transposition of the averages reverses their connection in the product P_L, and the scalar kernels commute with C. The remaining identities follow by products and inversion. Invertible matrices have analytic inverses on their open determinant domains.

Define the fine-to-L average and its embedding by

$$\mathcal{Q}_k^+(B) = \begin{cases} Q(B)\mathcal{Q}_k(B), & k \geq 1, \\ Q(B)J^*, & k = 0, \end{cases} \qquad \mathcal{Q}_k^{+,\sharp}(B) = \mathcal{Q}_k^+(-B)^T.$$

They obey $\mathcal{Q}_k^+\mathcal{Q}_k^{+,\sharp} = 1$. Set $\bar{b}_k = b_{k+1}/L$ and

$$F_k^+(B) = D_B + \mu + \bar{b}_k\mathcal{Q}_k^{+,\sharp}\mathcal{Q}_k^+, \qquad \Sigma_k^+(B) = F_k^+(B)^{-1}. \tag{13.5}$$

For $k = 0$ this is F_0 and its inverse Γ_0. For $k \geq 1$ it is the fine resolvent operator at $y = 0$, since $\bar{b}_k = b_k b_L/(b_k + b_L)$.

To verify existence and the identities needed below, write $\mathcal{B}_k = (b_k + b_L P_L)^{-1}$ for $k \geq 1$ and consider

$$\begin{pmatrix} b_k + b_L P_L & -b_k\mathcal{Q}_k \\ -b_k\mathcal{Q}_k^\sharp & F_\infty \end{pmatrix}.$$

Its lower-right Schur complement is K_k. Since P_L is idempotent and scalar in spin, $\mathcal{B}_k = b_k^{-1}(1 - P_L) + (b_k + b_L)^{-1}P_L$. The other Schur complement is

$$F_\infty - b_k^2\mathcal{Q}_k^\sharp\mathcal{B}_k\mathcal{Q}_k = D_B + \mu + \frac{b_k b_L}{b_k + b_L}\mathcal{Q}_k^\sharp P_L\mathcal{Q}_k = F_k^+.$$

The two assumed inverses make the block matrix invertible, and hence F_k^+ is invertible. Equating the upper-left and lower-left blocks of its inverse gives

$$\Gamma_k = \mathcal{B}_k + b_k^2\mathcal{B}_k\mathcal{Q}_k\Sigma_k^+\mathcal{Q}_k^\sharp\mathcal{B}_k, \qquad H_k\Gamma_k = b_k\Sigma_k^+\mathcal{Q}_k^\sharp\mathcal{B}_k. \tag{13.6}$$

This is the same Schur calculation as Proposition 11.2, retaining the off-diagonal block for the mean field.

Let

$$D_{k+1}^0(B) = b_L - b_L^2\mathcal{A}_k(B)\Gamma_k(B)\mathcal{A}_k^\sharp(B). \tag{13.7}$$

For $k \geq 1$, $QP_L = Q$ and $P_L Q^T(-B) = Q^T(-B)$ give $Q\mathcal{B}_k = (b_k + b_L)^{-1}Q$ and $\mathcal{B}_k Q^T(-B) = (b_k + b_L)^{-1}Q^T(-B)$. Insert these in (13.6) to obtain

$$\begin{aligned} D_{k+1}^0(B) &= \bar{b}_k - \bar{b}_k^2\mathcal{Q}_k^+\Sigma_k^+\mathcal{Q}_k^{+,\sharp}, \\ b_L H_k(B)\Gamma_k(B)\mathcal{A}_k^\sharp(B) &= \bar{b}_k\Sigma_k^+\mathcal{Q}_k^{+,\sharp}. \end{aligned} \tag{13.8}$$

For $k = 0$ both formulas follow directly from $\bar{b}_0 = b_L$, $H_0 = 1$, $\Sigma_0^+ = \Gamma_0$, and $\mathcal{Q}_0^+ = \mathcal{A}_0$.

Let R_L denote the scaling $f_L(c, g) = L^{-1}f(c/L, g)$ on chamber spinors and $v_L(z') = L^{-1}v(z'/L)$ on unit spinors. These are bijections between the corresponding incoming and outgoing spaces. Scaling preserves the label and the nested block centers, and $e_k\eta_k L^{-1/2} = e_{k+1}\eta_{k+1}$ preserves each transport phase. Consequently

$$(D_B + \mu)\mathsf{R}_L = L^{-1}\mathsf{R}_L(D_{B'} + L\mu), \qquad \mathcal{Q}_k^+(B)\mathsf{R}_L = \mathsf{R}_L\mathcal{Q}_{k+1}(B'),$$

and the corresponding embedding identity holds. For the average, its L^{-3} factor cancels the L^3 change of the chamber index measure. Thus

$$F_k^+(B)\mathsf{R}_L = L^{-1}\mathsf{R}_L F_\infty(B'), \qquad \Sigma_k^+(B)\mathsf{R}_L = L\mathsf{R}_L S_{k+1}(B'). \tag{13.9}$$

In particular the outgoing inverse exists. Equations (13.8) and (13.9) now imply

$$D_{k+1}^0(B)\mathsf{R}_L = L^{-1}\mathsf{R}_L D_{k+1}(B'), \qquad b_L H_k(B)\Gamma_k(B)\mathcal{A}_k^\sharp(B)\mathsf{R}_L = \mathsf{R}_L H_{k+1}(B'). \tag{13.10}$$

All expressions in the second identity act on the outgoing four-component field.

For independent outgoing Grassmann fields define

$$\Psi^c = b_L \Gamma_k(B) \mathcal{A}_k^\sharp(B) \Psi'_L, \qquad \bar\Psi^c = b_L \Gamma_k(B)^T \mathcal{A}_k(B)^T \bar\Psi'_L. \tag{13.11}$$

Then, with $\psi' = H_{k+1}(B')\Psi'$ and $\bar\psi' = \bar H_{k+1}(B')\bar\Psi'$,

$$H_k(B)\Psi^c = \psi'_L, \qquad \bar H_k(B)\bar\Psi^c = \bar\psi'_L. \tag{13.12}$$

The first relation is (13.10). For the second use (13.4) and apply the first relation at $-B$ to $\mathsf{C}\bar\Psi'$; the constant matrix C commutes with scaling and the averages. The covariant Hölder quotients of these induced fields are also the scaled outgoing quotients: the transports agree and d_X is divided by L, so the quotient carries the factor $L^{-1-\alpha}$. This holds for every pair, including a pair inside one cluster. Applying $T_k H_k = 1$ and its barred version shows that the means in (13.11) are also $T_k(B)\psi'_L$ and $\bar T_k(B)\bar\psi'_L$. At level zero these maps are identities on the chamber field.

The quadratic term in the Gaussian block formula is therefore exactly

$$\langle \bar\Psi'_L, D^0_{k+1}(B)\Psi'_L\rangle_L = \langle \bar\Psi', D_{k+1}(B')\Psi'\rangle_1,$$

since the pairings of scaled fields multiply by L. Together with (11.28), this identifies the entire outgoing quadratic density.

Proposition 13.2. *On the invertibility domain just specified,*

$$\det D_{k+1}(B') = b^{4n_{k+1}} \frac{\det \mathcal{D}_k(B)}{\delta Z_k(B)}, \qquad n_k = \ell^3, \quad n_{k+1} = (\ell/L)^3. \tag{13.13}$$

In particular, with $c_k = b^{-4n_{k+1}}$,

$$\frac{c_k}{\det D_k(0)} = \frac{1}{\det D_{k+1}(0)\delta Z_k(0)}. \tag{13.14}$$

Proof. Apply Sylvester's determinant identity to $b_L \mathcal{A}_k \Gamma_k$ and $\mathcal{A}_k^\sharp$ on the two different spaces. It gives

$$b_L^{-4n_{k+1}} \det D^0_{k+1}(B) = \det(1 - b_L \mathcal{A}_k^\sharp \mathcal{A}_k \Gamma_k) = \det(\mathcal{D}_k \Gamma_k) = \frac{\det \mathcal{D}_k(B)}{\delta Z_k(B)}.$$

The incoming determinant at $k = 0$ has size $192 n_0$; the exponent of b_L is the outgoing dimension $4n_1$, as in every other step. Equation (13.10) gives $\det D^0_{k+1}(B) = L^{-4n_{k+1}} \det D_{k+1}(B')$. Using $L b_L = b$ proves (13.13). At the zero connection all denominators are nonzero by Proposition 13.1, so (13.14) follows. □

13.3. **The density with an external connection.** For this subsection assume $0 < \mu \le 1/L$ and $e_{k+1} = L^{1/2} e_k \le e_0$, with the hypotheses of Propositions 9.7, 11.2 and 10.4. We also impose (10.42) and (12.6). The inverse domain $\widetilde G_k$ is the domain $\widetilde{\mathcal{G}}_k$ of the propagator estimates. Let $A' \in \mathrm{Dom}_{k+1}$ be a real block field, with arbitrary harmonic part, and let $\mathscr{C}' \in \mathfrak{X}_{k+1}$. Put

$$\mathscr{A}' = H^{\mathrm{ph}}_{k+1} A', \qquad \mathscr{A} = \mathscr{A}'_L, \qquad \mathscr{C} = \mathscr{C}'_L, \qquad \mathscr{T}' = \mathscr{A}' + \mathscr{C}', \qquad \mathscr{T} = \mathscr{T}'_L.$$

The photon block objects are

$$\begin{aligned} A_k(\zeta) &= A_k^{\min}(A') + \mathrm{Sq}_k \zeta, \\ \mathscr{Z}(\zeta) &= H_k^{\mathrm{ph}} \mathrm{Sq}_k \zeta, \qquad \mathscr{B}_k = H_k^{\mathrm{ph}} A_k^{\min}(A') = \mathscr{A} - d\omega_k. \end{aligned} \tag{13.15}$$

Here ω_k is real and periodic, as in (10.29). The truncated photon measure has the normalization

$$\mu_I(\mathrm{Box}_k) = \exp(-n_k \delta \varepsilon_k^0), \qquad \widehat\mu_k = \frac{\mathbf{1}_{\mathrm{Box}_k}}{\mu_I(\mathrm{Box}_k)} \mu_I, \tag{13.16}$$

proved in (10.23). Its denominator is positive: $e_k < 1$ makes $p_{0,k} = (-\log e_k)^{p_0} > 0$, and the box has nonempty interior in the photon coordinate space.

All connections read by the step have their required domains. Indeed, scaling gives $\mathscr{A} \in \frac{1}{4}L^{-3/4}\widetilde{R}_k \subset \widetilde{R}_k^-$ and $\mathscr{C} \in \mathfrak{X}_k$. Proposition 10.4 gives $A_k(\zeta) \in \mathrm{Dom}_k$ for every $\zeta \in \mathrm{Box}_k$. The real-gauge identity in (13.15) yields

$$H_k^{\mathrm{ph}} A_k(\zeta) = \mathscr{A} + \mathscr{Z}(\zeta) - d\omega_k.$$

Since $\frac{1}{4}\widetilde{R}_k$ is invariant under real gauges, $\mathscr{A}+\mathscr{Z}(\zeta)$ also lies in this domain. Thus both connection pairs $(\mathscr{A},\mathscr{C})$ and $(\mathscr{A}+\mathscr{Z},\mathscr{C})$ lie in $\widetilde{R}_k \times \mathfrak{X}_k$. By (12.7), $\mathscr{T}$, $\mathscr{T}+\mathscr{Z}$ and $\mathscr{T}+\mathscr{Z}-d\omega_k$, and their negatives, belong to $\widetilde{G}_k$. The fine and coarse inverse estimates consequently apply to them. The outgoing operators at $\mathscr{T}'$ then exist by (13.9).

For a total connection B let ev_B insert

$$\psi = H_k(B)\Psi, \qquad \bar\psi = \bar H_k(B)\bar\Psi, \qquad \chi = \delta_{\alpha,B} H_k(B)\Psi, \qquad \bar\chi = \delta_{\alpha,-B}\bar H_k(B)\bar\Psi.$$

At $k=0$ the two field maps are identities. For complex $m, \varepsilon^{\mathrm{v}}$ and $\mathscr{E} \in K_k^{x,\mathbb{C}}$, with $\mathscr{E}=0$ at $k=0$, define

$$\begin{aligned} I_k^x[m,\mathscr{E}](A,C;\Psi) &= \mathrm{ev}_{A+C}\left(-m\langle\bar\psi,\psi\rangle + \sum_X \mathscr{E}(X;A,C)\right), \\ \mathrm{Act}_k^x[m,\mathscr{E}](A,C;\Psi) &= -\langle\bar\Psi, D_k(A+C)\Psi\rangle + I_k^x[m,\mathscr{E}](A,C;\Psi). \end{aligned} \tag{13.17}$$

The mass pairing is always the chamber pairing $\eta_k^3\, 48^{-1}\sum_{c,g,\beta}\bar\psi_\beta(c,g)\psi_\beta(c,g)$. The kinetic pairing uses the measure of $\mathscr{V}_k$. For an arbitrary finite family of scalar functions $\mathfrak{P}_k(X,C)$ on $\mathfrak{X}_k$, set

$$\mathscr{F}_k^x[\varepsilon^{\mathrm{v}},m,\mathscr{E}](A,C;\Psi) = \det D_k(0)^{-1}\exp\left\{\mathrm{Act}_k^x[m,\mathscr{E}](H_k^{\mathrm{ph}}A,C;\Psi) - \varepsilon^{\mathrm{v}} n_k + \sum_X \mathfrak{P}_k(X,C)\right\}. \tag{13.18}$$

The arguments here are $A \in \mathrm{Dom}_k$ and $C \in \mathfrak{X}_k$. The fields Ψ and $\bar\Psi$ are independent Grassmann variables.

Define the block transformation on an even density F whose coefficients are integrable along $A_k(\zeta)$ by

$$\begin{aligned} (\mathbf{T}_k^x F)(A',\mathscr{C}';\Psi') &= c_k \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta) \int \mathrm{D}\Psi\, e^{\mathcal{Q}_{\mathrm{bl}}} F(A_k(\zeta),\mathscr{C};\Psi), \\ \mathcal{Q}_{\mathrm{bl}} &= -b_L\big\langle e^{-ie_k\omega_k}\bar\Psi'_L - \mathcal{A}_k(-\mathscr{B}_k-\mathscr{C})\bar\Psi,\, e^{ie_k\omega_k}\Psi'_L - \mathcal{A}_k(\mathscr{B}_k+\mathscr{C})\Psi\big\rangle_L. \end{aligned} \tag{13.19}$$

The gauge function in the two outgoing phases is evaluated at the L-spaced sites. The Berezin functional is the weighted functional of (11.22); its incoming mode measure is 1 for $k \geq 1$ and $1/48$ for $k=0$.

Its normalization is

$$\int \mathrm{D}\Psi'\,(\mathbf{T}_k^x F)(A',\mathscr{C}';\Psi') = \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta) \int \mathrm{D}\Psi\, F(A_k(\zeta),\mathscr{C};\Psi). \tag{13.20}$$

To prove this, integrate the outgoing family first. Reciprocal multiplication of its barred and unbarred generators removes the two phases and leaves the top coefficient unchanged. Translation by the two averaged incoming fields likewise leaves it unchanged: every translated term that contains an external odd field has lower degree in the outgoing family. The remaining outgoing Gaussian exponent is $-b_L\langle\bar\Psi'_L,\Psi'_L\rangle_L = -b\langle\bar\Psi',\Psi'\rangle_1$. Its integral is $b^{4n_{k+1}} = c_k^{-1}$ by (11.25). The two Berezin integrations commute, since both extract an even top-degree coefficient; integration of the finite-dimensional coefficient functions in ζ also commutes with them. This proves (13.20).

For the density (13.18) the coefficient integrability follows from the domain statements above. The bond entries of all inverse operators are analytic on their invertibility domains, and the

activity kernels are holomorphic. Composing them with the affine map $\zeta \mapsto H_k^{\mathrm{ph}} A_k(\zeta)$ gives continuous coefficients on the compact photon box. Their even Grassmann exponentials are also continuous there.

Proposition 13.3. *Let the data and domains be as above. Put*

$$m' = m + m(\mathscr{E}), \qquad \mathscr{E}^r = R\mathscr{E}, \qquad m^+ = Lm', \qquad \varepsilon^{\mathrm{v},+} = L^3\big(\varepsilon^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(\mathscr{E}) + \delta\varepsilon_k^0\big).$$

At $k = 0$ the activity and its extracted terms vanish. Define the outgoing inserted fields $\psi' = H_{k+1}(\mathscr{T}')\Psi'$, $\bar\psi' = \bar H_{k+1}(\mathscr{T}')\bar\Psi'$ and their covariant Hölder quotients, and let $\psi^{0,\#}$ denote their scaled fine fields, including the barred ones. Let the means $\Psi^c, \bar\Psi^c$ be those of (13.11) *at $B = \mathscr{T}$, and put*

$$\Psi^f = \Psi^c + \Gamma_k(\mathscr{T})W, \qquad \bar\Psi^f = \bar\Psi^c + \bar W.$$

With $\mathscr{Z} = H_k^{\mathrm{ph}}\mathrm{Sq}_k\zeta$, set

$$\begin{aligned} V_k^x &= \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A} + \mathscr{Z}, \mathscr{C}; \Psi^f) - \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi^f) \\ &\quad + I_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi^f) - I_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi^c), \\ \Xi_k^x &= \int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}_k(\zeta)\, e^{V_k^x}. \end{aligned} \tag{13.21}$$

Then the exact identity is

$$\begin{aligned} &(\mathbf{T}_k^x \mathscr{F}_k^x[\varepsilon^{\mathrm{v}}, m, \mathscr{E}])(A', \mathscr{C}'; \Psi') \\ &\quad = \det D_{k+1}(0)^{-1} \exp\Big\{ - \langle\bar\Psi', D_{k+1}(\mathscr{T}')\Psi'\rangle_1 - m^+\langle\bar\psi', \psi'\rangle - \varepsilon^{\mathrm{v},+} n_{k+1} \\ &\qquad\qquad + \sum_X \mathscr{E}^r(X; \mathscr{A}, \mathscr{C}; \psi^{0,\#}) + \sum_X \mathfrak{P}_k(X, \mathscr{C})\Big\} \frac{\delta Z_k(\mathscr{T})}{\delta Z_k(0)} \Xi_k^x. \end{aligned} \tag{13.22}$$

All integrals in this formula are finite coefficient integrals. Its right side depends on A' through $H_{k+1}^{\mathrm{ph}} A'$.

Proof. Extraction gives

$$I_k^x[m, \mathscr{E}](A, C; \Psi) = -\varepsilon_{\mathrm{v}}(\mathscr{E}) n_k + I_k^x[m', \mathscr{E}^r](A, C; \Psi).$$

This is the image of (12.16) under the field evaluation homomorphism. At $k = 0$ it holds with zero activity and $m' = m$.

For a fixed ζ, change the integrated variables by $\Psi \mapsto e^{ie_k\omega_k}\Psi$ and $\bar\Psi \mapsto e^{-ie_k\omega_k}\bar\Psi$. This leaves the weighted Berezin functional unchanged, since each mode's factors cancel in its top monomial, also for the $192n_0$ modes of weight $1/48$. The transport identity

$$U_{B - d\omega_k}(c, c') e^{ie_k\omega_k(c')} = e^{ie_k\omega_k(c)} U_B(c, c')$$

intertwines the Dirac operator and every average. Products and inverses preserve this intertwining, so it also holds for H_k and D_k. Its covariant difference identity uses the phase at the first photon site, including for a pair inside one cluster. The barred identity has the inverse phase. Gauge covariance of the activity and invariance of the mass pairing consequently turn its first connection argument $\mathscr{A} + \mathscr{Z} - d\omega_k$ into $\mathscr{A} + \mathscr{Z}$. The same change removes the gauge from the block Gaussian factor. The integral in (13.19) is now

$$\int \mathrm{D}\Psi\, \exp\big\{-b_L\langle\bar\Psi'_L - \mathcal{A}_k(-\mathscr{T})\bar\Psi,\ \Psi'_L - \mathcal{A}_k(\mathscr{T})\Psi\rangle_L + \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A} + \mathscr{Z}, \mathscr{C}; \Psi)\big\},$$

multiplied by $\det D_k(0)^{-1} \exp\{-(\varepsilon^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(\mathscr{E}))n_k + \sum_X \mathfrak{P}_k(X, \mathscr{C})\}$.

Separate the background quadratic term by writing

$$\mathcal{R}(\Psi) = I_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi) \\ + \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A} + \mathscr{Z}, \mathscr{C}; \Psi) - \mathrm{Act}_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi).$$

The last exponent above is then the background Gaussian exponent plus $\mathcal{R}(\Psi)$. Translation by the means (13.11) and the block formula (11.29) give

$$\delta Z_k(\mathscr{T}) e^{-\langle \bar\Psi', D_{k+1}(\mathscr{T}')\Psi'\rangle_1} \int \mathrm{d}\mu_I(W)\, e^{\mathcal{R}(\Psi^f)}.$$

The change of variables in this formula uses the same mode measure for Ψ and W. At $k = 0$ it is the identity

$$\int \mathrm{D}\Psi\, e^{-\langle \bar\Psi, F_0(\mathscr{T})\Psi\rangle^{\mathrm{ch}}} G(\bar\Psi, \Psi) = \det F_0(\mathscr{T}) \int \mathrm{d}\mu_I(W)\, G(\bar W, \Gamma_0(\mathscr{T})W),$$

on the entire $192n_0$-mode space. The Gaussian measure is $e^{-\langle \bar W, W\rangle^{\mathrm{ch}}}\mathrm{D}W$, with the weight $\sqrt{48}$ in its coefficient bound.

By definition of V_k^x, $\mathcal{R}(\Psi^f) = I_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi^c) + V_k^x$. Its first summand is independent of W and ζ, so its even exponential factors out. The induced-field identity (13.12) gives

$$I_k^x[m', \mathscr{E}^r](\mathscr{A}, \mathscr{C}; \Psi^c) = -m'\langle \bar\psi'_L, \psi'_L\rangle + \sum_X \mathscr{E}^r(X; \mathscr{A}, \mathscr{C}; \psi^{0,\#}).$$

The chamber measure changes by L^3 and each field by L^{-1}. Hence $\langle \bar\psi'_L, \psi'_L\rangle = L\langle \bar\psi', \psi'\rangle$, giving $m^+ = Lm'$. Normalizing the photon-box restriction in (13.16) adds $-\delta\varepsilon_k^0 n_k$ to the vacuum term. Since $n_k = L^3 n_{k+1}$, this term has exactly the stated outgoing vacuum coefficient. Finally (13.14) turns the remaining constant factor into $\det D_{k+1}(0)^{-1}\delta Z_k(0)^{-1}$, proving (13.22).

The fluctuation exponent is even and continuous on the photon box, by the same coefficient argument as for the input density. Thus its integral is defined, and its two finite integrations commute. Every background in the final expression is formed from $\mathscr{A}'$ and $\mathscr{C}'$; the remaining photon maps and measure are independent of A'. The minimum and the real gauge have disappeared from that expression, proving the last assertion. □

13.4. **Local substitution of the mean field.** For $k \geq 1$, let Y be a union of partition M-cubes and allow independent unit generators $\Psi, \bar\Psi$ over Y, together with the ordinary and Hölder generators. Using the two operators from (12.23), define the substitution that fixes the latter generators and sets, at the unit sites of Y,

$$\mathrm{sub}_{Y,B}\,\Psi = T_Y^{(1)}(B)\psi + T_Y^{(2)}(B)\chi, \\ \mathrm{sub}_{Y,B}\,\bar\Psi = \bar T_Y^{(1)}(B)\bar\psi + \bar T_Y^{(2)}(B)\bar\chi. \tag{13.23}$$

The images are odd linear combinations of independent generators and therefore anticommute with one another and with the fixed generators. They determine a unique Grassmann homomorphism. Its new ordinary legs lie over Y and its new pair legs are boundary pairs of the second kind with first point over Y. Its coefficients read B on bonds meeting Y, by the row support proved in Subsection 12.4. Since a bar index is preserved, the substitution preserves balance and parity.

If an evaluation has $\chi = \delta_{\alpha,B}\psi$ and the barred relation on these boundary pairs, then (12.24) turns (13.23) into $T_k(B)\psi$ and $\bar T_k(B)\bar\psi$. If in addition $\psi = H_k(B)\Phi$ and $\bar\psi = \bar H_k(B)\bar\Phi$, the left-inverse identity gives Φ and $\bar\Phi$ over Y. Thus this local substitution represents the exact mean field under evaluation.

The row covariances yield

$$\begin{aligned} a_g \,\mathrm{sub}_{gY,gB} &= \mathrm{sub}_{Y,B}\, a_g, \\ G_\lambda \,\mathrm{sub}_{Y,B^1} &= \mathrm{sub}_{Y,B}\, G_\lambda, \qquad B^1 = B - d\lambda \text{ on bonds meeting } Y, \\ \mathrm{Cg}\,\mathrm{sub}_{Y,-B} &= \mathrm{sub}_{Y,B}\,\mathrm{Cg}, \\ \mathbf{K}\,\mathrm{sub}_{Y,-\overline{B}} &= \mathrm{sub}_{Y,B}\,\mathbf{K}. \end{aligned} \tag{13.24}$$

The final relation uses real μ. Both sides fix the old generators according to the stated symmetry, so it suffices to compare their images of $\Psi, \bar\Psi$. The lattice and real-gauge relations follow from (12.32) and (12.31); for a local gauge equality the rows read only the bonds on which that equality is assumed. For charge conjugation use $T_Y^{(i)}(-B)\mathsf{C} = \mathsf{C}\bar T_Y^{(i)}(B)$ and its barred counterpart. For the antilinear map, (12.33) with $-B$ gives $\overline{T_Y^{(i)}(-\overline{B})} = \varkappa^{-1} T_Y^{(i)}(B)\varkappa$, and likewise for the barred row. These identities give (13.24) on each new generator.

Assume the constants and charge conditions of Proposition 12.2, including (12.6), with $0 < \mu \le 1$ and $0 < e_k \le e_0$. For $B \in \widetilde{R}_k + \mathfrak{X}_k$, the row estimate gives the following norm statement. Assign weights w_1, w_2 to the ordinary and pair legs and weight w_3 to the independent unit generators:

$$\| \mathrm{sub}_{Y,B}\, F\|_{w_1,w_2} \le \|F\|_{w_1,w_2;w_3} \quad \text{if } C_T(w_1+w_2) \le w_3. \tag{13.25}$$

Indeed the image of a Ψ generator has norm at most $C_T w_1 + C_T w_2$, and the original generators retain their weights. The weighted linear-substitution bound in Proposition 11.4 applies on the subalgebra whose unit generators lie over Y. The measures for the two row sums are $\eta_k^3/48$ and $(\eta_k^3/48)^2$, respectively; their identity kernels include the reciprocal column measure. Thus this estimate has the same normalization as the reconstruction bound.

13.5. **The first-step vertex.** At $k = 0$ write the charge as q. There is no input activity, and $H_0(B) = \bar H_0(B) = 1$. Consequently the exponent in (13.21) is

$$V_0^x = -\langle \bar\Psi^f, (D_{\mathscr{T}+\mathscr{Z}} - D_{\mathscr{T}})\Psi^f\rangle^{\mathrm{ch}} - m\left(\langle \bar\Psi^f, \Psi^f\rangle^{\mathrm{ch}} - \langle \bar\Psi^c, \Psi^c\rangle^{\mathrm{ch}}\right). \tag{13.26}$$

The terms with μ and m cancel in the connection difference of the actions; the second difference of interactions gives the displayed mass term. Here $\Psi^c = \psi'_L$, $\bar\Psi^c = \bar\psi'_L$, $\Gamma_0(\mathscr{T}) = F_0(\mathscr{T})^{-1}$, and $\Psi^f = \Psi^c + \Gamma_0(\mathscr{T})W$. All these are chamber fields on T_0^{f}. The identity measure in W has mode measure $1/48$, and its two-point kernel is 48 times the identity matrix of the modes.

For a bond fluctuation Z and an M-cube $\square$, define its local vertex by

$$\begin{aligned} \mathcal{V}_0(Z,\square) = \frac{1}{2}\int_\square \mathrm{d}x \Big\{ &\bar\psi(x)^T M_3(x)(e^{iqZ(c,c_3)} - 1)[\psi(x) + \chi(x, x^{(3)})] \\ &+ [\bar\psi(x) + \bar\chi(x, x^{(3)})]^T M_3(x^{(3)})(e^{-iqZ(c,c_3)} - 1)\psi(x)\Big\}, \end{aligned} \tag{13.27}$$

where $x = (c, g)$, $c_3 = c + ge_1$ and $\int_\square \mathrm{d}x = 48^{-1}\sum_{c\in\square}\sum_g$. The pair $(x, x^{(3)})$ has $d_X = 1$ and is admissible for $\square$ of the second kind. Thus the vertex has ordinary legs over the cube and precisely the boundary pair legs allowed in the activity algebra. Its coefficients are entire functions of Z on the bonds meeting the cube.

Proposition 13.4. *Evaluate $\chi = \delta_{\alpha,\mathscr{T}} f$, $\bar\chi = \delta_{\alpha,-\mathscr{T}}\bar f$, $\psi = f$ and $\bar\psi = \bar f$ at any pair of antiperiodic odd chamber fields. Then*

$$\mathrm{ev}\sum_\square \mathcal{V}_0(Z,\square) = -\langle \bar f, (D_{\mathscr{T}+Z} - D_{\mathscr{T}})f\rangle^{\mathrm{ch}}. \tag{13.28}$$

The two halves have equal images after summing over cubes. The vertex has the lattice, real-gauge, charge-conjugation and reality symmetries

$$\begin{aligned} a_g \mathcal{V}_0(gZ, g\square) &= \mathcal{V}_0(Z, \square), \\ G_\lambda \mathcal{V}_0(Z, \square) &= \mathcal{V}_0(Z, \square), \\ \mathrm{Cg}\mathcal{V}_0(-Z, \square) &= \mathcal{V}_0(Z, \square), \\ \mathbf{K}\mathcal{V}_0(-\overline{Z}, \square) &= \mathcal{V}_0(Z, \square). \end{aligned} \tag{13.29}$$

For any generator weights (w_1, w_2)*,*

$$\|\mathcal{V}_0(Z, \square)\|_{w_1,w_2} \le 12(1+\rho)M^3 w_1(w_1+w_2) \sup_{b \text{ oriented and meeting } \square} |e^{iqZ(b)} - 1|. \tag{13.30}$$

If $q \le 1$ *and* $q|Z(b)| \le \frac{1}{2}q^{1-\varepsilon/2}$ *on these bonds, then*

$$\|\mathcal{V}_0(Z, \square)\|_{h_0/2} \le 6(1+\rho)M^3 q^{1/2-\varepsilon/2} \le 12M^3 q^{1/2-\varepsilon/2}. \tag{13.31}$$

Proof. At a nearest-neighbor pair the six paths reduce to the bond, so the quotient relations read

$$f(x) + \chi(x, x^{(3)}) = u_{\mathscr{T}}(c, c_3) f(x^{(3)}), \qquad \bar{f}(x) + \bar{\chi}(x, x^{(3)}) = u_{\mathscr{T}}(c, c_3)^{-1} \bar{f}(x^{(3)}).$$

The first half of the summed vertex is therefore one half of the integral of

$$\bar{f}(x)^T M_3(x)(e^{iqZ(c,c_3)} - 1) u_{\mathscr{T}}(c, c_3) f(x^{(3)}).$$

Reindexing the second half by the measure-preserving involution $x \mapsto x^{(3)}$ gives the same integrand: both Z and $\mathscr{T}$ change sign on the reversed bond. Only the type-3 hop depends on the connection, and $u_{\mathscr{T}+Z} = e^{iqZ} u_{\mathscr{T}}$. The difference of the two Dirac operators is thus the negative of this hopping coefficient, proving (13.28). Every product contains two antiperiodic fields, so the reindexing is legitimate on the torus.

For lattice symmetry use $g^{-1}(x^{(3)}) = (g^{-1}x)^{(3)}$ and $S^{-1}M_3(x)S = M_3(g^{-1}x)$ for the spin lift S. Each half transforms separately, and reindexing maps $g\square \times O_{\mathrm{cub}}$ onto $\square \times O_{\mathrm{cub}}$ with the same measure. For a real gauge, the two generators in each term have the same first photon site and opposite bar indices, so their phases cancel. Charge conjugation interchanges the two halves. Its algebraic rule on the first half gives

$$\mathrm{Cg}[\bar{\psi}(x)^T m(\psi(x) + \chi(x, x^{(3)}))] = (\bar{\psi}(x) + \bar{\chi}(x, x^{(3)}))^T (\mathsf{C}^{-1} m^T \mathsf{C}) \psi(x).$$

The sign from exchanging the odd factors cancels the minus sign in the image of $\bar{\psi}$. As

$$\mathsf{C}^{-1} M_3(x)^T \mathsf{C} = M_3(x^{(3)}),$$

the first half at $-Z$ becomes the second at Z; the reverse relation gives the other half. Finally

$$\varkappa^T \overline{M_3(x)} \varkappa = M_3(x), \qquad \overline{e^{\pm iq(-\overline{Z})} - 1} = e^{\pm iqZ} - 1.$$

The antilinear map consequently preserves each half in its reality identity. This proves (13.29).

Put $\delta_x = e^{iqZ(c,c_3)} - 1$ and $\delta'_x = e^{-iqZ(c,c_3)} - 1$. Collecting the ordinary bilinear in (13.27), its coefficient is $[M_3(x)\delta_x + M_3(x^{(3)})\delta'_x]/96$. The other two products have coefficients $M_3(x)\delta_x/96$ and $M_3(x^{(3)})\delta'_x/96$, and weights $w_1 w_2$. The ordinary product has weight w_1^2. The chamber sites in the cube and the ordered pairs with these first sites are distinct, so these are exactly the coefficients, up to ordering signs. Each endpoint matrix satisfies $\sum_{\beta,\beta'} |M_3(\cdot)_{\beta\beta'}| \le 12(1+\rho)$, since $M_3 = 3(\rho \mp \gamma_\mu)$ and γ_μ has four entries of modulus one. There are $48M^3$ chamber sites in the cube. The triangle inequality for the collected bilinear therefore gives (13.30).

Under the smallness condition, $|e^{iqZ(b)} - 1| \le 2q|Z(b)| \le q^{1-\varepsilon/2}$. For $w = h_0/2$, $w_1(w_1 + w_2) = \frac{1}{4}q^{-1/2}(1 + q^{\varepsilon}) \le \frac{1}{2}q^{-1/2}$. Inserting these bounds proves (13.31). □

For $Z = \mathscr{Z}$ and the evaluation at $\Psi^f, \bar{\Psi}^f$, the first term of (13.26) is the image of the summed vertex. Under the additional charge condition $96M^3q^{1/4} \le 1$, its bound becomes $\|\mathcal{V}_0(\mathscr{Z}, \square)\|_{h_0/2} \le q^{1/4-5\varepsilon}$ whenever the displayed bond smallness condition holds. These statements apply before any expansion of the finite integral.

The first-step density identity retains every vacuum and determinant factor:

$$\begin{aligned}(\mathbf{T}_0^x \mathscr{F}_0^x[\varepsilon^{\mathrm{v}}, m, 0])(A', \mathscr{C}'; \Psi') \\ = \det D_1(0)^{-1} \exp\Big\{ &- \langle \bar{\Psi}', D_1(\mathscr{T}')\Psi'\rangle_1 - Lm\langle \bar{\psi}', \psi'\rangle \\ &- L^3(\varepsilon^{\mathrm{v}} + \delta\varepsilon_0^0)n_1 + \sum_X \mathfrak{P}_0(X, \mathscr{C})\Big\} \\ \times \frac{\det F_0(\mathscr{T})}{\det F_0(0)} &\int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}_0(\zeta)\, e^{V_0^x}.\end{aligned} \tag{13.32}$$

The two determinants of F_0 have size $192n_0$. The original level-zero density, integrated before the block transformation, is

$$\int \mathrm{D}\Psi\, \mathscr{F}_0^x[\varepsilon^{\mathrm{v}}, m, 0](A, C; \Psi) = e^{-\varepsilon^{\mathrm{v}} n_0 + \sum_X \mathfrak{P}_0(X,C)} \frac{\det(D_{A+C} + \mu + m)}{\det D_0(0)}. \tag{13.33}$$

This follows directly from the weighted determinant identity. The numerator need not be invertible; the nonzero denominator uses $\mu > 0$.

14. The nonlinear small-field step

The finite fluctuation integral in (13.21) has a logarithm which is a sum of local activities. Its construction has two localizations. The first resolves the change of background and reconstructs the coarse mean; the second resolves the fermion fluctuations. After these finite operations, a connected-polymer expansion gives the logarithm. The estimates below apply also to the first step, where the fluctuation generators have chamber measure $1/48$.

14.1. **Contour domains and localization.** Keep the admissible sides, mass and charge of Section 13, and assume in addition $\ell \ge Mr_k$. Put $e = e_k$, $h = (h_{k,1}, h_{k,2})$. The weakened photon and fermion primitives are taken on the polydisk

$$\mathrm{Pd}_k = \{s = (s', s'') : |s_{\mathrm{pc}}| \le M^{\alpha_0}\}.$$

Here $s' = (s'_M, s'_{LM})$ contains the photon reconstruction and square-root parameters, of kinds π_M, π_{LM} in (10.36); s'' contains the fermion M-cube parameters π_k of Section 11.1. We assume their analytic kernel and Hölder estimates, component restriction, the row-vanishing properties stated for each family, and coefficient support with common constants $(\alpha_0, \gamma_w, \kappa_w; C_w)$, where $\kappa_w = \kappa + \kappa_0 + 3$. The photon hypotheses are those of Proposition 10.3, (10.41) and the component restriction following them. The fermion hypotheses are those of Propositions 11.1 and 11.3, including (11.14), (11.15), and their covariance identities. At parameter value one all families equal the exact operators. Their gauge, lattice, transpose and reality actions are the actions of those exact operators with the parameters transformed accordingly. At level zero we retain the stated diagonal families and the chamber covariance with measure $1/48$.

The fixed parameters satisfy (12.61), (12.35) and (12.40), $M > 4L$, $L \ge 4^8$, $0 < 1 - \alpha < \varepsilon \le 1/200$, $C_w, c_\gamma \ge 1$, and $C_w \ge 4\exp(2\gamma_w)M^{27\alpha_0}$. The parameter radius satisfies $M^{\alpha_0} - 1 \ge \exp(N_c\kappa + \log(6c_X))$, where $N_c = (2L+1)^3$. For later use put $c_S = 10^5 c_X^5 K_0^5 \exp(6\kappa)$. The running charge lies in the fixed range on which

$$\begin{aligned}&p_{0,k} \le e^{-\varepsilon/4}, \qquad 12\exp(-p_{0,k}^2/2) \le e^7, \qquad 2e^{5\varepsilon/2} \le 1, \qquad c_{\mathfrak{fl}} e^{\varepsilon} \le 1, \qquad 3Me^{\varepsilon} \le 1, \\ &2c_C C_w^2 c_\gamma^2 e^{\varepsilon/4} \le 1, \qquad 16 c_X c_\gamma C_w C_T^2 M^3 e^{\varepsilon/2} \le 1, \qquad 96M^3 e^{1/4} \le 1,\end{aligned}$$

$$32C_wC_Te^{3\varepsilon/2}\le 1,\qquad 88M^3e^{1/4-6\varepsilon}\le 1,\qquad 112C_w^2c_\gamma e^{2\varepsilon}\le 3,$$
$$c_Se^{1/4-5\varepsilon}\le 1,\qquad c_SL^3e^{\varepsilon}\le 1,\qquad e^{\varepsilon}\le \tfrac14,\qquad 8c_{\mathrm{rb}}L^3C_dM^3e^{2\varepsilon}\le 1. \tag{14.1}$$

The density identity also uses the photon domain-preservation conditions of (10.42) and the determinant conditions of (11.37). All are imposed by the fixed choice of the charge ceiling.

For $\zeta\in \mathrm{Box}_k$, write

$$Z(s')=H_k^{\mathrm{ph}}(s')CC_k^{1/2}(s')\zeta. \tag{14.2}$$

In this formula C is the real photon coordinate map of (10.20). The diagonal photon family is used at $k=0$. For a bond field Z set

$$|Z|_*=\max\{\|Z\|_\infty,\|\partial Z\|_\infty,\|\delta_\alpha\partial Z\|_\infty\}.$$

The three thresholds defining R_k are $\theta_i=e^{-3/4+i\varepsilon}$, $i=1,2,3$. Thus $\widetilde{R}_k^-$ consists of the permitted real-gauge part plus a field satisfying the three strict bounds $(1-e^{\varepsilon})\theta_i$. Introduce the radii

$$r_1=\tfrac12 e^{-3/4+9\varepsilon/2},\qquad r_2=e^{-3/4+7\varepsilon},\qquad r_4=e^{-1/4+3\varepsilon}. \tag{14.3}$$

Proposition 14.1. *For every $s\in \mathrm{Pd}_k$ and $\zeta\in\mathrm{Box}_k$,*

$$|Z(s')|_*\le \tfrac12 e^{-\varepsilon/2}.$$

The radii satisfy $2\le r_2,r_4\le r_1$ and $(r_i-1)^{-1}\le 2/r_i$, $i=2,4$. If $A\in\widetilde{R}_k^-$, $\mathscr{C}\in\mathfrak{X}_k$, and $|Z|_*\le\frac12 e^{-\varepsilon/2}$, then

$$A+tZ\in\widetilde{R}_k,\qquad A+\mathscr{C}+tZ\in\widetilde{\mathcal{G}}_k\quad(|t|\le r_1).$$

Proof. The square-root kernel is bounded by $C_w\exp(-\gamma_w d(\Upsilon,\Upsilon'))$. The summability constant c_γ and the coordinate bound $|\zeta(\Upsilon')|\le p_{0,k}\le e^{-\varepsilon/4}$ give

$$\sup_{\Upsilon}|(C_k^{1/2}(s')\zeta)(\Upsilon)|\le C_wc_\gamma e^{-\varepsilon/4}.$$

The coordinate map is real. Its bound $\|Cv\|_\infty\le c_C\sup_\Upsilon|v(\Upsilon)|$ also gives the absolute row bound $\sum_\Upsilon|C(b,\Upsilon)|\le c_C$: apply the real bound to the signs of the entries of a fixed row. Therefore

$$\|CC_k^{1/2}(s')\zeta\|_\infty\le c_CC_wc_\gamma e^{-\varepsilon/4}.$$

Each of the reconstruction, derivative and Hölder derivative kernels has the same summable exponential bound. Consequently

$$|Z(s')|_*\le c_CC_w^2c_\gamma^2e^{-\varepsilon/4}\le\tfrac12e^{-\varepsilon/2},$$

where the last inequality is $2c_CC_w^2c_\gamma^2e^{\varepsilon/4}\le 1$. At level zero the polynomial diagonal family has the same row bounds: its constant is $4\exp(2\gamma_w)M^{27\alpha_0}\le C_w$.

Since $\varepsilon\le 1/200$,

$$\frac{r_2}{r_1}=2e^{5\varepsilon/2}\le 1,\qquad \frac{r_4}{r_1}=2e^{1/2-3\varepsilon/2}\le 2e^{5\varepsilon/2}\le 1.$$

The small-charge condition $88M^3e^{1/4-6\varepsilon}\le 1$ gives $r_4^{-1}=e^{1/4-3\varepsilon}\le 1/88$ and $r_2^{-1}\le r_4^{-1}$. In particular both radii are at least 2, and their stated inverse bounds follow.

For $|t|\le r_1$ the three norms of tZ are at most $\frac14e^{-3/4+4\varepsilon}$, strictly less than $e^{\varepsilon}\theta_3$. Adding this field fills less than the margin in each of the three defining bounds of $\widetilde{R}_k^-$. The permitted gauge part is unchanged. Hence $A+tZ\in\widetilde{R}_k$. Finally $\widetilde{R}_k+\mathfrak{X}_k\subset\widetilde{\mathcal{G}}_k$ gives the second inclusion. □

The first-step vertex estimate (13.31) now applies to $Z(s')$ for every weakening parameter. In particular, $96M^3e^{1/4} \le 1$ gives its bound $e^{1/4-5\varepsilon}$ in the $h/2$ norm.

Let $B = A + \mathscr{C}$. The arguments of $H_k^{\#}$ are independent pairs $(\Phi, \bar\Phi)$; in $H_k^{\#}\Phi$ the single symbol denotes this pair. Write $\psi^{\#} = (\psi, \chi; \bar\psi, \bar\chi)$ for the ordinary, pair, barred ordinary and barred pair generators, respectively. Then

$$\begin{aligned} H_k^{\#}(s'', B)\Phi = (&H_k(s'', B)\Phi, \delta_{\alpha,B}H_k(s'', B)\Phi; \\ &\bar H_k(s'', B)\bar\Phi, \delta_{\alpha,-B}\bar H_k(s'', B)\bar\Phi), \\ \bar H_k(s'', B) &= \mathsf{C}^{-1}H_k(s'', -B)\mathsf{C}. \end{aligned}$$

Define $J_k^{\#}(s'', B, Z) = H_k^{\#}(s'', B + Z) - H_k^{\#}(s'', B)$ and

$$\widetilde\Gamma_k(s'', B)(W, \bar W) = (\Gamma_k(s'', B)W, \bar W), \qquad W_k^{\#}(s'', B) = H_k^{\#}(s'', B)\big(\Gamma_k(s'', B)W, \bar W\big).$$

The notation $J_k, \bar J_k$ denotes the components without Hölder quotient; $\mathrm{Wc}, \overline{\mathrm{Wc}}$ denotes the corresponding components of $W_k^{\#}$. The auxiliary generators W have weight $\mathsf{w}_W = 1$ for $k \ge 1$ and $\sqrt{48}$ for $k = 0$.

These fields are continuous in all closed polydisk and box variables and analytic in the open parameter polydisk and the connection variables. For the Hölder rows this follows directly from

$$(\delta_{\alpha,B}H_k)((x, x'), u) = d_X(x, x')^{-\alpha}\big(U_B(x, x')H_k(x', u) - H_k(x, u)\big).$$

The transport is entire in B. Proposition 14.1 places every connection on the closed contours inside the open domain of the primitives. Thus compositions at $B + tZ(s')$ have the stated regularity also when the contour variable is on its circle.

In the following localization, cubes and polymers are sets of photon sites. Containment refers to all their lattice sites, and two cubes meet when they share a lattice site. Let $P_0(X)$ be the parameter cubes contained in X. For P outside $P_0(X)$ let $Y'(X, P)$ be the union of the M-cubes meeting X or a cube of $P_0(X) \cup P$. For a continuous analytic family $G(s, X)$ define

$$\mathcal{T}_P G(X) = \int_{[0,1]^P} \mathrm{d}s_P\, \partial_P G(s, X)\big|_{s=1 \text{ on } P_0(X),\, s=0 \text{ outside } P_0(X)\cup P}, \tag{14.4}$$

$$\mathrm{Loc}\, G(Y) = \sum_{(X,P):Y'(X,P)=Y} \mathcal{T}_P G(X). \tag{14.5}$$

Repeated one-variable integration gives the useful identity

$$\mathcal{T}_P G(X) = \sum_{Q\subseteq P} (-1)^{|P|-|Q|} G(\mathbf{1}_{P_0(X)\cup Q}, X). \tag{14.6}$$

In particular, coefficient locality can be checked at indicator families; no statement about the convergence of the later logarithm is used here.

Lemma 14.2. *Suppose $\|G(s, X)\| \le B_0 \exp(-\kappa' d_M(X))$ on the parameter polydisk, where $\kappa_0 \le \kappa' \le \kappa$. Assume $\mathcal{T}_P G(X) = 0$ whenever $Y'(X, P)$ is disconnected. Then*

$$\sum_Y \mathrm{Loc}\, G(Y) = \sum_X G(1, X), \tag{14.7}$$

$$\|\mathrm{Loc}\, G(Y)\| \le 4c_X K_0 B_0 \exp\big(-(\kappa' - \kappa_0 - 1)d_M(Y)\big). \tag{14.8}$$

Proof. Applying $f(1) = f(0) + \int_0^1 f'(t)\mathrm{d}t$ in each parameter outside $P_0(X)$ gives $G(1, X) = \sum_P \mathcal{T}_P G(X)$. The sums are finite. After removing the vanishing disconnected terms, each term belongs to exactly one polymer $Y'(X, P)$. This proves (14.7).

Put $R = M^{\alpha_0}$. Cauchy's estimate on a polydisk of radius less than $R-1$ about each integration point, followed by the limit of these radii to $R-1$, gives

$$\|\mathcal{T}_P G(X)\| \le B_0 \exp(-\kappa' d_M(X))(R-1)^{-|P|}.$$

If $Y'(X,P) = Y$, then $X \subseteq Y$ and $|Y \setminus X|_M \le N_c|P|$, since every added M-cube meets a parameter cube and each parameter cube meets at most $N_c = (2L+1)^3$ such cubes. To compare their tree lengths, order the cubes of $Y \setminus X$ by graph distance from X. Each added cube is adjacent to a preceding one. Extend a net by a segment from a point of that preceding closed cube to their common face, of length at most M. Starting from an arbitrarily near-minimal net for X gives $d_M(Y) \le d_M(X) + |Y \setminus X|_M$. Conversely every net for Y is a net for X. Thus $0 \le d_M(Y) - d_M(X) \le N_c|P|$. Writing $a_X = \log(6c_X)$, the condition $R - 1 \ge \exp(N_c\kappa + a_X)$ implies

$$\|\mathcal{T}_P G(X)\| \le B_0 \exp(-(\kappa' - \kappa_0) d_M(Y)) \exp(-\kappa_0 d_M(X)) \exp(-a_X|P|).$$

At most $3|Y|_M$ parameter cubes are contained in Y. Hence

$$\sum_P \exp(-a_X|P|) \le (1 + (6c_X)^{-1})^{3|Y|_M} \le \exp((d_M(Y)+1)/2).$$

The polymer counting and size estimates give

$$\sum_{X \subseteq Y} \exp(-\kappa_0 d_M(X)) \le K_0^{|Y|_M} \le 2c_X K_0 \exp(d_M(Y)/2).$$

The product of the last two bounds is at most $4c_X K_0 \exp(d_M(Y))$. Summing the estimate for $\mathcal{T}_P G(X)$ proves (14.8). □

14.2. **The two finite localizations.** For a function analytic near the closed disk $|t| \le r$, $r > 1$, write

$$\mathcal{C}_r f = \frac{1}{2\pi i} \int_{|t|=r} \frac{f(t)}{t(t-1)}\, \mathrm{d}t.$$

This is $f(1) - f(0)$ and satisfies $\|\mathcal{C}_r f\| \le (r-1)^{-1} \sup_{|t|=r} \|f(t)\|$. Let $(m, \mathscr{E}) \in \mathrm{Ball}_k^x(4)$ and set $m' = m + m(\mathscr{E})$, $\mathscr{E}^r = R\mathscr{E}$. For $k \ge 1$ define

$$G_1(s,X) = \mathcal{C}_{r_1} \langle \bar\Psi, M_k(X, B + tZ(s'))\Psi\rangle_1, \tag{14.9}$$

$$G_2(s,X) = \mathcal{C}_{r_2}\, \mathscr{E}^r(X; A + tZ(s'), \mathscr{C}, \psi^\# + tJ_k^\#(s'', B, Z(s'))\Psi), \tag{14.10}$$

$$G_3(s,\square) = \mathcal{C}_{r_2}(-m')\langle \bar\psi + t\bar J_k(s'', B, Z(s'))\bar\Psi, \psi + tJ_k(s'', B, Z(s'))\Psi\rangle_\square. \tag{14.11}$$

The symbols $M_k(X,\cdot)$ denote the local Dirac primitives. A function defined for one cube is zero on the other polymers. Put

$$E_i^*(Y) = \mathrm{Loc}\, G_i(Y), \qquad E_i(Y) = \mathrm{sub}_{Y,B}\, E_i^*(Y), \qquad E_{\le 3} = \sum_{i=1}^3 E_i.$$

The substitution is (13.23). At $k = 0$ there is no Ψ and no substitution: $G_1(s,\square) = \mathcal{V}_0(Z(s'),\square)$, $G_2 = G_3 = 0$, and $E_{\le 3} = \mathrm{Loc}\, G_1$. All parameter kinds occur in this first localization.

For the second localization, which uses only π_k, let $v = W_k^\#(s'', B)$ and define

$$G_{\le 3}^W(s'', X) = E_{\le 3}(X; A, \mathscr{C}, \zeta, \psi^\# + v), \tag{14.12}$$

$$G_4(s'', X) = \mathcal{C}_{r_4} \mathscr{E}^r(X; A, \mathscr{C}, \psi^\# + tv), \tag{14.13}$$

$$G_5(s'', \square) = \mathcal{C}_{r_4}(-m')\langle \bar\psi + t\overline{\mathrm{Wc}}, \psi + t\mathrm{Wc}\rangle_\square, \tag{14.14}$$

$$F(Y) = \mathrm{Loc}(G_{\le 3}^W + G_4 + G_5)(Y). \tag{14.15}$$

At $k = 0$ the term G_4 is zero and $m' = m$.

Proposition 14.3. *All finite expressions just defined exist on $\widetilde{R}_k^- \times \mathfrak{X}_k \times \mathrm{Box}_k$. The element $F(Y)$ is even, has ordinary legs over Y and pair legs admissible for Y, and has fluctuation generators only at unit indices in Y. Its coefficients read $A, \mathscr{C}$ only on bonds meeting Y and read ζ only through the coordinates $\zeta(\Upsilon)$ with $\operatorname{supp}\Upsilon \subset Y$. They are continuous in these arguments and analytic in the connection variables. The same assertions hold for $E_{\le 3}(Y)$, which has no fluctuation generators.*

The disconnected terms in both localizations vanish, and

$$\sum_Y E_i^*(Y) = \sum_X G_i(1,X), \qquad \sum_Y F(Y) = \sum_X (G_{\le 3}^W + G_4 + G_5)(1,X). \tag{14.16}$$

The finite expressions on Y depend on the datum only through m' and the $\mathscr{E}^r(X)$ with $X \subseteq Y$.

Proof. We give the component argument which is responsible for locality. Let $\mathrm{Ps} = P_0(X) \cup P$, let $Y' = Y'(X,P)$ and let Y_1' be its component containing X. Set the weakening parameters to zero outside Ps. For each of the three parameter kinds, any link-component Lnk of Ps is contained in the polymer formed by the M-cubes meeting it. This polymer lies in one component of Y'. If a site of Y_1' lies in Lnk, or in the enlarged parameter cube of a photon kind, that polymer has a cube in common with Y_1', and hence lies entirely in Y_1'.

The row-vanishing property of the weakened photon families and their component restriction give, on any fine bond b meeting Y_1',

$$Z(s')(b) = \sum_{b', \Upsilon, \Upsilon' \subset Y_1'} H_k^{\mathrm{ph}}(s')(b,b') C(b', \Upsilon) C_k^{1/2}(s')(\Upsilon, \Upsilon') \zeta(\Upsilon'). \tag{14.17}$$

Indeed, the reconstruction row is zero unless b lies in one link-component; its columns lie in that component. The coordinate map has columns supported in the L-blocks of the ends of b', and each block lies in the M-cube of the corresponding end. The square-root row then has columns in its own link-component. Each component involved is attached to Y_1' by the preceding argument. These rows are unchanged if the parameters outside Y_1' are changed while their common zero set outside Ps is kept.

For a chamber site $x = (c, g)$ at distance at most 1 from Y_1', the fermion row is zero unless every parameter cube meeting K_c is present. These cubes form one link-component. When the row is nonzero, that component contains c and a site of Y_1', so it lies in Y_1'. The row of $H_k(s'', B)$, and its barred counterpart, then has columns in that component and reads B only on bonds meeting it. At a column of this row, the corresponding row of $\Gamma_k(s'', B)$ has columns in the same component. These conclusions also hold at level zero: the diagonal factor is

$$\prod_{\mathrm{pc} \in \pi_0 : \mathrm{pc} \cap K_c \neq \varnothing} s''_{\mathrm{pc}},$$

and the kernel of that multiplication operator is $48 \mathbf{1}_{x=x'}$ times this factor. Its composition with Γ_0 uses the measure $1/48$.

An endpoint of a pair admissible for X is at distance at most $\eta_k \le 1$ from X. Its transport reads only bonds contained in X, or the one type-3 bond meeting X. Thus the preceding row argument applies to every row used in $J_k^{\#}$ and $W_k^{\#}$. The Hölder row has only the two columns x, x' and the entire transport coefficient displayed above. Finite compositions preserve this dependence: the columns of $O_2 O_1$ are the union of the columns of O_1 at the finitely many columns of O_2, and its read bonds are the union of their read bonds. Entries of the transport and reconstruction rows are entire in the connections, and finite compositions of these algebraic rows retain this property. Compositions involving weakened primitives are holomorphic on their stated primitive domains. These are the finite-row properties used here; the weakened operators themselves are not asserted to have finite range.

For G_1 at $k \geq 1$, the local primitive $M_k(X,\cdot)$ has both indices in X and reads only bonds meeting X. For G_2, G_3, the incoming activity or mass form has ordinary legs in X and admissible pair legs; the substitution by $J_k^{\#}\Psi$ introduces Ψ only in Y_1'. At $k=0$ the vertex has precisely these ordinary and type-3 pair legs and reads Z only on bonds meeting its cube. Equation (14.17) therefore proves the same component dependence for every first-localization function.

If Y' is disconnected, choose an M-cube outside Y_1'. It cannot meet X or a parameter cube contained in X, so it meets a cube of P outside Y_1'. The integrand is independent of that parameter. Pairing the subsets in (14.6) according to whether they contain this cube proves that the term is zero. For a term assigned to a connected Y, all parameter cubes in $P_0(X) \cup P$ and X lie in Y. Equation (14.6) now gives the claimed locality of E_i^*.

The reconstruction substitution adds only ordinary legs over Y and boundary pairs $(x, x^{(3)})$ admissible for Y. Its finite rows at a unit site y read bonds contained in Δ_y or type-3 bonds from this block. These bonds meet Y since $\Delta_y \subset Y$. The barred rows have the same support. Thus it preserves the claimed coefficient locality and removes Ψ. For the second localization, the row argument for $H_k^{\#}\widetilde{\Gamma}_k W$ gives fluctuation generators only in Y_1'. It applies to each of the three functions in (14.15). The same disconnected parameter cancellation proves the second assertion. Lemma 14.2 gives both sum identities.

Regularity follows from the established contour domains, the primitive regularity, and finite sums and substitutions. Contour integrals preserve analyticity: their Riemann sums converge uniformly on compact sets. The same argument applies to finite parameter integrals. The incoming activities and mass forms are even, and each substitution sends generators to odd linear combinations, so every expression remains even. Finally G_1 is independent of the datum; G_2, G_4 use $\mathscr{E}^r(X)$ and G_3, G_5 use m'. Tracing the two localizations shows that no incoming polymer outside Y enters the result. □

The proof identifies the primitive entries determining a finite expression on Y. With parameters zero outside Y, the photon field reads only $H_k^{\mathrm{ph}}(s')(b,b')$ for b meeting Y, $b' \subset Y$, and $C_k^{1/2}(s')(\Upsilon,\Upsilon')$ with both supports in Y. Its remaining factor is the fixed coordinate map. The fermion fields use $H_k, \bar{H}_k$ at chamber sites over Y, with unit columns in Y, at $B + Z(s')$ and B, and $\Gamma_k(s'', B)$ with both unit indices in Y. At zero, the kernel sums have measure $1/48$ and the diagonal rows have the factor 48 already recorded. The Dirac primitive has both indices in $X \subseteq Y$ and connection $B + tZ(s')$ on $|t| = r_1$. Hölder transports read only bonds meeting Y. The reconstruction substitution contains only the fixed finite $T_k^{(i)}, \bar{T}_k^{(i)}$ rows at unit sites of its own subpolymer.

Consequently two systems with the stated hypotheses on this torus give the same finite expressions on Y if these entries agree for every parameter family zero outside Y, at the connections formed with their respective photon fields. The restricted photon sum first gives equality of the fields on bonds meeting Y; the transports and incoming activities there then agree by their locality. The listed fermion and Dirac entries give equal first functions, their localizations and reconstruction substitutions. They give equal second functions and localizations in turn, and then equal activities $K(Y)$ and $H(Y)$ in (14.26) and (14.27), and each term of the connected series (14.28). Agreement of the photon entries and agreement of the fermion entries as functions on the full operator domain is sufficient: if the two composed connections differ outside Y, local row dependence removes that difference first. The same comparison passes to the convergent logarithm once its convergence has been established.

14.3. **Bounds before fluctuation integration.** Set

$$\kappa_w = \kappa + \kappa_0 + 3, \qquad a_1 = \kappa - \kappa_0 - 1, \qquad a_2 = \kappa - 2\kappa_0 - 2,$$
$$C_W = 28 C_w^2 c_\gamma, \qquad H_{\mathrm{cl}} = 36 c_X^2 K_0^2 e^{1/4 - 5\varepsilon}.$$

The notation H_{cl} is a scalar cluster bound. The finite primitive bounds imply

$$\|\langle\bar\Psi, M_k(X,B)\Psi\rangle_1\|_{w_3} \le 4c_X c_\gamma C_w M^3 w_3^2 \exp(-(\kappa_w-1)d_M(X)), \quad k\ge 1, \tag{14.18}$$

$$|H_k^{\#}(s'',B)|_{1,\infty} \le 4C_w, \tag{14.19}$$

$$|J_k^{\#}(s'',B,Z)|_{1,\infty} \le 16 C_w e^{3/4-9\varepsilon/2}, \tag{14.20}$$

$$\mathsf{w}_W |H_k^{\#}(s'',B)\widetilde\Gamma_k(s'',B)|_{1,\infty} \le C_W. \tag{14.21}$$

Here $B = A + \mathscr{C}$ and $|Z|_* \le \frac12 e^{-\varepsilon/2}$. To see the first bound, count $4M^3|X|_M$ rows and sum the exponential kernel in its second index, giving $c_\gamma C_w \exp(-\kappa_w d_M(X))$ per row. Then $|X|_M \le c_X \exp(d_M(X))$ gives (14.18). For $H_k^{\#}$ the primitive bound controls the ordinary and short-pair rows by C_w. A long pair, $d_X \ge 1$, has its Hölder quotient bounded by four times the ordinary sup norm. At level zero every pair is long. The monomial matrices $\mathsf{C}^{\pm1}$ preserve row norms, so the barred rows have the same bounds. Apply $\mathcal{C}_{r_1}$ to $H_k^{\#}(s'', B+tZ)$ to obtain the bound for $J_k^{\#}$. Finally $|\Gamma_k|_{1,\infty} \le C_w c_\gamma$ by kernel summation, using $1/48$ at level zero. Submultiplicativity and $\mathsf{w}_W \le 7$ give (14.21).

Proposition 14.4. *For the data of Proposition 14.3,*

$$\|E_{\le 3}(Y)\|_{h/2} \le 8 c_X K_0 e^{1/4-5\varepsilon} \exp(-a_1 d_M(Y)), \tag{14.22}$$

$$\|F(Y)\|_{(h/4;\mathsf{w}_W)} \le H_{\mathrm{cl}} \exp(-a_2 d_M(Y)). \tag{14.23}$$

The element $F(Y)$ is the sum of a term independent of $(m,\mathscr{E})$ and a term complex linear in these data.

Proof. Extraction gives

$$\|\mathscr{E}^r(X)\|_h \le 2\|\mathscr{E}\|_k^x \exp(-\kappa d_M(X)) \le 12 e^{1/4-7\varepsilon}\exp(-\kappa d_M(X)), \qquad |m'| \le 8e^{3/4-8\varepsilon}.$$

The mass bound uses $c_{\mathfrak{N}} e^{\varepsilon} \le 1$. With chamber measure $\eta_k^3/48$, direct counting gives

$$\|\langle\bar\psi,\psi\rangle_\square\|_{(w_1,w_2)} = 4M^3 w_1^2.$$

The number of monomials is $4\cdot 48\eta_k^{-3}M^3$ and their common coefficient is $\eta_k^3/48$.

In the first localization give Ψ the weight $w_3 = C_T h_{k,1}$ and the old generators the weights $h/2$. The contour bound and (14.18) give

$$\|G_1(s,X)\| \le 16 c_X c_\gamma C_w C_T^2 M^3 e^{1/4-9\varepsilon/2}\exp(-(\kappa_w-1)d_M(X)) \le e^{1/4-5\varepsilon}\exp(-\kappa d_M(X)).$$

The last step uses

$$16 c_X c_\gamma C_w C_T^2 M^3 e^{\varepsilon/2} \le 1.$$

At $k=0$ the weighted vertex instead gives $12M^3 e^{1/2-\varepsilon/2} \le e^{1/4-5\varepsilon}$. Moreover,

$$r_2 |J_k^{\#}|_{1,\infty} w_3 \le 16 C_w C_T e^{3\varepsilon/2} h_{k,2} \le \tfrac12 h_{k,2}.$$

The substitution bound therefore takes the integrand norm of G_2 to the h norm of $\mathscr{E}^r$, and that of G_3 to the h norm of the mass form. It gives

$$\|G_2(s,X)\| \le 24 e^{1-14\varepsilon}\exp(-\kappa d_M(X)), \qquad \|G_3(s,\square)\| \le 64 M^3 e^{1-15\varepsilon}.$$

The condition $88M^3 e^{1/4-6\varepsilon} \le 1$ implies

$$24 e^{3/4-9\varepsilon} + 64 M^3 e^{3/4-10\varepsilon} \le 1.$$

Thus $\sum_{i\le 3} G_i$ is bounded by $2e^{1/4-5\varepsilon}\exp(-\kappa d_M(X))$. Apply Lemma 14.2, using the disconnected cancellation in Proposition 14.3. The reconstruction substitution costs no norm, since $C_T(h_{k,1}+h_{k,2})/2 \le w_3$. This proves (14.22).

For the second localization, the condition $112C_w^2 c_\gamma e^{2\varepsilon} \le 3$ gives

$$r_4 C_W \le \tfrac{3}{4} h_{k,2}, \qquad C_W \le \tfrac{1}{4} h_{k,2}.$$

Consequently addition of $W_k^{\#}$ maps the $h/4$ norm to the $h/2$ norm at $t = 1$ and to the h norm on the contour $|t| = r_4$. The three input bounds are

$$\begin{aligned} \|G^W_{\le 3}(s'', X)\| &\le 8 c_X K_0 e^{1/4 - 5\varepsilon} \exp(-a_1 d_M(X)), \\ \|G_4(s'', X)\| &\le 24 e^{1/2 - 10\varepsilon} \exp(-\kappa d_M(X)), \\ \|G_5(s'', \square)\| &\le 64 M^3 e^{1/2 - 11\varepsilon}. \end{aligned}$$

Their sum is at most $9 c_X K_0 e^{1/4-5\varepsilon} \exp(-a_1 d_M(X))$, again by $88 M^3 e^{1/4-6\varepsilon} \le 1$. Since $\kappa_0 \le a_1 \le \kappa$, resummation gives (14.23). At level zero $G_4 = 0$; all the other inequalities include the factor w_W already, so they apply unchanged. Linearity follows from the definitions: G_1 is fixed, G_2, G_4 are linear in $R\mathscr{E}$, G_3, G_5 in m', and localization, the reconstruction substitution and the contour functionals are linear. □

14.4. **Connected polymers and the logarithm.** Two polymers are incompatible when their union is connected through faces. In particular a polymer is incompatible with itself. For an ordered tuple $(X_1, \dots, X_n)$ define $u^T(X_1) = 1$ and

$$u^T(X_1, \dots, X_n) = \sum_{G \text{ connected}} (-1)^{|G|}, \quad n \ge 2,$$

where G ranges over spanning subgraphs of the incompatibility graph of the tuple. The coefficient is zero when that graph is disconnected.

We will use the formal connected-graph identity

$$\sum_{\{X_i\} \text{ compatible}} \prod_i z_{X_i} = \exp\left\{ \sum_{n \ge 1} \frac{1}{n!} \sum_{(X_1, \dots, X_n)} u^T(X_1, \dots, X_n) \prod_i z_{X_i} \right\} \tag{14.24}$$

and the tree bound

$$|u^T(X_1, \dots, X_n)| \le \#\{\text{spanning trees of the incompatibility graph}\}. \tag{14.25}$$

Here the identity is one of formal series in commuting variables, before substitution into a normed algebra. For its proof, express the left side as an ordered sum with factor $1/n!$ and multiply by $\prod_{i<j}(1 - \mathbf{1}_{\{X_i, X_j\} \text{ incompatible}})$. Expanding this product sums over all graphs. A graph is uniquely the union of its connected components. Grouping by their vertex sets gives the exponential on the right: an ordered list of component sizes $n_1, \dots, n_r$ has $n!/(n_1! \cdots n_r!)$ vertex assignments, and division by $r!$ removes the order of its components. Only finitely many terms contribute to a fixed degree.

For (14.25), let $\mathbf{G}$ be the incompatibility graph and suppose it is connected. For a connected spanning subgraph G, root at 1 and, for $i \ne 1$, choose as its parent the least labelled neighbor at distance $d_G(i) - 1$ from the root. These edges form a tree $T(G)$ with the same root distances as G. Given a spanning tree τ, let $R(\tau)$ contain its edges, all edges of $\mathbf{G}$ within one distance level, and all edges from i to the preceding level whose other endpoint exceeds the parent of i in τ. Then

$$T(G) = \tau \quad \Longleftrightarrow \quad \tau \subseteq G \subseteq R(\tau).$$

In one direction the distances and least parents force every edge of G to belong to $R(\tau)$. Conversely, along any edge of $R(\tau)$ the tree distance changes by at most one. Thus a graph between τ

and $R(\tau)$ has the tree's distances, and the least neighbor in the preceding level is its tree parent. The connected subgraphs are therefore partitioned into these intervals, and their signed sum is

$$\sum_{\tau}(-1)^{n-1}\sum_{A\subseteq R(\tau)\setminus\tau}(-1)^{|A|}=(-1)^{n-1}\#\{\tau : R(\tau)=\tau\}.$$

This proves the tree bound, including the cases of a disconnected graph and $n=1$.

Proposition 14.5. *Let $V(X)$ be even elements in the old and fluctuation generators, continuous on Box_k, with*

$$\|V(X)\|_{(w;\mathsf{w}_W)}\le H_{\mathrm{cl}}\exp(-a_2 d_M(X)).$$

Assume that $V(X)$ has fluctuation indices only in X and depends on ζ only through coordinates supported in X. For any union Y of M-cubes put

$$K(Y)=\sum_{\substack{\{X_i\}\ distinct\\ \bigcup_i X_i=Y}}\prod_i(\exp V(X_i)-1),\qquad K(\varnothing)=1, \tag{14.26}$$

$$H(Y)=\int \mathrm{d}\mu_I(W)\,\mathrm{d}\widehat{\mu}(\zeta)\,K(Y). \tag{14.27}$$

For a connected Y define

$$E^{\#}(Y)=\sum_{n\ge1}\frac{1}{n!}\sum_{\bigcup_i Y_i=Y}u^T(Y_1,\dots,Y_n)\prod_i H(Y_i). \tag{14.28}$$

The last series converges absolutely in the w norm and

$$\int \mathrm{d}\mu_I(W)\,\mathrm{d}\widehat{\mu}(\zeta)\exp\left\{\sum_X V(X)\right\}=\exp\left\{\sum_Y E^{\#}(Y)\right\}, \tag{14.29}$$

$$\|E^{\#}(Y)\|_w\le c'_S H_{\mathrm{cl}}\exp(-a_3 d_M(Y)), \tag{14.30}$$

Here $a_3=\kappa-5\kappa_0-4$ and $c'_S=14c_X^2K_0^2\exp(3\kappa)$. The sum of the norms of the terms with $n>N$ in (14.28) *is at most 2^{-N} times the right side of* (14.30).

Proof. Set

$$a_4=\kappa-3\kappa_0-3,\quad a_5=\kappa-3\kappa_0-2,\quad a_6=\kappa-5\kappa_0-3,$$
$$H_1=7c_XK_0\exp(3a_5)H_{\mathrm{cl}},\qquad z_S=28H_1\exp(\kappa_0)K_0\exp(3a_6).$$

The fixed small-charge choice

$$c_S e^{1/4-5\varepsilon}\le1,\qquad c_S=10^5c_X^5K_0^5\exp(6\kappa)$$

implies

$$H_{\mathrm{cl}}\le1,\qquad 4H_{\mathrm{cl}}K_0c_X\exp(3a_5)\le1,\qquad z_S\le\tfrac12. \tag{14.31}$$

Indeed the second left side is $144c_X^3K_0^3\exp(3a_5)e^{1/4-5\varepsilon}$. For the third, $2z_S=14112c_X^3K_0^4\exp(3a_5+3a_6+\kappa_0)e^{1/4-5\varepsilon}$, and $3a_5+3a_6+\kappa_0\le6\kappa$.

Even elements commute, so expanding their exponential gives $\exp(\sum_X V(X))=\sum_Y K(Y)$. Moreover $K(Y)$ factors over the connected components of Y: every polymer in its defining set belongs to exactly one component. If distinct polymers $X_1,\dots,X_n$ have connected union Y, their incompatibility graph is connected. Otherwise two nonempty groups would partition their union; connectedness of Y would give coincident or adjacent cubes in different groups and hence an edge between them. For each incompatibility-tree edge, choose coincident or face-adjacent cubes in the two polymers and points where their nets meet these cubes. The two points lie in a box of side

lengths $M, M, 2M$, so they can be joined by a segment of length at most $\sqrt{6}M < 3M$. Joining near-minimal nets along the tree proves

$$d_M(Y) \le \sum_i d_M(X_i) + 3(n-1).$$

Using $\|\exp V - 1\| \le 2\|V\|$ and $a_2 = a_5 + \kappa_0$ therefore bounds the product by

$$\exp(-a_5 d_M(Y))\big(2H_{\mathrm{cl}} \exp(3a_5)\big)^n \prod_i \exp(-\kappa_0 d_M(X_i)).$$

Replace distinct sets by ordered tuples with factor $1/n!$, and use the counting bound for subpolymers of Y. The result is

$$\|K(Y)\| \le \exp(-a_5 d_M(Y)) x_Y \exp(x_Y), \qquad x_Y = 2H_{\mathrm{cl}} K_0 \exp(3a_5)|Y|_M.$$

By (14.31),

$$x_Y \le (d_M(Y)+1)/2,$$
$$x_Y \le 4H_{\mathrm{cl}} K_0 c_X \exp(3a_5) \exp(d_M(Y)/2).$$

Since $4\exp(1/2) \le 7$ and $a_5 - 1 = a_4$, we obtain

$$\|K(Y)\|_{(w;\mathsf{w}_W)} \le H_1 \exp(-a_4 d_M(Y)) =: \mu(Y). \tag{14.32}$$

Weighted Berezin integration costs no norm here: each paired moment at level zero is $\pm 48^{|J|/2}$, which is exactly covered by the weight $(\sqrt{48})^{|J|}$ of its generators. The corresponding moment is ± 1 at positive levels. The photon measure is a probability measure on its compact box. Hence $\|H(Y)\|_w \le \mu(Y)$. These integrals produce even elements, since a nonzero Berezin moment uses an even number of generators. For compatible polymers the fluctuation modes and supported photon coordinates are disjoint. The two product measures therefore factor, with the external old generators left as coefficients. It follows that

$$\int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}(\zeta) e^{\sum_X V(X)} = \sum_{\{Y_i\} \text{ compatible}} \prod_i H(Y_i). \tag{14.33}$$

We now bound the infinite connected series. Let

$$S_n(Y) = \sum_{\bigcup_i Y_i = Y} |u^T(Y_1, \dots, Y_n)| \prod_i \mu(Y_i).$$

For $n \ge 2$, apply (14.25). For every contributing tree the union estimate and $a_4 = a_6 + 2\kappa_0$ give

$$\prod_i \exp(-a_4 d_M(Y_i)) \le \exp(-a_6 d_M(Y)) \exp(3a_6(n-1)) \prod_i \exp(-2\kappa_0 d_M(Y_i)).$$

Drop the prescribed union, retaining $Y_i \subseteq Y$, and root the tree at 1. Write q_j for the number of children at j. A polymer incompatible with Y' contains one of at most $7|Y'|_M$ cubes consisting of its cubes and their face neighbors. Since $|Z|_M \le c_X(d_M(Z)+1) \le \kappa_0(d_M(Z)+1)$ and $a^q \le q! \exp(a)$, the counting estimate gives

$$\sum_{Z \text{ incompatible with } Y'} \exp(-2\kappa_0 d_M(Z))|Z|_M^q \le 7q! \exp(\kappa_0) K_0 |Y'|_M,$$
$$\sum_{Z \subseteq Y} \exp(-2\kappa_0 d_M(Z))|Z|_M^q \le q! \exp(\kappa_0) K_0 |Y|_M.$$

Summing from leaves to root thus costs at most $(7\exp(\kappa_0)K_0)^n|Y|_M\prod_j q_j!$. For prescribed children counts, Cayley's degree formula [8, Proposition 4] counts $(n-2)!/((q_1-1)!\prod_{j\geq 2}q_j!)$ trees, where $q_1\geq 1$ and $\sum_j q_j = n-1$. Consequently

$$\sum_{\text{trees}}\prod_j q_j! = (n-2)!\sum_{(q_j)} q_1 \leq (n-1)!\binom{2n-2}{n-1} \leq (n-1)!4^{n-1}.$$

Collecting the factors, also for $n=1$, gives

$$\frac{S_n(Y)}{n!} \leq 7H_1\exp(\kappa_0)K_0|Y|_M\exp(-a_6 d_M(Y))z_S^{n-1}. \tag{14.34}$$

Since $z_S\leq 1/2$, sum the geometric series and use $|Y|_M\leq c_X\exp(d_M(Y))$ and $a_6-1=a_3$. The bound becomes

$$98c_X^2K_0^2\exp(\kappa_0+3a_5)H_{\mathrm{cl}}\exp(-a_3 d_M(Y)) \leq c_S' H_{\mathrm{cl}}\exp(-a_3 d_M(Y)),$$

because $\kappa_0+3a_5 = 3\kappa-8\kappa_0-6$. Summing only $n>N$ gives the additional factor 2^{-N}. This proves absolute convergence and the claimed estimates.

It remains to justify the logarithm identity after substitution in (14.24). The even subalgebra with the coefficient norm is a commutative Banach algebra, and the polymer set is finite. The total norm sum of the connected terms is finite by (14.34). In the exponential of their sum, the norm sum of all products of connected terms is bounded by the exponential of that total norm sum. Regrouping is therefore permitted. Regroup by monomials in the formal variables z_Y. The coefficients are those of (14.24), namely one for compatible sets and zero for all other monomials. Its substitution $z_Y = H(Y)$ gives (14.33). Grouping the connected terms by their union gives $\sum_Y E^{\#}(Y)$ and proves (14.29). □

There are also polynomial approximants to this logarithm which converge uniformly for all families satisfying the preceding bound, at fixed side. Replace $\exp V - 1$ by $p_N(V) = \sum_{1\leq j\leq N} V^j/j!$ in K, H, and stop the connected series at $n=N$; denote the result $E_N^{\#}$. It is a polynomial in integrals of products of $V(X_i)$, $X_i\subseteq Y$, and has the same bound (14.30). To verify uniform convergence, write $D = |\mathcal{P}_k|$. Since $\|V(X)\|\leq 1$,

$$\|\exp V(X) - 1 - p_N(V(X))\| \leq \frac{\exp(1)}{(N+1)!}.$$

Telescoping each finite product in K gives

$$\|H(Y) - H_N(Y)\| \leq \frac{4^D D\exp(1)}{(N+1)!}.$$

Thus this difference is at most $\delta_N\mu(Y)$, where

$$\delta_N = \frac{4^D D\exp(1)}{(N+1)!\min_{Y\in\mathcal{P}_k}\mu(Y)} \longrightarrow 0.$$

Replacing the n factors in a connected term costs at most $n\delta_N\prod_i\mu(Y_i)$. Summing (14.34) with this factor, using $\sum_{n\geq 1} n2^{-(n-1)} = 4$, and adding the tail yields

$$\sup_Y \|E^{\#}(Y) - E_N^{\#}(Y)\| \leq 28\delta_N H_1\exp(\kappa_0)K_0(\ell/M)^3 + 2^{-N}c_S' H_{\mathrm{cl}}.$$

This convergence is uniform in the connections and the allowed data. The displayed error depends on the side; no volume-independent approximation is claimed.

14.5. **Symmetries of the local logarithm.** Apply Proposition 14.5 with $V = F$, $w = h/4$ and denote its output by $E_k^{\#}(m, \mathscr{E})$. Each connected term assigned to Y is a product of integrated activities on subpolymers of Y. Thus ordinary and admissible pair support, local connection dependence, and dependence only on m' and $\mathscr{E}^r(X)$, $X \subseteq Y$, pass to the sum. The uniform bound and the polynomial approximants prove analyticity of its coefficients in $(A, \mathscr{C})$.

Let $\mathbb{G}_k^L$ denote the chamber symmetry group with translations in $LM\mathbb{Z}^3$, so that it preserves every parameter partition. Its actions and the gauge, charge-conjugation and reality actions are those of Section 12.1.

Proposition 14.6. *The local logarithm satisfies*

$$a_{\tilde{g}} E_k^{\#}(\tilde{g}Y; \tilde{g}A, \tilde{g}\mathscr{C}) = E_k^{\#}(Y; A, \mathscr{C}), \qquad \tilde{g} \in \mathbb{G}_k^L, \tag{14.35}$$

$$G_{\lambda+\lambda'} E_k^{\#}(Y; A^1, \mathscr{C}^1) = E_k^{\#}(Y; A, \mathscr{C}), \tag{14.36}$$

$$\mathrm{Cg}\, E_k^{\#}(Y; -A, -\mathscr{C}) = E_k^{\#}(Y; A, \mathscr{C}). \tag{14.37}$$

In the gauge identity both pairs belong to the domain and $A^1 = A - d\lambda$, $\mathscr{C}^1 = \mathscr{C} - d\lambda'$ on all bonds at distance at most 1 from Y, with real λ, λ'. Here $d\lambda$ denotes the lattice gradient, as in the preceding gauge conventions. For real m and $\mathscr{E} \in K_k^x$ one also has

$$\mathbf{K} E_k^{\#}(Y; -\overline{A}, -\overline{\mathscr{C}}) = E_k^{\#}(Y; A, \mathscr{C}). \tag{14.38}$$

Every monomial is balanced.

Proof. We first record how transformations pass through the finite operations. Each of the four algebra maps is a continuous ring automorphism, linear for the first three and antilinear for $\mathbf{K}$. If an odd field transforms like the generators it replaces, the two compositions of the field substitution with the automorphism agree on each generator; hence they agree on the algebra. The same observation applies to adding a field. Real-coefficient finite differences (14.6) then commute with these actions. A lattice symmetry also permutes the parameter cubes and localization summands. It preserves face connectedness, incidence relations and the Ursell coefficient.

The contour functional commutes with the linear actions. For the antilinear action, parametrize $t = r\exp(i\theta)$:

$$\mathcal{C}_r f = \frac{1}{2\pi} \int_0^{2\pi} \frac{f(r\exp(i\theta))}{r\exp(i\theta) - 1}\, \mathrm{d}\theta.$$

Conjugation of coefficients, together with $t \mapsto \bar{t}$ and $\theta \mapsto -\theta$, gives the same integral at the transformed other arguments. The integration over W commutes with every linear or antilinear action which sends $W \mapsto UW$, $\bar{W} \mapsto (U^{-1})^T \bar{W}$. Indeed Wick's rule preserves every paired moment under this change, and all these moments are real. This applies to the lattice and gauge actions, to $\mathbf{K}$ with $U = \varkappa$, and to charge conjugation extended by the identity on W. The last extension will require a separate covariance argument below.

For a lattice symmetry the primitive covariance gives

$$Z(\tilde{g}s')[\tilde{g}\zeta] = \tilde{g} Z(s')[\zeta], \qquad H_k(\tilde{g}s'', \tilde{g}B)\tilde{g} = \tilde{g} H_k(s'', B), \qquad \Gamma_k(\tilde{g}s'', \tilde{g}B)\tilde{g} = \tilde{g}\Gamma_k(s'', B).$$

The barred action is $(S^{-1})^T$ on spinors; $(S^{-1})^T = \pm \mathsf{C}^{-1} S \mathsf{C}$ gives the same covariance for $\bar{H}_k$. Hölder quotients commute with the simultaneous transformation of their endpoints and connection. Thus $J_k^{\#}\Psi$ and $W_k^{\#}$ transform like their ordinary and pair generators. The quadratic primitive satisfies $M_k(\tilde{g}X, \tilde{g}B) = \tilde{g} M_k(X, B)\tilde{g}^{-1}$, so its bilinear form is invariant. The mass form is invariant by the change of chamber site and spin indices, and $R\mathscr{E}$ has the incoming lattice symmetry. The reconstruction substitution commutes with the action by (13.24). The box measure is invariant under $\zeta \mapsto \tilde{g}\zeta$. Apply these identities to G_1, G_2, G_3, the first localization and

substitution, then to the second localization, K, H, and the convergent connected sum. They prove (14.35).

For the gauge identity set $\Lambda = \lambda + \lambda'$. In a summand assigned to Y, all parameters outside Y vanish. The local row result proves that B^1 may be replaced by $B - d\Lambda$ in every row used in that summand. The latter field is still in the operator domain, since real gauge shifts preserve it. Telescoping the gradient along each path gives

$$U_{B-d\Lambda}(x,x') = U_B(x,x')\exp\big(-ie(\Lambda(c')-\Lambda(c))\big).$$

The ordinary primitive row obeys

$$H_k(s'',B^1)(x,\cdot) = \exp(ie\Lambda(c))H_k(s'',B)(x,\cdot)\exp(-ie\Lambda^u),$$

when it is nonzero, where Λ^u is the function on the unit indices. The barred row has the inverse phases. In the Hölder row the phase at its second point cancels that from the transport, leaving exactly $\exp(ie\Lambda(c))$, the phase of the first point. The row of Γ_k has the corresponding two-sided gauge covariance. Its composition with $H_k^{\#}$ thus gives the same covariance for $W_k^{\#}$, including the rows which vanish outside Y. The primitive bilinear and the local mass form are gauge invariant, and so is $R\mathscr{E}$ under the separate shifts of its two slots. The reconstruction substitution also commutes with this local gauge action. The photon fields and photon measure are unchanged. The finite operations and integration therefore prove (14.36).

For reality, the photon primitives are real on real parameter families. Their analytic dependence extends this identity to $\overline{O(\bar{s}')} = O(s')$ on the full polydisk. Since C is real,

$$Z(\bar{s}')[-\zeta] = -\overline{Z(s')[\zeta]}.$$

Transform $s \mapsto \bar{s}$, $t \mapsto \bar{t}$, $\zeta \mapsto -\zeta$ and $(A,\mathscr{C}) \mapsto (-\bar{A}, -\overline{\mathscr{C}})$. The fermion identities are

$$\overline{H_k(\bar{s}'',-\bar{B})} = \varkappa^{-1}H_k(s'',B)\varkappa, \qquad \overline{\Gamma_k(\bar{s}'',-\bar{B})} = \varkappa^{-1}\Gamma_k(s'',B)\varkappa.$$

For the barred field use $\varkappa\mathsf{C} = -\mathsf{C}\varkappa$. The transport relation $\overline{U_{-\bar{B}}(x,x')} = U_B(x,x')$ then gives the same identities for the Hölder rows. The local quadratic primitive obeys the corresponding identity, and $\varkappa^T\varkappa = 1$ preserves the mass form. The number m' is real for the present data, and $R\mathscr{E}$ and the reconstruction substitution have the reality symmetry. The contour argument just given, invariance of the box under sign reversal, and weighted integration prove (14.38).

Charge conjugation gives the first-localization symmetry by

$$H_k(s'',-B)\mathsf{C} = \mathsf{C}\bar{H}_k(s'',B), \qquad \bar{H}_k(s'',-B)\mathsf{C} = \mathsf{C}H_k(s'',B),$$
$$M_k(X,-B)^T = \mathsf{C}^{-1}M_k(X,B)\mathsf{C}.$$

Anticommuting the two transformed legs verifies the bilinear and mass-form identities. The first-step vertex has this symmetry by its explicit two-half formula. It follows that

$$\mathrm{Cg}\,E_{\le 3}(X;-A,-\mathscr{C},-\zeta) = E_{\le 3}(X;A,\mathscr{C},\zeta).$$

Direct covariance of the second substitution is not available, since its unbarred and barred fluctuations use different factors. We instead compare their Wick covariances.

Introduce independent generators $\phi^{\#}$ indexed like $\psi^{\#}$, with their own charge-conjugation action, and let ad add them to the old generators. Put

$$\mathcal{V}(X) = \mathrm{ad}\,E_{\le 3}(X) + \mathrm{ad}\,\mathscr{E}^r(X) - \mathscr{E}^r(X)$$
$$- \mathbf{1}_{X=\square}m'\big(\mathrm{ad}\langle\bar{\psi},\psi\rangle_\square - \langle\bar{\psi},\psi\rangle_\square\big).$$

It has charge-conjugation symmetry under $(A,\mathscr{C},\zeta) \mapsto (-A,-\mathscr{C},-\zeta)$. Let $I_{s'',B}$ fix the old generators and replace $\phi^{\#}$ by $W_k^{\#}(s'',B)$. The second-localization input is exactly $I_{s'',B}\mathcal{V}(X)$,

since the polynomial contours in G_4, G_5 give their value at 1 minus their value at 0. For any union U, interpolation restricted to cubes of U, with the disconnected terms removed, gives

$$\sum_{Y\subseteq U} F(Y) = I_{\mathbf{1}_U,B} \sum_{X\subseteq U} \mathcal{V}(X). \tag{14.39}$$

At $s'' = \mathbf{1}_U$ define

$$\phi_\pm = H_k(\mathbf{1}_U, \pm B)\Gamma_k(\mathbf{1}_U, \pm B)W, \qquad \bar\phi_\pm = \bar H_k(\mathbf{1}_U, \pm B)\bar W, \qquad \phi' = \mathsf{C}\bar\phi_-, \quad \bar\phi' = \mathsf{C}^{-1}\phi_-.$$

The substitution by $(\phi', \bar\phi')$ and their Hölder quotients, denoted I', satisfies $\mathrm{Cg}\, I_{\mathbf{1}_U,-B} = I'\mathrm{Cg}$ on every generator, when Cg fixes W. The only nonzero ordinary pair expectations for the positive substitution are the kernel of

$$\mathcal{G}_U(B) = H_k(\mathbf{1}_U, B)\Gamma_k(\mathbf{1}_U, B)\bar H_k(\mathbf{1}_U, B)^T.$$

The identity moment has kernel $\mathbf{1}_{u=u'}/\nu(u)$, with $\nu = 1/48$ at zero. This precisely cancels one of the two measures in the kernel composition, so the same formula holds at every level. The primitive transpose relation gives

$$\mathcal{G}_U(-B)^T = -\mathsf{C}^{-1}\mathcal{G}_U(B)\mathsf{C}^{-1}.$$

Anticommuting $\bar\phi_-$ and ϕ_- therefore gives

$$\int \mathrm{d}\mu_I(W)\, \phi'(x)\bar\phi'(x')^T = \mathsf{C}[-\mathcal{G}_U(-B)^T(x,x')]\mathsf{C} = \mathcal{G}_U(B)(x,x').$$

Same-bar pair expectations vanish in both substitutions. The pair generators are obtained by the identical linear Hölder maps at $B, -B$, so their pair expectations also agree. Wick's theorem now proves $\int \mathrm{d}\mu_I(W) I'Q = \int \mathrm{d}\mu_I(W) I_{\mathbf{1}_U,B} Q$ for every polynomial Q in the old and $\phi^\#$ generators: each monomial integral is the Pfaffian of its pair expectations, with old generators as coefficients.

Apply this equality to the exponential of $\sum_{X\subseteq U}\mathcal{V}(X)$ and use $\zeta \mapsto -\zeta$. By (14.39), the integrated subvolume expression $\Xi_U = \int \mathrm{d}\mu_I(W)\mathrm{d}\widehat{\mu}(\zeta)\exp(\sum_{Y\subseteq U} F(Y))$ has charge-conjugation symmetry. But $\Xi_U = \sum_{Z\subseteq U} H(Z)$, by the finite Mayer expansion. Inverting this finite subset sum gives

$$H(Z) = \sum_{U\subseteq Z} (-1)^{|Z\setminus U|_M}\Xi_U.$$

Thus each $H(Z)$ has the symmetry, and the real coefficients and absolute convergence of the connected series prove (14.37).

Finally take a constant λ and $\lambda' = 0$ in (14.36). A monomial with n_u unbarred and n_b barred legs acquires the factor $\exp(ie\lambda(n_u - n_b))$. Invariance for every real λ and independence of the basis monomials force its coefficient to vanish unless $n_u = n_b$. □

14.6. **The scaled correction.** Let S_L be the field scaling of (12.45). The nonlinear correction is

$$E_k^*(m,\mathscr{E})(Y;\mathscr{A},\mathscr{C}) = \sum_{\bar X^L = LY} S_L E_k^\#(m,\mathscr{E})(X;\mathscr{A}_L,\mathscr{C}_L). \tag{14.40}$$

This formula uses field scaling and summation over fine polymers. Quadratic adjustment enters the incoming activity term in the full step below. The domain inclusions (12.44) ensure $(\mathscr{A}_L,\mathscr{C}_L) \in \widetilde{R}_k^- \times \mathfrak{X}_k$.

Proposition 14.7. *For* $(m,\mathscr{E}) \in \mathrm{Ball}_k^x(4)$,

$$E_k^*(m,\mathscr{E}) \in K_{k+1}^{x,\mathbb{C}}, \qquad \|E_k^*(m,\mathscr{E})\|_{k+1}^x \le e^{1/4-6\varepsilon}.$$

For real data its value belongs to K^x_{k+1}. On any affine complex line in the data space its value is norm continuous on the part in the closed ball and norm analytic in its interior. For two data in $\mathrm{Ball}^x_k(1)$*,*

$$\|E^*_k(m_1,\mathscr{E}_1)-E^*_k(m_2,\mathscr{E}_2)\|^x_{k+1}\le e^{1/4-6\varepsilon}\left(\frac{|m_1-m_2|}{w^{\mathrm{m}}_k}+\frac{\|\mathscr{E}_1-\mathscr{E}_2\|^x_k}{w^{\mathrm{E}}_k}\right). \tag{14.41}$$

The second quotient is absent at $k=0$. Restriction to $\mathscr{C}=0$ gives the correction formed without an external slot from $(m,\mathscr{E}|_0)$.

Proof. The scaling identity gives

$$\|S_LQ\|_{h_{k+1}}=\|Q\|_{(L^{-1}h_{k+1,1},L^{-1-\alpha}h_{k+1,2})}\le\|Q\|_{h_k/4},$$

since $h_{k+1,1}=L^{-1/8}h_{k,1}$, $h_{k+1,2}=L^{-1/8+\varepsilon/2}h_{k,2}$ and $L\ge4$. The polymer sum in (14.40) uses $Ld_M(Y)\le d_M(X)$ and $X\subseteq LY$. Splitting $a_3=(a_3-\kappa_0)+\kappa_0$ therefore yields

$$\begin{aligned}\sum_{\bar X^L=LY}\exp(-a_3d_M(X))&\le\exp(-L(a_3-\kappa_0)d_M(Y))\sum_{X\subseteq LY}\exp(-\kappa_0d_M(X))\\&\le c_XK_0L^3\exp(-[L(\kappa-6\kappa_0-4)-1]d_M(Y))\le c_XK_0L^3\exp(-\kappa d_M(Y)).\end{aligned}$$

Together with (14.30) this gives

$$\|E^*_k\|^x_{k+1}\le c_XK_0L^3c'_SH_{\mathrm{cl}}=504c_X^5K_0^5\exp(3\kappa)L^3e^{1/4-5\varepsilon}\le e^{1/4-6\varepsilon},$$

using $c_SL^3e^{\varepsilon}\le1$.

All generators of a summand lie over X or at a pair admissible for X. Scaling maps the ordinary sites into Y; it maps each of the six paths into the corresponding scaled path and maps $(x,x^{(3)})$ to $(x/L,(x/L)^{(3)})$. These pairs are admissible for Y, including intracluster pairs. A bond meeting X scales to one meeting Y, giving local connection dependence. The scaled-domain inclusion and coefficient analyticity give analytic kernels. The map $(S,a)\mapsto(S,a/L)$ identifies $\mathbb{G}^L_k$ with $\mathbb{G}_{k+1}$ and intertwines its generator actions with S_L. For gauge transformations use $\Lambda_L(c)=L^{1/2}\Lambda(c/L)$ and $(d\Lambda)_L=d(\Lambda_L)$. A gauge relation on the unit neighborhood of Y then holds on the unit neighborhood of every $X\subseteq LY$. Charge conjugation and reality also commute with the real scaling factors. Proposition 14.6 thus gives all the defining symmetries of $K^{x,\mathbb{C}}_{k+1}$, and the real space assertion follows from its reality clause.

For norm analyticity, take data $(m_a+tm_b,\mathscr{E}_a+t\mathscr{E}_b)$ in the closed ball and denote their parameter set by T. It is closed and convex. The finite activity $F_t(X)$ is affine in t by Proposition 14.4. The polynomial approximants to $E^{\#}_k$ are therefore polynomials in t. Reblock them by (14.40); the resulting polynomials $f_N(t)$ converge uniformly on T in the full activity norm, since both finite polymer sets are fixed and the error bound of the previous subsection is uniform. Their values have bounded full norm. If T has more than one point, it contains infinitely many points. The coefficients of each polynomial are finite numerical linear combinations of its values at distinct points of T, by polynomial interpolation. Hence these coefficients have finite full norm, and the uniform limit is norm continuous on T. If T is empty or a singleton there is no continuity or interior analyticity assertion to check.

For a closed disk $|t-t_0|\le r$ in T, take its coefficientwise Cauchy integrals

$$f_j=\frac{1}{2\pi i}\int_{|t-t_0|=r}f(t)(t-t_0)^{-j-1}\,\mathrm{d}t.$$

They have analytic connection kernels, and the support, symmetry and balance conditions pass through this complex linear integral. Moreover $\|f_j\|^x_{k+1}\le e^{1/4-6\varepsilon}r^{-j}$. Coefficientwise Cauchy's formula, with this summable norm bound, gives $f(t)=\sum_{j\ge0}f_j(t-t_0)^j$ in activity norm for $|t-t_0|<r$. This proves the stated analyticity.

To obtain (14.41), vary first the mass, keeping $\mathscr{E}_1$ fixed. With $\Delta m = m_1 - m_2 \neq 0$, the line $(m_2 + t\Delta m, \mathscr{E}_1)$ remains in the large ball for $|t| \leq r = 3w_k^{\mathrm{m}}/|\Delta m|$. Here $r \geq 3/2$. The contour formula on any circle $1 < r' < r$ gives

$$\|f(1) - f(0)\|_{k+1}^x \leq \frac{e^{1/4-6\varepsilon}}{r'-1}.$$

Let $r' \uparrow r$ and use $r - 1 \geq r/3$. The bound is the first quotient on the right of (14.41). Vary the activity with mass m_2 fixed and radius $3w_k^{\mathrm{E}}/\|\mathscr{E}_1 - \mathscr{E}_2\|_k^x$ to obtain the second quotient. The zero differences require no contour.

Finally restriction commutes with extraction: $m(\mathscr{E}) = m(\mathscr{E}|_0)$ and $(R\mathscr{E})|_0 = R(\mathscr{E}|_0)$. Every fermion operator uses the sum of its first connection and $\mathscr{C}$; photon operations and measures do not use $\mathscr{C}$. At $\mathscr{C} = 0$ the two constructions therefore have identical finite expressions, connected series and scaled corrections. Analyticity without a slot is understood in the first connection alone. □

The possible Hölder legs can also be read directly. In G_2, G_4 an existing pair generator is changed by addition of a fluctuation field; this creates no new pair index. The mass contours contain no pair generators. The reconstruction substitution and the first-step vertex introduce only pairs $(x, x^{(3)})$. Every later sum, product and integral preserves this set of available indices. Thus the correction has only scaled incoming pair indices and these scaled type-3 indices. At $k = 0$ it has only the latter, of length η_1, since the incoming activity is zero.

14.7. **The step identity and bounds.** For the local determinant terms of Proposition 11.5, put

$$\delta E_k^{\det}(X; \mathscr{A}, \mathscr{C}) = E_k^d(X, \mathscr{A} + \mathscr{C}) - E_k^d(X, \mathscr{C}), \qquad E_k^{\det}(X, \mathscr{C}) = E_k^d(X, \mathscr{C}) - E_k^d(X, 0).$$

These are scalar activities. We use reblocking $\mathcal{L}$ from (12.49). The adjustment acts on the incoming extracted activity. Define

$$\mathscr{E}^+ = \mathcal{L}(R\mathscr{E})^{\mathrm{nat}} + \mathcal{L}\delta E_k^{\det} + E_k^*(m, \mathscr{E}), \tag{14.42}$$

$$m^+ = L(m + m(\mathscr{E})), \tag{14.43}$$

$$\varepsilon^{\mathrm{v},+} = L^3(\varepsilon^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(\mathscr{E}) + \delta\varepsilon_k^0), \tag{14.44}$$

$$\mathfrak{P}_{k+1}(Y, \mathscr{C}') = \sum_{\bar{X}^L = LY} \big(\mathfrak{P}_k(X, \mathscr{C}'_L) + E_k^{\det}(X, \mathscr{C}'_L)\big). \tag{14.45}$$

Here $\mathfrak{P}_k$ is any given family of scalar functions of the external slot.

The reblocked determinant term belongs to K_{k+1}^x. Indeed the scaled arguments lie in its inverse domain, and each summand is analytic and reads only fine bonds meeting X. These scale to bonds meeting Y. There are no legs. The lattice symmetry permutes the reblocking summands; local gauge shifts of the two slots become the corresponding real shifts on these fine read bonds, where determinant locality and gauge invariance apply. Evenness of E_k^d gives charge conjugation, and its conjugation identity gives reality. The pointwise bound (11.38) and (A.5) show

$$\|\mathcal{L}\delta E_k^{\det}\|_{k+1}^x \leq 8c_{\mathrm{rb}}L^3C_dM^3e_k^{1/4} \leq e_k^{1/4-2\varepsilon} \leq e_{k+1}^{1/4-2\varepsilon}. \tag{14.46}$$

Here $L(\kappa - \kappa_0 - 1) \geq \kappa$ controls the remaining exponential weight, and the fixed charge condition is $8c_{\mathrm{rb}}L^3C_dM^3e_k^{2\varepsilon} \leq 1$. Restriction to $\mathscr{C} = 0$ gives the same result for $\mathcal{L}E_k^{\det}$.

Theorem 14.8. *Let $\varepsilon^{\mathrm{v}}, m$ be real, $\mathscr{E} \in K_k^x$, and $(m, \mathscr{E}) \in \mathrm{Ball}_k^x(1)$, with $\mathscr{E} = 0$ at $k = 0$. For $A' \in \mathrm{Dom}_{k+1}$ with arbitrary harmonic part and $\mathscr{C}' \in \mathfrak{X}_{k+1}$, the exact density satisfies*

$$\mathbf{T}_k^x \mathscr{F}_k^x[\varepsilon^{\mathrm{v}}, m, \mathscr{E}] = \mathscr{F}_{k+1}^x[\varepsilon^{\mathrm{v},+}, m^+, \mathscr{E}^+], \tag{14.47}$$

with external scalar families $\mathfrak{P}_k$ and $\mathfrak{P}_{k+1}$ on the respective sides. Its integral satisfies

$$y^x_{k+1}(A';\mathscr{C}') = \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta)\, y^x_k(A^{\min}_k(A') + \mathrm{Sq}_k\zeta;\mathscr{C}'_L), \tag{14.48}$$

where $y^x_j = \int \mathrm{D}\Psi\, \mathscr{F}^x_j$ with the data and scalar family of that level. Moreover $\mathscr{E}^+ \in K^x_{k+1}$ and

$$\|\mathscr{E}^+\|^x_{k+1} \le c_{\mathrm{fl}} L^{-1/4+2\varepsilon}\|\mathscr{E}\|^x_k + e^{1/4-2\varepsilon}_{k+1} + e^{1/4-6\varepsilon}_k, \tag{14.49}$$

$$|m^+ - Lm| \le L c_{\mathrm{fl}} e^{1/2}_k \|\mathscr{E}\|^x_k, \tag{14.50}$$

$$|\varepsilon^{\mathrm{v},+} - L^3\varepsilon^{\mathrm{v}}| \le L^3(c_{\mathrm{fl}}\|\mathscr{E}\|^x_k + e^7_k). \tag{14.51}$$

At level zero this formula integrates all $192n_0$ incoming pairs of generators, with chamber measure $1/48$, and leaves $4n_1$ outgoing pairs. Its data are

$$\mathscr{E}^+ = \mathcal{L}\delta E^{\det}_0 + E^*_0(m,0), \quad m^+ = Lm, \quad \varepsilon^{\mathrm{v},+} = L^3(\varepsilon^{\mathrm{v}} + \delta\varepsilon^0_0).$$

In particular $0 \le \varepsilon^{\mathrm{v},+} - L^3\varepsilon^{\mathrm{v}} \le L^3 q^7$.

Proof. Use the notation of Proposition 13.3, with $A = \mathscr{A}'_L$, $B = A + \mathscr{C}$ and $Z = Z(1)$ the full-strength photon fluctuation. The primitive families satisfy their exact identities at full strength: $H_k(1,B) = H_k(B)$, $\Gamma_k(1,B) = \Gamma_k(B)$, and $\sum_X M_k(X,B) = b_k - D_k(B)$ for $k \ge 1$. The induced fields obey

$$\psi^{0,\#} = H^\#_k(B)\Psi^c, \qquad H^\#_k(B)\Psi^f = \psi^{0,\#} + W^\#_k(1,B).$$

Define three evaluation homomorphisms. Let ev send old generators to $\psi^{0,\#}$ and fix W; let ev_1 send them to $H^\#_k(B)\Psi^f$; for $k \ge 1$, let ev_f additionally send $\Psi, \bar\Psi$ to $\Psi^f, \bar\Psi^f$. At zero take $\mathrm{ev}_f = \mathrm{ev}_1$. These maps exist since their images are odd and have the required antiperiodicity. They commute with the finite coefficient integrals; ev also commutes with Berezin integration, since it fixes W.

On the first contours $\mathcal{C}_r f = f(1) - f(0)$. For G_1, summing the local primitives gives the Dirac difference $-\langle\bar\Psi, (D_k(B+Z) - D_k(B))\Psi\rangle_1$. For G_2, G_3, evaluating $\psi^\# + J^\#_k(1,B,Z)\Psi$ gives $H^\#_k(B+Z)\Psi^f$. The evaluations of their contour differences therefore give

$$\mathrm{ev}_f \sum_X \sum_{i\le 3} G_i(1,X) = \mathrm{Act}^x_k[m',\mathscr{E}^r](A+Z,\mathscr{C};\Psi^f) - \mathrm{Act}^x_k[m',\mathscr{E}^r](A,\mathscr{C};\Psi^f). \tag{14.52}$$

At zero the explicit first vertex gives this same identity with the chamber pairing: $D_0(B) = D_B + \mu$ and the mass form has no connection dependence.

The reconstruction left inverse gives, on every local first expression, $\mathrm{ev}_1 \mathrm{sub}_{Y,B} = \mathrm{ev}_f$. This follows on the Ψ generators from $T_k(B)H_k(B) = 1$ and on the old generators from their definition. Adding $W^\#_k(1,B)$ to $\psi^\#$ in a polynomial and then applying ev gives ev_1 applied to that polynomial. Apply these identities to (14.16). The evaluated $G^W_{\le 3}$ sum is (14.52). The evaluated polynomial contours G_4, G_5 give

$$I^x_k[m',\mathscr{E}^r](A,\mathscr{C};\Psi^f) - I^x_k[m',\mathscr{E}^r](A,\mathscr{C};\Psi^c).$$

Consequently

$$\mathrm{ev}\sum_Y F(Y) = V^x_k, \qquad \Xi^x_k = \exp\left\{\sum_X \mathrm{ev}\, E^\#_k(m,\mathscr{E})(X;A,\mathscr{C})\right\}.$$

The second equality is (14.29) evaluated by ev; its evaluation commutes with the integrals as just noted.

The other remaining factor is the determinant quotient. Equation (11.34) gives

$$\frac{\delta Z_k(B)}{\delta Z_k(0)} = \exp\left\{\sum_X \delta E_k^{\det}(X; A, \mathscr{C}) + \sum_X E_k^{\det}(X, \mathscr{C})\right\}.$$

On inserted fields adjustment changes no value: $\operatorname{ev} R\mathscr{E}(X) = \operatorname{ev}(R\mathscr{E})^{\mathrm{nat}}(X)$. The equality is the telescoping reconstruction identity for a small polymer and the definition for a large one. Finally $\operatorname{ev} = \operatorname{ev}' S_L$, where ev' inserts the outgoing fields. Each fine polymer has exactly one outgoing reblocking polymer. Thus the three activity sums combine into the outgoing $\mathscr{E}^+$, and the two scalar external sums combine into (14.45). Insert these identities in (13.22). All exponents are even and commute, so their sums give (14.47), with exactly its previously derived mass, vacuum and determinant normalization.

Integrate over the outgoing generators and apply the exact marginal (13.20). This gives (14.48). The integrand is admissible throughout the photon box: Proposition 10.4 places $A_k^{\min}(A') + \mathrm{Sq}_k\zeta$ in Dom_k, and its reconstructed connection is an affine, continuous function of ζ. The density coefficients are analytic in this connection on its operator and activity domains, hence continuous and integrable on the compact box. The external scalar factor is fixed during this integral. No condition on the harmonic part of A' enters.

The three terms of (14.42) belong to K_{k+1}^x by Proposition 12.5, (14.46), and Proposition 14.7. Their respective norms are bounded by the three terms in (14.49). Equation (12.62) bounds the first term with the stronger exponent $3\varepsilon/2$. The mass and vacuum extraction bounds give (14.50) and (14.51), since

$$0 \le \delta\varepsilon_k^0 \le 12\exp(-p_{0,k}^2/2) \le e_k^7.$$

At zero all incoming extracted terms vanish. The exact algebra uses all $192n_0$ pairs with the weight $1/48$ and integrates to the $4n_1$-pair outgoing density. There is no removed multiplicity or additional vacuum normalization. The displayed zero-level formulas and bounds follow. □

The restriction to zero external connection commutes with all three terms of (14.42). The determinant difference there becomes $E_k^d(X, \mathscr{A}) - E_k^d(X, 0)$, and the external scalar map preserves $\mathfrak{P}_k(\cdot, 0) = 0$. Thus (14.47) and (14.48) also give the step without a slot, with incoming activity $\mathscr{E}|_0$.

The displayed norm estimates depend on the running charge and the fixed constants, and on no side or harmonic part. They were proved for any primitive system with the stated hypotheses on the present torus. Identifying the outputs from different systems or different volumes requires the separate comparison arguments; it does not follow from these bounds.

15. Covers and comparison of local steps

We construct the primitive systems on one large torus and project them to smaller tori through odd covering maps. Projection preserves the estimates with a fixed loss in the common constant and one loss in polymer decay. The comparison of local steps then uses supported rows near a common lift, where the signed image sum has only one surviving term.

15.1. **Odd covers and projected kernels.** Fix $k \ge 0$, $0 < \mu \le 1$, $0 < e_k \le e_0$, and admissible sides $\ell_2 = J_c\ell_1$, where J_c is odd. Write $T_i = \mathbb{L}_k/\ell_i\mathbb{Z}^3$ and $p : T_2 \to T_1$ for the cover. Let pr_i denote the covering-lattice projections. Pullback p^* is inclusion of functions on the covering index sets. They have character $\chi(w) = 1$ for photon fields, coordinates and parameters, and $\chi(w) = (-1)^{w_1+w_2+w_3}$ for spinors and pair fields. Oddness gives $\chi(J_c w) = \chi(w)$, so a field on T_1 is a field on T_2 with the same character. The image is

$$\mathrm{Per} = \{f : f(t + \ell_1 w) = \chi(w) f(t),\ w \in \mathbb{Z}^3\}.$$

Translations move the photon site and preserve the chamber label. In particular pullback commutes with J, J^*, complex conjugation and constant spin matrices.

The index weight is 1 on unit sites, unit bonds and photon coordinate indices, η_k^3 on fine photon sites and bonds, and $\eta_k^3/48$ on chamber sites. For fields with a common index set and character,

$$\|p^* f\|_\infty = \|f\|_\infty, \qquad \langle p^* f', p^* f\rangle_{T_2} = J_c^3 \langle f', f\rangle_{T_1}. \tag{15.1}$$

Indeed a fundamental set for T_2 is the union of $\mathrm{Fnd}_1 + \ell_1 w$, over representatives $w \in (\mathbb{Z}/J_c\mathbb{Z})^3$. The two characters cancel in each paired summand, and the weight is the same on both tori. At zero the chamber weight remains $1/48$.

Pullback maps the parameter polydisk, including its interior, into the corresponding larger polydisk, since $(p^* s)_{\mathrm{pc}} = s_{p\,\mathrm{pc}}$. It also maps $\widetilde{\mathcal{G}}_k(T_1)$ into $\widetilde{\mathcal{G}}_k(T_2)$. To check the latter assertion, write the domain decomposition $A = A_0 + A_1$. Pullback preserves reality of A_0 and the sup norm of A_1. Each five-cube neighborhood has a common nonwrapping lift of side $11M < \ell_1$. The real local gauge representative of A_0 on its image in T_1 pulls back on this lift to the corresponding neighborhood in T_2, with its same sup bound and gradient identity. This is precisely the local real-gauge condition in the inverse domain.

Every exact finite fermion operator used below intertwines with pullback. For the chamber Dirac operator this follows term by term from its four neighbors and bond phases. For $\mathcal{Q}_k = Q_k J^*$ and its reverse it follows from the same block sums and the same six paths on the cover; J^* averages the same 48 labels. The penalty and the finite matrices multiplying the resolvent have the same finite block formulas. Sums and products preserve intertwining. If $O_2 p^* = p^* O_1$ and both operators are invertible, then $O_2 p^* O_1^{-1} = p^*$ and hence $O_2^{-1} p^* = p^* O_1^{-1}$. The domain preservation just proved makes the already established inverse bounds applicable on both tori. It follows, from the formulas in Section 11, that the fine resolvents, $H_k, \bar{H}_k, D_k, M_k, \Gamma_k$ intertwine at the same connection:

$$O_2(p^* A)p^* = p^* O_1(A). \tag{15.2}$$

At zero use the entire chamber Dirac stencil and the inverse of F_y, on all chamber components. The exact photon operators have the corresponding identity (10.45).

Let a primitive system on T_2 be given. For a parameter family $O_2(s, A)$ put $\widetilde{O}(s, A) = O_2(p^* s, p^* A)$. For a polymer family put

$$\widetilde{O}(X, A) = \sum_{X' \in \mathcal{P}_k(T_2) : pX' = X} O_2(X', p^* A).$$

This is applied to M_k for $k \geq 1$ and to $S_{k,y}$ at every level. A projected polymer is connected: adjacent M-cubes have equal or adjacent images. Moreover

$$d_M(pX') \leq d_M(X'). \tag{15.3}$$

A projected net meets all image cubes. Projecting its chosen finite segment representation preserves the lengths of the segments; taking the infimum gives (15.3). There are only finitely many polymers on either torus.

Proposition 15.1. *If the system has translation covariance under $\ell_1\mathbb{Z}^3$, its projection is well defined by $p^* O_1 = \widetilde{O} p^*$ and has kernel*

$$O_1(\cdot)(t, t') = \sum_{w \in (\mathbb{Z}/J_c\mathbb{Z})^3} \chi(w) \widetilde{O}(\cdot)(t, t' + \ell_1 w). \tag{15.4}$$

This formula uses the same index weights on both tori. The signed summand depends only on the class of w. Transposition of an intertwining operator also intertwines:

$$O_2^T p^* = p^* O_1^T. \tag{15.5}$$

Proof. The pulled-back parameters and connection are invariant under ℓ_1 translations. Parameter-family covariance therefore says that $\widetilde{O}$ commutes with those translations. For polymer families, translation permutes the polymers X' with $pX' = X$, so the same commutation holds after their finite sum. An operator commuting with these translations maps the character eigenspace Per into itself. Since p^* is injective onto that space, this defines exactly one projected operator.

Sum its action over the fundamental sets used in (15.1) and insert $f(t' + \ell_1 w) = \chi(w) f(t')$. The coefficient of $\nu(t') f(t')$ is (15.4). Replacing w by $w + J_c v$ multiplies the character by $\chi(v)$ and the large-torus kernel by the same character, so the product is unchanged. This proves the asserted kernel formula, including the weights $\eta_k^3/48$ or $1/48$ on chamber columns.

Translations have transpose $T_v^T = T_{-v}$ for the displayed bilinear pairings. Hence O_2^T also commutes with ℓ_1 translations and maps Per into itself. Write $O_2^T p^* f' = p^* g$. For every source field f, intertwining and (15.1) give

$$J_c^3 \langle g, f \rangle_{T_1} = \langle O_2^T p^* f', p^* f \rangle_{T_2} = J_c^3 \langle f', O_1 f \rangle_{T_1}.$$

Nondegeneracy gives $g = O_1^T f'$. □

15.2. **The uniform projection estimate.** The supports of primitive row and column indices have sup diameter at most L. Assume $\gamma_w M \geq 1$. Then

$$\sum_{w \in (\mathbb{Z}/J_c\mathbb{Z})^3} \exp\big(-\gamma_w d_2(t, t' + \ell_1 w)\big) \leq 64 \exp\big(-\gamma_w d_1(t, t')\big). \tag{15.6}$$

To prove this, translate the column lift by an ℓ_1 period so that the distance of the two support sets equals $d_1 \leq \ell_1/2$. For each class of w, choose a representative w' which realizes its distance on T_2. Distinct classes have distinct representatives. Their distances are at least d_1; for $|w'|_\infty = j \geq 2$ they are at least

$$\ell_1 j - d_1 - 2L \geq d_1 + (\ell_1 - 2L)(j - 1).$$

This uses the support diameters and $d_1 \leq \ell_1/2$. Since $\ell_1 \geq 13LM$, we have $\gamma_w(\ell_1 - 2L) \geq 6$. There are 27 lattice vectors of norm at most 1 and at most $26j^2$ of norm j. The sum is therefore bounded by

$$\exp(-\gamma_w d_1) \left(27 + \sum_{j \geq 2} 26 j^2 \exp(-6(j-1)) \right).$$

The latter series starts below 0.269 and its successive ratios are at most $(9/4)\exp(-6) < 1/100$. Thus the bracket is below $27.3 < 64$, proving (15.6).

There is a second, polymer summation estimate. For a unit or chamber row index t, let $\square_t$ be its M-cube on T_2. By (15.3) and counting,

$$\sum_{\substack{pX' = X \\ X' \supseteq \square_t}} \exp\big(-(\kappa_w + \kappa_0) d_M(X')\big) \leq K_0 \exp\big(-\kappa_w d_M(X)\big). \tag{15.7}$$

Indeed the factor with rate κ_w is at most the factor on the right; the remaining sum is bounded by $\sum_{X' \supseteq \square_t} \exp(-\kappa_0 d_M(X')) \leq K_0$. The additional top-system decay κ_0 pays for the polymer image sum.

Proposition 15.2. *Suppose the system on T_2 has the primitive properties with parameters*

$$(\alpha_0, \gamma_w, \kappa_w + \kappa_0; C_w^{(1)}), \qquad C_w^{(1)} \geq 4 \exp(2\gamma_w) M^{27\alpha_0}.$$

Its projection has the corresponding properties on T_1 with

$$C_w = 64 K_0 C_w^{(1)} \tag{15.8}$$

and parameters $(\alpha_0, \gamma_w, \kappa_w; C_w)$*. In particular it satisfies the primitive hypotheses of Theorem 14.8. At zero the resolvent estimate retains* $(1+y)^{-1}$*.*

Proof. We verify the identities, estimates and local structure needed for the step. At full strength the exact intertwining identities and the primitive sum identities on T_2 give

$$H^{\mathrm{ph}}_{k,1}(1) = H^{\mathrm{ph}}_{k,1}, \quad C^{1/2}_{k,1}(1) = C^{1/2}_{k,1}, \quad H_{k,1}(1, A) = H_{k,1}(A), \quad \Gamma_{k,1}(1, A) = \Gamma_{k,1}(A).$$

For the two polymer families, sum first over all X: each X' occurs once, under $X = pX'$. Equation (15.2) then identifies the sum with the exact operator on T_1. All projected entries are finite sums of original entries composed with the linear pullback. Continuity on the closed parameter polydisk, holomorphy in its interior and in the connection for any fixed parameter, and Borel dependence on the resolvent parameter follow from these finite sums.

For the photon reconstruction, its first and Hölder derivative kernels have the same image-sum formula: the difference acts in the row variable, and short lifted differences agree on the two tori. Equation (15.6) bounds each by $64C^{(1)}_w \exp(-\gamma_w d_1)$. It gives the same bound for the square-root kernel and for Γ_k. The H_k sup and short-pair Hölder bounds instead follow by the isometric pullback in (15.1); their constant is $C^{(1)}_w$. A short pair and its six paths lift identically. An intracluster quotient is $\eta_k^{-\alpha}(f(c, g') - f(c, g))$ on each torus.

For $M_k(X, A)$, only X' containing $\square_t$ contribute to its row, by original polymer support. Combine (15.4), (15.6) and (15.7) to get

$$|M_{k,1}(X, A)(t, t')|_c \le C_w \exp(-\gamma_w d_1(t, t') - \kappa_w d_M(X)).$$

For a positive-level fine resolvent, the pullback of a unit-block indicator is the disjoint sum of its J_c^3 translated large-torus block indicators. The same two estimates bound its block operator norm. At zero use its chamber kernel directly: the original factor $(1+y)^{-1}$ is common to all terms of both image sums and remains on the right. All column weights are unchanged, so no further factor 48 appears.

For component locality, an index of one parameter kind whose support is contained in its active union belongs to exactly one link-component. Here is the geometric point. Site supports are singletons. For a bond of length $a \in \{\eta_k, 1\}$, cubes of radius M meeting its support either share an endpoint or would have centers differing in that direction by $2M + a$. The latter is impossible since the centers differ by a multiple of M and $0 < a < M$. For a coordinate index, its support has diameter at most L. The centers of two cubes of radius LM meeting it differ coordinatewise by less than $3LM$, and are multiples of LM; choosing the coordinate midpoint when the difference is $2LM$, and either center when it is LM, exhibits a lattice site in both cubes. Thus all cubes meeting the support are linked. The same argument applies on either torus by lifting the support and these cubes into a nonwrapping neighborhood.

Pull back the active parameter cubes on T_1. A link-component on T_2 maps into one component on T_1, since the covering map preserves links. For a row in a small-torus component, the unique large-torus component containing its lift maps into that component. Original component restriction confines all surviving columns there. Changing the parameters and connection outside the small component leaves these original rows unchanged; the signed image sum therefore preserves their row dependence. The identical argument applied to each lifted column preserves column component restriction. Each row-vanishing property stated for an original family passes to its projection. In particular, if a cube meeting K_c is missing, its unique lift meeting the lifted K_c is missing too, so the H_k row vanishes. For $k \ge 1$, the parameter-independent term $\mathcal{B}_{k,0}$ in Γ_k is retained; it preserves every union of partition M-cubes, by Proposition 11.3 and its proof. For a projected polymer family a nonzero term has both original indices in X', so both projected indices lie in X. Equality of connections on bonds meeting X gives equality on bonds meeting every such X', preserving the polymer's local dependence.

Gauge covariance passes through pullback because the local gradients, restriction to unit sites and phases commute with it. Lattice covariance follows by lifting a symmetry (S, a) with one representative $a \in LM\mathbb{Z}^3$ to the same affine map on both tori; it permutes the image polymers. A change of representative multiplies source and target spinors by the same sign, which cancels in the operator relation. Transposition passes by (15.5); the constant C commutes with pullback. Complex conjugation and $\varkappa$ also commute with it. These facts prove the charge and reality relations for the projected primitives.

At zero, the photon and fermion diagonal factors are unchanged by projection. Cubes containing a given bond, and cubes meeting K_c, have unique corresponding lifts, so their parameter products are the same on both tori. The fermion factor is the same for all 48 labels at c. Their constants are bounded by $4\exp(2\gamma_w)M^{27\alpha_0} \le C_w^{(1)}$. The remaining zero-level properties are those just proved for the square root, chamber covariance and local resolvent. This completes the system verification with the common constant (15.8). □

15.3. **A common top-torus system.** On a chosen top torus use the photon families (10.36) and the local fermion families (11.14). In the fermion walk construction take the polymer decay rate $\kappa' = \kappa_w + \kappa_0$. The formulas do not depend on this rate; it specifies the parameter restrictions and the constants in their estimates. We write $\gamma^F = \gamma$ for the fermion decay rate in (9.29). Define

$$\alpha_0 = 1/54, \qquad \gamma_w = \min(\gamma^F, \gamma^P), \quad \gamma^P = \gamma_P/2,$$

$$C^P = 116 C_P^{\mathrm{ph}} M^{402\alpha_0}, \qquad C_w^{(1)} = \max\{C^F, C^P, 4\exp(2\gamma_w)M^{27\alpha_0}\}, \qquad C_w = 64 K_0 C_w^{(1)}. \tag{15.9}$$

Here C^F, γ^F are the constants of the fermion construction at the indicated polymer rate. The fixed M satisfies both its fermion condition and (10.37); the charge satisfies the inverse and walk conditions of those constructions.

These families form a system with $(\alpha_0, \gamma_w, \kappa_w + \kappa_0; C_w^{(1)})$. This follows by matching their proved properties. The photon kernel multipliers are polynomials in their parameter cubes; at full strength the partition of unity and the products give the exact kernels. Their path products confine rows and columns to their active link-components. The photon derivative and square-root bounds have constants C^P and rate γ^P. The fermion families are the convergent weakened walks and the localized block expressions, with their proved support, covariance, analyticity and Hölder estimates. Replacing their rates γ^F, γ^P by the smaller γ_w and their constants by the larger $C_w^{(1)}$ preserves every inequality. The row and column restriction identities contain no decay constant and are unchanged.

At level zero the two diagonal families are used. The photon derivative bound has constant $4\exp(2\gamma_w)M^{27\alpha_0}$, and the fermion diagonal factor is bounded by $M^{8\alpha_0}$, below that constant. There is no primitive short pair with $0 < d_X < 1$ at $\eta_0 = 1$. The covariance and local resolvent are chamber kernels with measure $1/48$, and the resolvent bound includes $(1+y)^{-1}$. A unit block has total chamber measure 1, since $(\eta_k^3/48)(48\eta_k^{-3}) = 1$. Consequently the positive-level block estimates and zero-level chamber estimates use precisely the common constants (15.9). Projecting this system by Proposition 15.2 gives a system with these primitive bounds and the same C_w on every odd smaller torus. The top system itself has these weaker bounds as well. None of these constants depends on the level, side, mass, charge or harmonic background.

15.4. **Common lifts and local algebras.** For the rest of the comparison assume

$$\ell_1 > (31L + 16)M. \tag{15.10}$$

Put $d_U = 2L + 1$, $c_l = 2L$. For a finite nonempty union $\widetilde{X}$ of covering M-cubes, let $\mathrm{box}(\widetilde{X})$ be its smallest rectangular union of cubes and let $\mathrm{diam}(\widetilde{X})$ be that box's largest side. Define

$$N(\widetilde{X}) = \mathrm{box}(\widetilde{X})^{\sim 6d_U}.$$

We call $\widetilde{X}$ admissible when $\operatorname{diam}(\widetilde{X}) \le \ell_1 - 14 d_U M$; a polymer is flat when it is the image of a connected admissible $\widetilde{X}$. These are the sets on which the two torus algebras will be identified.

Proposition 15.3. *For admissible $\widetilde{X}$, projection is injective on $N(\widetilde{X})$ and on every supported index contained there. An index of either torus supported in the image of a subset of $N(\widetilde{X})$ has exactly one lift contained in that subset. Pairs have their shortest lifts, with the same d_X and six paths. Local incidences of total support diameter below $2d_U M$ are preserved in both directions.*

There is a norm-preserving algebra isomorphism $\iota_{\widetilde{X}}$ between the two subalgebras generated by ordinary fields over $N(\widetilde{X})$, pair fields with both endpoints there, and their unit fluctuation and auxiliary fields there. It identifies covering generator indices. At zero it includes the chamber W fields and no auxiliary Ψ.

Proof. Every side of $N(\widetilde{X})$ is at most $\ell_1 - 2d_U M$. Since faces contain no fine lattice sites, any two of its sites satisfy

$$|u - u'|_\infty \le \ell_1 - 2d_U M - \eta_k.$$

For $w \neq 0$ this implies

$$|u - u' - \ell_i w|_\infty \ge 2d_U M + \eta_k. \tag{15.11}$$

It proves disjointness of $N(\widetilde{X})$ from its nonzero ℓ_1 translates, injectivity, and uniqueness of any representative with distance below $2d_U M$.

To lift an index supported in the projected subset, translate one of its lifts so that one support point lies in that subset. Every remaining support point is congruent to a point of the subset and is within $2LM < 2d_U M$ of the first. Equation (15.11) forces the congruence translation for each point to be zero. Thus the whole index lies in the subset; injectivity gives uniqueness. In particular the shortest pair has its endpoints there, distance at most M, and its same covering paths.

For an incidence relation, lift a tuple realizing the projected relation and translate its first index to the prescribed lift. Each other lifted index differs from the prescribed one by an ℓ_i period. Its support is at distance below $2d_U M$ from the first one. Equation (15.11) again forces the period to vanish. This proves the reverse incidence implication; the forward one is projection. The relations used below meet this range condition: nearest neighbors and chamber neighbors, short pairs and their paths, block and cube membership, common faces, linked parameter cubes, cube meeting, and coordinate support. The largest diameter is $4LM < 2d_U M$, for two linked photon parameter cubes. They are all invariant under $LM\mathbb{Z}^3$ translations.

No two covering generator indices in $N(\widetilde{X})$ have the same period class on either torus. Relative to any fundamental generator system their images are therefore independent generators, up to their antiperiodic signs. Identifying the same covering indices extends uniquely to a free Grassmann algebra isomorphism. It preserves bar indices and degrees and sends basis monomials to basis monomials, up to the same coefficient signs. All generator weights and index measures are common, so it preserves norms. The maps for nested admissible sets restrict to one another. □

A local activity on $X_i = \operatorname{pr}_i \widetilde{X}$ belongs to this identified algebra. Ordinary sites lift to $\widetilde{X}$; an admissible pair lifts to $\widetilde{X}^{\sim 1}$. For its first kind the two sites and paths lie in $\widetilde{X}$. For its second kind the pair is $(x, x^{(3)})$ with an endpoint over $\widetilde{X}$. Each assertion follows from the incidence comparison. Two local families F_1, F_2 agree on flat polymers when

$$\iota_{\widetilde{X}} F_1(X_1; A_1, \mathscr{C}_1) = F_2(X_2; A_2, \mathscr{C}_2) \tag{15.12}$$

whenever the two admissible connection pairs have equal lifts on $N(\widetilde{X})$. It suffices to check $A_2 = p^* A_1$, $\mathscr{C}_2 = p^* \mathscr{C}_1$: for other pairs with equal lifts, locality of F_2 identifies them with these

pullbacks on all bonds meeting $\widetilde{X}$. Thus once agreement is proved, equality holds already under equality on those read bonds.

Every small polymer is flat. The net diameter bound gives $\operatorname{diam}(\widetilde{X}) \le M(d_M(\widetilde{X}) + 3) \le M(L+3)$ for a small connected lift. By (15.10), $M(L+3) + 14 d_U M < \ell_1$. The nonwrapping lift proposition then identifies small torus polymers containing a chosen cube with small covering polymers containing its chosen lift. This identification is the same on both tori. Conversely, if an admissible connected lift projects to a small polymer, the small nonwrapping lift is its period translate: start at one common cube and propagate equality along adjacent cubes using the incidence result. Hence smallness, tree length and volume coincide for its two images.

Subpolymers and parameter cubes contained in an admissible $\widetilde{Y}$ correspond bijectively on the two tori. Common faces and links are preserved, so incompatibility is preserved. The definitions by cube meeting give

$$P_0(\operatorname{pr}_i \widetilde{X}) = \operatorname{pr}_i P_0(\widetilde{X}), \qquad \operatorname{pr}_i Y'(\widetilde{X}, \widetilde{P}) = Y'(\operatorname{pr}_i \widetilde{X}, \operatorname{pr}_i \widetilde{P}).$$

When $\widetilde{X}$ and all cubes of $\widetilde{P}$ are contained in $\widetilde{Y}$, the resulting Y' is contained there too, since the M-cubes partition sites. Reblocking satisfies $\overline{\widetilde{X}}^{L} \subset \widetilde{Y}^{\sim(L-1)}$ and commutes with projection. These larger sets still lie in $N(\widetilde{Y})$, so their incidences have the same unique lifts.

15.5. **Supported rows near a flat set.** The preceding identification applies to the weakened primitives because setting parameters to zero separates their components. We first record the component argument. Let s vanish at every parameter cube not contained in a union Z of M-cubes. If Z_1 is a face-connected component of Z and $\square(t)^{\sim 2L} \subset Z_1$, then

$$\begin{aligned} O(s,A)(t,t') &= 0 \quad \text{unless } t' \text{ is contained in } Z_1, \\ O(s,A)(t,\cdot) &= O(s\mathbf{1}_{Z_1}, A')(t,\cdot). \end{aligned} \tag{15.13}$$

Here O is a parameter primitive, and A' is any connection in the primitive's domain equal to A on bonds meeting Z_1; the photon primitives have no connection argument. The set $\square(t)$ is the union of the M-cubes meeting the support of t.

To prove (15.13), choose a parameter cube containing that support inside $\square(t)^{\sim 2L}$. For a chamber site or a unit site it is its M-cube. For a fine bond it is the parameter cube of radius M about the M-cube center of one end: both ends lie in it, and it is contained in $\square(t)^{\sim 1}$. For a photon coordinate index choose the parameter cube of radius LM about the LM-cube center of one point of its support. The support diameter is at most L and $M \ge 2$, so this cube contains the entire support. If z_c is the M-cube center of that point and z'' is the M-cube center of any site in the parameter cube, then

$$|z'' - z_c|_\infty < (3L/2 + 1)M \le (2L+1)M.$$

These centers differ by a multiple of M, hence the cube is contained in $\square(t)^{\sim 2L}$.

Let $\mathscr{Y}$ be all parameter cubes contained in Z, and let $\mathscr{Y}_1$ be their link component containing the cube just chosen. The M-cubes meeting $\bigcup \mathscr{Y}_1$ form a face-connected polymer contained in Z. It meets $\square(t) \subset Z_1$, and therefore is contained in the component Z_1. The component restriction property of the weakened primitives, proved in the photon and fermion constructions, now confines the row to $\bigcup \mathscr{Y}_1$. Replacing s by $s\mathbf{1}_{Z_1}$ leaves all its parameters there unchanged; replacing A by A' leaves all its read bonds unchanged. This proves (15.13).

Proposition 15.4. *Let $\widetilde{Y}$ be admissible, $Y_i = \operatorname{pr}_i \widetilde{Y}$, and $0 \le j \le 5d_U$. Let s_1 vanish outside Y_1. Define s_2 by giving the corresponding parameter cubes contained in $\widetilde{Y}$ the same values and setting all other entries to zero. Suppose $A_i \in \widetilde{\mathcal{G}}_k(T_i)$ have equal lifts on the bonds contained in*

$\widetilde{Y}^{\sim(j+d_U)}$. *For each parameter primitive and each covering row* t *contained in* $\widetilde{Y}^{\sim j}$, *the two rows, written with covering indices in* $N(\widetilde{Y})$, *coincide:*

$$O_1(s_1, A_1)(t, t') = O_2(s_2, A_2)(t, t'). \tag{15.14}$$

Both rows have columns contained in $Y_i^{\sim(j+c_l)}$, *and read the connection only on bonds contained in* $Y_i^{\sim(j+d_U)}$.

For a connected nonempty $\widetilde{X} \subset \widetilde{Y}$ *and connections with equal lifts on bonds meeting* $\widetilde{X}$, *the local polymer families also satisfy*

$$O_1(\mathrm{pr}_1 \widetilde{X}, A_1)(t, t') = O_2(\mathrm{pr}_2 \widetilde{X}, A_2)(t, t') \tag{15.15}$$

at rows and columns over $\widetilde{X}$. *Here* $O = M_k$ *for* $k \geq 1$, *or* $O = S_{k,y}$ *for any* k *and* $y \geq 0$.

Proof. Put $\widetilde{B} = \widetilde{Y}^{\sim(j+c_l)}$. Its one-cube enlargement is $\widetilde{Y}^{\sim(j+d_U)} \subset N(\widetilde{Y})$. For the row t, the set $\square(t)$ is one M-cube or two face-adjacent M-cubes. This follows directly for site and bond indices; a coordinate support lies in one L-block or is the support of one bond. Thus $\square(t)^{\sim c_l}$ is a connected box contained in $\widetilde{B}$. Let $\widetilde{B}_1$ be the component of $\widetilde{B}$ containing this box.

Its image is a component of $\mathrm{pr}_1 \widetilde{B}$. On T_2 its image is a component both of $\mathrm{pr}_2 \widetilde{B}$ and of $p^{-1}(\mathrm{pr}_1 \widetilde{B})$. Indeed a cube adjacent to its image has a lift adjacent to a cube of $\widetilde{B}_1$. This lift lies in $\widetilde{B}^{\sim 1} \subset N(\widetilde{Y})$. If it belongs to a period translate of $\widetilde{B}$, separation (15.11) forces that translate to be zero. It then lies in $\widetilde{B}$ and in the same component. This proves maximality of the images as components.

Apply (15.13) on T_1. It confines the row to $\mathrm{pr}_1 \widetilde{B}_1$ and lets the connection be changed off the bonds meeting that component. Such bonds lift into $\widetilde{B}^{\sim 1}$, proving the claimed read set. On the parameter cubes contained in $\mathrm{pr}_2 \widetilde{B}_1$, the families $p^* s_1$ and s_2 agree. To see this even at cubes not contained in $\widetilde{Y}$, observe that each such cube in $N(\widetilde{Y})$ has a unique lift; its image cannot also be the image of a cube in $\widetilde{Y}$. Both entries are therefore zero. The connections $p^* A_1$ and A_2 agree on bonds meeting $\mathrm{pr}_2 \widetilde{B}_1$. Applying (15.13) twice on T_2 gives

$$O_2(p^* s_1, p^* A_1)(t, \cdot) = O_2(s_2, A_2)(t, \cdot),$$

with columns confined to $\mathrm{pr}_2 \widetilde{B}_1$.

In the signed image formula (15.4), a nonzero summand at $t' + \ell_1 w$ would have a lift in $\widetilde{B}_1$. Both this lift and t' belong to $N(\widetilde{Y})$; their difference is an ℓ_1 period. Separation forces $w = 0$ in $(\mathbb{Z}/J_c\mathbb{Z})^3$. That summand has character 1, proving (15.14).

For the polymer families, every polymer X' on T_2 with $pX' = \mathrm{pr}_1 \widetilde{X}$ is $\mathrm{pr}_2(\widetilde{X} + v)$ for one $v \in \ell_1\mathbb{Z}^3/\ell_2\mathbb{Z}^3$. Distinct classes give disjoint cube sets. To verify the assertion for an arbitrary X', lift two adjacent cubes as cubes of $\widetilde{X} + v_1$ and $\widetilde{X} + v_2$. Their adjacency places a translate of the second cube in $\widetilde{X}^{\sim 1} \subset N(\widetilde{Y})$. Separation forces $v_1 = v_2$ modulo $\ell_2\mathbb{Z}^3$. Connectedness gives one class for every cube of X', and projection onto the whole image gives equality with that entire copy.

The projected kernel is consequently a double sum over v and the column image w. The local polymer support requires both its row and column to be over $\mathrm{pr}_2(\widetilde{X} + v)$. The common row lift forces $v = 0$; the common column lift then forces $w = 0$. The remaining term is $O_2(\mathrm{pr}_2 \widetilde{X}, p^* A_1)(t, t')$. Locality on the bonds meeting this polymer permits $p^* A_1$ to be replaced by A_2.

The barred mean field satisfies $\bar{H}_k(s, A) = \mathsf{C}^{-1} H_k(s, -A)\mathsf{C}$, so its comparison follows at the connections $-A_i$. At zero the mean-field kernels are $48\delta_{x,x'}\mathbf{1}_4$ times their parameter products; the coordinate and chamber measures are the same on the two tori, and the chamber label is preserved. There is no M_0 family. The proof applies to Γ_0 and $S_{0,y}$ on the full chamber indices. □

15.6. **Finite rows and the local operations.** For finite operators, equality of rows follows directly from their covering formulas. Suppose a row has all its columns in $N(\widetilde{Y})$, and its connection and polymer membership data agree on the bonds and sites which that row reads. The two formulas are then the same finite sum on either torus. There are no repeated column classes by Proposition 15.3. Writing the row as $\sum_{t'} r(t,t')f(t')$, its kernel is

$$O(t,t') = \nu(t')^{-1} r(t,t'). \tag{15.16}$$

Thus equality holds for the kernels as well, with the same column weight.

We need this observation at connections B_i having equal lifts on bonds meeting a connected $\widetilde{U} \subset \widetilde{Y}$. The photon coordinate map C reads no connection: a noncentral face bond is its own coordinate, and the central bond is the negative sum of the other face bonds and the interior straight-path contributions $n_c(b)/L$. All columns are supported in the one or two L-blocks of the row bond's ends. Hence its rows at bonds in $\widetilde{U}^{\sim(1+2d_U)}$ agree and have range at most $2L-1$.

The covariant difference row at an admissible pair is

$$d_X(x,x')^{-\alpha}\big(U_B(x,x')f(x') - f(x)\big).$$

Its only columns are x, x', and it reads exactly the six pair paths. These paths are contained in $\widetilde{U}$ for the first kind, or are the bond meeting $\widetilde{U}$ for the second kind. The range is M; inside a chamber cluster the transport is 1 and no bond is read. These rows therefore agree, also at $-B_i$.

For $k \geq 1$ the reconstruction rows $T^{(1)}, T^{(2)}$ and their barred versions are the two-piece formulas of the reconstruction estimate. At $y \in \widetilde{U}$ they have columns over Δ_y and at $x^{(3)}$, or at the pair $(x, x^{(3)})$ across its boundary. They read the bonds in Δ_y and the bonds (c, c_3) starting there. All meet $\widetilde{U}$. Their membership indicators are the same because

$$u \in \widetilde{U} + \ell_i \mathbb{Z}^3 \quad \Longleftrightarrow \quad u \in \widetilde{U} \qquad (u \in \widetilde{U}^{\sim 1}),$$

again by separation. Every site involved is at distance at most $(1+\eta_k)/2 < 1$ from y. This proves equality of both pieces, including the boundary term.

The finite rows of $Q_k J^*$ and $JQ_k^T(-B)$ have their columns and read bonds in a single Δ_y. The row of $B_{k,r}$, where $r \geq 0$ is the resolvent parameter, has its columns in one L-block and reads the paths from its center to its sites. These paths lie in that block. Their kernels are

$$\begin{aligned}
(Q_k J^*)(y,x) &= \mathbf{1}[c \in \Delta_y]\varphi_B(y,c)\mathbf{1}_4, \\
(JQ_k^T(-B))(x,y) &= \mathbf{1}[c \in \Delta_y]\varphi_B(y,c)^{-1}\mathbf{1}_4, \\
B_{k,r} &= (b_k + \mathrm{i}\gamma_5 r)^{-1} I \\
&\quad + \big((b_k + b_L + \mathrm{i}\gamma_5 r)^{-1} - (b_k + \mathrm{i}\gamma_5 r)^{-1}\big) P_B.
\end{aligned}$$

The projection P_B has kernel L^{-3} times the two block transports when its indices lie in the same L-block, and zero otherwise. The first two rows have range below 1, the third range at most $L-1$. At rows in $\widetilde{U}$ their columns and read bonds stay in its M-cube. They agree on both tori.

At zero, F_0 has only its diagonal, the four chamber neighbors, and its block-penalty columns. The hopping part of its kernel is

$$48\Big(\big(\mu + \tfrac{\rho}{2}\sum_{i=0}^{3} w_i\big)\delta_{x,x'}\mathbf{1}_4 - \sum_{j=0}^{3} \delta_{x',x^{(j)}} M_j(x) U_B(x, x^{(j)})\Big).$$

The factor 48 is the inverse column weight. Only the fourth neighbor reads a bond meeting $\widetilde{U}$; the penalty reads paths inside the L-block of c. All columns lie over $\widetilde{U}^{\sim 1}$ and its range is at most $L-1$. Thus this entire row agrees at zero.

These equalities also identify the mass form with the covering sum:

$$\langle\bar\psi,\psi\rangle_{\mathrm{pr}_i\,\widetilde{\square}} = (\eta_k^3/48)\sum_{c\in\widetilde{\square},g,\beta}\bar\psi_\beta(c,g)\psi_\beta(c,g).$$

At zero the two halves of the first-step vertex have the common coefficient $1/96$, the same two boundary transports, and the same generators over the cube and at its outward pairs. Hence the first vertex also agrees under $\iota_{\widetilde{Y}}$.

The remaining local operations can be compared in this common free algebra. Equality of elements gives equality of their constant coefficients and of their antisymmetric quadratic kernels: expand in ordered covering monomials and use the common measure $\eta_k^3/48$. For a small polymer, the adjustment $\Theta_{X,B}$ uses averages over shortest chains of M-cubes in X. Common faces, path membership and transports are identical on its lift, as are its volume and the coefficients d_X^α. Therefore

$$\iota_{\widetilde{X}}\Theta_{X_1,B_1} = \Theta_{X_2,B_2}\iota_{\widetilde{X}}. \tag{15.17}$$

If $\widetilde{U}$ is admissible at level $k+1$, then $L\widetilde{U}$ is admissible at level k. Its box diameter is $L\,\mathrm{diam}(\widetilde{U})$, and $N(L\widetilde{U})\subset LN(\widetilde{U})$. The scaling rules on the common generators are $\psi(x)\mapsto L^{-1}\psi(x/L)$ and $\chi(x,x')\mapsto L^{-1-\alpha}\chi(x/L,x'/L)$. They preserve the chamber label and the period signs. Thus

$$S_L\iota_{L\widetilde{U}} = \iota_{\widetilde{U}}S_L. \tag{15.18}$$

Finally, the integrations agree on any local element. For W, expand in monomials. A monomial moment is zero unless its modes are paired; otherwise it is its ordering sign times $\nu^{-\#\mathrm{pairs}}$. The mode indices and the weight ν agree. At zero this weight is $1/48$, so the factor is $48^{\#\mathrm{pairs}}$; other modes integrate to 1. This proves

$$\iota_{\widetilde{Y}}\int \mathrm{d}\mu_I(W)\,F = \int \mathrm{d}\mu_I(W)\,\iota_{\widetilde{Y}}F. \tag{15.19}$$

The restricted photon law is a product of the same normalized Gaussian on $[-p_{0,k},p_{0,k}]$ in each coordinate. Consequently its marginal on common coordinates is identical. With $d_V = 2-2L^{-3}$, its vacuum contribution on either torus is

$$\delta\varepsilon_k^0 = -d_V\log\left((2\pi)^{-1/2}\int_{-p_{0,k}}^{p_{0,k}}\exp(-z^2/2)\mathrm{d}z\right). \tag{15.20}$$

This number depends on e_k and L, and is independent of the side.

15.7. **Extraction and determinant comparison.** Suppose $\mathscr{E}_i\in K_k^x(T_i)$ agree in the sense of (15.12). For a small covering polymer $\widetilde{X}$, their values at $(0,0)$ have the same constant and quadratic coefficients. Their local extractions (12.10) are therefore the same: their integrated kernels and their common volume are the same. Fixing a covering cube identifies all small polymers containing its image on either torus. The sums (12.11) consequently give

$$m(\mathscr{E}_1) = m(\mathscr{E}_2),\qquad \varepsilon_{\mathrm{v}}(\mathscr{E}_1) = \varepsilon_{\mathrm{v}}(\mathscr{E}_2). \tag{15.21}$$

Subtracting the common constant and mass forms preserves local agreement of $R\mathscr{E}_i$. On a small polymer, its quadratic part also agrees; (15.17) proves agreement of $(R\mathscr{E}_i)^{\mathrm{nat}}$. On other polymers neither of these operations changes the relevant element.

The determinant terms agree on every flat polymer, not just on small ones. Let $X_i=\mathrm{pr}_i\,\widetilde{X}$ and suppose B_i have equal lifts on bonds meeting $\widetilde{X}$. For $k\ge 1$, set

$$O_1 = B_{k,r}\gamma_5 B_{k,r},\qquad O_2 = Q_kJ^*S_{k,r}(X_i,B_i)JQ_k^T(-B_i).$$

The trace in the determinant integral is

$$\mathrm{Tr}(O_1O_2) = \sum_{z_1,z_2\in X_i\cap\mathbb{Z}^3} \mathrm{tr}\,\big(O_1(z_2,z_1)O_2(z_1,z_2)\big). \tag{15.22}$$

Here and below Tr is the trace on the whole finite function space, while tr is the spin trace. Indeed

$$O_2(z_1,z_2) = \int_{\Delta_{z_1}} \mathrm{d}x \int_{\Delta_{z_2}} \mathrm{d}x'\, \varphi_{B_i}(z_1,c) S_{k,r}(X_i,B_i)(x,x') \varphi_{B_i}(z_2,c')^{-1}.$$

Local polymer support confines x, x' to X_i, and Δ_z lies in the M-cube of z. Thus both unit sites in this trace lie in X_i. The rows of $B_{k,r}$ stay in their own L-block, so the intermediate sum in $O_1(z_2,z_1)$ also stays in X_i. All sums may now be formed on the lift $\widetilde{X}$; the two antiperiodic kernel signs cancel in each trace summand. The finite-row comparison and (15.15) make every factor the same on both tori.

At zero the required traces are

$$\mathrm{Tr}(F_0(B_i)S_{0,r}(X_i,B_i)) = \int_{X_i} \mathrm{d}x \int_{X_i} \mathrm{d}x'\ \mathrm{tr}\,\big(F_0(B_i)(x,x')S_{0,r}(X_i,B_i)(x',x)\big), \tag{15.23}$$

$$\mathrm{Tr}(\gamma_5 S_{0,r}(X_i,B_i)) = \int_{X_i} \mathrm{d}x\ \mathrm{tr}\,\big(\gamma_5 S_{0,r}(X_i,B_i)(x,x)\big). \tag{15.24}$$

Support again restricts all arguments to X_i. Both are the same covering sums with measure $1/48$ and all 48 chamber labels. The local F_0 and $S_{0,r}$ kernels agree, so the traces agree for every $r \ge 0$.

These pointwise identities can be integrated in r. The required domination follows from the primitive estimates. Since $P_B^2 = P_B$,

$$B_{k,r}\gamma_5 B_{k,r} = \gamma_5\big((b_k + \mathrm{i}\gamma_5 r)^{-2}(I-P_B) + (b_k+b_L+\mathrm{i}\gamma_5 r)^{-2}P_B\big).$$

In the component matrix norm, $|(a+\mathrm{i}\gamma_5 r)^{-1}|_c \le \sqrt{2}(a^2+r^2)^{-1/2}$ for $a>0$. Hence the first trace is bounded by a finite B_i, X_i dependent constant times $(b_k^2+r^2)^{-1}$. The other factor is uniformly bounded by the local resolvent estimate and the two finite block transports. This majorant is integrable because $b_k > 0$. At zero, $|S_{0,r}(X_i,B_i)(x,x')|_c \le C_w(1+r)^{-1}$. The first trace in (15.23) is integrated against $\mathrm{d}r/r$ on $[1,\infty)$, and (15.24) against $\mathrm{d}r$ on $[0,1]$; both are integrable. The kernels are Borel in r, so these are ordinary absolutely convergent scalar integrals.

The two unsymmetrized determinant primitives agree. Applying the argument at $-B_i, \bar{B}_i, -\bar{B}_i$ identifies all four terms in its symmetrization. It gives

$$E_k^{d,(1)}(X_1,B_1) = E_k^{d,(2)}(X_2,B_2). \tag{15.25}$$

Subtracting the values at $\mathscr{C}_i$ or at zero proves the corresponding equality for $\delta E_k^{\mathrm{det}}$ and E_k^{det}.

15.8. **Comparison of the local fluctuation sums.** Let m be real with $|m| \le w_k^{\mathrm{m}}$, let $\|\mathscr{E}_i\|_k^x \le w_k^{\mathrm{E}}$, and assume $\mathscr{E}_i = 0$ at zero and local agreement as above. Fix $A_1 \in \widetilde{R}_k^-(T_1)$ and $\mathscr{C}_1 \in \mathfrak{X}_k(T_1)$, and put $(A_2,\mathscr{C}_2) = p^*(A_1,\mathscr{C}_1)$. All small-charge and primitive hypotheses of the nonlinear step are imposed on both tori. In particular the contour radii satisfy $1 < r_2 \le r_1$, $r_4 > 1$, and the domain inclusions of Proposition 14.1 hold. We compare the terms of that step on a connected nonempty $\widetilde{U} \subset \widetilde{Y}$.

Assign common coordinate values $z_\Upsilon \in [-p_{0,k}, p_{0,k}]$ to the photon basis elements contained in $\widetilde{U}$, and set all other coordinate values to zero on each torus. Assign common parameter values of modulus at most M^{α_0} to the parameter cubes contained in $\widetilde{U}$, and set all other parameters to zero. These data will be denoted by ζ_i, s_i. They belong to the respective box and polydisk because indices on the lift are injective.

The fine fluctuation fields have the same lifts on $\widetilde{U}^{\sim(1+d_U)}$:

$$Z_1(s_1')(b) = Z_2(s_2')(b). \tag{15.26}$$

To check the successive supports in their composition, write it in kernels,

$$Z_i(s_i')(b) = \sum_{b',\Upsilon,\Upsilon'} H_k^{\mathrm{ph},(i)}(s_i')(b,b')C_i(b',\Upsilon)C_k^{1/2,(i)}(s_i')(\Upsilon,\Upsilon')\zeta_i(\Upsilon').$$

At such a row b, Proposition 15.4 with $j = 1 + d_U$ confines b' to $\widetilde{U}^{\sim 2d_U}$ and identifies the row kernels. The coordinate row has columns in the M-cubes of the two ends of b', hence in $\widetilde{U}^{\sim(1+2d_U)}$; those finite rows agree. The square-root row, with $j = 1 + 2d_U$, confines Υ' to $\widetilde{U}^{\sim 3d_U}$. All these sets lie in $N(\widetilde{U})$, and the last coordinate values agree there. The two compositions are therefore the same finite covering sum. At zero the first row is diagonal and the sum over b' has one term.

Put $B_i = A_i + \mathscr{C}_i$ and $B_i(t) = B_i + tZ_i(s_i')$. For $|t| \le r_1$ these fields lie in $\widetilde{\mathcal{G}}_k(T_i)$, and $A_i + tZ_i(s_i') \in \widetilde{R}_k(T_i)$. They have equal lifts on the bonds just compared. The fermion mean-field rows at sites at distance at most η_k from $\widetilde{U}$ consequently agree, as do the Γ_k rows at unit sites in $\widetilde{U}$ and the $M_k(X_i, B_i(t))$ kernels for lifted subpolymers $\widetilde{X} \subset \widetilde{U}$. Here the parameters vanish outside U_i. The row-vanishing and component argument in the proof of Proposition 14.3 confines their columns to U_i itself: a nonzero mean-field row has its active parameter component attached to that set; the Γ_k row starting in this component has the same confined columns. For $k \ge 1$, the parameter-independent term $\mathcal{B}_{k,0}$ in Γ_k also preserves U_i, a union of partition M-cubes. Thus all these fields are polynomials in the common generators over $\widetilde{U}$.

For example, at a relevant chamber site,

$$(H_k^{(i)}(s_i'', B_i(t))\Psi)_\beta(x) = \sum_{y\in\widetilde{U}\cap\mathbb{Z}^3,\beta'} H_k^{(i)}(s_i'', B_i(t))(x,y)_{\beta\beta'}\Psi(y,\beta',0)$$

for $k \ge 1$, with the analogous barred sum. Their coefficients agree. At every pair admissible for $\widetilde{X}$, applying the common finite difference row preserves this equality. The differences between these fields at $t = 1$ and $t = 0$ are precisely the ordinary and pair components of $J_k^{\#}\Psi$ used in G_2, G_3. Both values are in the disk because $r_1 > 1$. Moreover locality of $R\mathscr{E}_i$ gives equality of its values at the pairs $(A_i + tZ_i, \mathscr{C}_i)$ on each subpolymer. These pairs need only agree on its read bonds, by the criterion following (15.12).

It follows that the three first-localization contour integrands agree under $\iota_{\widetilde{U}}$. For G_1, this is the common local M_k bilinear. For G_2, addition of the common odd field $tJ_k^{\#}\Psi$ to each generator commutes with an algebra homomorphism; apply it to the equal local activity. For G_3, the same substitution applies to the equal mass form, and $m' = m + m(\mathscr{E}_i)$ is common by (15.21). Their contour radii and scalar contour measures are the same. Integration thus gives

$$\iota_{\widetilde{U}} G_a^{(1)}(s_1, X_1) = G_a^{(2)}(s_2, X_2), \qquad a = 1, 2, 3. \tag{15.27}$$

At zero $G_2 = G_3 = 0$ and no auxiliary field occurs. G_1 is the first-step vertex; its two factors $\exp(\pm \mathrm{i}qZ_i(c, c_3)) - 1$ agree by (15.26), with its common coefficient $1/96$ and the common outward pair. This proves (15.27) at that level too.

The finite localization commutes with this equality. Indeed the pairs (X, P) with $Y'(X, P) = U_i$ correspond bijectively to lifted pairs $(\widetilde{X}, \widetilde{P})$ inside $\widetilde{U}$ having $Y'(\widetilde{X}, \widetilde{P}) = \widetilde{U}$. If a torus pair has that union, its polymer and all its parameter cubes are contained in U_i, so the unique lifts give the inverse correspondence. The identities for P_0, Y' proved above show that it preserves the parameter prescriptions. Its summand is therefore

$$\int_{[0,1]^{\widetilde{P}}} \mathrm{d}u \prod_{\mathrm{pc}\in\widetilde{P}} \partial_{u_{\mathrm{pc}}} G_i(s_i[u], \mathrm{pr}_i\, \widetilde{X}), \tag{15.28}$$

where $s_i[u]$ is 1 on $P_0(\widetilde{X})$, u on $\widetilde{P}$, and zero elsewhere. Identity (15.27) holds throughout the parameter polydisk. Since $M^{\alpha_0} > 1$, the integration cube is in its interior, and coefficient differentiation and integration preserve the equality. This proves equality of $\operatorname{Loc} G_a(U_i)$.

The reconstruction substitution replaces each $\Psi(y)$ by the same two-piece row $T^{(1)}\psi + T^{(2)}\chi$. Both rows, including their boundary membership indicators, have already been compared. Their resulting ordinary generators are over $\widetilde{U}$, and their boundary pairs are admissible for it. Equality on generators proves equality of the substitution homomorphisms. We obtain

$$\iota_{\widetilde{U}} E^{(1)}_{\le 3}(U_1) = E^{(2)}_{\le 3}(U_2). \tag{15.29}$$

At zero no reconstruction substitution is made.

For the second localization, the common fermion field is

$$\mathrm{Wc} = H_k(s'', B)\Gamma_k(s'', B)W, \qquad \overline{\mathrm{Wc}} = \bar{H}_k(s'', B)\bar{W}.$$

For $k \ge 1$ both are sums over the common unit sites of $\widetilde{U}$. For $k = 0$ they are

$$\mathrm{Wc}(x) = d(c)\frac{1}{48}\sum_{x' \text{ over } \widetilde{U}} \Gamma_0(s'', B)(x, x')W(x'), \qquad \overline{\mathrm{Wc}}(x) = d(c)\bar{W}(x),$$

where the common diagonal factor $d(c)$ vanishes outside U_i. These formulas preserve all chamber labels and the weight $1/48$. The kernel equalities proved above, followed by the common pair difference rows, identify $W_k^{\#}$ at every index used by a subpolymer.

The first-localized activity on a subpolymer reads only photon coordinates supported there. Hence (15.29), proved with coordinates supported on that subpolymer, still holds when coordinates are supported on the larger $\widetilde{U}$. Adding the common $W_k^{\#}$ to its generators proves equality of $G^W_{\le 3}$. The same addition to $R\mathscr{E}_i$ and to the mass form proves equality of the contour integrands of G_4, G_5. The second localization uses only π_k; the correspondence and derivative argument (15.28) apply with this one kind. Consequently

$$\iota_{\widetilde{U}} F_1(U_1) = F_2(U_2). \tag{15.30}$$

We now pass to the local connected logarithm. Its finite Mayer activity is

$$K_i(X_i) = \sum_{\{\widetilde{X}_a\}:\bigcup_a \widetilde{X}_a = \widetilde{X}} \prod_a \big(\exp(F_i(\mathrm{pr}_i\, \widetilde{X}_a)) - 1\big),$$

where the $\widetilde{X}_a$ are distinct connected subpolymers of $\widetilde{X}$. The products are even. Their index sets and factors are the same on both tori by (15.30); coordinate locality allows their evaluation with common coordinates on $\widetilde{X}$. For the integrated activity $H_i(X_i)$, (15.19) identifies the Wick integrals. The other photon coordinates integrate to 1, leaving the same product of restricted normal laws over the coordinates supported in $\widetilde{X}$. Hence $\iota_{\widetilde{X}} H_1(X_1) = H_2(X_2)$.

The logarithm has the expansion

$$E_k^{\#,(i)}(X_i) = \sum_{n \ge 1} \frac{1}{n!} \sum_{\bigcup_{a=1}^n \widetilde{X}_a = \widetilde{X}} u^T(\mathrm{pr}_i\, \widetilde{X}_1, \dots, \mathrm{pr}_i\, \widetilde{X}_n) \prod_{a=1}^n H_i(\mathrm{pr}_i\, \widetilde{X}_a).$$

Projection preserves incompatibility inside this lift, so the graph determining u^T is the same. Every summand agrees under the common algebra map. Both series converge absolutely in the weighted norm by Proposition 14.5. The map $\iota_{\widetilde{X}}$ preserves that norm, so equality passes to their sums:

$$\iota_{\widetilde{X}} E_k^{\#,(1)}(X_1; A_1, \mathscr{C}_1) = E_k^{\#,(2)}(X_2; A_2, \mathscr{C}_2). \tag{15.31}$$

15.9. **Agreement of the step after reblocking.**

Theorem 15.5. *Suppose the two sides at levels k and $k+1$ are admissible and satisfy $\ell_1/L > (31L+16)M$. Use the projected and top systems constructed above, with the hypotheses of Theorem 14.8 on both tori. Let the real mass m be common and let the incoming activities agree on flat polymers, with $\mathscr{E}_i = 0$ at zero. Then their extracted mass and vacuum contributions are equal. Their outgoing activities agree on flat polymers at level $k+1$. The external scalar recursion* (14.45) *preserves agreement of scalar families on such polymers. All these assertions hold for the activities without an external slot as well.*

Proof. The scalar conclusions are (15.21) and (15.20). For the outgoing activity it suffices to compare a level $k+1$ connection pair with its pullback. Let $\widetilde{U}$ be connected and admissible there, put $Y_i = \mathrm{pr}_i' \widetilde{U}$, and set $\widetilde{Y} = L\widetilde{U}$. The latter is a connected union of LM-cubes, hence also of M-cubes, admissible at level k. Its neighborhood is contained in $LN(\widetilde{U})$.

If $(\mathscr{A}', \mathscr{C}')$ satisfies the outgoing-domain hypotheses at level $k+1$, its scaled pair lies in $\widetilde{R}_k^- \times \mathfrak{X}_k$ by (12.44). Scaling and pullback commute on covering bonds: both give $L^{-1/2}B'(b/L)$. Denote the scaled pairs by $(A_i, \mathscr{C}_i)$. They satisfy exactly the hypotheses used in (15.31).

The polymers with $\bar{X}^L = LY_i$ are precisely $\mathrm{pr}_i \widetilde{X}$, where $\widetilde{X}$ is connected in $\widetilde{Y}$ and its LM-closure is $\widetilde{Y}$. Indeed $X \subset \bar{X}^L$ gives its unique lift inside $\widetilde{Y}$. Closure commutes with projection, and lies in $\widetilde{Y}^{\sim(L-1)}$. Projection is injective on that enlargement, so equality of its projected closure with $\mathrm{pr}_i \widetilde{Y}$ forces equality of the lifted closure with $\widetilde{Y}$. This also proves the converse and shows that the reblocking index set is common.

In that index set the outgoing activity is

$$\mathscr{E}_i^+(Y_i; \mathscr{A}_i', \mathscr{C}_i') = \sum_{\substack{\widetilde{X} \subset \widetilde{Y} \text{ connected} \\ \bar{\widetilde{X}}^L = \widetilde{Y}}} S_L \Big[(R\mathscr{E}_i)^{\mathrm{nat}}(\mathrm{pr}_i \widetilde{X}; A_i, \mathscr{C}_i) + \delta E_k^{\mathrm{det},(i)}(\mathrm{pr}_i \widetilde{X}; A_i, \mathscr{C}_i) + E_k^{\#,(i)}(\mathrm{pr}_i \widetilde{X}; A_i, \mathscr{C}_i) \Big]. \tag{15.32}$$

Each of the three entries agrees under $\iota_{\widetilde{Y}}$, by extraction and adjustment, (15.25), and (15.31), respectively. The entries belong to its ordinary and pair subalgebra. Equation (15.18) identifies their scaled values, so the sums agree under $\iota_{\widetilde{U}}$. Locality of the outgoing family, already proved in the step theorem, then extends the comparison from pullback pairs to pairs with equal local lifts.

The external scalar recursion has the same reblocking index set. Its incoming scalar term agrees by hypothesis, and its local determinant term agrees by (15.25) at $(\mathscr{C}_1, \mathscr{C}_2)$ and at $(0,0)$. Its two sums therefore agree.

The argument without an external slot uses precisely these local operations with $B_i = A_i$. The determinant difference is then $E_k^{\mathrm{det}}(X_i, A_i) = E_k^d(X_i, A_i) - E_k^d(X_i, 0)$. Extraction, addition of odd fields, finite localization and integration do not require that slot. Their same domain and convergence assertions were established for this case in the nonlinear-step construction. Thus the three bracketed terms in (15.32) agree in that case too, and scaling gives the conclusion. Restriction of a family with a slot to $\mathscr{C} = 0$ also commutes with every operation in the displayed formula.

At $k = 0$ the incoming activity and its extracted terms are zero. The comparison is for the local determinant difference and the first correction, with chamber generators, traces and Wick moments of weight $1/48$, as explicitly included above. □

16. The tuned flow

The mass recursion expands by a factor L, while the normalized activity recursion contracts. We determine the masses backwards from a zero terminal mass and the activities forwards from zero initial activity. This gives a contraction on the whole finite sequence. The vacuum is then fixed by a scalar sum. Comparing local steps transfers the pair chosen on the top torus to every smaller torus used in the construction.

16.1. **Physical scales and stopping levels.** Fix $\bar m \in (0,1]$ and put $h = L^{\lfloor \log_L(1/\bar m)\rfloor}$. For a cutoff $N \ge 0$, define

$$\eta = L^{-N}, \qquad K = N + \log_L h, \qquad a_j = L^{j-N}, \qquad e_j = e\sqrt{a_j}, \qquad \mu_j = \bar m a_j. \tag{16.1}$$

Thus a_j is the physical length of one unit at level j, and $a_K = h$. At zero the charge is written $q = e\sqrt{\eta}$; the fixed charge ceilings e_0, e_1 retain their earlier meanings. A formula with variable index j uses e_j for the running charge, including q when $j = 0$. For $0 \le j \le K$ set

$$v_j = -\log e_j, \qquad r_j = \min\{L^i : i \in \mathbb{N}\cup\{0\},\ L^i \ge v_j^r\}, \qquad p_j = v_j^p, \quad p_{0,j} = v_j^{p_0}.$$

The integers r, p, p_0 and the constants of the preceding estimates are fixed. Here $r \ge 2$, L is odd with $L \ge 13$, and M is a power of L with $M \ge L^2$. We assume the hypotheses of the primitive and step estimates, and use a ceiling $e_1 \le \min(e_0, 1/3)$ such that, for all $v \ge -\log e_1$,

$$\begin{gathered} v \ge 8, \qquad \left(1 + \frac{3\log L}{2v}\right)^r < L, \qquad v^r > 31L + 16, \\ (c_{\mathrm{fl}} + 2)\exp(-\varepsilon v) \le \frac{1}{16}(1 - L^{-\varepsilon/2}). \end{gathered} \tag{16.2}$$

In addition $0 < \varepsilon \le 1/200$, $c_{\mathrm{fl}} \ge 1$, $e_0^{\varepsilon} \le 1/4$, $c_{\mathrm{fl}} L^{-3/8+6/200} \le 1/16$, and $L^{-5/8} \le 1/4$. The construction below applies whenever $0 < e\sqrt{h} \le e_1$.

The physical top side and reference side are

$$R_{\mathrm{top}} = hMr_K, \qquad R^* = R_{\mathrm{top}}/L = hL^{T^*(e)}, \qquad T^*(e) = \log_L(Mr_K) - 1. \tag{16.3}$$

They are independent of N, since $v_K = -\log(e\sqrt{h})$. The admissible physical sides for the flow are

$$\mathcal{T}(e) = \{hL^T : T \in \mathbb{N}\cup\{0\},\ hL^T \le R_{\mathrm{top}}\}.$$

For $R \in \mathcal{T}(e)$ put $\ell_j(R) = R/a_j$, $T_j(R) = \mathbb{L}_j/\ell_j(R)\mathbb{Z}^3$ and $\mathfrak{n}_j = \log_L(\ell_j/(Mr_j))$. The allowed unit sides are $\mathrm{Cl} = \{\ell : \ell/(LM) \text{ is an odd integer at least } 13\}$. Their occurrence at every required scale is part of the following arithmetic argument.

Proposition 16.1. *For each $R \in \mathcal{T}(e)$ there is a finite integer $N_{\min}(R,e) \ge 0$ such that, when $N \ge N_{\min}(R,e)$, the first level with $\mathfrak{n}_j = 0$ exists and satisfies $1 \le k_*(R) \le K$. Its physical quantities $a_*, e_*, \mu_*, \ell_*, r_*$ are independent of N in this range. For $j \le k_*(R)$,*

$$\begin{gathered} \ell_j \in \mathrm{Cl}, \qquad \ell_j \ge Mr_j, \qquad r_j \ge r_* > 31L + 16, \\ v_j \ge v_* \ge v_K, \qquad 0 < e_j \le e_* \le e_K \le e_1, \qquad 0 < \mu_j \le \mu_* \le \bar m h \le 1. \end{gathered} \tag{16.4}$$

For $j < k_*(R)$ one has $\ell_j \ge LMr_j$, $\ell_{j+1} = \ell_j/L \in \mathrm{Cl}$, $e_{j+1} = \sqrt{L}e_j \le e_0$ and $0 < \mu_j = \mu_{j+1}/L \le 1/L$. At the stop,

$$\ell_* = Mr_*, \qquad Mv_*^r \le \ell_* < LMv_*^r, \qquad \mu_*^{-1} < L\ell_*, \qquad e_*^2\ell_* = e^2 R. \tag{16.5}$$

If $R \le R'$ in $\mathcal{T}(e)$, then $N_{\min}(R',e) \le N_{\min}(R,e)$ and $k_*(R) \le k_*(R')$ for a common cutoff above both stated thresholds.

Proof. Use the depth $s = K - j$. Its quantities are

$$\widehat{v}_s = v_K + \tfrac{s}{2}\log L, \qquad \widehat{r}_s = \min\{L^i : i \geq 0,\ L^i \geq \widehat{v}_s^r\}, \qquad \widehat{n}_s(R) = T + s - \log_L(M\widehat{r}_s).$$

They do not depend on N. At depth $s = K - j$ they equal $v_j, r_j, \mathfrak{n}_j$, while $a_j = hL^{-s}$ and $\ell_j = L^{T+s}$. Since $\widehat{v}_s \geq v_K$, the scalar condition gives

$$\widehat{v}_{s+1}^r = \widehat{v}_s^r \left(1 + \frac{\log L}{2\widehat{v}_s}\right)^r \leq L\widehat{v}_s^r \leq L\widehat{r}_s.$$

Thus $\widehat{r}_{s+1} \in \{\widehat{r}_s, L\widehat{r}_s\}$ and $\widehat{n}_{s+1} - \widehat{n}_s \in \{0, 1\}$. Also $\widehat{n}_0 \leq 0$ because $R \leq R_{\mathrm{top}}$. Minimality of the rounded power gives $\widehat{r}_s < L\widehat{v}_s^r$ and hence

$$\widehat{n}_s > T + s - 1 - \log_L M - r\log_L(v_K + \tfrac{s}{2}\log L) \longrightarrow \infty.$$

The integer sequence therefore has a largest zero. Denote its index by $s_*(R, e)$. It is nonpositive up to that index and at least 1 afterwards.

The condition $\ell_0 \geq LMr_0$ is $\widehat{n}_K \geq 1$, equivalently $K > s_*$. Its exact threshold is

$$N_{\min}(R, e) = \max\{0, s_*(R, e) + 1 - \log_L h\}, \qquad k_*(R) = K - s_*(R, e). \tag{16.6}$$

This gives $1 \leq k_* \leq K$ and $a_* = hL^{-s_*}$, proving cutoff independence of the terminal quantities. Increasing T increases every $\widehat{n}_s$ by the same integer; thus it decreases s_*, decreases $N_{\min}$ and increases k_*.

For $j \leq k_*$, the depth satisfies $K - j \geq s_*$. Monotonicity gives the asserted ranges of $\mathfrak{n}_j, r_j, v_j, e_j$, and the running mass increases with j. Moreover $\bar{m}h = L^{\lfloor \log_L(1/\bar{m})\rfloor - \log_L(1/\bar{m})} \in (1/L, 1]$. Since $\mu_*\ell_* = \bar{m}R \geq \bar{m}h > 1/L$, we obtain $\mu_*^{-1} < L\ell_*$. The rounded-power inequality at v_* gives the two bounds for ℓ_* in (16.5); its last identity uses $e_*^2 = e^2 a_*$.

Finally $r_* \geq v_*^r > 31L + 16 > L$ implies $r_* \geq L^2$, as it is a power of L. At every required level, $\ell_j/(LM) = r_j L^{\mathfrak{n}_j - 1}$ is a positive power of the odd integer $L \geq 13$. This proves $\ell_j \in \mathrm{Cl}$. Before the stop $\mathfrak{n}_j \geq 1$, giving the stronger side bound. The mass and charge bounds for the step follow at $j + 1$. □

On R^* define the reference mesh threshold

$$N_{\mathrm{ref}}(e) = N_{\min}(R^*(e), e). \tag{16.7}$$

The stop on this torus has depth $s_* \in \{1, 2\}$. Indeed, for $s = 1, 2, 3$ the stronger inequality in (16.2) gives $\widehat{r}_s \in \{r_K, Lr_K\}$. On R^*, $\widehat{n}_s = s - 1 - \log_L(\widehat{r}_s/r_K)$. Its values at $s = 0, 1, 2, 3$ are respectively -1, an element of $\{-1, 0\}$, an element of $\{0, 1\}$, and an element of $\{1, 2\}$. The zero-step-or-one-step rule forces the last zero to be 1 or 2. On R_{top} the sequence is larger by one, so its last zero is 0 or 1. Consequently

$$\begin{gathered} k_*(R^*) \in \{K - 1, K - 2\}, \qquad a_* \in \{h/L, h/L^2\}, \qquad r_* \in \{r_K, Lr_K\}, \\ k_{\mathrm{top}} \in \{K, K - 1\}, \qquad L^{-3} < \mu_* \leq L^{-1} \quad \text{on } R^*. \end{gathered} \tag{16.8}$$

All these physical terminal values are independent of $N \geq N_{\mathrm{ref}}(e)$.

The scale proposition verifies every side, mass and charge hypothesis of the step theorem on R_{top} and on the smaller tori it covers with odd degree. At each level use the top system of Section 15.3; on a smaller torus use its projection from Proposition 15.2. These are the actual mean-field, covariance, local resolvent and photon families constructed earlier, with the same common estimates. The ratio R_{top}/R is a power of the odd integer L, and $k_*(R) \leq k_{\mathrm{top}}$. Thus the required top families exist whenever a smaller-volume step is used. Furthermore $\ell_{j+1}(R) \geq Mr_{j+1} > (31L + 16)M$; the next-side hypothesis of the cover comparison holds at every such step.

16.2. **Continuation within the mass range.** Write the mass and activity bounds as

$$w_j^{\mathrm{m}} = e_j^{3/4-8\varepsilon}, \qquad w_j^{\mathrm{E}} = e_j^{1/4-7\varepsilon}.$$

For a real pair $(m_{\mathrm{in}}, \varepsilon_{\mathrm{in}}^{\mathrm{v}})$, the initial vacuum is $\varepsilon_{\mathrm{in}}^{\mathrm{v}}$, the initial member of the mass sequence is m_{in}, and $E_0 = \mathscr{E}_0 = \mathfrak{P}_0 = 0$. The variable-index notation m_j includes this initial member. The recursion is

$$m_{j+1} = L(m_j + m(E_j)), \qquad \varepsilon_{j+1}^{\mathrm{v}} = L^3(\varepsilon_j^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(E_j) + \delta\varepsilon_j^0), \tag{16.9}$$

$$E_{j+1} = \mathcal{L}(RE_j)^{\mathrm{nat}} + \mathcal{L}E_j^{\mathrm{det}} + E_j^*(m_j, E_j), \tag{16.10}$$

$$\mathscr{E}_{j+1} = \mathcal{L}(R\mathscr{E}_j)^{\mathrm{nat}} + \mathcal{L}\delta E_j^{\mathrm{det}} + E_j^*(m_j, \mathscr{E}_j). \tag{16.11}$$

The scalar family $\mathfrak{P}_j$ follows (14.45). In the last line the correction has the external slot; in (16.10) it has none. These are the corresponding maps proved earlier. The trajectory exists up to a level if every preceding $(m_j, \mathscr{E}_j)$ lies in $\mathrm{Ball}_j^x(4)$. It is controlled when it exists to $k_*(R)$ and $|m_j| \le w_j^{\mathrm{m}}$ throughout, including the stop.

The normalized estimates used below are

$$\begin{gathered} L^{-1}\frac{w_{j+1}^{\mathrm{m}}}{w_j^{\mathrm{m}}} = L^{-5/8-4\varepsilon} \le \tfrac{1}{4}, \qquad c_{\mathrm{fl}} e_j^{1/2}\frac{w_j^{\mathrm{E}}}{w_j^{\mathrm{m}}} = c_{\mathrm{fl}} e_j^{\varepsilon}, \\ c_{\mathrm{fl}} L^{-1/4+2\varepsilon}\frac{w_j^{\mathrm{E}}}{w_{j+1}^{\mathrm{E}}} = c_{\mathrm{fl}} L^{-3/8+11\varepsilon/2} \le \tfrac{1}{16}, \\ \frac{e_{j+1}^{1/4-2\varepsilon}}{w_{j+1}^{\mathrm{E}}} \le e_{j+1}^{\varepsilon}, \qquad \frac{e_j^{1/4-6\varepsilon}}{w_{j+1}^{\mathrm{E}}} = L^{-1/8+3\varepsilon} e_{j+1}^{\varepsilon} \le e_{j+1}^{\varepsilon}. \end{gathered} \tag{16.12}$$

Their first and third bounds use the fixed inequalities stated above; their last two use $e_{j+1} < 1$ and $\varepsilon \le 1/200$. The charge condition also gives

$$\sum_{j=0}^{k_*} (c_{\mathrm{fl}} + 2) e_j^{\varepsilon} \le (c_{\mathrm{fl}} + 2) e_*^{\varepsilon} (1 - L^{-\varepsilon/2})^{-1} \le \tfrac{1}{16}. \tag{16.13}$$

In particular $c_{\mathrm{fl}} e_j^{\varepsilon} \le 1/16$, $2e_j^{\varepsilon} \le 1/16$, and $e_j \le e_j^{\varepsilon} \le 1/48$.

Proposition 16.2. *If the recursively produced masses satisfy $|m_i| \le w_i^{\mathrm{m}}$ for $i < j \le k_*(R)$, then the trajectory is defined to j and*

$$\|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le \frac{32}{15} e_j^{\varepsilon} w_j^{\mathrm{E}} \le 4 e_j^{1/4-6\varepsilon} \le w_j^{\mathrm{E}}. \tag{16.14}$$

The data satisfy $E_j = \mathscr{E}_j|_{\mathscr{C}=0}$, $\mathfrak{P}_j(X, 0) = 0$, and belong to their real mass, vacuum and activity spaces.

Proof. At zero all activities and scalar families vanish. Assume the assertion through $i < j$. The mass condition and $4e_i^{\varepsilon} \le 1$ put the input in the unit ball, so Theorem 14.8 applies at i. The scaled connection pair lies in $\widetilde{R}_i^- \times \mathfrak{X}_i$, so every local determinant and correction in the recursion is defined. The scalar external sum is finite and uses $\mathscr{C}_L \in \mathfrak{X}_i$. The step theorem preserves the real spaces and commutes with restriction to $\mathscr{C} = 0$. Consequently the coupling maps computed from E_i or $\mathscr{E}_i$ coincide. The scalar recursion vanishes at zero because each $E_i^{\mathrm{det}}(X, 0)$ vanishes.

The activity bound of the current step, divided by w_{i+1}^{E} and combined with (16.12), gives

$$\frac{\|\mathscr{E}_{i+1}\|_{i+1}^x}{w_{i+1}^{\mathrm{E}}} \le \frac{1}{16}\frac{\|\mathscr{E}_i\|_i^x}{w_i^{\mathrm{E}}} + 2e_{i+1}^{\varepsilon}.$$

Starting from zero, the bound $(32/15)e_i^{\varepsilon}$ is preserved since $e_i \le e_{i+1}$ and $(32/15)/16 + 2 = 32/15$. Restriction does not increase the activity norm. This proves (16.14) and the induction.

At the first step the formulas use $192\ell_0^3$ incoming pairs with weight $1/48$ and $4\ell_1^3$ outgoing pairs, as proved in the exact density identity. Its coupling maps are $m_1 = Lm_{\mathrm{in}}$ and $\varepsilon_1^{\mathrm{v}} = L^3(\varepsilon_{\mathrm{in}}^{\mathrm{v}} + \delta\varepsilon_0^0)$; its activity is the local determinant term plus the first correction. The normalized estimate has zero incoming-activity term and the same two source bounds. Thus the induction includes that step. □

16.3. **The terminal conditions and the fixed point.**

Theorem 16.3. *On any $R \in \mathcal{T}(e)$ with $N \ge N_{\min}(R, e)$, there is exactly one real controlled pair whose trajectory has $m_{k_*} = 0$ and $\varepsilon_{k_*}^{\mathrm{v}} = 0$. Along its trajectory,*

$$|m_j| \le \tfrac{1}{8} w_j^{\mathrm{m}}, \qquad \|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le 4e_j^{1/4-6\varepsilon}, \qquad |\varepsilon_j^{\mathrm{v}}| \le 5c_{\mathfrak{fl}} e_j^{1/4-6\varepsilon}. \tag{16.15}$$

The vacuum at every level is

$$\varepsilon_j^{\mathrm{v}} = -\sum_{i=j}^{k_*-1} L^{-3(i-j)} \big(\varepsilon_{\mathrm{v}}(E_i) + \delta\varepsilon_i^0\big). \tag{16.16}$$

Proof. Take the finite product of the real mass coordinates and real activity Banach spaces, with norm

$$\|\mathbf{z}\|_{\mathfrak{fl}} = \max_{0\le j\le k_*} \max\left(\frac{|m_j|}{w_j^{\mathrm{m}}}, \frac{\|E_j\|_j}{w_j^{\mathrm{E}}}\right), \qquad \mathbf{z} = (m_j, E_j)_{0\le j\le k_*}.$$

The closed subset $\mathcal{B}$ defined by these quotients being at most 1 and $E_0 = 0$ is nonempty and complete. Define a self-map by

$$\widehat{m}_j = L^{-1} m_{j+1} - m(E_j) \quad (j < k_*), \qquad \widehat{m}_{k_*} = 0,$$

$$\widehat{E}_0 = 0, \qquad \widehat{E}_{j+1} = \mathcal{L}(RE_j)^{\mathrm{nat}} + \mathcal{L}E_j^{\mathrm{det}} + E_j^*(m_j, E_j). \tag{16.17}$$

All maps here are the actual maps without an external slot; their inputs belong to the unit ball. Extraction, adjustment followed by reblocking, the determinant activity and the nonlinear correction are defined on that ball and preserve the real space.

We verify that the map preserves $\mathcal{B}$. The extraction estimate implies $|m(E_j)| \le c_{\mathfrak{fl}} e_j^{1/2} \|E_j\|_j$. Equations (16.12) and (16.13) give

$$\frac{|\widehat{m}_j|}{w_j^{\mathrm{m}}} \le \tfrac{1}{4} + c_{\mathfrak{fl}} e_j^{\varepsilon} \le \tfrac{5}{16}.$$

For the activity, the contraction, determinant and correction estimates give

$$\frac{\|\widehat{E}_{j+1}\|_{j+1}}{w_{j+1}^{\mathrm{E}}} \le \tfrac{1}{16} + 2e_{j+1}^{\varepsilon} \le \tfrac{1}{8}.$$

The fixed end components are zero. Hence the map takes $\mathcal{B}$ into itself.

For two sequences, let $d_j^{\mathrm{m}}, d_j^{\mathrm{E}}$ be the corresponding normalized differences. Linearity of extraction gives

$$\widehat{d}_j^{\mathrm{m}} \le \tfrac{1}{4} d_{j+1}^{\mathrm{m}} + c_{\mathfrak{fl}} e_j^{\varepsilon} d_j^{\mathrm{E}} \le \tfrac{5}{16} \|\mathbf{z} - \widetilde{\mathbf{z}}\|_{\mathfrak{fl}}.$$

Adjustment followed by reblocking is linear in the activity at fixed connection. The local determinant term is independent of the sequence. The correction's coupling estimate (14.41) therefore gives

$$\widehat{d}_{j+1}^{\mathrm{E}} \le \tfrac{1}{16} d_j^{\mathrm{E}} + e_{j+1}^{\varepsilon} (d_j^{\mathrm{m}} + d_j^{\mathrm{E}}) \le \tfrac{1}{8} \|\mathbf{z} - \widetilde{\mathbf{z}}\|_{\mathfrak{fl}}.$$

The whole map contracts by $5/16$. Banach's fixed point theorem [9, Theorem 8] gives a unique fixed point in $\mathcal{B}$. Its equations are precisely the mass and activity recursions with $m_{k_*} = 0$.

Let m_{in} be its initial mass coordinate and choose

$$\varepsilon^{\mathrm{v}}_{\mathrm{in}} = -\sum_{i=0}^{k_*-1} L^{-3i}\big(\varepsilon_{\mathrm{v}}(E_i) + \delta\varepsilon^0_i\big).$$

This pair is real. Its full trajectory, including the external slot, agrees with the fixed-point mass and activity at every level: agreement at zero is the initial condition; at the next level Proposition 16.2 makes the full step defined, and the fixed-point equations identify its mass and restricted activity. The induction reaches k_* and proves control. Solving the scalar vacuum recursion backwards gives

$$\varepsilon^{\mathrm{v}}_j = L^{-3(k_*-j)}\varepsilon^{\mathrm{v}}_{k_*} - \sum_{i=j}^{k_*-1} L^{-3(i-j)}\big(\varepsilon_{\mathrm{v}}(E_i) + \delta\varepsilon^0_i\big).$$

At $j = 0$ its comparison with the chosen input forces $\varepsilon^{\mathrm{v}}_{k_*} = 0$, giving (16.16). Conversely any controlled pair with the two terminal zeros has its mass/activity sequence in $\mathcal{B}$ by the continuation proposition. It is the same fixed point. Its initial vacuum is then the same scalar sum. This proves uniqueness of the pair.

For the sharper mass bound put $B_j = |m_j|/w^{\mathrm{m}}_j$. The backward mass equation and the unit activity bound imply $B_j \le B_{j+1}/4 + 1/16$, while $B_{k_*} = 0$. Summing gives $B_j \le (1/16)\sum_{s\ge 0} 4^{-s} = 1/12 \le 1/8$. The activity bound in (16.15) is the continuation estimate.

It remains to bound the vacuum sum. Extraction gives $|\varepsilon_{\mathrm{v}}(E_i)| \le 4c_{\mathrm{fl}} e_i^{1/4-6\varepsilon}$. For $i \ge j$ the associated summand obeys

$$L^{-3(i-j)} e_i^{1/4-6\varepsilon} = e_j^{1/4-6\varepsilon} L^{-(i-j)(23/8+3\varepsilon)} \le e_j^{1/4-6\varepsilon} 16^{-(i-j)}.$$

Thus its sum is at most $(64/15)c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}$. The box normalization satisfies $0 \le \delta\varepsilon^0_i \le e_i^7$ by the step estimate. Its remaining sum is bounded by

$$\sum_{i=j}^{k_*-1} L^{-3(i-j)} e_i^7 = e_j^6 \sum_{i=j}^{k_*-1} e_i \le e_j^6 e_* (\sqrt{L}-1)^{-1} \le \tfrac{1}{48} e_j^{1/4-6\varepsilon}.$$

Here $\sqrt{L} \ge 2$ and $e_* \le 1/48$. Since $c_{\mathrm{fl}} \ge 1$, $64/15 + 1/48 = 343/80 < 5$, giving the claimed bound. The sums are empty at the stop.

At zero $E_0 = 0$. The backward mass equation reads $m_{\mathrm{in}} = L^{-1} m_1$, and the forward activity is $\mathcal{L}E_0^{\mathrm{det}} + E_0^*(m_{\mathrm{in}}, 0)$. The determinant and correction estimates used above include their full chamber trace and the first weighted fluctuation integral. The initial vacuum sum contains $\delta\varepsilon^0_0$. No additional coupling is selected by the fixed point. □

16.4. **One pair on all tori of the construction.** For $N \ge N_{\min}(R_{\mathrm{top}}, e)$ define

$$P_N(e) = \big(m_{0,N}(e), \varepsilon^{\mathrm{v}}_{0,N}(e)\big)$$

to be the pair of Theorem 16.3 on the prescribed side $R_{\mathrm{top}}(e)$. The dependence on the fixed $\bar{m}$ and fixed construction parameters is suppressed. For lower cutoffs set $m_{0,N}(e) = 0$.

Theorem 16.4. *The pair $P_N(e)$ is controlled on every $R \in \mathcal{T}(e)$ with $N \ge N_{\min}(R, e)$. Up to its stop, its mass and vacuum sequence agree with those on the top torus; its activity and external scalar families agree there on flat polymers. For every such level,*

$$|m_j| \le \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \qquad \|E_j\|_j \le \|\mathscr{E}_j\|^x_j \le 4e_j^{1/4-6\varepsilon}, \qquad |\varepsilon^{\mathrm{v}}_j| \le 5c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}. \tag{16.18}$$

The real mass sequence is independent of the side datum and of the twist. At all $N \ge 0$,

$$|m_{0,N}(e)| \le \tfrac{1}{8} q^{3/4-8\varepsilon} < \tfrac{1}{8}, \qquad q = eL^{-N/2}. \tag{16.19}$$

The total lattice mass is $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e)$.

Proof. The pair is defined because $N_{\min}(R_{\mathrm{top}}, e) \le N_{\min}(R, e)$. The two trajectories have the same initial couplings and zero initial activities. Suppose they agree through $j < k_*(R)$. The top mass is bounded by $w_j^{\mathrm{m}}/8$; the same bound on the smaller side permits continuation to $j+1$ by Proposition 16.2. Both inputs are now in the unit balls, are real, and have locally agreeing activity and scalar families. The scale proposition verifies their primitive, domain and next-side comparison hypotheses. Theorem 15.5 gives the same extracted mass and vacuum, including the same $\delta\varepsilon_j^0$, and local agreement of the outgoing activity and scalar family. The scalar recursions therefore give the same $m_{j+1}, \varepsilon_{j+1}^{\mathrm{v}}$. Restriction to zero external slot preserves agreement of E_{j+1}. This proves the induction up to $k_*(R)$.

The common mass bound follows from the top trajectory. The local comparison alone is not an estimate of the entire activity on the smaller torus; that bound follows from its own continuation estimate. The vacuum is a common scalar and satisfies the top bound at this same level. Thus (16.18) follows. At a smaller-volume stop the couplings need not vanish; the zeros defining $P_N(e)$ are imposed at k_{top}.

The fixed-point map on the top torus is determined by $(N, e, \bar{m})$, the fixed construction parameters and photon coordinate basis, and the top primitive families constructed from them. The selected top side is itself a function of $e, \bar{m}$ and these parameters. The smaller side R enters no part of this definition. Nor does a twist: the twist is a harmonic part of a connection inserted in the density identity, whereas the coupling maps are maps of entire local activity families and contain no such insertion. The full forward activity recursion retains the external slot as a variable, whereas the fixed-point map uses its restriction to zero. No external connection or probe is inserted when selecting the pair. This proves the asserted independence within the fixed construction.

For a cutoff on which the pair is defined, (16.19) is its initial mass bound on the top torus. At the other finitely many cutoffs it holds because the assigned mass is zero. Finally $h \ge 1$ implies $e \le e_1 \le 1/3$, so $q \le 1/3$ and its positive power is below 1. The sequence is real in both cases. The sign of $\boldsymbol{\mu}_N$ is not fixed by this estimate; the lattice construction admits a real mass. □

On the reference torus, for $N \ge N_{\mathrm{ref}}(e)$, the outputs needed at the terminal scale are

$$a_* \in \{h/L, h/L^2\}, \qquad |m_*| \le \tfrac{1}{8} e_*^{3/4-8\varepsilon}, \qquad \|E_*\|_* \le \|\mathscr{E}_*\|_*^x \le 4 e_*^{1/4-6\varepsilon}. \tag{16.20}$$

The primitive estimates along this flow have the same fixed constants on every torus and at every cutoff. The remaining terminal density and partition estimates concern the integration of these fields.

17. A positive continuum fourth coefficient

The fourth-order response calculation produces the coefficient $-6\,\mathrm{tr}[(S_{\bar{m}} i\gamma \cdot d^*\beta)^4]$. Its sign is not apparent from this expression. We determine the leading term at large mass and then dilate a compactly supported probe. The large-mass calculation is local: a gauge choice removes the constant connection, and quartic homogeneity makes the leading term depend only on the antisymmetric first derivative. We compute that coefficient in a constant field.

For a real $a \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ put

$$V_a = i\gamma \cdot a, \qquad S_M = (M + \gamma \cdot \partial)^{-1}, \qquad q_{4,M}(a) = -\tfrac{1}{4}\,\mathrm{tr}[(S_M V_a)^4], \quad M > 0.$$

The trace is on $L^2(\mathbb{R}^3; \mathbb{C}^4)$; V_a denotes multiplication. We use Euclidean norms in this section.

Proposition 17.1. *The operator $(S_M V_a)^4$ is trace class and its trace is real. There is $C_a < \infty$ such that, for $M \geq 1$,*

$$\left| q_{4,M}(a) - \frac{1}{240\pi M^5} \int_{\mathbb{R}^3} |\operatorname{curl} a|^4 \, \mathrm{d}x \right| \leq C_a M^{-7}. \tag{17.1}$$

In particular, for every $\bar{m} > 0$ there is a real $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ for which $f = dd^*\beta$ satisfies $b(f) > 0$ in (1.1).

17.1. **Kernels and differentiation in the mass.** The kernel and Fourier multiplier of S_M are

$$S_M(z) = (M - \gamma \cdot \partial) \frac{e^{-M|z|}}{4\pi|z|}, \tag{17.2}$$

$$\widehat{S}_M(p) = \frac{M - i\gamma \cdot p}{M^2 + |p|^2}, \qquad |\widehat{S}_M(p)| = (M^2 + |p|^2)^{-1/2}. \tag{17.3}$$

In particular,

$$|S_M(z)| \leq \frac{e^{-M|z|}(2M|z| + 1)}{4\pi|z|^2}, \qquad S_M(z) = M^2 S_1(Mz), \qquad \|S_M\|_{L^1} \leq 3/M.$$

The operator $G = S_M V_a S_M$ is Hilbert–Schmidt. Indeed, its momentum kernel, with measure $(2\pi)^{-3}\mathrm{d}p$, is $\widehat{S}_M(p) i\gamma \cdot \widehat{a}(p-q) \widehat{S}_M(q)$. Writing $u(p) = (M^2 + |p|^2)^{-1}$, the matrix norm bound and Young's inequality give

$$\|G\|_{\mathrm{HS}}^2 \leq 8(2\pi)^{-6} \int u(p) (|\widehat{a}|^2 * u)(p) \, \mathrm{d}p \leq 8(2\pi)^{-3} \|u\|_2^2 \|a\|_2^2 = \frac{\|a\|_2^2}{\pi M}.$$

Here $\|u\|_2^2 = \pi^2/M$. Since $(S_M V_a)^4 = G V_a G V_a$,

$$|q_{4,M}(a)| \leq \frac{\|a\|_2^2 \|a\|_\infty^2}{4\pi M}. \tag{17.4}$$

The matrix $\gamma_5 = \left(\begin{smallmatrix} 0 & 1_2 \\ 1_2 & 0 \end{smallmatrix}\right)$ conjugates S_M to S_M^* and V_a to $V_a^* = -V_a$. Conjugation of the fourth power and cyclicity of the trace therefore give a real trace. Cyclicity here involves a bounded factor and the trace-class product GV_aG, so it does not presume that $S_M V_a$ itself is trace class.

We will evaluate the mass derivative on the diagonal of a kernel with five propagators. This requires continuity at the diagonal, including for the linear connections used below. For either a compact smooth connection or a linear connection, $|a(z)| \leq c_a(1 + |z|)$ and a is Lipschitz. Choose $s(z) = K|z|^{-2} e^{-M|z|/2}$ so that $|S_M(z)|(1 + |z|)^{24} \leq s(z)$. Its convolution powers obey

$$s^{*2}(z) \leq C|z|^{-1} e^{-M|z|/2}, \quad s^{*2}(z) \leq C|z|^{-3/2}, \quad s^{*3}(z) \leq C|z|^{-1/2}. \tag{17.5}$$

For example, the bound for the second convolution follows by factoring out $e^{-M|z|/2}$ and scaling $\int |z - w|^{-2} |w|^{-2} \, \mathrm{d}w = C|z|^{-1}$; the third follows by convolving $|z|^{-2}$ and $|z|^{-3/2}$. Thus $s^{*2} \in L^2$, $s^{*4} = s^{*2} * s^{*2}$ is bounded and continuous, and so is $s^{*5} = s * s^{*4}$.

Let $R_a^{(j)}(x, y)$ be the integral kernel of $(-1)^j (S_M V_a)^j S_M$, initially off the diagonal, for $0 \leq j \leq 4$. Distributing the factors $1 + |z_l|$ along the chain of propagators gives

$$|R_a^{(j)}(x, y)| \leq [c_a(1 + |x|)]^j s^{*(j+1)}(x - y). \tag{17.6}$$

These integrals act as the indicated operators on Schwartz spinors; absolute integration against such a spinor justifies every composition. The fourth kernel extends continuously to all (x, y). To see this, use the four independent increments of the first three and last propagators. Changing (x, y) then changes four Lipschitz factors and the middle propagator. On $|x|, |y|, |x'|, |y'| \leq r$, the resulting difference is bounded by

$$C_{a,r}\big((|x - x'| + |y - y'|) \|s^{*5}\|_\infty \tag{17.7}$$

$$+ \|s^{*4}\|_\infty \|S_M - S_M(\,\cdot + (x'-y') - (x-y))\|_{L^1}\big).$$

This tends to zero. Define $r_4(x;M,a) = \operatorname{tr} R_a^{(4)}(x,x)$ using this extension.

For compact a, differentiation in M now gives

$$\frac{\mathrm{d}}{\mathrm{d}M} q_{4,M}(a) = \int r_4(x;M,a)\,\mathrm{d}x, \qquad q_{4,M}(a) \longrightarrow 0 \quad (M \longrightarrow \infty). \tag{17.8}$$

To justify this differentiation, write the trace as the absolutely convergent four-vertex integral

$$-\tfrac{1}{4}\int \operatorname{tr}\big[S_M(z_4-z_1)V_a(z_1)S_M(z_1-z_2)V_a(z_2)S_M(z_2-z_3)V_a(z_3)S_M(z_3-z_4)V_a(z_4)\big]\,\mathrm{d}z_1\cdots\mathrm{d}z_4.$$

Its absolute integral is at most $4\|a\|_\infty^3\|a\|_{L^1}s^{*4}(0)$ before the factor $1/4$. On any compact interval of positive masses, S_M and $\partial_M S_M$ have a common majorant of the same form. Since $\partial_M S_M = -S_M * S_M$, the four differentiated terms have equal integrals by cyclicity. The factor $-1/4$ and the minus sign in this derivative cancel. Inserting the convolution yields the diagonal five-propagator integral in (17.8); its absolute integral is bounded by $4\|a\|_\infty^3\|a\|_{L^1}s^{*5}(0)$. This also proves continuous differentiability. The limit at infinity follows from (17.4).

17.2. **The local large-mass expansion.** Write

$$V_j(x;w) = i\gamma\cdot\sum_{|\alpha|=j}\frac{w^\alpha}{\alpha!}\partial^\alpha a(x).$$

The substitution $z_l = x + M^{-1}w_l$ in the diagonal kernel gives

$$\begin{aligned} r_4(x;M,a) = M^{-2}\operatorname{tr}\int\, & S_1(-w_1)V_a(x+M^{-1}w_1)S_1(w_1-w_2)V_a(x+M^{-1}w_2) \\ & \cdot S_1(w_2-w_3)V_a(x+M^{-1}w_3)S_1(w_3-w_4)V_a(x+M^{-1}w_4)S_1(w_4)\,\mathrm{d}w. \end{aligned} \tag{17.9}$$

Let $\rho_j(x;a)$ be the sum of these integrals with the four connection factors replaced by $V_{j_l}(x;w_l)$ and $j_1+\cdots+j_4 = j$. Taylor expansion through total order five gives

$$r_4(x;M,a) = \sum_{j=0}^{5} M^{-2-j}\rho_j(x;a) + O_a(M^{-8}), \qquad M \geq 1, \tag{17.10}$$

uniformly in x. The remainder contains four factors bounded by $C\|a\|_{C^6}(1+|w_l|)^6$ and at least the factor M^{-6}. With $w_0 = 0$, $\prod_l(1+|w_l|)^6 \leq \prod_l(1+|w_l - w_{l-1}|)^{24}$. Use the weighted majorant at unit mass for the five factors S_1. Its fifth convolution at zero bounds the remainder integral by a constant independent of M. The coefficients vanish off $\operatorname{supp} a$. There is also a spatial tail bound

$$\int_{\operatorname{dist}(x,\operatorname{supp} a)\geq 1} |r_4(x;M,a)|\,\mathrm{d}x \leq C_a e^{-M}. \tag{17.11}$$

Indeed, the first and last propagators each travel at least $d = \operatorname{dist}(x,\operatorname{supp} a)$; their bound is Me^{-Md} for $d \geq 1$. Integrating the three intervening propagators gives

$$|r_4(x;M,a)| \leq 108\|a\|_\infty^4 |\operatorname{supp} a| M^{-1}e^{-2Md}.$$

For $M, d \geq 1$, $2Md \geq M + d$, which proves (17.11).

The reduction to field strength must be justified at the level of this diagonal kernel. Let $\lambda \in C_c^\infty(\mathbb{R}^3)$ be real. On Schwartz spinors, as a formal power series in u,

$$(M + \gamma\cdot\partial + ui\gamma\cdot(a + d\lambda))^{-1} = e^{-iu\lambda}(M + \gamma\cdot\partial + ui\gamma\cdot a)^{-1}e^{iu\lambda}.$$

All coefficients preserve Schwartz space. Comparing the coefficient of u^4 gives, almost everywhere off the diagonal,

$$R^{(4)}_{a+d\lambda}(x,y) = \sum_{j=0}^{4} R^{(j)}_a(x,y)\frac{[-i(\lambda(x)-\lambda(y))]^{4-j}}{(4-j)!}. \tag{17.12}$$

For $j = 0, 1, 2, 3$, the bounds in (17.5)–(17.6) show that these terms are respectively $O(|x-y|^2)$, $O(|x-y|^2)$, $O(|x-y|^{3/2})$, and $O(|x-y|)$ locally in x. Continuity of the fourth kernel therefore implies

$$r_4(x; M, a + d\lambda) = r_4(x; M, a).$$

Uniqueness of the finite asymptotic expansion in (17.10) gives the same invariance for ρ_j, $0 \le j \le 5$.

Fix x. If $p^{(k)}(z)$ is the homogeneous part of degree k of the fifth-order Taylor polynomial of $a(x+z)$, choose a compact λ equal near x to

$$\lambda(x+z) = -\sum_{k=0}^{5} \frac{z \cdot p^{(k)}(z)}{k+1}.$$

Euler's identity gives $z \cdot (a + d\lambda)^{[5]}(x+z) = 0$ for the Taylor polynomial. Thus the transformed constant term vanishes, and its first derivative is the antisymmetric matrix $F_{\nu\mu}(x)/2$, where $F_{\nu\mu} = \partial_\nu a_\mu - \partial_\mu a_\nu$. Each term defining ρ_j has four connection factors. Hence it vanishes for $j < 4$, and at order four only $(j_1, j_2, j_3, j_4) = (1, 1, 1, 1)$ remains. Put

$$A_{F,\mu}(z) = \tfrac{1}{2}\sum_{\nu} z_\nu F_{\nu\mu}, \qquad \Psi(F) = r_4(0; 1, A_F).$$

Then $\rho_4(x; a) = \Psi(F(x))$. The odd coefficients vanish as well: substitution $w_l \mapsto -w_l$ and conjugation by γ_5 leave the trace unchanged and multiply it by $(-1)^j$. In particular $\rho_5 = 0$. Consequently

$$r_4(x; M, a) = M^{-6}\Psi(F(x)) + O_a(M^{-8}). \tag{17.13}$$

17.3. **The constant-field coefficient.** Rotation of the variables and conjugation by the spin lift in $SU(2)$ show that $\Psi(F)$ depends on the vector $B_l = \frac{1}{2}\sum_{\nu,\mu}\varepsilon_{l\nu\mu}F_{\nu\mu}$ only through $|B|$. Take $B = (0, 0, b)$, $b \ge 0$, and $A = (-bz_2/2, bz_1/2, 0)$. Set $D_u = \partial + iuA$ and

$$h_u = 1 - \Delta + uW_1 + u^2W_2, \qquad W_1 = -2iA\cdot\partial + b\sigma_3, \quad W_2 = |A|^2.$$

The Clifford relations give

$$(1-\gamma\cdot D_u)(1+\gamma\cdot D_u) = \operatorname{diag}(h_u, h_u), \qquad (1+\gamma\cdot D_u)^{-1} = (1-\gamma\cdot D_u)\operatorname{diag}(h_u^{-1}, h_u^{-1}). \tag{17.14}$$

These are identities of formal series acting on Schwartz spinors; no expansion of an operator with an unbounded connection is being assumed in operator norm.

We compute the fourth coefficient of h_u^{-1} from the kernel

$$k_u(t; x, y) = (4\pi t)^{-3/2}e^{-|x-y|^2/(4t)}\frac{\tau}{\sinh\tau}\exp\left(-\frac{(\tau\coth\tau - 1)|x_p - y_p|^2}{4t} + \frac{iub}{2}x\wedge y\right)e^{-\tau\sigma_3}, \tag{17.15}$$
$$\tau = ubt, \quad x_p = (x_1, x_2), \quad x\wedge y = x_1y_2 - x_2y_1.$$

It is analytic in u for $|u|bt < \pi$. The kernel satisfies $(\partial_t + h_u - 1)k_u = 0$. The planar verification uses $\alpha = (ub/4)\coth(ubt)$ and $\phi = -\alpha|x_p - y_p|^2 + iub(x\wedge y)/2$:

$$\frac{\check{h}e^{\phi}}{e^{\phi}} = 4\alpha - (4\alpha^2 - u^2b^2/4)|x_p - y_p|^2, \qquad \alpha' = -(4\alpha^2 - u^2b^2/4),$$

where $\check{h} = -\partial_1^2 - \partial_2^2 - iub(x_1\partial_2 - x_2\partial_1) + u^2b^2|x_p|^2/4$. Differentiation of the prefactor supplies -4α; differentiation of $e^{-\tau\sigma_3}$ supplies the spin term. The longitudinal Gaussian satisfies the one-dimensional heat equation.

Here only the coefficients of (17.15) are needed. Write $k_u = \sum_{n\ge 0} u^n k^{(n)}$. Each coefficient is the Gaussian times a finite polynomial in $t, x-y, y$, whose monomials $t^j(x-y)^\alpha y^\beta$ have $2j+|\alpha| \ge n$. The factor $e^{iub(x-y)\wedge y/2}$ has weight at least one per power of u, and the other factors have weight at least two, if t has weight two and $x-y$ weight one. Thus

$$K^{(n)}(x,y) = \int_0^\infty e^{-t} k^{(n)}(t;x,y)\,\mathrm{d}t$$

converges off the diagonal and, for $n \ge 1$, on the diagonal. Its terms are y^β times

$$m_{j,\alpha}(x-y) = \int_0^\infty e^{-t} t^j (x-y)^\alpha (4\pi t)^{-3/2} e^{-|x-y|^2/(4t)}\,\mathrm{d}t.$$

Their Fourier transforms are $(i\partial_p)^\alpha[j!(1+|p|^2)^{-j-1}]$, so they define operators preserving Schwartz space.

Coefficientwise integration of the heat equation identifies these operators with the formal inverse of h_u. Indeed, for a Schwartz spinor φ, the zeroth heat coefficient converges to φ as $t \downarrow 0$. The nth coefficient for $n \ge 1$ tends to zero uniformly, since every polynomial term is bounded after integration against φ by $C_\varphi t^{j+|\alpha|/2}$. Integrating from ϵ to infinity and sending ϵ to zero therefore gives

$$(1-\Delta)K^{(n)} + W_1 K^{(n-1)} + W_2 K^{(n-2)} = \delta_{n0}1.$$

The convergence can be taken in tempered distributions; the polynomial differential operators are continuous there. Since the constant operator $1-\Delta$ is invertible on Schwartz space, the right inverse is the unique formal inverse.

To evaluate the diagonal, the kernel of its fourth coefficient must agree with the continuous kernel already constructed. The functions $m_{j,\alpha}$ are continuous when $2j+|\alpha| \ge 2$ and continuously differentiable when $2j+|\alpha| \ge 3$. For this, write $m_{j,\alpha}(z) = z^\alpha \mu_j(|z|)$ with

$$\mu_j(s) = \int_0^\infty e^{-t} t^j (4\pi t)^{-3/2} e^{-s^2/(4t)}\,\mathrm{d}t.$$

For $j \ge 1$, μ_j is bounded and continuous at zero and $\mu_j'(s) = -s\mu_{j-1}(s)/2$ for $s > 0$; for $j = 0$ use $\mu_0(s) = e^{-s}/(4\pi s)$. These formulas give the asserted regularity at zero. Consequently $K^{(3)}, K^{(4)}$ and $\partial_x K^{(4)}$ are continuous. From (17.14) the fourth Dirac coefficient has kernel

$$(1-\gamma\cdot\partial_x)\,\mathrm{diag}(K^{(4)},K^{(4)})(x,y) - i\gamma\cdot A(x)\,\mathrm{diag}(K^{(3)},K^{(3)})(x,y).$$

It equals $R_A^{(4)}$ because both continuous kernels give the same operator on Schwartz space. At $x=y=0$, $A(0)=0$, and the derivative term has zero trace: the two Pauli blocks have opposite signs. It follows that

$$\Psi(F) = 2\int_0^\infty e^{-t}\,\mathrm{tr}\,k^{(4)}(t;0,0)\,\mathrm{d}t \tag{17.16}$$

$$= -\frac{4}{45}(4\pi)^{-3/2}\Gamma(7/2)b^4 = -\frac{1}{48\pi}|B|^4. \tag{17.17}$$

In the second line we used $\mathrm{tr}\,k_u(t;0,0) = 2(4\pi t)^{-3/2}\tau\coth\tau$ and $\tau\coth\tau = 1+\tau^2/3-\tau^4/45+O(\tau^6)$.

17.4. **Integration and dilation.** Integrate (17.13) over the unit neighbourhood of $\operatorname{supp} a$, and use (17.11) on its complement. Equations (17.8) and (17.17) give

$$\frac{\mathrm{d}}{\mathrm{d}M} q_{4,M}(a) = -\frac{1}{48\pi M^6} \int |\operatorname{curl} a|^4 \,\mathrm{d}x + O_a(M^{-8}).$$

Integration from M to infinity proves (17.1).

Finally let $\varphi(x) = e^{-1/(1-|x|^2)}$ for $|x| < 1$ and zero otherwise, set $\beta_{12} = \varphi$, $\beta_{13} = \beta_{23} = 0$, and $a = d^*\beta = (\partial_2\varphi, -\partial_1\varphi, 0)$. The third component of $\operatorname{curl} a$ is $-(\partial_1^2 + \partial_2^2)\varphi$ and has value $4e^{-1}$ at zero. Thus $I = \int |\operatorname{curl} a|^4 \,\mathrm{d}x > 0$. For $\ell \geq 1$ put $\beta^{(\ell)}(x) = \ell^2\beta(x/\ell)$, $a_\ell(x) = \ell a(x/\ell)$, and $f_\ell = dd^*\beta^{(\ell)}$. The unitary dilation $U_\ell\psi(x) = \ell^{-3/2}\psi(x/\ell)$ gives

$$U_\ell^{-1} S_{\bar m} V_{a_\ell} U_\ell = \ell^2 S_{\bar m \ell} V_a, \qquad q_{4,\bar m}(a_\ell) = \ell^8 q_{4,\bar m\ell}(a).$$

For $\bar m\ell \geq 1$, therefore,

$$b(f_\ell) = 24 q_{4,\bar m}(a_\ell) = \frac{\ell^3 I}{10\pi \bar m^5} + O_a(\ell \bar m^{-7}). \tag{17.18}$$

This is positive for sufficiently large ℓ and completes the proof of Proposition 17.1.

18. Free response coefficients

The coefficient controlling the fourth cumulant is a derivative of the free chamber determinant in a smooth external connection. The lattice derivative has contact terms, because one bond phase can be differentiated several times. We first show that these terms vanish in the continuum limit, then compare the trace with four first-order vertices to the continuum Dirac trace. The second-order coefficient requires a different cancellation: the two-vertex trace and its contact term must be combined before their limit is taken.

18.1. **The probe and the finite determinant.** Fix $0 < \rho < 1$, the standard chamber weights, and $0 < \bar m \leq 1$. Let β be a real nonzero compactly supported smooth two-form, and put $a = d^*\beta$, $f = dd^*\beta$. Thus $a_\nu = -\sum_{\mu\neq\nu} \partial_\mu \beta_{\mu\nu}$. Let D_β be the least side of a closed coordinate cube Q_β containing its support. In this section D_β measures the support itself, without a buffer. For $0 < \eta \leq 1$ use the seminorm

$$|\beta|_{C^j} = \max_{\mu<\nu,\, |\alpha|\leq j} \|\partial^\alpha \beta_{\mu\nu}\|_\infty.$$

Define

$$J_\eta v(y) = \eta^{-3} \int_{[-\eta/2,\eta/2]^3} v(y+z)\,\mathrm{d}z, \qquad \delta_{\eta,\mu} v(y) = \frac{v(y+\eta e_\mu/2) - v(y-\eta e_\mu/2)}{\eta}.$$

The interpolating functions

$$a^0_{\eta,\nu}(y) = -\sum_{\mu\neq\nu} \delta_{\eta,\mu} J_\eta \beta_{\mu\nu}(y) \tag{18.1}$$

are defined without periodization. If $n = R/\eta \in \mathbb{N}$, $n \geq 2$, the lattice one-cochain on the physical torus is

$$a_{\eta,\nu}(\eta c) = \sum_{z\in\mathbb{Z}^3} a^0_{\eta,\nu}(\eta c + \eta e_\nu/2 + Rz). \tag{18.2}$$

It is the adjoint lattice coboundary of the cube-averaged probe at plaquette centers. In lattice units the external connection is $B_t(c, c+e_\nu) = t\eta a_{\eta,\nu}(\eta c)$, extended with the orientation sign to reversed bonds. The free mass in these units is $m = \bar m\eta$. The cochains are periodic on the oriented nearest-neighbor edges of the covering lattice. When $n = 2$, distinct edge orbits under period translations remain distinct even when their quotient endpoints coincide.

On antiperiodic chamber spinors let

(18.3)
$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[B_t] + m)}{\det(D^{\mathrm{ch}}[0] + m)}, \qquad c_j = N_0^{(j)}(0), \qquad a_{\mathrm{ch}} = (\log N_0)''(0), \quad b_{\mathrm{ch}} = (\log N_0)^{(4)}(0).$$

The logarithm is the holomorphic branch equal to zero at the origin. Its denominator is nonzero: the real part of $D^{\mathrm{ch}}[0]+m$ is at least $m > 0$. For comparison with the charge convention, $q = e\eta^{1/2}$ and the normalized probe is $\widehat{a} = \eta^{1/2} a_\eta$, so $B_t = q(t/e)\widehat{a}$. The free ratio and its coefficients are independent of e.

18.2. **The singular part of the Bloch inverse.** Use the fiber $V = \ell^2(O_{\mathrm{cub}}; \mathbb{C}^4)$ with measure $48^{-1}\sum_g$ and the constant-label isometry J. Its dimension is 192, and traces use this full dimension; the label measure does not multiply the matrix trace by $1/48$. Write $\mathcal{D}_m(p)$ for the free symbol of §9.2, and $R_m(p) = \mathcal{D}_m(p)^{-1}$. For real p it is invertible when $m > 0$.

We need a bound on the difference between the full inverse and its single light Dirac part, including mixed and heavy blocks. There is $C_\rho < \infty$ such that

$$R_m(p) = J(m + i\gamma \cdot p)^{-1} J^* + \mathcal{E}_m(p), \qquad \|\mathcal{E}_m(p)\| \le C_\rho, \qquad p \in [-\pi, \pi]^3, \quad 0 < m \le 1. \tag{18.4}$$

To prove this, use the block inversion formula (9.14). The off-diagonal blocks vanish at zero momentum, and are $O(|p|)$ because only the external hop varies with p. The heavy inverse is bounded by (9.8). The light Schur complement therefore has the form

$$B_m(p) = m + i\gamma \cdot p + r_m(p), \qquad \|r_m(p)\| \le C_\rho |p|^2.$$

For $|p|$ below a fixed ρ-dependent radius, the product $r_m(p)(m + i\gamma \cdot p)^{-1}$ has norm at most one half. Its Neumann inverse gives

$$\|B_m(p)^{-1} - (m + i\gamma \cdot p)^{-1}\| \le 2C_\rho |p|^2 (m^2 + |p|^2)^{-1} \le 2C_\rho, \qquad |p| \|B_m(p)^{-1}\| \le 2.$$

In the other four terms of the block inverse, each mixed block supplies a factor $|p|$ beside this light inverse. The heavy terms are bounded. This proves (18.4) near zero. Away from zero, the real-part and high-momentum ellipticity bounds of §9.2 bound the full inverse uniformly, and the continuum inverse is bounded by $1/|p|$. This proves the estimate on the whole cube. In particular

$$\|R_m(p)\| \le C_\rho + (m^2 + |p|^2)^{-1/2} \le \frac{C'_\rho}{m + |p|}. \tag{18.5}$$

Define external-hop matrices

$$(T_\nu^{\pm} u)(g) = 1_{\{ge_1 = \pm e_\nu\}} 3(\rho \mp \gamma_\nu) u(g r_{e_1}).$$

Then

$$\begin{aligned} \mathcal{D}_m(p) &= \mathcal{D}_{m,\mathrm{loc}} - \sum_\nu (e^{ip_\nu} T_\nu^+ + e^{-ip_\nu} T_\nu^-), \\ V_\nu(p) &:= \partial_\nu \mathcal{D}_m(p) = -i(e^{ip_\nu} T_\nu^+ - e^{-ip_\nu} T_\nu^-), \\ W_\nu(p) &:= \partial_\nu^2 \mathcal{D}_m(p) = e^{ip_\nu} T_\nu^+ + e^{-ip_\nu} T_\nu^-. \end{aligned} \tag{18.6}$$

Mixed second derivatives vanish. The disjoint rows indexed by $ge_1 = \pm e_\nu$ imply $\|\sum z_\nu^\pm T_\nu^\pm\| \le 3(1+\rho)\max|z_\nu^\pm|$. Since eight labels contribute to each direction,

$$J^* T_\nu^\pm J = \tfrac{1}{2}(\rho \mp \gamma_\nu), \qquad J^* V_\nu(0) J = i\gamma_\nu. \tag{18.7}$$

All nonzero derivatives of the symbol are bounded by six on real momenta. Differentiating $R_m \mathcal{D}_m = 1$ gives

$$\partial^\alpha R_m = -\sum_{0 < \delta \le \alpha} \binom{\alpha}{\delta} (\partial^{\alpha-\delta} R_m)(\partial^\delta \mathcal{D}_m) R_m.$$

Starting with (18.5), induction bounds each summand by a constant times $(\operatorname{dist}(p, 2\pi\mathbb{Z}^3) + m)^{-1-|\alpha|}$; the remaining factor of the distance has nonnegative exponent and is bounded by a power of seven. Thus

$$\|\partial^\alpha R_m(p)\| \le C_{\alpha,\rho}(\operatorname{dist}(p, 2\pi\mathbb{Z}^3) + m)^{-1-|\alpha|}. \tag{18.8}$$

At $m = \bar{m}\eta$, (18.4) and the first two differentiated inverse identities give

$$\eta^{1+|\alpha|}(\partial^\alpha R_{\bar{m}\eta})(\eta P_\eta) \longrightarrow J\partial^\alpha(\bar{m} + i\gamma\cdot P)^{-1}J^*, \quad |\alpha| \le 2, \quad P_\eta \to P. \tag{18.9}$$

In the second derivative the contact term has two inverse factors and one bounded W_ν, so its rescaled norm has an extra factor η and vanishes.

The involution $(\mathcal{U}u)(g) = \gamma_5 u(-g)$ satisfies

$$\mathcal{U}\mathcal{D}_m(p)\mathcal{U} = \mathcal{D}_m(-p), \qquad \mathcal{U}V_\nu(p)\mathcal{U} = -V_\nu(-p). \tag{18.10}$$

Indeed $-g$ reverses every bond direction, and $\gamma_5\gamma_\nu\gamma_5 = -\gamma_\nu$ restores the hopping matrix. The external phase then changes p to $-p$. This parity will cancel the first momentum derivative of the polarization kernel.

18.3. **Derivative and contact terms.** Let $\mathcal{P}_n = \frac{2\pi}{n}(\mathbb{Z} + \frac{1}{2})^3 \cap (-\pi, \pi]^3$. The normalized Bloch vectors are $n^{-3/2}e^{ip\cdot c}$, and the lattice free propagator G is the multiplier $R_m(p)$ on these antiperiodic momenta. Write $A_\nu(c) = a_{\eta,\nu}(\eta c)$ and $A^-_\nu(c) = A_\nu(c - e_\nu)$, and let $U^\pm_\nu\psi(c) = \psi(c \pm e_\nu)$. The bond values are $b^+_\nu = \eta A_\nu$, $b^-_\nu = -\eta A^-_\nu$. The exact connection derivatives are

$$D_j = -\sum_{\nu,\pm}(ib^\pm_\nu)^j T^\pm_\nu U^\pm_\nu, \quad D^{\mathrm{ch}}[B_t] - D^{\mathrm{ch}}[0] = \sum_{j\ge1}\frac{t^j}{j!}D_j, \quad Y_j = GD_j. \tag{18.11}$$

The shifts act first. These D_j are derivatives, with the factorials outside the vertices.

For small complex t, $N_0(t) = \det(1 + \sum_{j\ge1} t^j Y_j/j!)$ and its zero-at-zero logarithm is the convergent finite-matrix trace series $\sum_{l\ge1}(-1)^{l+1}\operatorname{Tr}(\sum_j t^j Y_j/j!)^l/l$. The tail $l \ge 5$ is $O(t^5)$, so it contributes no fourth derivative. Cyclicity of the trace in the terms $l \le 4$ gives

$$\begin{aligned} b_{\mathrm{ch}} = {} & \operatorname{Tr} Y_4 - 4\operatorname{Tr}(Y_1Y_3) - 3\operatorname{Tr} Y_2^2 \\ & + 12\operatorname{Tr}(Y_1^2Y_2) - 6\operatorname{Tr} Y_1^4. \end{aligned} \tag{18.12}$$

For example, the cyclically equal orders $(1,1,2)$, $(1,2,1)$ and $(2,1,1)$ contribute $24\cdot\frac{1}{3}\cdot\frac{3}{2} = 12$. The quadratic identity is $a_{\mathrm{ch}} = \operatorname{Tr} Y_2 - \operatorname{Tr} Y_1^2$. The determinant series is used only near zero; the determinant itself is entire in t.

We bound all four contact traces in (18.12) uniformly in R and the position of the probe. The interpolating periodic field is

$$\widetilde{A}_\nu(y) = \sum_z a^0_{\eta,\nu}(y + \eta e_\nu/2 + Rz), \qquad A_\nu(c) = \widetilde{A}_\nu(\eta c).$$

When $R \ge D_\beta + 1$, averaging translates of the derivatives of β shows

$$\begin{gathered} \|\widetilde{A}\|_\infty + \|\partial\widetilde{A}\|_\infty \le C^{\mathrm{free}}_\beta, \qquad \|\widetilde{A} - a^R\|_\infty \le C^{\mathrm{free}}_\beta\eta, \\ \eta^3|\{c : A_\nu(c) \ne 0 \text{ for some } \nu\}| \le (D_\beta + 3)^3, \qquad \|\eta A_\nu\|_{\ell^4} \le C^{\mathrm{free}}_\beta\eta^{1/4}, \\ \|\eta(A^-_\nu - A_\nu)\|_{\ell^4} \le C^{\mathrm{free}}_\beta\eta^{5/4}, \qquad \|\widetilde{A} - a^R\|_{L^4(T_R)} \le C^{\mathrm{free}}_\beta\eta. \end{gathered} \tag{18.13}$$

Every averaging displacement has size at most η in sup norm. The translates of $\operatorname{supp}\beta$ are disjoint, so their derivative sup norms are unchanged. The supports of the interpolants lie in $Q_\beta + [-\eta, \eta]^3$ modulo the periods. Its volume and number of lattice points give the last estimates, including the union with its one-step translate for the difference.

Let $\widehat{A^j_\nu}(k) = n^{-3}\sum_c A_\nu(c)^j e^{-ik\cdot c}$ on the finite momentum group $\mathcal{G}_n = \frac{2\pi}{n}\mathbb{Z}^3/2\pi\mathbb{Z}^3$, and set $F_j(k) = \sum_\nu |\widehat{A^j_\nu}(k)|$. There are constants $C_{\mathrm{Four},1}, C_{\mathrm{Four},0}$ depending only on β with

$$\|F_j\|_{\ell^1(\mathcal{G}_n)} \le C^j_{\mathrm{Four},1}, \qquad R^3\|F_j\|_\infty \le C^{j-1}_{\mathrm{Four},1} C_{\mathrm{Four},0}. \tag{18.14}$$

The transform of the centered difference and the cube average in (18.1) is

$$-\sum_{\mu\neq\nu} iP_\mu \operatorname{sinc}(\eta P_\mu/2)\prod_l \operatorname{sinc}(\eta P_l/2)\widehat{\beta}_{\mu\nu}(P),$$

where $\operatorname{sinc}(z) = \sin z/z$ and $\operatorname{sinc}(0) = 1$. The midpoint shift multiplies this transform by a phase of modulus one. The shifted transform has modulus at most $C^{\mathrm{free}}_\beta(1+|P|)^{-4}$. Poisson summation gives

$$\widehat{A}_\nu(k) = R^{-3}\sum_{z\in\mathbb{Z}^3} a^0_{\eta,\nu}(\widehat{\cdot + \eta e_\nu}/2)((k+2\pi z)/\eta).$$

Summing over representatives k and z gives a single sum over $\frac{2\pi}{R}\mathbb{Z}^3$, bounded after multiplication by R^{-3} by its integrable radial majorant, uniformly for $R \ge 1$. For $|k|_\infty \le \pi$, nonzero z have $|(k+2\pi z)/\eta| \ge \pi|z|_\infty/\eta$; the convergent sum $\sum_{z\neq 0}|z|_\infty^{-4}$ gives the sup bound. Powers of A_ν give convolutions on $\mathcal{G}_n$; $\ell^1 * \ell^1 \to \ell^1$ and $\ell^1 * \ell^\infty \to \ell^\infty$ give (18.14) for all j.

The momentum blocks of a vertex are

$$D_j(p,p') = -\eta^j \sum_\nu \widehat{A^j_\nu}(p-p')[i^j e^{ip'_\nu}T^+_\nu + (-i)^j e^{-ip_\nu}T^-_\nu], \qquad \|D_j(p,p')\| \le 6\eta^j F_j(p-p').$$

For $l \le 3$ and $j_1 + \cdots + j_l = 4$, apply these bounds and (18.5) to the cyclic trace. With differences $k_i = p_i - p_{i+1}$, the sum is parametrized by $p_1, k_1, \ldots, k_{l-1}$. Hölder in p_1 bounds the product of its l translates of $(m+|p|)^{-1}$ by $\sum_p (m+|p|)^{-l}$. Bound the final F_{j_l} in sup norm and the others in ℓ^1. Thus

$$|\operatorname{Tr}(Y_{j_1}\cdots Y_{j_l})| \le 192(6C'_\rho)^l C^3_{\mathrm{Four},1} C_{\mathrm{Four},0}\,\eta\left(n^{-3}\sum_{p\in\mathcal{P}_n}(m+|p|)^{-l}\right). \tag{18.15}$$

The factor η is $\eta^4 n^3/R^3$.

For antiperiodic momenta, comparison with their disjoint cubes of side $2\pi/n$ gives, for a decreasing radial function h, $n^{-3}\sum_p h(|p|) \le \pi^{-3}\int h(|y|)\,\mathrm{d}y$. Indeed in the cube of p one has $|y| \le 2|p|$, and a rescaling by two gives the factor π^{-3}. On the bounded momentum cube this yields constants for $l = 1, 2$ and $\log(1+5/m)$ for $l = 3$. Inserting the multiplicities $1, 4, 3, 12$ from (18.12) proves

$$\begin{aligned}|\operatorname{Tr} Y_4| + 4|\operatorname{Tr}(Y_1Y_3)| + 3|\operatorname{Tr} Y_2^2| + 12|\operatorname{Tr}(Y_1^2Y_2)| \\ \le C_{\mathrm{ct}}(\rho,\beta)\,\eta\,(1+\log(1+5/(\bar m\eta)))\,.\end{aligned} \tag{18.16}$$

The constant does not depend on $\bar m$, η, R or the position of $\operatorname{supp}\beta$.

18.4. **The fourth-order continuum coefficient.** Write $s_m(p) = (m + i\gamma\cdot p)^{-1}$. On the continuum antiperiodic torus its momentum set is $\mathcal{P}_R = \frac{2\pi}{R}(\mathbb{Z}+\frac12)^3$, and S^R is the multiplier $s_{\bar m}(P)$ there. For $R \ge D_\beta + 1$ let $a^R = \sum_z a(\cdot + Rz)$, and put

$$b_R(f) = -6\operatorname{Tr}(S^R i\gamma\cdot a^R)^4, \qquad b(f) = -6\operatorname{Tr}(S_{\bar m} i\gamma\cdot a)^4. \tag{18.17}$$

Both traces will be justified below. The compactly supported potential $a = d^*\beta$ is determined by $f = da$: two such potentials differ by a compactly supported smooth one-form which is closed and coclosed, hence harmonic and zero. Thus the notation $b(f)$ and $b_R(f)$ depends only on f.

Proposition 18.1. *The operators $S^R i\gamma\cdot a^R$ and $S_{\bar m} i\gamma\cdot a$ belong to the Schatten class of order four; their fourth-power traces are real. There are constants $C_1(\rho,\bar m,\beta)$ and $C_2(\bar m,\beta)$ such that*

$$|b_{\mathrm{ch}}(\eta,R)-b_R(f)|\le C_1\eta^{1/4},$$
$$|b_R(f)-b(f)|\le C_2 e^{-\bar m(R-D_\beta)},\qquad R\ge D_\beta+1,\quad 0<\eta\le 1,\quad R/\eta\in\mathbb{N}. \tag{18.18}$$

The constants are unchanged by every real translate of the probe. In particular $b_{\mathrm{ch}}(\eta,R)$ tends to $b(f)$ whenever $\eta\to 0$ and $R\to\infty$.

Proof. We first establish the trace bounds used in the comparisons. For a lattice multiplier M_h with fiber symbol $h(p)$ and a scalar multiplication operator v, one has

$$\|M_h v\|_4\le\left(n^{-3}\sum_{p\in\mathcal{P}_n}\|h(p)\|_4^4\right)^{1/4}\|v\|_{\ell^4}. \tag{18.19}$$

To prove it put $U=M_{h^*h}$ and $V=|v|^2$. Expanding the nonnegative trace of $(UV-VU)^*(UV-VU)$ gives $\mathrm{Tr}(UVUV)\le\mathrm{Tr}(U^2V^2)$. The left side is $\|M_hv\|_4^4$, and each normalized Bloch vector has constant squared modulus n^{-3}. Evaluating the right side in that basis gives (18.19). On the continuum torus the same proof, first for a finite set of momenta, gives

$$\|M_h v\|_4\le\left(R^{-3}\sum_{P\in\mathcal{P}_R}\|h(P)\|_4^4\right)^{1/4}\|v\|_{L^4(T_R)}. \tag{18.20}$$

Truncation in momentum converges in the Schatten norm when the sum is finite, and its operator-norm limit is M_hv, proving the general form.

The cube comparison used above also gives

$$\begin{aligned} &n^{-3}\sum_{p\in\mathcal{P}_n}(m^2+|p|^2)^{-2}\le\frac{1}{\pi m},\qquad R^{-3}\sum_{P\in\mathcal{P}_R}(\bar m^2+|P|^2)^{-2}\le\frac{1}{\pi\bar m},\\ &R^{-3}\sum_{|P|_\infty\le\pi/(2\eta)}|P|^{-2}\le C\eta^{-1},\qquad R^{-3}\sum_{|P|_\infty>\pi/(2\eta)}|P|^{-4}\le C\eta. \end{aligned} \tag{18.21}$$

For the last bound use the decreasing radial majorant $\min((2\eta/\pi)^4,|P|^{-4})$; its integral is $O(\eta)$. The preceding bounds and $\|s_{\bar m}(P)\|_4^4=4(\bar m^2+|P|^2)^{-2}$ show that $S^R i\gamma\cdot a^R$ has a bounded Schatten-four norm, uniformly in $R\ge D_\beta+1$. The same estimate applies to the interpolating field $\widetilde A$.

On $\mathbb{R}^3$, let

$$Y_{\bar m}(x)=\frac{e^{-\bar m|x|}}{4\pi|x|},\qquad S_{\bar m}(x)=(\bar m-\gamma\cdot\partial)Y_{\bar m}(x).$$

The scalar Fourier transform is $(\bar m^2+|P|^2)^{-1}$; the derivative has no point mass, since its boundary term on a sphere of radius r is $O(r)$. Its Fourier transform therefore gives $s_{\bar m}(P)$. These kernels are integrable, so their convolutions are the corresponding bounded Fourier multipliers on L^2. Also $\|Y_{\bar m}\|_{L^2}^2=(8\pi\bar m)^{-1}$. For bounded compactly supported v, the kernel of $v^*S_{\bar m}^*S_{\bar m}v$ is $\overline{v(x)}Y_{\bar m}(x-y)v(y)1_4$. Tonelli and Cauchy–Schwarz in the translated factors give

$$\|S_{\bar m}v\|_4^4\le 4\|Y_{\bar m}\|_{L^2}^2\|v\|_{L^4}^4=\frac{\|v\|_{L^4}^4}{2\pi\bar m}.$$

For the matrix multiplication $i\gamma\cdot a$, the Clifford identity gives the same proof with $|a|$ in place of $|v|$. Thus

$$\|S_{\bar m}i\gamma\cdot a\|_4^4\le(2\pi\bar m)^{-1}\int_{\mathbb{R}^3}|a|^4\,\mathrm{d}x. \tag{18.22}$$

The traces in (18.17) exist by Schatten Hölder.

To check reality, treat $S = S_{\bar m}$ or S^R and $v = i\gamma\cdot a$ or $i\gamma\cdot a^R$ together. Factor $v = |a|^{1/2}(i\gamma\cdot a/|a|)|a|^{1/2}$, with the middle factor zero at its zeros. The preceding estimates put SvS in the Hilbert–Schmidt class. One has $\gamma_5 S\gamma_5 = S^*$ and $\gamma_5 v\gamma_5 = v^*$. Hence conjugating $\operatorname{Tr}[(SvS)v(SvS)v]$ by γ_5 gives its complex conjugate after cyclically moving the bounded v past a trace-class product of the two Hilbert–Schmidt factors. This proves reality.

We now compare the lattice fourth-power trace with its continuum torus value. The four-power inequality used throughout is

$$|\operatorname{Tr} X^4 - \operatorname{Tr} Z^4| \le 4\max(\|X\|_4, \|Z\|_4)^3 \|X - Z\|_4. \tag{18.23}$$

It follows by expanding $X^4 - Z^4$ into four products with one $X - Z$ factor and applying Schatten Hölder. All the operators compared below have uniformly bounded Schatten-four norms; each bound follows from (18.19) or (18.20). In particular (18.5) gives $n^{-3}\sum_p \|R_m(p)\|_4^4 \le C_\rho(1+m^{-1})$, which, with $\|\eta A_\nu\|_{\ell^4} \le C_\beta^{\text{free}}\eta^{1/4}$, proves $\|Y_1\|_4 \le C_{\rho,\bar m,\beta}$.

Removing the bounded part of the inverse. Let $\chi(p) = 1_{\{|p|_\infty \le \pi/2\}}$ and S_c be the lattice multiplier $\chi(p)s_m(p)$ on four-component antiperiodic spinors. The complementary inverse $R_m(p) - J\chi(p)s_m(p)J^*$ is uniformly bounded: (18.4) controls the whole cube, and s_m is bounded on $1-\chi$. Thus

$$\|Y_1 - JS_cJ^*D_1\|_4 \le C_{\rho,\beta}\eta^{1/4}.$$

The two norms in (18.23) are bounded by the same estimates. Since $J^*J = 1$, cyclicity reduces the fourth-power trace of the light operator to that of $S_c w_1$, where

$$w_1 = J^*D_1J = -\frac{i\eta}{2}\sum_\nu [A_\nu(\rho - \gamma_\nu)U_\nu^+ - A_\nu^-(\rho+\gamma_\nu)U_\nu^-]. \tag{18.24}$$

This comparison costs $C\eta^{1/4}$ in the trace. The compression uses (18.7), and therefore has one four-component light species.

Replacing the vertex by multiplication. Move shifts to the left by $A_\nu U_\nu^+ = U_\nu^+ A_\nu^-$ and $A_\nu^- U_\nu^- = U_\nu^- A_\nu$. Then

$$\begin{aligned} w_1 - i\eta\gamma\cdot A = -\frac{i\eta}{2}\sum_\nu \big[&(\rho-\gamma_\nu)(U_\nu^+ - 1)A_\nu^- - (\rho+\gamma_\nu)(U_\nu^- - 1)A_\nu \\ &+ (\rho-\gamma_\nu)(A_\nu^- - A_\nu)\big]. \end{aligned} \tag{18.25}$$

The multiplier $\chi(p)s_m(p)(\rho \mp \gamma_\nu)(e^{\pm ip_\nu} - 1)$ has norm at most two, since $|e^{ip_\nu} - 1| \le |p_\nu|$. Its Schatten-four multiplier average is consequently bounded independently of the mass. The first two terms of (18.25) then cost $O(\eta^{1/4})$. The last term costs $O(m^{-1/4}\eta^{5/4}) = O(\eta)$ by (18.13). Therefore $\|S_c(w_1 - i\eta\gamma\cdot A)\|_4 \le C\eta^{1/4}$, and (18.23) gives the same trace error.

Identifying the truncated continuum momenta. The map $p \mapsto P = p/\eta$ identifies the retained lattice momenta with $\mathcal{P}_R^< = \{P \in \mathcal{P}_R : |P|_\infty \le \pi/(2\eta)\}$, and $\eta s_m(p) = s_{\bar m}(P)$. The operator $S_c i\eta\gamma\cdot A$ has range in this truncated space. Its fourth trace is thus that of its compression to the space. In the normalized momentum basis this compression has blocks

$$T_{\text{lat}}(P,P') = s_{\bar m}(P) i\sum_\nu \gamma_\nu n^{-3}\sum_c A_\nu(c)e^{-i\eta(P-P')\cdot c}.$$

Let $\widetilde A$ be the smooth interpolant and $\widehat A_R(k) = R^{-3}\int_{T_R}\widetilde A(y)e^{-ik\cdot y}\,\mathrm{d}y$. Absolute Fourier convergence, using four derivatives of the interpolant, gives the exact alias identity

$$n^{-3}\sum_c A_\nu(c)e^{-i\eta k\cdot c} = \widehat A_{R,\nu}(k) + \sum_{z\ne 0}\widehat A_{R,\nu}(k + 2\pi z/\eta). \tag{18.26}$$

For $k = P - P'$, $|k|_\infty \le \pi/\eta$. Four integrations by parts give $|\widehat{A}_{R,\nu}(k')| \le C_\beta^{\mathrm{free}} R^{-3} |k'|_\infty^{-4}$; the nonzero aliases in (18.26) are therefore bounded by $C_\beta^{\mathrm{free}} R^{-3}\eta^4$.

Replacing the discrete matrix elements by $\widehat{A}_R$ gives the compression T_{int} of $\chi S^R i\gamma\cdot\widetilde{A}$. Its norm is bounded. For the difference, sum the squared Hilbert–Schmidt norms of its blocks. There are at most $R^3\eta^{-3}$ choices of P', while $\sum_{P\in\mathcal{P}_R^<} |P|^{-2} \le CR^3\eta^{-1}$. Since $\|\gamma\cdot z\|_2^2 = 4|z|^2$, the alias bound gives

$$\|T_{\mathrm{lat}} - T_{\mathrm{int}}\|_2^2 \le C_\beta^{\mathrm{free}} (R^3\eta^{-1})(R^3\eta^{-3})(R^{-3}\eta^4)^2 = C_\beta^{\mathrm{free}}\eta^4.$$

All powers of R cancel. The Schatten-four norm is at most the Hilbert–Schmidt norm, so the trace error in this comparison is $O(\eta^2)$.

Replacing the interpolant and removing the cutoff. The difference between the compressions of $\chi S^R i\gamma\cdot\widetilde{A}$ and $\chi S^R i\gamma\cdot a^R$ has Schatten-four norm $O(\eta)$ by (18.20) and (18.13). For an operator X with $X = \chi X$, cyclicity gives $\operatorname{Tr} X^4 = \operatorname{Tr}(\chi X\chi)^4$; the same holds for $X = X\chi$. Thus we can remove the compression and compare $\chi S^R i\gamma\cdot a^R$ directly with $S^R i\gamma\cdot a^R$. The removed momentum tail in (18.21) gives

$$\|(1-\chi)S^R i\gamma\cdot a^R\|_4 \le C_\beta^{\mathrm{free}}\eta^{1/4}.$$

These are respectively the fourth and fifth comparisons; their trace errors are $O(\eta)$ and $O(\eta^{1/4})$. The preceding three comparisons, together with (18.16), prove the first line of (18.18). Here $\eta(1+\log(1+5/(\bar m\eta))) \le C_{\bar m}\eta^{1/4}$.

It remains to compare torus and infinite-space traces. Because the multiplication by a is supported in Q_β, both traces can be computed on $L^2(Q_\beta;\mathbb{C}^4)$, using extension by zero to $\mathbb{R}^3$ or antiperiodic extension to T_R. The same compression identity proves this localization. Convolution by the integrable kernel $S_{\bar m}(x)$ on antiperiodic functions agrees with S^R on its momentum basis and hence on L^2. After restricting its two variables to Q_β, its difference from the infinite-space kernel is

$$W_R(x,y) = \sum_{z\in\mathbb{Z}^3\setminus\{0\}} (-1)^{z_1+z_2+z_3} S_{\bar m}(x-y-Rz).$$

For $|u| \ge 1$ the Yukawa formula gives $\|S_{\bar m}(u)\| \le (3/(4\pi))e^{-\bar m|u|}$. Moreover $|x-y-Rz| \ge R|z|_\infty - D_\beta \ge (R-D_\beta) + |z|_\infty - 1 \ge 1$. Counting $24j^2+2$ images at $|z|_\infty = j$ yields

$$\|W_R(x,y)\| \le C_{\bar m} e^{-\bar m(R-D_\beta)}, \qquad x,y\in Q_\beta.$$

The series converges absolutely and uniformly there. Multiplication by $i\gamma\cdot a(y)$ then gives a Hilbert–Schmidt difference with this same exponential bound, since Q_β has finite volume. The infinite localized operator has bounded Schatten-four norm, and the torus one is bounded by it and the difference. Applying (18.23) proves the second line of (18.18).

The constants above involve only ρ, $\bar m$, the side of the support cube, derivative sup and L^1 norms of β, and moduli of its Fourier transform. All are unchanged by a real translation. This gives the asserted uniformity and the limit. □

18.5. **Evenness and placement of the probe.** The coefficients in (18.3) are real, and

$$c_{2j+1} = 0, \qquad c_2 = a_{\mathrm{ch}}, \qquad c_4 = b_{\mathrm{ch}} + 3c_2^2. \tag{18.27}$$

We verify the determinant identities for complex external connections: the later comparison of coefficients uses them on a complex circle.

For a real connection, each chamber bond involution with its phase, $R_i(B)$, is self-adjoint and unitary. The direction of a bond changes sign at its other endpoint. Consequently

$$D^{\mathrm{ch}}[B] = \frac{\rho}{2}\sum_i w_i(1-R_i(B)) + \frac{1}{2}\sum_i w_i(\gamma\cdot\widehat{d}_i)R_i(B),$$

where the first sum is nonnegative and the second is anti-self-adjoint. The two spin sectors $\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu)$ give adjoint restrictions, as in (2.18). Thus $\det(D^{\mathrm{ch}}[B] + m)$ is a squared absolute value. The real part of $D^{\mathrm{ch}}[B] + m$ is at least $m > 0$, so the determinant is strictly positive. In particular $N_0(t) > 0$ for real t.

The transpose identity extends charge symmetry to complex connections. Let C be the charge-conjugation matrix of §2.4. Transposing a phased bond reverses its direction and changes its phase from B to $-B$. The identity $C\gamma_v^\top C^{-1} = -\gamma_v$ changes the transposed matrix at the other endpoint back to the matrix at the original one. The diagonal part is scalar; hence

$$C(D^{\mathrm{ch}}[B] + m)^\top C^{-1} = D^{\mathrm{ch}}[-B] + m.$$

The transpose here is for the bilinear chamber pairing. In the site basis this pairing is $1/48$ times the identity, so this is the ordinary matrix transpose and preserves the determinant. Every matrix entry is a finite sum of constants and exponentials of linear functions of t. Therefore N_0 is entire, and the transpose identity gives $N_0(-t) = N_0(t)$ for all $t \in \mathbb{C}$. Expanding its logarithm at zero proves (18.27).

We also record exact covariance under lattice translations and cubic reflections. Let $g = (v, r)$, where $v \in \mathbb{Z}^3$ and r is a signed permutation. A reflection with normal $\widehat{n}$ has the spin lift $S_{\widehat{n}} = (\gamma \cdot \widehat{n})\gamma_5$, for which

$$S_{\widehat{n}}(\gamma \cdot z)S_{\widehat{n}}^{-1} = \gamma \cdot (z - 2(z \cdot \widehat{n})\widehat{n}), \qquad z \in \mathbb{C}^3.$$

Choose a word representing r in the three linear chamber reflections and let S_g be the corresponding product. It follows that $S_g(\gamma \cdot z)S_g^{-1} = \gamma \cdot rz$. The site map $(c, h) \mapsto (rc + v, rh)$ commutes with each bond involution and carries $\widehat{d}_i(c, h)$ to $r\widehat{d}_i(c, h)$. Therefore, for

$$(U_g\psi)(x) = S_g\psi(g^{-1}x), \qquad (g_*B)(c, c') = B(g^{-1}c, g^{-1}c'),$$

the hopping formula gives

$$D^{\mathrm{ch}}[g_*B] + m = U_g(D^{\mathrm{ch}}[B] + m)U_g^{-1}, \qquad B \text{ complex}. \tag{18.28}$$

This identity descends to the antiperiodic torus. Indeed $r^{-1}e_\mu = \pm e_\nu$, and either translation by ne_ν or by $-ne_\nu$ changes an antiperiodic spinor's sign. Thus U_g and its inverse preserve the spinor space. Taking determinants in (18.28) shows that N_0, c_j, a_{ch} and b_{ch} are unchanged when the external cochain is replaced by g_*B_t.

For a physical translation $v \in \eta\mathbb{Z}^3$, cube averaging and centered differences commute with translation. Periodization in (18.2) then gives $a_\eta[\beta(\,\cdot - v)](y) = a_\eta[\beta](y - v)$. This is exactly the cochain action with lattice translation v/η. The same determinant coefficients can therefore be used at each lattice placement of the probe and at each of its reflected images.

18.6. **The two-photon coefficient.** The coefficient is

$$a_{\mathrm{ch}} = \mathrm{Tr}(GD_2) - \mathrm{Tr}(GD_1GD_1).$$

Estimating the two traces separately would lose a power of the mesh. We keep their cancellation by expressing their difference as a polarization kernel. Its integral vanishes at zero momentum; parity removes the first momentum derivative.

Define the bosonic momentum representatives

$$\mathcal{K}_{\eta,R} = \left\{ \frac{2\pi}{R} j : j \in \mathbb{Z}^3,\ -n/2 < j_\nu \le n/2 \right\}, \qquad n = R/\eta,$$

and the midpoint transform of the unperiodized probe

$$h_{\eta,\nu}(K) = \eta^3 \sum_{y \in \eta\mathbb{Z}^3 + \eta e_\nu/2} a^0_{\eta,\nu}(y) e^{-iK\cdot y}. \tag{18.29}$$

The sum is finite. For real p, k put

$$\begin{aligned} L_{\mu\nu}(p,k) &= \delta_{\mu\nu}\,\mathrm{tr}[R_m(p)W_\nu(p)] \\ &\quad - \mathrm{tr}[R_m(p)V_\mu(p+k/2)R_m(p+k)V_\nu(p+k/2)], \end{aligned} \tag{18.30}$$

$$\Pi^{\eta,R}_{\mu\nu}(K) = \eta^{-1}n^{-3}\sum_{p\in\mathcal{P}_n} L_{\mu\nu}(p,\eta K),$$

$$\Pi^{\eta}_{\mu\nu}(K) = \eta^{-1}\int_{[-\pi,\pi]^3} L_{\mu\nu}(p,\eta K)\frac{\mathrm{d}p}{(2\pi)^3}. \tag{18.31}$$

Only $\Pi^{\eta,R}$ is a finite-volume polarization; the integral Π^{η} retains the lattice symbol but has no volume parameter. The exact coefficient is

$$a_{\mathrm{ch}}(\eta,R) = R^{-3}\sum_{K\in\mathcal{K}_{\eta,R}}\sum_{\mu,\nu}\overline{h_{\eta,\mu}(K)}h_{\eta,\nu}(K)\Pi^{\eta,R}_{\mu\nu}(K). \tag{18.32}$$

To check the midpoint and normalization in this formula, write $b_\nu(c) = \eta A_\nu(c)$ and $\widehat{b}_\nu(k) = n^{-3}\sum_c b_\nu(c)e^{-ik\cdot c}$. The derivative vertices of §18.3 give the blocks

$$\begin{aligned} D_1[p+k,p] &= \sum_\nu \widehat{b}_\nu(k)e^{-ik_\nu/2}V_\nu(p+k/2), \\ D_1[p,p+k] &= \sum_\mu \overline{\widehat{b}_\mu(k)e^{-ik_\mu/2}}V_\mu(p+k/2), \\ D_2[p,p] &= \sum_\nu\sum_k |\widehat{b}_\nu(k)|^2 W_\nu(p). \end{aligned}$$

Here k ranges over $\eta\mathcal{K}_{\eta,R}$; the last identity uses Parseval for the real function b_ν. For example, the positive hop in $D_1[p+k,p]$ is $-i\widehat{b}_\nu(k)e^{ip_\nu}T^+_\nu$, and the negative hop is $i\widehat{b}_\nu(k)e^{-i(p_\nu+k_\nu)}T^-_\nu$. Their sum is the stated midpoint vertex. Under $k_\nu \mapsto k_\nu + 2\pi$ both the phase and that vertex change sign, so their product is well-defined on momentum representatives.

Unfolding the periodization gives, for $k = \eta K$,

$$\widehat{b}_\nu(k)e^{-ik_\nu/2} = n^{-3}\eta\sum_{c\in\mathbb{Z}^3} a^0_{\eta,\nu}(\eta c + \eta e_\nu/2)e^{-iK\cdot(\eta c+\eta e_\nu/2)} = \eta R^{-3}h_{\eta,\nu}(K).$$

Substitute these three block identities into the two traces. The scalar normalization is $\eta^2R^{-6} = R^{-3}\eta^{-1}n^{-3}$, which proves (18.32). This identity requires no separation of the probe from its periodic images.

Proposition 18.2. *There is $C_\rho < \infty$, independent of $0 < \bar{m} \le 1$, $0 < \eta \le 1$, $R/\eta \in \mathbb{N}$, $R/\eta \ge 2$, and $K \in \mathbb{R}^3$, such that*

$$|\Pi^{\eta,R}_{\mu\nu}(K) - \Pi^{\eta}_{\mu\nu}(K)| \le C_\rho\bar{m}^{-3}R^{-4}, \tag{18.33}$$

$$|\Pi^{\eta}_{\mu\nu}(K)| \le C_\rho\bar{m}^{-1}|K|^2, \qquad |\nabla_K\Pi^{\eta}_{\mu\nu}(K)| \le C_\rho\bar{m}^{-1}|K|. \tag{18.34}$$

Proof. We first estimate the quadrature error. If F is a C^4 periodic function on the momentum torus, its Fourier coefficients satisfy

$$|\widehat{F}(z)| \le |z|_\infty^{-4}\max_j\int_{[-\pi,\pi]^3}|\partial_j^4F(p)|\frac{\mathrm{d}p}{(2\pi)^3}, \qquad z \ne 0.$$

This follows by four integrations by parts in a coordinate attaining $|z|_\infty$. The coefficients are absolutely summable, since $\sum_{z\ne0}|z|_\infty^{-4} < 42$. Their Fourier series equals F: its difference from F has zero Fourier coefficients, and density of trigonometric polynomials in continuous periodic

functions makes the difference zero. The antiperiodic mean selects the coefficients at nz with sign $(-1)^{z_1+z_2+z_3}$. Consequently

$$\left| n^{-3} \sum_{p \in \mathcal{P}_n} F(p) - \int_{[-\pi,\pi]^3} F(p) \frac{\mathrm{d}p}{(2\pi)^3} \right| \leq 42 n^{-4} \max_j \int_{[-\pi,\pi]^3} |\partial_j^4 F(p)| \frac{\mathrm{d}p}{(2\pi)^3}. \tag{18.35}$$

Set $x(p) = \operatorname{dist}(p, 2\pi\mathbb{Z}^3) + m$. On a period cube $x(p) = |p| + m$ and $x(p) < 7$. The inverse derivative estimate (18.8), used up to order four, and the bounded symbol derivatives give

$$|\partial_j^4 L_{\mu\nu}(p,k)| \leq C_\rho [x(p)^{-6} + x(p+k)^{-6}]. \tag{18.36}$$

For the contact term, each Leibniz summand has at most five powers of $x(p)^{-1}$; the remaining power is supplied by $x(p) < 7$. For the bubble term, a summand with a derivatives on the first inverse and b on the second has the factor $x(p)^{-1-a} x(p+k)^{-1-b}$, with $a + b \leq 4$. The other derivatives fall on bounded vertices. Since $u^{-s} v^{-t} \leq u^{-s-t} + v^{-s-t}$ for $s, t > 0$, and $x < 7$, the factor is bounded by a constant times the right side of (18.36). This bound is uniform in the shift k.

For $j > 3$, periodicity and the radial integral give

$$\int_{[-\pi,\pi]^3} x(p+u)^{-j} \, \mathrm{d}p \leq 4\pi \int_0^\infty \frac{r^2 \, \mathrm{d}r}{(r+m)^j} = \frac{8\pi m^{3-j}}{(j-1)(j-2)(j-3)}, \qquad u \in \mathbb{R}^3. \tag{18.37}$$

Indeed the integral over a translated period cube is unchanged, and on the original cube it is bounded by the integral over $\mathbb{R}^3$. The case $j = 6$ bounds the integral in (18.35) by $C_\rho m^{-3}$. The definitions of the two polarizations then give the first estimate, because

$$\eta^{-1} n^{-4} m^{-3} = \eta^{-1} (\eta/R)^4 (\bar{m}\eta)^{-3} = \bar{m}^{-3} R^{-4}.$$

For the quadratic bound, use the symmetric bubble

$$F_{\mu\nu}(p,k) = \operatorname{tr}[R_m(p - k/2) V_\mu(p) R_m(p + k/2) V_\nu(p)].$$

It is periodic in p. Translating the period cube by $-k/2$ shows that its integral is the integral of the bubble in (18.30). At $k = 0$ the whole integrand is a total derivative:

$$L_{\mu\nu}(p,0) = \partial_\mu \operatorname{tr}[R_m(p) V_\nu(p)]. \tag{18.38}$$

In fact $\partial_\mu R_m = -R_m V_\mu R_m$ and $\partial_\mu V_\nu = \delta_{\mu\nu} W_\nu$. Its integral therefore vanishes, including the contact term. We obtain

$$\Pi^\eta_{\mu\nu}(K) = -\eta^{-1} \int_{[-\pi,\pi]^3} [F_{\mu\nu}(p, \eta K) - F_{\mu\nu}(p,0)] \frac{\mathrm{d}p}{(2\pi)^3}. \tag{18.39}$$

Conjugation by the involution in (18.10) changes (p,k) to $(-p,-k)$ in $F_{\mu\nu}$; the signs of its two vertices cancel. Thus $F_{\mu\nu}(-p,-k) = F_{\mu\nu}(p,k)$. Its first k derivative at zero is odd in p, and hence

$$\int_{[-\pi,\pi]^3} (\partial_{k_a} F_{\mu\nu})(p,0) \, \mathrm{d}p = 0. \tag{18.40}$$

This removes the linear term as well as the constant term removed by (18.38).

The Hessian has two derivatives in total on the inverse factors; the midpoint vertices no longer depend on k. With $p_\pm = p \pm k/2$ and the vertices evaluated at p, it is

$$\begin{aligned} \partial_{k_a} \partial_{k_b} F_{\mu\nu} = \tfrac{1}{4} \big(& \operatorname{tr}[(\partial_a \partial_b R_m)(p_-) V_\mu R_m(p_+) V_\nu] \\ & - \operatorname{tr}[(\partial_a R_m)(p_-) V_\mu (\partial_b R_m)(p_+) V_\nu] \\ & - \operatorname{tr}[(\partial_b R_m)(p_-) V_\mu (\partial_a R_m)(p_+) V_\nu] \\ & + \operatorname{tr}[R_m(p_-) V_\mu (\partial_a \partial_b R_m)(p_+) V_\nu] \big). \end{aligned}$$

Apply (18.8) through order two and the same product inequality used above. It follows that

$$|\partial_{k_a}\partial_{k_b}F_{\mu\nu}(p,k)| \le C_\rho[x(p-k/2)^{-4} + x(p+k/2)^{-4}],$$

and (18.37) with $j=4$ gives

$$\int_{[-\pi,\pi]^3} |\partial_{k_a}\partial_{k_b}F_{\mu\nu}(p,k)|\frac{\mathrm{d}p}{(2\pi)^3} \le C_\rho m^{-1}, \qquad k\in\mathbb{R}^3. \tag{18.41}$$

Taylor's formula with integral remainder, followed by (18.40), now makes the two momentum powers explicit:

$$\Pi^\eta_{\mu\nu}(K) = -\eta\int_{[-\pi,\pi]^3}\int_0^1 (1-s)\sum_{a,b} K_aK_b(\partial_{k_a}\partial_{k_b}F_{\mu\nu})(p,s\eta K)\,\mathrm{d}s\,\frac{\mathrm{d}p}{(2\pi)^3}. \tag{18.42}$$

This gives the bound $C_\rho(\eta/m)|K|^2$. For the gradient, differentiate (18.39), subtract its zero-momentum derivative using parity, and apply the first-order integral remainder:

$$\partial_{K_a}\Pi^\eta_{\mu\nu}(K) = -\eta\int_{[-\pi,\pi]^3}\int_0^1 \sum_b K_b(\partial_{k_a}\partial_{k_b}F_{\mu\nu})(p,s\eta K)\,\mathrm{d}s\,\frac{\mathrm{d}p}{(2\pi)^3}.$$

All differentiations are under an integral over a compact cube of a smooth function, since $m>0$. Using (18.41) and $\eta/m=\bar m^{-1}$ proves (18.34). □

18.7. **Uniform bounds on the determinant coefficients.** We finish by pairing the polarization bounds with decay of the probe transform. This step needs no continuum evaluation of the two-photon kernel.

For each integer $j\ge 0$ there is $C_{\beta,j}<\infty$ such that

$$|h_{\eta,\nu}(K)| \le C_{\beta,j}(1+|K|)^{-j}, \qquad |K|_\infty \le \pi/\eta, \quad 0<\eta\le 1. \tag{18.43}$$

The constants are unchanged by any real translation of β. Here is the discrete integration by parts that makes this uniform in the mesh. Suppose a smooth function v is supported in a cube of side ℓ, and sum it over a translate of $\eta\mathbb{Z}^3$. The backward difference $\Delta_{\eta,\mu}v(y)=\eta^{-1}(v(y)-v(y-\eta e_\mu))$ satisfies $\|\Delta^j_{\eta,\mu}v\|_\infty \le \|\partial^j_\mu v\|_\infty$, by its integral representation. Its support is enlarged by at most $j\eta$ in one direction, and the number of sampling points, multiplied by η^3, is at most $(\ell+j+1)^3$. Summation by parts gives

$$[\eta^{-1}(1-e^{-i\eta K_\mu})]^j\eta^3\sum_y v(y)e^{-iK\cdot y} = \eta^3\sum_y \Delta^j_{\eta,\mu}v(y)e^{-iK\cdot y}.$$

On $|\eta K_\mu|\le 3\pi/2$ the modulus of the bracket is at least $0.3|K_\mu|$. Choose a coordinate attaining $|K|_\infty$. For $|K|\ge 1$ this is at least $(1+|K|)/12$; for $|K|<1$ use the point-count estimate directly. We thus obtain, on that momentum cube,

$$\left|\eta^3\sum_y v(y)e^{-iK\cdot y}\right| \le 12^j(\ell+j+1)^3\max(\|v\|_\infty, \max_\mu\|\partial^j_\mu v\|_\infty)(1+|K|)^{-j}.$$

Apply this to $v=a^0_{\eta,\nu}$ on $\eta\mathbb{Z}^3+\eta e_\nu/2$. Cube averaging has norm one on derivative sup norms, and a centered difference is an average of the corresponding first derivative. Consequently $\|\partial^j_\mu a^0_{\eta,\nu}\|_\infty \le 2|\beta|_{C^{j+1}}$, while its support has side at most $D_\beta+2\eta$. One may take

$$C_{\beta,j} = 2\cdot 12^j(D_\beta+j+3)^3|\beta|_{C^{j+1}}.$$

This proves (18.43), with position-independent constants and no assumption on R.

Proposition 18.3. *For each fixed ρ, $\bar{m}$ and β as above,*

$$|c_2| \le C_{\rho,\bar{m},\beta}, \qquad 0 < \eta \le 1, \quad R \ge 1, \quad R/\eta \in \mathbb{N}, \quad R/\eta \ge 2.$$

If in addition $R \ge D_\beta + 1$, then $|c_4| \le C_{\rho,\bar{m},\beta}$. The bounds are uniform in all real translations of β, and the coefficients have no dependence on e.

Proof. Proposition 18.2 gives

$$|\Pi^{\eta,R}_{\mu\nu}(K)| \le C_\rho(\bar{m}^{-3}R^{-4} + \bar{m}^{-1}|K|^2).$$

In (18.32) use the decay estimate with $j = 2$ for the quadrature term and $j = 3$ for the quadratic term. Since $R \ge 1$, both contributions are bounded by a constant times $(1 + |K|)^{-4}$; hence

$$|c_2| = |a_{\mathrm{ch}}| \le 9C_\rho(\bar{m}^{-3}C^2_{\beta,2} + \bar{m}^{-1}C^2_{\beta,3})R^{-3} \sum_{K\in\mathcal{K}_{\eta,R}} (1 + |K|)^{-4}.$$

The last sum is bounded uniformly for $R \ge 1$. To see this, tile $\mathbb{R}^3$ with cubes of side $2\pi/R$ centered at $K \in (2\pi/R)\mathbb{Z}^3$. On each cube, $1 + |P| \le (1 + \sqrt{3}\pi)(1 + |K|)$, so its volume times $(1 + |K|)^{-4}$ is bounded by a constant times the integral of $(1 + |P|)^{-4}$ there. Sum over the cubes; that integral over $\mathbb{R}^3$ is finite. Restricting the sum to $\mathcal{K}_{\eta,R}$ only decreases it. This proves the uniform second-order bound.

For the fourth coefficient, use (18.27) and Proposition 18.1. The infinite-space trace $b(f)$ is finite by (18.22), and the two errors in (18.18) are bounded for $R \ge D_\beta + 1$ and $0 < \eta \le 1$. Thus $|b_{\mathrm{ch}}|$ is bounded, and adding $3c_2^2$ gives the asserted bound for c_4. All constants in the proof depend on support sizes and derivative norms of β, not on its position. Independence of e follows directly from the external connection convention. □

19. Probes on the reference cells

The external-source identity uses a finitely supported bond cochain. The continuum cumulant uses the test $f = dd^*\beta$. We identify the discrete source with a nearby smooth test and place it strictly inside one reference cell. Copies in distinct cells then have orthogonal curvatures, which makes the normal-ordering factor separate over cells.

19.1. **The discrete source and the smooth test.** Fix a real nonzero $\beta \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$. Let D_β retain the unbuffered support-cube side of Section 18.1, and write $D^{\mathrm{buf}}_\beta = D_\beta + 4$. Choose a cube $Q_\beta = z_\beta + [-D^{\mathrm{buf}}_\beta/2, D^{\mathrm{buf}}_\beta/2]^3$ containing $\operatorname{supp}\beta + [-2, 2]^3$. Use the centred differences $\delta_{\eta,\mu}$ and cube average J_η of that section. Put

$$a'_{\eta,\nu} = -\sum_{\mu\neq\nu} \delta_{\eta,\mu}\beta_{\mu\nu},$$

$$f^\eta_{\mu\nu} = \delta_{\eta,\mu}a'_{\eta,\nu} - \delta_{\eta,\nu}a'_{\eta,\mu}. \tag{19.1}$$

All these functions are smooth and real. Since each centred difference enlarges support by at most $\eta/2$ in its direction,

$$\operatorname{supp} f^\eta, \ \operatorname{supp} J_\eta a'_\eta \subset \operatorname{supp}\beta + [-\eta, \eta]^3.$$

In particular their supports lie in Q_β for $0 < \eta \le 1$.

On the physical lattice $\eta\mathbb{Z}^3$ set

$$b_{\eta,\mu\nu}(c) = (J_\eta\beta_{\mu\nu})(c + \eta(e_\mu + e_\nu)/2), \qquad a^o_\eta = d^*_\eta b_\eta.$$

Here d_η is the mesh-normalized forward cochain differential and d^*_η its adjoint for the pairing $\eta^3\sum$. The adjoint formula gives

$$a^o_{\eta,\nu}(c) = (J_\eta a'_{\eta,\nu})(c + \eta e_\nu/2). \tag{19.2}$$

Thus $J_\eta a'_\eta$ is the interpolating vector field a^0_η already used in the free response. Transport to the unit lattice and define

$$\hat{\mathfrak{a}}^o_\eta(x, x+e_\nu) = \eta^{1/2} a^o_{\eta,\nu}(\eta x).$$

Its coboundary uses unscaled unit-lattice differences. The midpoint identities yield

$$d\hat{\mathfrak{a}}^o_\eta(\sigma) = f^\eta_{[\eta]}(\sigma). \tag{19.3}$$

Indeed, the bond values in the curl are evaluated at the two points displaced by $\pm\eta e_\mu/2$ from the plaquette centre. Their difference is $\eta\delta_{\eta,\mu} J_\eta a'_{\eta,\nu}$. Commutation of the average and differences then gives $\eta^{3/2} J_\eta f^\eta$ at that centre, which is the right side of (19.3).

Periodize the cochain on a torus $C(n)$ of physical side $R = n\eta$. Formula (6.3) gives the exact observable identity

$$\langle d\hat{\mathfrak{a}}_\eta, F^{\mathrm{u}}\rangle = F^\sharp(f^\eta). \tag{19.4}$$

The source is transverse: $a^o_\eta = d^*_\eta b_\eta$ is orthogonal to every closed bond cochain, by summation by parts. This applies to both the covering cochain and its periodization.

The continuum approximation has the uniform bound

$$\|f^\eta - f\|_{H^s} \le \eta^2 \|\beta\|_{H^{s+4}} \qquad (s \in \mathbb{R},\ 0 < \eta \le 1). \tag{19.5}$$

To see this, the difference symbol is $is_\mu(p)$ with $s_\mu(p) = 2\eta^{-1}\sin(\eta p_\mu/2)$. One has $|s_\mu| \le |p_\mu|$ and $|s_\mu - p_\mu| \le \eta^2 |p_\mu|^3/24$, so

$$|s_\mu s_\nu - p_\mu p_\nu| \le \eta^2 |p|^4/12.$$

Each component of $f^\eta - f$ contains four such terms. Cauchy–Schwarz on these terms, followed by summation over the three two-form components, gives $|\widehat{f^\eta - f}|^2 \le (\eta^2/3)^2 |p|^8 |\hat\beta|^2$. Plancherel's theorem proves (19.5).

19.2. **Reference geometry and placement.** Let $T^* \ge 1$ be the reference exponent furnished by the construction, and put

$$R^* = hL^{T^*}, \qquad n_* = R^*/\eta, \qquad p = n_*/L, \qquad w = p/2, \qquad \ell_c = \eta w = R^*/(2L).$$

Since h and η^{-1} are powers of the odd integer L, p is an odd integer, $w \in \mathbb{Z} + 1/2$, and the open unit-lattice reference cells are $Q^\circ_c = w(c + (0,1)^3)$. The reference torus $C(n_*)$ has $2L$ cells per side and $N_* = 8L^3$ in all. The second reference torus has side Ln_*. Every cutoff torus $n = 8hL^{T+N}$ with $T \ge T^* + 1$ is an integer multiple of Ln_*. These are in the range of Proposition 25.1, with $s_0 = 2L$ and $j = L$.

The probe fits the reference cells if

$$D^{\mathrm{buf}}_\beta + 3\eta \le \ell_c. \tag{19.6}$$

A placement is a vector $v \in \eta\mathbb{Z}^3$ satisfying

$$|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \le \eta/2.$$

Such a vector always exists by coordinatewise rounding. It translates Q_β into $[\eta, \ell_c - \eta]^3$: the displacement of its centre is at most $\eta/2$, and the half-side is at most $(\ell_c - 3\eta)/2$. Hence

$$\operatorname{supp}(\tau_v \beta) \subset [2 + \eta, \ell_c - 2 - \eta]^3. \tag{19.7}$$

Form the source $\hat{\mathfrak{a}}^o$ with this translated probe.

Every endpoint of a bond in the source support is within $3\eta/2$ in sup distance of $\operatorname{supp}(\tau_v\beta)$. By (19.7), its lattice coordinates therefore lie in $[2/\eta - 1/2, w - 2/\eta + 1/2]$. If a plaquette has nonzero $d\hat{\mathfrak{a}}^o$, at least one of its boundary bonds belongs to that support. Its other two corners are one unit away. Thus the whole plaquette lies in the box

$$B_0 = (w/2)(1,1,1) + [-r_0, r_0]^3, \qquad r_0 = w/2 - 2/\eta + 3/2. \tag{19.8}$$

Condition (19.6) gives $r_0 \geq 3$. The distance of this box from each bounding wall is $2/\eta - 3/2 \geq 1/2$, so $B_0 \subset Q_0^\circ$. Its reflected copies lie strictly inside the corresponding cells. This is the support margin needed by the local source identity.

19.3. **Orthogonal sources and translated observables.** Let g_c be the wall-group element carrying Q_0° to Q_c°. Put $\hat{\mathfrak{a}}_c^o = (g_c)_* \hat{\mathfrak{a}}^o$ on the covering lattice and periodize it on $C(n)$, where p divides n and $n \geq n_*$, to obtain $\hat{\mathfrak{a}}_c$. The period lattice identifies cell indices modulo $(2n/p)\mathbb{Z}^3$. The coboundary commutes with these actions and periodization, and its support lies in $B_c = g_c B_0$. For cell representatives on this torus,

$$\langle d\hat{\mathfrak{a}}_c, d\hat{\mathfrak{a}}_{c'} \rangle_n = v_\eta \, \mathbf{1}_{\{c=c'\}}, \qquad v_\eta = \|d\hat{\mathfrak{a}}^o\|^2. \tag{19.9}$$

The covering norm is the unweighted ℓ^2 norm, and the torus pairing is the counting pairing.

To prove the identity, unfold one factor:

$$\langle d\hat{\mathfrak{a}}_c, d\hat{\mathfrak{a}}_{c'} \rangle_n = \sum_\sigma \sum_{z \in \mathbb{Z}^3} d\hat{\mathfrak{a}}_c^o(\sigma) d\hat{\mathfrak{a}}_{c'}^o(\sigma + nz).$$

A nonzero summand has $\sigma \subset Q_c^\circ$ and $\sigma + nz \subset Q_{c'}^\circ$. The open cells are disjoint, so necessarily $c' = c + (2n/p)z$. For distinct torus cells there is no such z. For the same cell exactly one translate can contribute; its summands are the squares of $d\hat{\mathfrak{a}}_c^o$. The wall action preserves the counting norm, which proves (19.9). Changing the placement by a vector in $\eta\mathbb{Z}^3$ translates the covering source by an integer vector. Thus v_η depends only on β and η, and is independent of the torus and the placement.

For a closed cochain u define $X_c(u) = \langle d\hat{\mathfrak{a}}_c, u \rangle_n$. Every polynomial in this observable is read by one and the same local integrand on all these tori. On the cover its local expression is

$$\widehat{A} \longmapsto P\left(\sum_{\sigma \subset B_c} d\hat{\mathfrak{a}}_c^o(\sigma) (d\widehat{A})(\sigma) \right).$$

This is a scalar gauge-invariant polynomial. Cauchy–Schwarz bounds its argument by

$$\sqrt{v_\eta} \left(\sum_{\sigma \subset B_c} |d\widehat{A}(\sigma)|^2 \right)^{1/2}.$$

The polynomial is therefore bounded by

$$C_{\mathrm{pol}} \exp\left(\frac{1}{4} \sum_{\sigma \subset B_c} |d\widehat{A}(\sigma)|^2 \right)$$

for a finite constant C_{pol} depending on the polynomial and the covering source. This is the growth condition for $\mathcal{A}(Q_c^\circ)$.

The local boxes embed in the torus. Indeed, writing $p = 2m_p + 1$, condition (19.6) gives $m_p \geq 7$, and

$$2r_0 + 2 = w - 4/\eta + 5 \leq w + 1 < p \leq n.$$

For each c, the translates of $B_c + [-1,1]^3$ by nz, $z \in \mathbb{Z}^3$, are consequently disjoint. On a local lift, $d\widehat{A} = F^{\mathrm{u}}$; unfolding the finite cochain sum reads the polynomial as $P(X_c)$. For a real polynomial its reflected copy from the base cell is exactly $P(X_c)$, since conjugation fixes its coefficients. This applies to the centered Wick polynomials.

The variance in (19.9) approaches the continuum normal-ordering variance:

$$\left| \sqrt{v_\eta} - \|f\|_2 \right| \leq 21\eta |\beta|_{C^3} (D_\beta^{\mathrm{buf}})^{3/2} + \eta^2 \|\beta\|_{H^4}. \tag{19.10}$$

To see this, first (19.3) gives $v_\eta = |(f^\eta)_\eta|_\eta^2$. Let $\Pi_\eta f^\eta$ be its componentwise cell-average step function on a period cube. Its L^2 norm is $\sqrt{v_\eta}$. Each first derivative of $f_{\mu\nu}^\eta$ is bounded by $4|\beta|_{C^3}$,

from the four centred-difference terms in (19.1) and the identity $\delta_{\eta,\mu}g = \int_{-1/2}^{1/2}(\partial_\mu g)(\,\cdot + t\eta e_\mu)\,\mathrm{d}t$. The pointwise error of the cell average is at most $12\eta|\beta|_{C^3}$. It vanishes outside the periodic images of the buffered cube, whose measure in one period is $(D_\beta^{\mathrm{buf}})^3$. Summing the three component squares bounds the L^2 error by $21\eta|\beta|_{C^3}(D_\beta^{\mathrm{buf}})^{3/2}$. The norm triangle inequality and (19.5) prove (19.10).

Finally the even cell indices give pure translations. For $y \in (R^*/L)\mathbb{Z}^3 = p\eta\mathbb{Z}^3$ set $c(y) = 2y/(p\eta) \in 2\mathbb{Z}^3$. Then $g_{c(y)}$ is the unit-lattice translation by y/η, and

$$X_{c(y)} = F^\sharp(\tau_{v+y}f^\eta). \tag{19.11}$$

Two such cells agree on $C(n)$ exactly when their physical translations agree modulo $R\mathbb{Z}^3$. Any fixed finite set of distinct translations therefore occupies distinct cells once the volume is sufficiently large. Lattice translation invariance removes the common placement v from the expectation of any polynomial in these observables. Its integrability follows from Gaussian domination. Thus the reference-cell product estimates give estimates for the translates $\tau_y f^\eta$ used in Section 26.

20. Exact external-response identities

We express the generating function of a curvature observable as a determinant in an external connection. The identity is first proved in each integer flux sector. In the zero-flux sector an unfolding of the integer cochains replaces the compact photon integral by a Gaussian integral over transverse cochains and an average over the three twists. We compute its normalization before applying the source shift.

20.1. **The determinant at complex connections.** Use lattice units and fix $q > 0$, $r_q = 2\pi/q$ and a real lattice mass μ_{lat}. Put $M_{\mathrm{ch}} = \mu_{\mathrm{lat}} + \frac{\rho}{2}\sum_i w_i$. For the cutoff laws of Theorem 1.1, $\mu_{\mathrm{lat}} = \boldsymbol{\mu}_N$. The weights are the standard chamber weights, and $0 < \rho < 1$. For an integer triple $n = (n_1, n_2, n_3)$ let

$$V = |n| = n_1n_2n_3, \qquad B_n = \prod_{\mu=1}^{3}\{0, \dots, n_\mu - 1\}, \qquad \mathcal{L}_n = B_n \times \{1, 2, 3\}.$$

In this subsection each n_μ may be as small as one. Cochains are periodic functions on the oriented covering cells, so the $3V$ link orbits remain distinct even at those small sides. The chamber spinors are antiperiodic in each direction and have dimension $192V$. Write D_B for the massless chamber operator with phases $\exp(iqB)$, and $\det_n$ for its finite-dimensional determinant. In the phase convention of §18.1, $D_B = D^{\mathrm{ch}}[qB]$. The weight $1/48$ in the spinor pairing does not rescale this determinant.

The determinant is a polynomial in the $6V$ variables $\exp(iqB(b))$ and $\exp(-iqB(b))$, of degree at most 32 in each. In particular it is entire in the complex link values. To see the degree bound, choose the site basis on $B_n \times O_{\mathrm{cub}}$ and include the antiperiodic sign whenever a neighbor is reduced into B_n. Each of the four rows at a site (c, g) has constant entries except for the one external-hop block. If $ge_1 = e_\mu$, this block contains $\exp(iqB_\mu(c))$; if $ge_1 = -e_\mu$, it contains $\exp(-iqB_\mu(c - e_\mu))$. For a fixed oriented photon link there are eight labels at each end. Thus each phase variable occurs in exactly $8 \cdot 4 = 32$ rows. The Leibniz determinant chooses one entry in each row, proving the bound. The five site labels of the diagonal and the four hops are distinct, even if a side is one, so no other phase is hidden in one of these blocks.

The two invariances needed below hold for complex cochains:

$$\det_n(D_{B+r_qk} + \mu_{\mathrm{lat}}) = \det_n(D_B + \mu_{\mathrm{lat}}), \qquad \det_n(D_{B+d\lambda} + \mu_{\mathrm{lat}}) = \det_n(D_B + \mu_{\mathrm{lat}}), \tag{20.1}$$

where k is integer valued and λ is a periodic complex zero-cochain. The first identity follows directly from the phases. For the second, multiplication by $\exp(iq\lambda(c))$ preserves antiperiodicity and has inverse multiplication by $\exp(-iq\lambda(c))$. On an external hop the factor

$$e^{iq(B-d\lambda)(c,c')}e^{iq\lambda(c')} = e^{iq\lambda(c)}e^{iqB(c,c')}$$

gives $D_{B-d\lambda} = e^{iq\lambda}D_B e^{-iq\lambda}$. Internal hops lie over one photon site and give the same conjugation. This proves the determinant identity.

For $s \geq 0$ define

$$K_n(s) = \left(|M_{\mathrm{ch}}| + \frac{1+\rho}{2}\left(\sum_i w_i - 6 + 6e^{qs} \right) \right)^{192V},$$

$$|\det_n(D_B + \mu_{\mathrm{lat}})| \leq K_n(\|\operatorname{Im} B\|_\infty). \tag{20.2}$$

Indeed each chamber bond involution permutes the site representatives. For an internal hop its phase has modulus one; for an external hop the modulus is at most $e^{q\|\operatorname{Im} B\|_\infty}$. The hopping matrix has norm $w_i(1+\rho)/2$. The operator norm is therefore bounded by the base of the power in (20.2), and a determinant in dimension $192V$ is bounded by that norm to the same power. This estimate is uniform in the real part of the connection.

The twist connection is

$$H_\theta = \sum_\mu \frac{\theta_\mu}{qn_\mu}\mathbf{1}_\mu, \qquad \mathbf{1}_\mu = \sum_{c\in B_n} \mathbf{1}_{(c,\mu)}. \tag{20.3}$$

Its determinant has period 2π in each complex twist variable. For later use we need the cochain implementing this period. Set

$$k_\nu^{(\mu)}(c) = \delta_{\mu\nu}1_{\{c_\mu = n_\mu - 1\}}, \qquad \lambda^{(\mu)}(c) = -\frac{r_q}{n_\mu}c_\mu, \qquad c \in B_n,$$

with periodic extension. On links not crossing the μ seam, $d\lambda^{(\mu)} = -r_q/n_\mu$ in direction μ and zero in the other directions. On a seam link its value is $r_q - r_q/n_\mu$. Thus

$$H_{2\pi e_\mu} = r_q k^{(\mu)} - d\lambda^{(\mu)}. \tag{20.4}$$

The two invariances in (20.1) prove the asserted twist period. For $n_\mu = 1$, $\lambda^{(\mu)} = 0$ and $k^{(\mu)} = \mathbf{1}_\mu$, so the same formula holds.

20.2. **Real and integer cochains on the torus.** From now on $n_\mu \geq 4$. The counting pairings on cochains are extended bilinearly when their arguments are complex. Use

$$V_{\mathrm{g}} = dC^0(n), \qquad V_{\mathrm{h}} = \operatorname{span}_{\mathbb{R}}\{\mathbf{1}_1, \mathbf{1}_2, \mathbf{1}_3\}, \qquad V_{\mathrm{t}} = (V_{\mathrm{g}} + V_{\mathrm{h}})^\perp.$$

We recall the cochain arguments needed for the unfolding and for its Jacobian. In particular the integer condition in the flux sum requires an integer primitive, not merely a real one.

Let

$$\mathcal{P}_n^{\mathrm{per}} = \{p : p_\mu = 2\pi j_\mu/n_\mu,\ 0 \leq j_\mu < n_\mu\}, \qquad e_p(c) = V^{-1/2}e^{ip\cdot c}, \qquad \xi_\mu(p) = e^{ip_\mu} - 1.$$

The characters e_p are orthonormal. At each momentum p the coboundary on exterior-valued cochains is exterior multiplication by $\xi(p)$. If $\iota_{\bar\xi}$ denotes contraction by $\bar\xi$, then

$$\iota_{\bar\xi}(\xi \wedge u) + \xi \wedge \iota_{\bar\xi}u = |\xi|^2 u.$$

For $p \neq 0$ a closed coefficient u is consequently $\xi \wedge (|\xi|^{-2}\iota_{\bar\xi}u)$, and is exact. At zero momentum the cochains are constant and the coboundary vanishes. Nonzero and zero modes are orthogonal, and taking real parts of a complex primitive gives a real primitive for a real cochain. Therefore

$$\ker(d : C^1 \to C^2) = V_{\mathrm{g}} \oplus V_{\mathrm{h}}, \qquad C^1 = V_{\mathrm{g}} \oplus V_{\mathrm{h}} \oplus V_{\mathrm{t}}, \qquad \ker(d : C^2 \to C^3) = dC^1 \oplus \mathrm{Harm}^2. \tag{20.5}$$

Here Harm^2 is the space of constant two-cochains. The same Fourier formula gives $d^2 = 0$.

For a closed integer two-cochain z, define $w_{\mu\nu}(z)$ as its sum over one coordinate (μ, ν) two-torus. The sum is independent of the transverse coordinate: sum $dz = 0$ over the layer of cubes between two consecutive planes. Differences in the two tangential directions cancel by periodicity; the remaining difference is that of the two plane sums. The fluxes give

$$z \in dC^1(n;\mathbb{Z}) \quad \Longleftrightarrow \quad w_{\mu\nu}(z) = 0 \text{ for all } \mu < \nu. \tag{20.6}$$

The integer coefficients follow from the cellular cohomology of the product of the three subdivided circles. Its cochain complex is precisely the integer complex used here. The three coordinate two-tori give a basis of $H_2(T^3;\mathbb{Z})$, and $H_1(T^3;\mathbb{Z})$ is free. The universal coefficient theorem identifies $H^2(T^3;\mathbb{Z})$ with $\mathrm{Hom}(H_2(T^3;\mathbb{Z}),\mathbb{Z})$ by evaluation. The class of z evaluates to $w_{\mu\nu}(z)$ on each basis two-torus, so it vanishes exactly when these sums vanish. Vanishing of this cellular class is membership in $dC^1(n;\mathbb{Z})$. [7, Theorems 3.2, 3.5 and 3B.6]

We next describe the closed integer one-cochains. If $d\lambda$ is integer valued, its sum along a path from 0 to c shows that $\lambda(c) - \lambda(0)$ is integer valued. If $d\lambda = 0$, the same path argument shows that λ is constant. It follows that

$$C^1(n;\mathbb{Z}) \cap V_{\mathrm{g}} = dC^0(n;\mathbb{Z}).$$

Each seam cochain $k^{(\mu)}$ in (20.4) is closed, because its multiple by r_q lies in $V_{\mathrm{g}} + V_{\mathrm{h}}$. For an arbitrary closed integer one-cochain k, (20.5) gives $k = d\lambda + \sum_\mu t_\mu \mathbf{1}_\mu$. Its sum along a periodic μ loop is $n_\mu t_\mu$, an integer j_μ. Subtracting $\sum_\mu j_\mu k^{(\mu)}$ leaves an integer exact cochain, so

$$k = d\lambda' + \sum_\mu j_\mu k^{(\mu)}, \qquad \lambda' \in C^0(n;\mathbb{Z}), \quad j \in \mathbb{Z}^3. \tag{20.7}$$

The harmonic projection of k determines j, and then determines $d\lambda'$. Moreover

$$r_q k = \left(r_q d\lambda' + \sum_\mu j_\mu d\lambda^{(\mu)} \right) + H_{2\pi j}. \tag{20.8}$$

The first term is exact and the second is harmonic. Their coupling through j will matter when choosing a fundamental domain.

The scalar Laplacian $\Delta_0 = d^*d$ has eigenvalues $|\xi(p)|^2 = \sum_\mu (2 - 2\cos p_\mu)$ and a one-dimensional kernel. Write

$$\det{}'\Delta_0 = \prod_{p \in \mathcal{P}_n^{\mathrm{per}} \setminus \{0\}} |\xi(p)|^2, \qquad \mathcal{L}(n) = \log \det{}'\Delta_0. \tag{20.9}$$

At each nonzero momentum the transverse one-cochains are the two-dimensional space $\xi(p)^\perp$. For $v \in \mathbb{C}^3$,

$$\sum_{\mu<\nu} |\xi_\mu v_\nu - \xi_\nu v_\mu|^2 = |\xi|^2 |v|^2 - |\bar{\xi} \cdot v|^2.$$

Thus d^*d acts on that transverse space as multiplication by $|\xi(p)|^2$, and

$$\dim V_{\mathrm{t}} = 2V - 2, \qquad \det(d^*d|_{V_{\mathrm{t}}}) = (\det{}'\Delta_0)^2. \tag{20.10}$$

Diagonalizing this positive real form gives its Gaussian integral

$$\mathcal{G}_n := \int_{V_{\mathrm{t}}} e^{-\|dA\|^2/2} \, \mathrm{d}\sigma(A) = \frac{(2\pi)^{V-1}}{\det{}'\Delta_0}. \tag{20.11}$$

The measure σ is Euclidean Lebesgue measure for the counting pairing, not a coordinate measure chosen along a nonorthogonal basis.

There is one more determinant needed for the exact directions. The cochains $d\mathbf{1}_c$, $c \in B_n \setminus \{0\}$, form both a real basis of V_{g} and an integer basis of $dC^0(n;\mathbb{Z})$. Indeed one subtracts the value at

zero from a zero-cochain; independence follows because the resulting zero-cochain has value zero there and cannot be a nonzero constant. Their Gram determinant is

$$\det\left(\langle d\mathbf{1}_c, d\mathbf{1}_{c'}\rangle\right)_{c,c'\neq 0} = \frac{\det{}'\Delta_0}{V}. \tag{20.12}$$

To prove this formula, let S be the scalar Laplacian matrix in the site basis. The coefficient of t in $\det(t+S)$ is $\det{}'\Delta_0$, by its eigenvalues. Multilinearity of the determinant in columns identifies that coefficient with the sum of the V principal minors obtained by deleting one row and column. All these minors have the same determinant, since site translations conjugate S by permutations. The minor at zero is the Gram matrix above.

20.3. **Unfolding the zero-flux sector.** Let Φ be a Borel function on $C^2(n)$ with $|\Phi(u)| \le C_\Phi e^{C_\Phi\|u\|}$. Let W be a bounded continuous complex function on $C^1(n)$, invariant under addition of real exact cochains and of r_q times integer one-cochains. Then

$$\begin{aligned}&\sum_{\substack{z\in C^2(n;\mathbb{Z}),\, dz=0\\ w(z)=0}} \int_{[0,r_q)^{\mathcal{L}_n}} \Phi(dA + r_q z) e^{-\|dA+r_q z\|^2/2} W(A)\,\mathrm{d}A \\ &\qquad = \mathcal{N}_n \int_{[-\pi,\pi]^3} \frac{\mathrm{d}\theta}{(2\pi)^3} \int_{V_{\mathrm{t}}} \Phi(dA) e^{-\|dA\|^2/2} W(A+H_\theta)\,\mathrm{d}\sigma(A), \end{aligned}\tag{20.13}$$

$$\mathcal{N}_n = r_q^{V+2} (\det{}'\Delta_0)^{1/2}. \tag{20.14}$$

Both sides converge absolutely. The coefficient comes from the lattice of exact gauge transformations and the three periods of the harmonic directions; we give the unfolding with both contributions.

First, the Gaussian sum (2.5) gives

$$\sum_{z\in C^2(n;\mathbb{Z})} \int_{[0,r_q)^{\mathcal{L}_n}} e^{-\epsilon\|dA+r_q z\|^2}\,\mathrm{d}A \le (r_q C_q(\epsilon))^{3V}.$$

Here the integer closedness and flux restrictions have been dropped. Completing the scalar square bounds $|\Phi(u)|e^{-\|u\|^2/2}$ by $C_\Phi e^{C_\Phi^2} e^{-\|u\|^2/4}$. Together with boundedness of W, this proves absolute convergence of the left side.

Let $\mathcal{Z}^1_{\mathbb{Z}} = \ker d \cap C^1(n;\mathbb{Z})$ and choose representatives $\mathcal{R}$ of $C^1(n;\mathbb{Z})/\mathcal{Z}^1_{\mathbb{Z}}$. By (20.6), $k \mapsto dk$ is a bijection from $\mathcal{R}$ onto the closed zero-flux integer two-cochains. Put

$$P(A) = \Phi(dA) e^{-\|dA\|^2/2} W(A), \qquad \mathcal{D}_1 = \bigcup_{k\in\mathcal{R}} ([0,r_q)^{\mathcal{L}_n} + r_q k).$$

The two invariances of W turn the left side of (20.13) into $\int_{\mathcal{D}_1} P(A)\,\mathrm{d}A$. The boxes are disjoint. Every A has a unique decomposition into its coordinatewise remainder in $[0,r_q)$ and r_q times an integer cochain. Decomposing that cochain into its chosen representative and a closed one shows that $\mathcal{D}_1$ is a Borel fundamental domain for $r_q\mathcal{Z}^1_{\mathbb{Z}}$. The function P is invariant under that group.

We replace this domain by

$$\mathcal{F}_{\mathrm{g}} = \left\{ r_q \sum_{c\neq 0} t_c\, d\mathbf{1}_c : 0 \le t_c < 1 \right\}, \qquad \mathcal{F}_{\mathrm{h}} = \{H_\theta : \theta \in [-\pi,\pi)^3\}, \qquad \mathcal{D}_2 = \mathcal{F}_{\mathrm{g}} + \mathcal{F}_{\mathrm{h}} + V_{\mathrm{t}}.$$

To check that it is a fundamental domain for the same group, write $A = A_{\mathrm{g}} + A_{\mathrm{h}} + A_{\mathrm{t}}$. Formula (20.8) shows that subtracting $r_q k$ changes the harmonic part by $H_{2\pi j}$ and the exact part by $\sum_\mu j_\mu d\lambda^{(\mu)} + r_q d\lambda'$. There is exactly one j placing the harmonic part in $\mathcal{F}_{\mathrm{h}}$. For this j there is exactly one $d\lambda'$ placing the exact part in $\mathcal{F}_{\mathrm{g}}$, by the integer basis used in (20.12). The transverse

part is unchanged. Uniqueness in (20.7) gives uniqueness of k. The orthogonal decomposition identifies $\mathcal{D}_2$ with a product of Borel sets, so it is Borel.

For an integrable periodic Borel function, the integral over two such fundamental domains is the same. Indeed partition $\mathcal{D}_1$ into its intersections with $\mathcal{D}_2 + h$, $h \in r_q \mathcal{Z}^1_{\mathbb{Z}}$, and translate each intersection by $-h$. They partition $\mathcal{D}_2$, and invariance of both the measure and the function gives equality. Tonelli proves it first for nonnegative functions; applying it to the modulus then justifies the complex case by absolute integrability. Thus $\int_{\mathcal{D}_1} P = \int_{\mathcal{D}_2} P$.

The two finite-dimensional volumes are

$$\operatorname{vol}(\mathcal{F}_{\mathrm{g}}) = r_q^{V-1} (\det{}' \Delta_0 / V)^{1/2}, \tag{20.15}$$

$$\mathrm{d}A_{\mathrm{h}} = V^{1/2} q^{-3}\, \mathrm{d}\theta = r_q^3 V^{1/2} \frac{\mathrm{d}\theta}{(2\pi)^3}. \tag{20.16}$$

The first follows from the square root of the Gram determinant. For the second, the constant cochains $V^{-1/2}\mathbf{1}_\mu$ are orthonormal, and the matrix of $\theta \mapsto H_\theta$ in this basis has diagonal entries $V^{1/2}/(q n_\mu)$. Their product is $V^{1/2} q^{-3}$. Their combined factor is precisely $\mathcal{N}_n$. Since $P(A_{\mathrm{g}} + A_{\mathrm{h}} + A_{\mathrm{t}}) = \Phi(dA_{\mathrm{t}}) e^{-\|dA_{\mathrm{t}}\|^2/2} W(A_{\mathrm{t}} + A_{\mathrm{h}})$, it is independent of A_{g}. Fubini over the orthogonal decomposition and the two displayed volume formulas prove (20.13), including absolute convergence of its right side.

Apply this identity to $W(A) = \det_n(D_A + \mu_{\mathrm{lat}})$. Its hypotheses follow from (20.1) and (20.2). If $Z_n^{(0)}[\Phi]$ is the zero-flux part of the configuration integral, we have proved

$$Z_n^{(0)}[\Phi] = \mathcal{N}_n \int \frac{\mathrm{d}\theta}{(2\pi)^3} \int_{V_{\mathrm{t}}} \Phi(dA) e^{-\|dA\|^2/2} \det_n(D_{A+H_\theta} + \mu_{\mathrm{lat}})\, \mathrm{d}\sigma(A). \tag{20.17}$$

The twist integral always runs over $[-\pi, \pi]^3$. For real A and θ the determinant is nonnegative. Also $Z_n^{(0)}[1] > 0$ at every real μ_{lat}: the lower bound in Proposition 2.3 was obtained from the single term $z = 0$, which belongs to this sector. Dividing (20.17) by its value at $\Phi = 1$ identifies the zero-flux curvature law with the law of dA under the harmonic-averaged transverse measure.

20.4. **The Gaussian normalization and matter density.** Let $F_n = dC^1(n) = dV_{\mathrm{t}}$ and let γ_n be the standard Gaussian probability measure on this Euclidean space. The map $d_{\mathrm{t}} = d|_{V_{\mathrm{t}}}$ is a linear bijection onto F_n, by (20.5); write $A(u) = d_{\mathrm{t}}^{-1} u$. Changing variables along this map gives

$$\int_{V_{\mathrm{t}}} G(dA) e^{-\|dA\|^2/2}\, \mathrm{d}\sigma(A) = \mathcal{G}_n \int G(u)\, \mathrm{d}\gamma_n(u). \tag{20.18}$$

Indeed the Jacobian is $|\det d_{\mathrm{t}}|^{-1}$, and $\dim F_n = 2V - 2$. The Gaussian density contributes $(2\pi)^{V-1}$; setting $G = 1$ identifies the constant with (20.11). The formula holds for nonnegative Borel functions, and for the exponentially bounded complex functions used above by their Gaussian integrability.

Define the photon factor and normalized matter factor by

$$Z_n^{\mathrm{ph}} = \mathcal{N}_n \mathcal{G}_n = r_q^{V+2} (2\pi)^{V-1} e^{-\mathcal{L}(n)/2}, \tag{20.19}$$

$$z_{r,n}(q, \mu_{\mathrm{lat}}) = \int \frac{\mathrm{d}\theta}{(2\pi)^3} \int \det_n(D_{A(u)+H_\theta} + \mu_{\mathrm{lat}})\, \mathrm{d}\gamma_n(u). \tag{20.20}$$

Then (20.17) and (20.18) imply

$$Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n}, \qquad 0 < z_{r,n} \le K_n(0). \tag{20.21}$$

Positivity follows from that of $Z_n^{(0)}[1]$; the upper bound follows because both integration measures in (20.20) are probability measures.

On a cubic torus $V = n^3$, fix a positive reference mass $\bar{m}$, mesh η and a real pair $(m_{\mathrm{in}}, \varepsilon^{\mathrm{v}}_{\mathrm{in}})$, with $\mu_{\mathrm{lat}} = \bar{m}\eta + m_{\mathrm{in}}$. The density with external cochain a is

$$\mathrm{dens}^{\theta,a}(u) = e^{-\varepsilon^{\mathrm{v}}_{\mathrm{in}} n^3} \frac{\det_n(D_{A(u)+H_\theta+a} + \mu_{\mathrm{lat}})}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)}. \tag{20.22}$$

The denominator is positive, since the real part of $D^{\mathrm{ch}}[0] + \bar{m}\eta$ is at least $\bar{m}\eta > 0$. At $a = 0$ this is a nonnegative bounded continuous density. Let $\nu^{\theta,0}$ be its measure against γ_n and let $\bar{\nu}$ be the twist average. Equations (20.17) and (20.18) give

$$\bar{\nu}(\Phi) = \frac{e^{-\varepsilon^{\mathrm{v}}_{\mathrm{in}} n^3}}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta) Z^{\mathrm{ph}}_n} Z^{(0)}_n[\Phi], \tag{20.23}$$

$$\bar{\nu}(1) = \frac{e^{-\varepsilon^{\mathrm{v}}_{\mathrm{in}} n^3} z_{r,n}}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)} \in (0, \infty). \tag{20.24}$$

Thus $\bar{\nu}/\bar{\nu}(1)$ is exactly the zero-flux curvature law. The vacuum normalization cancels in this quotient. The quantities Z^{ph}_n and $z_{r,n}$ depend only on $(n, q, \mu_{\mathrm{lat}})$ and the fixed chamber parameters; the auxiliary split $\mu_{\mathrm{lat}} = \bar{m}\eta + m_{\mathrm{in}}$ does not change them.

20.5. **Holomorphic external partition functions.** Write $\mathfrak{F}(n)$ for the exponentially bounded Borel functions Φ used above. For $w \in \mathbb{Z}^3$ let $\mathcal{M}_w$ be the closed integer two-cochains of flux w. For a complex one-cochain a define

$$Z^{\mathrm{x},(w)}_n(a;\Phi) = \sum_{z \in \mathcal{M}_w} \int_{[0,r_q)^{\mathcal{L}_n}} \Phi(F) e^{-\|F\|^2/2} \det_n(D_{A+a} + \mu_{\mathrm{lat}})\, \mathrm{d}A, \qquad F = dA + r_q z, \tag{20.25}$$

$$\mathbf{Z}^{\mathrm{x}}_n(a;\Phi) = \int \frac{\mathrm{d}\theta}{(2\pi)^3} \int_{V_{\mathrm{t}}} \Phi(dA) e^{-\|dA\|^2/2} \det_n(D_{A+H_\theta+a} + \mu_{\mathrm{lat}})\, \mathrm{d}\sigma(A). \tag{20.26}$$

When $\Phi = 1$ we omit it from the notation. These are partition functions in an external connection; the curvature F in their photon weight is not shifted.

Proposition 20.1. *For every $\Phi \in \mathfrak{F}(n)$ both expressions converge absolutely, locally uniformly in a, and are entire in its $3V$ complex link values. The sum over all sectors converges absolutely, locally uniformly as well. Moreover*

$$Z^{\mathrm{x},(0)}_n(a;\Phi) = \mathcal{N}_n \mathbf{Z}^{\mathrm{x}}_n(a;\Phi), \qquad Z^{\mathrm{x},(w)}_n(0;\Phi) = Z^{(w)}_n[\Phi]. \tag{20.27}$$

In particular $\mathbf{Z}^{\mathrm{x}}_n(0) > 0$.

Proof. For real A, θ the imaginary parts of $A + a$ and $A + H_\theta + a$ are both $\operatorname{Im} a$. On a polydisc $\max_b |a(b)| \le s$ the determinant is bounded by $K_n(s)$. If $|\Phi(u)| \le C_\Phi e^{C_\Phi \|u\|}$, the absolute value of each integrand is consequently bounded by

$$K_n(s) C_\Phi e^{C_\Phi^2} e^{-\|F\|^2/4} \quad \text{or} \quad K_n(s) C_\Phi e^{C_\Phi^2} e^{-\|dA\|^2/4}.$$

The first majorant can be summed over every integer two-cochain and integrated over the compact box, giving $K_n(s) C_\Phi e^{C_\Phi^2} (r_q C_q(1/4))^{3V}$. The second is integrable by (20.11): rescaling A by $\sqrt{2}$ gives

$$\int_{V_{\mathrm{t}}} e^{-\|dA\|^2/4}\, \mathrm{d}\sigma(A) = 2^{V-1} \mathcal{G}_n.$$

These bounds also show local uniform absolute convergence of the sector sum.

We recall the parameter-integral argument to justify entireness and all later differentiations. Let (Y, ν) be a σ-finite measure space. Suppose $v(y, a)$ is measurable in (y, a), entire in $a \in \mathbb{C}^d$ for

each y, and bounded on each polydisc by a ν-integrable $M_s(y)$. If $v_\alpha(y)$ are its Taylor coefficients at zero, integration over the product of circles of radius s gives

$$v_\alpha(y) = s^{-|\alpha|} \int_{[0,2\pi]^d} v(y, se^{i\theta}) e^{-i\alpha\cdot\theta} \frac{\mathrm{d}\theta}{(2\pi)^d}, \qquad |v_\alpha(y)| \le s^{-|\alpha|} M_s(y).$$

The integral formula makes the coefficients measurable. For $\max_j |a_j| \le s$, applying the bound at radius $2s$ yields

$$\sum_\alpha \int |v_\alpha(y) a^\alpha| \,\mathrm{d}\nu(y) \le \sum_\alpha 2^{-|\alpha|} \int M_{2s}(y) \,\mathrm{d}\nu(y) = 2^d \int M_{2s}(y) \,\mathrm{d}\nu(y) < \infty.$$

Thus the Taylor series may be integrated term by term, and its integral is an entire function with coefficients $\int v_\alpha$. This also proves differentiation under the integral at zero. Translating the parameter to an arbitrary a^0 proves it there, using the majorant $M_{s+\|a^0\|_\infty}$. This argument applies to a counting measure times Lebesgue measure as well as to the transverse integral.

Apply this argument to the two displayed majorants. The dependence on a in each integrand is a finite sum of exponentials of linear forms, by the determinant polynomial. The other factors are Borel and independent of a, so the hypotheses hold also when Φ is merely Borel. This proves the asserted holomorphy. For fixed complex a, the function $W_a(A) = \det_n(D_{A+a} + \mu_{\mathrm{lat}})$ on real cochains is bounded by $K_n(\|\operatorname{Im} a\|_\infty)$, continuous, and has the two invariances required in (20.13). That formula gives (20.27); the value at zero follows from the definitions. Its positive zero value follows from $Z_n^{(0)}[1] > 0$ and $\mathcal{N}_n > 0$. □

The complex densities in (20.22) define finite complex measures $\nu^{\theta,a}$ and their twist average $\bar\nu^a$ against γ_n. Their absolute values are bounded by

$$|\mathrm{dens}^{\theta,a}(u)| \le \frac{e^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3} K_n(\|\operatorname{Im} a\|_\infty)}{\det_n(D^{\mathrm{ch}}[0] + \bar m\eta)}. \tag{20.28}$$

This bound is independent of (θ, u). Fubini therefore writes the averaged measure as a bounded Borel density against γ_n. Applying (20.18) at each twist, and then (20.27), gives

$$\bar\nu^a(\Phi) = \frac{e^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3}}{\det_n(D^{\mathrm{ch}}[0] + \bar m\eta)\mathcal{G}_n} \mathbf{Z}_n^{\mathrm{x}}(a;\Phi) = \frac{e^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3}}{\det_n(D^{\mathrm{ch}}[0] + \bar m\eta) Z_n^{\mathrm{ph}}} Z_n^{\mathrm{x},(0)}(a;\Phi). \tag{20.29}$$

The exponential majorants in the proof above justify both changes of integration for $\Phi \in \mathfrak{F}(n)$. In particular, for every Borel set $E \subset F_n$, $a \mapsto \bar\nu^a(E)$ is entire and

$$|\bar\nu^a(E)| \le \frac{e^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3} K_n(\|\operatorname{Im} a\|_\infty)}{\det_n(D^{\mathrm{ch}}[0] + \bar m\eta)}.$$

Indeed F_n is a closed linear subspace, so its Borel set indicator extends to a bounded Borel function on $C^2(n)$. Composition with any linear parameter map $t \mapsto a(t)$ remains entire. These statements concern the unnormalized complex measures; the positive normalization at $a = 0$ is (20.24).

20.6. **The source shift and Wick derivatives.**

Proposition 20.2. *For every flux w and complex one-cochain a,*

$$Z_n^{(w)}[e^{\langle da, F\rangle}] = e^{\langle da, da\rangle/2} Z_n^{\mathrm{x},(w)}(a). \tag{20.30}$$

Both sides, and their sums over w, converge absolutely. Consequently

$$Z_n[e^{\langle da, F\rangle}] = e^{\langle da, da\rangle/2} \sum_w Z_n^{\mathrm{x},(w)}(a). \tag{20.31}$$

The quadratic form in the prefactor is bilinear; there is no complex conjugation of da.

Proof. First let a be real. On real cochains define

$$P_a(A) = \sum_{z \in \mathcal{M}_w} e^{-\|dA + r_q z\|^2/2 + \langle da, dA + r_q z\rangle} \det_n(D_A + \mu_{\mathrm{lat}}).$$

This is a nonnegative Borel function. Adding $r_q \mathbf{1}_b$ to A adds $d\mathbf{1}_b$ to the integer two-cochain in its curvature. This map permutes $\mathcal{M}_w$: its image is closed and has the same flux, since each exact plaquette cochain has zero plane sums. The determinant is unchanged. Hence P_a is invariant under $r_q C^1(n;\mathbb{Z})$. Both the compact box and its translate by a are fundamental domains for this group. The domain argument in §20.3 gives

$$\int_{[0,r_q)^{\mathcal{L}_n}} P_a(A)\,\mathrm{d}A = \int_{[0,r_q)^{\mathcal{L}_n}} P_a(A+a)\,\mathrm{d}A.$$

On the right the exponent completes to

$$-\tfrac{1}{2}\|F + da\|^2 + \langle da, F + da\rangle = -\tfrac{1}{2}\|F\|^2 + \tfrac{1}{2}\|da\|^2.$$

Tonelli now identifies the two sides with (20.30) for real a. Their finiteness also follows from the Gaussian majorant used below.

For complex a, the right side of (20.30) is entire by Proposition 20.1. The left side is entire by the same parameter-integral argument. In fact, on $\max_b |a(b)| \le s$,

$$\|d\,\mathrm{Re}\,a\| \le 4s\sqrt{3V} =: c_s, \qquad \left| e^{-\|F\|^2/2 + \langle da, F\rangle} \det_n(D_A + \mu_{\mathrm{lat}}) \right| \le K_n(0) e^{c_s^2} e^{-\|F\|^2/4}.$$

This can be summed and integrated over all integer two-cochains, so it supplies an integrable polydisc majorant. Both sides agree on the real cochain space. An entire function vanishing there has all complex partial derivatives zero there: at a real point these derivatives can be taken along real increments. Repeating this for all derivatives shows that every Taylor coefficient at zero vanishes. Thus the two entire functions agree for all complex a. Finally the same Gaussian majorant permits summation over sectors; the sector sum on the right converges absolutely by Proposition 20.1. □

Let $a^{(1)}, \dots, a^{(r)}$ be real cochains and $X_j = \langle da^{(j)}, F\rangle$. Dividing (20.30) at $w = 0$ by its positive zero-source value yields the normalized generating identity

$$\mathbb{E}_n^{\mathrm{av}} \exp\left(\sum_j z_j X_j - \tfrac{1}{2} \sum_{i,j} z_i z_j \langle da^{(i)}, da^{(j)}\rangle \right) = \frac{\mathbf{Z}_n^{\mathrm{x}}(\sum_j z_j a^{(j)})}{\mathbf{Z}_n^{\mathrm{x}}(0)}, \qquad z \in \mathbb{C}^r. \tag{20.32}$$

The expectation $\mathbb{E}_n^{\mathrm{av}}$ is the zero-flux law identified in (20.17). The Gaussian factor on the left is determined by the counting pairing of the curvatures of the source cochains.

Use the reference cells of Section 19 on the torus n_*. Their index set is $\mathcal{C}_* = \{0, \dots, 2L-1\}^3$, and $\widehat{a}_c$ is the periodization of $\hat{\mathfrak{a}}_c^o$. These are real periodic cochains. Equation (19.9) gives

$$\langle d\widehat{a}_c, d\widehat{a}_{c'}\rangle = v_\eta\, 1_{\{c=c'\}}. \tag{20.33}$$

The variance $v_\eta = \|d\hat{\mathfrak{a}}_\eta^o\|^2$ is independent of the cell and the placement. This is the cell-geometry input needed here: the external-response proof uses neither the cell width nor its offsets otherwise. Put

$$a(t) = e^{-1} \sum_{c \in \mathcal{C}_*} t_c \widehat{a}_c, \qquad \bar{Z}^{\mathrm{ext}}(t) = \int \frac{\mathrm{d}\theta}{(2\pi)^3} \int \det_{n_*}(D_{A(u) + H_\theta + a(t)} + \mu_{\mathrm{lat}})\,\mathrm{d}\gamma_{n_*}(u).$$

The reference torus has $n_* \ge 4$, and $e > 0$ is the physical charge in this convention. Equation (20.18) gives

$$\bar{Z}^{\mathrm{ext}}(t) = \mathcal{G}_{n_*}^{-1} \mathbf{Z}_{n_*}^{\mathrm{x}}(a(t)), \qquad \mathring{\Phi}(t) := \frac{\bar{Z}^{\mathrm{ext}}(t)}{\bar{Z}^{\mathrm{ext}}(0)} = \frac{\mathbf{Z}_{n_*}^{\mathrm{x}}(a(t))}{\mathbf{Z}_{n_*}^{\mathrm{x}}(0)}. \tag{20.34}$$

Its zero value in the denominator is positive, and $\mathring{\Phi}$ is entire. With $X_c = \langle d\widehat{a}_c, F\rangle$, (20.32) becomes

$$\mathbb{E}^{\mathrm{av}}_{n_*} \exp \sum_c (z_c X_c - \tfrac{1}{2} v_\eta z_c^2) = \mathring{\Phi}(ez). \tag{20.35}$$

Define the Hermite polynomials with this variance by $H_j(y) = \partial_z^j e^{zy - v_\eta z^2/2}|_{z=0}$. In particular

$$H_2(y) = y^2 - v_\eta, \qquad H_4(y) = y^4 - 6v_\eta y^2 + 3v_\eta^2.$$

To differentiate (20.35) under the expectation, on a polydisc $\max_c |z_c| \le s$ use

$$\left| \exp \sum_c (z_c X_c - \tfrac{1}{2} v_\eta z_c^2) \right| \le \exp\left(s|\mathcal{C}_*|\sqrt{v_\eta}\|F\| + \tfrac{1}{2}|\mathcal{C}_*| s^2 v_\eta \right).$$

This is exponentially bounded in the real curvature and is integrable under the zero-flux law, by the Gaussian majorant. The parameter-integral argument therefore applies to this probability measure. The resulting products of H_2 or H_4 are also integrable: $|X_c| \le \sqrt{v_\eta}\|F\|$ bounds them by a finite polynomial in $\|F\|$, which is bounded by a constant times $e^{\|F\|}$.

For $m = 2$ or 4 and a subset $S \subset \mathcal{C}_*$, differentiate in precisely those variables. The factor ez on the right supplies $e^{m|S|}$, giving

$$\mathbb{E}^{\mathrm{av}}_{n_*} \prod_{c\in S} e^{-m} H_m(X_c) = \left(\prod_{c\in S} \partial_{t_c}^m \mathring{\Phi} \right)(0).$$

Expanding the product over subsets proves, for every real c_m,

$$\mathbb{E}^{\mathrm{av}}_{n_*} \prod_{c\in\mathcal{C}_*} (e^{-m} H_m(X_c) - c_m) = \left(\prod_{c\in\mathcal{C}_*} (\partial_{t_c}^m - c_m) \mathring{\Phi} \right)(0). \tag{20.36}$$

The derivative operators commute because $\mathring{\Phi}$ is entire. This identity is the finite-volume quantity to which the terminal interacting estimates will be applied; all its normalization factors have already been included in (20.34).

21. Terminal operators and Gaussian bounds

The terminal estimates use the fine and coarse field maps, their weighted Grassmann norms, and Gaussian domination in the zero-flux sector. We establish these inputs before the partition comparison and the response estimates.

21.1. **Reference scales and the probe.** We use the reference torus of the tuned flow. Fix $\bar{m} \in (0,1]$, put $h = L^{\lfloor \log_L(1/\bar{m})\rfloor}$, and assume $0 < e\sqrt{h} \le e_1$ and $N \ge N_{\mathrm{ref}}(e)$. By (16.1) and (16.7),

$$\begin{aligned} \eta = L^{-N}, \qquad K = N + \log_L h, \qquad a_j = L^{j-N}, \\ e_j = e\sqrt{a_j}, \qquad \eta_j = L^{-j} = \eta/a_j, \qquad \ell_j = R^*/a_j. \end{aligned} \tag{21.1}$$

Thus a point with level-j coordinates c has physical coordinates $a_j c$, and $T_j = \eta_j \mathbb{Z}^3/\ell_j\mathbb{Z}^3$. At $j = 0$ the charge is $q = e\sqrt{\eta}$. The symbols e_0, e_1 without a running index retain their meanings as charge ceilings. For a bond cochain a on the fine unit torus, its rescaling is

$$a^{[j]}(b) = L^{j/2} a(L^j b). \tag{21.2}$$

The physical scale a_j is a number, while a denotes a connection. On the reference torus, (16.8) and (16.4) give

$$k_* \in \{K-1, K-2\}, \qquad a_* \in \{h/L, h/L^2\}, \qquad \ell_* = Mr_*, \qquad r_* > 31L + 16.$$

These stop-scale quantities are independent of $N \geq N_{\mathrm{ref}}(e)$. Write r_{loc} for the localization power in Section 16.

Let $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ be real and nonzero. Use the cube $Q_\beta = z_\beta + [-D_\beta^{\mathrm{buf}}/2, D_\beta^{\mathrm{buf}}/2]^3$ of Section 19, where $D_\beta^{\mathrm{buf}} = D_\beta + 4$. In this section a placement $v \in \eta\mathbb{Z}^3$ is admitted if, in physical coordinates,

$$v + Q_\beta \subset [\ell_c/8, 7\ell_c/8]^3, \qquad \ell_c = R^*/(2L). \tag{21.3}$$

A centered placement has $|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \leq \eta/2$. It is admitted when $D_\beta^{\mathrm{buf}}/2 + \eta/2 \leq 3\ell_c/8$. This follows by adding the displacement of the cube center to its half-side.

Form $\widehat{\mathfrak{a}}$ with $\tau_v\beta$ as in (19.2), and periodize it on $C(n_*)$, $n_* = R^*/\eta$. For $c \in \mathcal{C}_* = \{0,\dots,2L-1\}^3$, write $\widehat{\mathfrak{a}}_c$ for its cell copy. In physical coordinates the wall map carrying the first cell to cell c acts in coordinate μ by translation through $2i\ell_c$ if $c_\mu = 2i$, and by reflection $y_\mu \mapsto (2i+2)\ell_c - y_\mu$ if $c_\mu = 2i+1$. Define the marked connections by

$$a(t) = e^{-1}\sum_c t_c \widehat{\mathfrak{a}}_c, \qquad \widehat{c}_c = e^{-1}\widehat{\mathfrak{a}}_c^{[k_*]}, \qquad \mathscr{C}(t) = a(t)^{[k_*]} = \sum_c t_c \widehat{c}_c. \tag{21.4}$$

For a level-j bond function A, ∂A uses forward differences divided by η_j, and

$$|\delta_\alpha \partial A|_\infty = \sup_{\mu,\nu} \sup_{0<d(x,x')\leq 1} \frac{|\partial_\nu A_\mu(x') - \partial_\nu A_\mu(x)|}{d(x,x')^\alpha}.$$

The external norm of a fine connection is $|a|_{\mathrm{ext}} = |a^{[k_*]}|_\infty + |\partial a^{[k_*]}|_\infty + |\delta_\alpha \partial a^{[k_*]}|_\infty$.

We define the uniform probe constants over both possible stop scales. For $\eta' = L^{-N'}$ put

$$g_\nu^{\eta'}[\beta'] = -\sum_{\mu\neq\nu} \delta_{\eta',\mu} J_{\eta'} \beta'_{\mu\nu}.$$

This is the smooth vector field $J_{\eta'}a'_{\eta'}$ in the midpoint formula. Let $\mathcal{I}_\beta$ consist of triples (N', a_{s}, r') with $N' \geq 0$, $a_{\mathrm{s}} \in \{h/L, h/L^2\}$, $N' + \log_L a_{\mathrm{s}} \geq 1$, and $r' \in \{\pm 1\}^3$. r' is identified with its diagonal orthogonal matrix, and $U_{r'}\beta(x) = (\wedge^2 r')\beta(r'^{-1}x)$. At such a triple set $\eta'' = \eta'/a_{\mathrm{s}}$, $g^0 = g^{\eta'}[U_{r'}\beta]$, and

$$G_\nu(c) = \sqrt{a_{\mathrm{s}}}\, g_\nu^0(a_{\mathrm{s}}c + \eta' e_\nu/2) \quad (c \in \eta''\mathbb{Z}^3), \qquad W_\nu(x) = \sqrt{\eta'}\, g_\nu^0(\eta' x + \eta' e_\nu/2) \quad (x \in \mathbb{Z}^3).$$

The three norms of G are taken at mesh η''. Set

$$C_\beta = \sup_{\mathcal{I}_\beta}(|G|_\infty + |\partial G|_\infty + |\delta_\alpha \partial G|_\infty), \qquad \mathbf{D} = \sup_{\mathcal{I}_\beta} \|dW\|^2_{\ell^2(\mathrm{plaquettes})}. \tag{21.5}$$

These suprema depend on $\bar{m}, L, \alpha, \beta$, and contain no charge, torus, or placement. They have the finite bounds

$$C_\beta \leq 2\sqrt{h}(1 + h + 3h^2)|\beta|_{C^3}, \qquad \mathbf{D} \leq 48|\beta|^2_{C^2}(D_\beta^{\mathrm{buf}} + 3)^3. \tag{21.6}$$

Indeed the centered difference is the average of the corresponding derivative along its segment, and $J_{\eta'}$ has norm one on each derivative sup norm. Each component of g^0 has two summands, hence $|D^i g_\nu^0|_\infty \leq 2|\beta|_{C^{i+1}}$ for $i = 0,1,2$. This gives $|G|_\infty \leq 2a_{\mathrm{s}}^{1/2}|\beta|_{C^1}$. A first lattice difference of G is $a_{\mathrm{s}}^{3/2}$ times an average of $\partial_\mu g_\nu^0$, so $|\partial G|_\infty \leq 2a_{\mathrm{s}}^{3/2}|\beta|_{C^2}$. The difference of these averages at c, c' is bounded by $6a_{\mathrm{s}}|c-c'|\,|\beta|_{C^3}$. Since $|c-c'| \leq 1$ and $\alpha < 1$, the third norm is at most $6a_{\mathrm{s}}^{5/2}|\beta|_{C^3}$. Sum these bounds and use $a_{\mathrm{s}} \leq h$.

For the energy, each curl component of W is a difference of two first differences, and has modulus at most $4(\eta')^{3/2}|\beta|_{C^2}$. A supporting plaquette has a supporting bond, whose endpoints in physical coordinates lie in $r'Q_\beta$. Its basepoint is therefore in that cube enlarged by η' in each negative coordinate direction. There are at most $(D_\beta^{\mathrm{buf}}/\eta' + 2)^3$ basepoints and three plaquette directions. Squaring the pointwise curl bound gives $48|\beta|^2_{C^2}(D_\beta^{\mathrm{buf}} + 2\eta')^3$, which implies (21.6).

At the reference stop, $(N, a_*, 1) \in \mathcal{I}_\beta$. The midpoint formula and (21.2) give

$$\widehat{\mathfrak{a}}_\nu^{[k_*]}(c) = \sum_{z\in\mathbb{Z}^3} G_\nu(c - v/a_* + \ell_* z).$$

Supporting bond endpoints lie in $a_*^{-1}(v+Q_\beta)$. Distinct periodic copies of that cube have distance at least $\ell_* - 3\ell_*/(8L) > \ell_*/2 > 1 + 2\eta_*$. Each functional defining one of the three norms reads only bonds with endpoint distances at most $1 + 2\eta_*$. It therefore reads at most one periodic copy. Taking its supremum identifies each torus norm with the corresponding norm of G. Translations and coordinate reflections preserve these norms, including the one-bond shift that reverses an oriented component. The same argument applies to the cell copies.

The image of $v + Q_\beta$ under each wall map lies in $\prod_\mu [c_\mu \ell_c + \ell_c/8, (c_\mu + 1)\ell_c - \ell_c/8]$. For distinct cells, a separating coordinate on the periodic circle gives distance at least $\ell_c/4$: any joining arc contains the two margins of size $\ell_c/8$. Thus supporting endpoints of distinct $\widehat{c}_c$ have stop-coordinate distance at least $\ell_*/(8L)$, which is larger than $1 + 2\eta_*$. Again each external-norm functional reads at most one copy, so

$$|a(t)|_{\mathrm{ext}} \le C_\beta |t|_\infty / e, \qquad v_\eta = \|d\widehat{\mathfrak{a}}\|^2 \le \mathbf{D}. \tag{21.7}$$

For the second bound, the covering curvature dW is supported in the cube of side $D_\beta^{\mathrm{buf}}/\eta + 2 < n_*$ obtained by enlarging $\eta^{-1}(v + Q_\beta)$. Its periodic translates do not overlap, so periodization preserves its counting norm. Distinct cell curvatures also have disjoint plaquette supports: a common plaquette would have vertices in both endpoint sets at physical distance at most $\eta \le 1$, whereas $\ell_c/4 > 1$. This gives (19.9) for every admitted placement.

Let P_c be the unit-block centers whose blocks contain an endpoint of a supporting bond of $\widehat{c}_c$, and let P_c^+ be their $(M + 1)$-neighborhood. For $y \in P_c$, $y' \in P_{c'}$ choose the corresponding endpoints in $\Delta_y, \Delta_{y'}$. Each endpoint is at distance less than $1/2$ from its center. Consequently

$$d_{\mathrm{sep}} := \min_{c\neq c'} d(P_c, P_{c'}) \ge \ell_*/(8L) - 1 = M r_*/(8L) - 1, \tag{21.8}$$

$$|P_c^+| \le p_\beta := (D_\beta^{\mathrm{buf}}/a_* + 2M + 4)^3. \tag{21.9}$$

For the count, the endpoints have a lift in a cube of side $D_\beta^{\mathrm{buf}}/a_*$. A center adds less than $1/2$ to each coordinate bound, and its neighborhood adds $M + 1$. The resulting open interval has length $D_\beta^{\mathrm{buf}}/a_* + 2M + 3$ and contains fewer than one plus that many integers. Multiply the three counts. If the lattice probe vanishes, all these sets are empty and $d_{\mathrm{sep}} = \infty$.

21.2. **Terminal operators and local insertions.** At the stop level write $\eta_* = L^{-k_*}$, $T_* = \eta_* \mathbb{Z}^3/\ell_*\mathbb{Z}^3$ and $T_*^0 = \mathbb{Z}^3/\ell_*\mathbb{Z}^3$. The fine chamber spinors have counting measure $\eta_*^3\, 48^{-1} \sum_{c,g}$, and the coarse spinors have four components per unit site and weight one. We use

$$k_* \ge 1, \qquad \ell_* > 2M + 2, \qquad \gamma^F M \ge 16,$$
$$1 - L^{-1} =: b_- \le b_* \le 1, \qquad L^{-3} =: \mu_- < \mu_* \le L^{-1}.$$

The running mass μ_* is positive; the bare mass coupling is treated separately. Let

$$F_*(A) = D_A + \mu_* + b_* J P_*(A) J^*, \qquad S_*(A) = F_*(A)^{-1},$$
$$H_*(A) = b_* S_*(A) J Q_*^T(-A), \qquad D_*(A) = b_* - b_* Q_*(A) J^* H_*(A). \tag{21.10}$$

These are the blocked operators already constructed in (11.8) and (11.9), at $k = k_*$. The symbol F_* here is an operator on spinors, distinct from the curvature variable in the external partition function.

For a unit site y let Δ_y be its fine block. The averaging phases have the form $\varphi_A(y,c) = e^{i\phi_A(y,c)}$, where $\phi_A = e_*\eta_*\tau_{k_*}A$ is linear in A. The nested block centers give

$$|\phi_B(y,c)| \le \tfrac{3}{2}e_*|B|_\infty, \qquad |\upsilon_B(c,c')| \le 3|c-c'|e_*|B|_\infty. \tag{21.11}$$

The first phase uses only bonds inside Δ_y; the second, which defines $U_A(c,c') = e^{i\upsilon_A(c,c')}$, uses only bonds in the coordinate box between its endpoints. For the first bound, the path lengths at block scales $L^j\eta_*$ add up to $\frac{3}{2}(L-1)\eta_*\sum_{j<k_*}L^j \le \frac{3}{2}$. Each of the six coordinate-order paths between c and c' has length at most $3|c-c'|$, which gives the second bound.

For a complex direction B, let P_B be the unit sites y whose block Δ_y contains an endpoint of a bond in $\operatorname{supp} B$. Define

$$\epsilon_B = e_*|B|_\infty, \qquad P_B^+ = \{y : d(y,P_B) \le M+1\}.$$

The distance is the periodic sup distance on the unit lattice. For an operator O between fine, coarse or pair spaces, localize each fine spinor in its unit block and each pair at its first point, and put

$$\begin{aligned} |O|(y,y') &= \sup_{\|v\|_\infty\le 1} \|\mathbf{1}_y O \mathbf{1}_{y'} v\|_\infty, \\ \|O\|_\gamma &= \max\left\{ \sup_y \sum_{y'} |O|(y,y')e^{\gamma d(y,y')}, \sup_{y'}\sum_y |O|(y,y')e^{\gamma d(y,y')} \right\}. \end{aligned} \tag{21.12}$$

The spinor sup norm uses the maximum of component moduli. Inserting the unit-block partition between two factors gives $|O_1O_2| \le |O_1||O_2|$ and $\|O_1O_2\|_\gamma \le \|O_1\|_\gamma\|O_2\|_\gamma$. For coarse spinor operators,

$$\|O\|_{\ell^2\to\ell^2} \le 2\|O\|_0, \qquad \|O\|_1 \le 4\sum_{y,y'}|O|(y,y'). \tag{21.13}$$

Indeed the matrix row sums are bounded by $\|O\|_0$ and the column sums by $4\|O\|_0$, so Schur's test gives the first bound. Polar decomposition bounds the trace norm by the sum of all entry moduli, and the entries in one block have total modulus at most four times its maximum row sum.

We use the fixed constants C^F, γ^F of the fermion construction, with its calibration $a' \ge 1$ and

$$C^F \ge 608\exp(1)\max(1,b)a', \qquad C^F \ge 64\exp(1)b^2a', \qquad b = 1.$$

The first inequality also dominates the component norm bounds $64a'$ and $608a'$ for the fine inverse and its short covariant quotients. The charge ceiling satisfies $6Le_*^\varepsilon \le 1$ and $3Me_*^\varepsilon \le 1$; on $\widetilde{G}_*$, $\mathfrak{i}(A) < e_*^\varepsilon$. The phase estimates therefore give $|\varphi_A^{\pm1}|, |U_A^{\pm1}| \le \exp(1)$, including every allowed long pair. The unweakened estimate (9.41), at its spectral parameter $y = \infty$, and the block phases give the following terminal bounds. On the outer field domain $\widetilde{G}_*$ the inverse S_* is analytic, and, with $E(y,y') = e^{-\gamma^F d(y,y')}$,

$$|S_*|, \quad |\delta_{\alpha,A}S_*|, \quad |H_*|, \quad |\delta_{\alpha,A}H_*| \le C^F E. \tag{21.14}$$

The two pair majorants are restricted to $d_X < 1$, including pairs inside a chamber cluster. The field domain

$$\mathcal{W} = \{A_r + \mathscr{C} : A_r \in \widetilde{R}_* \text{ real}, \ \mathscr{C} \in \mathfrak{X}_*\}$$

is contained in $\widetilde{G}_*$, is invariant under $A \mapsto -A$, and satisfies $\mathfrak{i}(A) := e_*|\operatorname{Im} A|_\infty < 1/16$ and $3M\mathfrak{i}(A) < 1/4$. For long pairs, $|\delta_{\alpha,A}v(x,x')|_c \le \mathrm{e}|v(x')|_c + |v(x)|_c$. The constants $C^F \ge 1$ and $0 < \gamma^F \le 1$ are uniform in the mesh and reference torus. These are the stop-level forms of the preceding walk and local inverse estimates. To obtain the bounds for H_*, write $H_* = b_*S_*JQ_*^T(-A)$. The last factor has Euclidean spinor sup norm at most $\exp(1)$, preserves each unit block, and has no chamber factor: it repeats the same coarse spinor over the 48 labels,

multiplied by the block phase. Composition with the localized inverse bounds gives constants $32\exp(1)ba'$ and $304\exp(1)ba'$. Since $|v|_c \le |v|_2 \le 2|v|_c$ on $\mathbb{C}^4$, their component sup norms are at most twice these constants, hence at most C^F.

We shall also need the global component row bound, without an exponential weight:

$$|H_*^{\#}(A)|_{1,\infty} \le C^F, \qquad |\bar{H}_*^{\#}(A)|_{1,\infty} \le C^F. \tag{21.15}$$

Here the rows include every allowed pair, and the columns are the four coarse components at the unit sites. Indeed, the unweighted consequences of (9.41) give

$$\|H_*(A)v\|_{\mathrm{su}} \le 32\exp(1)ba'\|v\|_{\mathrm{su}}, \qquad \|\delta_{\alpha,A}H_*(A)v\|_{\mathrm{su}} \le 304\exp(1)ba'\|v\|_{\mathrm{su}}$$

for short pairs. These include distinct labels at the same photon site. For a long pair, $1 \le d_X(x,x') \le M$, so $d_X^{-\alpha} \le 1$ and $|U_A| \le \exp(1)$. Consequently

$$|(\delta_{\alpha,A}H_*(A)v)(x,x')|_2 \le (1+\exp(1))\|H_*(A)v\|_{\mathrm{su}} \le 128\exp(1)ba'\|v\|_{\mathrm{su}}.$$

Thus $304\exp(1)ba'$ bounds the Euclidean sup norm for all pair rows. Conversion to the component sup norm costs a factor of two, and the calibration of C^F gives (21.15). To identify this operator norm with the row sum, fix one output component and prescribe each input component to have the conjugate phase of its matrix entry. The coarse spinor values on a fundamental domain can be prescribed independently; their antiperiodic extension does not affect the entry moduli. All unit sites have weight one.

The barred bound follows from $\bar{H}_*(A) = \mathsf{C}^{-1}H_*(-A)\mathsf{C}$. The charge-conjugation matrix C is a signed permutation of the four components, and commutes with the scalar phases in the covariant quotient. It preserves both spinor norms. Applying the preceding argument at $-A$ proves the second inequality in (21.15). Finally, localization of the long-pair argument gives the bound $C^F\exp(-\gamma^F\min\{d([x],y'),d([x'],y')\})$ for the coefficient at a coarse site y'. The intermediate Euclidean constant is $128\exp(1)ba'$, and its component constant $256\exp(1)ba'$ is again at most C^F.

Put

$$\mathcal{S} = C^F(24/\gamma^F)^3, \qquad h_{\mathrm{L}} = 6\mathcal{S}, \qquad C_H = 203M(C^F)^2, \qquad C_K = 226\mathcal{S}^2.$$

The lattice sum $\sum_{y'} e^{-\gamma d(y,y')} \le (12/\gamma)^3$, $0 < \gamma \le 6$, follows by choosing nearest periodic representatives and summing $e^{-\gamma|z|_\infty} \le e^{-\gamma|z|_1/3}$ over $\mathbb{Z}^3$. The one-dimensional sum is $\coth(\gamma/6) \le 12/\gamma$.

For a fine spinor or Hölder-pair index u, let $\mathrm{Leg}_A(u;y',\beta')$ be the coefficient of $\Psi_{\beta'}(y')$ after substituting $H_*^{\#}(A)\Psi$ in that generator. Barred generators use the corresponding barred coefficient. The set $\mathrm{pos}(u)$ consists of the unit block containing the first endpoint, and, for a pair with $d_X(x,x') \ge 1$, also the block containing its second endpoint. Thus it has at most two elements.

Proposition 21.1. *For $A \in \mathcal{W}$, the blocked field maps and their Hölder maps are analytic in A. The first coarse-operator derivative satisfies*

$$|\partial_B D_*(A)|(y,y') \le 226(C^F)^2\epsilon_B \sum_{z\in P_B} E(y,z)E(z,y'),$$

$$\|\partial_B D_*(A)\|_{\gamma^F/2} \le C_K\epsilon_B, \qquad \sum_{y,y'} |\partial_B D_*(A)|(y,y') \le C_K\epsilon_B|P_B|. \tag{21.16}$$

If $d(P_B,P_{B'}) \ge M+2$, then

$$|\partial_B\partial_{B'}D_*(A)|(y,y') \le C_H^2\epsilon_B\epsilon_{B'} \sum_{z\in P_B, z'\in P_{B'}} \big(E(y,z)E(z,z')E(z',y') + E(y,z')E(z',z)E(z,y')\big). \tag{21.17}$$

The corresponding fine-field legs satisfy

$$|\operatorname{Leg}_A|(u;y') \le 3C^F \sum_{w\in\operatorname{pos}(u)} E(w,y'),$$

$$|\partial_B \operatorname{Leg}_A|(u;y') \le C_H\epsilon_B \sum_{w\in\operatorname{pos}(u),z\in P_B^+} E(w,z)E(z,y'), \tag{21.18}$$

$$|\partial_B\partial_{B'} \operatorname{Leg}_A|(u;y') \le C_H^2\epsilon_B\epsilon_{B'} \sum_{\substack{w\in\operatorname{pos}(u)\\ z\in P_B^+,z'\in P_{B'}^+}} \big(E(w,z)E(z,z')E(z',y') + E(w,z')E(z',z)E(z,y')\big). \tag{21.19}$$

The last bound has the same separation hypothesis. These estimates hold for barred and unbarred legs.

Proof. The block averages and their transposes are diagonal in unit blocks. From (21.11) and $\mathfrak{i}(A) \le 1/4$ their majorants are at most $3/2$. Their directional derivatives are bounded by

$$(9/4)\epsilon_B 1_{\{y\in P_B\}}.$$

Differentiating the only phased Dirac hop gives

$$(\partial_B D_A v)(x) = -ie_*B(c,c_3)M_3(x)u_A(c,c_3)v(x^{(3)}).$$

Its majorant is at most $8\epsilon_B$, vanishes unless the unit blocks coincide or are neighbors, and vanishes unless both belong to P_B. The derivative of $JP_*(A)J^*$ has diagonal majorant at most $7\epsilon_B$. Thus for $F_B = \partial_B F_*(A)$,

$$|F_B|(z,z') \le \epsilon_B(15\,1_{\{z=z'\}} + 8\,1_{\{z,z' \text{ neighbors}\}})1_{\{z,z'\in P_B\}}.$$

Summing the six neighboring blocks gives $\|F_B\|_\gamma \le 146\epsilon_B$ for $0 \le \gamma \le 1$. Also, since $E(z',y') \le \mathrm{e}E(z,y')$ for neighboring z, z',

$$\sum_{z,z'} E(y,z)|F_B|(z,z')E(z',y') \le 146\epsilon_B \sum_{z\in P_B} E(y,z)E(z,y'). \tag{21.20}$$

If the supports are separated by $M+2$, the mixed derivatives of D_A, the averages, JP_*J^*, F_* and $\delta_{\alpha,A}$ vanish. Each entry is an exponential of a linear phase supported in a unit block, a single external bond or a coordinate box of diameter at most M. Two separated directions cannot both occur in one phase. For JP_*J^* the remaining product terms have disjoint unit-block supports.

The inverse derivative and the blocked-field definition give

$$\partial_B H_* = -S_*F_BH_* + b_*S_*\partial_B(JQ_*^T(-A)).$$

Use (21.14) and (21.20) in the first term. The diagonal derivative in the second is absorbed as the summand $z = y'$. The combined coefficient is at most $149(C^F)^2$:

$$|\partial_B H_*|(y,y') \le 149(C^F)^2\epsilon_B \sum_{z\in P_B} E(y,z)E(z,y'). \tag{21.21}$$

The same estimate holds for $\delta_{\alpha,A}\partial_B H_*$ on short pairs, since the first factor S_* can be replaced by $\delta_{\alpha,A}S_*$.

For separated directions differentiating once more leaves four terms:

$$\begin{aligned}\partial_B\partial_{B'}H_* = {}& S_*F_BS_*F_{B'}H_* + S_*F_{B'}S_*F_BH_* \\ & - b_*S_*F_BS_*\partial_{B'}(JQ_*^T(-A)) - b_*S_*F_{B'}S_*\partial_B(JQ_*^T(-A)).\end{aligned}$$

Apply (21.20) twice to the first two terms. Each gives an ordered three-link exponential chain with coefficient $146^2(C^F)^3$. The diagonal last factors in the other two terms give the same chains,

with coefficient $(657/2)(C^F)^2$. Each ordered chain therefore has coefficient at most $21645(C^F)^3$. Again the first factor may be replaced by the short-pair Hölder inverse.

Differentiate $D_* = b_* - b_* Q_* J^* H_*$. The derivative of the average contributes $(9/4)C^F$ to the first exponential chain and the derivative of H_* contributes $(3/2)149(C^F)^2$. Their sum is at most $226(C^F)^2$, proving the first line of (21.16). Weighted row and column sums give two factors $(24/\gamma^F)^3$, hence C_K; without weights the same sum retains one factor $|P_B|$. For the mixed derivative the three product terms contain one derivative on each factor, or the mixed derivative of H_*. Each chain has coefficient at most $(1341/4)(C^F)^2 + (64935/2)(C^F)^3 \le 32803(C^F)^3 \le C_H^2$. This proves (21.17).

A leg is a component row of H_* or of $\delta_{\alpha,A} H_*$. For a short pair its position is its first unit block; for a long pair its position set contains both endpoint blocks. The sum over spin components of the row is bounded by the relevant majorant. For long pairs use $|\delta_{\alpha,A} v| \le \mathrm{e}|v(x')| + |v(x)|$; for short pairs use the Hölder input directly. This gives the first line of (21.18), and its weighted row sum is at most h_{L}.

The derivative of a Hölder leg has two terms: $(\partial_B \delta_{\alpha,A}) H_* + \delta_{\alpha,A} \partial_B H_*$. The first is bounded by $4 d_X^{1-\alpha} \epsilon_B |H_* v(x')|$, and vanishes unless the pair is within M of P_B. It vanishes for pairs inside a cluster, since their phase is one. For short pairs this adds at most $11 C^F$ to $149(C^F)^2$; for long pairs it adds $4MC^F$ to $(4/3)149(C^F)^2$. Both are at most C_H, after enlarging P_B to P_B^+. This proves the derivative bound in (21.18).

For its mixed derivative the second derivative of the Hölder phase vanishes by separation. There remain $(\partial_B \delta_{\alpha,A}) \partial_{B'} H_*$ and its interchange, and $\delta_{\alpha,A} \partial_B \partial_{B'} H_*$. On short pairs each ordered chain has coefficient at most $1639(C^F)^2 + 21645(C^F)^3$. On long pairs the coefficient at the second endpoint is at most $596M(C^F)^2 + 28860(C^F)^3$; the first endpoint has the smaller coefficient $21645(C^F)^3$. All are at most C_H^2, proving (21.19). Barred legs follow by the monomial charge conjugation matrix and the replacement $A \mapsto -A$, which preserve component row sums; the domain is invariant under that replacement. □

21.3. **Complex neighborhoods and the real-field covariance.** Choose

$$c_\tau = \min\{1/8, \mu_-/48, (292\mathcal{S})^{-1}\}, \qquad C'_K = 1328\mathcal{S}^2, \qquad \mathcal{W}^+ = \{A + A_1 : A \in \mathcal{W},\ e_* |A_1|_\infty \le c_\tau\}.$$

Every point of $\mathcal{W}^+$ has imaginary part $\mathfrak{i} < 1/4$. The elementary averages and F_* therefore satisfy the same local derivative bounds as above, although the exponential propagator input is initially given only on $\mathcal{W}$. The enlargement preserves invertibility:

$$\|S_*(A + A_1)\|_{\gamma^F/2} \le 2\mathcal{S}. \tag{21.22}$$

To prove this, integrate $\partial_{A_1} F_*$ along the segment to get $\|F_*(A + A_1) - F_*(A)\|_{\gamma^F/2} \le 146 c_\tau$. Multiplying by $S_*(A)$ gives norm at most $146 c_\tau \mathcal{S} \le 1/2$. The Neumann inverse in the submultiplicative majorant norm proves (21.22). Since its finite-dimensional determinant is nonzero, S_* is analytic in a neighborhood of each point of $\mathcal{W}^+$, as is D_*.

Differentiate $D_* = b_* - b_*^2 Q_* J^* S_* J Q_*^T(-A)$. The three terms have majorant-norm coefficients

$$\tfrac{9}{4}(2\mathcal{S})\tfrac{3}{2}, \qquad \tfrac{3}{2}(2\mathcal{S})146(2\mathcal{S})\tfrac{3}{2}, \qquad \tfrac{3}{2}(2\mathcal{S})\tfrac{9}{4}.$$

Their sum is $\frac{27}{2}\mathcal{S} + 1314\mathcal{S}^2 \le C'_K$. Keeping the support indicator of either the average derivative or F_B gives, respectively,

$$\|\partial_B D_*(A)\|_{\gamma^F/2} \le C'_K \epsilon_B, \qquad \sum_{y,y'} |\partial_B D_*(A)|(y, y') \le C'_K \epsilon_B |P_B|, \qquad A \in \mathcal{W}^+. \tag{21.23}$$

In the middle term, sum first the two outer S_* factors using their row and column bounds $2\mathcal{S}$, and then sum the local derivative over its first endpoint in P_B. Integrating (21.23) along any segment inside $\mathcal{W}^+$ gives the same two bounds for $D_*(A + B) - D_*(A)$.

The covariance at a real field requires no small-field hypothesis. If A_r is real, $R = D_{A_r} + \mu_*$ has real part at least μ_*. The averaging map $Q = Q_*(A_r)J^*$ satisfies $Q^* = JQ_*^T(-A_r)$ and $QQ^* = 1$, because its phases have modulus one and every block has fine measure one. Thus $F_*(A_r) = R + b_*Q^*Q$ also has real part at least μ_*. Set $C = QR^{-1}Q^*$ and $Y = (b_*^{-1} + C)^{-1}$. The inverse R^{-1} is accretive, so $b_*^{-1} + C$ is invertible. Multiplication verifies

$$S_* = R^{-1} - R^{-1}Q^*YQR^{-1}, \qquad QS_*Q^* = b_*^{-1}(1 - b_*^{-1}Y).$$

The definition of D_* therefore gives

$$D_*(A_r)^{-1} = b_*^{-1} + QR^{-1}Q^*, \qquad \|D_*(A_r)^{-1}\|_2 \le b_*^{-1} + \mu_*^{-1}. \tag{21.24}$$

Its block decay follows by conjugating the fine inverse with a weight. Fix y' and put $\psi(c) = \vartheta\, d(c, \Delta_{y'})$ on the periodic fine lattice, with $\vartheta = \mu_*/(12\mathrm{e}) \le 1$. This weight is zero on $\Delta_{y'}$, is ϑ-Lipschitz, and on Δ_y is at least $\vartheta(d(y, y') - 1)$. Only the external Dirac hop changes under conjugation by e^{ψ}, since the internal hops stay over one photon site. Consequently

$$\|e^{\psi} D_{A_r} e^{-\psi} - D_{A_r}\|_2 \le 6\eta_*^{-1}(e^{\vartheta\eta_*} - 1) \le 6\mathrm{e}\vartheta = \mu_*/2.$$

The conjugated R has real part at least $\mu_*/2$, so its inverse has norm at most $2/\mu_*$. Apply it between Q^* times unit spinors at y and y'; these have norm one and supports in their respective blocks. Moving the weight across the scalar product gives

$$\|D_*(A_r)^{-1}(y, y')\|_2 \le b_*^{-1} 1_{\{y=y'\}} + \frac{2\mathrm{e}}{\mu_*} e^{-\mu_* d(y,y')/(12\mathrm{e})}. \tag{21.25}$$

The factor e accounts for the unit block diameter in the lower bound for ψ.

Set

$$g = (b_-^{-1} + \mu_-^{-1})^{1/2}, \qquad h_G = 1 + g, \qquad g_1 = b_-^{-1} + 4/\mu_-, \qquad \theta_{\mathrm{CT}} = \mu_-/(12\mathrm{e}),$$

$$\vartheta_0 = \min\{\theta_{\mathrm{CT}}/2, \gamma^F/8, 1/M\}, \qquad \alpha_G = 2b_-^{-1} + 4\mathrm{e}\mu_-^{-1}(24/\theta_{\mathrm{CT}})^3.$$

A component row or column of a 4×4 matrix has sum at most twice its Hilbert-space norm. Using (21.25) and the lattice exponential sum, every row and column of $G = D_*(A_r)^{-1}$ therefore satisfies

$$\sum_{u'} |G(u, u')| e^{\vartheta_0 d(u,u')} \le \alpha_G, \qquad \sum_{u} |G(u, u')| e^{\vartheta_0 d(u,u')} \le \alpha_G, \qquad \|G\|_2 \le g^2. \tag{21.26}$$

These constants depend on L, C^F, γ^F, M, not on the size of the real connection.

For later complex twists we also need the coarse inverse away from real fields. Suppose $A = A_r + A_1$, with A_r real, $e_*|A_1|_\infty \le \mu_*/24$, and $F_*(A)$ invertible. The external-hop phase difference gives $\|D_A - D_{A_r}\|_2 \le 12e_*|A_1|_\infty \le \mu_*/2$. Hence $D_A + \mu_*$ is accretive with inverse norm at most $2/\mu_*$. Its two average maps have norm at most $\sqrt{2}$, because $|\varphi_A^{\pm 1}| \le e^{1/16} \le \sqrt{2}$. The same finite-rank inversion calculation then gives

$$D_*(A)^{-1} = b_*^{-1} + Q_*(A)J^*(D_A + \mu_*)^{-1}JQ_*^T(-A), \qquad \|D_*(A)^{-1}\|_2 \le g_1. \tag{21.27}$$

Here invertibility of $b_*^{-1} + QR^{-1}Q'$ follows from that of $F_*(A)$ and R: $\det(1 + b_*R^{-1}Q'Q) = \det(1 + b_*QR^{-1}Q')$. This determinant identity follows by the two block-triangular factorizations of $\left(\begin{smallmatrix} 1 & -b_*R^{-1}Q' \\ Q & 1 \end{smallmatrix}\right)$. The inverse formula is then checked by multiplication exactly as above. Each covariance entry is bounded by g_1, so each row has sum at most $4\ell_*^3 g_1$.

21.4. **The terminal Grassmann norms.** The terminal integral is over four coarse components per unit site, with Grassmann weight one. Set

$$a_1 = e_*^{-1/4}, \qquad a_2 = C_0^{-1} e_*^{-1/4+\varepsilon}, \qquad h_* = (e_*^{-1/4}, e_*^{-1/4+\varepsilon}).$$

By (21.15), the substitution $H_*^{\#}(A)\Psi$ has row norm at most $C^F \le C_0$, for both spinor and Hölder legs, including all long and cluster pairs. The coarse kernel $b_* - D_*(A) = b_*^2 Q_* J^* S_* J Q_*^T(-A)$ is bounded by $C^F E(y,y')$: block compression gives Euclidean constant $32\exp(1)b^2a'$ by (11.17), and conversion to the component norm doubles it. The calibration of C^F applies.

The weighted Grassmann substitution estimate applies with $C_0 a_2 = h_{*,2} \le h_{*,1}$. Therefore

$$|F(H_*^{\#}(A)\Psi)|_{a_2} \le \|F\|_{h_*}. \tag{21.28}$$

For the mass form before substitution, the fine measure sums to ℓ_*^3 and there are four spin components. Its norm at weight $C_0 a_2$ is exactly $4\ell_*^3 (C_0 a_2)^2$. In particular

$$|\langle \bar\psi, \psi\rangle|_{a_2} \le 4 e_*^{-1/2+2\varepsilon} \ell_*^3. \tag{21.29}$$

The 48 labels are canceled by the fine measure $1/48$; there is no factor 192 in this estimate. Similarly the coarse bilinear form has norm a_1^2 times the sum of its matrix entry moduli. Its diagonal term contributes $4b_*$ per unit site, and its decaying kernel contributes at most $4C^F \sum_{y'} E(y,y')$. For $c(\gamma) = \sum_{z\in\mathbb{Z}^3} e^{-\gamma|z|_\infty}$, the earlier lattice-sum estimate gives $c(\gamma) \le (12/\gamma)^3$ when $0 < \gamma \le 6$. The chosen $C_0 \ge 4 + 16 C^F c(\gamma^F)$ therefore gives

$$|\langle \bar\Psi, D_*(A)\Psi\rangle|_{a_1} \le C_0 e_*^{-1/2} \ell_*^3. \tag{21.30}$$

These estimates hold on $\widetilde{G}_*$. For the external activity evaluated at $\mathscr{T} = \mathscr{A} + \mathscr{C}$, the same substitution applies because $\widetilde{R}_* + \mathfrak{X}_* \subset \widetilde{G}_*$.

21.5. **Gaussian bounds in the zero-flux sector.** The complement estimate below concerns the normalized zero-flux law, denoted $\bar\mu$. The all-flux Gaussian bound does not directly give a bound for this conditional law. We therefore retain the sector projection in the coordinate-slice trace.

On $C(n_1,n_2,n_3)$ use the spaces H^s, H^l, P_G, E, and M_0 of (3.35) and (3.37). Let P_k project onto the magnetic-flux summand k. Choose coordinate slice cycles S_2, S_3, through a fixed slice site, and for $\xi \in [0,1]^2$ set $b_\xi = q(\xi_2 \mathbf{1}_{S_2} + \xi_3 \mathbf{1}_{S_3})$. For bounded magnetic factors u_j and electric factors v_j with sub-Gaussian growth, let K_j^ξ have the periodized convolution kernel

$$v_j(A'-A) e^{i\langle b_\xi, A'-A\rangle} \prod_b g_1(A'(b) - A(b)).$$

The zero-flux trace formula is

$$Z^{(0)}\left[\prod_j u_j(F_j^{\mathrm{mag}}) v_j(F_j^{\mathrm{el}})\right] = (2\pi)^{n_1 n_2 n_3} \int_{[0,1]^2} \mathrm{tr}\left(P_0 \prod_{j=n_1-1,\dots,0} r_q^{n_2 n_3} K_j^\xi P_G M_0 u_j(F^{\mathrm{mag}}) M_0^* \right) \mathrm{d}\xi. \tag{21.31}$$

To derive it, sum the electric curvature along S_2, S_3 and over time. The sums are $r_q k_{12}$ and $r_q k_{13}$. The added electric phases are therefore $e^{2\pi i(\xi_2 k_{12} + \xi_3 k_{13})}$. Restricting the first magnetic slice to flux k_{23} inserts $P_{k_{23}}$ in the word trace, since the sum of $d_s A$ on the slice vanishes. All operators preserve that summand. The flux sum is absolutely convergent by the curvature Gaussian bound on the modulus of the integrand. Set $k_{23} = 0$ and integrate over ξ; Fourier orthogonality keeps precisely $k_{12} = k_{13} = 0$. This proves (21.31).

Proposition 21.2. *For the harmonic-average zero-flux law at every real lattice mass and every coordinate side $n_\mu \geq 4$,*

$$\bar{\mu}(e^{\langle g, F^{\mathrm{u}}\rangle}) \leq e^{\|\Pr_{F_N} g\|^2/2}. \tag{21.32}$$

In particular, for $\phi \in F_N$ with $\|\phi\| \leq 1$, $\bar{\mu}\{|\langle \phi, F^{\mathrm{u}}\rangle| > a\} \leq 2e^{-a^2/2}$.

Proof. Put $\mathcal{Z}(\beta) = Z^{(0)}[e^{-\langle \beta, F\rangle - \|\beta\|^2/2}]$. It is finite by the all-flux bound and is positive because the sector contains the $z = 0$ term, which has positive integral; the shifted photon weight is strictly positive. For time direction 1, split β into electric parts β_j^{el} and magnetic parts β_j^{mag}. Take the corresponding exponential u_j and v_j in (21.31), replacing u_j first by $u_j^\delta = \min(u_j, \delta^{-1})$. The bounded magnetic factors make the word trace applicable.

Completing the electric Gaussian gives

$$K_j^\xi = z_j U_j e^{-|E - b_\xi|^2/2}, \qquad U_j = e^{i\langle \beta_j^{\mathrm{el}}, E\rangle}, \qquad |z_j| = 1.$$

Let $K_\xi = r_q^{n_2 n_3} e^{-|E-b_\xi|^2/2} P_G$. It is positive, and its square root is $r_q^{n_2 n_3/2} e^{-|E-b_\xi|^2/4} P_G$. The operator $K_\xi^{1/2} M_0$ is Hilbert–Schmidt. The periodized square-root Gaussian has the additional phase $e^{i\langle b_\xi, A' - A\rangle}$; its modulus is bounded by the kernel majorant used in (3.38). The same real-slice integral and flux sum therefore proves this assertion uniformly in ξ. Multiplication by bounded $(u_j^\delta)^{1/2}$ preserves it.

Put $Y_j = K_\xi^{1/2} M_0 (u_j^\delta)^{1/2}$. Cyclicity turns the trace into $\prod_j z_j$ times the closed chain

$$\operatorname{tr} \prod_j U_j P_0 Y_j (P_0 Y_j)^*.$$

All factors preserve the magnetic-flux summand; U_j is unitary. Schatten Hölder with $2n_1$ factors of exponent $2n_1$ bounds this trace by

$$\prod_j \left(\operatorname{tr}_{H^s} P_0 (Y_j^* Y_j)^{n_1}\right)^{1/n_1}.$$

Hölder in ξ and the Bloch trace formula identify its jth integrated factor with the zero-flux functional having the same magnetic factor at every slice and no electric factor. Let $\beta^{(j)}$ denote that purely magnetic two-cochain. Monotone convergence as $\delta \downarrow 0$ gives

$$\mathcal{Z}(\beta) \leq \prod_j \mathcal{Z}(\beta^{(j)})^{1/n_1}.$$

Rotate the torus by an orientation-preserving cyclic coordinate permutation. Zero flux and the pairing are preserved, and $\beta^{(j)}$ becomes purely electric for the new time direction. The same inequality then bounds $\mathcal{Z}(\beta^{(j)})$ by $\mathcal{Z}(0)$. Hence $\mathcal{Z}(\beta) \leq Z^{(0)}[1]$.

The zero-sector identity (20.17) identifies its normalized curvature law with $\bar{\mu}$. Since this law is supported on F_N, the pairing with g equals that with $\Pr_{F_N} g$. Apply the preceding inequality with the negative of that projection and divide by the positive zero-sector partition function. This proves (21.32). Exponential Markov applied to the two signs and optimized at parameter a gives the stated tail bound. □

22. The terminal integral and partition bounds

We first put the curvature Gaussian in coordinates adapted to all block steps. The energy splits into the terminal photon energy and independent unit Gaussian fluctuation coordinates. Iterating the exact density identity in these coordinates then gives the terminal fermion integral on the small-field set.

22.1. **The downward photon coordinates.** Fix a flow side and cutoff as in Proposition 16.1. Put $k = k_*(R)$ and $n = R/\eta = L^k \ell_*$. On the unit torus of side $\ell_j = n/L^j$ use the fluctuation space V_j and its fixed orthonormal basis, and the maps $A_j^{\min}, \mathrm{Sq}_j$ of the photon construction. For a real terminal unit bond field A and $\zeta = (\zeta_j)_{j<k}$ define

$$A_k^{\downarrow} = A, \qquad A_j^{\downarrow} = A_j^{\min}(A_{j+1}^{\downarrow}) + \mathrm{Sq}_j \zeta_j, \qquad j = k-1, \dots, 0. \tag{22.1}$$

The last field is denoted $A_0^{\downarrow}(A, \zeta)$. These fields are linear in their data.

For any fine unit field A_0 define its block fields by $A_j = (\mathbf{Q}^{(j)} A_0)^{[j]}$, where $\mathbf{Q}^{(j)}$ is the j-fold photon average on the original unit torus and $B^{[j]}(b) = L^{j/2} B(L^j b)$. The block-averaging/scaling identities give $(A_{j+1})_L = \mathbf{Q} A_j$. Let Ax_k be the fields A_0 for which $\tau A_j = 0$ for $j < k$. For each unit torus put $V_{\mathrm{t}} = \delta C^2$, the real coclosed bond fields with mean zero. On the fine torus let $F_N = dC^1(n)$ with its counting inner product and standard Gaussian γ_N. The slice is

$$\mathrm{Sl}_k = \{A_0 \in \mathrm{Ax}_k : A_k \in V_{\mathrm{t}}(T_k^0)\}.$$

We will use the elementary cochain identities

$$C^1 = dC^0 \oplus \{\text{constant bond fields}\} \oplus \delta C^2, \qquad \ker d = dC^0 \oplus \{\text{constant bond fields}\}. \tag{22.2}$$

The sum is orthogonal. Indeed $dd = 0$, so dC^0 and δC^2 are orthogonal, and their common orthogonal complement is $\ker(d\delta + \delta d)$. Commuting lattice differences make this operator the scalar Laplacian in each component. Its kernel consists of constants: the energy identity forces every nearest-neighbor difference to vanish. This proves (22.2). Consequently d is injective on V_{t} and, on a unit torus with ℓ^3 sites, $\dim V_{\mathrm{t}} = \dim dC^1 = 2\ell^3 - 2$.

Proposition 22.1. *The map $(A, \zeta) \mapsto A_0^{\downarrow}(A, \zeta)$ is a linear bijection from $C^1(T_k^0) \times \prod_{j<k} V_j$ to Ax_k. The block fields of its image are exactly $A_j^{\downarrow}$. The coboundary is a linear bijection $d : \mathrm{Sl}_k \to F_N$. Thus curvature has coordinates*

$$\mathcal{J}_{\mathrm{ph}} : F \longmapsto (A_*; \zeta_0, \dots, \zeta_{k-1}), \qquad F = dA_0^{\downarrow}(A_*, \zeta), \tag{22.3}$$

where $A_* \in V_{\mathrm{t}}(T_k^0)$. They satisfy the exact energy identity

$$\|F\|^2 = \|dH_k^{\mathrm{ph}} A_*\|_2^2 + \sum_{j<k} \|\zeta_j\|^2. \tag{22.4}$$

The image of γ_N under $\mathcal{J}_{\mathrm{ph}}$ is

$$\mathsf{dm}_*(A_*) \prod_{j<k} \mathrm{d}\mu_I(\zeta_j), \tag{22.5}$$

where the terminal centered Gaussian on $V_{\mathrm{t}}(T_k^0)$ is nondegenerate and has density proportional to

$$\exp(-\|dH_k^{\mathrm{ph}} A\|_2^2/2).$$

There are orthonormal vectors $\phi_{j,\Upsilon} \in F_N$ such that $\zeta_j(\Upsilon) = \langle \phi_{j,\Upsilon}, F \rangle$. For each terminal unit plaquette p there is $\phi_p \in F_N$ with

$$(dA_*)(p) = \langle \phi_p, F \rangle, \qquad \|\phi_p\| \le 1. \tag{22.6}$$

There are $3\ell_*^3$ such plaquettes, and

$$\dim V_j = (2 - 2L^{-3})\ell_j^3 \le 2\ell_*^3 L^{3(k-j)}. \tag{22.7}$$

Proof. The one-step minimizer and fluctuation maps obey $\mathbf{Q} A_j^{\min}(A') = A'_L$, $\tau A_j^{\min}(A') = 0$, $\mathbf{Q}\mathrm{Sq}_j = 0$ and $\tau \mathrm{Sq}_j = 0$. Apply $\mathbf{Q}$ to (22.1); the injective scaling map then identifies the next block field with $A_{j+1}^{\downarrow}$. Starting from $A_0 = A_0^{\downarrow}$ proves this at all levels, including $\tau A_j = 0$. Conversely, for $A_0 \in \mathrm{Ax}_k$ its difference $A_j - A_j^{\min}(A_{j+1})$ is annihilated by $\mathbf{Q}$ and τ. The coordinate

bijection (10.20) and the invertible square root give exactly one ζ_j with this difference equal to $\mathrm{Sq}_j\zeta_j$. Downward recursion recovers A_0. This proves the first bijection and its inverse.

The orthogonality (10.27) and the Gaussian fluctuation decomposition give at each level

$$\|dH_j^{\mathrm{ph}}A_j\|_2^2 = \|dH_{j+1}^{\mathrm{ph}}A_{j+1}\|_2^2 + \|\zeta_j\|^2.$$

Summing, and using $H_0^{\mathrm{ph}} = I$, proves (22.4). The dimension of the slice is

$$2\ell_k^3 - 2 + \sum_{j<k}(2\ell_j^3 - 2\ell_{j+1}^3) = 2n^3 - 2 = \dim F_N.$$

If $dA_0 = 0$ on the slice, the energy identity forces every $\zeta_j = 0$ and $dH_k^{\mathrm{ph}}A_k = 0$. Since $\mathbf{Q}_k H_k^{\mathrm{ph}} = I$ and $d\mathbf{Q}_k = \mathbf{Q}_k^{(2)}d$, this implies $dA_k = 0$. The injectivity of d on V_{t} gives $A_k = 0$, whence $A_0 = 0$ by the downward bijection. Equality of dimensions therefore proves that $d : \mathrm{Sl}_k \to F_N$ is bijective and gives (22.3).

Equip the coordinate space with the scalar product

$$\langle (A,\zeta),(A',\zeta')\rangle_{\mathrm{ph}} = \langle A, \Delta_k A'\rangle + \sum_{j<k}\langle \zeta_j, \zeta_j'\rangle.$$

Its terminal part is positive definite: $\langle A, \Delta_k A\rangle = \|dH_k^{\mathrm{ph}}A\|_2^2 \geq \|dA\|_2^2 > 0$ for nonzero $A \in V_{\mathrm{t}}$. Here the block curvature contraction follows from (10.2). The energy identity and polarization make $\mathcal{J}_{\mathrm{ph}}$ a linear isometry for this scalar product. It maps orthonormal bases to orthonormal bases; the Jacobian is therefore 1 for their Euclidean volume measures. These are the measures in the normalized standard Gaussians. In an orthogonal direct sum, that volume measure and the Gaussian density factor into those of the summands. This proves (22.5), including its normalization. The volume measure for the terminal scalar product differs from the ordinary Lebesgue measure on V_{t} by one constant factor. Thus its Gaussian has exactly the stated density. Its quadratic form is positive definite, so it is nondegenerate.

For a basis vector Υ in V_j, let $\mathbf{e}_{j,\Upsilon}$ be the coordinate vector with that single nonzero entry. Its inverse image under the isometry is $\phi_{j,\Upsilon}$. The coordinate vectors are orthonormal, proving orthonormality and the required coordinate formula. The terminal curvature functional satisfies

$$|(dA_*)(p)| \leq \|dA_*\|_2 = \|\mathbf{Q}_k^{(2)}dH_k^{\mathrm{ph}}A_*\|_2 \leq \|dH_k^{\mathrm{ph}}A_*\|_2 \leq \|F\|.$$

It is consequently represented by a vector of norm at most 1 in the finite Euclidean space F_N, giving (22.6). Finally (10.18) and $\ell_j = L^{k-j}\ell_*$ give (22.7). □

If a terminal field A^c is closed, linearity gives the difference of downward fields as $A_0^{\downarrow}(A^c, 0)$. For a closed input the background identity (10.29) gives zero curvature energy at the preceding level. Applying the average shows that its block minimizer is closed. The identity $\mathbf{Q}A_j^{\min}(A') = A_L'$ and preservation of means by the block average give $\mathrm{mean}(A_j^{\min}(A')) = L^{-1/2}\,\mathrm{mean}(A')$. Its final mean is $L^{-k/2}\,\mathrm{mean}(A^c)$. Equation (22.2) therefore gives

$$A_0^{\downarrow}(A + A^c, \zeta) = A_0^{\downarrow}(A,\zeta) + L^{-k/2}\,\mathrm{mean}(A^c) + d\lambda. \tag{22.8}$$

In particular adding a coarse gradient adds a fine gradient. For the harmonic twist $c_{\theta,\mu} = \theta_\mu/(e_*\ell_*)$, $L^{-k/2}c_\theta = H_\theta$, where $H_{\theta,\mu} = \theta_\mu/(qn)$. Thus its downward shift is $H_\theta + d\lambda$.

22.2. **The terminal density with an external connection.** Let $(m_{\mathrm{in}}, \varepsilon_{\mathrm{in}}^{\mathrm{v}})$ be a controlled real pair on R, with the trajectory of Section 16. An external connection is a complex fine unit bond field a satisfying $a^{[j]} \in \mathfrak{X}_j$ for every $j \leq k$; here $a^{[j]}(b) = L^{j/2}a(L^j b)$. Write $\mathscr{C}_* = a^{[k]}$. For

$A \in \mathrm{Dom}_k$ put $\mathscr{A} = H_k^{\mathrm{ph}} A$, $B = \mathscr{A} + \mathscr{C}_*$, and insert the fermion fields $H_k^{\#}(B)\Psi$ and their barred partners. The terminal function is

$$Y_*^x(A;\mathscr{C}_*) = \det D_k(0)^{-1} \exp\left(-\varepsilon_k^{\mathrm{v}} \ell_*^3 + \sum_X \mathfrak{P}_k(X,\mathscr{C}_*)\right)$$
$$\times \int \mathrm{D}\Psi \, \exp\left(-\langle\bar\Psi, D_k(B)\Psi\rangle - m_k\langle\bar\psi,\psi\rangle + \sum_X \mathscr{E}_k(X;\mathscr{A},\mathscr{C}_*,\psi,\chi)\right). \tag{22.9}$$

Since $k \geq 1$, this Berezin integral has $4\ell_*^3$ pairs with weight 1. The inserted chamber fields and kernels retain their fine weight $\eta_k^3/48$. The inverse and activity domains in the preceding sections make every term well defined, and $\det D_k(0) \neq 0$. At zero external slot, $\mathfrak{P}_k(X,0) = 0$ and $\mathscr{E}_k|_0 = E_k$; we write $Y_*(A) = Y_*^x(A;0)$.

For the original fine unit field let

$$w_N(B) = \exp(-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3) \frac{\det(D^{\mathrm{ch}}[qB] + \bar m\eta + m_{\mathrm{in}})}{\det(D^{\mathrm{ch}}[0] + \bar m\eta)}. \tag{22.10}$$

The matrices have size $192n^3$. The bracket argument of D^{ch} is the bond phase, as in (18.3); an unabsorbed unit connection B therefore has argument qB. The denominator is nonzero by the free positive-mass coercivity estimate; no invertibility of the numerator is required. The weighted Gaussian Berezin identity gives the initial integrated density exactly as $w_N(A_0 + a)$. In particular all initial measure and vacuum factors are included in (22.10).

This fine weight is entire in B and is invariant under real gauge shifts. Its entries involve only the four chamber neighbors, with link phases in the fourth neighbor. Multiplication by $\exp(-\mathrm{i}q\lambda(c))$ on all chamber components over c conjugates the operator at B to the operator at $B + d\lambda$: the endpoint phases cancel the gradient in the fourth hopping phase, and the other hops stay over the same c. The determinants therefore agree. The same stencil bounds every row sum at $B_0 + a$, with B_0 real, by $|\boldsymbol{\mu}| + 41 + 6\exp(q\|\operatorname{Im} a\|_\infty)$, where $\boldsymbol{\mu} = \bar m\eta + m_{\mathrm{in}}$. Indeed the three internal hopping block norms sum with the diagonal to less than $|\boldsymbol{\mu}| + 41$, and the fourth has row norm at most $6\exp(q\|\operatorname{Im} a\|_\infty)$. The determinant expansion bounds its modulus by the product of the absolute row sums: its permutation sum is a subsum of the sum over all column choices. Consequently

$$|w_N(B_0 + a)| \leq B_N(a) := \frac{\exp(-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3)}{|\det(D^{\mathrm{ch}}[0] + \bar m\eta)|} \big(|\boldsymbol{\mu}| + 41 + 6\exp(q\|\operatorname{Im} a\|_\infty)\big)^{192n^3} < \infty. \tag{22.11}$$

This bound is used for integrability at a fixed cutoff.

Let $A(F) \in V_{\mathrm{t}}$ be the transverse fine potential of $F \in F_N$, and define the finite complex measure

$$\mathrm{d}\nu^{\theta,a}(F) = w_N(A(F) + H_\theta + a)\mathrm{d}\gamma_N(F).$$

The small-field event and the terminal indicator are

$$\mathfrak{G} = \{F : |\zeta_j(\Upsilon)| \leq p_{0,j} \ (j < k), \ \|dA_*\|_\infty \leq p_*\}, \qquad \chi_*(A) = \mathbf{1}[\|dA\|_\infty \leq p_*]. \tag{22.12}$$

Here A_*, ζ are the actual coordinates (22.3). Products with χ_* are extended by zero off its support.

Theorem 22.2. *For every $A \in \mathrm{Dom}_k$,*

$$Y_*^x(A;\mathscr{C}_*) = \int \prod_{j<k} \big(\mathbf{1}_{\mathrm{Box}_j}(\zeta_j)\mathrm{d}\mu_I(\zeta_j)\big) w_N(A_0^{\downarrow}(A,\zeta) + a). \tag{22.13}$$

The integrand is continuous on the compact product of boxes. If $A, A + d\lambda \in \mathrm{Dom}_k$, with λ real, their terminal functions agree. For every bounded Borel function Φ on $V_{\mathrm{t}}(T_k^0)$ and every real twist

θ,

$$\nu^{\theta,a}(\mathbf{1}_{\mathfrak{G}}\Phi(A_*)) = \int \mathsf{dm}_*(A)\,\Phi(A)\chi_*(A)Y_*^x(A+c_\theta;\mathscr{C}_*). \tag{22.14}$$

Its right-hand density is bounded and Borel, and $Y_*^x(A+c_\theta;\mathscr{C}_*)$ is jointly continuous in (θ,A) on $\{\chi_*=1\}$.

Proof. The photon domain conditions (10.42) hold at each running scale by the fixed charge choice. Proposition 10.4 gives two consequences used here: a real unit field with curvature bounded by p_j lies in Dom_j; and a field in Dom_{j+1} descends to Dom_j after addition of any box fluctuation. Downward induction therefore puts every field of (22.1) in its respective domain whenever its top field lies in Dom_k and all fluctuations lie in their boxes.

Let $y_j^x(A;a^{[j]})$ be the integrated density at level j along this trajectory. The exact integrated step gives

$$y_{j+1}^x(A';a^{[j+1]}) = \int_{\mathrm{Box}_j} \mathrm{d}\mu_I(\zeta_j)\, y_j^x(A_j^{\min}(A') + \mathrm{Sq}_j\zeta_j;a^{[j]}).$$

Its external slot is the required one because $(a^{[j+1]})_L = a^{[j]}$, directly from the two scaling factors. Its coupling maps are the actual trajectory maps, and its vacuum includes the box normalization. The first such identity integrates the full $192n^3$ weighted chamber pairs into $4\ell_1^3$ coarse pairs.

For a top field in Dom_j, the function $w_N(A_0^{\downarrow}(A,\zeta)+a)$ is continuous in the finitely many fluctuation coordinates. Every finite product of boxes is compact, and (22.11) bounds it. Thus all repeated integrals are absolutely integrable. Starting with the initial identity $y_0^x(A_0;a) = w_N(A_0+a)$, use the one-step formula, the downward recursion and Fubini at each level. This proves (22.13) at $j=k$, with its unnormalized factors $\mathbf{1}_{\mathrm{Box}_j}\mathrm{d}\mu_I$. The continuity assertion follows from the same finite-dimensional composition. Equation (22.8) turns a coarse real gradient into a fine gradient; gauge invariance of w_N inside this integral then proves gauge invariance of Y_*^x.

To integrate against an arbitrary bounded Borel observable, first express the fine density in the curvature coordinates. The slice potential $A_0^{\downarrow}(A_*,\zeta)$ has mean zero: its block average has the mean of the fine field, and its top field has mean zero. Its difference from $A(F)$ is closed and has mean zero, hence is a gradient by (22.2). The harmonic shift identity gives

$$w_N(A(F)+H_\theta+a) = w_N(A_0^{\downarrow}(A_*+c_\theta,\zeta)+a).$$

Under the tower isometry the Gaussian becomes (22.5), and the event $\mathfrak{G}$ has indicator

$$\chi_*(A_*)\prod_{j<k}\mathbf{1}_{\mathrm{Box}_j}(\zeta_j).$$

The integrand is a bounded Borel function with bound $\|\Phi\|_\infty B_N(a)$. The change-of-variables identity for its image measure and Fubini therefore give

$$\nu^{\theta,a}(\mathbf{1}_{\mathfrak{G}}\Phi(A_*)) = \int \mathsf{dm}_*(A)\,\Phi(A)\chi_*(A)\int\prod_{j<k}\big(\mathbf{1}_{\mathrm{Box}_j}\mathrm{d}\mu_I(\zeta_j)\big)w_N(A_0^{\downarrow}(A+c_\theta,\zeta)+a).$$

Where $\chi_*(A)=1$, the field $A+c_\theta$ has the same curvature bound and belongs to Dom_k. The inner integral is $Y_*^x(A+c_\theta;\mathscr{C}_*)$ by (22.13), proving (22.14).

The inner integral defines a function on all real (θ,A), whether or not $A+c_\theta\in\mathrm{Dom}_k$. Its modulus is at most $B_N(a)$. Its integrand is continuous and uniformly continuous on a compact neighborhood of any (θ,A) times the compact box product. Integrating this uniform bound proves joint continuity. Multiplying by χ_* gives a bounded Borel function, since $\{\chi_*=1\}$ is closed. On that set it is the asserted terminal density. □

22.3. **The large-field complement.** At zero external connection write $\nu^{\theta} = \nu^{\theta,0}$, and let $\bar{\nu}$ be its average over $[-\pi,\pi]^3$ with measure $\mathrm{d}\theta/(2\pi)^3$. The real chamber operator has two adjoint spin blocks; thus its determinant at every real lattice mass is the squared modulus of the determinant of one block. Consequently ν^{θ} is positive. The zero-sector identity (20.23) gives $0 < \bar{\nu}(1) < \infty$ and identifies $\bar{\mu} = \bar{\nu}/\bar{\nu}(1)$ with the zero-flux curvature probability law.

Fix $C_0 \geq 1$, which will also dominate the terminal field-map and Dirac-form constants in Section 21.4. Set

$$\varepsilon_3 = 6\ell_*^3\left(e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)}\right), \qquad \varepsilon_4 = 6\ell_*^3 e^{-p_*^2/(2C_0^2)}. \tag{22.15}$$

Proposition 22.3. *Suppose $v_* \geq 8$, the integer $p_0 \geq 1$ satisfies $p_0 v_*^{p_0-1} \geq 28$, and $L \geq 5$. Then, at every real initial pair,*

$$\bar{\mu}(\mathfrak{G}^c) \leq \varepsilon_3, \qquad \mathsf{m}_*(\chi_* = 0) \leq \varepsilon_4. \tag{22.16}$$

For fixed $(\bar{m}, e, R)$, these bounds are independent of the ultraviolet cutoff and of the initial pair.

Proof. The zero-flux domination estimate in Proposition 21.2 and the rows of (22.6) give

$$\bar{\mu}\{|\zeta_j(\Upsilon)| > p_{0,j}\} \leq 2e^{-p_{0,j}^2/2}, \qquad \bar{\mu}\{|(dA_*)(p)| > p_*\} \leq 2e^{-p_*^2/2}.$$

Using the coordinate counts in (22.7), the union bound is

$$\bar{\mu}(\mathfrak{G}^c) \leq 4\ell_*^3 \sum_{i=1}^{k} L^{3i} e^{-(v_* + i\log L/2)^{2p_0}/2} + 6\ell_*^3 e^{-p_*^2/2}. \tag{22.17}$$

Here $i = k - j$ and $v_j = v_* + i\log L/2$. The first two terms of the binomial expansion give

$$p_{0,j} = (v_* + i\log L/2)^{p_0} \geq p_{0,*} + 28i\log L/2.$$

Since $p_{0,j} \geq 1$, its square is at least $p_{0,j}$. It follows that

$$L^{3i} e^{-p_{0,j}^2/2} \leq e^{-p_{0,*}/2} L^{-4i}.$$

Summing the geometric series, whose sum is $(L^4 - 1)^{-1} \leq 1$, bounds (22.17) by $4\ell_*^3 e^{-p_{0,*}/2} + 6\ell_*^3 e^{-p_*^2/2}$. This is at most ε_3, since $C_0 \geq 1$.

The image of γ_N under $F \mapsto A_*$ is m_* by (22.5). For each plaquette the variable $\langle \phi_p, F\rangle$ is Gaussian with variance $\|\phi_p\|^2 \leq 1$. Its two-sided exponential Markov bound is $2e^{-p_*^2/2}$. Summing over the $3\ell_*^3$ plaquettes proves $\mathsf{m}_*(\chi_* = 0) \leq 6\ell_*^3 e^{-p_*^2/2} \leq \varepsilon_4$. Finally, ℓ_* and e_* are independent of N after the mesh threshold, by Proposition 16.1; hence the displayed errors have the asserted dependence. □

22.4. **Free pressure and twist dependence.** The partition comparison requires an error in the free pressure which depends on the physical size, rather than on the number of lattice sites. It also requires a twist estimate for the operator with a block penalty. The latter penalty removes the small-mass singularity before the fine mesh is taken to zero.

For $0 < m \leq 1$ let $\mathfrak{d}_m(p) = \det \mathcal{D}_m(p)$, where $\mathcal{D}_m$ is the 192-dimensional chamber symbol in Section 9.2, and set

$$p_{\mathrm{ch}}^0(m) = \int_{[-\pi,\pi]^3} \log \mathfrak{d}_m(p) \frac{\mathrm{d}p}{(2\pi)^3}. \tag{22.18}$$

The spin-block identity and positive real part imply $\mathfrak{d}_m(p) > 0$ for real p. In this subsection $|z| = |z|_\infty$ for $z \in \mathbb{Z}^3$. Write $\kappa_{(0)} = \kappa_{\mathrm{strip}}$ from Proposition 9.2.

Proposition 22.4. *There is a finite C_6, depending only on ρ, such that, with*

$$\varepsilon_5(x) = C_6 \log \frac{1}{1 - e^{-\kappa_{(0)} x}} \qquad (x > 0), \tag{22.19}$$

the free twisted determinant on a unit torus of side $n \geq 1$ satisfies

$$\left| \log \det(D^{\mathrm{ch}}[qH_\theta] + m) - n^3 p^0_{\mathrm{ch}}(m) \right| \leq \varepsilon_5(mn). \tag{22.20}$$

Here $H_{\theta,\mu} = \theta_\mu/(qn)$ and $\theta \in \mathbb{R}^3$. For $m = \bar{m}\eta$ and $n = R/\eta$ define

$$z^{\mathrm{free}}_{r,n} = \int_{[-\pi,\pi]^3} \det(D^{\mathrm{ch}}[qH_\theta] + \bar{m}\eta) \frac{\mathrm{d}\theta}{(2\pi)^3}.$$

Then $z^{\mathrm{free}}_{r,n} > 0$ and

$$\left| \log z^{\mathrm{free}}_{r,n} - n^3 p^0_{\mathrm{ch}}(\bar{m}\eta) \right| \leq \varepsilon_5(\bar{m}R). \tag{22.21}$$

Proof. The plane waves with momenta $P_{n,\theta} = \{(2\pi j + \pi \mathbf{1} + \theta)/n : j \in \{0, \dots, n-1\}^3\}$ diagonalize the translation part of the chamber operator. Inserting a plane wave in the fourth hop shifts its momentum by θ/n, whereas the internal hops are unchanged. Thus

$$\det(D^{\mathrm{ch}}[qH_\theta] + m) = \prod_{p \in P_{n,\theta}} \mathfrak{d}_m(p) > 0. \tag{22.22}$$

The phases contain $qH_\theta = \theta/n$, so the determinant is independent of q.

Factor the determinant by the light–heavy decomposition:

$$\log \mathfrak{d}_m(p) = f_H(p) + f_S(p), \qquad f_H(p) = \log\det D_{\mathrm{hh}}(p), \quad f_S(p) = \log\det B_m(p).$$

Both logarithms are real and smooth on real momenta. Their derivatives, which suffice for contour shifting, are

$$\partial_\mu f_H = \mathrm{tr}(D^{-1}_{\mathrm{hh}} \partial_\mu D_{\mathrm{hh}}), \qquad \partial_\mu f_S = \mathrm{tr}(B_m^{-1} \partial_\mu B_m).$$

The heavy block has dimension 188. On the strip of width θ_0 its inverse has norm at most $2/g_{\mathrm{h}}$, and the fourth-hop derivative has norm at most $3(1+\rho)e^{\theta_0} = 3(1+\rho) + g_{\mathrm{h}}/2$. Hence

$$|\partial_\mu f_H| \leq C_{\mathrm{lH}} := 188 \left(1 + \frac{6(1+\rho)}{g_{\mathrm{h}}} \right).$$

For $z \neq 0$ choose $|z_\mu| = |z|$. One integration by parts and a shift of that contour by $-\mathrm{i}\theta_0 \operatorname{sgn}(z_\mu)$ give

$$|\widehat{f}_H(z)| \leq C_{\mathrm{lH}} |z|^{-1} e^{-\theta_0 |z|}. \tag{22.23}$$

The derivative trace is holomorphic in a neighborhood of the closed strip. Its periodicity cancels the vertical sides of the contour rectangle; no choice of a complex logarithm is needed.

For the light factor put $r_m(p') = (|p'|_2^2 + m^2)^{1/2}$. The inverse derivative bounds (9.13) hold on $|\operatorname{Im} p_\nu| \leq \kappa_{(0)} m$. The bound K_s for B_m on the strip of width θ_0 and Cauchy's formula on polydiscs of radius $\theta_0/2$ give $\|\partial^\beta B_m\| \leq 24 K_s (2/\theta_0)^{|\beta|}$ for $1 \leq |\beta| \leq 4$. Leibniz's rule in the derivative trace therefore gives

$$|\partial^\alpha \partial_\mu f_S(p' + \mathrm{i}p'')| \leq C_{\mathrm{lS}}\, r_m(p')^{-1-|\alpha|} \leq C_{\mathrm{lS}} |p'|_2^{-1-|\alpha|} \quad (|\alpha| \leq 3,\ p' \neq 0), \tag{22.24}$$

where one may take

$$C_{\mathrm{lS}} = \max_{|\alpha| \leq 3} 96 K_s \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} C_\beta(\rho) (2/\theta_0)^{|\alpha|-|\beta|+1} (3\pi^2 + 1)^{(|\alpha|-|\beta|)/2}.$$

This constant is independent of m. Shift the differentiated Fourier integral by $-\mathrm{i}\kappa_{(0)} m \operatorname{sgn}(z_\mu) e_\mu$. The periodic Fourier estimate proved in Proposition 9.3, with singular order 1 and three derivatives, bounds this integral by $c_{\mathrm{F}} C_{\mathrm{lS}} (1+|z|)^{-2}$, where c_{F} is its constant at those orders. Thus

$$|\widehat{f_S}(z)| \le c_{\mathrm{F}} C_{\mathrm{lS}} |z|^{-1} (1+|z|)^{-2} e^{-\kappa_{(0)} m |z|}. \tag{22.25}$$

The strip is contained in the heavy strip since $\kappa_{(0)} m \le \theta_0/2$. The inverse is holomorphic near the closed strip, so the contour shift is legitimate.

The Fourier coefficients of $f = f_H + f_S$ are summable. For a summable periodic Fourier series, summing over $P_{n,\theta}$ keeps precisely the indices $z = nw$:

$$\sum_{p \in P_{n,\theta}} f(p) = n^3 \sum_{w \in \mathbb{Z}^3} \widehat{f}(nw) (-1)^{w_1+w_2+w_3} e^{\mathrm{i}\theta \cdot w}. \tag{22.26}$$

This follows by summing the geometric progression in each coordinate; absolute convergence permits the interchange. Its $w = 0$ term is $n^3 p^0_{\mathrm{ch}}(m)$. There are $24t^2 + 2 \le 26t^2$ indices with $|w| = t$. The contribution of (22.25) is at most

$$26 c_{\mathrm{F}} C_{\mathrm{lS}} \sum_{t \ge 1} t^{-1} e^{-\kappa_{(0)} m n t}.$$

For the heavy term use $\kappa_{(0)} m \le \theta_0/2$ and $x^2 e^{-\theta_0 x/2} \le 16/(\exp(1)\theta_0)^2$. Its contribution is at most

$$\frac{416 C_{\mathrm{lH}}}{(\exp(1)\theta_0)^2} \sum_{t \ge 1} t^{-1} e^{-\kappa_{(0)} m n t}.$$

Since $\sum_{t \ge 1} e^{-xt}/t = \log(1 - e^{-x})^{-1}$, (22.20) follows with $C_6 = 26[c_{\mathrm{F}} C_{\mathrm{lS}} + 16 C_{\mathrm{lH}}/(\exp(1)\theta_0)^2]$. Finally every twisted determinant lies between $\exp(n^3 p^0_{\mathrm{ch}}(m) \pm \varepsilon_5(mn))$. Its harmonic average lies in the same positive interval. Taking logarithms proves (22.21). □

We next retain the block penalty. Let $\mathfrak{N} \ge 3$ be odd and put $\tau = \mathfrak{N}^{-1}$. On the fine torus of side $\ell \in \mathbb{N}$ let $\mathcal{H}_\theta$ be the chamber spinors with mesh τ and boundary multipliers $-e^{\mathrm{i}\theta_\mu}$. The scalar product uses $\tau^3/48$. The coarse space has four components per unit site and counting scalar product. The average Q takes the mean over each unit block and its 48 labels, and Q^* extends a coarse field constantly on that block. Thus $QQ^* = 1$, $\|Q\| = \|Q^*\| = 1$, and $\Pi_{\mathrm{bl}} = Q^* Q$ is an orthogonal projection. For $0 < \mu \le 1$ and $b_- \le b \le b_+$ set

$$T_\theta = D + \mu + b\Pi_{\mathrm{bl}}, \qquad D^{\mathrm{bl}}_\theta = b - b^2 Q T_\theta^{-1} Q^*. \tag{22.27}$$

All determinants below are determinants of linear maps. The weighting of a scalar product changes their matrix representations, but does not add a determinant factor.

Proposition 22.5. *The determinants of T_θ, $D + \mu$, and D^{bl}_θ are positive, and*

$$(D^{\mathrm{bl}}_\theta)^{-1} = b^{-1} + Q(D+\mu)^{-1} Q^*, \tag{22.28}$$

$$\det D^{\mathrm{bl}}_\theta = b^{4\ell^3} \frac{\det(D+\mu)}{\det T_\theta}. \tag{22.29}$$

There are $C_\Sigma < \infty$ and $0 < \gamma_\Sigma \le 1$, depending only on (b_-, b_+, ρ), such that

$$\left| \log \frac{\det T_\theta}{\det T_0} \right| \le C_\Sigma \ell^{-1} e^{-\gamma_\Sigma \ell}. \tag{22.30}$$

For the level $k \ge 1$ data

$$(\mathfrak{N}, \ell, \mu, b) = (L^k, \ell_*, \mu_*, b_k), \qquad c_{\theta,\mu} = \theta_\mu/(e_* \ell_*),$$

put $R_\theta^{\mathrm{tw}} = \det D_k(c_\theta)/\det D_k(0)$. *Then*

$$R_\theta^{\mathrm{tw}} = \frac{\det(D^{\mathrm{ch}}[qH_\theta] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)} \frac{\det T_0}{\det T_\theta} > 0. \tag{22.31}$$

Proof. The real part of $D + \mu$ is at least μ. The same holds for T_θ, since its penalty is positive. Both operators have two adjoint spin blocks; their determinants are therefore positive. Put $G = (D+\mu)^{-1}$, $S = QGQ^*$ and $Z = (b^{-1} + S)^{-1}$. Their inverses exist since $\operatorname{Re} G \geq 0$ and $\operatorname{Re}(b^{-1} + S) \geq b^{-1}$. Multiplication gives $T_\theta^{-1} = G - GQ^*ZQG$ and $D_\theta^{\mathrm{bl}} = Z$, proving (22.28). Sylvester's determinant identity, applied to $-bQ$ and $T_\theta^{-1}Q^*$, gives (22.29). Its right side is positive.

We prove (22.30) with constants independent of τ and μ. Let

$$c_{\mathrm{nz}} = \min\left\{\frac{\rho}{\pi}, \frac{g_{\mathrm{h}}\rho^2}{49\sqrt{3}\,\pi^3}\right\} > 0.$$

For real $q' \in [-\pi, \pi]^3$,

$$\|\mathcal{D}_0(q')u\| \geq c_{\mathrm{nz}}|q'|_2\|u\|. \tag{22.32}$$

Normalize $\|u\| = 1$ and write $u = Pu + \Pi_{\mathrm{h}}u$ for its light and heavy parts, where $\Pi_{\mathrm{h}} = 1 - P$ is the cluster-heavy projection. If $d = \|\mathcal{D}_0(q')u\|$, then $g_{\mathrm{h}}\|\Pi_{\mathrm{h}}u\|^2 \leq d$. The light compression is $W(q') = \rho\sum_\mu(1-\cos q'_\mu) + \mathrm{i}\gamma\cdot\sin q'$; its squared modulus is at least $(2\rho/\pi)^2|q'|_2^2$. Also $J^*\mathcal{D}_0(0) = 0$, and the fourth-hop difference has norm at most $6|q'|_2$. Projecting $\mathcal{D}_0(q')u$ onto the light space gives

$$d \geq |q'|_2\left(\frac{2\rho}{\pi} - 7\|\Pi_{\mathrm{h}}u\|\right).$$

If $\|\Pi_{\mathrm{h}}u\| \leq \rho/(7\pi)$ this is at least $(\rho/\pi)|q'|_2$. Otherwise $d \geq g_{\mathrm{h}}\rho^2/(49\pi^2)$; since $|q'|_2 \leq \sqrt{3}\pi$, this proves (22.32).

On one unit cell use the Bloch space $\mathcal{H}(p) = \{f(c + e_\mu, g) = e^{\mathrm{i}p_\mu}f(c,g)\}$. Its modes are $p_j = p + 2\pi j$, with $j \in I_{\mathfrak{N}}^3$, $I_{\mathfrak{N}} = \{-(\mathfrak{N}-1)/2, \dots, (\mathfrak{N}-1)/2\}$. The map $\{u_j\} \mapsto \sum_j e^{\mathrm{i}p_j\cdot c}u_j(g)$ is unitary for the one-cell scalar product $\tau^3/48$. It diagonalizes D with blocks $X_j = \tau^{-1}\mathcal{D}_0(\tau p_j)$. Define

$$u_1(t) = \mathfrak{N}^{-1}\sum_{s\in I_{\mathfrak{N}}} e^{-\mathrm{i}ts/\mathfrak{N}}, \qquad u(p) = \prod_\mu u_1(p_\mu).$$

The block projection in this representation has kernel $u(p_j)u(p_{j'})P$. This follows by averaging the plane waves on the centered unit block; discrete Fourier orthogonality expands the constant block extension. For $p \in [-\pi,\pi]^3$ one has $\tau p_j \in [-\pi,\pi]^3$ and $|p_j|_2 \geq \pi|j|$ when $j \neq 0$. Hence (22.32) gives $\|X_ju\| \geq c_{\mathrm{nz}}|p_j|_2\|u\|$.

The inverse on $\mathcal{H}(p)$ is uniformly bounded. To see this, let $\|f\| = 1$ and $d = \|(D+\mu+b\Pi_{\mathrm{bl}})f\| \leq 1$. The real-part estimate gives

$$\mu \leq d, \qquad \|\Pi_{\mathrm{bl}}f\| \leq (d/b_-)^{1/2}, \qquad \|\Pi_{\mathrm{h}}f\| \leq (d/(3g_{\mathrm{h}}))^{1/2}, \qquad \|Df\| \leq (2 + \sqrt{b_+})\sqrt{d}.$$

Separate the mode $j = 0$ from f^{r}, the remaining modes. The bound on X_j gives $\|f^{\mathrm{r}}\| \leq \|Df\|/(\pi c_{\mathrm{nz}})$. Moreover $u_1(t) = \sin(t/2)/[\mathfrak{N}\sin(t/(2\mathfrak{N}))] \geq 2/\pi$ for $|t| \leq \pi$, with value 1 at zero. Thus the block projection of the zero mode controls its light part by the factor $8/\pi^3$. Using $\|\Pi_{\mathrm{bl}}\| = 1$ and subtracting the other modes,

$$\sqrt{d/b_-} \geq \frac{8}{\pi^3}(1 - \|\Pi_{\mathrm{h}}f\|) - (1 + 8/\pi^3)\|f^{\mathrm{r}}\|.$$

It follows that $8/\pi^3 \le K_{\mathrm{iv}}\sqrt{d}$, where

$$K_{\mathrm{iv}} = b_-^{-1/2} + \frac{8}{\pi^3}(3g_{\mathrm{h}})^{-1/2} + (1+8/\pi^3)\frac{2+\sqrt{b_+}}{\pi c_{\mathrm{nz}}}.$$

Consequently the inverse norm is at most $C_{\mathrm{iv}} = \max\{1, (\pi^3 K_{\mathrm{iv}}/8)^2\}$.

For complex momenta use the periodic gauge $f(c,g) \mapsto e^{\mathrm{i}p\cdot c} f(c,g)$ to work on the fixed periodic one-cell space. Let $D(p)$ and $\Pi(p)$ be the resulting operators, so $\mathcal{T}(p) = D(p) + \mu + b\Pi(p)$. The fourth hopping phase is $e^{\mathrm{i}\tau p\cdot g e_1}$, and the block kernel is $e^{-\mathrm{i}p\cdot(c-c')}$. A shift of p_ν by 2π conjugates these operators by multiplication by $e^{2\pi \mathrm{i} c_\nu}$. Thus their determinant and derivative traces are periodic. For $p = p' + \mathrm{i}te_\nu$, $|t| \le 1$, and $1 \le j \le 4$, direct differentiation gives

$$\|\partial_\nu^j D(p)\| \le 9\tau^{j-1}, \qquad \|\partial_\nu^j \Pi(p)\| \le \exp(1), \qquad \operatorname{rank} \partial_\nu^j \Pi(p) \le 4(j+1). \tag{22.33}$$

The last bound follows by expanding $(c_\nu - c'_\nu)^j$ into $j+1$ products of a function of c and a function of c', each with four spin components. The norm estimates use $|c_\nu - c'_\nu| < 1$ in one block, $\tau \le 1/3$, and $\|M_3\| \le 6$. The difference from the real momentum has norm at most $(6+b_+)|t|e^{|t|}$. With

$$\gamma_\Sigma = \min\{1, [2\exp(1)(6+b_+)C_{\mathrm{iv}}]^{-1}\},$$

a Neumann series gives $\|\mathcal{T}(p)^{-1}\| \le 2C_{\mathrm{iv}}$ on this strip. Its singular values are at most twice those at p'.

The growing dimension of the fine cell cannot be used in a trace bound. Instead write $G(p) = \mathcal{T}(p)^{-1}$ and compare at real p' with $(D(p')+1)^{-1}$. The resolvent identity, the bound just proved, and the singular-value multiplication inequality [10, slides 3–9] imply

$$s_j(G(p)) \le C_{\mathrm{sv}} s_j((D(p')+1)^{-1}), \qquad C_{\mathrm{sv}} = 2[1 + (1+b_+)C_{\mathrm{iv}}].$$

The comparison operator is the direct sum of $(X_j+1)^{-1}$. Since $\operatorname{Re} X_j \ge 0$, each of its 192 singular values is at most $(1 + c_{\mathrm{nz}}^2 |p'_j|_2^2)^{-1/2}$. Set

$$S_4 = \sum_{j\in\mathbb{Z}^3\setminus\{0\}} |j|^{-4} = 4\pi^2 + \pi^4/45, \quad C_{\mathrm{uv}} = (1+3\pi^2 c_{\mathrm{nz}}^2)^{1/2}, \quad S_{\mathrm{jf}} = 1 + S_4/(\pi c_{\mathrm{nz}})^4.$$

For $1 \le s \le 4$ we obtain

$$\tau^{4-s}\|G(p)\|_{\mathfrak{S}_s}^s \le 192 C_{\mathrm{sv}}^s C_{\mathrm{uv}}^{4-s} S_{\mathrm{jf}}. \tag{22.34}$$

Indeed $\tau^2(1 + c_{\mathrm{nz}}^2|p'_j|_2^2) \le C_{\mathrm{uv}}^2$, so multiplication by τ^{4-s} makes each summand a multiple of $(1+c_{\mathrm{nz}}^2|p'_j|_2^2)^{-2}$. The nonzero indices are bounded by $(\pi c_{\mathrm{nz}}|j|)^{-4}$ and have the summable majorant displayed above.

Let $F(p) = \log\det \mathcal{T}(p)$ on real momenta. The fourth derivative is the trace expression

$$\begin{aligned} F_4(p) = {} & \operatorname{tr}(GT_4) - 4\operatorname{tr}(GT_1GT_3) - 3\operatorname{tr}(GT_2GT_2) \\ & + 12\operatorname{tr}(GT_1GT_1GT_2) - 6\operatorname{tr}((GT_1)^4), \end{aligned} \tag{22.35}$$

where $T_j = \partial_\nu^j \mathcal{T}$. This follows by differentiating $\partial_\nu F = \operatorname{tr}(GT_1)$ and $\partial_\nu G = -GT_1G$, using trace cyclicity. The expression is holomorphic near the closed strip in the ν coordinate. Each trace in (22.35) has $s \le 4$ inverse factors and derivatives of total order four. Expand every T_j as $D_j + b\Pi_j$. For the term with only D_j, Schatten Hölder and (22.33) give the bound $9^s\tau^{4-s}\|G\|_{\mathfrak{S}_s}^s$. Equation (22.34) bounds this by $9^4\, 192 C_{\mathrm{sv}}^4 C_{\mathrm{uv}}^3 S_{\mathrm{jf}}$. In each other term estimate one $b\Pi_j$ in trace norm: its rank bound gives $\|b\Pi_j\|_{\mathfrak{S}_1} \le 20\exp(1)b_+$. The other factors have operator norms at most $2C_{\mathrm{iv}}$ and $\max\{9, \exp(1)b_+\}$. There are at most 15 such terms in any trace. Since the five numerical coefficients have absolute sum 26, this proves $|F_4(p)| \le C_{\mathrm{Flg}}$, where

$$C_{\mathrm{Flg}} = 26\left[9^4\, 192 C_{\mathrm{sv}}^4 C_{\mathrm{uv}}^3 S_{\mathrm{jf}} + 300\exp(1)b_+(2C_{\mathrm{iv}})^4 \max\{9, \exp(1)b_+\}^3\right].$$

All Schatten applications are on a finite-dimensional space, with s factors of exponent s and reciprocal exponents summing to one [10, slides 54–56].

For $z \neq 0$ choose $|z_\nu| = |z|$. Four integrations by parts and a contour shift by $-\mathrm{i}\gamma_\Sigma \operatorname{sgn}(z_\nu) e_\nu$ now give $|\widehat{F}(z)| \le C_{\mathrm{Flg}} |z|^{-4} e^{-\gamma_\Sigma |z|}$. The trace expression is periodic by the preceding conjugacy and is holomorphic near the closed strip. Decomposition into unit-cell Bloch spaces gives $\log\det T_\theta = \sum_{p \in P_{\ell,\theta}} F(p)$. Using (22.26) at θ and at zero, and subtracting, proves

$$\left| \log \frac{\det T_\theta}{\det T_0} \right| \le 2\ell^3 \sum_{w \neq 0} C_{\mathrm{Flg}} (\ell |w|)^{-4} e^{-\gamma_\Sigma \ell |w|} \le 2 S_4 C_{\mathrm{Flg}} \ell^{-1} e^{-\gamma_\Sigma \ell}.$$

Thus $C_\Sigma = 2 S_4 C_{\mathrm{Flg}}$.

Finally constant connections are twists by multiplication by $e^{\mathrm{i}\theta \cdot c/\ell_*}$. Every rectilinear path contributing to the average has the same constant-connection integral, so its phase is $e^{\mathrm{i}\theta\cdot(c-y)/\ell_*}$. Consequently the covariant chamber averages and fine inverse are conjugate to Q, Q^* and T_θ^{-1}, and $D_k(c_\theta)$ is conjugate to D_θ^{bl}. Dilating the mesh τ to one gives

$$\det(D + \mu \text{ on } \mathcal{H}_\theta) = \tau^{-192 n^3} \det(D^{\mathrm{ch}}[q H_\theta] + \tau\mu), \qquad n = \ell_* \mathfrak{N}.$$

Here $\tau\mu_* = \bar{m}\eta$. Divide (22.29) at the two twists. The factors $b^{4\ell_*^3}$ and $\tau^{-192 n^3}$ cancel, proving (22.31). □

22.5. **The terminal remainder.** Choose C_0 to bound the photon row sums, the field maps and the Dirac form in Section 21.4. Set

$$c_\kappa = \sup_{\square} \sum_{X \supset \square} e^{-\kappa d_M(X)} \le K_0.$$

In addition to the primitive and flow hypotheses, impose the following inequalities at the stop:

$$\begin{aligned} 8 C_0^2 \ell_*^6 (b_-^{-1} + \mu_*^{-1}) &\le e_*^{-\varepsilon/2}, \\ 4(2C_0 + c_\kappa + 5 c_{\mathrm{fl}} + 6) \ell_*^3 &\le e_*^{-\varepsilon}, \\ e_*^{1/4 - 9\varepsilon} \le \tfrac{1}{8}, \qquad & 4 C_0 \ell_*^2 p_* < e_*^{-\varepsilon}, \\ 12 \ell_*^3 \left(e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)} \right) \le e_*, \qquad & C_\Sigma e^{-\gamma_\Sigma \ell_*} \le e_*. \end{aligned} \tag{22.36}$$

These are the terminal conditions used in the proof. The choice in Proposition 24.1 is required to satisfy them: the scale estimates give $M v_*^r \le \ell_* < L M v_*^r$ and $\mu_*^{-1} < L \ell_*$. Substitute the latter two bounds in the first four inequalities; use the lower bound in the last exponential inequality. Thus the conditions are uniform in the cutoff.

Put

$$\varepsilon_1 = \tfrac{1}{2} e_*^{1/4 - 9\varepsilon}, \qquad \varepsilon_2 = C_\Sigma \ell_*^{-1} e^{-\gamma_\Sigma \ell_*}. \tag{22.37}$$

Together with (22.15), (22.36) gives $\varepsilon_1 \le 1/16$, $\varepsilon_2 \le e_*$, and $\varepsilon_4 \le \varepsilon_3 \le e_*/2$.

Proposition 22.6. *For a controlled real pair, every mean-zero real coarse field A with $\|dA\|_\infty \le p_*$ and every $\theta \in \mathbb{R}^3$ satisfy*

$$Y_*(A + c_\theta) = e^{-\varepsilon_k^{\mathrm{Y}} \ell_*^3} R_\theta^{\mathrm{tw}} (1 + r_*(\theta, A)), \qquad |r_*(\theta, A)| \le \varepsilon_1. \tag{22.38}$$

The remainder is continuous in (θ, A) on this set.

Proof. The gauge construction in (10.31)–(10.32) gives a real λ such that $A^g = A - d\lambda$ satisfies $\|A^g\|_\infty \le 2\ell_*^2 p_*$. The curvature bound puts both $A + c_\theta$ and $A^g + c_\theta$ in Dom_k, by Proposition 10.4;

Theorem 22.2 gives equality of their terminal densities. Set $\mathscr{A}^0 = H_k^{\mathrm{ph}} A^g$. Summing each of the three photon kernel bounds gives a norm at most $C_0\|A^g\|_\infty$; hence

$$\|\mathscr{A}^0\|_{\mathrm{each}} \le 2C_0\ell_*^2 p_* < \tfrac{1}{2}e_*^{-\varepsilon}.$$

The photon minimizer preserves constants, so the background is $\mathscr{A} = \mathscr{A}^0 + c_\theta$. For $\rho_* = e_*^{-3/4+4\varepsilon} \ge 8$ and $|t| < 2\rho_*$, each norm of $t\mathscr{A}^0$ is less than $e_*^{-3/4+3\varepsilon}$, the smallest analytic-domain threshold. Invariance under real constant shifts gives

$$\mathscr{A}^{(t)} = t\mathscr{A}^0 + c_\theta \in \widetilde{R}_k \subset \widetilde{G}_k \qquad (|t| < 2\rho_*). \tag{22.39}$$

Let $D^{\mathrm{tw}} = D_k(c_\theta)$ and $C_\theta = (D^{\mathrm{tw}})^{-1}$. The free block formula and its twist conjugacy give $\|C_\theta\|_{\mathrm{op}} \le b_-^{-1} + \mu_*^{-1}$. At the connection $\mathscr{A}$, after the fermion field substitution, put

$$\begin{aligned} V_* = &-\langle\bar\Psi, [D_k(\mathscr{A}) - D_k(c_\theta)]\Psi\rangle - m_k\langle\bar\psi,\psi\rangle \\ &+ \sum_X E_k(X;\mathscr{A},\psi,\chi). \end{aligned} \tag{22.40}$$

All terms are even. Completing the Grassmann Gaussian in (22.9), at zero external slot, gives

$$Y_*(A + c_\theta) = e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}} \int \mathrm{d}\mu_{C_\theta}(\Psi) e^{V_*}. \tag{22.41}$$

This is the weight-one identity of the terminal coarse space; the fine fields retain their $\eta_k^3/48$ pairing in V_*.

We estimate the three terms before integrating. Let $a_1 = e_*^{-1/4}$ and $a_2 = C_0^{-1}e_*^{-1/4+\varepsilon} \le a_1$. The analytic quadratic form $f(t) = \langle\bar\Psi, D_k(\mathscr{A}^{(t)})\Psi\rangle$ has $|f(t)|_{a_1} \le C_0 e_*^{-1/2}\ell_*^3$ on (22.39), by (21.30). Coefficientwise Cauchy's formula with kernel $[t(t-1)]^{-1}$ on $|t| = \rho_*$ gives

$$|f(1) - f(0)|_{a_2} \le |f(1) - f(0)|_{a_1} \le \frac{C_0 e_*^{-1/2}\ell_*^3}{\rho_* - 1} \le 2C_0 e_*^{1/4-4\varepsilon}\ell_*^3.$$

The mass-form and activity-substitution bounds (21.29) and (21.28) give

$$|m_k\langle\bar\psi,\psi\rangle|_{a_2} \le 4e_*^{1/4-8\varepsilon}\ell_*^3, \qquad |E_k(X;\mathscr{A},\psi,\chi)|_{a_2} \le e^{-\kappa d_M(X)} w_k^{\mathrm{E}}.$$

Here $|m_k| \le e_*^{3/4-8\varepsilon}$ and $w_k^{\mathrm{E}} = e_*^{1/4-7\varepsilon}$. These norm estimates include the ordered pairs within one chamber cluster. The mass pairing uses the fine site measure $\eta_k^3/48$; the four spin components in the mass form therefore contribute $4\ell_*^3$, rather than $192\ell_*^3$. Every polymer contains an M-cube, so $\sum_X e^{-\kappa d_M(X)} \le (\ell_*/M)^3 c_\kappa \le c_\kappa\ell_*^3$. Consequently

$$|V_*|_{a_2} \le (2C_0 + c_\kappa + 4)e_*^{1/4-8\varepsilon}\ell_*^3. \tag{22.42}$$

The covariance integral is the unit-covariance integral after the one-sided substitution $\Psi \mapsto C_\theta\Psi$, $\bar\Psi \mapsto \bar\Psi$. Each covariance row has $4\ell_*^3$ entries of modulus at most $b_-^{-1} + \mu_*^{-1}$. Its row norm, together with the identity on barred variables, is bounded by

$$4\ell_*^3(b_-^{-1} + \mu_*^{-1}) \le \frac{e_*^{-\varepsilon/2}}{2C_0^2\ell_*^3} = a_2\,\frac{e_*^{1/4-3\varepsilon/2}}{2C_0\ell_*^3} \le a_2.$$

The weighted substitution estimate therefore sends V_* to V_*' with $|V_*'|_1 \le |V_*|_{a_2}$. The second inequality of (22.36) and (22.42) imply $|V_*'|_1 \le e_*^{1/4-9\varepsilon}/4 \le 1/32$. The unit-covariance Wick integral is bounded by the weight-one Grassmann norm and integrates 1 to 1. Thus

$$\left|\int \mathrm{d}\mu_{C_\theta} e^{V_*} - 1\right| \le |e^{V_*'} - 1|_1 \le e^{|V_*'|_1} - 1 \le 2|V_*'|_1 \le \varepsilon_1.$$

Together with (22.41), this proves (22.38). The density is jointly continuous by Theorem 22.2. The twist determinant is a continuous positive function, so its quotient in (22.38) is continuous on the asserted set. □

22.6. **Partition bounds and comparison of torus sizes.** Let $z_{r,n}$ be the matter factor in (20.20), formed at the real mass $\bar{m}\eta + m_{\mathrm{in}}$. The fine normalization is

$$z_{r,n} = e^{\varepsilon^{\mathrm{v}}_{\mathrm{in}} n^3} \det(D^{\mathrm{ch}}[0] + \bar{m}\eta)\, \bar{\nu}(1) > 0. \tag{22.43}$$

Both signs of the total lattice mass are allowed here. The positive free reference mass $\bar{m}\eta$ is used only in the determinant normalization.

Proposition 22.7. *For every controlled real pair on R, the following two-sided estimates hold:*

$$e^{-\varepsilon^{\mathrm{v}}_k \ell_*^3} R^{\mathrm{tw}}_\theta (1-\varepsilon_1)(1-\varepsilon_4) \le \nu^\theta(\mathfrak{G}) \le e^{-\varepsilon^{\mathrm{v}}_k \ell_*^3} R^{\mathrm{tw}}_\theta (1+\varepsilon_1), \tag{22.44}$$

$$(1-\varepsilon_3)\bar{\nu}(1) \le \bar{\nu}(\mathfrak{G}) \le \bar{\nu}(1). \tag{22.45}$$

In particular

$$\left|\log z_{r,n} - \log z^{\mathrm{free}}_{r,n} - \varepsilon^{\mathrm{v}}_{\mathrm{in}} n^3 + \varepsilon^{\mathrm{v}}_k \ell_*^3\right| \le 2\varepsilon_1 + \varepsilon_2 + 2\varepsilon_3 + 2\varepsilon_4 \le 2e_*^{1/4-9\varepsilon}, \tag{22.46}$$

$$\left|\log z_{r,n} - [p^0_{\mathrm{ch}}(\bar{m}\eta) + \varepsilon^{\mathrm{v}}_{\mathrm{in}}] n^3 + \varepsilon^{\mathrm{v}}_k \ell_*^3\right| \le \varepsilon_5(\bar{m}R) + 2e_*^{1/4-9\varepsilon}. \tag{22.47}$$

Proof. Apply (22.14) at $a = 0$ and $\Phi = 1$, then insert (22.38). The result is

$$\nu^\theta(\mathfrak{G}) = e^{-\varepsilon^{\mathrm{v}}_k \ell_*^3} R^{\mathrm{tw}}_\theta \int \chi_*(A)(1 + r_*(\theta, A))\, \mathrm{d}\mathsf{m}_*(A).$$

The left side and the prefactor are real, and the prefactor is positive. The last integral therefore equals its real part. The remainder bound and $1 - \varepsilon_4 \le \mathsf{m}_*(\chi_*) \le 1$ give (22.44). The first inequality of (22.16) gives $\bar{\nu}(\mathfrak{G}^c) \le \varepsilon_3 \bar{\nu}(1)$. This and positivity prove (22.45).

Multiply (22.44) by $\det(D^{\mathrm{ch}}[0] + \bar{m}\eta) > 0$. The determinant identity (22.31) and the penalty twist bound replace this product by the fine twisted free determinant, with factors between $e^{-\varepsilon_2}$ and e^{ε_2}. Integrating in θ gives

$$\begin{aligned} &e^{-\varepsilon^{\mathrm{v}}_k \ell_*^3} z^{\mathrm{free}}_{r,n} e^{-\varepsilon_2} (1-\varepsilon_1)(1-\varepsilon_4) \\ &\qquad \le \det(D^{\mathrm{ch}}[0] + \bar{m}\eta)\bar{\nu}(\mathfrak{G}) \le e^{-\varepsilon^{\mathrm{v}}_k \ell_*^3} z^{\mathrm{free}}_{r,n} e^{\varepsilon_2}(1+\varepsilon_1). \end{aligned}$$

All integrands are continuous in θ, and the factors being compared are positive. Use (22.45) and (22.43), then take logarithms. Writing the expression on the left of (22.46) without absolute values as Q_n, we obtain

$$-\varepsilon_2 + \log(1-\varepsilon_1) + \log(1-\varepsilon_4) \le Q_n \le \varepsilon_2 + \log(1+\varepsilon_1) - \log(1-\varepsilon_3).$$

The errors lie in $[0, 1/2]$. The inequalities $-\log(1-x) \le 2x$ and $\log(1+x) \le x$ therefore give the first bound in (22.46). Its right side is at most $e_*^{1/4-9\varepsilon} + 3e_*$. Since $e_*^{1/4} \le 1/8$, we have $3e_* \le e_*^{1/4} \le e_*^{1/4-9\varepsilon}$, which gives the second bound. Adding (22.21) proves (22.47). □

We now use the pair $P_N(e)$ tuned on the top torus in Theorem 16.4. Define the common pressure

$$p_F(N, e, \bar{m}) = p^0_{\mathrm{ch}}(\bar{m}\eta) + \varepsilon^{\mathrm{v}}_{0,N}(e). \tag{22.48}$$

The pair and this pressure are independent of R within the fixed torus scheme.

Theorem 22.8. *For every $R \in \mathcal{T}(e)$ and $N \ge N_{\min}(R,e)$, the tuned matter factor satisfies*

$$\Big|\log z_{r,n} - \log z_{r,n}^{\mathrm{free}} - \varepsilon_{0,N}^{\mathrm{v}}(e) n^3\Big| \le 3e_*^{1/4-9\varepsilon},$$

$$\Big|\log z_{r,n} - p_F n^3\Big| \le \varepsilon_P(R) := \varepsilon_5(\bar m R) + 3e_*^{1/4-9\varepsilon} \le \varepsilon_5(1/L) + 3e_K^{1/4-9\varepsilon}. \tag{22.49}$$

In particular let $R_* = R_{\mathrm{top}}/L$, $n_* = R_*/\eta$, and $N \ge N_{\mathrm{ref}}(e)$. The two tori R_* and LR_* use the same mass and the same pressure. Their matter comparison satisfies

$$\begin{aligned} |D_{\mathrm{mat}}| &:= \big|L^3 \log z_{r,n_*} - \log z_{r,Ln_*}\big| \le L^3 \varepsilon_P(R_*) + \varepsilon_P(LR_*) \\ &\le (L^3+1)\Big[\varepsilon_5(\bar m R_*) + 3e_K^{1/4-9\varepsilon}\Big] \le C_{\mathrm{mat}}, \end{aligned}$$

$$C_{\mathrm{mat}} = (L^3+1)\big(\varepsilon_5(1/L)+1\big). \tag{22.50}$$

The displayed bounds are uniform in N; C_{mat} depends only on L, ρ and the fixed chamber weights.

Proof. The common pair is controlled on every torus in the asserted range, by Theorem 16.4. Its terminal vacuum obeys $|\varepsilon_k^{\mathrm{v}}| \le 5c_{\mathrm{fl}} e_*^{1/4-6\varepsilon}$. The second inequality of (22.36) therefore gives

$$|\varepsilon_k^{\mathrm{v}}|\ell_*^3 \le \tfrac14 e_*^{1/4-7\varepsilon} \le e_*^{1/4-9\varepsilon}.$$

Remove this term from (22.46) and (22.47). This proves the first two estimates in (22.49). The function ε_5 is decreasing. The scheme's scale bounds give $\bar m R \ge \mu_K > 1/L$ and $e_* \le e_K$. Since $1/4 - 9\varepsilon > 0$, they prove its last estimate.

Both R_* and $LR_* = R_{\mathrm{top}}$ belong to $\mathcal{T}(e)$. The mesh threshold is monotone: $N_{\min}(LR_*, e) \le N_{\min}(R_*, e) = N_{\mathrm{ref}}(e)$. Thus the common pair and pressure are available on both tori at every cutoff in the theorem. The bulk pressure cancels exactly:

$$D_{\mathrm{mat}} = L^3(\log z_{r,n_*} - p_F n_*^3) - (\log z_{r,Ln_*} - p_F (Ln_*)^3).$$

The pressure bounds give the first inequality in (22.50). Monotonicity of ε_5 and $e_* \le e_K$ give the second. Finally $\bar m R_* > 1/L$ and $e_K^{1/4-9\varepsilon} \le 1/8$ show that the bracket is at most $\varepsilon_5(1/L)+1$. The terminal scales, R_* and e_K, are independent of N; this proves the stated uniformity. □

23. Interacting response on the reference torus

The free coefficients do not by themselves control the reference Wick products. We must compare the normalized interacting generating function with its one-cell free factors, including mixed derivatives in different cells. The quantity to be estimated is

$$\Theta_m^{(0)} = \left| \left(\prod_{c \in \mathcal{C}_*} (\partial_{t_c}^m - c_m) \mathring{\Phi} \right)(0) \right|^{1/N_*}, \qquad m = 2, 4, \qquad N_* = |\mathcal{C}_*| = 8L^3, \tag{23.1}$$

where $c_m = N_0^{(m)}(0)$. The exact relation to Wick products is (20.36). The proof uses decay between separate insertions, an analytic logarithm for the terminal fermion integral, and a separate estimate of the complement of the small-field set.

23.1. **The terminal determinant ratio.** For the marked connections of (21.4), set

$$C_b = 1 + \sqrt{h}\, C_\beta, \qquad t_x = \frac{e_*^{3\varepsilon}}{C_b}, \qquad t_x^+ = \frac{e_*^{3\varepsilon}}{\sqrt{h}\, C_\beta}, \qquad r = t_x/4. \tag{23.2}$$

If $C_\beta = 0$, interpret $t_x^+ = \infty$. For t in the polydisc $\mathbb{D}(t_x^+) = \{|t_c| < t_x^+ \text{ for every } c\}$, all the rescaled connections $a(t)^{[j]}$ belong to $\mathfrak{X}_j$ and $e_* |\mathscr{C}(t)|_\infty < e_*^{3\varepsilon}$. Indeed, (21.7) and $e_* \le e\sqrt{h}$ give $|a(t)|_{\mathrm{ext}} < e_*^{-1+3\varepsilon}$. This bounds the stop sup and first-difference norms by the defining thresholds of $\mathfrak{X}_*$. For $i = k_* - j$ the rescaling multiplies the sup norm by $L^{-i/2}$ and the first differences

by $L^{-3i/2}$, while $e_j = L^{-i/2}e_*$. Since $\varepsilon < 1/3$, the rescaled bounds are strictly below $e_j^{-1+2\varepsilon}$ and $e_j^{-1+3\varepsilon}$, respectively.

We first control the determinant activities accumulated along the flow. Recall the local determinant function $E_j^d(X,A)$ of Proposition 11.5 and set $E_j^{\det}(X,A) = E_j^d(X,A) - E_j^d(X,0)$. Its analyticity, locality, real gauge invariance, and evenness will all be used below; the relevant bound is

$$|E_j^d(X,A)| \le C_d M^3 e^{-(\kappa+1)d_M(X)} \qquad (A \in \widetilde{G}_j). \tag{23.3}$$

Let $\mathfrak{P}_0 = 0$ and

$$\mathfrak{P}_{j+1}(Y,C) = \sum_{\overline{X}^{\,L} = LY} \big(\mathfrak{P}_j(X,C_L) + E_j^{\det}(X,C_L)\big). \tag{23.4}$$

Then $\mathfrak{P}_j(X,\cdot)$ is analytic on $\mathfrak{X}_j$, vanishes at zero, and depends only on bonds meeting X. To see locality under reblocking, a bond meeting X has, after division by L, an endpoint in Y whenever $\overline{X}^{\,L} = LY$. Analyticity follows from $C_L \in \mathfrak{X}_j$.

For $d_M(Y) \ge 1$, the reblocking estimate gives

$$\sum_{\overline{X}^{\,L} = LY} e^{-\kappa d_M(X)} \le c_{\mathrm{rb}} L^3 e^{-L(\kappa-\kappa_0-1)d_M(Y)}.$$

With $L(\kappa-\kappa_0-1) \ge \kappa$ and $c_{\mathrm{rb}}L^3 e^{-[L(\kappa-\kappa_0-1)-\kappa]} \le 1/2$, the weighted supremum q_j of $\mathfrak{P}_j$ over $d_M(X) \ge 1$ satisfies $q_{j+1} \le (q_j + 2C_dM^3)/2$. Since $q_0 = 0$,

$$|\mathfrak{P}_j(X,C)| \le 2C_d M^3 e^{-\kappa d_M(X)} \qquad (d_M(X) \ge 1). \tag{23.5}$$

At $j = 0$ the determinant activity includes all 192 chamber components. The primitive bound and reblocking estimate have the same constants at that first step.

The determinant identity (13.13) gives

$$\frac{\det(D^{\mathrm{ch}}[qa] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)} = \exp\left(\sum_X \mathfrak{P}_*(X, a^{[k_*]})\right) \frac{\det D_*(a^{[k_*]})}{\det D_*(0)}, \tag{23.6}$$

$$\sum_X \mathfrak{P}_*(X, a^{[k_*]}) = \sum_{j<k_*} \sum_X E_j^{\det}(X, a^{[j]}). \tag{23.7}$$

Here $a^{[j]} \in \mathfrak{X}_j$ for every j. The fine determinant uses the phase-argument convention of (18.3): its connection argument is $qa(t) = \sqrt{\eta}\sum_c t_c \widehat{\mathfrak{a}}_c$. For a single cell this is precisely the phase connection B_t in (18.3). The one-step identity reads

$$\det D_{j+1}(a^{[j+1]}) = b^{4n_{j+1}} \det D_j(a^{[j]})/\delta Z_j(a^{[j]}).$$

Divide it by its zero-field version and use (11.34). The resulting ratio acquires the factor

$$\exp\left(-\sum_X E_j^{\det}(X, a^{[j]})\right)$$

at each step, which proves the first identity by iteration. Summing (23.4) over Y proves the second: each fine polymer has exactly one reblocked image. The zero-field determinants are nonzero by the real covariance formula. At $j = 0$ the incoming determinant has dimension $192n_0$, the outgoing one dimension $4n_1$; the factor b^{4n_1} cancels in the ratio.

Fix a coarse real photon field A with $\chi_*(A) = 1$, and a real twist θ. The real gauge construction in (10.31)–(10.32) gives a representative A^g with $|A^g|_\infty \le 2\ell_*^2 p_*$. The curvature bound and

Proposition 10.4 put both $A + c_\theta$ and $A^g + c_\theta$ in Dom_{k_*}. Write

$$\mathscr{A}^0 = H_*^{\mathrm{ph}} A^g, \qquad c_{\theta,\mu} = \frac{\theta_\mu}{e_* \ell_*}, \qquad \mathscr{A} = \mathscr{A}^0 + c_\theta.$$

Summing the three photon kernel estimates in (10.16) bounds each field norm of $\mathscr{A}^0$ by $2C_0 \ell_*^2 p_* < e_*^{-\varepsilon}/2$; the strict inequality is the corresponding condition in (22.36). The local photon domains and their invariance under real constants then give $s\mathscr{A}^0 + c_\theta \in \frac{1}{4}\widetilde{R}_*$ for $0 \le s \le 1$. In particular all connections needed below belong to $\mathcal{W}$.

The flow bound (16.20) gives an activity with even Grassmann kernels, with equal barred and unbarred degrees, and

$$\|\mathscr{E}_*\|_*^x \le 4 e_*^{1/4-6\varepsilon}, \qquad |m_*| \le e_*^{3/4-8\varepsilon}/8. \tag{23.8}$$

The coefficient norm includes the weights $\eta_*^3/48$ for a fine spinor index and $(\eta_*^3/48)^2$ for a pair index, with all admissible pairs, including pairs inside a cluster. Its polymer factor is $e^{\kappa d_M(X)}$. For $(A, C) \in \widetilde{R}_* \times \mathfrak{X}_*$ define

$$Z_{\mathrm{in}}(A, C) = \int D\Psi \, \exp\left\{ -\langle \bar{\Psi}, D_*(A + C)\Psi \rangle - m_* \langle \bar{\psi}, \psi \rangle + \sum_X \mathscr{E}_*(X; A, C, \psi, \chi) \right\}, \tag{23.9}$$

where $\psi, \bar{\psi}$ are substituted using $H_*^{\#}(A + C)\Psi$ and their barred partners, and $\chi, \bar{\chi}$ using the corresponding covariant Hölder quotients. The inclusion $\widetilde{R}_* + \mathfrak{X}_* \subset \widetilde{G}_*$ makes all these maps analytic; the activity kernels are analytic in (A, C). Berezin integration is coefficient extraction, and the exponential is its scalar exponential times a finite polynomial in the positive-degree coefficients. Thus Z_{in} is analytic on this two-field domain. Set

$$Z_{\mathrm{tm}}(t) = Z_{\mathrm{in}}(\mathscr{A}, \mathscr{C}(t)), \tag{23.10}$$

$$f_{\mathrm{int}}(t) = \frac{Z_{\mathrm{tm}}(t)}{\det D_*(c_\theta + \mathscr{C}(t))}. \tag{23.11}$$

The Berezin measure has one pair for each of the $4\ell_*^3$ coarse spinor coordinates. The two-field analyticity gives analyticity of Z_{tm} on $\mathbb{D}(t_x^+)$.

Proposition 23.1. *Suppose*

$$4C_0 \ell_*^3 g_1 \le e_*^{-1/4+\varepsilon}, \qquad 4\ell_*^3(2C_K + K_0 + 4) \le e_*^{-2\varepsilon}, \qquad e_*^{1/4-9\varepsilon} \le 1/8, \tag{23.12}$$

$$48 e_*^{2\varepsilon} \le \mu_-. \tag{23.13}$$

Then f_{int} is analytic and

$$|f_{\mathrm{int}}(t) - 1| \le \frac{1}{2} e_*^{1/4-9\varepsilon} \le \frac{1}{16}. \tag{23.14}$$

In particular Z_{tm} has no zero on the polydisc. The normalized terminal ratio

$$R_{\mathrm{tm}}(t; \theta, A) = \frac{Y_*^{\mathrm{x}}(A + c_\theta; \mathscr{C}(t))}{Y_*^{\mathrm{x}}(A + c_\theta; 0)} = \exp\left(\sum_X \mathfrak{P}_*(X, \mathscr{C}(t)) \right) \frac{Z_{\mathrm{tm}}(t)}{Z_{\mathrm{tm}}(0)} \tag{23.15}$$

is nonzero, continuous in (t, θ, A) on $\mathbb{D}(t_x^+) \times \mathbb{R}^3 \times \{\chi_* = 1\}$, and analytic in t. It has a unique continuous logarithm $\mathcal{G}(t; \theta, A)$ that vanishes at zero, and this logarithm is analytic in t.

Proof. Use the covariance

$$C_t = D_*(c_\theta + \mathscr{C}(t))^{-1}.$$

The inverse exists by (21.27), because $e_* |\mathscr{C}(t)|_\infty < e_*^{3\varepsilon} < \mu_*/24$. Each covariance row has sum at most $4\ell_*^3 g_1$. Put $a_w = C_0^{-1} e_*^{-1/4+\varepsilon}$. The first condition in (23.12) gives $a_w \ge 4\ell_*^3 g_1 \ge 1$.

Separate the quadratic form with covariance C_t from the remaining even interaction V_t. Normalized Gaussian Grassmann integration gives

$$|f_{\mathrm{int}}(t) - 1| \le e^{|V_t|_{a_w}} - 1. \tag{23.16}$$

To justify this particular norm bound, replace the unbarred generators by $C_t\Psi$ and leave the barred generators fixed. Their row sums are at most a_w, so this substitution maps the a_w coefficient norm into the unit coefficient norm. Integration against the identity covariance has norm one in that latter norm. Apply this to $e^{V_t} - 1$, and use the submultiplicativity of the coefficient norm.

There are three terms in V_t. The difference between the two coarse quadratic forms has entry sum at most $C_K e_*^{1-\varepsilon}\ell_*^3/2$, by (21.16) integrated along $c_\theta + \mathscr{C}(t) + s\mathscr{A}^0$. A quadratic Grassmann form has coefficient norm at most $4a_w^2$ times this block sum, so its contribution is bounded by $2C_K\ell_*^3 e_*^{1/2+\varepsilon}$. The terminal mass estimate and (23.8) give respectively

$$\frac{1}{2}\ell_*^3 e_*^{1/4-6\varepsilon}, \qquad K_0\ell_*^3 e_*^{1/4-7\varepsilon}.$$

For the activity sum, substitute the terminal field maps in the coefficient norm, then sum $e^{-\kappa d_M(X)}$ by assigning each polymer one of its M-cubes and using the polymer sum bound K_0. We have used $4e_*^{\varepsilon} \le 1$. Consequently

$$|V_t|_{a_w} \le \ell_*^3(2C_K + K_0 + 4)e_*^{1/4-7\varepsilon} \le \frac{1}{4}e_*^{1/4-9\varepsilon}.$$

The inequality $e^x - 1 \le 2x$ proves (23.14). The denominator in (23.11) is analytic and nonzero, so f_{int} is analytic.

By (22.9) the terminal density has the form

$$Y_*^{\mathrm{x}}(A'; C) = (\det D_*(0))^{-1} \exp\left(-\varepsilon_*^{\mathrm{v}}\ell_*^3 + \sum_X \mathfrak{P}_*(X, C)\right) Z_{\mathrm{in}}(H_*^{\mathrm{ph}}A', C).$$

Its gauge invariance in Theorem 22.2 permits replacing A by A^g. The photon minimizer preserves real constants. Dividing the density at $C = \mathscr{C}(t)$ by its value at zero proves (23.15); the quotient also shows that the result is independent of the chosen gauge representative. For continuity, use the ungauged linear map $A + c_\theta \mapsto H_*^{\mathrm{ph}}(A + c_\theta)$ in the displayed formula. The terminal Grassmann integral is analytic in its two field arguments, and its zero-mark denominator is nonzero.

The required logarithm is

$$\mathcal{G}(t; \theta, A) = \int_0^1 \sum_c t_c \frac{\partial_{t_c} R_{\mathrm{tm}}(st; \theta, A)}{R_{\mathrm{tm}}(st; \theta, A)}\, \mathrm{d}s. \tag{23.17}$$

Local Cauchy formulas show that the derivatives in this integrand are jointly continuous; integration over $[0, 1]$ preserves continuity and analyticity. Along a ray the integrand is the logarithmic derivative of $R_{\mathrm{tm}}(st)$. Differentiating the product of this ratio with the exponential of minus the integrated logarithmic derivative gives zero, and its value at $s = 0$ is one. Hence $e^{\mathcal{G}} = R_{\mathrm{tm}}$. Any two continuous logarithms differ by $2\pi i$ times an integer-valued continuous function on the connected polydisc. Their common value at zero proves uniqueness. □

23.2. **The one-cell free logarithm and twists.** For a fixed cell c and real twist θ, let

$$g_D^\theta(t) = \mathrm{tr}\log\left(1 + D_*(c_\theta)^{-1}[D_*(c_\theta + t\widehat{c}_c) - D_*(c_\theta)]\right). \tag{23.18}$$

The logarithm here is the matrix power series. Set

$$C_{\mathrm{fr}} = 2592 C_d K_0 C_\beta^2 \left(M^2 (D_\beta^{\mathrm{buf}})^3 L^4 h^{-2} + 8M^5 h\right),$$
$$B_{\mathrm{wr}} = 16\exp(6) C_d M^3 K_0 p_\beta e^{-r_*}, \qquad B_1 = 16 g_1 C'_K p_\beta e_*^{2\varepsilon},$$

$$B_0 = C_{\mathrm{fr}} e_*^{4\varepsilon} + B_{\mathrm{wr}} + B_1. \tag{23.19}$$

Proposition 23.2. *Suppose* (23.12)–(23.13) *hold,* $r_* > 31L + 16$, *and*

$$2g_1 C_K' e_*^{2\varepsilon} \le 1/2, \quad B_0 \le 1/4, \quad \tau_s := c_\tau \ell_* \ge 1, \quad 52 B_1 e^{-\tau_s} \le e_*^{1/4-9\varepsilon}. \tag{23.20}$$

The free ratio N_0 *has the logarithm*

$$\ell_0(t) = \sum_X \mathfrak{P}_*(X, t\widehat{c}_c) + g_D^0(t), \qquad N_0(t) = e^{\ell_0(t)}, \tag{23.21}$$

analytic on $|t| < t_x^+$. *It is independent of* c *and of the admitted placement, is even and real on the real axis, and satisfies*

$$\left| \sum_X \mathfrak{P}_*(X, t\widehat{c}_c) \right| \le C_{\mathrm{fr}} e_*^{-2\varepsilon} |t|^2 + B_{\mathrm{wr}}, \tag{23.22}$$

$$|\ell_0(t)| \le B_0 \qquad (|t| \le t_x). \tag{23.23}$$

The one-cell interacting error $w_c(t) = \mathcal{G}(te_c; \theta, A) - \ell_0(t)$ *is analytic, vanishes at zero, and obeys*

$$|w_c(t)| \le 3 e_*^{1/4-9\varepsilon} \qquad (|t| < t_x^+). \tag{23.24}$$

Proof. Write $K = D_*(c_\theta + t\widehat{c}_c) - D_*(c_\theta)$. The real covariance and the integrated insertion bounds give

$$\|D_*(c_\theta)^{-1} K\|_2 \le 2g_1 C_K e_*^{2\varepsilon} \le 1/2, \qquad \|D_*(c_\theta)^{-1} K\|_1 \le 4 g_1 C_K p_\beta e_*^{2\varepsilon}.$$

Thus the series in (23.18) converges, is analytic, vanishes at zero, and has modulus at most B_1. Triangularizing the matrix identifies its exponential with $\det D_*(c_\theta + t\widehat{c}_c)/\det D_*(c_\theta)$. Combine this with (23.6). The lattice symmetries carrying one cell probe to another preserve the fine determinant, so the resulting fine ratio is N_0.

The bound (23.22) requires a cancellation before summing the levels. Put $i = k_* - j$ and $B_j = e^{-1}\widehat{\mathfrak{a}}_c^{[j]}$. Rescaling gives

$$e_j |\partial B_j|_\infty \le a_j^2 a_*^{-3/2} C_\beta. \tag{23.25}$$

Let $\mathcal{Q}_j$ be the M-cubes containing endpoints of the supporting bonds of B_j. The buffered support cube yields

$$|\mathcal{Q}_j| \le 4 \left((D_\beta^{\mathrm{buf}})^3 M^{-3} a_j^{-3} + 8 \right), \qquad |\mathcal{Q}_j| \le L^{3i} p_\beta. \tag{23.26}$$

By locality only polymers containing a cube of $\mathcal{Q}_j$ contribute to (23.7).

For such a polymer X, put $d = d_M(X)$ and $s_X = M(d+3)$. Suppose first that $2s_X < \ell_j$. All bonds meeting X lie in a coordinate cube of radius $M(d+2)+1$ around a chosen site $\tilde{c}$ of X. The strict inequality $2[M(d+2)+1] + \eta_j < 2s_X < \ell_j$ makes this cube a lift to the lattice. Define a real function there by $\lambda(c) = \sum_\nu B_{j,\nu}(\tilde{c})(c - \tilde{c})_\nu$, and extend it by zero outside. On bonds meeting X, its gradient is the constant value $B_j(\tilde{c})$. If $[\,\cdot\,]_X$ restricts a connection to those bonds, then

$$B_X = [B_j - d\lambda]_X, \qquad |B_X|_\infty \le 3 s_X |\partial B_j|_\infty.$$

The last inequality follows by integrating the first differences along coordinate paths inside the lifted cube.

Real gauge invariance must be used before continuing to the complex gauge parameter. On the convex open set where $|[tB_j - zd\lambda]_X|_\infty < e_j^{-1+\varepsilon}$, the analytic function $z \mapsto E_j^d(X, [tB_j - zd\lambda]_X)$ equals $E_j^d(X, tB_j)$ for real z near zero: the unrestricted connection is related to tB_j by a real gauge transformation, and locality permits the restriction. The identity theorem gives this equality throughout that set. In particular it holds at $z = t$ whenever $|t| < \rho_X$, where $\rho_X = e_j^{-1+\varepsilon}/(3 s_X |\partial B_j|_\infty)$.

The function $z \mapsto E_j^d(X, zB_X) - E_j^d(X, 0)$ is even, bounded by $2C_d M^3 e^{-(\kappa+1)d}$ on $|z| < \rho_X$, and has a double zero at zero. Apply the maximum principle after dividing by z^2. For $|t| \geq \rho_X$ the original primitive bound gives the same inequality with the factor $(|t|/\rho_X)^2 \geq 1$. Thus, in both cases,

$$|E_j^{\det}(X, tB_j)| \leq 18 C_d M^5 C_\beta^2 (d+3)^2 e^{-(\kappa+1)d} |t|^2 e_j^{-2\varepsilon} a_j^4 a_*^{-3}. \tag{23.27}$$

When $\partial B_j = 0$, $B_X = 0$ and this contribution vanishes. This proves the cancellation also at the chamber first step.

Since $(d+3)^2 e^{-(\kappa+1)d} \leq 9e^{-\kappa_0 d}$, summing (23.27) over cubes and polymers gives at level j the bound

$$648 C_d K_0 C_\beta^2 |t|^2 e_j^{-2\varepsilon} a_*^{-3} \left(M^2 (D_\beta^{\mathrm{buf}})^3 a_j + 8 M^5 a_j^4 \right).$$

For the remaining polymers $2s_X \geq \ell_j$, $d \geq r_* L^i/2 - 3$. The extra exponential in the primitive bound then gives at level j

$$2 \exp(6) C_d M^3 K_0 p_\beta L^{3i} e^{-r_* L^i}.$$

Sum over $i \geq 1$, using $a_j = L^{-i} a_*$, $e_j^{-2\varepsilon} = e_*^{-2\varepsilon} L^{i\varepsilon}$, $\sum_i L^{-i(1-\varepsilon)} \leq 1$, and $a_*^{-2} \leq L^4 h^{-2}$, $a_* \leq h$. For $r_* > 31L + 16$, the second sum is bounded by a constant times e^{-r_*}. The stated definitions of C_{fr} and B_{wr} give (23.22). Together with $|g_D^0| \leq B_1$ this proves (23.23).

Evenness and reality follow from the corresponding fine determinant symmetries. Both $\ell_0(t)$ and $\ell_0(-t)$ vanish at zero and have exponential $N_0(t)$, so they coincide. For real t, the nonzero real ratio $N_0(t)$ is positive by continuity from one at zero; the normalized logarithm is therefore real. Cell and placement independence follows in the same way from the determinant translation and reflection symmetries.

It remains to control the twist. For $|\operatorname{Im} \theta_\mu| \leq \tau_s$ the connections $c_\theta + st\widehat{c}_c$ lie in $\mathcal{W}^+$, and $\|D_*(c_\theta)^{-1}\|_2 \leq g_1$. There is a positive strip margin $\ell_*(e_*^{2\varepsilon} - e_*^{3\varepsilon})$: split the imaginary constant connection into a part bounded by c_τ/e_* and a remaining part absorbed into $\mathfrak{X}_*$. The latter has vanishing first differences. The mass condition bounds its total imaginary part by $\mu_*/24$. On this larger open strip, (21.23) gives the same matrix power-series bound $|g_D^\theta| \leq B_1$. The function is holomorphic there and 2π-periodic in each twist variable. For real twists, periodicity is conjugation by the periodic gauge factor $\exp(-2\pi i c_\mu/\ell_*)$; the identity theorem extends it to the strip.

We recall the short Fourier estimate to make the size of this suppression explicit. A bounded periodic holomorphic function g in a closed strip of width $\tau \geq 1$ has $|\widehat{g}(m)| \leq B e^{-\tau |m|_1}$: shift each Fourier contour towards $-i\tau \operatorname{sgn}(m_\mu)$, first using an interior width and then taking its limit. The vertical sides cancel by periodicity. The nonconstant Fourier sum is bounded by

$$B \left[\left(\frac{1 + e^{-\tau}}{1 - e^{-\tau}} \right)^3 - 1 \right] \leq 25 B e^{-\tau}.$$

The last inequality follows by expanding the bracket divided by $e^{-\tau}$ as a power series with positive coefficients and evaluating at $e^{-\tau} = e^{-1}$. Thus the difference between two real values is at most $52 B e^{-\tau}$. Apply this to $g_D^\theta(t)$ to obtain

$$|g_D^\theta(t) - g_D^0(t)| \leq 52 B_1 e^{-\tau_s} \leq e_*^{1/4 - 9\varepsilon}. \tag{23.28}$$

Finally the normalized logarithm identity and (23.11) give

$$w_c(t) = g_D^\theta(t) - g_D^0(t) + \log f_{\mathrm{int}}(te_c) - \log f_{\mathrm{int}}(0).$$

The principal logarithm of f_{int} is analytic and has modulus at most $e_*^{1/4 - 9\varepsilon}$. Both sides vanish at zero and exponentiate to the same ratio; uniqueness of the logarithm justifies the identity. Equation (23.28) proves (23.24). □

Let $\phi_0(t) = \sum_{i=0}^4 N_0^{(i)}(0)t^i/i!$. Cauchy's estimate applied to $|N_0(t) - 1| \le 2B_0$ on $|t| = t_x = 4r$ gives

$$|c_2| \le r^{-2}/16, \qquad |c_4| \le r^{-4}/16, \qquad \|\phi_0 - 1\|_r \le 1/6, \qquad \|\phi_0\|_r \le 2. \tag{23.29}$$

The odd coefficients vanish. Expanding $\log N_0$ through degree four gives $\ell_0^{(4)}(0) = c_4 - 3c_2^2$; hence this is the chamber fourth coefficient estimated in the free-response section.

23.3. **Derivatives in separated cells.** To estimate mixed logarithmic derivatives, we use the real covariance $G = D_*(\mathscr{A})^{-1}$. On the coarse Grassmann algebra, write an even element without constant term as $F = \sum_{p\ge2}(p!)^{-1}\sum F_p(u_1,\dots,u_p)\Psi(u_1)\cdots\Psi(u_p)$ with antisymmetric kernels. For a symmetric weight $w \ge 1$, define

$$\rho_G(F;w) = \sum_{p\ge2}\frac{h_G^p}{(p-1)!}\sup_{u_1}\sum_{u_2,\dots,u_p}|F_p(u_1,\dots,u_p)|w,$$

$$N_G(F;w) = \sum_{p\ge2}\frac{h_G^p}{(p-1)!}\sum_{u_1,\dots,u_p}|F_p(u_1,\dots,u_p)|w. \tag{23.30}$$

The supremum in ρ_G is taken separately at each degree. The indices include site, spin, and barred or unbarred status; their distance is the distance between the sites. We use

$$w_\delta(u_1,\dots,u_p) = e^{\vartheta_0 \max_{i,j} d(u_i,u_j)}, \qquad w_P(u_1,\dots,u_p) = e^{\vartheta_0 \max_i d(u_i,P)}.$$

The constants $h_G, \vartheta_0, \alpha_G$ are those of (21.26). Antisymmetrizing a kernel that is not antisymmetric bounds N_G by the same expression for that kernel. For ρ_G, replace the supremum at the first argument by the largest supremum over all argument positions.

In the fine algebra absorb the terminal mass into each single-cube activity:

$$E^a(X;t) = \mathscr{E}_*(X;\mathscr{A},\mathscr{C}(t)) - \mathbf{1}_{\{X \text{ is an } M\text{-cube}\}} m_*\langle\bar\psi,\psi\rangle_X.$$

Let $E^a_{nm}(X;t)$ denote its fine spinor/pair kernels. The antisymmetric mass kernel has norm $8M^3$: its two fine-index weights multiply $48\eta_*^{-3}$, its diagonal Kronecker factors leave $48(M/\eta_*)^3$ chamber sites, and there are four spin components and two orders. It follows from (23.8) that, when $a_w \le e_*^{-1/4+\varepsilon}$ and $(4 + M^3/2)e_*^\varepsilon \le 1$,

$$\sum_{n,m}\frac{a_w^{n+m}}{n!m!}\|E^a_{nm}(X;t)\| \le w_*^{\mathrm{E}} e^{-\kappa d_M(X)}, \qquad w_*^{\mathrm{E}} = e_*^{1/4-7\varepsilon} \quad (|t|_\infty \le t_x). \tag{23.31}$$

Cauchy's formula on the circles of radius $t_x/2$ in one or two distinct mark variables gives, on $|t|_\infty \le t_x/2$, the same bound for $\partial_c E^a_{nm}$ and $\partial_c\partial_{c'}E^a_{nm}$ multiplied respectively by $q_x = 2/t_x$ and q_x^2. The circle integrals precede the finite weighted kernel sums, so the coefficient norm is bounded without replacing it by a supremum of individual coefficients. Locality gives $\partial_c E^a(X;t) = 0$ unless $X \cap P_c \ne \varnothing$.

After substituting the terminal field maps, put

$$V(t) = -\langle\bar\Psi, K(t)\Psi\rangle + \sum_X E^a(X;t)[H_*^\#(\mathscr{A}+\mathscr{C}(t))\Psi],$$

$$K(t) = D_*(\mathscr{A}+\mathscr{C}(t)) - D_*(\mathscr{A}), \qquad V_0(t) = \sum_X E^a_{00}(X;t), \qquad V_+(t) = V(t) - V_0(t). \tag{23.32}$$

It is even and analytic, and V_+ has no constant term. Gaussian integration gives

$$Z_{\mathrm{tm}}(t) = \det D_*(\mathscr{A}) e^{V_0(t)} \int \mathrm{d}\mu_G\, e^{V_+(t)}. \tag{23.33}$$

The choice of the real covariance will allow the exponential weights in the tree expansion.

Proposition 23.3. *Suppose*

(23.34)

$$5h_{\mathrm{L}}h_G \le e_*^{-1/4+\varepsilon}, \qquad (4+M^3/2)e_*^{\varepsilon} \le 1, \qquad \alpha_G \bar{r} \le 1/2, \quad \bar{r} = 4h_G^2 C_K e_*^{2\varepsilon} + 650 K_0 e_*^{1/4-7\varepsilon}.$$

Let $d_{\mathrm{sep}} = \min_{c \neq c'} d(P_c, P_{c'})$ *and suppose* $d_{\mathrm{sep}} \ge 6M+8$. *With* $s_F = (24/\gamma^F)^3$ *and* $C_V^{\mathrm{ins}} = 10^3 C_H^2 s_F^4 K_0$, *we have*

$$\rho_G(V_+(t); w_\delta) \le \bar{r} \qquad (|t|_\infty \le t_x),$$

$$N_G(\partial_c V_+(t); w_{P_c}) \le q_x C_V^{\mathrm{ins}} h_G^2 p_\beta, \quad \rho_G(\partial_c V_+(t); w_{P_c}) \le q_x C_V^{\mathrm{ins}} h_G^2,$$

(23.35)

$$N_G(\partial_c \partial_{c'} V_+(t); 1) + |\partial_c \partial_{c'} V_0(t)| \le q_x^2 C_V^{\mathrm{ins}} h_G^2 p_\beta e^{-\vartheta_0(d_{\mathrm{sep}} - 4M - 6)}.$$

The last two lines hold on $|t|_\infty \le t_x/2$, *with* $c \neq c'$ *in the last line. If the lattice probe vanishes, all marked derivatives vanish.*

Proof. We give the kernel summation which preserves the marked supports. For a polymer X, let X^+ be its unit-site one-neighborhood. Every fine index in an activity supported on X has its position set in X^+. This includes both chamber sites in an internal pair and the first site of an external nearest-neighbor pair whose other endpoint lies over X. The polymer diameter estimate gives

$$\vartheta_0 \operatorname{diam}(X^+) \le d_M(X) + 3. \tag{23.36}$$

The weighted kernel after substituting the field maps is, at degree $p = n+m$,

$$K_{nm}^X(u_1, \dots, u_p) = \int E_{nm}^a(X; t; v_1, \dots, v_p) \prod_{i=1}^p \mathrm{Leg}(v_i; u_i)\, \mathrm{d}v.$$

Here $\mathrm{d}v$ includes the fine weights already specified. Passing from the bidegree expression to a degree-p kernel produces the factor $p!/(n!m!)$; in the tree norms its remaining factor is $p h_G^p/(n!m!)$.

Suppose a weight is bounded by $c_w \prod_i e^{\vartheta_0 d(u_i, \mathrm{pos}(v_i))}$. Each summed leg contributes at most h_{L}. If one argument u is fixed, its leg contributes at most $h_{\mathrm{L}} e^{-(\gamma^F/4) d(u, X^+)}$. Thus the full kernel sum is at most $c_w h_{\mathrm{L}}^p \|E_{nm}^a(X;t)\|$, and the sum with one fixed argument retains the extra exponential in $d(u, X^+)$. For w_δ, the triangle inequality between the two farthest arguments and their position sets, followed by (23.36), permits $c_w = e^{d_M(X)+3}$.

The needed rooted polymer sum is

$$\sum_X e^{-(\gamma^F/4) d(u, X^+)} e^{-(\kappa-1) d_M(X)} \le 32 K_0. \tag{23.37}$$

Indeed the sum of $e^{-(\gamma^F/4) d(u, \square^+)}$ over M-cubes is at most 32. The neighboring 27 cubes contribute at most 27; a cube j layers away has distance at least $(j-1)M$ and there are at most $(2j+1)^3$ such cubes. Use $\gamma^F M/4 \ge 4$ to sum the remaining geometric tail. Assigning a cube of X and applying the anchored polymer sum K_0 proves (23.37).

Apply (23.31) with $a_w = 5h_{\mathrm{L}}h_G$. The factor 5^{-p} absorbs the polynomial degree factors. The preceding full and rooted sums give

$$\rho_G\left(\sum_X E^a(X;t)[H_*^{\#}\Psi]_+; w_\delta\right) \le 650 K_0 w_*^{\mathrm{E}}.$$

More generally, for kernels bounded by $\mathcal{B} e^{-\kappa d_M(X)}$ at that weight, these sums give $650 K_0 \mathcal{B}$ in ρ_G and $\exp(3)\mathcal{B}\sum_X e^{-(\kappa-1)d_M(X)}$ in N_G. This follows by summing $\sum_{p \ge 2} p 5^{-p} = 9/80$ in the rooted case and using $p \le 5^p$ in the full case. The quadratic term contributes at most $4h_G^2 C_K e_*^{2\varepsilon}$, since $\vartheta_0 \le \gamma^F/2$. This proves the first line of (23.35).

For the first derivative, there are three types of term: the coarse quadratic derivative, the activity-kernel derivative, and a derivative of one leg. The first has respective bounds $8h_G^2C_K\epsilon_c p_\beta$ and $4h_G^2C_K\epsilon_c$ in N_G and ρ_G, with $\epsilon_c = e_*|\widehat{c}_c|_\infty \le q_x$. In its chain through $z \in P_c$, the weight w_{P_c} is absorbed by the exponential distances from the two coarse endpoints to z.

For the kernel derivative, X must meet P_c. The same weight comparison uses that meeting point and (23.36). The Cauchy bound multiplies $\mathcal{B} = w_*^{\mathrm{E}}$ by q_x, and the sum of polymers meeting P_c is at most $K_0 p_\beta$. Its N_G bound is $\exp(3)K_0 p_\beta q_x w_*^{\mathrm{E}}$ and its ρ_G bound is $650K_0 q_x w_*^{\mathrm{E}}$.

For a differentiated leg at position j, the insertion estimate has a chain $w \in \mathrm{pos}(v_j) \to z \in P_c^+ \to u_j$. The weight obeys

$$w_{P_c} \le e^{d_M(X)+5} e^{\vartheta_0[d(w,z)+d(z,u_j)]} \prod_{i\neq j} e^{\vartheta_0 d(u_i,\mathrm{pos}(v_i))}.$$

The extra two in the exponent accounts for $d(P_c^+, P_c) \le M+1$. Summing the differentiated leg and the other legs leaves, before the degree sum,

$$2e^{d_M(X)+5} h_{\mathrm{L}}^{p-1} C_H \epsilon_c s_F \sum_{z\in P_c^+} e^{-(\gamma^F/2)d(z,X^+)} \|E^a_{nm}(X;t)\|.$$

There are p choices of the differentiated leg. The kernel norm contributes one further factor p, and $p^2 \le 5^p$ absorbs both. Use (23.37) with $\gamma^F/2$ in place of $\gamma^F/4$. The result is bounded by $9500 C_H s_F K_0 w_*^{\mathrm{E}} p_\beta \epsilon_c$ in N_G.

For ρ_G, if the fixed argument lies on the differentiated leg, split its two-edge exponent as $7\gamma^F/8 = 3\gamma^F/8 + \gamma^F/2$. Summing the intermediate site leaves $s_F e^{-(\gamma^F/4)d(u,X^+)}$. If the fixed argument lies on another leg, that leg supplies the same rooted exponential, and the differentiated leg contributes $2C_H\epsilon_c s_F^2$. Both cases therefore have the bound $9500 C_H s_F^2 K_0 w_*^{\mathrm{E}} \epsilon_c$. Here the degree sum is $\sum_{p\ge 2} p^2 5^{-p} = 43/160$. Adding the three types of term, using $w_*^{\mathrm{E}} \le 1/8$, $C_K \le C_H s_F^2$, and $C_H \ge 2$, proves the second line with $C_V^{\mathrm{ins}} = 10^3 C_H^2 s_F^4 K_0$.

For the mixed derivative, put

$$d_0 = d(P_c^+, P_{c'}^+), \quad d_1 = d(X^+, P_c^+), \quad d_2 = d(X^+, P_{c'}^+).$$

Then $d_0 \ge d_{\mathrm{sep}} - 2M - 2$. The differentiated coarse quadratic term is bounded in N_G by

$$16 h_G^2 C_H^2 s_F^3 p_\beta q_x^2 e^{-(\gamma^F/2)d_{\mathrm{sep}}}.$$

For an activity kernel with p legs, there are $(p+1)^2$ placements of the two derivatives. They may act on two legs, the same leg, a kernel and a leg, or both on the kernel. The summed first differentiated legs are bounded by $2C_H\epsilon_c s_F^2 e^{-(\gamma^F/2)d_1}$ and its c' version. The mixed leg is bounded by

$$2C_H^2 \epsilon_c \epsilon_{c'} s_F^3 e^{-(\gamma^F/2)d_0} \big(e^{-(\gamma^F/2)d_1} + e^{-(\gamma^F/2)d_2}\big).$$

If a derivative acts on the kernel, its corresponding distance d_i is zero by locality. Normalize the first kernel derivatives by q_x and the mixed one by q_x^2. Each of the four resulting kernel types satisfies (23.31). Every product-rule term is consequently at most $4q_x^2 C_H^2 s_F^4 h_{\mathrm{L}}^p \|E_{\mathrm{type}}\| \mathcal{D}(X)$, where

$$\mathcal{D}(X) = e^{-(\gamma^F/2)(d_1+d_2)} + e^{-(\gamma^F/2)d_0}\big(e^{-(\gamma^F/2)d_1} + e^{-(\gamma^F/2)d_2}\big).$$

The triangle inequality through X^+ gives $d_0 \le d_1 + d_2 + M d_M(X) + 2M + 2$. Use $\vartheta_0 \le \gamma^F/8$ and $\vartheta_0 M \le 1$ to obtain

$$\mathcal{D}(X) e^{-\kappa d_M(X)} \le 2e^{-\vartheta_0(d_0-2M-2)}\big(e^{-(\gamma^F/4)d_1} + e^{-(\gamma^F/4)d_2}\big) e^{-(\kappa-1)d_M(X)}.$$

Sum the two distance factors over the sites of P_c^+ and $P_{c'}^+$, then use (23.37). The degree factor $p(p+1)^2$ is at most 5^p for $p \ge 2$. At $p = 0$ there is one kernel term with no legs, and the same bound controls $\partial_c \partial_{c'} V_0$. The four kernel types contribute in total at most $256 C_H^2 s_F^4 K_0 p_\beta q_x^2 e^{-\vartheta_0(d_{\mathrm{sep}}-4M-4)}$.

Adding the quadratic term proves the last line of (23.35). If the lattice probe is zero, $\mathscr{C}(t) = 0$ and there is no mark dependence. □

We next turn these kernel estimates into a bound for the logarithm. The finite-dimensional tree formula is recalled with its proof, since the distance between its two marked vertices is the source of the mixed-cell decay.

For n copies of the generators, let $\Delta^{ij} = \sum_{x,y} G(x,y)\partial_{\bar\Psi^{(j)}(y)}\partial_{\Psi^{(i)}(x)}$ and $\Delta_{\{i,j\}} = \Delta^{ij} + \Delta^{ji}$. The derivatives are left Grassmann derivatives. For a forest T and edge parameters $\tau \in [0,1]^T$, let $s^T_{ij}(\tau)$ be the minimum parameter on the path from i to j, zero if there is no path, and $s^T_{ii} = 1$. Write $\Delta[s] = \sum_{i,j} s_{ij}\Delta^{ij}$ and let ϵ_0 extract the constant term. For even elements $F_1, \dots, F_n$ without constant term, define

$$T_n(F_1,\dots,F_n) = \sum_{\text{trees } T} \int_{[0,1]^T} \epsilon_0 \left\{ \prod_{\ell\in T} \Delta_\ell e^{\Delta[s^T(\tau)]} \prod_i F_i(\Psi^{(i)}) \right\} \mathrm{d}\tau. \tag{23.38}$$

For $n = 1$ this is the Gaussian expectation.

Proposition 23.4. *If $V_+(t)$ satisfies* (23.34) *and is analytic, then*

$$W(t) = V_0(t) + \sum_{n\ge 1} \frac{1}{n!} T_n(V_+(t),\dots,V_+(t))$$

is analytic on $\mathbb{D}(t_x)$ and $Z_{\mathrm{tm}}(t) = \det D_*(\mathscr{A}) e^{W(t)}$. For $c \neq c'$ and nonempty unit-site sets P, P',

$$\begin{aligned} |\partial_c\partial_{c'}(W - V_0)| &\le 2N_G(\partial_c\partial_{c'}V_+;1) \\ &\quad + 2\alpha_G N_G(\partial_c V_+; w_P)\rho_G(\partial_{c'}V_+; w_{P'}) e^{-\vartheta_0 d(P,P')}. \end{aligned} \tag{23.39}$$

Proof. The forest interpolation formula for a polynomial H is

$$H(\mathbf{1}) = \sum_F \int \left(\prod_{\ell\in F} \partial_{s_\ell} H \right)(s^F(\tau))\,\mathrm{d}\tau.$$

For a partition π of the vertices, keep fixed entries $s_{\mathrm{in}}(i,j)$ inside its blocks and put $X_\pi(u)_{ij} = u$ on pairs joining distinct blocks. The chain rule and the fundamental theorem of calculus give

$$G(X_\pi(u)) = G(X_\pi(0)) + \sum_{\substack{\ell=(i,j)\\ i,j \text{ in distinct blocks of } \pi}} \int_0^u (\partial_{s_\ell} G)(X_\pi(w))\,\mathrm{d}w. \tag{23.40}$$

When ℓ joins two blocks, merge them and assign the value w to every pair between those two blocks. The interpolation point is unchanged. Begin with singleton blocks, $G = H$ and $u = 1$, and iterate this identity on each integrand. Each derivative merges two components, so after at most $n-1$ steps the procedure ends. The successive parameters are ordered decreasingly; when two vertices merge, their common value is the minimum parameter on their eventual path. For a fixed forest, its edge orderings divide the parameter cube into these ordered simplices. Summing them gives the formula.

The matrix s^F is a Gram matrix of unit vectors: at $u \in [0,1]$, assign to vertex i the basis vector of its component among the edges with $\tau_\ell > u$, and integrate their scalar products in u. This scalar product is the length of the interval on which the two vertices remain connected, namely s^F_{ij}. Wick's formula and the Gram inequality therefore bound a residual monomial of degree p under $e^{\Delta[s^F]}$ by g^p, where $\|G\|_2 \le g^2$. Indeed, its determinant entries are $\langle \omega_i \otimes \delta_x, \omega_j \otimes G\delta_y\rangle$. These facts apply whether or not the interpolated covariance is invertible.

Let vertex i of a tree have degree k_i and let the chosen homogeneous part of F_i have degree p_i. The edge derivatives leave the factorial $(p_i - k_i)!^{-1}$ and $p_i - k_i$ uncontracted fields at that vertex;

terms with $p_i < k_i$ vanish. Wick's Gram bound supplies $g^{p_i-k_i}$. Sum contracted arguments from the leaves to vertex 1, using the weighted row and column bound α_G for every edge. If nonnegative symmetric weights w_i and a distance D satisfy

$$e^{\vartheta_0 D} \le \prod_i w_i(\text{arguments at } i) \prod_{\ell=(i,j)} e^{\vartheta_0 d(\text{edge arguments})},$$

this yields $e^{-\vartheta_0 D}\alpha_G^{n-1}$ times the full kernel sum at vertex 1 and the rooted sums at the other vertices. The number of trees with degrees $k_1, \dots, k_n$ is $(n-2)!/\prod_i (k_i-1)!$. Enlarge the sum over degrees and use

$$\sum_{k=1}^{p} \frac{g^{p-k}}{(k-1)!(p-k)!} = \frac{(1+g)^{p-1}}{(p-1)!} \le \frac{h_G^p}{(p-1)!}.$$

We obtain, for $n \ge 2$,

$$|T_n(F_1, \dots, F_n)| \le e^{-\vartheta_0 D}(n-2)!\alpha_G^{n-1} N_G(F_1; w_1) \prod_{i=2}^{n} \rho_G(F_i; w_i). \tag{23.41}$$

For $n = 1$ the bound is $N_G(F_1; 1)$. This also explains why the tree norms use $(p-1)!$ rather than $p!$.

Apply the forest formula to the replicated Gaussian expectation of V_+^n. Different forest components contain disjoint generator sets and their even factors commute, so their expectations factor. Grouping by component partitions gives

$$\int \mathrm{d}\mu_G \, V_+^n = \sum_{\{B_1,\dots,B_q\}} \prod_{a=1}^{q} T_{|B_a|}(V_+, \dots, V_+).$$

The exponential formula for partitions identifies the Gaussian integral with the exponential of the connected tree sum. The identity first holds as a formal series in a scalar multiplying V_+. Bound (23.41) and $\alpha_G \rho_G(V_+; 1) \le 1/2$ make the connected series absolutely convergent up to that scalar equal to one, so the formal identity is an analytic identity there. The tree terms are polynomial in the activity coefficients, and their series converges locally uniformly in t. Thus W is analytic and its derivatives are obtained term by term.

Two derivatives of the symmetric multilinear tree expression either act on the same vertex or on two different vertices. The first terms, including $n = 1$, sum to at most $2N_G(\partial_c \partial_{c'} V_+; 1)$. For the second terms put the two derivatives at vertices 1 and 2. The weights there are $w_P, w_{P'}$, and those at all other vertices are w_δ. Along the unique path from 1 to 2, the distance $d(P, P')$ is at most the sum of the two endpoint distances, all edge distances, and the argument diameters at intermediate vertices. Hence the weight condition for (23.41) holds with $D = d(P, P')$. The tree sum leaves $\alpha_G N_G(\partial_c V_+; w_P)\rho_G(\partial_{c'} V_+; w_{P'}) e^{-\vartheta_0 d(P,P')}$ times $\sum_{n\ge 2}(\alpha_G \rho_G(V_+; w_\delta))^{n-2} \le 2$. This proves (23.39). □

For a function f of the marks, let t^T keep the coordinates indexed by T and set all others to zero, and define its component with variable set S by

$$f_S(t) = \sum_{T \subset S} (-1)^{|S \setminus T|} f(t^T).$$

Then $f = \sum_S f_S$, $f_\varnothing = f(0)$, and f_S vanishes if any variable of S vanishes. The latter statement follows by pairing the terms with and without that variable. In particular a variable absent from f makes every component containing it zero. Two coordinate differences can be represented by two integrals of the mixed derivative. Consequently

$$\sup_{|t|_\infty \le R} |f_S(t)| \le 2^{|S|-2} R^2 \sup_{|t|_\infty \le R} |\partial_c \partial_{c'} f(t)| \qquad (c, c' \in S,\ c \ne c'). \tag{23.42}$$

Proposition 23.5. *For $|S| \ge 2$, the normalized logarithm satisfies*

$$\sup_{|t|_\infty \le t_x/2} |\mathcal{G}_S(t;\theta,A)| \le \varpi_{\mathrm{sep}} \le \varpi_{\mathrm{sep}}^{+}, \tag{23.43}$$

where

$$\begin{aligned} C_W^{\mathrm{mix}} &= 3C_V^{\mathrm{ins}} h_G^2 + 2\alpha_G (C_V^{\mathrm{ins}})^2 h_G^4 + 4\exp(6) C_d M^3 K_0, \\ \varpi_{\mathrm{sep}} &= 2^{N_*} C_W^{\mathrm{mix}} p_\beta e^{-\vartheta_0 (d_{\mathrm{sep}} - 4M - 6)}, \\ \varpi_{\mathrm{sep}}^{+} &= 2^{N_*} C_W^{\mathrm{mix}} p_\beta e^{-\vartheta_0 (M r_* /(8L) - 4M - 7)}. \end{aligned} \tag{23.44}$$

Proof. The tree representation and uniqueness of the normalized logarithm give

$$\mathcal{G}(t) = \sum_X \mathfrak{P}_*(X, \mathscr{C}(t)) + W(t) - W(0).$$

The component of a determinant activity is zero unless X meets every P_c, $c \in S$. Such a polymer has $d_M(X) \ge d_{\mathrm{sep}}/M - 2 \ge 1$. Use (23.5), sum the polymers anchored at one of at most p_β cubes, and keep $e^{-2d_M(X)}$ from $\kappa - \kappa_0 \ge 2$. Its total contribution is at most $2^{|S|} 2\exp(4) C_d M^3 K_0 p_\beta e^{-\vartheta_0 d_{\mathrm{sep}}}$.

Apply (23.39) with $P = P_c$, $P' = P_{c'}$ and (23.35). The bound for $\partial_c \partial_{c'} W$ is

$$q_x^2 p_\beta \left[2C_V^{\mathrm{ins}} h_G^2 e^{-\vartheta_0 (d_{\mathrm{sep}} - 4M - 6)} + 2\alpha_G (C_V^{\mathrm{ins}})^2 h_G^4 e^{-\vartheta_0 d_{\mathrm{sep}}} \right].$$

In (23.42), take $R = t_x/2$ and use $q_x R = 1$. Add the determinant contribution and $|S| \le N_*$ to obtain the first inequality. The placement estimate $d_{\mathrm{sep}} \ge M r_*/(8L) - 1$ proves the second. When the lattice probe is zero, $R_{\mathrm{tm}} = 1$, $\mathcal{G} = 0$ and the first separation error is interpreted as zero. □

23.4. **The small-field representation and its complement.** Proposition 22.1 identifies F_N with $(A_*; \zeta_j)_{j<k_*}$. Their scalar coordinates have the form $\zeta_j(\Upsilon) = \langle \phi_{j,\Upsilon}, F^{\mathrm{u}} \rangle$ with orthonormal $\phi_{j,\Upsilon}$; each coarse plaquette coordinate $(dA_*)(p)$ has the form $\langle \phi_p, F^{\mathrm{u}} \rangle$ with $\|\phi_p\| \le 1$. There are at most $2\ell_*^3 L^{3(k_*-j)}$ coordinates at level j and $3\ell_*^3$ coarse plaquettes. The stop Gaussian $\mathrm{d}\mathsf{m}_*$ is nondegenerate on the real mean-zero coclosed coarse fields. The row and count statements are (22.6) and (22.7); the Gaussian assertion is (22.5).

Let $\mathfrak{G}$ require $|\zeta_j(\Upsilon)| \le p_{0,j}$ and $|dA_*|_\infty \le p_*$, and let $\mathfrak{G}^-$ decrease each threshold by one. Put

$$\begin{aligned} \varpi_0 &= 18\ell_*^3 e^{-p_{0*}/2} + 6\ell_*^3 e^{-(p_*-1)^2/(2C_0^2)}, \\ t_2 &= e \min\left\{ N_*^{-1/2},\, C_\beta^{-1},\, (2\sqrt{N_* \mathbf{D}})^{-1} \right\}, \\ \varpi_L &= \sqrt{\varpi_0}\, e^{\mathbf{D}/2}, \qquad M_L = 2 + e^{\mathbf{D}/2}, \\ \lambda_L &= (\varpi_L^{1/3} M_L^{2/3})^{1/N_*} 5 (2N_* r/t_2)^4. \end{aligned} \tag{23.45}$$

A reciprocal with zero denominator is interpreted as $+\infty$. The number $\mathbf{D}$ is the uniform probe energy bound, so $v_\eta \le \mathbf{D}$.

Proposition 23.6. *Suppose the preceding conditions hold, and, for a fixed $0 < s < 1$, set $\bar\kappa_1 = 6e_*^{1/4 - 9\varepsilon}$. Assume*

$$2\varpi_{\mathrm{sep}}^{+} \le \bar\kappa_1^{N_*}, \quad N_* \bar\kappa_1^{1-s} \le 1/2, \quad \varpi_0 \le 1/4, \quad 8N_* t_2 \le t_x. \tag{23.46}$$

There is a finite positive measure v_{sf} on $[-\pi,\pi]^3 \times \{\chi_* = 1\}$, of mass in $[1 - \varpi_0, 1]$, such that

$$\frac{\bar\nu^{a(t)}(\mathfrak{G})}{\bar\nu(F_N)} = \int e^{\mathcal{G}(t;\theta,A)}\, v_{\mathrm{sf}}(\mathrm{d}\theta\, \mathrm{d}A). \tag{23.47}$$

The normalized complement $L(t) = \bar{\nu}^{a(t)}(F_N \setminus \mathfrak{G})/\bar{\nu}(F_N)$ is entire. Its fourth jet in each variable has coefficient norm at radius r at most $\lambda_L^{N_*}$.

Proof. First, the union bound and (21.32) give

$$\bar{\mu}(F_N \setminus \mathfrak{G}^-) \le \varpi_0. \tag{23.48}$$

For $i = k_* - j \ge 1$, write $v_j = v_* + i(\log L)/2$. The scale conditions $v_* \ge 8$, $p_0 v_*^{p_0-1} \ge 28$ give $p_{0,j} \ge p_{0*} + 14i \log L$. Since $(p_{0,j}-1)^2 \ge p_{0,j}$, the individual tail is at most $2e^{-p_{0*}/2}L^{-7i}$. Multiply by the coordinate count $2\ell_*^3 L^{3i}$ and sum L^{-4i}. The $3\ell_*^3$ coarse plaquettes give the second term in (23.45). These estimates yield (23.48), with room in the coefficient 18.

The exact density identity defines

$$v_{\mathrm{sf}}(\mathrm{d}\theta\,\mathrm{d}A) = \frac{\chi_*(A) Y_*^{\mathrm{x}}(A + c_\theta; 0)}{\bar{\nu}(F_N)} \frac{\mathrm{d}\theta}{(2\pi)^3} \mathrm{d}\mathsf{m}_*(A). \tag{23.49}$$

We check positivity of this density, which does not follow from the formal Grassmann representation. For real marks, the original fine law $\nu^{\theta,a(t)}$ is nonnegative. The identity (22.14) says, for every nonnegative bounded Borel Φ,

$$\int \Phi(A)\chi_*(A) Y_*^{\mathrm{x}}(A + c_\theta; \mathscr{C}(t))\,\mathrm{d}\mathsf{m}_*(A) = \nu^{\theta,a(t)}(\mathbf{1}_{\mathfrak{G}} \Phi(A_*)) \ge 0.$$

The terminal density is continuous. On the interior of the compact convex set $\{\chi_* = 1\} = \{|dA|_\infty \le p_*\}$, a nonzero imaginary part or negative real part would persist on a ball of positive Gaussian measure, contradicting this identity. At the boundary, approach along uA, $u \uparrow 1$; zero is an interior point. Thus the density is nonnegative everywhere on this set. Proposition 23.1 makes it nonzero there, hence strictly positive. This reasoning also proves $\mathcal{G}(t; \theta, A) \in \mathbb{R}$ for real t: its exponential is a positive ratio and its continuous value at zero is zero.

Integration of the exact terminal identity over θ, followed by (23.15), proves (23.47). At $t = 0$ the mass is $\bar{\mu}(\mathfrak{G})$, which lies in $[1 - \varpi_0, 1]$. All integrals here are well-defined Borel integrals: the density is continuous on the compact support and zero elsewhere.

For real $t = ez$ with $|t|_\infty \le t_2$, put $G_z = \sum_c z_c d\widehat{\mathfrak{a}}_c$. Disjoint supports imply $\|G_z\|^2 = v_\eta |z|_2^2 \le 1/4$. The real source shift gives

$$L(ez) = e^{-v_\eta |z|_2^2/2} \bar{\mu}\left(\mathbf{1}_{(F_N \setminus \mathfrak{G}) + G_z} e^{\langle G_z, F^{\mathrm{u}} \rangle}\right).$$

Every photon coordinate changes by at most $\|G_z\| \le 1$ under the shift. Thus the shifted complement is contained in $F_N \setminus \mathfrak{G}^-$. Cauchy–Schwarz, (23.48), and Gaussian domination at $2G_z$ give

$$0 \le L(t) \le \sqrt{\varpi_0}\, e^{v_\eta |z|_2^2/2} \le \varpi_L. \tag{23.50}$$

This is the only place where the lowered thresholds of $\mathfrak{G}^-$ are needed.

The source-shift identity also gives, for complex marks,

$$|\mathring{\Phi}(t)| \le \exp\left(\frac{v_\eta}{2e^2} \sum_c (\operatorname{Im} t_c)^2\right).$$

It follows directly by bounding the expectation of the real exponential and canceling its real quadratic normalization. On $|t|_\infty \le t_2$ this is at most $e^{\mathbf{D}/2}$. The small-field integral has modulus at most 2. Indeed, evenness and the double zero of ℓ_0 give $|\ell_0(t)| \le B_0 |t|^2/t_x^2$, while the zero of w_c gives $|w_c(t)| \le (3/8)|t|/t_x$. Their sum over cells at $t_2 \le t_x/(8N_*)$ is at most $1/(256N_*) + 3/64$. The remaining logarithm components sum to at most $2^{N_*} \varpi_{\mathrm{sep}}^{+} \le (2\bar{\kappa}_1)^{N_*}/2 \le N_*^{-N_*}/2$, by (23.46). Thus $|\mathcal{G}| \le 1/2$ and $|e^{\mathcal{G}}| \le 2$. Since L is the difference between the full generating function and the small-field integral, $|L| \le M_L$ on that polydisc. It is entire because the finite determinant

density is entire and is bounded on compact mark sets uniformly in the real photon field and real twist. The modulus bound in the external-response section supplies the required Gaussian integrable majorant also for either Borel event.

We use the following two-constants estimate. Let $n \geq 1$ be an integer. If a holomorphic function on a polydisc in $\mathbb{C}^n$ of radius R has modulus at most M, and at most $\delta \leq M$ on its real points, then its modulus is at most $\delta^{1/3}M^{2/3}$ on the polydisc of radius $R/(2n)$. On the upper half of the unit disc the harmonic function $\omega(z) = \frac{2}{\pi}\arg[(1+z)/(1-z)]$ is zero on the diameter and one on the semicircle. Comparing $\log|f(R'z)|$ with $(1-\omega)\log\delta + \omega\log M$, for $R' < R$, the maximum principle gives the same inequality in the half disc. To handle its endpoints, subtract $\epsilon\log[4/|1-z^2|]$ before applying the principle, then let $\epsilon \downarrow 0$. Its harmonic conjugate gives a holomorphic function to which the modulus principle applies. On $|z| \leq q$, $\omega(z) \leq (4/\pi)\arctan q$; reflection treats the lower half disc. Let $R' \uparrow R$. Apply this one-variable estimate successively with $q = 1/(2n)$. The remaining exponent of δ is at least $(1 - \frac{4}{\pi}\arctan(1/(2n)))^n \geq 1 - 2/\pi > 1/3$, proving the assertion. The case $\delta = 0$ follows by the identity theorem, and $M = 0$ is immediate.

Apply it to L, with $n = N_*$ and $R = t_2$. Cauchy's coefficient bound gives

$$|L_\alpha| \leq \varpi_L^{1/3} M_L^{2/3} (2N_*/t_2)^{|\alpha|}.$$

In the fourth jet, every exponent is at most four. Since $y = 2N_* r/t_2 \geq 1$,

$$\sum_{\alpha_c \leq 4} |L_\alpha| r^{|\alpha|} \leq \varpi_L^{1/3} M_L^{2/3} (1 + y + \cdots + y^4)^{N_*} \leq \lambda_L^{N_*}.$$

Reality on real marks gives real jet coefficients. □

23.5. **The fourth-jet error.** Work in the finite ring $\mathcal{J} = \mathbb{C}[t_c : c \in \mathcal{C}_*]/(t_c^5 : c \in \mathcal{C}_*)$, with coefficient norm $\|P\|_r = \sum_{\alpha_c \leq 4} |P_\alpha| r^{|\alpha|}$. This norm is submultiplicative, since discarding terms with an exponent at least five can only decrease the coefficient sum. The jet map is a ring homomorphism; it keeps degree at most four in each variable. For $S = \{c\}$ set $k_S = e^{w_c} - 1$ in this ring, and for $|S| \geq 2$ set $k_S = e^{\mathcal{G}_S} - 1$. Every monomial of k_S uses every variable of S and no other variable.

For a component f_S bounded on a polydisc of radius R, Cauchy's estimate at a smaller radius $R' \leq R/2$ gives

$$\|f_S\|_{R'} \leq \sup|f_S| \left(\sum_{j=1}^{4} (R'/R)^j \right)^{|S|} \leq \sup|f_S|.$$

Its monomials contain each variable of S because it vanishes on every corresponding coordinate hyperplane. Use this with $R = t_x/2$, $R' = r$, and use the one-cell estimate on radius t_x, to obtain

$$\|k_{\{c\}}\|_r \leq \bar{\kappa}_1, \qquad \|k_S\|_r \leq 2\varpi_{\mathrm{sep}}^{+} \leq \bar{\kappa}_1^{N_*} \quad (|S| \geq 2). \tag{23.51}$$

We used $\|w_c\|_r \leq 3e_*^{1/4-9\varepsilon} \leq 3/8$ and $\|e^P - 1\|_r \leq 2\|P\|_r$ for $\|P\|_r \leq 1/2$. The coefficients are continuous in (θ, A): express the logarithm coefficients by Cauchy integrals on a fixed small torus, and then take the finite polynomial expressions in those coefficients. They are real because the logarithm is real at real marks.

In $\mathcal{J}$ the normalized generating function has the exact representation

$$\mathring{\Phi} = \prod_c \phi_0(t_c) \int \prod_{\varnothing \neq S \subset \mathcal{C}_*} (1 + k_S)\, v_{\mathrm{sf}}(\mathrm{d}\theta\, \mathrm{d}A) + L^{[4]}. \tag{23.52}$$

To justify taking the jet inside the integral, use joint continuity, holomorphy, and $|e^{\mathcal{G}}| \leq 2$ on a fixed polydisc of radius at most t_2. Cauchy's coefficient formulas and Fubini then commute every required coefficient with the finite measure integral. Decompose $\mathcal{G}$ into its variable-set components;

its singleton components are $\ell_0(t_c)+w_c(t_c)$. Exponentiation, the jet homomorphism, and $e^{\ell_0}=N_0$ prove the representation.

Theorem 23.7. *Assume the terminal inputs and conditions above, and $\lambda_L \le \bar\kappa_1^s$. For $m=2,4$,*

$$\Theta_m^{(0)} \le m! r^{-m}\big(2\exp(1)\bar\kappa_1^s + \lambda_L\big) \le 8m! r^{-m}\bar\kappa_1^s. \tag{23.53}$$

With $\mathfrak{e} = s(1/4-9\varepsilon)-12\varepsilon > 0$,

$$\Theta_4^{(0)} \le 3\cdot 10^5 C_b^4 e_*^{\mathfrak{e}} \le 3\cdot 10^5 C_b^4 h^{1/8} e^{\mathfrak{e}},$$

$$(1+|c_2|)\Theta_2^{(0)} \le 3\cdot 10^4 C_b^4 e_*^{\mathfrak{e}} \le 3\cdot 10^4 C_b^4 h^{1/8} e^{\mathfrak{e}}. \tag{23.54}$$

The constants are independent of the ultraviolet cutoff, the twist, and the admitted placement.

Proof. On the one-variable jet space define $\ell_c(t_c^j) = m!\mathbf{1}_{\{j=m\}} - c_m \mathbf{1}_{\{j=0\}}$ and let $\ell = \bigotimes_c \ell_c$. Then $(\Theta_m^{(0)})^{N_*} = |\ell(\mathring\Phi)|$. The free jet satisfies $\ell_c(\phi_0)=0$. From (23.29), $|\ell(P)| \le (m!r^{-m})^{N_*}\|P\|_r$. These statements follow by evaluating the functionals on monomials with every exponent in $\{0,m\}$.

Expand the product in (23.52) over families $\mathcal{Y}$ of nonempty cell sets. If $\mathcal{Y}$ fails to cover a cell c, its error factors do not contain t_c; the factor $\phi_0(t_c)$ is therefore annihilated by ℓ_c. Only covering families contribute. Assign cost one to each singleton and cost N_* to each larger set, and let $a(\mathcal{Y})$ be the sum of these costs. A covering family has $a(\mathcal{Y}) \ge N_*$: either it contains a larger set or it contains every singleton. By (23.51) its coefficient norm is at most $\bar\kappa_1^{a(\mathcal{Y})}$. The free factors have norm at most 2^{N_*}. Keep $\bar\kappa_1^{sN_*}$ from the cost and sum the rest over all families. This sum is

$$(1+\bar\kappa_1^{1-s})^{N_*}(1+\bar\kappa_1^{(1-s)N_*})^{2^{N_*}-1-N_*} \le \exp\left(N_*\bar\kappa_1^{1-s} + 2^{N_*}\bar\kappa_1^{(1-s)N_*}\right) \le \exp(1).$$

The last step uses $N_*\bar\kappa_1^{1-s} \le 1/2$. The positive small-field measure has mass at most one. Its total contribution is thus bounded by $(m!r^{-m})^{N_*}\exp(1)(2\bar\kappa_1^s)^{N_*}$. The complement contributes at most $(m!r^{-m}\lambda_L)^{N_*}$. Taking the N_*th root proves (23.53).

Since $r^{-4} = 256 C_b^4 e_*^{-12\varepsilon}$ and $\bar\kappa_1^s \le 6 e_*^{s(1/4-9\varepsilon)}$, the $m=4$ bound has coefficient at most $8\cdot 24\cdot 1536 < 3\cdot 10^5$. For $m=2$, $1+|c_2| \le (17/16)r^{-2}$, so its coefficient is at most $17\cdot 1536 < 3\cdot 10^4$. Finally $0<\mathfrak{e}<1/4$ and $e_* \le e\sqrt{h}$ imply $e_*^{\mathfrak{e}} \le h^{1/8} e^{\mathfrak{e}}$. The scale and probe bounds entering this proof are uniform in the stated choices. □

23.6. **A positive charge threshold.** We now verify that the response conditions admit a common interval of charges, independent of the ultraviolet cutoff. In addition to (23.12), (23.13), (23.20), (23.34), and (23.46), require $r_* > 31L+16$ and

$$Mr_*/(8L) - 1 \ge 6M+8, \qquad \lambda_L \le \bar\kappa_1^s, \tag{23.55}$$

$$D_\beta^{\mathrm{buf}}/2 + 1/2 \le 3\ell_c/8, \qquad D_\beta^{\mathrm{buf}} + 3 \le \ell_c. \tag{23.56}$$

The construction of Section 16 fixes an odd $L \ge 13$, a power M of L with $M \ge L^2$, a localization power $r_{\mathrm{loc}} \ge 2$, and cutoff powers $p_0 \ge 2$, $p > 2p_0$. It also fixes $0<\varepsilon \le 1/200$ and $0<s<1$ such that

$$\mathfrak{e} = s(1/4-9\varepsilon) - 12\varepsilon, \qquad \sigma = \mathfrak{e} - \frac{2a_L}{N_*} > 0, \qquad a_L = \frac{L^3-1}{(L-1)^3}. \tag{23.57}$$

Only $\mathfrak{e}$ enters the terminal estimate; the loss $2a_L/N_*$ belongs to the subsequent finite-volume transfer.

Proposition 23.8. *For the fixed construction constants, $\bar m$, and probe β, there is $e_\sharp(\beta) > 0$ such that all response conditions hold for $0 < e \le e_\sharp(\beta)$ and every $N \ge N_{\mathrm{ref}}(e)$. The threshold depends on the probe only through $C_\beta, \mathbf{D}, D_\beta^{\mathrm{buf}}$. It is independent of N, the twist, and the admitted placement. All centered placements are admitted in this range.*

Proof. Write $v_K = -\log(e\sqrt{h})$ and $r_K = \min\{L^i : L^i \geq v_K^{r_{\mathrm{loc}}}\}$. Then $R^*(e) = hMr_K/L$ and $\ell_c = R^*(e)/(2L)$. Proposition 16.1 and (16.7) give a finite $N_{\mathrm{ref}}(e)$.

For $N \geq N_{\mathrm{ref}}(e)$ the stop level is $K-1$ or $K-2$ by (16.8). In particular its scales are independent of N and satisfy

$$\begin{gathered} v_K \leq v_*, \quad e_* = e^{-v_*}, \quad a_* \in \{h/L, h/L^2\}, \\ v_*^{r_{\mathrm{loc}}} \leq r_*, \quad Mv_*^{r_{\mathrm{loc}}} \leq \ell_* < LMv_*^{r_{\mathrm{loc}}}, \quad p_{0*} = v_*^{p_0}, \quad p_* = v_*^p, \\ p_\beta \leq p_\beta^+ := (D_\beta^{\mathrm{buf}} L^2/h + 2M + 4)^3, \qquad \ell_c \geq hMv_K^{r_{\mathrm{loc}}}/(2L^2). \end{gathered} \tag{23.58}$$

The strict wrapping condition $r_* > 31L + 16$ already follows from (16.4) whenever $e\sqrt{h} \leq e_1$. Thus every response condition is a condition on e alone once the construction and probe are fixed.

A bound $Av^d e^{-\gamma v}$ with $A, d \geq 0$ and $\gamma > 0$ tends to zero, uniformly on a sufficiently large half-line. Split the exponential in half and use $v^d e^{-\gamma v/2} \leq (2d/(\gamma \exp(1)))^d$. Likewise, cv^2 exceeds $bv + \log^+ A$ on a sufficiently large half-line for $c > 0$. We apply these two inequalities to the following groups of conditions.

The local tree smallness and terminal factor conditions follow from exponential decay with positive rates ε, 2ε, $1/4 - \varepsilon$, and $1/4 - 9\varepsilon$; the volume factors are bounded by $(LM)^3 v_*^{3r_{\mathrm{loc}}}$. For $\bar{r}$, use $1/4 - 7\varepsilon > 2\varepsilon$. The complex mass and coarse determinant conditions similarly have rate 2ε. The support count is bounded by p_β^+. All three summands of B_0 tend to zero: their rates are 4ε, 2ε, and the quadratic decay $e^{-r_*} \leq e^{-v_*^2}$. Also $\tau_s \geq c_\tau M v_*^2$, so both $\tau_s \geq 1$ and $52B_1 e^{-\tau_s} \leq e_*^{1/4-9\varepsilon}$ hold once v_* is large.

The separation condition follows from $r_* \geq v_*^2$. Its error obeys

$$2\varpi_{\mathrm{sep}}^+ \leq 2^{N_*+1} C_W^{\mathrm{mix}} p_\beta^+ e^{\vartheta_0(4M+7)} e^{-\vartheta_0 M v_*^2/(8L)}.$$

Since $\bar{\kappa}_1 \geq e^{-v_*/4}$, quadratic decay makes this at most $\bar{\kappa}_1^{N_*}$ for all sufficiently large v_*. The family summation condition $N_* \bar{\kappa}_1^{1-s} \leq 1/2$ uses the positive rate $(1-s)(1/4 - 9\varepsilon)$.

For the photon complement, $p > 2p_0$ implies $(p_* - 1)^2 \geq C_0^2 p_{0*}$ once v_* is large. Therefore

$$\varpi_0 \leq 24(LM)^3 v_*^{3r_{\mathrm{loc}}} e^{-v_*^2/2}. \tag{23.59}$$

This is eventually at most $1/4$. Moreover t_2/e is fixed and positive, while e is between $(L/h)^{1/2} e_*$ and $Lh^{-1/2} e_*$. Hence $8N_* t_2/t_x$ is bounded by a fixed constant times $e_*^{1-3\varepsilon}$, which tends to zero.

The potentially largest loss is the fourth derivative radius in λ_L. From (23.59), the definition of λ_L, and $r/t_2 \leq Ce^{v_*}$, we obtain

$$\lambda_L \leq A_8 \exp\left(-\frac{v_*^2}{12N_*} + \left(\frac{r_{\mathrm{loc}}}{2N_*} + 4 \right) v_* \right) \tag{23.60}$$

for a finite $A_8 > 0$ depending only on the fixed constants and probe bounds. For clarity one can take

$$A_8 = [24(LM)^3]^{1/(6N_*)} e^{\mathbf{D}/(6N_*)} M_L^{2/(3N_*)} 5 \left[\frac{N_* \sqrt{h}}{2C_b \sqrt{L}} (t_2/e)^{-1} \right]^4.$$

If $v_*^2/(12N_*) \geq (r_{\mathrm{loc}}/(2N_*) + 17/4)v_* + \log^+ A_8$, then $\lambda_L \leq e^{-v_*/4} \leq \bar{\kappa}_1^s$. Thus the complex continuation and the fourth derivatives preserve a positive small-charge interval. Finally the two placement inequalities hold when $\ell_c \geq (4/3) D_\beta^{\mathrm{buf}} + 3$; this follows from (23.58) as $v_K \to \infty$.

Choose a finite V beyond all the half-lines just specified and put $e_V = h^{-1/2} \min(e_1, e^{-V}) > 0$. Every $0 < e \leq e_V$ has $v_* \geq v_K \geq V$ and satisfies every response condition. Define $e_\sharp$ as one half of the supremum of the charges $e' \leq e_1 h^{-1/2}$ for which all conditions hold for every $0 < e \leq e'$. Its defining set contains e_V, so $e_\sharp \geq e_V/2 > 0$. For $e \leq e_\sharp$, a member of the defining set exceeds $e_\sharp$,

which ensures that all inequalities hold at e even if the scale functions have jumps. This avoids assuming monotonicity of the rounded stop scales.

The scale identities depend on e, h and the fixed construction constants; the probe enters through $C_\beta, \mathbf{D}, D_\beta^{\mathrm{buf}}$. They do not involve the twist or placement, and they are unchanged by the coordinate reflections used to form the cell probes. Centered placements satisfy the admission inequality by (23.56), since their centers differ from the cell center by at most $\eta/2 \le 1/2$. The same conditions give $D_\beta^{\mathrm{buf}} + 3\eta \le \ell_c$ and $R^*(e) \ge D_\beta + 1$. □

24. Choice of the construction constants

The estimates impose conditions on fixed constants and on the running charge. The constants are chosen before the charge, mesh, volume and twist. We first choose the polymer decay and the two geometric scales; the charge bounds are then imposed for every smaller charge. The local inverse constants are fixed before M, whereas the weakened kernel constants may depend on M. Neither group depends on the running charge. We verify this order before imposing the small-charge conditions.

Proposition 24.1. *Fix $0 < \rho < 1$ and the standard chamber weights. The constants of the photon and fermion primitives, small-field step, cover comparison and tuned flow can be chosen in the order described below, with positive charge ceilings e_0, e_1 independent of the mesh, volume, twist and reference mass. For every $0 < \bar m \le 1$, the physical ceiling e_G in (24.10) is positive. The terminal conditions (22.36) hold on the top and reference tori throughout $0 < e \le e_G$. For a fixed compact real probe with $b(dd^*\beta) > 0$, there is a positive $e_\beta \le e_G$ for which (25.31) holds with $\sigma > 0$ and $C_f e^\sigma \le b(dd^*\beta)/2$.*

24.1. **The geometric scales and the positive exponent.** *Proof.* Set $\bar\varepsilon = 1/200$, $c_X = 54$, $\kappa_0 = 54 \log 72$, $K_0 = 72^{54}/36$, $c_{\mathrm{rb}} = 54K_0$ and $c_{\mathrm{fl}} = 61 c_X K_0$. Choose

$$1 - \bar\varepsilon < \alpha < 1, \qquad \kappa \ge 7\kappa_0 + 7.$$

An odd integer $L \ge 5$ is required to satisfy

$$\begin{aligned} L^{1/8 - 3\bar\varepsilon/2} &\ge 2400, \qquad c_{\mathrm{fl}} L^{-3/8 + 6\bar\varepsilon} \le 1/16, \\ L(\kappa - 6\kappa_0 - 4) - 1 &\ge \kappa, \\ c_{\mathrm{rb}} L^3 \exp\bigl(-L(\kappa - \kappa_0 - 1) + \kappa\bigr) &\le 1/2. \end{aligned} \tag{24.1}$$

These are simultaneous lower-bound conditions on L. The powers $1/8 - 3\bar\varepsilon/2$ and $3/8 - 6\bar\varepsilon$ are positive, as are $\kappa - 6\kappa_0 - 4$ and $\kappa - \kappa_0 - 1$. The last expression tends to zero as $L \to \infty$. Thus every sufficiently large odd L satisfies (24.1).

After L is fixed, the primitive construction fixes a', C_V, γ_w and γ_P, with $a', C_V > 0$ and $\gamma_w, \gamma_P > 0$. Put $N_c = (2L+1)^3$ and $\widehat\kappa = N_c\kappa + \log(6c_X) > 0$. The photon and fermion constants have no dependence on M at this stage. Put $\alpha_0 = 1/54$, $\kappa' = \kappa_w + \kappa_0$ and

$$M_F = \max\{4L + 1,\ (108\pi C_V a')^2,\ 108\pi C_V a' \exp(27\kappa')\}.$$

Choose M to be a power of L satisfying

$$\begin{aligned} M \ge L^2, \qquad M > 4L, \qquad \gamma_w M \ge 16, \\ M^{\alpha_0} \ge 1 + \exp(\widehat\kappa), \qquad M \ge M_F, \qquad 804\alpha_0 \log M \le \gamma_P M. \end{aligned} \tag{24.2}$$

All these conditions hold for every sufficiently large power of L. For the last one, $\log x \le \sqrt{x}$ on $x \ge 1$ gives the sufficient lower bound $M \ge (804\alpha_0/\gamma_P)^2$. The definition of M_F gives the two fermion conditions in (11.3), because $27\alpha_0 = 1/2$. The last inequality gives (10.37). The first four give the geometric separation, decay and parameter-radius conditions used in localization.

The exponents can be fixed independently of the charge. Take $r = 2$, $p_0 = 9$, $p = 19$ and $1 - \alpha < \varepsilon \le \bar{\varepsilon}$, and take $s = 1/2$. Then $p_0 > 3r + 2$ and $p > 2p_0$, while

$$\mathfrak{e} = s(1/4 - 9\varepsilon) - 12\varepsilon \ge 17/400.$$

For $N_* = 8L^3$ and $a_L = (L^3 - 1)/(L - 1)^3$,

$$\frac{2a_L}{N_*} < \frac{1}{4(L-1)^3} \le \frac{1}{256} < \frac{17}{400}.$$

It follows that $\sigma = \mathfrak{e} - 2a_L/N_* > 0$. In particular the positive exponent is fixed before the charge is chosen.

24.2. **Calibration of the primitive and terminal bounds.** The chamber coercivity, heavy-sector gap and local inverse construction fix a' and $\gamma^F > 0$ before M is chosen; see (9.32) and (9.41). The photon construction likewise fixes C_P^{ph} and $\gamma_P > 0$ before M; its weakened families are considered only after (10.37) holds. The sums $c(\gamma) = \sum_{z \in \mathbb{Z}^3} \exp(-\gamma |z|_\infty)$ are finite for $\gamma > 0$. Thus the constants used to choose M in (24.2) have already been fixed.

Keep $b = 1$, $b_- = 1 - L^{-1}$ and $b_+ = 1$ from the inverse construction; these are already fixed before M. The coarse resolvent and its auxiliary block matrix give

$$C_{\mathrm{By}} = (1 + 2\exp(1))\sqrt{2}\max(1, b_-^{-1}), \qquad C_\Gamma = \frac{2(1 + 2\exp(1))}{b_-} + 128\exp(1)(1 + 2\exp(1))^2 a'.$$

The latter is the constant in (11.19). Define

$$\begin{aligned} C_1 = \max\{&304\exp(1)\max(1,b)a'M^{27\alpha_0}, C_\Gamma M^{27\alpha_0}, \\ &32\exp(1)\max(1,b^2)a'\exp(27\kappa')\}, \\ C^F = 2\max\{&C_1, C_{\mathrm{By}}, 304\exp(1)\max(1,b)a', C_\Gamma, \exp(1)b^2(32a')^2\}. \end{aligned} \tag{24.3}$$

The factor two converts Euclidean spinor bounds to component bounds. The walk and coarse-family estimates (11.4)–(11.5) and (11.18)–(11.19) are bounded by C^F in this convention. The last entry also bounds the compressed quadratic kernel. At level zero the normalized kernel has the same bound, and $M^{8\alpha_0} \le M^{27\alpha_0}$ bounds its diagonal field multiplier. The unweakened all-pair field bound (21.15) is at most $608\exp(1)\max(1,b)a' \le C^F$; at zero its constant is at most four, also below C^F.

Take the common primitive constants

$$C^P = 116 C_P^{\mathrm{ph}} M^{402\alpha_0}, \qquad C_w^{(1)} = \max\{C^F, C^P, 4\exp(2\gamma_w)M^{27\alpha_0}\}, \qquad C_w = 64 K_0 C_w^{(1)}.$$

These are precisely (15.9). The top systems and their projections therefore satisfy the same bounds at every required level and side. The determinant, photon-domain and localization constants are given in Sections 11.7, 10.8 and 14.1. In particular,

$$\begin{aligned} c_C = 3L^{9/2}, \qquad C_W = 28 C_w^2 c_\gamma, \\ C_d = 16 c_X C_w \max\{\tfrac{\pi}{2} b\exp(1)(1 + 2\exp(1)), 2 + 58\exp(1) + b\exp(1)\}. \end{aligned}$$

The kernel-sum constant c_γ is bounded by $2L^3\exp((2L+1)\gamma_w)c(\gamma_w)$; the local-gauge constant C_B depends only on $C_P^{\mathrm{ph}}, \gamma_P, L, M$. All these constants are finite after M is fixed.

For the terminal norms fix

$$C_0 = \max\{C_P^{\mathrm{ph}} c_{\mathrm{bd}}(\gamma_P), C^F, 4b_+ + 16 C^F c(\gamma^F)\}, \qquad c_{\mathrm{bd}}(\gamma) = 3\exp(3\gamma)c(\gamma). \tag{24.4}$$

The photon row sums, both field-map row norms and the terminal Dirac form in Section 21.4 are then bounded by this C_0. In particular $C_0 \ge 1$. The polymer sum has $1 \le c_\kappa \le K_0$. The free Bloch construction fixes the twist constants $C_\Sigma < \infty$ and $\gamma_\Sigma > 0$ independently of the running charge, mass in its prescribed range, mesh and side; see (22.30).

The differentiated-response constants are now fixed by the terminal inverse and field-map estimates: $\mathcal{S} = C^F(24/\gamma^F)^3$, $C_K = 226\mathcal{S}^2$, $C'_K = 1328\mathcal{S}^2$ and $C_H = 203M(C^F)^2$. The real covariance coefficients g, h_G, g_1, α_G and its rate ϑ_0 are defined with (21.26). The insertion and mixed-logarithm coefficients are $C_V^{\mathrm{ins}} = 10^3 C_H^2 (24/\gamma^F)^{12} K_0$ and C_W^{mix} in (23.44). Their inverse-decay rates are positive and their coefficients finite because $b_- > 0$, $\mu_- = L^{-3} > 0$ and $\gamma^F > 0$. They depend on L, M and the fixed chamber data; they have no dependence on the later charge or probe. The probe-dependent constants will be chosen last.

24.3. **Uniform running-charge bounds.** All coefficients of the small-charge conditions are fixed after the preceding choices. The elementary estimate used to meet them simultaneously is as follows. For finitely many A_i, d_i, λ_i with $A_i, d_i \geq 0$ and $\lambda_i > 0$, and for $b > 0$, put

$$V(A, d, \lambda; b) = \max_i \frac{2}{\lambda_i} \log^+ \left[\frac{nA_i}{b} \left(\frac{2d_i}{\lambda_i \exp(1)} \right)^{d_i} \right]. \tag{24.5}$$

Here n is the number of terms, $\log^+ 0 = 0$ and a factor with $d_i = 0$ is interpreted as one. For all $v \geq V(A, d, \lambda; b)$ with $v > 0$,

$$\sum_i A_i v^{d_i} \exp(-\lambda_i v) \leq b. \tag{24.6}$$

Indeed $v^d \exp(-\lambda v/2)$ has maximum $(2d/(\lambda \exp(1)))^d$ for $d > 0$. For $d = 0$ use $\exp(-\lambda v) \leq \exp(-\lambda v/2)$. The threshold makes each resulting term at most b/n.

The primitive domains, determinant, adjustment and nonlinear step require (10.42), (12.6), (12.35), (12.40), (11.37) and (14.1). Their charge conditions are obtained from the following finite families for $v = -\log e_k$. Write $T(A, d, \lambda) = Av^d \exp(-\lambda v)$. The polynomial-exponential conditions are

$$\begin{aligned}
T(1, p_0, \varepsilon/4) &\leq 1, & T(1, 0, 5\varepsilon/2) &\leq \tfrac{1}{2},\\
T(c_{\mathrm{fl}}, 0, \varepsilon) &\leq 1, & T(3M, 0, \varepsilon) &\leq 1,\\
T(2C_w^2 c_\gamma^2 c_C, 0, \varepsilon/4) &\leq 1, & T(16 c_X c_\gamma C_w C_T^2 M^3, 0, \varepsilon/2) &\leq 1,\\
T(96M^3, 0, 1/4) &\leq 1, & T(32 C_w C_T, 0, 3\varepsilon/2) &\leq 1,\\
T(88M^3, 0, 1/4 - 6\varepsilon) &\leq 1, & T(112 C_w^2 c_\gamma, 0, 2\varepsilon) &\leq 3,\\
T(c_S, 0, 1/4 - 5\varepsilon) &\leq 1, & T(c_S L^3, 0, \varepsilon) &\leq 1,\\
T(36 c_X^2 K_0^2, 0, 1/4 - 5\varepsilon) &\leq 1, & T(1, 0, \varepsilon) &\leq \tfrac{1}{4},\\
T(3M, 0, 1/4 + \varepsilon) + T(3M, 0, 2\varepsilon) &\leq 1, & & \\
T(1, 0, 1/4) + T(1, 0, \varepsilon) &\leq 1, & & \\
T(160 L^{1+3\varepsilon/2} M, 0, \varepsilon) &\leq 1, & T(90 c_X LM, 0, \varepsilon) &\leq 1,\\
T(9ML^\varepsilon, 0, 2\varepsilon) &\leq 1, & & \\
T(2M, 1, \varepsilon) + T(32M, 0, \varepsilon) &\leq 1, & & \\
T(8C_B, p, 3/4 - 3\varepsilon) &\leq 1, & T(8, 0, 3/4 - 7\varepsilon/2) &\leq 1,\\
T(6L, 0, \varepsilon) &\leq 1, & T(4 C_V a', 0, \varepsilon) &\leq 1,\\
T(4, 0, 1/4 + 2\varepsilon) + T(2, 0, 3\varepsilon) &\leq 1, & & \\
T(8 c_{\mathrm{rb}} L^3 C_d M^3, 0, 2\varepsilon) &\leq 1. & &
\end{aligned} \tag{24.7}$$

Here $c_S = 10^5 c_X^5 K_0^5 \exp(6\kappa)$. The inequality with $96M^3$ controls the first step; the one with $C_T^2 M^3$ controls the positive-level fluctuation substitution. The conditions with c_S and $36 c_X^2 K_0^2$

control the connected logarithm and its scaling. The domain conditions occur with target $1/4$, with rates $(1/4,\varepsilon)$, with coefficients $8C_B, 8$, and with coefficients $4, 2$. The remaining displayed charge hypotheses above are individual inequalities of this inventory; (14.31) follows from the condition containing c_S.

Every decay rate in the inventory is positive for $0 < \varepsilon \le 1/200$. The only additional running-charge restrictions in this group are

$$v^{2p_0}/2 - 7v \ge \log 12, \qquad v^2 \ge 10L, \qquad v \ge 9.$$

They hold respectively for $v \ge 15$, $v \ge \sqrt{10L}$ and $v \ge 9$. Taking the maximum of these thresholds and the finitely many values in (24.5) gives a finite V_0. Choose a running-charge ceiling $0 < e_0 < \exp(-V_0)$ so that every such condition holds for every $v \ge -\log e_0$, equivalently for every running charge in $(0, e_0]$. The set of permitted ceilings is downward closed.

The terminal-scale conditions require a second ceiling $e_1 \le \min(e_0, 1/3)$. Write $\ell(v) = LMv^r$. They follow from

$$\begin{aligned}
(c_{\mathrm{fl}} + 2)\exp(-\varepsilon v) &\le \tfrac{1}{16}(1 - L^{-\varepsilon/2}),\\
8C_0^2\ell(v)^6(b_-^{-1} + L\ell(v)) &\le \exp(\varepsilon v/2),\\
4(2C_0 + c_\kappa + 5c_{\mathrm{fl}} + 6)\ell(v)^3 &\le \exp(\varepsilon v),\\
\exp(-(1/4 - 9\varepsilon)v) &\le \tfrac{1}{8},\\
4C_0\ell(v)^2 v^p &< \exp(\varepsilon v),\\
12\ell(v)^3\left(\exp(-v^{p_0}/2) + \exp(-v^{2p}/(2C_0^2))\right) &\le \exp(-v),\\
C_\Sigma \exp(-\gamma_\Sigma M v^r) &\le \exp(-v),\\
L^2\exp(-2v)\ell(v) &\le 1,\\
6N_* M v^r \exp(-v)(v\exp(-v) + \tfrac{1}{2}) &\le \pi^2. \qquad (24.8)
\end{aligned}$$

These are the scalar hypotheses of (16.2), (22.36) and (25.26), after substituting (16.5). Every polynomial-exponential rate here is positive for $0 < \varepsilon \le 1/200$. The reference flux condition expands as

$$6N_* M v^{r+1}\exp(-2v) + 3N_* M v^r \exp(-v) \le \pi^2.$$

The superlinear tails reduce to these bounds: for $v \ge \max(4, 4C_0^2)$,

$$v - \tfrac{1}{2}v^{p_0} \le -v, \qquad v - \frac{v^{2p}}{2C_0^2} \le -v.$$

Hence

$$12\ell(v)^3\left[\exp\left(v - \tfrac{1}{2}v^{p_0}\right) + \exp\left(v - \frac{v^{2p}}{2C_0^2}\right)\right] \le 24(LM)^3 v^{3r}\exp(-v).$$

For the twist tail, if $v \ge \max(1, 2/(\gamma_\Sigma M), \log^+ C_\Sigma)$, then $C_\Sigma\exp(v - \gamma_\Sigma M v^r) \le 1$. Put $\lg = \frac{1}{2}\log L$. The scale-ratio condition follows for $v \ge 2r$:

$$\left(1 + \frac{3\lg}{v}\right)^r \le \exp\left(\frac{3r\lg}{v}\right) \le L^{3/4} < L. \qquad (24.9)$$

Together with $v \ge 8$, $v \ge 31L + 17$ and $p_0 v^{p_0 - 1} \ge 28$, these bounds give a finite threshold V_1. Its maximum includes $2r$. Let $\mathcal{E}_1$ consist of the numbers $e' \in (0, \min(e_0, 1/3)]$ for which the geometric conditions and (24.8) hold for every $v \ge -\log e'$. It contains $\min(e_0, 1/3, \exp(-V_1)) > 0$ and is downward closed. Fix

$$e_1 = \tfrac{1}{2}\sup\mathcal{E}_1 > 0.$$

There is an element of $\mathcal{E}_1$ larger than e_1; downward closure therefore gives $e_1 \in \mathcal{E}_1$. Thus all terminal-scale conditions hold for every $v \geq -\log e_1$.

24.4. **Physical and probe-dependent charge intervals.** Fix the reference mass and hence $h \geq 1$. Set

$$e_G = \begin{cases} e_1, & h = 1, \\ \min\{e_1/\sqrt{h},\, 1/(\sqrt{h}\log h)\}, & h > 1. \end{cases} \tag{24.10}$$

This is positive. For $0 < e \leq e_G$, every running charge up to level K is at most e_1, and $e\sqrt{h}\log h \leq 1$. The scale proposition therefore applies on the top and reference tori. The terminal inventory gives (22.36) at their respective stops, and (25.17) holds on the reference torus. The common pair of Theorem 16.4 is defined for $N \geq N_{\min}(R_{\mathrm{top}}, e)$. At lower cutoffs only its mass shift is assigned zero. For $N \geq N_{\mathrm{ref}}(e)$ the same pair is controlled on the reference torus. Its mass shift is real, independent of the cutoff volume, and has absolute value less than $1/8$.

Fix a compact real probe β. Its buffered support diameter, energy and uniform rescaled norms are finite by (21.6). The differentiated inverse and interaction constants have already been fixed; these probe constants are chosen last. Proposition 23.8 gives $e_\sharp(\beta) > 0$ for which all its factor, one-cell, tree, jet and placement conditions hold for every $0 < e \leq e_\sharp(\beta)$, uniformly in $N \geq N_{\mathrm{ref}}(e)$ and in the allowed placements. Its proof uses $a_* \in \{h/L, h/L^2\}$ and $v_* = v_K + O(1)$: polynomial factors have fixed degrees, the separated-cell errors decay exponentially in v_*^r, and the complement decays faster than the Cauchy-radius loss. This gives one interval of charges for all cutoffs.

For $0 < e \leq \min(e_G, e_\sharp(\beta))$, (22.50) and (23.54) supply the two inputs in (25.28). The probe placement inequalities imply (19.6) for $\eta \leq 1$. The bounds for the free coefficients and the constant C_U in (25.35) are finite and independent of e, N, T and placement. They depend only on the fixed construction and probe.

Suppose now that $b(dd^*\beta) > 0$ and put $C_f = 8C_U + 3C_U^2$. Choose

$$0 < e_\beta \leq \min\{e_G, e_\sharp(\beta), 1, (b(dd^*\beta)/(2C_f))^{1/\sigma}\}. \tag{24.11}$$

The minimum is positive, since $\sigma > 0$ and $C_f > 0$. Proposition 25.4 holds throughout $(0, e_\beta]$ and gives the cutoff hypotheses of Proposition 26.2. In particular $C_f e^\sigma \leq b(dd^*\beta)/2$. The construction constants were chosen before h and the probe; the physical charge interval precedes the probe-dependent restriction. □

25. Transfer from the reference volumes

The renormalization estimates are obtained on two reference tori. To use them for volume-first limits, we need bounds on products of local observables in arbitrarily large cutoff tori. The chessboard estimate reduces each such product to a folded one-cell observable. A separate convexity estimate controls the normalization by the partition functions of the two reference volumes.

25.1. **Folded observables and the partition defect.** Use unit-lattice lengths in this subsection. Fix the charge $q > 0$ and a real lattice mass. Choose coordinate cells of width $w \in \frac{1}{2}\mathbb{Z}_{>0}$ and offsets $o_\mu \in \frac{1}{2}\mathbb{Z}$:

$$Q_c^\circ = \{y : o_\mu + wc_\mu < y_\mu < o_\mu + w(c_\mu + 1),\ \mu = 1, 2, 3\}.$$

The walls are admissible coordinate mirrors. For even positive integers s_μ with $ws_\mu \geq 4$, let $C(ws)$ be the rectangular torus with these side lengths and antiperiodic fermions. It has $s_1 s_2 s_3$ cells. The wall-reflection group acts simply transitively on the covering cells. Write $O^{[c]}$ for the reflected copy of $O \in \mathcal{A}(Q_0^\circ)$ in Q_c°, and

$$O_{ws}^\sharp = \prod_{c \bmod s} O^{[c]}, \qquad \mathcal{Z}_s(O) = Z_{C(ws)}[O_{ws}^\sharp].$$

The copies use the antilinear reflection action, including conjugation of scalar coefficients.

These arrays satisfy the rectangular chessboard estimate (3.24). The half-integer width causes no change in its proof: all walls remain admissible, and the torus sides are integers at least four. For a support box $B = y + [-r, r]^3 \subset Q_c^\circ$ one has $2r < w$. A nonzero period has sup norm at least $2w > 4r$ and at least four. Thus its enlarged boxes $B + [-1, 1]^3$ are disjoint, by the same embedding estimate used in the proof of Theorem 3.3. The products therefore belong to the required slab classes.

Hold two entries of s fixed and vary s_μ. The product of the copies in the layer $c_\mu = 0$ gives a Hilbert–Schmidt slab operator Y independent of s_μ. Its reflected layer pairs fill the entire folded array. Equation (3.21) gives

$$\mathcal{Z}_s(O) = \operatorname{Tr}(Y^*Y)^{s_\mu/2}. \tag{25.1}$$

In particular $\mathcal{Z}_s(O) \geq 0$, and $s_\mu \mapsto \mathcal{Z}_s(O)^{1/s_\mu}$ is nonincreasing. If $Y \neq 0$, the logarithm of this folded partition function is convex in s_μ. The first assertion is the monotonicity of Schatten norms, and the second follows from Hölder's inequality for the trace of positive powers. The normalization of Y here is that of the slab operators in Section 3.3.

Choose an even reference cell count $s_0 \geq 2$ with $n_r = ws_0 \geq 4$, and an integer $j \geq 2$. Put

$$N_r = s_0^3, \qquad D_j = j^3 \log Z_{C(n_r)}[1] - \log Z_{C(jn_r)}[1], \qquad \delta_j = \frac{D_j}{(j-1)^3 N_r}. \tag{25.2}$$

For an observable in the base cell its normalized reference seminorm is

$$\|O\|_* = \left(\frac{\mathcal{Z}_{(s_0,s_0,s_0)}(O)}{Z_{C(n_r)}[1]} \right)^{1/N_r}. \tag{25.3}$$

The partition function is positive, and the numerator is nonnegative, so this is well-defined also when it vanishes.

Proposition 25.1. *One has $D_j \geq 0$. Let s be even, with either $s = s_0$ or $s \geq js_0$, and put $r = s/s_0$, $n = ws$. For every set $\mathcal{C}$ of distinct cells of $C(n)$ and $O_c \in \mathcal{A}(Q_0^\circ)$,*

$$\left| \mathbb{E}_{C(n)} \prod_{c \in \mathcal{C}} O_c^{[c]} \right| \leq \prod_{c \in \mathcal{C}} e^{\delta_j} \|O_c\|_*. \tag{25.4}$$

The right side uses only the two reference partition functions and the one-cell observables folded on $C(n_r)$.

Proof. Successively decreasing the three side counts in (25.1) gives, for $s \geq s_0$,

$$\mathcal{Z}_{(s,s,s)}(O) \leq \mathcal{Z}_{(s_0,s_0,s_0)}(O)^{r^3}. \tag{25.5}$$

Taking $O = 1$ and $r = j$ proves $D_j \geq 0$.

We next obtain the lower bound for the normalizing partition function. Define $p(r_1, r_2, r_3) = \log Z_{C(r_1 n_r, r_2 n_r, r_3 n_r)}[1]$ on ratios with integer side lengths at least four. The pressure trace (3.30) shows that p is convex in each argument and that p/r_μ is nonincreasing in that argument. For $r \geq j$ put $a = (r-1)/(j-1)$. Convexity gives $p(r, \cdot, \cdot) \geq ap(j, \cdot, \cdot) - (a-1)p(1, \cdot, \cdot)$. Apply this in each argument, retaining the positive coefficient a at the next step. Bound the three subtracted terms by the side monotonicity. The result is

$$\begin{aligned} p(r,r,r) &\geq a^3 p(j,j,j) \\ &\quad - (a-1)(a^2j^2 + ajr + r^2)p(1,1,1) \\ &= r^3 p(1,1,1) - a^3 D_j. \end{aligned}$$

The last equality uses $aj-(a-1)=r$ three times. The negative coefficients are why upper bounds are used on the three subtracted terms; no sign condition on p is needed. Thus, including the equality case $r=1$,

$$\log Z_{C(n)}[1] \geq r^3\left(\log Z_{C(n_r)}[1] - \frac{D_j}{(j-1)^3}\right). \tag{25.6}$$

Fill the cells outside $\mathcal{C}$ with the constant integrand. The normalized rectangular chessboard estimate then gives

$$\left|\mathbb{E}_{C(n)}\prod_{c\in\mathcal{C}} O_c^{[c]}\right| \leq \prod_{c\in\mathcal{C}}\left(\frac{\mathcal{Z}_{(s,s,s)}(O_c)}{Z_{C(n)}[1]}\right)^{1/(r^3N_r)}.$$

Use (25.5) in the numerator and (25.6) in the denominator. Each factor is at most $e^{\delta_j}\|O_c\|_*$, proving (25.4). □

25.2. **The photon part of the defect.** The zero-flux Gaussian factor involves the determinant of the unit-lattice scalar Laplacian with its constant mode removed. Write its logarithm as

$$\mathcal{L}(n_1,n_2,n_3) = \sum_{\substack{k\in\prod_\mu \mathbb{Z}/n_\mu\mathbb{Z}\\ k\neq 0}} \log\left(\sum_{\mu=1}^{3}\left(2-2\cos\frac{2\pi k_\mu}{n_\mu}\right)\right), \qquad \mathcal{L}(n)=\mathcal{L}(n,n,n).$$

For $j\geq 2$ put

$$G_j = \left(1+\frac{j}{3.2}\right)\frac{2j^2}{2.56}, \qquad K_j = (1+j+j^2)(\log j + jG_j).$$

The estimate used for the two reference volumes is

$$\frac{1}{2}\big(\mathcal{L}(jn)-j^3\mathcal{L}(n)\big) \leq -(j^3-1)\log n + K_j \qquad (n,j\in\mathbb{Z},\ n,j\geq 2). \tag{25.7}$$

Its logarithmic term cancels the corresponding scale dependence of the Gaussian normalization. We retain the finite-volume calculation since a bulk determinant bound would lose this term.

Sum first in the first coordinate. For a nonzero transverse momentum $k'=(k_2,k_3)$ define $\omega(k')>0$ by

$$2\cosh\omega(k') = 2+\sum_{\mu=2,3}\left(2-2\cos\frac{2\pi k_\mu}{n_\mu}\right).$$

Factoring over the n_1-th roots of unity gives

$$\prod_{k_1\in\mathbb{Z}/n_1\mathbb{Z}}\left(2\cosh\omega - 2\cos\frac{2\pi k_1}{n_1}\right) = 4\sinh^2(n_1\omega/2).$$

For $k'=0$, omit $k_1=0$; the corresponding product is n_1^2. Therefore

$$\mathcal{L}(n_1,n_2,n_3) = 2\log n_1 + 2\sum_{k'\neq 0}\log\big(2\sinh(n_1\omega(k')/2)\big). \tag{25.8}$$

For $z>0$ let $g(z)=-\log(1-e^{-2z})$. The identity $2\sinh z = e^z(1-e^{-2z})$ shows that

$$\log(2\sinh(jz)) - j\log(2\sinh z) \leq jg(z).$$

Consequently

$$\mathcal{L}(jn_1,n_2,n_3) - j\mathcal{L}(n_1,n_2,n_3) \leq 2\log(jn_1) - 2j\log n_1 + 2j\sum_{k'\neq 0} g(n_1\omega(k')/2). \tag{25.9}$$

In the three applications below, $n_1 = n$ and $n_2, n_3 \in \{n, jn\}$. Represent each transverse momentum by $|k_\mu| \le n_\mu/2$ and put $p_\mu = 2\pi k_\mu/n_\mu$. For $M(p) = \sum_{\mu=2,3}(2 - 2\cos p_\mu)$,

$$\frac{4}{\pi^2}|p|^2 \le M(p) \le 8, \qquad \omega = 2\,\mathrm{arcsinh}(\sqrt{M}/2) \ge 0.51|p|.$$

The last inequality follows from concavity of arcsinh on $[0, \sqrt{2}]$ and $\mathrm{arcsinh}\,\sqrt{2} > 0.81\sqrt{2}$. Thus $n\omega(k')/2 \ge c_j|k'|$, where $c_j = 1.6/j$. Since $g(z) \le e^{-2z}(1 + (2z)^{-1})$, and $|k'| \ge (|k_2| + |k_3|)/\sqrt{2}$, one obtains

$$\begin{aligned}\sum_{k'\neq 0} g(n\omega(k')/2) &\le \left(1 + \frac{1}{2c_j}\right)\left[\left(\sum_{i\in\mathbb{Z}} e^{-\sqrt{2}c_j|i|}\right)^2 - 1\right] \\ &= \left(1 + \frac{1}{2c_j}\right)\sinh(c_j/\sqrt{2})^{-2} \le \left(1 + \frac{1}{2c_j}\right)\frac{2}{c_j^2} = G_j.\end{aligned}$$

Enlargement to all of $\mathbb{Z}^2 \setminus \{0\}$ is valid because the chosen representatives are distinct.

Finally telescope the enlargement of the three sides:

$$\begin{aligned}\mathcal{L}(jn, jn, jn) - j^3\mathcal{L}(n, n, n) = {}& \big[\mathcal{L}(jn, jn, jn) - j\mathcal{L}(n, jn, jn)\big] \\ &+ j\big[\mathcal{L}(jn, n, jn) - j\mathcal{L}(n, n, jn)\big] \\ &+ j^2\big[\mathcal{L}(jn, n, n) - j\mathcal{L}(n, n, n)\big].\end{aligned}$$

Symmetry of $\mathcal{L}$ supplies this identity. Each bracket is bounded by $2\log j + 2(1 - j)\log n + 2jG_j$ by (25.9). Multiplication by $1, j, j^2$ and summation prove (25.7).

25.3. **The photon, matter and flux contributions.** Fix $q > 0$ and a real lattice mass. On a cubic torus with $n \ge 4$ sites per side, let $Z_n^{(w)}[1]$ be the partition function in flux sector $w \in \mathbb{Z}^3$, and put $P_n(w) = Z_n^{(w)}[1]/Z_n[1]$. The flux order is $(23, 13, 12)$. Define

$$\varsigma_n = \sum_{w\neq 0} \exp\left(-\frac{2\pi^2|w|_2^2}{q^2 n}\right). \tag{25.10}$$

The notation $z_{r,n}$ below is the positive matter factor of (20.20).

Proposition 25.2. *For $n \ge 4$ and an integer $j \ge 2$,*

$$Z_n[1] = \frac{Z_n^{\mathrm{ph}} z_{r,n}}{P_n(0)}, \qquad 0 < P_n(0) \le 1, \tag{25.11}$$

$$P_n(w) \le \exp\left(-\frac{2\pi^2|w|_2^2}{q^2 n}\right), \qquad P_n(0) \ge 1 - \varsigma_n. \tag{25.12}$$

If $q^2 n \le 1$, then

$$\varsigma_n \le 7\exp\left(-\frac{2\pi^2}{q^2 n}\right) < \frac{1}{8}. \tag{25.13}$$

The defect D_j of (25.2) decomposes as

$$D_j = D_j^{\mathrm{ph}} + D_j^{\mathrm{mat}} + D_j^{\mathrm{fl}}, \tag{25.14}$$

$$D_j^{\mathrm{mat}} = j^3\log z_{r,n} - \log z_{r,jn}, \qquad D_j^{\mathrm{fl}} = -j^3\log P_n(0) + \log P_{jn}(0),$$

$$\begin{aligned}D_j^{\mathrm{ph}} &= (j^3 - 1)\log\frac{2\pi}{q^2} + \tfrac{1}{2}[\mathcal{L}(jn) - j^3\mathcal{L}(n)] \\ &\le (j^3 - 1)\log\frac{2\pi}{q^2 n} + K_j.\end{aligned} \tag{25.15}$$

If $\varsigma_n < 1$, then

$$D_j^{\mathrm{fl}} \le -j^3 \log(1-\varsigma_n). \tag{25.16}$$

All identities use the same charge and lattice mass at the two sides; the mass can have either sign.

Proof. The zero-flux factorization (20.21) gives $Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n} > 0$. All sectors have nonnegative weights, and their absolutely convergent sum is $Z_n[1]$. Hence $0 < P_n(0) \le 1$ and (25.11).

For a configuration of flux w, let h_w be the constant two-cochain with value $r_q w_{\mu\nu}/n^2$ on plaquettes of orientation $\mu\nu$. The Fourier Hodge decomposition writes $r_q z = dB + h$ with h constant. Summing over a coordinate two-torus annihilates dB and gives $n^2 h_{\mu\nu} = r_q w_{\mu\nu}$. Thus $F^{\mathrm{u}} - h_w$ is exact. A constant cochain is orthogonal to every exact cochain, by periodic summation by parts. In particular $\langle h, F^{\mathrm{u}}\rangle = \langle h, h_w\rangle$ on sector w whenever h is constant.

Apply Gaussian domination (3.43) with $h = h_{w'}$ and split the expectation into its sectors:

$$1 \ge \sum_w \exp\bigl(\langle h_{w'}, h_w\rangle - \|h_{w'}\|^2/2\bigr) P_n(w) \ge \exp(\|h_{w'}\|^2/2) P_n(w').$$

Since $\|h_w\|^2 = r_q^2 |w|_2^2/n$, this proves the first bound in (25.12). The identity $\sum_w P_n(w) = 1$ proves the second.

Set $x = \exp(-2\pi^2/(q^2 n))$. Positivity permits the factorization

$$\varsigma_n = \left(\sum_{i\in\mathbb{Z}} x^{i^2}\right)^3 - 1 \le \left(1 + \frac{2x}{1-x}\right)^3 - 1.$$

For $q^2 n \le 1$, one has $x < 2^{-18}$. The bound $(1-x)^{-1} \le 1 + 2x$ gives $2x/(1-x) \le 2.001x < 10^{-5}$. Therefore $\varsigma_n \le 2.001x(3 + 3\cdot 10^{-5} + 10^{-10}) \le 7x < 1/8$. This also proves convergence of (25.10).

Taking logarithms in (25.11) at n and jn gives (25.14). By (20.19),

$$\log Z_n^{\mathrm{ph}} = (n^3+2)\log r_q + (n^3-1)\log(2\pi) - \mathcal{L}(n)/2.$$

Its bulk terms cancel in the defect. The remaining constant is $(j^3-1)\log(2\pi/q^2)$. Equation (25.7) proves (25.15). Finally $\log P_{jn}(0) \le 0$, and $P_n(0) \ge 1 - \varsigma_n > 0$ gives (25.16). □

25.4. **Wick products on arbitrary cutoff tori.** Use the cell sources of Section 19 and write

$$N_* = 8L^3, \qquad a_L = \frac{L^3-1}{(L-1)^3}, \qquad D^{\mathrm{mat}} = L^3 \log z_{r,n_*} - \log z_{r,Ln_*}, \qquad \delta = \frac{D_L}{(L-1)^3 N_*}.$$

Fix $0 < e \le 1/3$, with $q = e\sqrt{\eta}$ and $q^2 n_* = e^2 R^*$, and assume

$$6 N_* e^2 R^* \log(1/e) \le \pi^2. \tag{25.17}$$

The choice of scales ensuring this condition is given below. Let $\mathcal{C}_* = \{0,\dots,2L-1\}^3$ and use the probe variance v_η of (19.9). For real c_2, c_4 put

$$H_2(x;v) = x^2 - v, \qquad H_4(x;v) = x^4 - 6vx^2 + 3v^2,$$

$$O_m^{[c]} = e^{-m} H_m(X_c; v_\eta) - c_m, \qquad m = 2, 4, \tag{25.18}$$

$$\Theta_m^{(0)} = \left| \left[\prod_{c\in\mathcal{C}_*} (\partial_{t_c}^m - c_m) \right] \mathring{\Phi}(0) \right|^{1/N_*}. \tag{25.19}$$

$$M_4 = 600 N_*^2 v_\eta^2 e^{-4} + |c_4|, \qquad M_2 = 12 N_* v_\eta e^{-2} + |c_2|,$$

$$T_m = \Theta_m^{(0)} + 2e^6 M_m, \qquad u_m = \exp(\delta) T_m. \tag{25.20}$$

Here $\mathring{\Phi}$ is the normalized external partition function (20.34). The notation e^{-m} denotes an inverse power of the charge.

Proposition 25.3. *Assume the probe placement condition* (19.6). *For $n = n_*$, or $n \in p\mathbb{Z}$ with $n \ge Ln_*$, the all-flux law with the fixed real lattice mass obeys*

$$0 \le \delta \le \frac{a_L}{N_*} \log \frac{2\pi}{e^2 R^*} + \frac{K_L + D^{\mathrm{mat}} + 1}{(L-1)^3 N_*}, \tag{25.21}$$

$$\left| \mathbb{E}_n^{\mathrm{af}} \prod_{c \in \mathcal{C}} O_m^{[c]} \right| \le u_m^{|\mathcal{C}|} \tag{25.22}$$

for every set $\mathcal{C}$ of distinct cells and $m = 2, 4$. In particular, if κ_j denotes the cumulant of X_0 on that torus, then

$$\left| e^{-2}(\kappa_2 - v_\eta) - c_2 \right| \le u_2, \tag{25.23}$$

$$\left| e^{-4}\kappa_4 - (c_4 - 3c_2^2) \right| \le u_4 + 3u_2(2|c_2| + u_2). \tag{25.24}$$

The bounds are uniform in n. Their right sides are formed on the two reference tori; $\Theta_m^{(0)}$, and therefore u_m, may depend on the placement.

Proof. Condition (25.17), $N_* \ge 1000$ and $\log(1/e) \ge \log 3 > 1$ imply $e^2 R^* < 1$. For $x = \exp(-2\pi^2/(e^2 R^*))$ they also give

$$x \le e^{12N_*}, \qquad x^{1/(2N_*)} \le e^6.$$

Apply Proposition 25.2 at n_* and $j = L$. Since $\varsigma_{n_*} \le 7x < 1/8$,

$$D_L^{\mathrm{fl}} \le -L^3 \log(1 - \varsigma_{n_*}) \le 8L^3 x = N_* x \le N_* 3^{-12N_*} \le 1.$$

The photon estimate and the defect decomposition now prove the upper bound in (25.21); its lower bound is Proposition 25.1.

All the polynomials in (25.18) are the same local integrands on every carrying torus, by Section 19. They are real elements of the cell observable class. Write $O_m^\sharp = \prod_{c \in \mathcal{C}_*} O_m^{[c]}$ on $C(n_*)$. Its all-flux expectation is nonnegative, by the fold trace (25.1). We estimate it by splitting flux zero from the other sectors.

First, Gaussian domination and the moment bound give, for every positive integer i,

$$\mathbb{E}_{n_*}^{\mathrm{af}} X_c^{2i} \le (2i \exp(1) v_\eta)^i.$$

Minkowski's inequality at exponent $2N_*$ yields

$$\begin{aligned} \|H_4(X_c; v_\eta)\|_{2N_*} &\le [64 \exp(2) N_*^2 + 24 \exp(1) N_* + 3] v_\eta^2 \le 600 N_*^2 v_\eta^2, \\ \|H_2(X_c; v_\eta)\|_{2N_*} &\le [4 \exp(1) N_* + 1] v_\eta \le 12 N_* v_\eta. \end{aligned}$$

Thus $\|O_m^{[c]}\|_{2N_*} \le M_m$. The finite moments justify all uses of Hölder and Cauchy–Schwarz below, also when $v_\eta = 0$.

On the zero-flux sector, (20.36) gives

$$\left| \mathbb{E}_{n_*}^{\mathrm{af}} [O_m^\sharp \mathbf{1}_{\{\mathrm{flux}=0\}}] \right| = P_{n_*}(0) \left| \mathbb{E}_{n_*}^{\mathrm{av}} O_m^\sharp \right| \le (\Theta_m^{(0)})^{N_*}.$$

On the remaining sectors, Cauchy–Schwarz followed by Hölder with N_* factors gives

$$\begin{aligned} \left| \mathbb{E}_{n_*}^{\mathrm{af}} [O_m^\sharp \mathbf{1}_{\{\mathrm{flux}\neq 0\}}] \right| &\le \varsigma_{n_*}^{1/2} \big(\mathbb{E}_{n_*}^{\mathrm{af}} |O_m^\sharp|^2 \big)^{1/2} \\ &\le \varsigma_{n_*}^{1/2} \prod_{c \in \mathcal{C}_*} \|O_m^{[c]}\|_{2N_*} \\ &\le \big(\varsigma_{n_*}^{1/(2N_*)} M_m \big)^{N_*}. \end{aligned}$$

Since $\varsigma_{n_*}^{1/(2N_*)} \le 2e^6$, the nonnegative N_*-th root satisfies

$$\|O_m\|_* = \big(\mathbb{E}^{\mathrm{af}}_{n_*} O_m^\sharp\big)^{1/N_*} \le \Theta_m^{(0)} + 2e^6 M_m = T_m.$$

For nonnegative a, b we used $(a^{N_*} + b^{N_*})^{1/N_*} \le a + b$. Proposition 25.1, with $s_0 = 2L$ and $j = L$, proves (25.22).

For one cell the same estimate gives $|e^{-m}\mathbb{E}H_m(X_0; v_\eta) - c_m| \le u_m$. Charge conjugation makes the odd moments zero. Writing $y = e^{-2}\mathbb{E}H_2(X_0; v_\eta)$, one has

$$e^{-4}\kappa_4 = e^{-4}\mathbb{E}H_4(X_0; v_\eta) - 3y^2.$$

The second-order conclusion follows at once. For the fourth, use $|y - c_2| \le u_2$ and $|y + c_2| \le 2|c_2| + u_2$ to obtain (25.24). □

25.5. **Reference scales and the loss of charge exponent.** The reference side is chosen from the logarithmic charge scale. Fix a power M of L with $M \ge L^2$ and an integer $r \ge 2$. With h as in (1.2), set

(25.25)

$$e_K = e\sqrt{h}, \qquad v_K = -\log e_K, \qquad r_K = \min\{L^i : L^i \ge v_K^r,\ i \in \mathbb{Z}_{\ge 0}\}, \qquad R^* = \frac{hMr_K}{L}.$$

Assume $v_K \ge 8$, $e \le 1/3$, $e_K \log h \le 1$, and

$$6N_* M v_K^r \exp(-v_K)\left(v_K \exp(-v_K) + \tfrac{1}{2}\right) \le \pi^2. \tag{25.26}$$

These are finite charge conditions after h, L, M, r are fixed. They hold on a sufficiently small charge interval: $v_K \to \infty$ as $e \to 0$, and the polynomial times exponential in (25.26) tends to zero.

Minimality gives $v_K^r \le r_K < L v_K^r$. Consequently $T^* = \log_L(Mr_K) - 1$ is an integer at least one, $R^* = hL^{T^*}$, and $\ell_c = R^*/(2L) \to \infty$ as $e \to 0$. Moreover $1/L < \bar{m}h \le 1$, so

$$\bar{m}R^* > \frac{Mr_K}{L^2} \ge r_K \ge v_K, \qquad \exp(-\bar{m}R^*) \le e\sqrt{h}. \tag{25.27}$$

The same upper bound on r_K gives

$$\begin{aligned} e^2 R^* \log(1/e) &< M v_K^r \exp(-2v_K)(v_K + \tfrac{1}{2}\log h) \\ &\le M v_K^r \exp(-v_K)(v_K \exp(-v_K) + \tfrac{1}{2}). \end{aligned}$$

Thus (25.26) implies (25.17). The probe fits every sufficiently small charge and sufficiently fine mesh, since D_β^{buf} is fixed and $\ell_c \to \infty$.

We now isolate the two reference estimates used in the passage to arbitrary volumes. Their dependence on the charge matters: a bounded partition defect still loses a small power of e through its photon contribution. Let C_{mat} and C_{ref} be finite constants independent of the charge, mesh and placement. Assume that for $N \ge N_{\mathrm{ref}}(e)$ the mass assignment and reference response satisfy

$$\begin{aligned} D^{\mathrm{mat}} &\le C_{\mathrm{mat}}, \\ \Theta_4^{(0)} &\le C_{\mathrm{ref}} e^{\mathfrak{e}}, \qquad (1 + |c_2|)\Theta_2^{(0)} \le C_{\mathrm{ref}} e^{\mathfrak{e}}, \end{aligned} \tag{25.28}$$

where

$$\mathfrak{e} = s(1/4 - 9\varepsilon) - 12\varepsilon, \qquad \sigma = \mathfrak{e} - \frac{2a_L}{N_*} > 0, \qquad 0 < s < 1, \quad 0 < \varepsilon \le 1/200. \tag{25.29}$$

Here $N_{\mathrm{ref}}(e)$ is a fixed finite mesh threshold. For the chamber construction, (22.50) and (23.54) give (25.28) for $0 < e \le \min(e_G, e_\sharp(\beta))$. The constants and cutoff thresholds are those fixed in Section 24.

On the reference torus take the free coefficients $c_m = N_0^{(m)}(0)$ from (18.27), using the source associated with the same placed probe. Proposition 18.3 and the support fit give a constant, enlarged into C_{ref} if necessary, such that

$$|c_2| + |c_4|^{1/2} \le C_{\mathrm{ref}}, \qquad |c_4 - 3c_2^2 - b(f)| \le C_{\mathrm{ref}}(e + \eta^{1/4}). \tag{25.30}$$

Indeed $R^* \ge D_\beta + 1$, and (25.27) bounds the finite-volume error in Proposition 18.1. The coefficients are real and independent of placement.

Proposition 25.4. *Fix the probe and the scale data above, and assume* (19.6) *and* (25.28)*. There is a finite constant C_U, independent of e, N, T and placement, such that for all the stated reference charges and $N \ge N_{\mathrm{ref}}(e)$,*

$$u_4 \le C_U e^\sigma, \qquad (1 + |c_2|)u_2 \le C_U e^\sigma, \qquad |c_4 - 3c_2^2 - b(f)| \le C_U(e^\sigma + \eta^{1/4}). \tag{25.31}$$

On every cutoff torus with $T \ge T^* + 1$, every finite set $Y \subset (R^*/L)\mathbb{Z}^3$ of points incongruent modulo $R\mathbb{Z}^3$ satisfies

$$\left| \mathbb{E}_{N,T} \prod_{y \in Y} W_m(F^\sharp(\tau_y f^\eta); v_\eta, c_m) \right| \le u_m^{|Y|}, \qquad m = 2, 4, \tag{25.32}$$

where $W_m(x; v, c) = e^{-m} H_m(x; v) - c$. The fourth cumulant of the fixed smooth test obeys

$$|\kappa_4(F^\sharp(f)) - e^4 b(f)| \le (8C_U + 3C_U^2)e^{4+\sigma} + C_U(\eta^2 + e^4\eta^{1/4}). \tag{25.33}$$

The tests f^η have one fixed compact support, $f^\eta \to f$ in L^2, and $v_\eta \to \|f\|_2^2$. Thus (25.31) *and* (25.32) *give exactly the cutoff hypotheses of Proposition 26.2.*

Proof. Write

$$s_\beta = \|f\|_2 + \|\beta\|_{H^4}, \qquad C_\Gamma^{\mathrm{tr}} = (2\pi)^{a_L/N_*} \exp\left(\frac{K_L + C_{\mathrm{mat}} + 1}{(L-1)^3 N_*} \right).$$

Since $R^* \ge 1$, (25.21) gives

$$\exp(\delta) \le C_\Gamma^{\mathrm{tr}} e^{-2a_L/N_*}. \tag{25.34}$$

The inverse power here is a power of the charge. The source-curvature identity and Jensen's inequality for the disjoint averaging cubes give $v_\eta^{1/2} \le \|f^\eta\|_2 \le s_\beta$. Use this and (25.30) in (25.20):

$$\begin{aligned} T_4 &\le C_{\mathrm{ref}} e^{\mathfrak{e}} + 1200 N_*^2 s_\beta^4 e^2 + 2C_{\mathrm{ref}}^2 e^6 \\ &\le (C_{\mathrm{ref}} + 1200 N_*^2 s_\beta^4 + 2C_{\mathrm{ref}}^2) e^{\mathfrak{e}}, \\ (1 + |c_2|)T_2 &\le C_{\mathrm{ref}} e^{\mathfrak{e}} + 2e^6(12 N_* s_\beta^2 e^{-2} + C_{\mathrm{ref}})(1 + C_{\mathrm{ref}}) \\ &\le [C_{\mathrm{ref}} + 2(12 N_* s_\beta^2 + C_{\mathrm{ref}})(1 + C_{\mathrm{ref}})] e^{\mathfrak{e}}. \end{aligned}$$

Here $0 < \mathfrak{e} < 1/4$, from (25.29). One possible constant is

$$\begin{aligned} C_U = \max\{ & C_\Gamma^{\mathrm{tr}}(C_{\mathrm{ref}} + 1200 N_*^2 s_\beta^4 + 2C_{\mathrm{ref}}^2), \\ & C_\Gamma^{\mathrm{tr}}[C_{\mathrm{ref}} + 2(12 N_* s_\beta^2 + C_{\mathrm{ref}})(1 + C_{\mathrm{ref}})],\ C_{\mathrm{ref}},\ 60 s_\beta^4 \}. \end{aligned} \tag{25.35}$$

Multiplication by (25.34) proves the first two bounds in (25.31). The last follows from (25.30), since $e \le e^\sigma$.

For a cutoff side $n = 8hL^{T+N}$ with $T \ge T^* + 1$, one has $n = 8L^{T-T^*} n_*$, a multiple of p with $n \ge L n_*$. For $y \in (R^*/L)\mathbb{Z}^3$ use the even cell index $c(y)$ of (19.11). Incongruent points give distinct cells. Proposition 25.3 applies to their product, and lattice translation invariance removes

the common placement. This proves (25.32). For a fixed finite set of distinct points it applies as soon as $R > 2\max_{y\in Y}|y|_\infty$.

It remains to replace the discrete smooth test f^η by f in the cumulant. We give the quantitative multilinear estimate used for this purpose. For a real two-cochain g, charge conjugation and Gaussian domination give

$$\mathbb{E}\cosh(t\langle g, F^{\mathrm{u}}\rangle) \le \exp(t^2\|g\|^2/2).$$

Extract its nonnegative term of degree $2i$ and choose $t^2 = 2i/\|g\|^2$. This gives

$$\mathbb{E}\langle g, F^{\mathrm{u}}\rangle^{2i} \le (2i)!\left(\frac{\|g\|^2}{2i}\right)^i \exp(i).$$

For $i = 2$ the bound is $(3/2)\exp(2)\|g\|^4$. Hölder bounds the four-field moment by

$$\tfrac{3}{2}\exp(2)\prod_a \|g_a\|.$$

The variance bound bounds each of its three pair subtractions by $\prod_a \|g_a\|$. Since $(3/2)\exp(2)+3 < 15$, the symmetric multilinear forms K_2, K_4 obey

$$|K_2(g_1,g_2)| \le \|g_1\|\|g_2\|, \qquad |K_4(g_1,g_2,g_3,g_4)| \le 15\prod_a \|g_a\|. \tag{25.36}$$

For $g = 0$ the corresponding moment is zero.

Let $g = \eta^{3/2} f_\eta$ and $g' = \eta^{3/2}(f^\eta)_\eta$ on the cutoff torus. The common support cube has side less than R. The averaging bound and (19.5) give

$$\|g\|, \|g'\| \le s_\beta, \qquad \|g - g'\| \le \eta^2\|\beta\|_{H^4}.$$

The observables paired with these cochains are $F^\sharp(f)$ and $F^\sharp(f^\eta)$. Telescope the four slots of K_4 and the two slots of K_2 in (25.36). This proves

$$\begin{aligned} |\kappa_4(F^\sharp(f)) - \kappa_4(F^\sharp(f^\eta))| &\le 60 s_\beta^4 \eta^2, \\ |\kappa_2(F^\sharp(f)) - \kappa_2(F^\sharp(f^\eta))| &\le 2 s_\beta^2 \eta^2. \end{aligned} \tag{25.37}$$

One-cell transfer and (25.31) bound the fourth cumulant of $F^\sharp(f^\eta)$ by

$$|\kappa_4(F^\sharp(f^\eta)) - e^4(c_4 - 3c_2^2)| \le (7C_U + 3C_U^2)e^{4+\sigma}.$$

Adding the free-coefficient error and the first bound in (25.37) proves (25.33). Finally the compact support, L^2 convergence and variance convergence are respectively (19.1), (19.5) and (19.10). □

26. Fourth cumulants of the limiting components

The fourth-cumulant estimate for a mixed limiting law does not by itself control each ergodic component. We use the stronger estimate on products of translated Wick polynomials. It bounds their translation-invariant conditional expectations, and hence the second and fourth moments of almost every component.

26.1. **From distinct-point products to the invariant part.** Let $U(y)$, $y \in \mathbb{Z}^d$, be the operators of a measure-preserving action on a probability space, where $d \ge 1$, with $U(y)G = G \circ T_{-y}$. Let P_{lat} be the orthogonal projection in L^2 onto the vectors invariant under this action.

Lemma 26.1. *Suppose that G is real and belongs to every L^p, $p < \infty$. If $u \ge 0$ and*

$$\left|\mathbb{E}\prod_{a=1}^J U(y_a)G\right| \le u^J \tag{26.1}$$

for every finite family of pairwise distinct points of $\mathbb{Z}^d$, then $|P_{\mathrm{lat}}G| \le u$ almost surely.

Proof. For $B_n = \{0, \dots, n-1\}^d$ put $A_n = n^{-d} \sum_{y \in B_n} U(y)$. These contractions converge strongly in L^2 to P_{lat}. Indeed, the orthogonal complement of the invariant space is the closed span of $(U(y) - 1)H$. On such a vector, cancellation of the common terms in B_n and $y + B_n$ gives

$$\|A_n(U(y) - 1)H\|_2 \le 2|y|_1 n^{-1} \|H\|_2.$$

On invariant vectors the averages are the identity, proving the asserted convergence by the orthogonal decomposition.

For an even integer $J \ge 2$, expand $\mathbb{E}(A_n G)^J$. The distinct-index terms obey (26.1). There are at most $\binom{J}{2} n^{d(J-1)}$ remaining tuples: each has a repeated pair of indices. Hölder's inequality bounds their individual expectations in modulus by $\|G\|_J^J$. Therefore

$$\mathbb{E}|A_n G|^J \le u^J + \binom{J}{2} n^{-d} \|G\|_J^J.$$

Along a subsequence the L^2 convergence is almost sure. Fatou's lemma gives $\mathbb{E}|P_{\mathrm{lat}} G|^J \le u^J$ for every even J. For $t > u$, Markov's inequality now gives $\Pr(|P_{\mathrm{lat}} G| > t) \le (u/t)^J$. Let $J \to \infty$ and then $t \downarrow u$ through a countable sequence. □

Let a translation-invariant law μ on tempered fields have the decomposition $\mu = \int \mu_\xi \, \mathrm{d}\nu(\xi)$. Write I for its translation-invariant sigma algebra and $P = \mathbb{E}_\mu[\,\cdot \mid I]$. Apply the lemma to translations on $\lambda\mathbb{Z}^3$, with $\lambda > 0$ fixed. The range of P is contained in that of P_{lat}, so $P = P P_{\mathrm{lat}}$. Positivity of conditional expectation therefore gives

$$|PG| \le P|P_{\mathrm{lat}} G| \le u, \qquad |\mu_\xi(G)| \le u \quad \text{for } \nu\text{-almost every } \xi. \tag{26.2}$$

The second assertion uses the conditional-law representation of the decomposition. Integrability under the conditional laws and the representation for this integrable G follow by truncating its positive and negative parts.

26.2. **Passing the Wick-product estimate to a limit law.** Fix the charge $e > 0$ and a real compactly supported test f. For $v \ge 0$ define

$$H_2(x; v) = x^2 - v, \qquad H_4(x; v) = x^4 - 6vx^2 + 3v^2, \qquad W_m(x; v, c) = e^{-m} H_m(x; v) - c \quad (m = 2, 4).$$

Here e^{-m} is the inverse power of the charge. In the application v is a variance, while the exponent in the small-charge error is denoted by σ.

Take a joint cutoff sequence defining a limit law μ. Suppose that the lattice estimates give compactly supported real tests f_i, all supported in one fixed cube, and real numbers

$$v_i, \; c_{2,i}, \; c_{4,i}, \; u_{2,i}, \; u_{4,i},$$

with $v_i \ge 0$, such that

$$\|f_i - f\|_2 \to 0, \qquad v_i \to \|f\|_2^2, \qquad \sup_i (|c_{2,i}| + |c_{4,i}|) < \infty. \tag{26.3}$$

For a fixed $\lambda > 0$, assume that

$$\left| \mathbb{E}_i \prod_{y \in Y} W_m(F^\sharp(\tau_y f_i); v_i, c_{m,i}) \right| \le u_{m,i}^{|Y|}, \qquad m = 2, 4, \tag{26.4}$$

for every finite $Y \subset \lambda\mathbb{Z}^3$ and all sufficiently large i depending on Y. This is the translated-cell form of the transfer estimate; the volume condition makes the points of Y incongruent modulo the torus. The reference estimates also give

$$\begin{aligned} 0 \le u_{4,i} \le C e^\sigma, \qquad 0 \le (1 + |c_{2,i}|) u_{2,i} \le C e^\sigma, \\ |c_{4,i} - 3c_{2,i}^2 - b(f)| \le C(e^\sigma + \eta_i^{1/4}), \end{aligned} \tag{26.5}$$

with C independent of the cutoff, $\sigma > 0$ and $0 < e \le 1$. For the chamber laws, Proposition 25.4 supplies these hypotheses throughout the charge interval (24.11).

Pass to a subsequence on which the four bounded sequences $c_{2,i}, c_{4,i}, u_{2,i}, u_{4,i}$ converge, with limits c_2, c_4, u_2, u_4. The weak limit remains μ. Equation (26.5) gives

$$u_4 \le Ce^{\sigma}, \quad (1 + |c_2|)u_2 \le Ce^{\sigma}, \quad |c_4 - 3c_2^2 - b(f)| \le Ce^{\sigma}. \tag{26.6}$$

Set $G_m = W_m(F(f); \|f\|_2^2, c_m)$. Then (26.4) passes to

$$\left| \mathbb{E}_{\mu} \prod_{y \in Y} U(y) G_m \right| \le u_m^{|Y|} \qquad (Y \subset \lambda\mathbb{Z}^3 \text{ finite}). \tag{26.7}$$

To justify this passage for the varying tests, expand the finite product of Wick polynomials into monomials. For a monomial of positive total degree J, telescope the replacement of each $F^{\sharp}(\tau_y f_i)$ by $F^{\sharp}(\tau_y f)$. All these tests have support in one open R_i-cube once i is large. The test-map variance and product-moment bounds then give an error at most

$$J c_J B^{J-1} \|f_i - f\|_2, \qquad B = \sup_i \max(\|f_i\|_2, \|f\|_2).$$

This tends to zero. The moments of the fixed tests converge by Section 6, and the polynomial coefficients converge by (26.3). There are only finitely many monomials for each fixed Y. This proves (26.7).

26.3. **The cumulant in an ergodic component.**

Proposition 26.2. *Under* (26.3)–(26.5)*, suppose that almost every translation-ergodic component of μ satisfies Gaussian domination. Then*

$$\left|\kappa_4^{\mu_\xi}(f) - e^4 b(f)\right| \le (8C + 3C^2)e^{4+\sigma} \tag{26.8}$$

for almost every component. In particular, if $b(f) > 0$ and $(8C + 3C^2)e^{\sigma} \le b(f)/2$, its fourth cumulant is positive.

Proof. Gaussian domination of μ makes the real polynomials G_2, G_4 belong to every $L^p(\mu)$. Apply Lemma 26.1 to (26.7), and then use (26.2). On one set of full component measure,

$$|d_m| \le u_m, \qquad d_m = \mathbb{E}_{\mu_\xi} G_m, \qquad m = 2, 4.$$

Intersect with the full-measure set of components having Gaussian domination. For a component in this intersection, put $X = F(f)$ and $v = \|f\|_2^2$. Gaussian domination gives $\mathbb{E}_{\mu_\xi} X = 0$ and all moments finite. Thus

$$\kappa_4^{\mu_\xi}(f) = \mathbb{E}_{\mu_\xi} H_4(X; v) - 3\big(\mathbb{E}_{\mu_\xi} H_2(X; v)\big)^2.$$

The definition of G_m gives the exact identity

$$e^{-4}\kappa_4^{\mu_\xi}(f) = c_4 + d_4 - 3(c_2 + d_2)^2. \tag{26.9}$$

Consequently (26.6) implies

$$\begin{aligned} |e^{-4}\kappa_4^{\mu_\xi}(f) - b(f)| &\le |c_4 - 3c_2^2 - b(f)| + u_4 + 6|c_2|u_2 + 3u_2^2 \\ &\le 8Ce^{\sigma} + 3C^2 e^{2\sigma} \le (8C + 3C^2)e^{\sigma}. \end{aligned}$$

This proves the assertion uniformly over the full-measure set of components. A centered normal variable has zero fourth cumulant, so positivity also proves that these laws are non-Gaussian. □

27. Tensor Schwinger functions

The Gaussian bound has already defined a continuous map $\Psi \mapsto Q_\Psi$ from $\mathcal{S}_n$ to $L^2(\lambda)$. To use law-level reflection positivity for arbitrary tensor tests, we must show that support in a half-space makes this random polynomial measurable with respect to the corresponding half-space sigma algebra. The Hermite expansion used for the growth bound does not preserve supports. We therefore use a compact Fourier expansion for this step.

27.1. **Products and supported tensor tests.** For $\Psi \in \mathcal{S}_k$ and $\Phi \in \mathcal{S}_l$, the tensor evaluation obeys

$$Q_\Psi Q_\Phi = Q_{\Psi\otimes\Phi} \quad \text{in } L^1(\lambda). \tag{27.1}$$

Both sides are separately continuous bilinear maps to L^1: for the left side use the L^2 bounds and Cauchy–Schwarz, and for the right side use continuity of the tensor product in Schwartz seminorms. They agree for elementary tensors. Their linear span is dense, so approximation in each variable proves (27.1). The same argument gives

$$Q_{\overline{\Psi}} = \overline{Q_\Psi}, \qquad Q_{g\cdot\Psi} = \widehat{g} Q_\Psi \quad \text{if } g_*\lambda = \lambda. \tag{27.2}$$

Here the second identity uses the exterior-square action, including the tensor indices.

We recall the needed supported expansion. Let $O \subset \mathbb{R}^3$ be open and $\Psi \in C_c^\infty(O^n)$. Its support projects into compact sets $K_a \subset O$. Choose $\chi_a \in C_c^\infty(O)$ equal to one on K_a, and a cube of side s containing all these supports in its interior. Extend Ψ periodically from this cube in all its variables. Its Fourier coefficients $c_{\mathbf{k}}$ obey, for every integer p,

$$|c_{\mathbf{k}}| \le C_{p,s,\Psi}(1+|\mathbf{k}|^2)^{-p}, \qquad \mathbf{k} = (k_1,\dots,k_n) \in (\mathbb{Z}^3)^n.$$

This follows by applying $(1-\Delta)^p$ and integrating by parts; there are no boundary terms. Multiplication of the Fourier series by $\prod_a \chi_a(y_a)$ gives the identity on all space

$$\Psi = \sum_{\mathbf{k}} c_{\mathbf{k}} \bigotimes_{a=1}^{n} \left(\chi_a(y_a) e^{2\pi i k_a\cdot y_a/s} \right). \tag{27.3}$$

The seminorm q_m of each tensor factor product grows at most as $(1+|\mathbf{k}|^2)^{m/2}$. Taking $p = m+3n$ makes the series absolutely convergent in that seminorm. The same argument applies componentwise to two-form tensor tests. It also proves the density of elementary tensors used above; arbitrary Schwartz tests are approximated by growing compact cutoffs.

Each elementary tensor in (27.3) has factors supported in O, so its random evaluation is Σ_O-measurable and belongs to $L^2(\lambda)$ by the moment bound. Continuity of Q and closedness of $L^2(\Sigma_O,\lambda)$ give

$$Q_\Psi \in L^2(\Sigma_O,\lambda) \qquad (\Psi \in C_c^\infty(O^n)). \tag{27.4}$$

This extends to tests supported in the closed half-space $H = \{x : v\cdot x \ge c\}$, with $|v| = 1$. Choose a fixed compact cutoff χ equal to one near zero and put

$$\Psi_j(y) = \prod_{a=1}^{n} \chi(y_a/j)\, \Psi(y_1 - v/j, \dots, y_n - v/j).$$

Each Ψ_j has compact support in O^n, $O = \{x : v\cdot x > c\}$. Translation is continuous in Schwartz space, and the growing cutoff tends to one in its action on each Schwartz function. More explicitly, Leibniz' rule gives

$$q_m(\Psi_j - \Psi) \le C_{m,n,\chi} j^{-1} q_{m+1}(\Psi).$$

Consequently $Q_{\Psi_j} \to Q_\Psi$ in $L^2(\lambda)$, and

$$Q_\Psi \in L^2(\Sigma_O,\lambda) \qquad (\operatorname{supp}\Psi \subset H^n). \tag{27.5}$$

Thus tests meeting the boundary plane belong to the same positive-side L^2 space.

27.2. **The five conditions.**

Proposition 27.1. *Let λ be a Gaussian-dominated law of real tempered two-forms. Suppose it is invariant under Euclidean motions and reflection positive at every plane through the origin. Then its tensor Schwinger functions satisfy* (E0′)–(E3) *of Theorem 1.1. If, for every $A, B \in L^2(\lambda)$,*

$$\mathbb{E}_\lambda[\overline{A}\, U(a) B] \longrightarrow \overline{\mathbb{E}_\lambda A}\, \mathbb{E}_\lambda B \quad \text{as } |a| \longrightarrow \infty,$$

they also satisfy (E4).

Proof. The Hermite estimate proves (E0′). Covariance follows from (27.2) and invariance of expectation. Permutation symmetry holds on elementary tensors because the random field evaluations commute; continuity gives (E3) on all tensor tests.

For (E2), let r be the stated reflection and H the chosen closed half-space. By (27.5), $A = \sum_{k=0}^{M} Q_{\Psi_k}$ lies in its positive-side L^2 space, where $Q_{\Psi_0} = \Psi_0$. Reflection positivity extends from bounded characters to this whole space by their density and the boundedness of the reflection form. Equations (27.1)–(27.2) give

$$\overline{\mathcal{R}_r Q_{\Psi_k}} = Q_{\Theta_k \Psi_k}, \qquad Q_{\Theta_k \Psi_k} Q_{\Psi_l} = Q_{\Theta_k \Psi_k \otimes \Psi_l} \quad \text{in } L^1.$$

Therefore

$$0 \le \mathbb{E}_\lambda[(\overline{\mathcal{R}_r A})A] = \sum_{k,l=0}^{M} S_{k+l}(\Theta_k \Psi_k \otimes \Psi_l).$$

This is the tensor reflection form with the conjugation and signs in the first-page statement.

Finally, (27.1) and translation covariance identify the expression in (E4) with $\mathbb{E}_\lambda[Q_\Psi U(a) Q_\Phi]$. Both random polynomials are in $L^2(\lambda)$. Apply the stated clustering with $A = \overline{Q_\Psi}$ and $B = Q_\Phi$ to obtain $\mathbb{E}_\lambda Q_\Psi\, \mathbb{E}_\lambda Q_\Phi$. □

For the component laws constructed above, qualitative clustering has already been obtained from the cone and the invariant projections. Once rotation invariance is proved for each component, the nine-mirror reflection form is transported to every plane, and the proposition applies.

28. Completion of the field-strength construction

Proof of Theorem 1.1. Fix the construction constants of Section 24, in their stated order. In particular $\sigma > 0$ in (25.29). Take the positive physical ceiling e_G in (24.10). Theorem 16.4 supplies real bare mass shifts $m_{0,N}(e)$, independent of the volume and twist, for $0 < e \le e_G$; below its defining mesh threshold they are set to zero. Their bound (16.19) gives $|m_{0,N}(e)| < 1/8$ for every N. The all-flux laws at (1.2) are therefore defined at every cutoff, even when the total lattice mass has either sign. The physical ceiling in the first-page statement is $e_{\mathrm{phys}} = e_G$. Proposition 17.1 proves the existence of the probes in (1.1). Fix any such probe and $0 < e \le e_\beta$, with e_β as in (24.11).

For this probe, Proposition 23.8 provides $e_\sharp(\beta) > 0$. The terminal partition bound (22.50) and the response estimate (23.54) supply both parts of (25.28). The free-coefficient bound (25.30) supplies its remaining reference input. Thus Proposition 25.4 applies throughout this interval, with one C_U independent of the cutoff, volume, placement and limit law. Its estimates hold whenever $N \ge N_{\mathrm{ref}}(e)$ and $T \ge T^*(e) + 1$. These are finite thresholds for each fixed e.

Volume-first laws and their Euclidean components. The tightness estimate (6.11) shows that every fixed-spacing volume-limit set $\mathcal{V}_N$ is nonempty and that the collection of these limits is tight on H_{-4}. Consequently volume-first limit points exist. Let μ be any one of them, and write

$\mu^{(N_i)} \to \mu$, where $N_i \to \infty$ and $\mu^{(N_i)} \in \mathcal{V}_{N_i}$. For each i, choose a volume index T_i along a defining subsequence for $\mu^{(N_i)}$ such that

$$T_i \geq i, \qquad d_{\mathrm{w}}(P_{N_i,T_i}, \mu^{(N_i)}) \leq 1/i,$$

where d_{w} metrizes weak convergence on H_{-4}. Then $d_{\mathrm{w}}(P_{N_i,T_i}, \mu) \leq 1/i + d_{\mathrm{w}}(\mu^{(N_i)}, \mu) \to 0$. This is the joint sequence in Proposition 6.1; both indices diverge, and all fixed finite-family moments converge along it.

Proposition 6.1 gives closedness, translation invariance, Gaussian domination with constant one, and invariance and reflection positivity at all planes in the nine mirror families. Section 7.5 transfers these properties to a measurable set of full measure in the translation-ergodic decomposition. The Gaussian bound holds there for all real Schwartz tests at once, by the countable-density and Fatou argument in (7.15). The same section transfers each additional mirror by transporting the conditional laws and using uniqueness of the decomposition. Intersecting the nine full-measure sets gives all mirror forms in each component; these reflections also generate the cubic group.

Theorem 8.6 now applies separately to μ and to every component in this full-measure set. It proves their invariance under the full Euclidean group and reflection positivity at every plane. For an ergodic component, the sharp cone and (7.10) identify the translation-invariant projection with the constant projection. Proposition 7.3 therefore gives clustering for all L^2 observables as $|a| \to \infty$. Proposition 27.1 gives (E0′)–(E3) for the parent law and (E0′)–(E4) for these components, with exactly the tensor action and growth bound of the first-page statement. In particular, supported tensor evaluations belong to the positive-side L^2 space even when their supports meet the reflection plane, by (27.5). For clustering the relevant identity is

$$S^{\xi}_{k+l}(\Psi \otimes \tau_a^{\otimes l}\Phi) = \mathbb{E}_{\mu_\xi}[Q_\Psi\, U(a) Q_\Phi], \qquad Q_\Psi, Q_\Phi \in L^2(\mu_\xi).$$

It gives (1.6) on all tensor Schwartz tests, using $A = \overline{Q_\Psi}$ in the L^2 clustering formula.

Fourth cumulants in the same components. Fix e in the interval above and put $\lambda = R^*(e)/L$. This spacing of translated cells is independent of N. Along the joint sequence just chosen, the reference numbers $c_{2,i}, c_{4,i}, u_{2,i}, u_{4,i}$ are bounded and depend on N_i and the chosen probe placement, rather than on the outer volume or on a set of translated cells. Pass to one subsequence on which these four numbers converge. Its weak limit is still μ.

For any finite $Y \subset \lambda\mathbb{Z}^3$, all large i satisfy the two cutoff thresholds above. They also satisfy $R_i > 2(\max_{y\in Y} |y|_\infty + r_f)$, where a fixed cube $[-r_f, r_f]^3$ contains the supports of f and f^{η_i}. Thus the points of Y are incongruent modulo $R_i\mathbb{Z}^3$, and all the translated tests lie in the interior of one period cube. Equation (25.32) applies:

$$\left| \mathbb{E}_{N_i,T_i} \prod_{y\in Y} W_m(F^\sharp(\tau_y f^{\eta_i}); v_i, c_{m,i}) \right| \leq u_{m,i}^{|Y|}, \qquad m = 2, 4.$$

For each fixed Y, expansion into monomials and the test replacement estimate in Section 26 pass this inequality to (26.7). More explicitly, the replacement error for a monomial of degree $J > 0$ is at most $J c_J B^{J-1} \|f^{\eta_i} - f\|_2 \to 0$; the remaining fixed-test moments converge along the joint sequence. The variances and polynomial coefficients converge as well. This works on the same coefficient subsequence for every finite Y in the countable lattice: the index beyond which support containment holds may depend on Y, while the limiting coefficients and constants are common.

The limits obey (26.6). Lemma 26.1, followed by (26.2), gives $|d_m| \leq u_m$ almost surely in the component parameter, where $d_m = \mathbb{E}_{\mu_\xi} W_m(F(f); \|f\|_2^2, c_m)$. Intersect this set with the full-measure set used for the OS conditions. Proposition 26.2 applies there. Its exact identity is

$$e^{-4}\kappa_4^{\mu_\xi}(f) = c_4 + d_4 - 3(c_2 + d_2)^2,$$

and gives $|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \le C_f e^{4+\sigma}$ with $C_f = 8C_U + 3C_U^2$. The coefficient limits may depend on the defining subsequence; this constant does not. Finally (24.11) gives $C_f e^\sigma \le b(f)/2$ throughout the interval. Then $\kappa_4^{\mu_\xi}(f) \ge e^4 b(f)/2 > 0$. This proves (1.7) uniformly in the chosen limit law and almost every component. A centered Gaussian variable has zero fourth cumulant, so every component in this full-measure set is non-Gaussian. □

Appendix A. Polymer geometry

Let L be odd, M a power of L, and ℓ a multiple of LM with $\ell \ge 13LM$. The M-cubes have centres in $M\mathbb{Z}^3/\ell\mathbb{Z}^3$. They partition the lattice sites; their closed continuum versions may meet along faces and edges. A polymer is a nonempty union of M-cubes connected through faces. Denote its number of cubes by $|X|_M$. A net in the continuum torus is a connected finite union of images of specified closed straight segments in $\mathbb{R}^3$, a point being allowed as a segment. Its length is the sum of the Euclidean lengths of those segments. Set

$$M d_M(X) = \inf\{\text{length}(N) : N \text{ is a net meeting every closed cube of } X\},$$

where the infimum also extends over the finite segment representations. The same definition is used for finite face-connected unions of M-cubes in $\mathbb{R}^3$, with nets in $\mathbb{R}^3$. Distances are sup distances.

A.1. **Size, diameter, and reblocking.** Put

$$c_X = 54, \qquad \kappa_0 = 54 \log 72, \qquad K_0 = e^{\kappa_0}/36, \qquad c_{\mathrm{rb}} = c_X K_0.$$

The following estimates are uniform in M and ℓ:

$$d_M(X) \le |X|_M - 1, \qquad |X|_M \le c_X(d_M(X) + 1), \tag{A.1}$$

$$d(p,q) \le M(d_M(X) + 2) \qquad (p, q \text{ in the closed cubes of } X), \tag{A.2}$$

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le K_0, \qquad \sum_{X \subset Y} e^{-\kappa_0 d_M(X)} \le K_0 |Y|_M. \tag{A.3}$$

Here Y is any union of M-cubes. If ℓ/L satisfies the same side conditions stated above, let $\bar{X}^L$ be the union of LM-cubes containing cubes of X. There is a unique polymer Y on the rescaled torus with $\bar{X}^L = LY$, and

$$L d_M(Y) \le d_M(X), \tag{A.4}$$

$$\sum_{X : \bar{X}^L = LY} e^{-\kappa' d_M(X)} \le c_{\mathrm{rb}} L^3 e^{-L(\kappa' - \kappa_0 - 1) d_M(Y)} \quad (\kappa' \ge \kappa_0). \tag{A.5}$$

To prove the first inequality of (A.1), order the cubes along a spanning tree of their adjacency graph. A net meeting one cube can be extended to an adjacent cube by a segment of length at most M perpendicular to their common face. Starting with one point and repeating this construction gives a net of length at most $M(|X|_M - 1)$.

For the reverse inequality, subdivide the segments of any net N at their intersections and discard overlapping pieces. Its total length does not increase. Doubling the edges of the resulting connected graph gives a closed traversal of length at most $2\,\text{length}(N)$. Cut it into at most $2\,\text{length}(N)/M + 1$ arcs of length at most M. Each arc lifts to a curve in $\mathbb{R}^3$ whose range in each coordinate has length at most M. Such an interval meets at most three closed cube intervals, so an arc meets at most 27 closed M-cubes. Every cube of X meets one of these arcs. Taking the infimum over N gives

$$|X|_M \le 27(2d_M(X) + 1) \le c_X(d_M(X) + 1).$$

For (A.2), take a net of length less than $Md_M(X)+\delta$ and points p', q' of the net in the respective cubes of p, q. The path in the net from p' to q' has length at most that of the net. Hence $d(p,q) \le M + (Md_M(X)+\delta) + M$; let $\delta \downarrow 0$. The same argument holds in $\mathbb{R}^3$.

A polymer with n cubes containing a fixed cube determines a closed walk of length $2(n-1)$ starting there: double a spanning tree of its adjacency graph and traverse it. Choose one such walk for each polymer. This assignment is injective, since the visited cubes recover the polymer, and there are at most $6^{2(n-1)} = 36^{n-1}$ possible walks. By the size estimate, $d_M(X) \ge n/c_X - 1$. Summing over n gives

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le e^{\kappa_0} \sum_{n \ge 1} 36^{n-1} 72^{-n} = K_0.$$

Sum this bound over the cubes of Y to obtain the second inequality of (A.3).

Every M-cube belongs to one LM-cube, since L is odd, and adjacent M-cubes belong to the same or to adjacent LM-cubes. Thus $\bar{X}^L$ is connected through faces. Projection by $c \mapsto c/L$ gives its unique rescaled polymer Y. A net for X meets every LM-cube of $\bar{X}^L$; its rescaling has length divided by L. This proves (A.4). For (A.5), split the decay into rates κ_0 and $\kappa' - \kappa_0$ and use (A.4). All terms satisfy $X \subset LY$, which has $L^3|Y|_M$ fine cubes, so

$$\sum_{X : \bar{X}^L = LY} e^{-\kappa_0 d_M(X)} \le K_0 L^3 |Y|_M \le c_X K_0 L^3 e^{d_M(Y)}.$$

Combining the two factors proves the stated bound.

A.2. **Nonwrapping lifts.** Let $\mathrm{pr} : \mathbb{R}^3 \to \mathbb{R}^3/\ell\mathbb{Z}^3$ be the projection.

Proposition A.1. *If X is a polymer and*

$$2M(d_M(X)+2) < \ell, \tag{A.6}$$

then $X = \mathrm{pr}(\widetilde{X})$ for a finite face-connected union of M-cubes in $\mathbb{R}^3$. The projection is injective on $\widetilde{X}$, preserves adjacency of its cubes in both directions, and $d_M(\widetilde{X}) = d_M(X)$. The lift is unique up to a translation in $\ell\mathbb{Z}^3$.

Proof. Put $D = M(d_M(X)+2)$, fix p_0 in a closed cube of X, and fix its lift $\widetilde{p}_0$. The projection is injective on the closed sup ball $\widetilde{B} = \{|p - \widetilde{p}_0|_\infty \le D\}$, since two points there differ in every coordinate by at most $2D < \ell$. It maps this ball onto the torus ball $B = \{p : d(p, p_0) \le D\}$. Its inverse $\varphi : B \to \widetilde{B}$ is continuous, and is locally isometric. The diameter estimate places every closed cube of X in B. A segment or a closed M-cube lying in B lifts under φ to a segment of the same length or a closed M-cube: the translates $\widetilde{B} + \ell n$ are pairwise disjoint, and a connected lift lies in one of them. The same inverse maps shared faces to shared faces. Thus $\widetilde{X} = \varphi(X)$ has the asserted injectivity, adjacency, and connectedness.

Any other face-connected injective lift can be translated so that it contains the chosen lift of the cube of p_0. There are no two different neighbours of one lifted cube with the same projection: their centres differ by at most $2M < \ell$. Propagate this observation along a chain of adjacent cubes. Every cube of the other lift equals the corresponding cube of $\widetilde{X}$, proving uniqueness.

Projecting a net for $\widetilde{X}$ gives a net for X with no greater length, so $d_M(X) \le d_M(\widetilde{X})$. Conversely, let N be a net for X of length at most $Md_M(X) + M$, and choose its point q_0 in the cube of p_0. Every point of N is joined to q_0 by a curve of length at most that of N, and $d(q_0, p_0) \le M$. Therefore $N \subset B$. Its lift $\varphi(N)$ has the same length and meets every cube of $\widetilde{X}$. Taking an infimizing sequence of these nets proves the reverse inequality. □

For the determinant estimate, suppose in addition that $M(d_M(X)+4) < \ell$ and $M \ge 2$. The closed cubes of $\widetilde{X}$ have coordinate ranges of length at most $M(d_M(X)+2)$ by (A.2). Enlarge these intervals by one mesh $\eta_k \le 1$ at each end and intersect with $\eta_k\mathbb{Z}^3$. The resulting box W contains

every bond meeting the lifted polymer and has each side of length at most $M(d_M(X)+3)$. This length, increased by one mesh, is less than ℓ. Thus the projection is injective on W, and every torus bond meeting X has exactly one lift in W.

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